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(12) **United States Patent**
Crowe, Jr. et al.

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(45) **Date of Patent:** **Sep. 9, 2025**

(54) **HUMAN MONOCLONAL ANTIBODIES TO ENTEROVIRUS D68**

(71) Applicant: **VANDERBILT UNIVERSITY**,
Nashville, TN (US)

(72) Inventors: **James E. Crowe, Jr.**, Nashville, TN (US); **Matthew R. Vogt**, Nashville, TN (US)

(73) Assignee: **VANDERBILT UNIVERSITY**,
Nashville, TN (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 862 days.

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Related U.S. Application Data

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(51) **Int. Cl.**

C07K 16/10 (2006.01)

A61K 39/00 (2006.01)

A61K 39/42 (2006.01)

A61P 31/14 (2006.01)

G01N 33/569 (2006.01)

(52) **U.S. Cl.**

CPC **C07K 16/1009** (2013.01); **A61K 39/42** (2013.01); **A61P 31/14** (2018.01); **C07K 16/10** (2013.01); **G01N 33/56983** (2013.01); **A61K 2039/505** (2013.01); **A61K 2039/55** (2013.01); **C07K 2317/21** (2013.01); **C07K 2317/34** (2013.01); **G01N 2333/085** (2013.01); **G01N 2469/10** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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Primary Examiner — Thomas J. Visone

Assistant Examiner — Katherine A. Willard

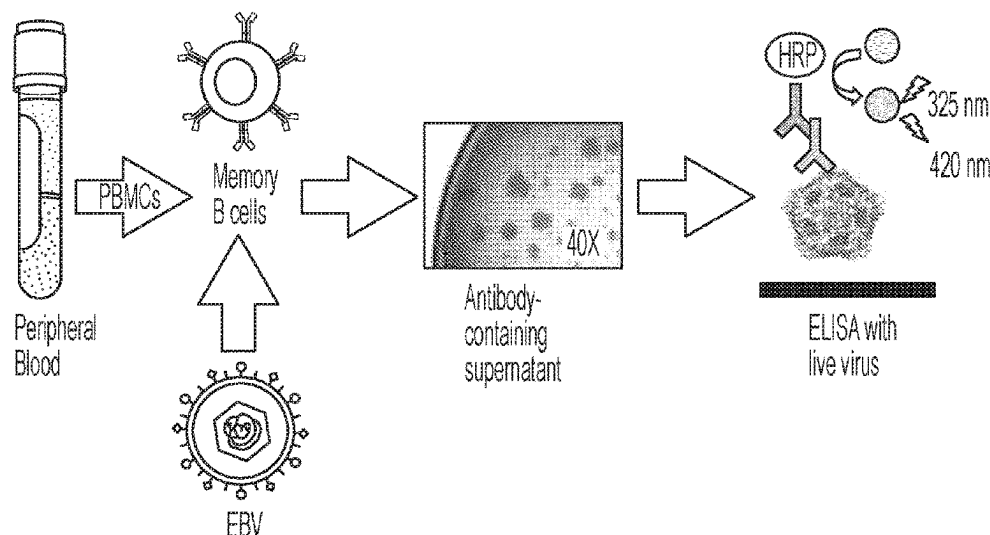
(74) *Attorney, Agent, or Firm* — pH IP Law

(57) **ABSTRACT**

The present disclosure is directed to human and engineered antibodies and fragments thereof binding to enterovirus D68 (EV-D68), engineered cells producing the same, and of using the antibodies and fragments thereof in methods for both diagnosing and treating EV-D68 infections. Also disclosed are vaccine formulations comprising one of more of the antibody and fragments thereof, as well as vectors encoding the same.

26 Claims, 91 Drawing Sheets

Specification includes a Sequence Listing.



(56)

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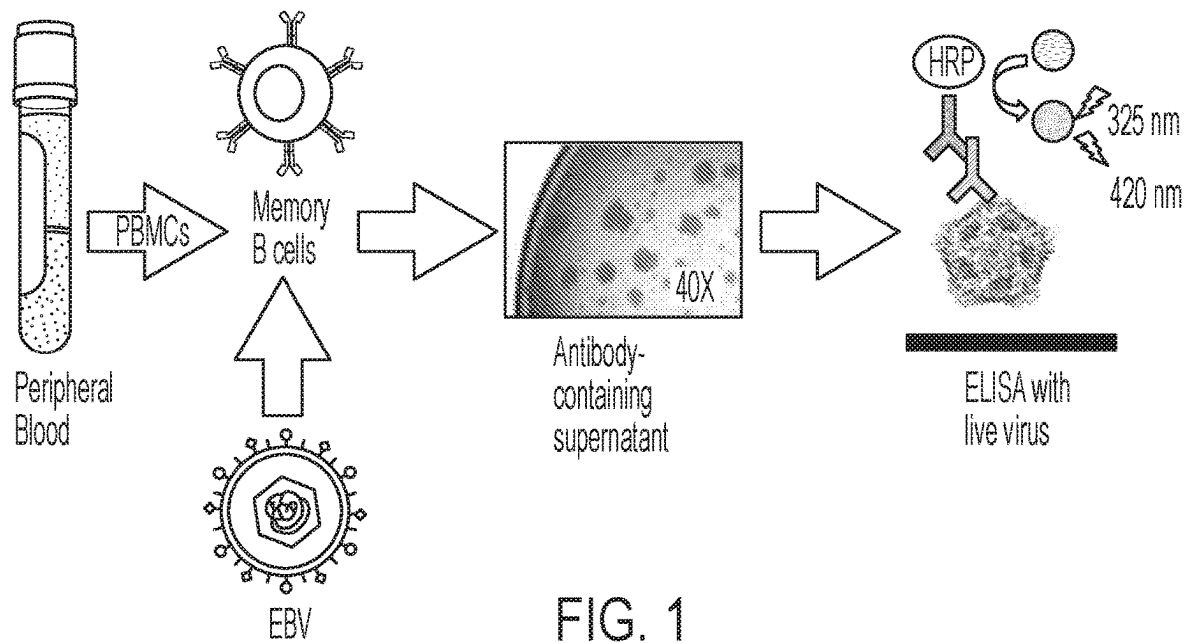


FIG. 1

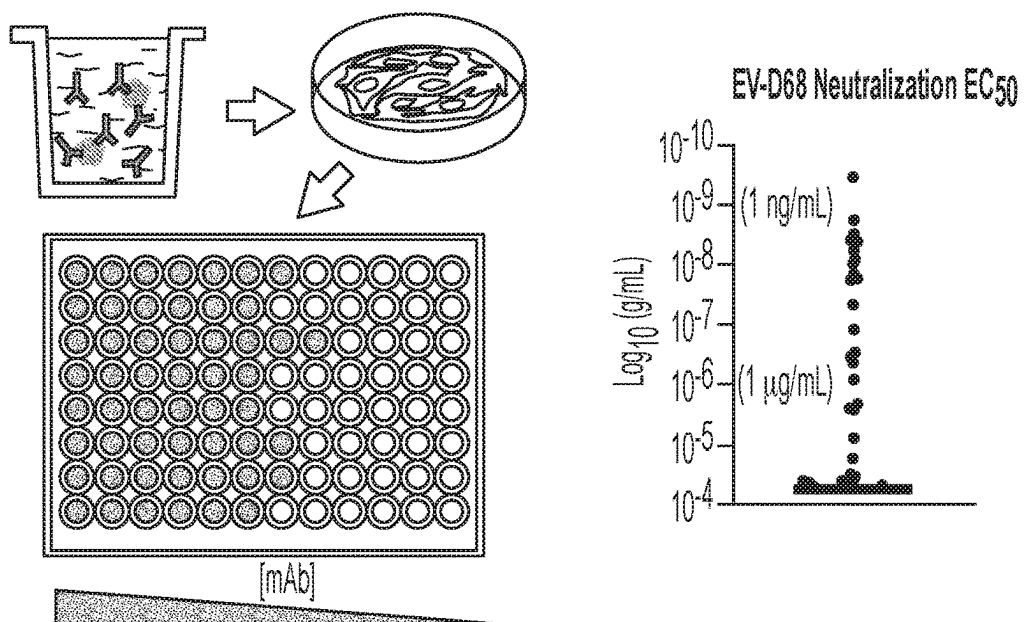


FIG. 2

Cross-reactive binding among clades is variable

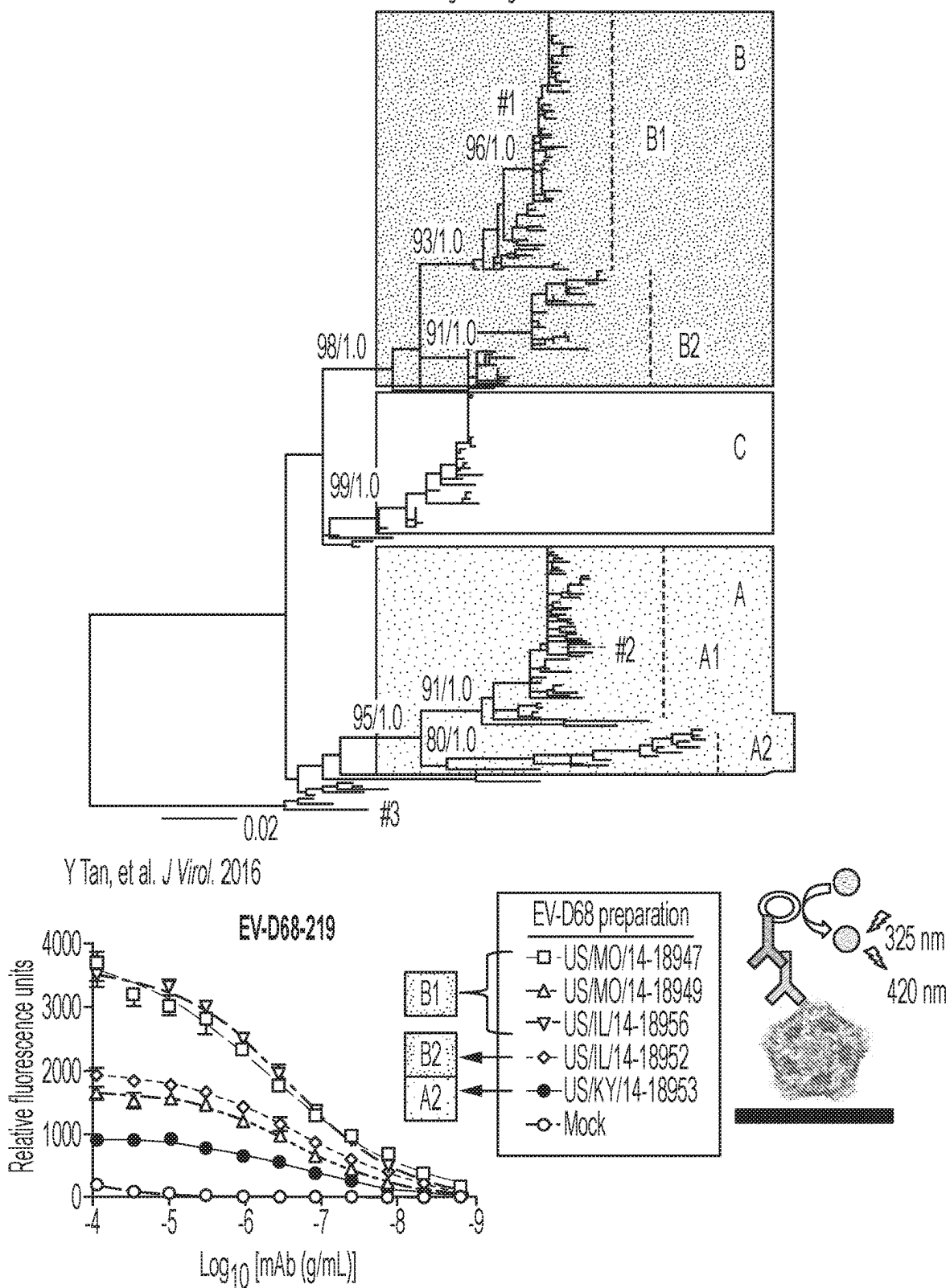


FIG. 3

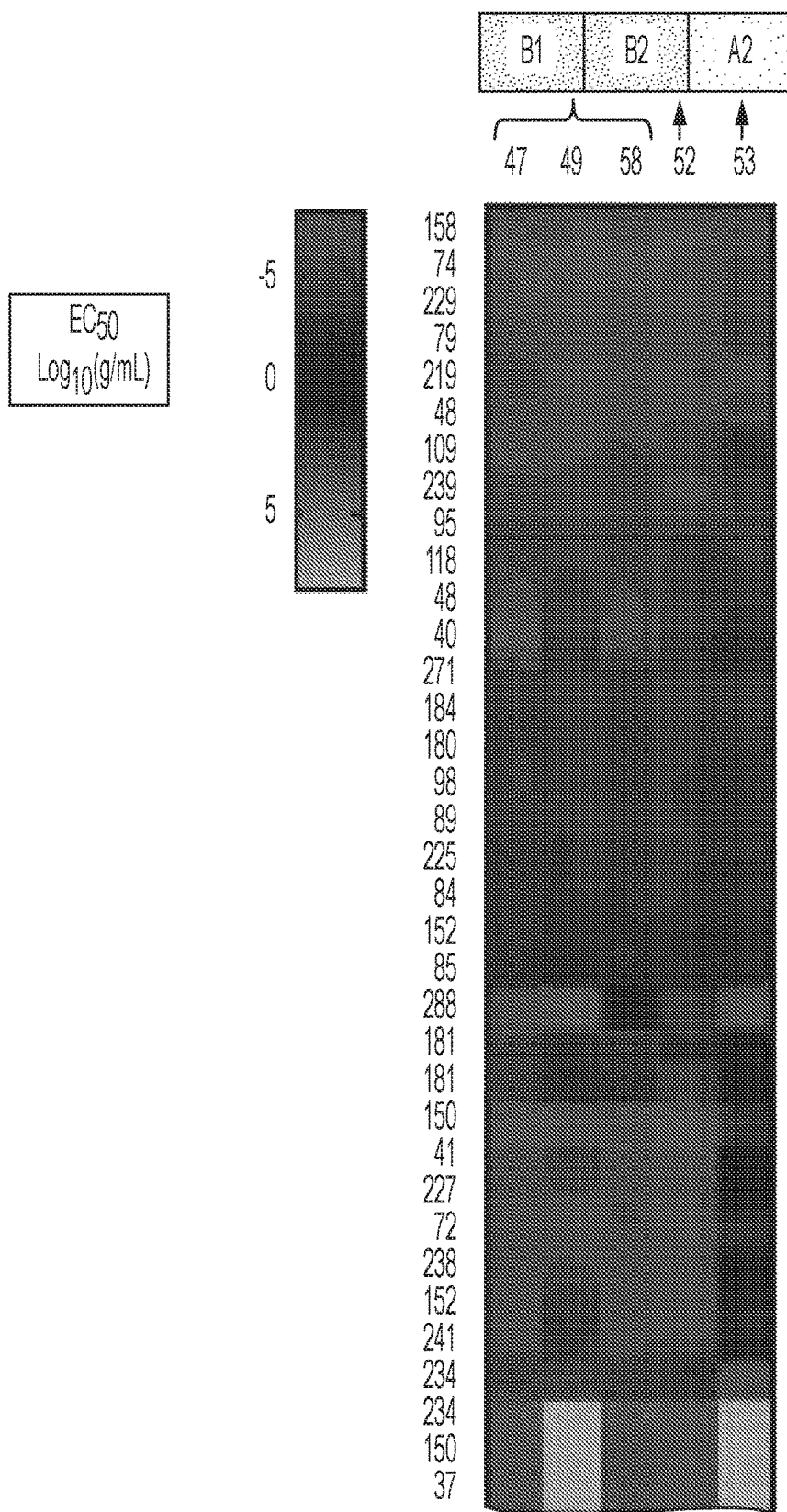


FIG. 3
CONTINUED

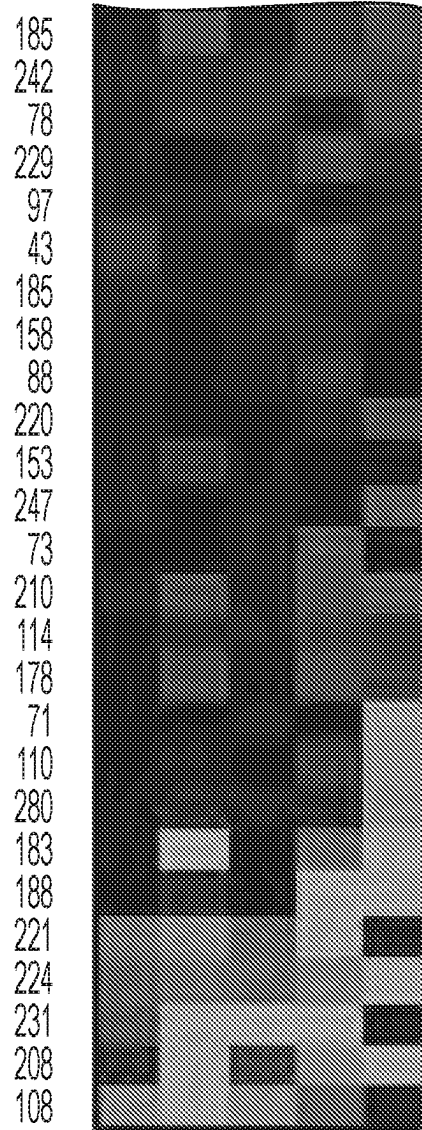


FIG. 3
CONTINUED

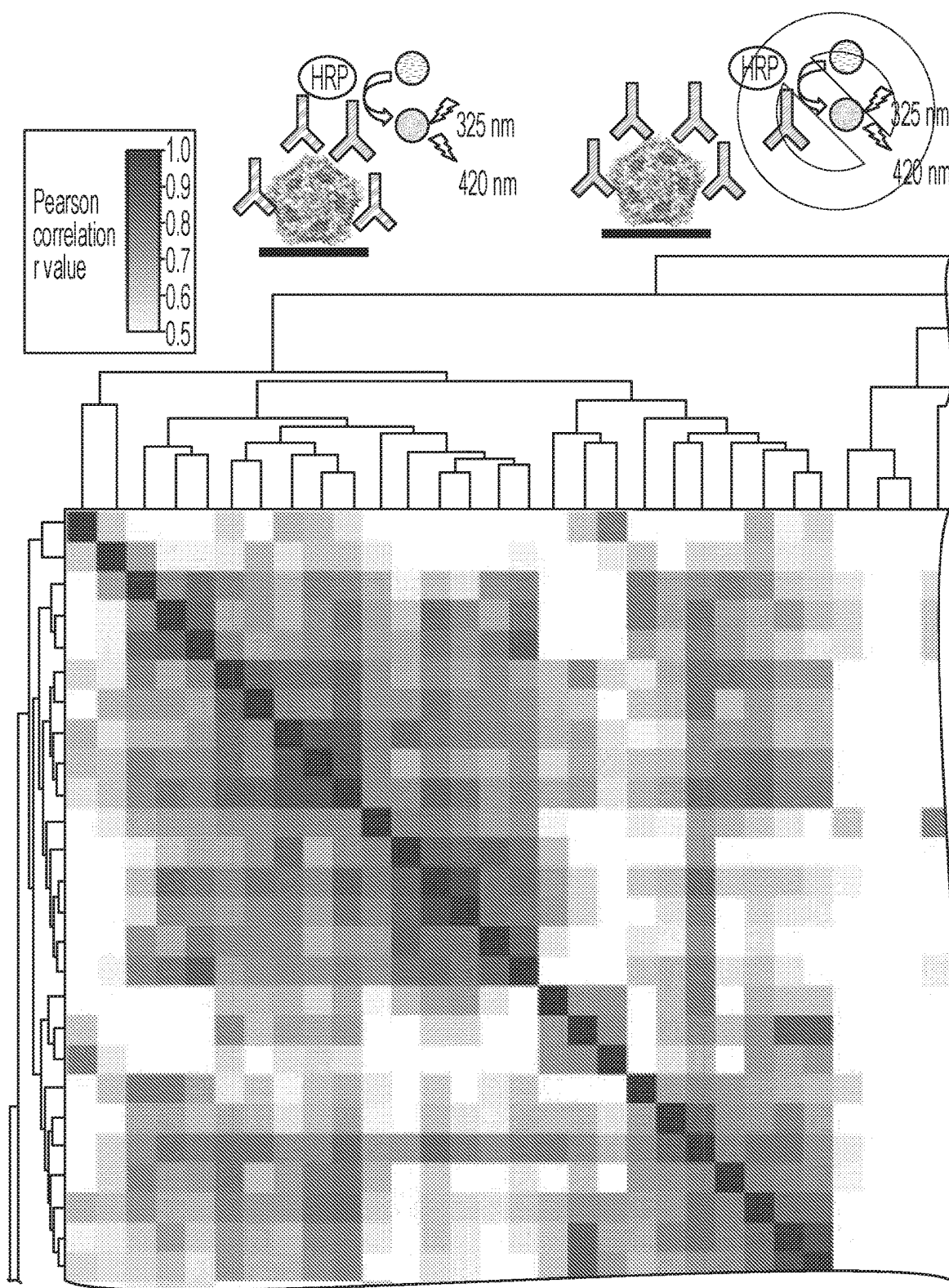


FIG. 4

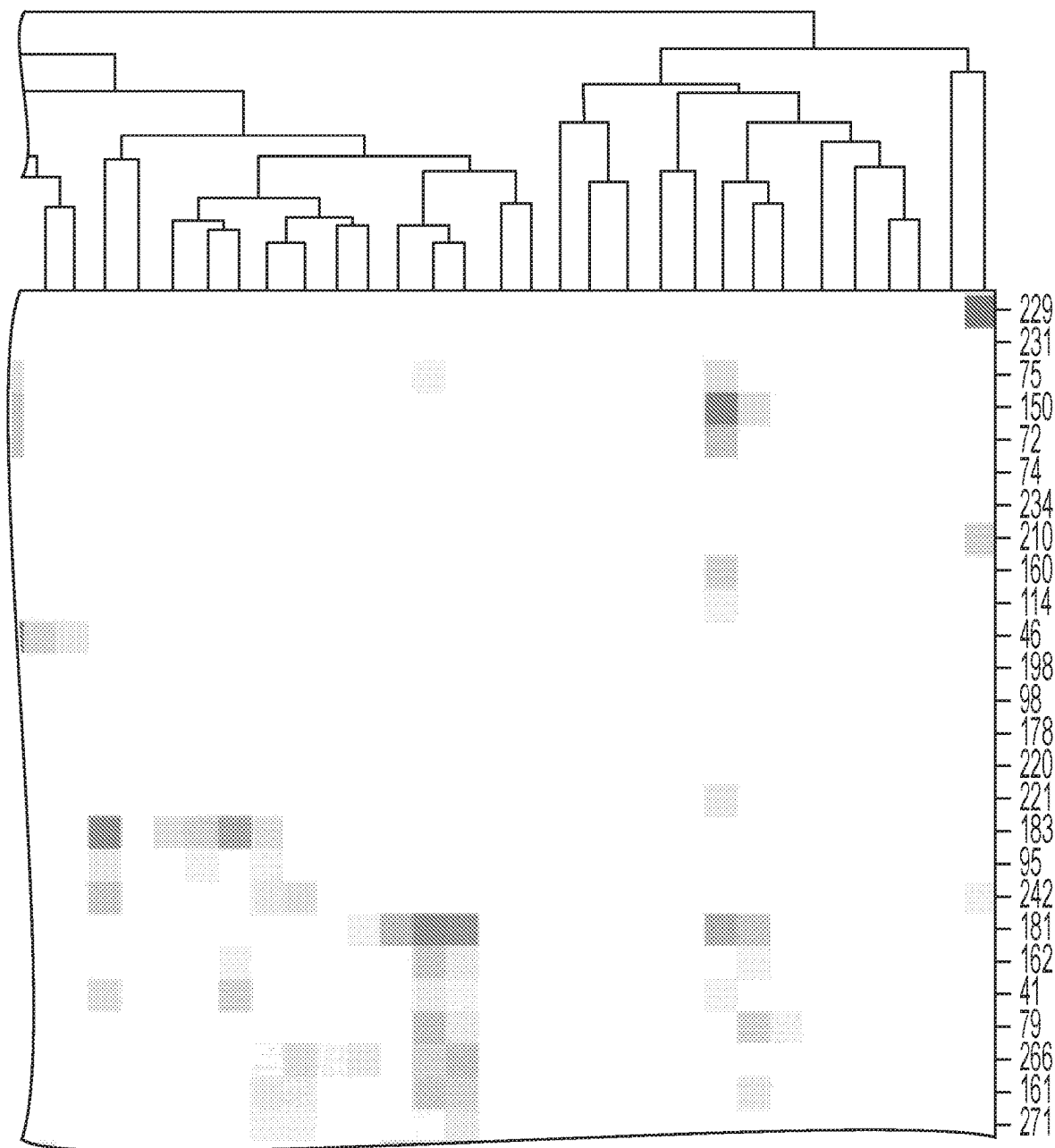


FIG. 4
CONTINUED

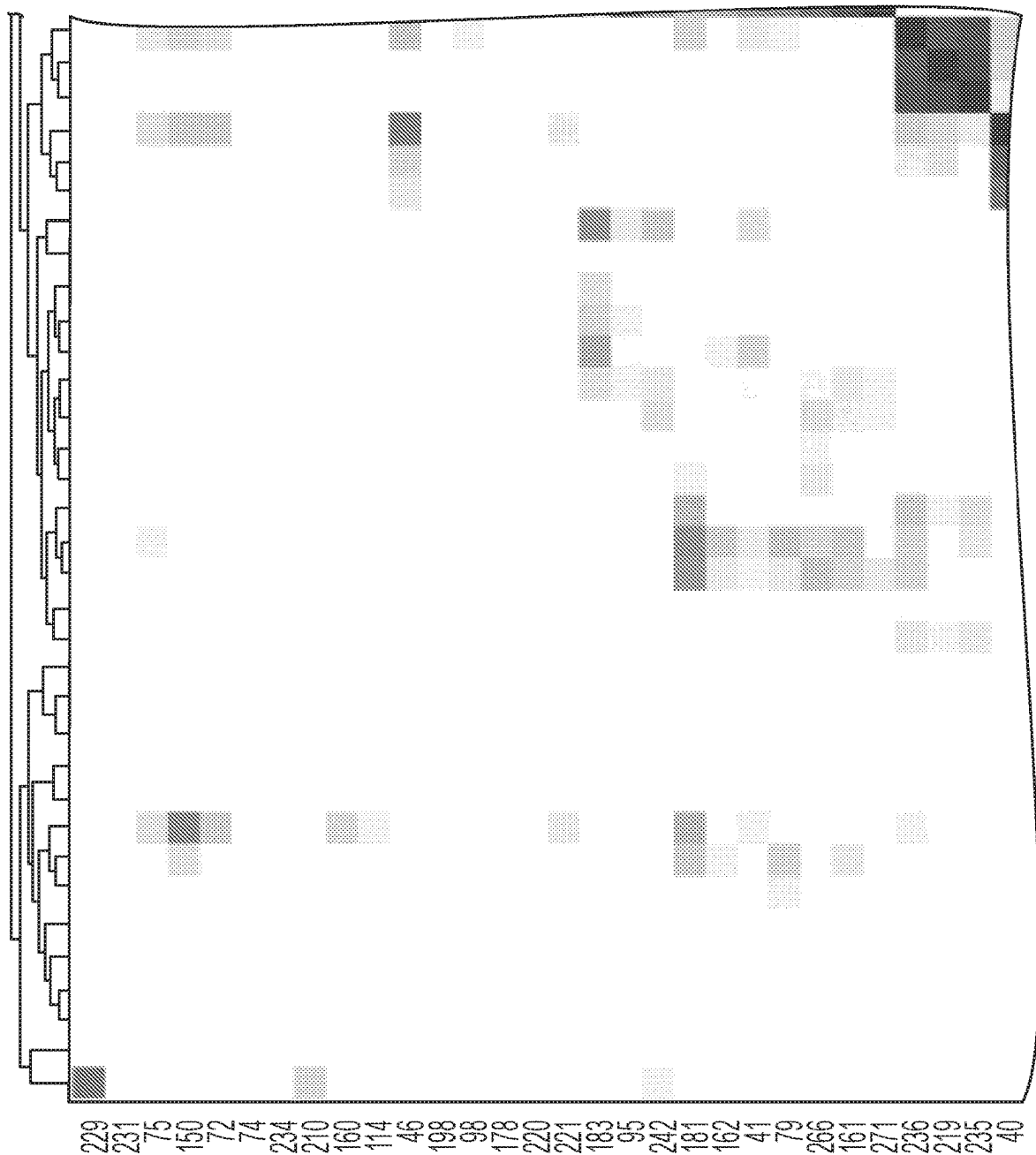


FIG. 4
CONTINUED

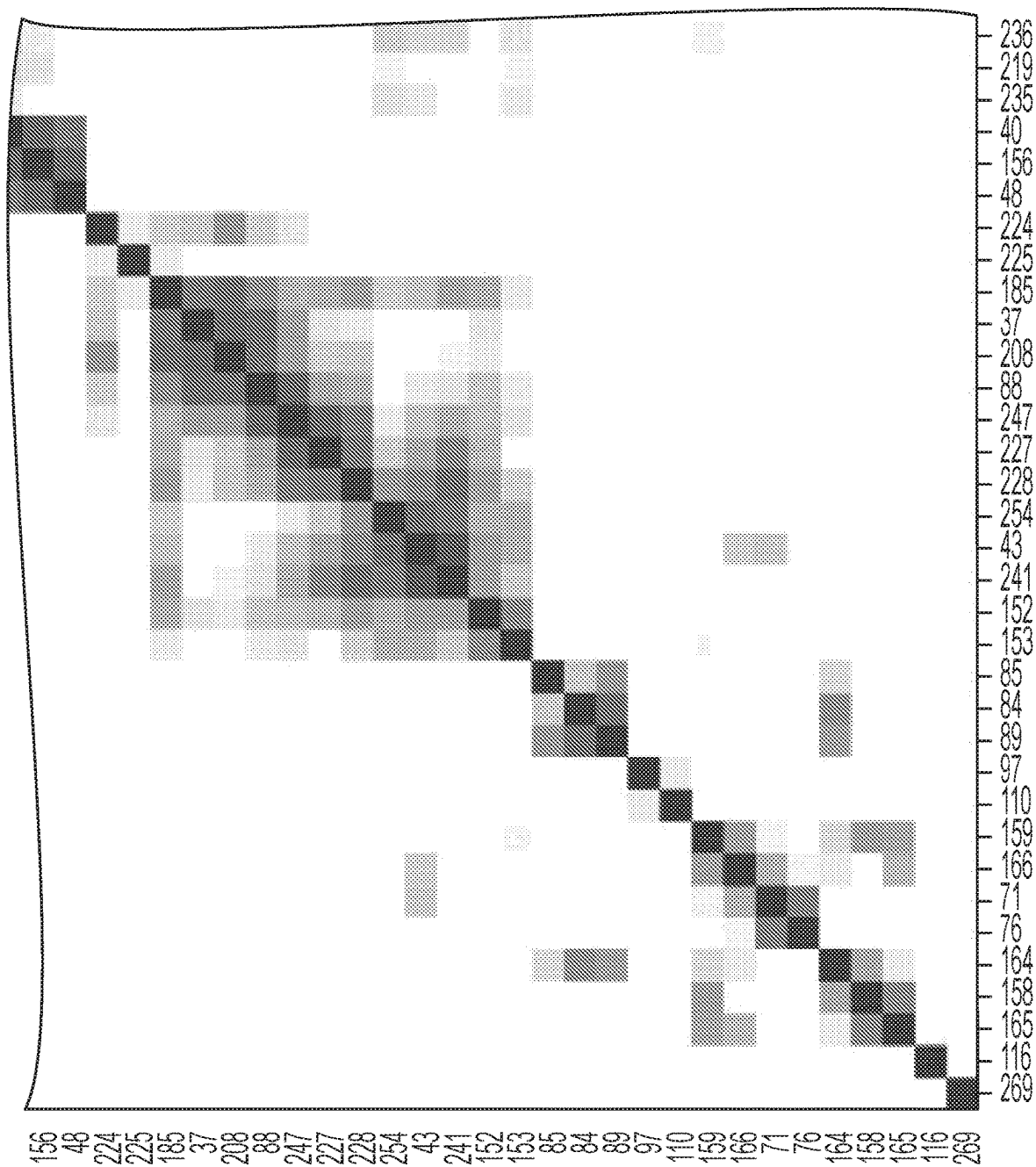


FIG. 4
CONTINUED

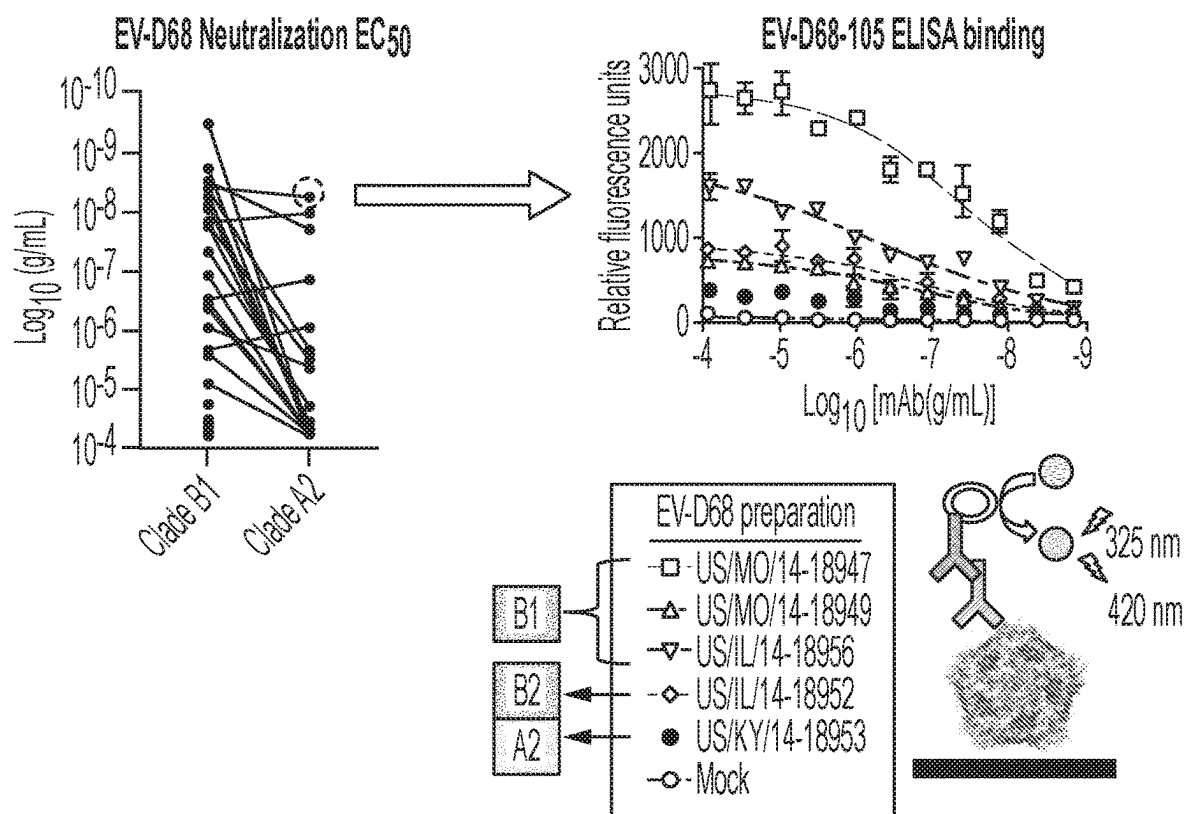


FIG. 5

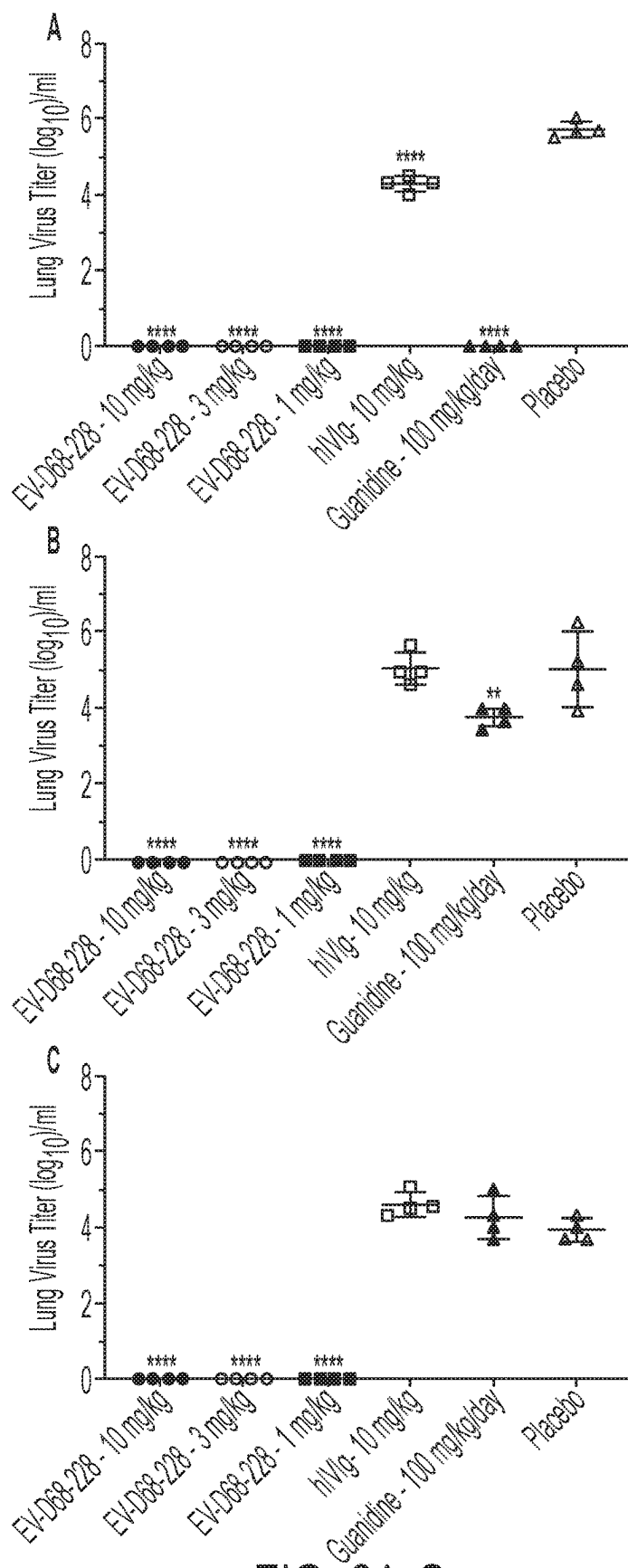


FIG. 6A-C

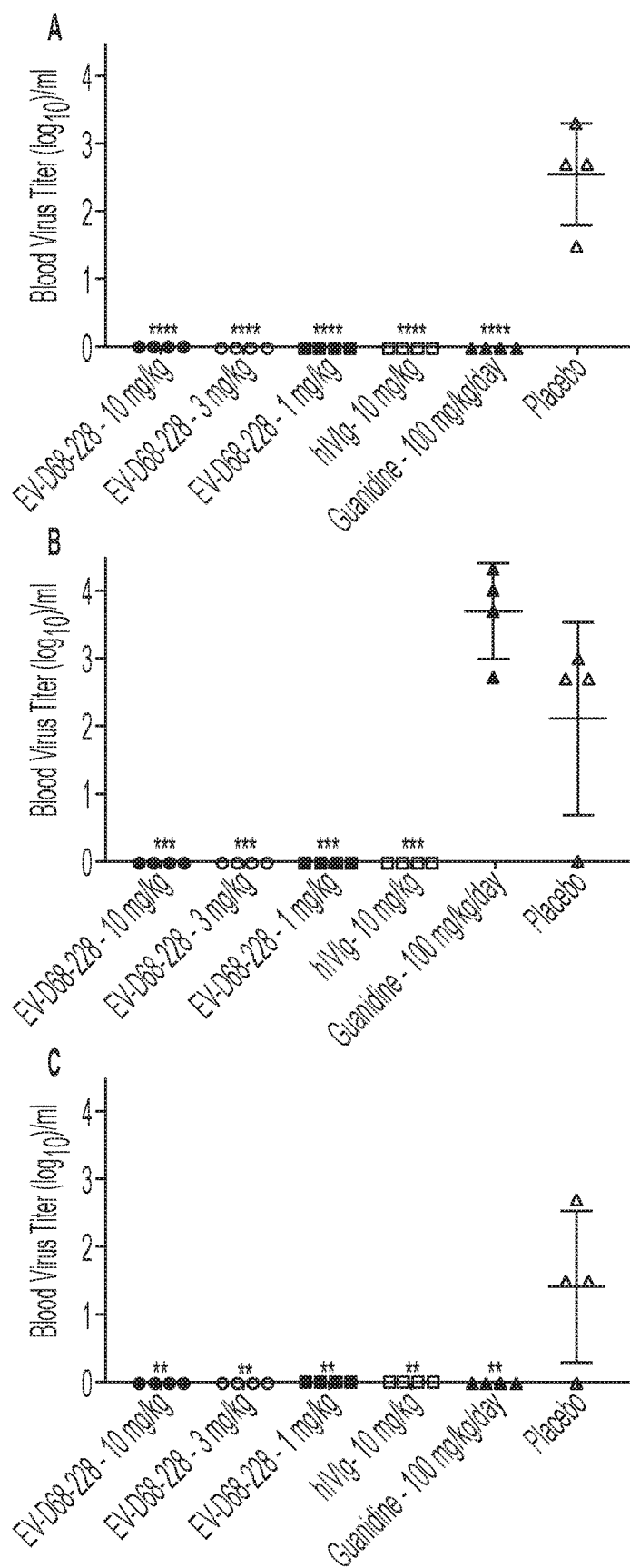


FIG. 7A-C

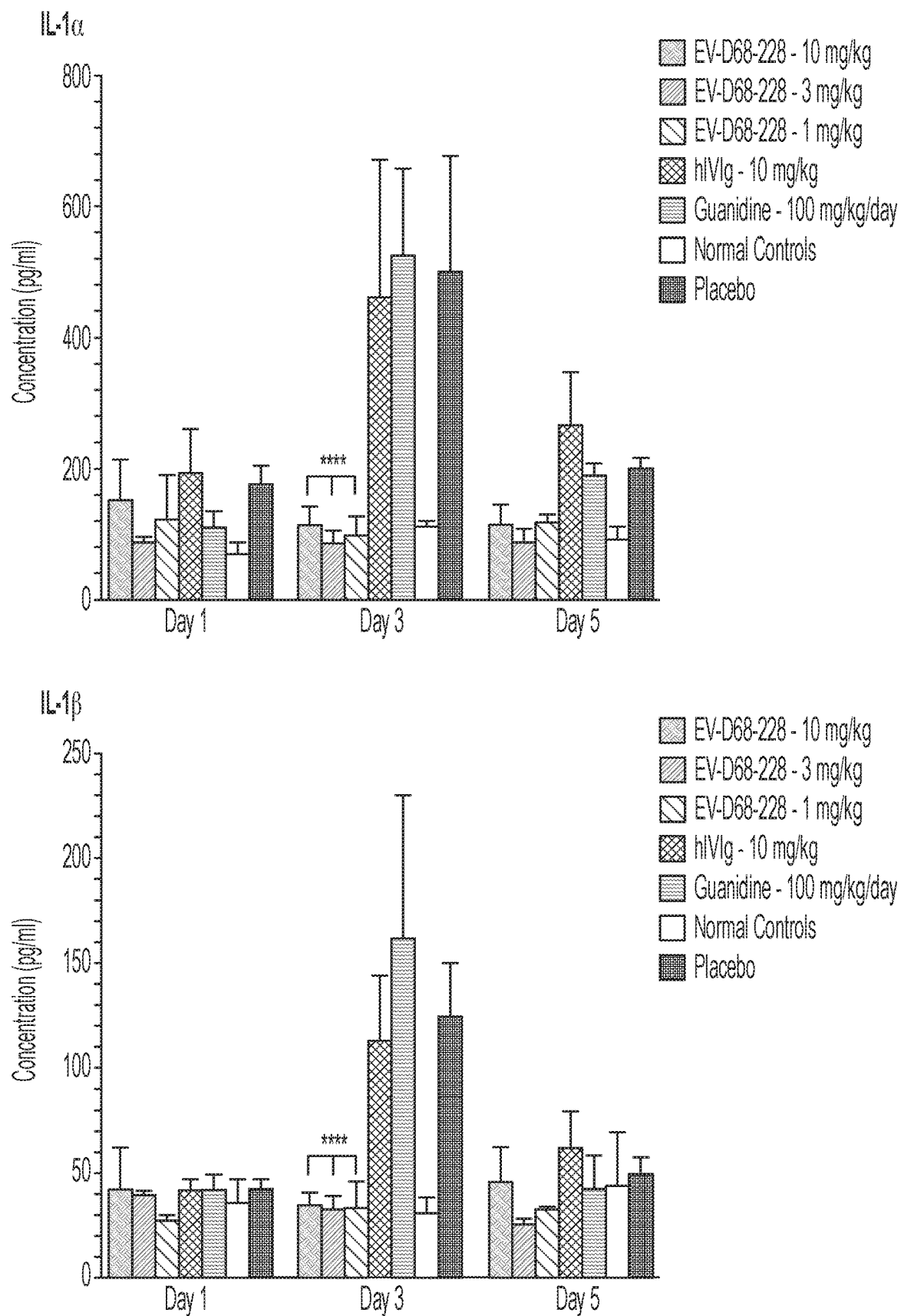


FIG. 8

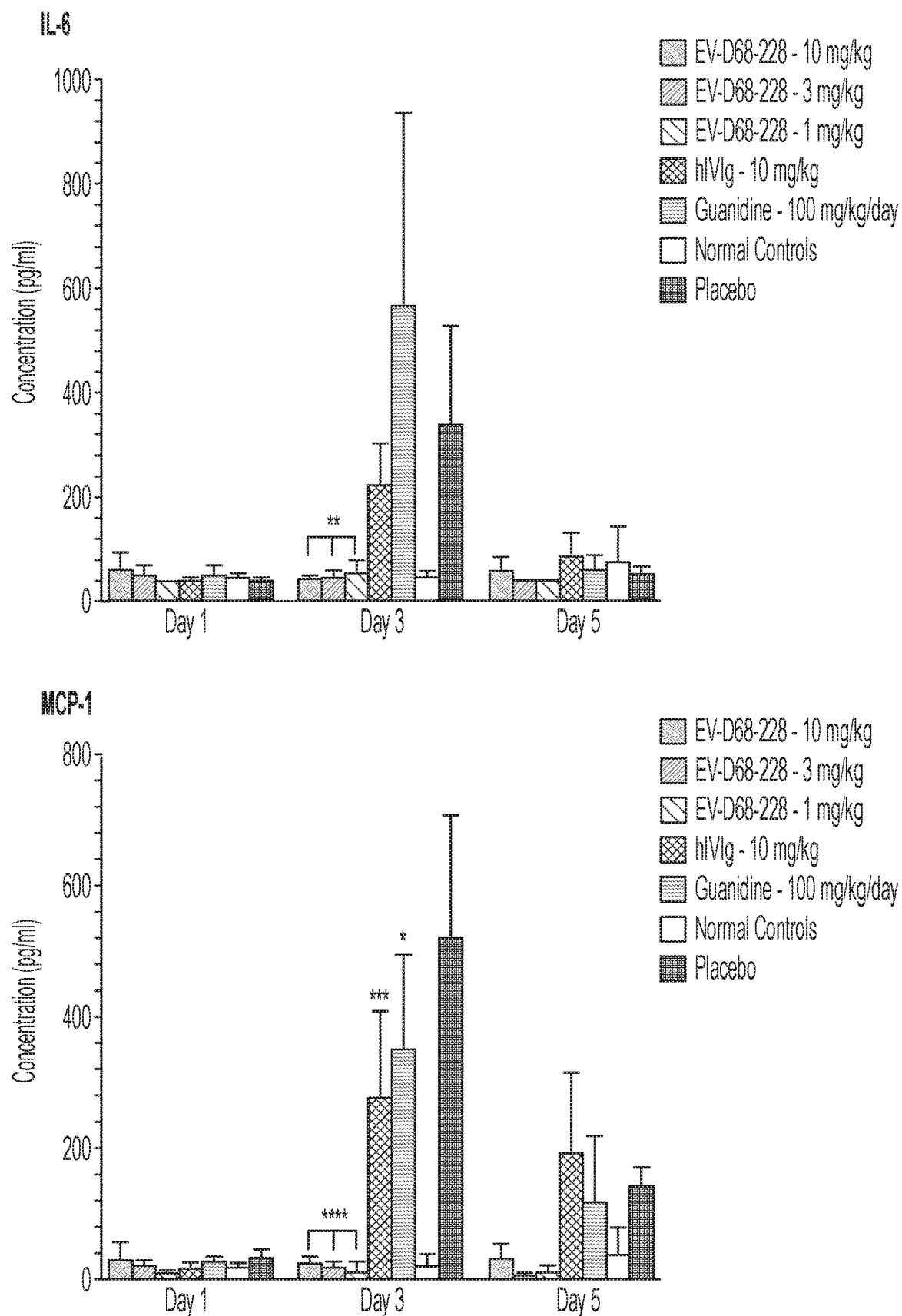


FIG. 9

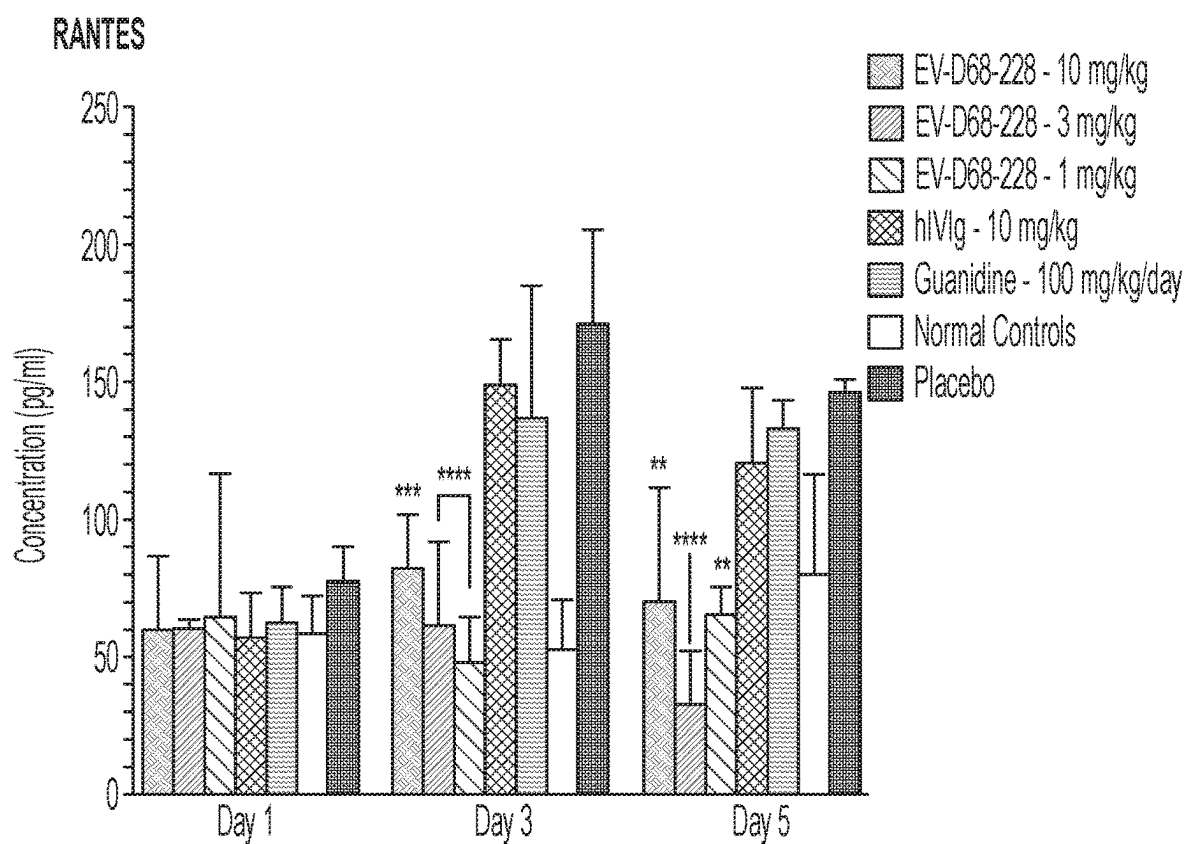


FIG. 10

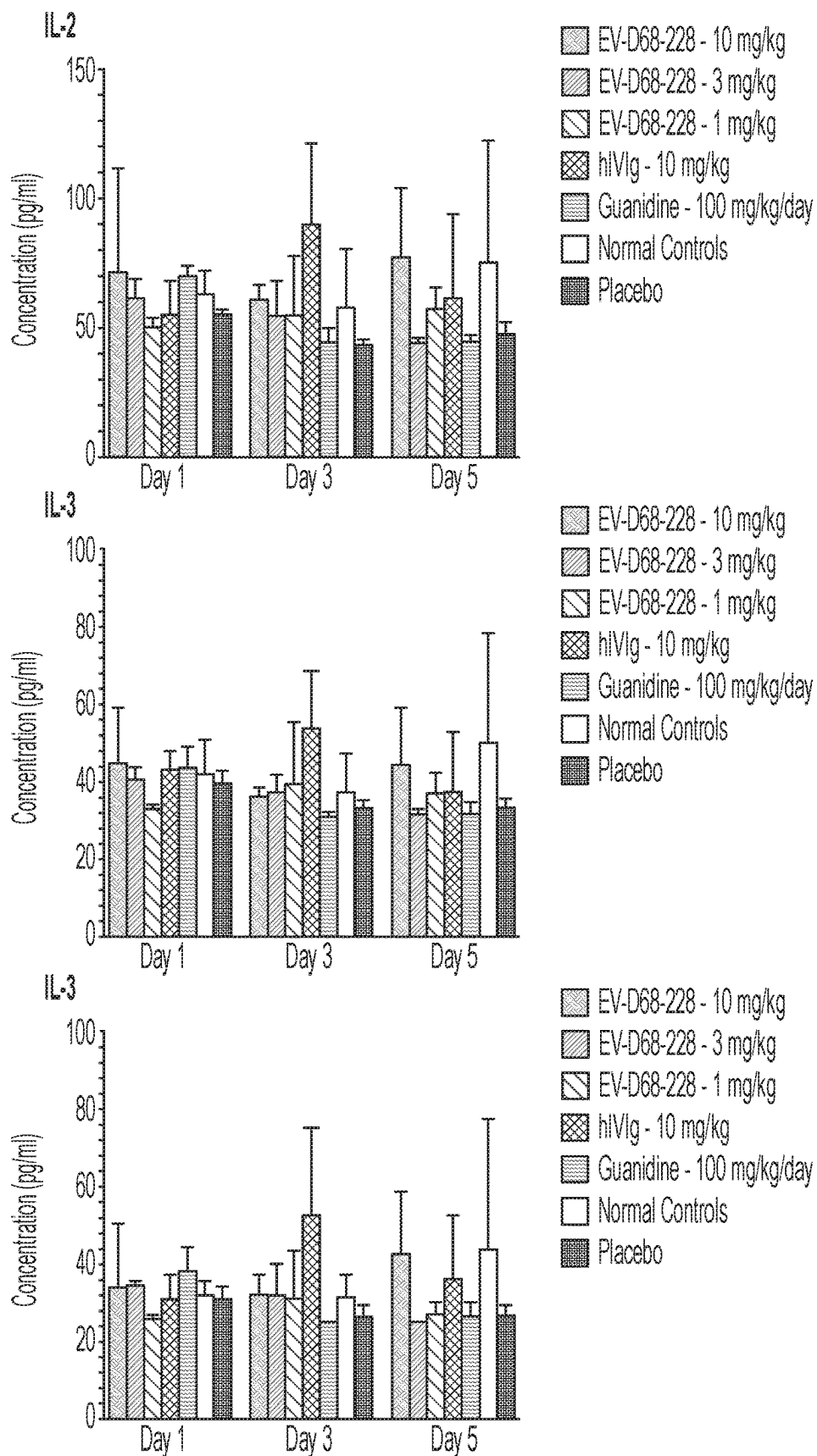


FIG. 11

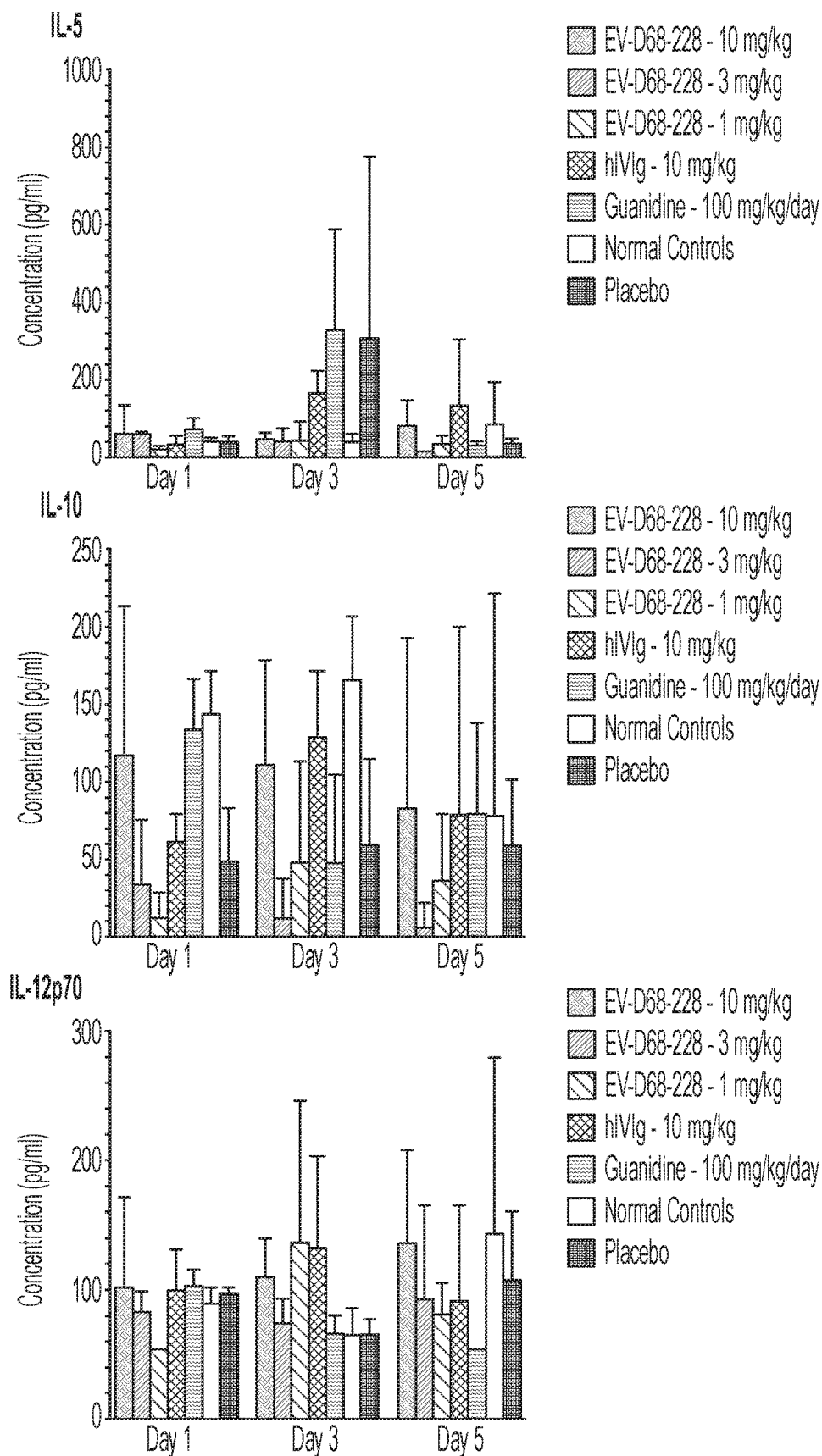


FIG. 12

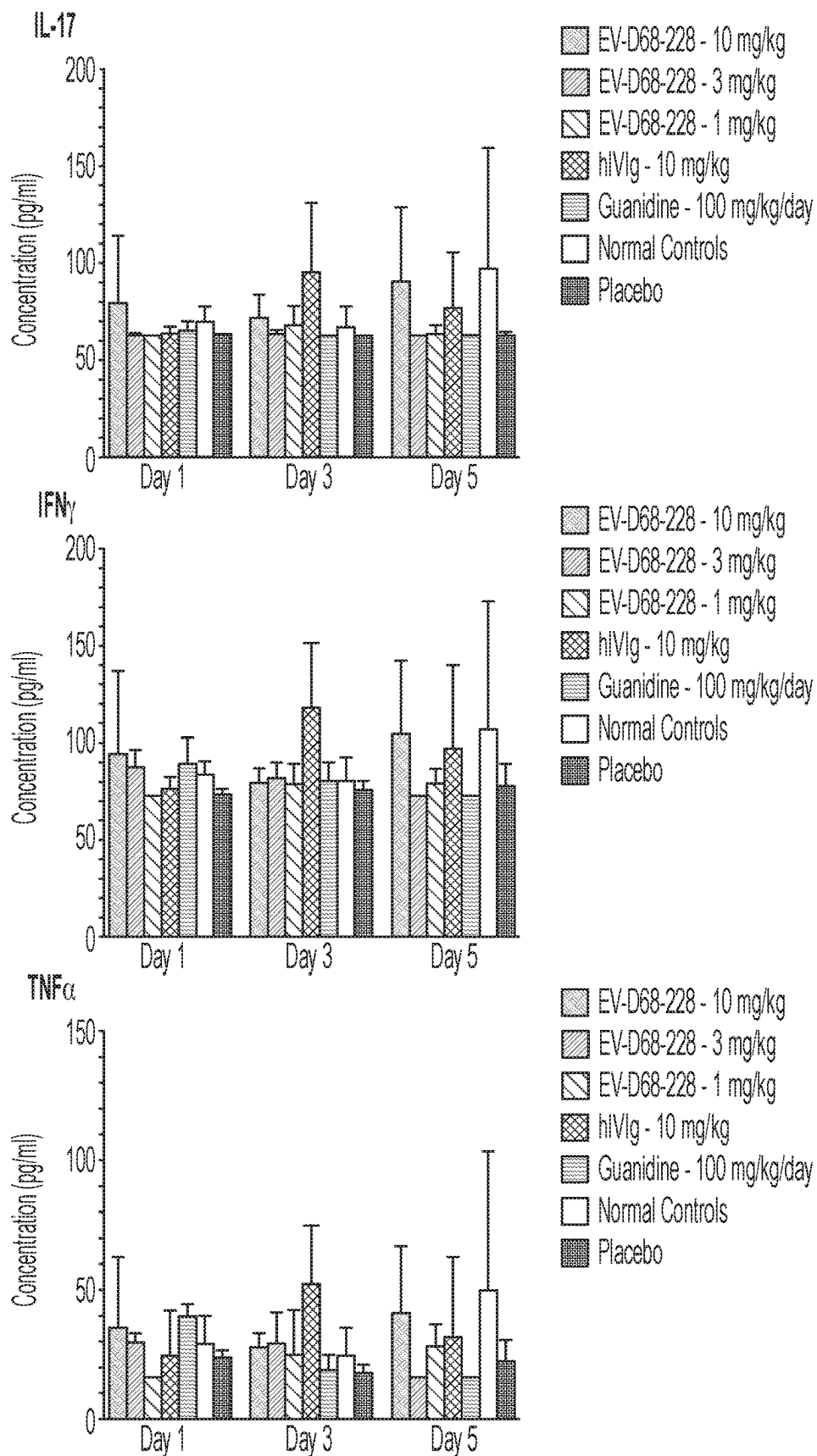


FIG. 13

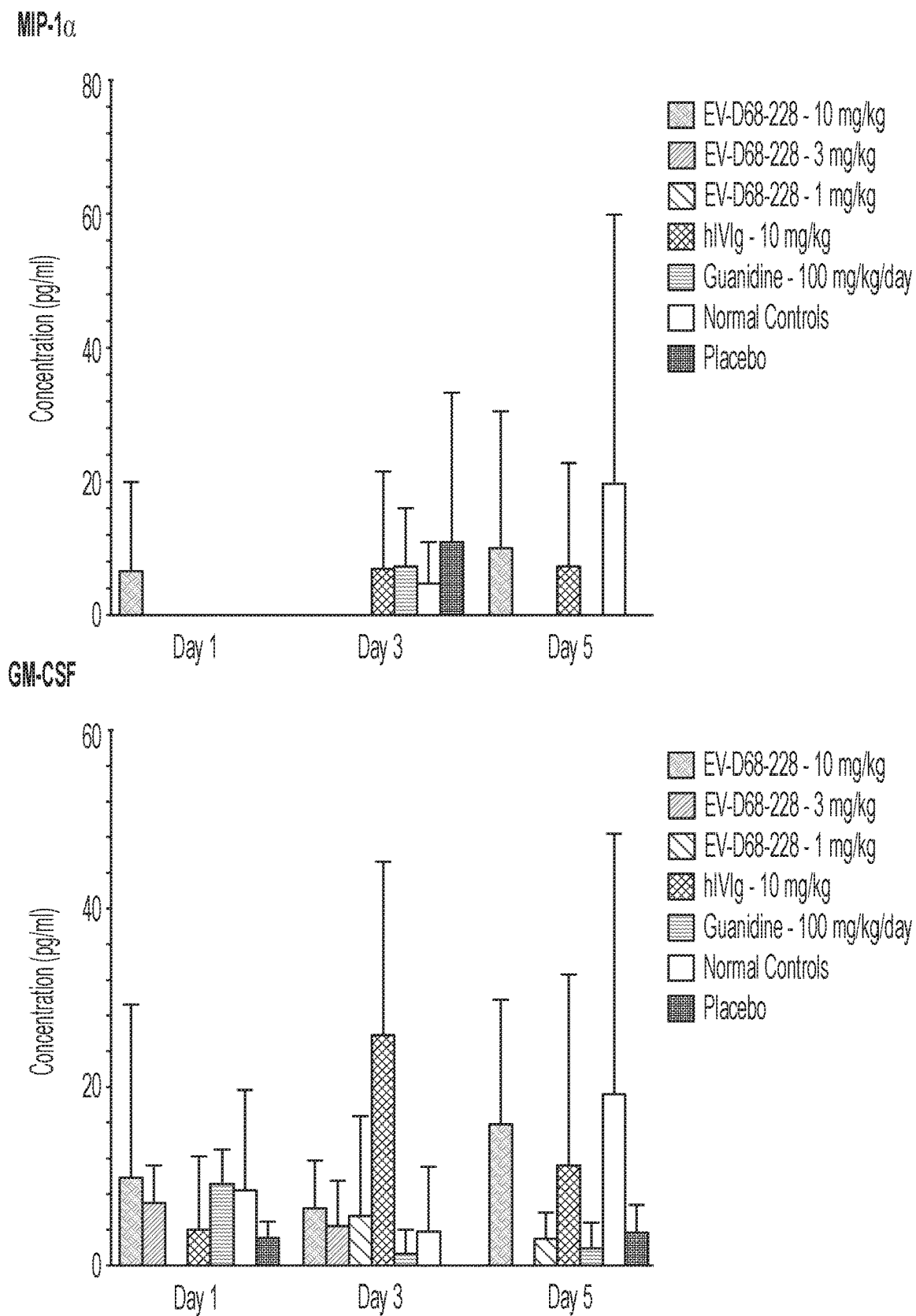


FIG. 14

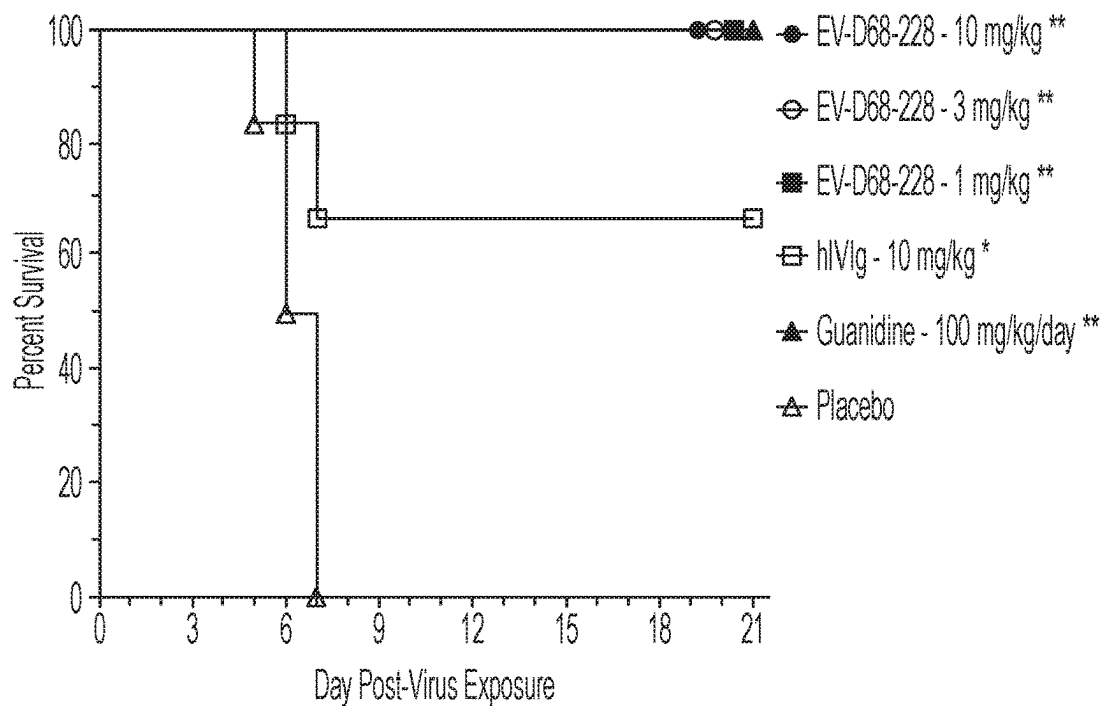


FIG. 15

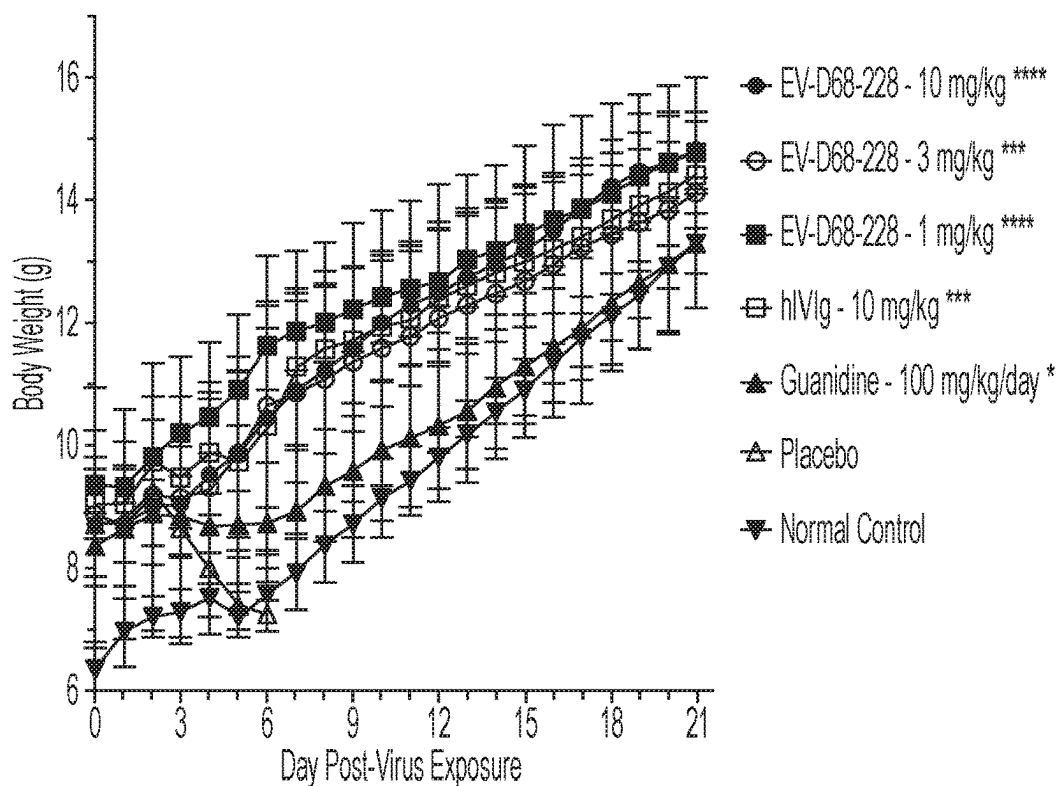


FIG. 16

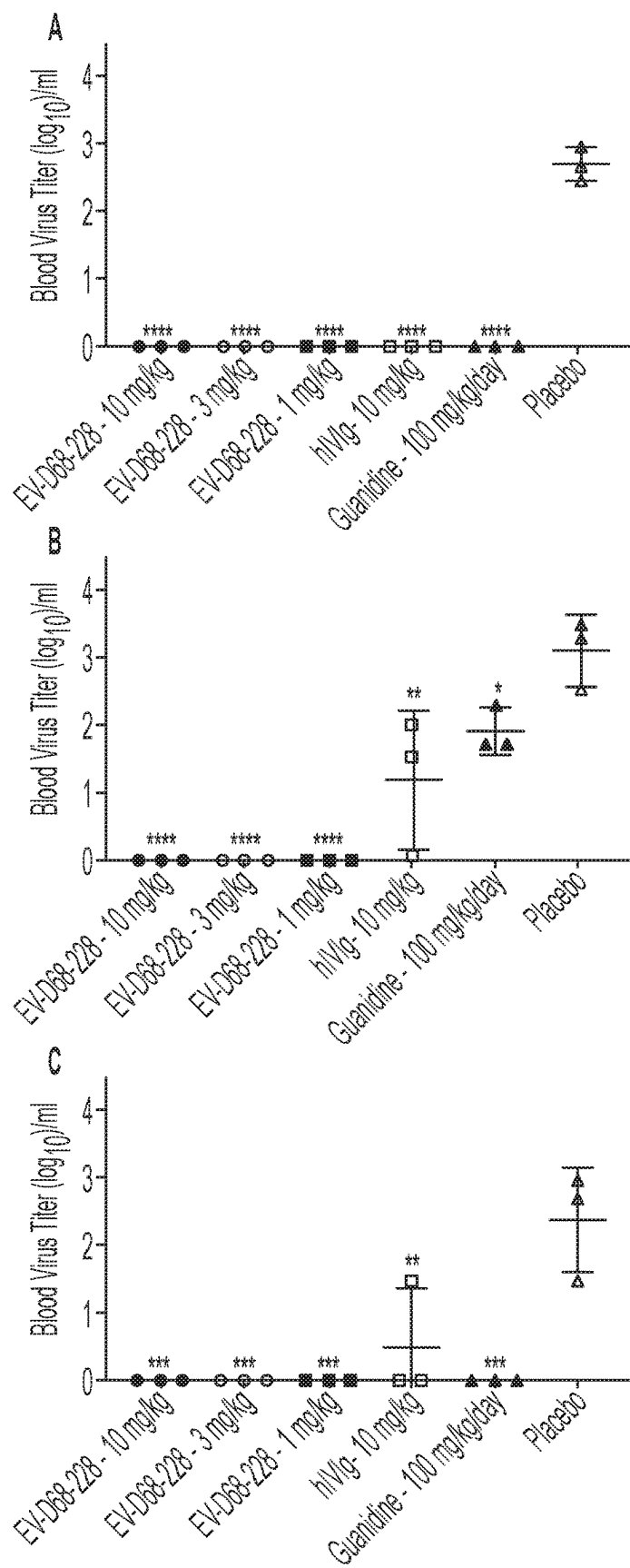


FIG. 17A-C

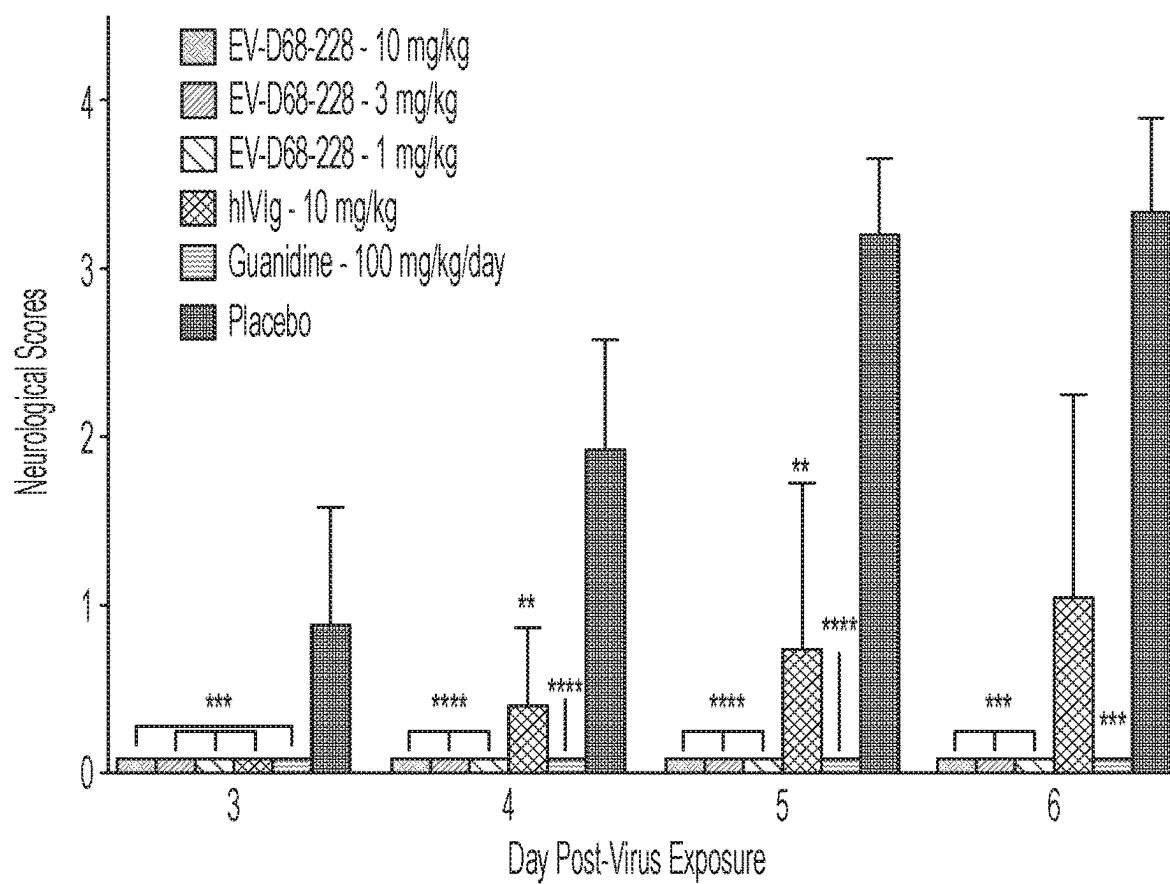


FIG. 18

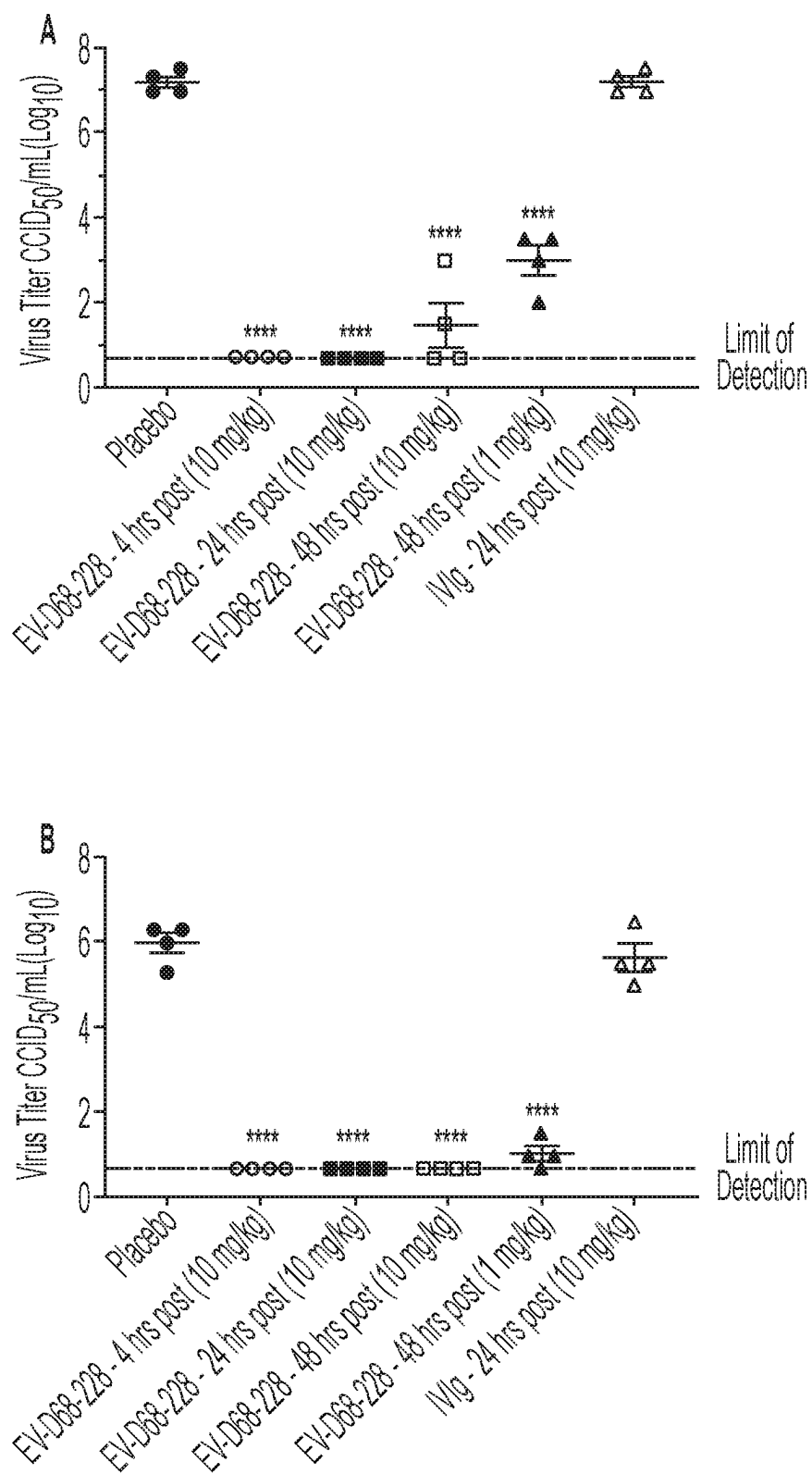


FIG. 19A-B

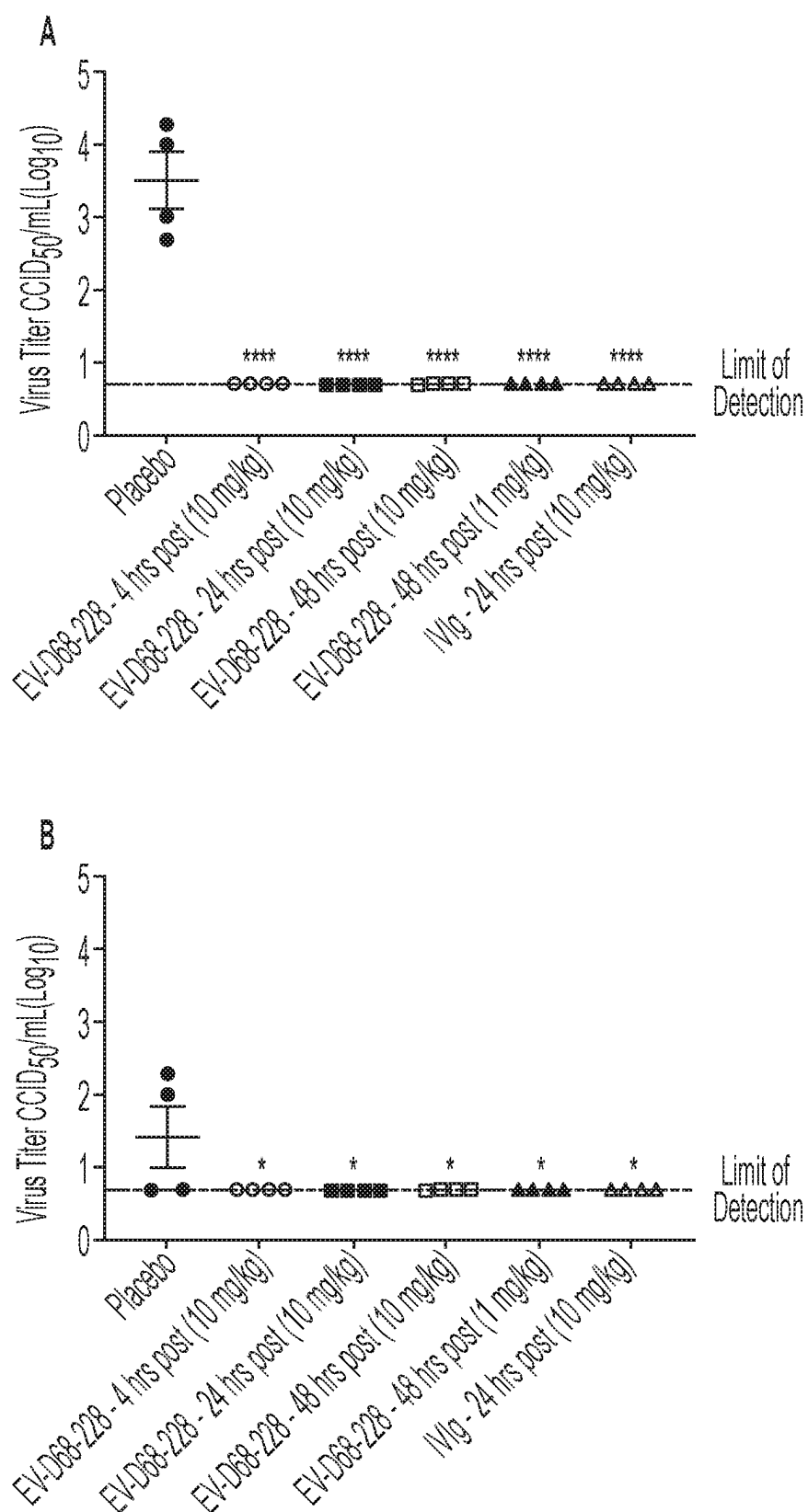


FIG. 20A-B

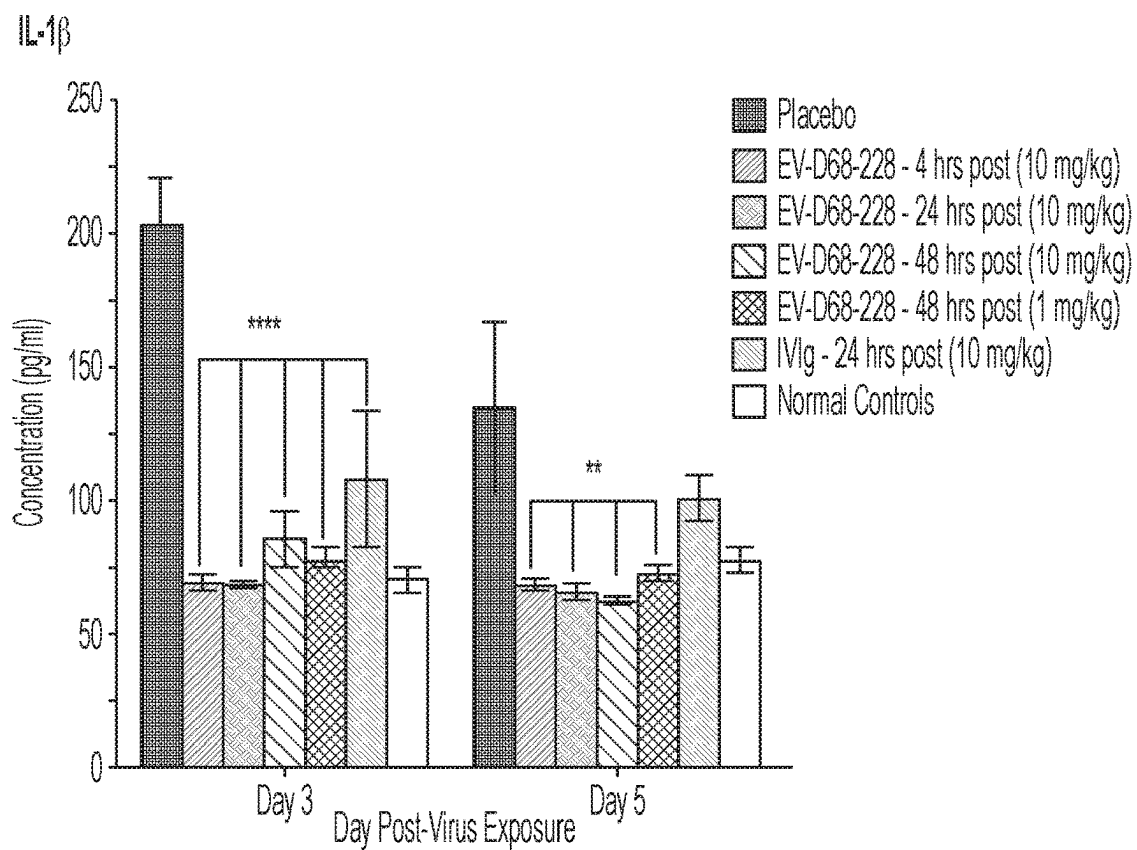
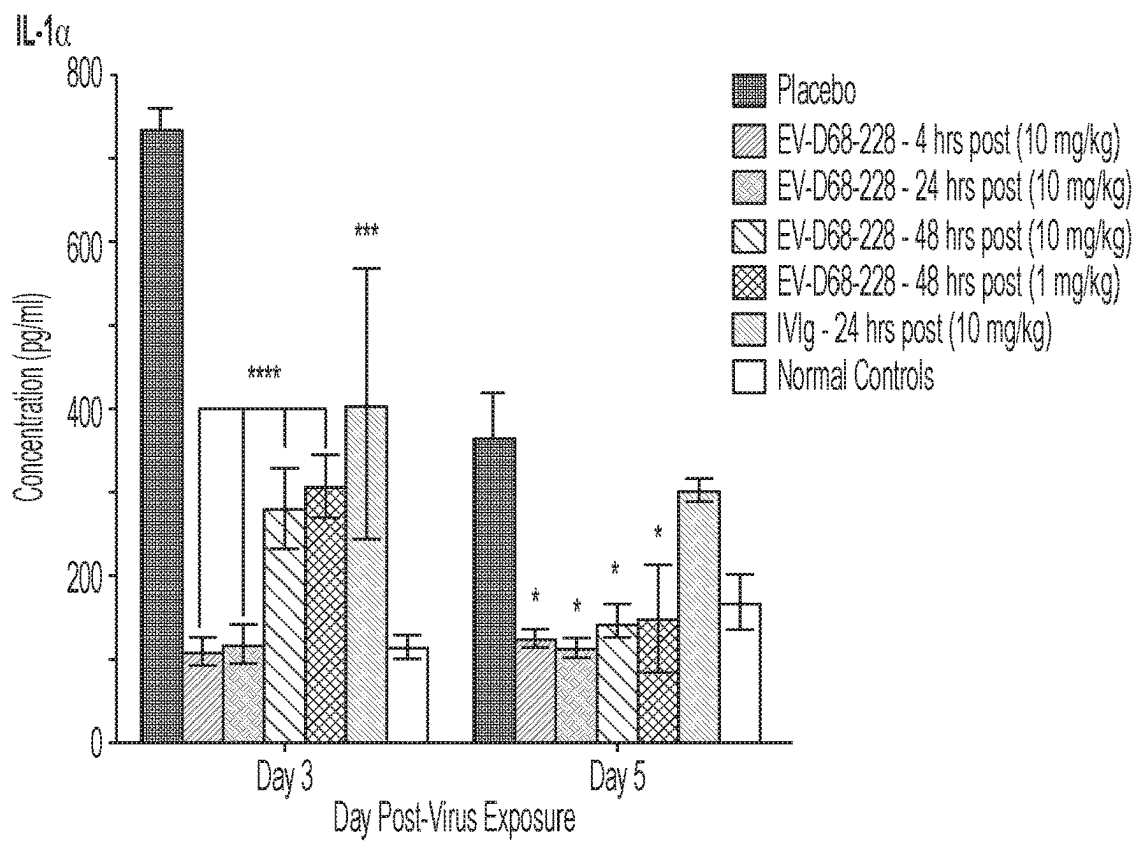


FIG. 21

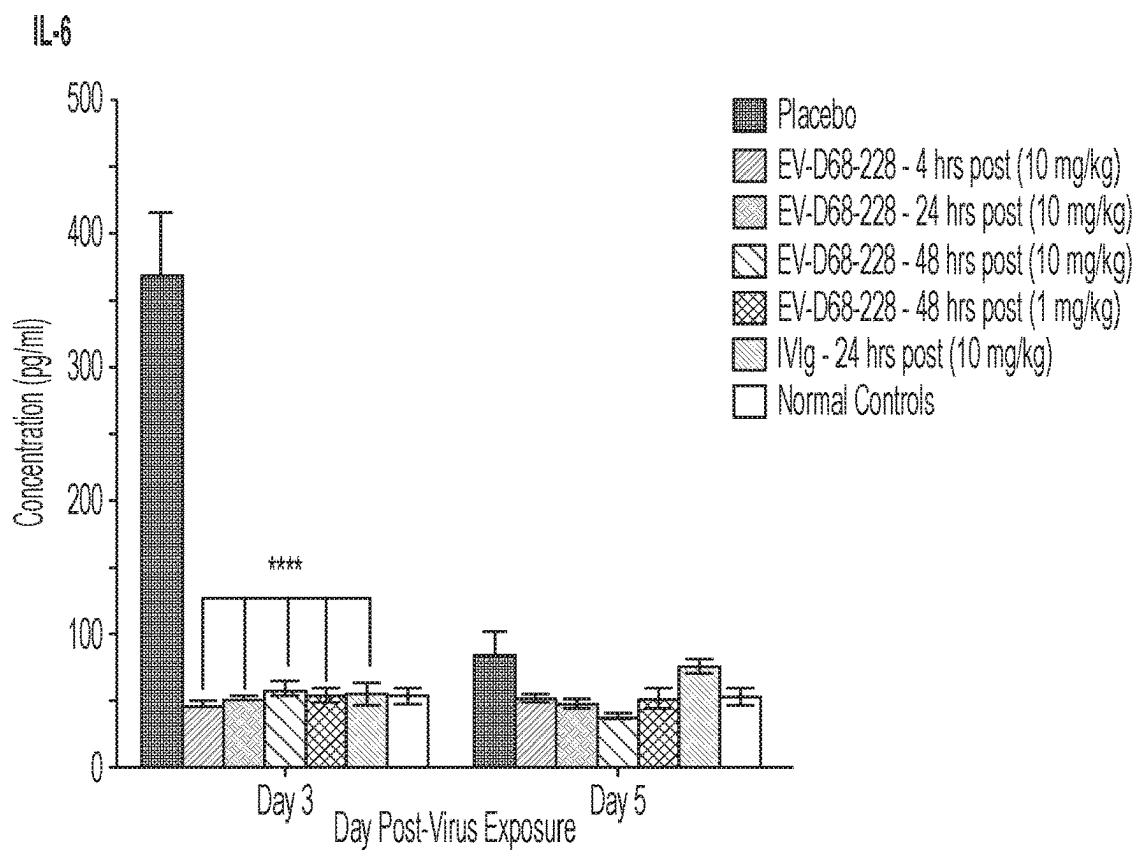
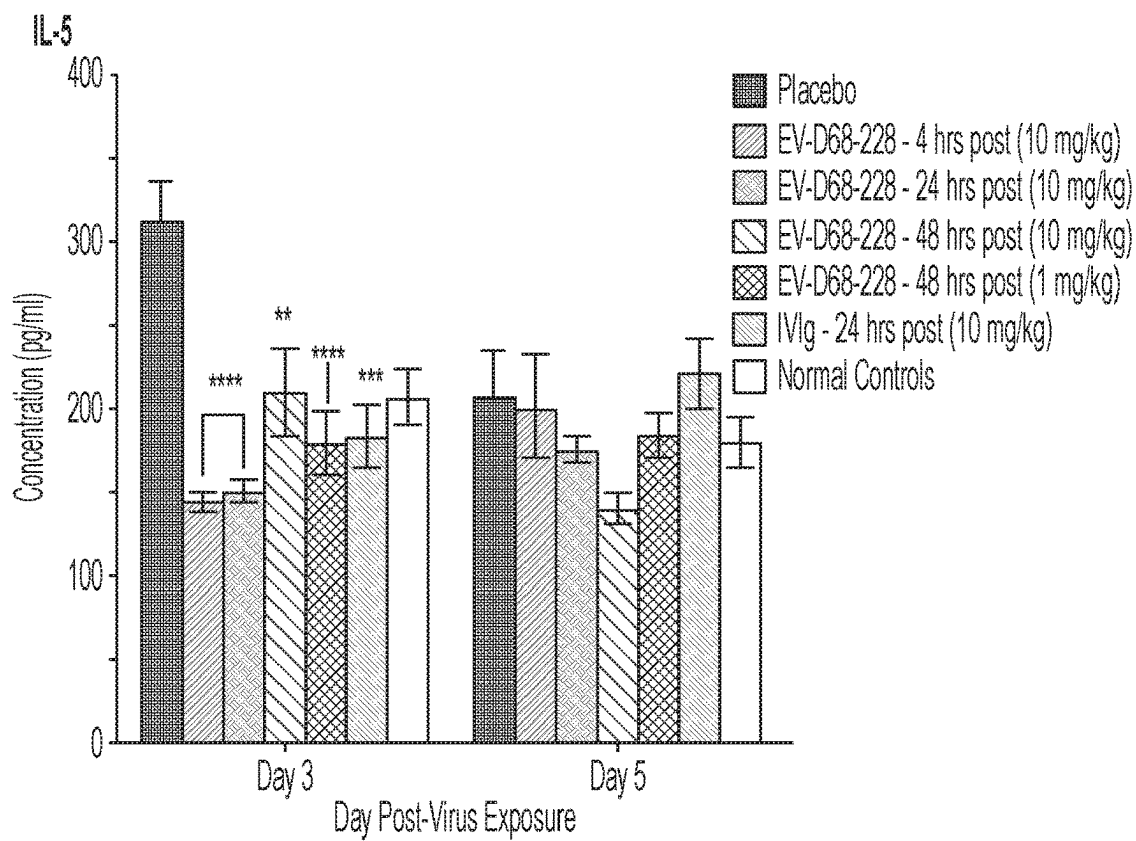


FIG. 22

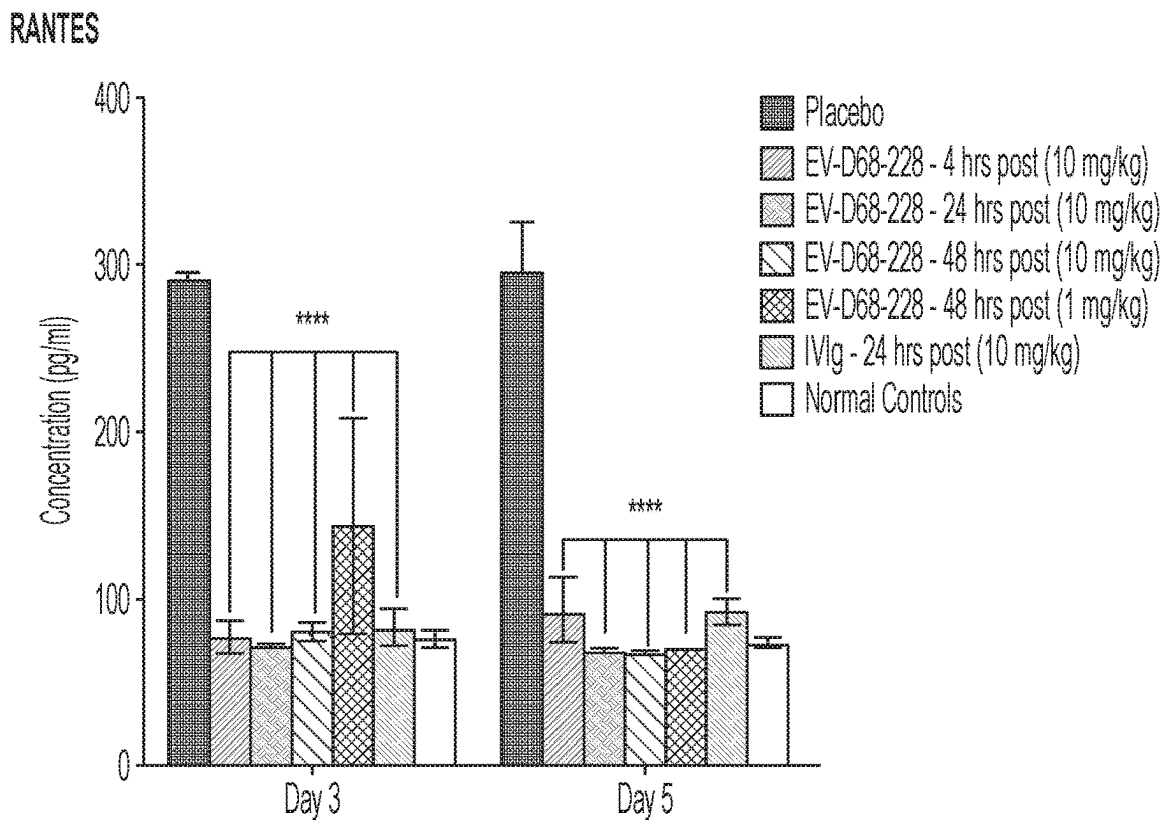
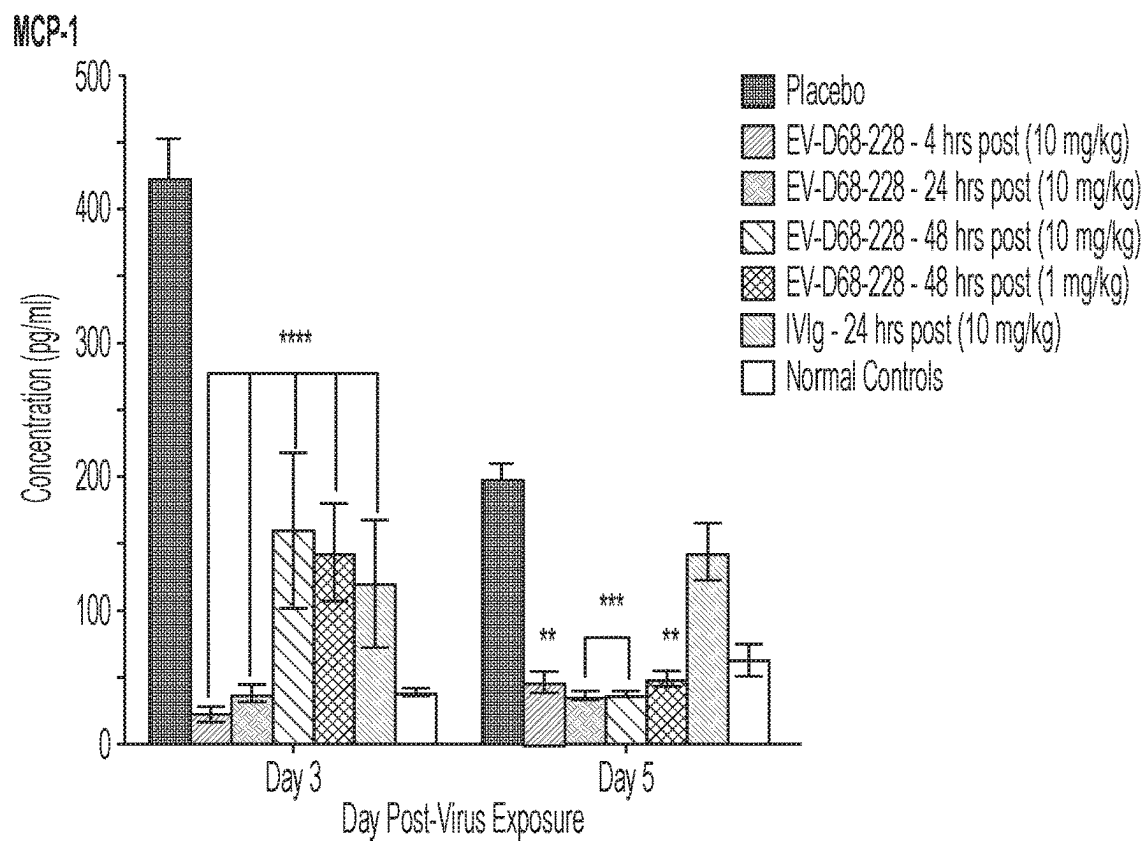


FIG. 23

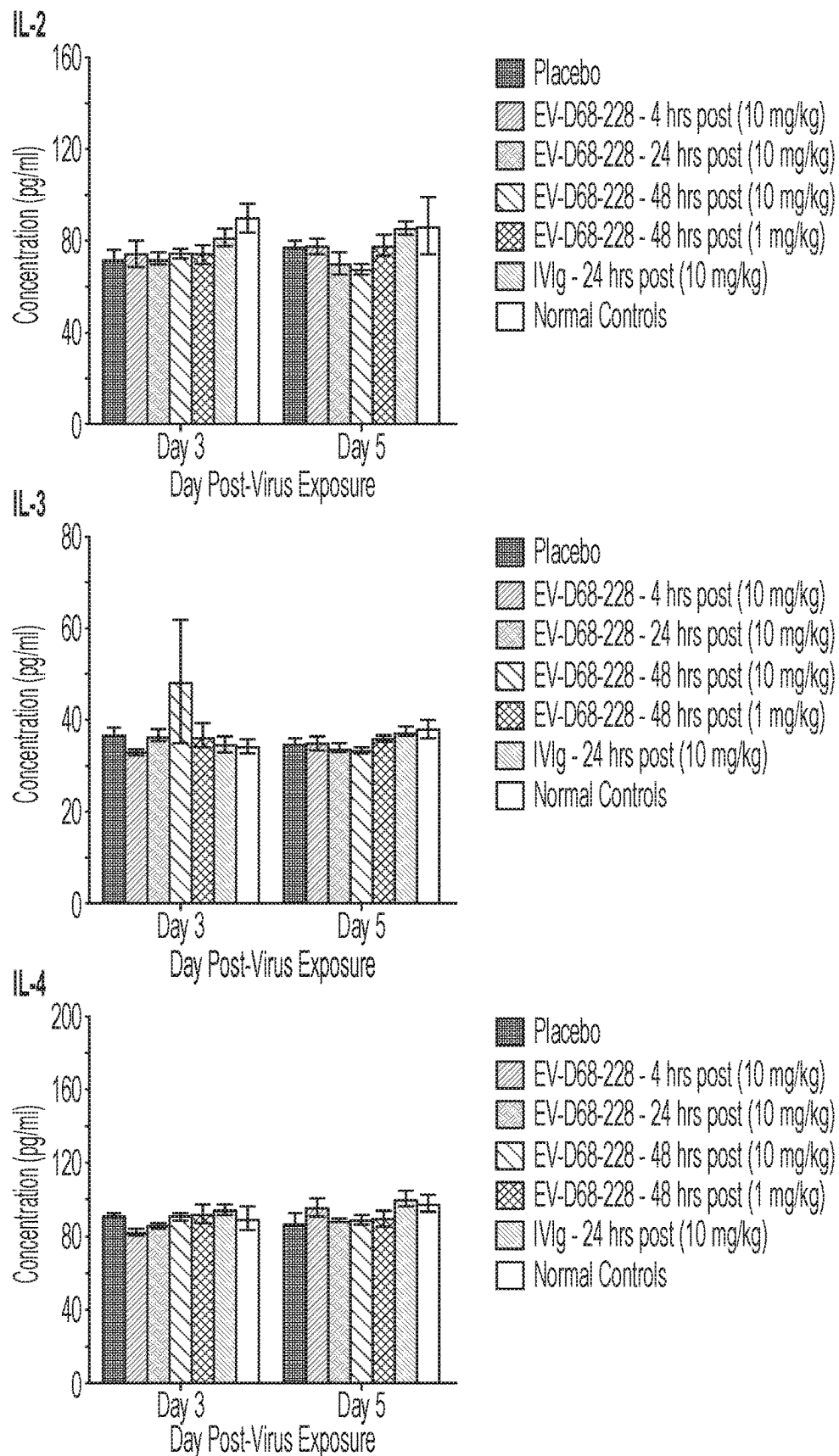


FIG. 24

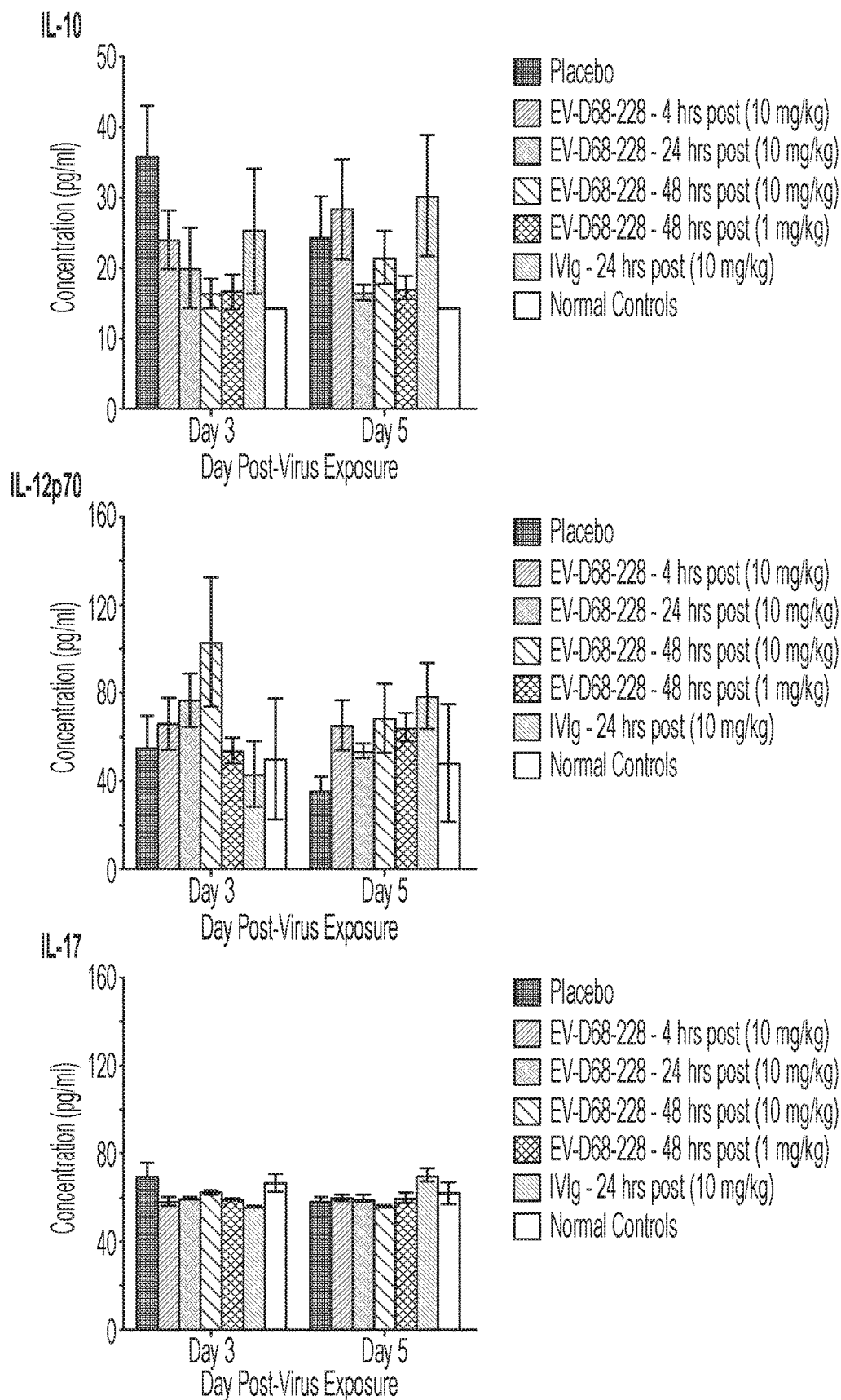


FIG. 25

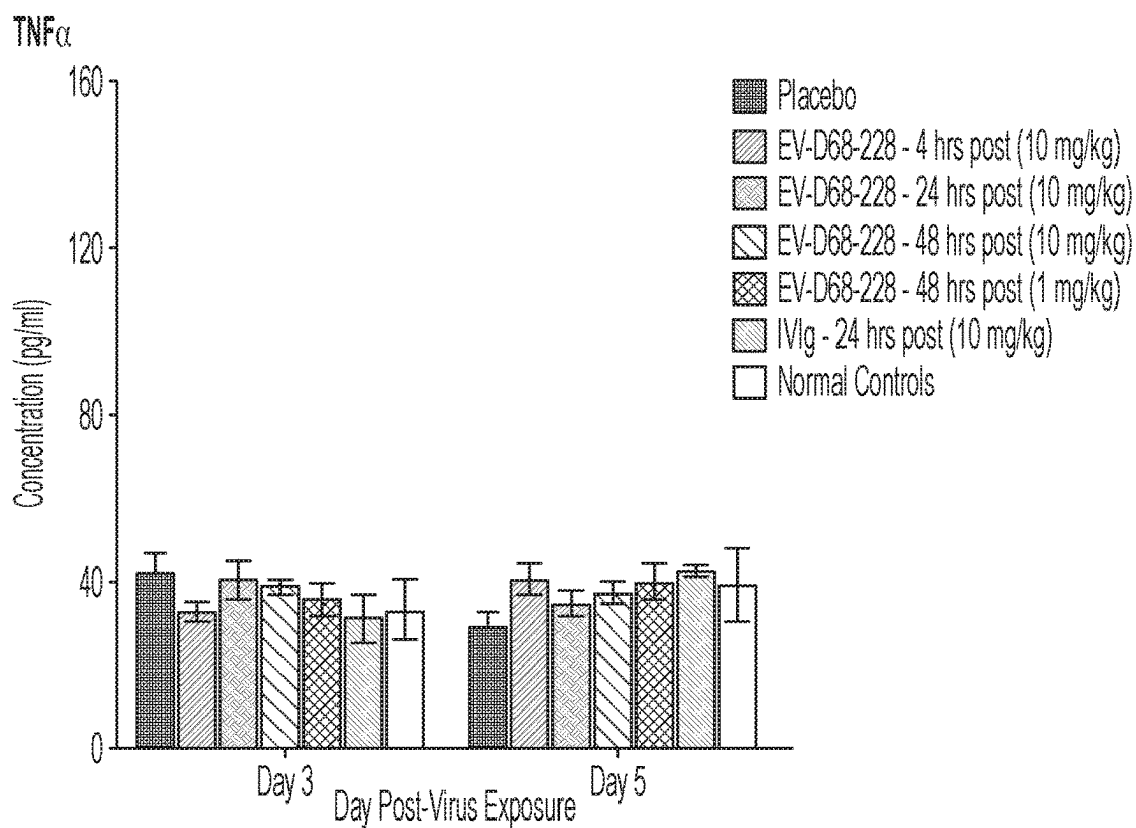
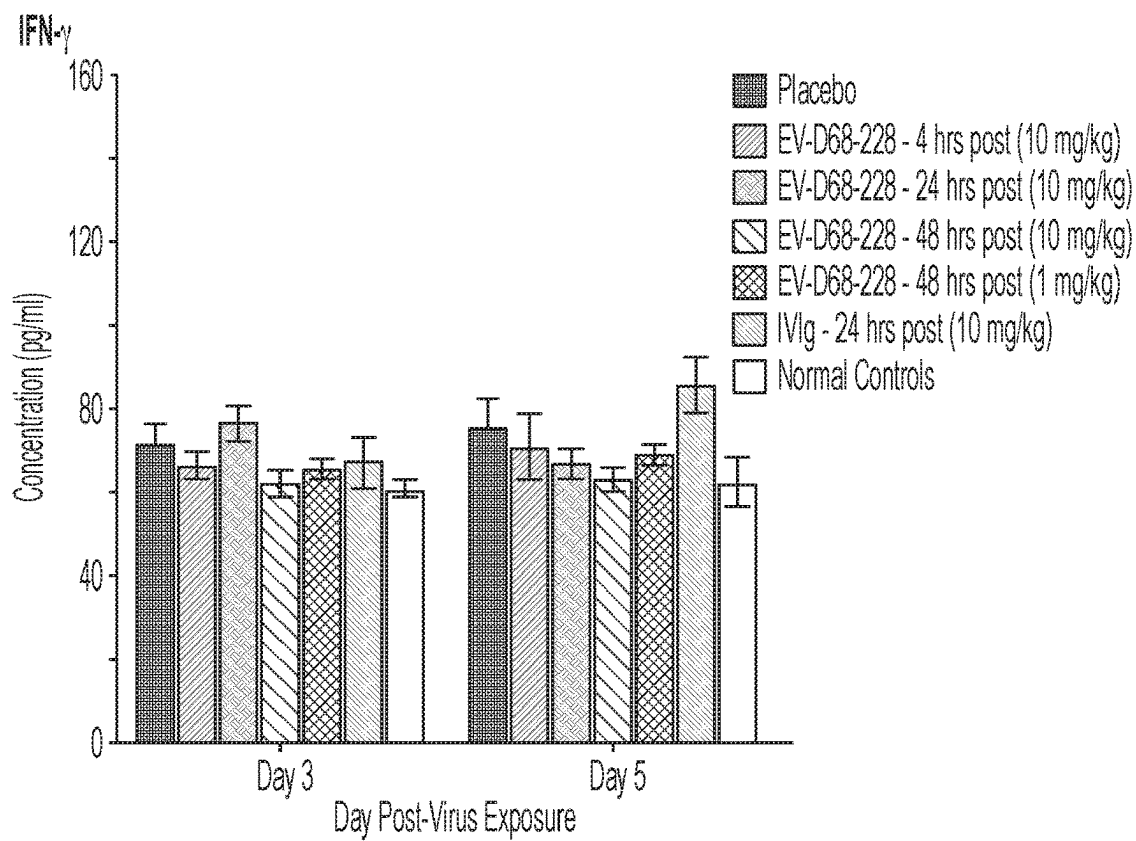


FIG. 26

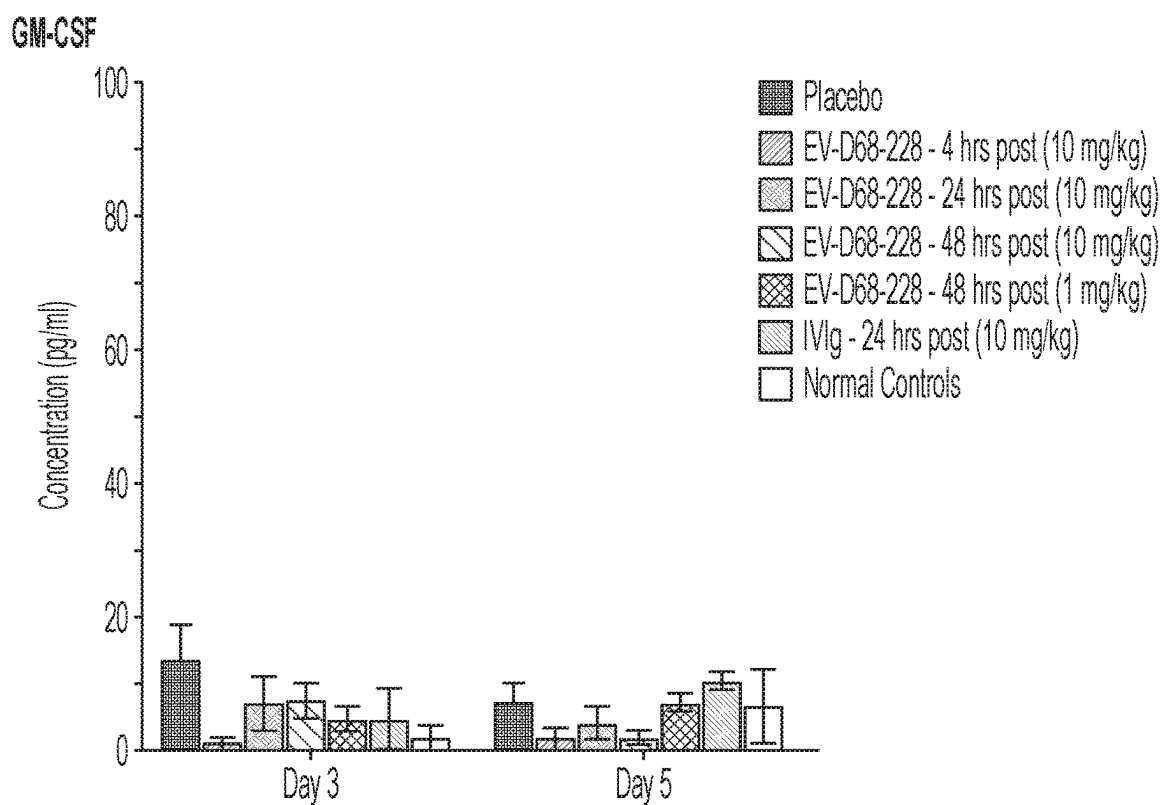
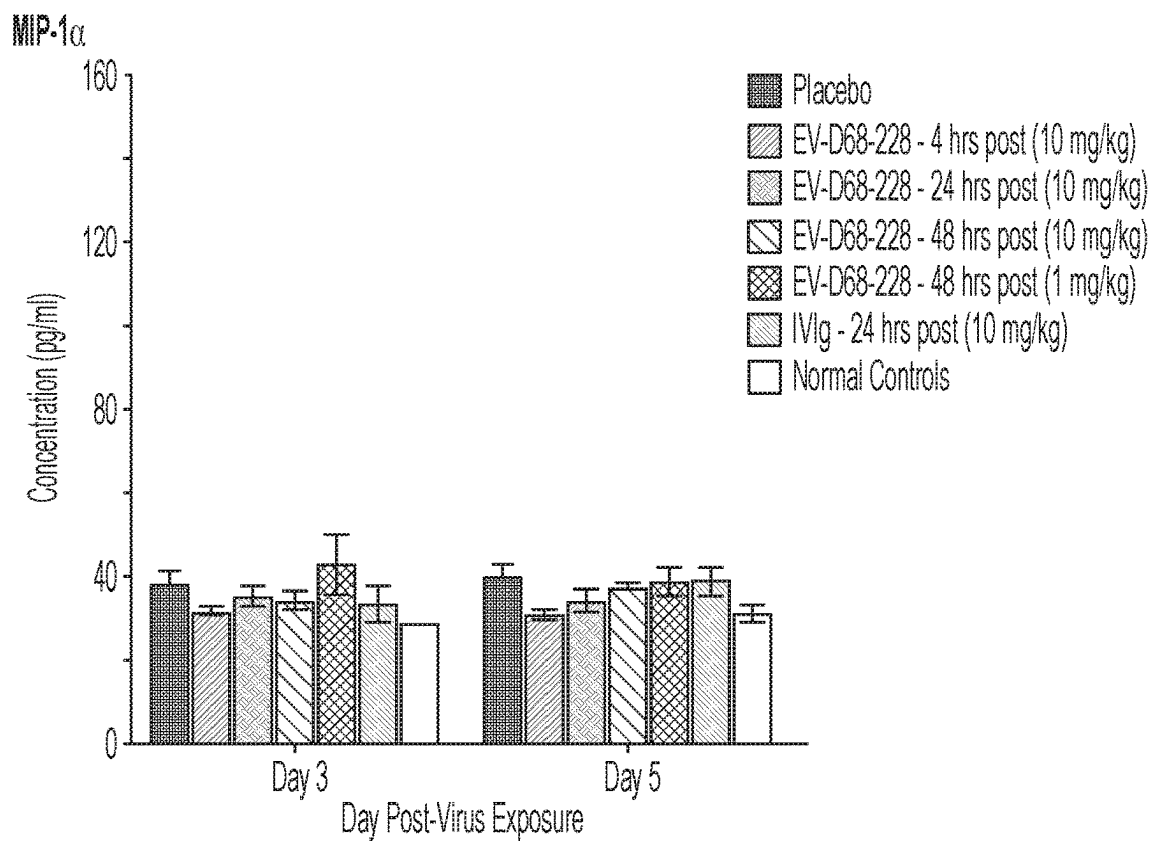


FIG. 27

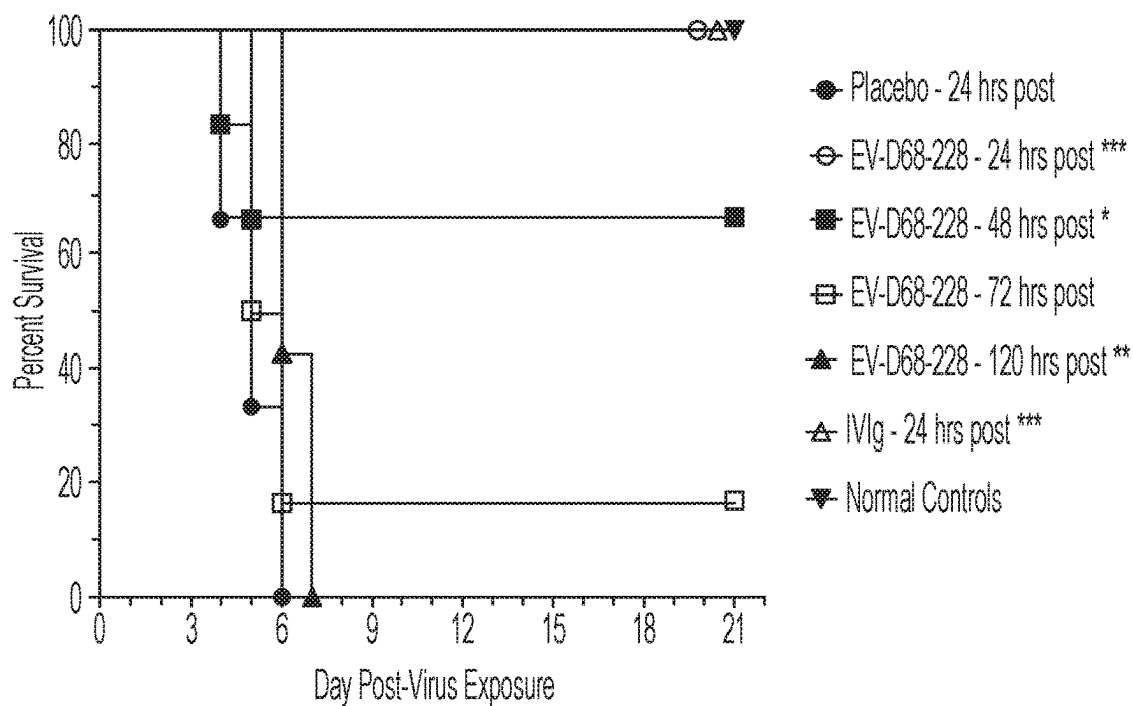


FIG. 28

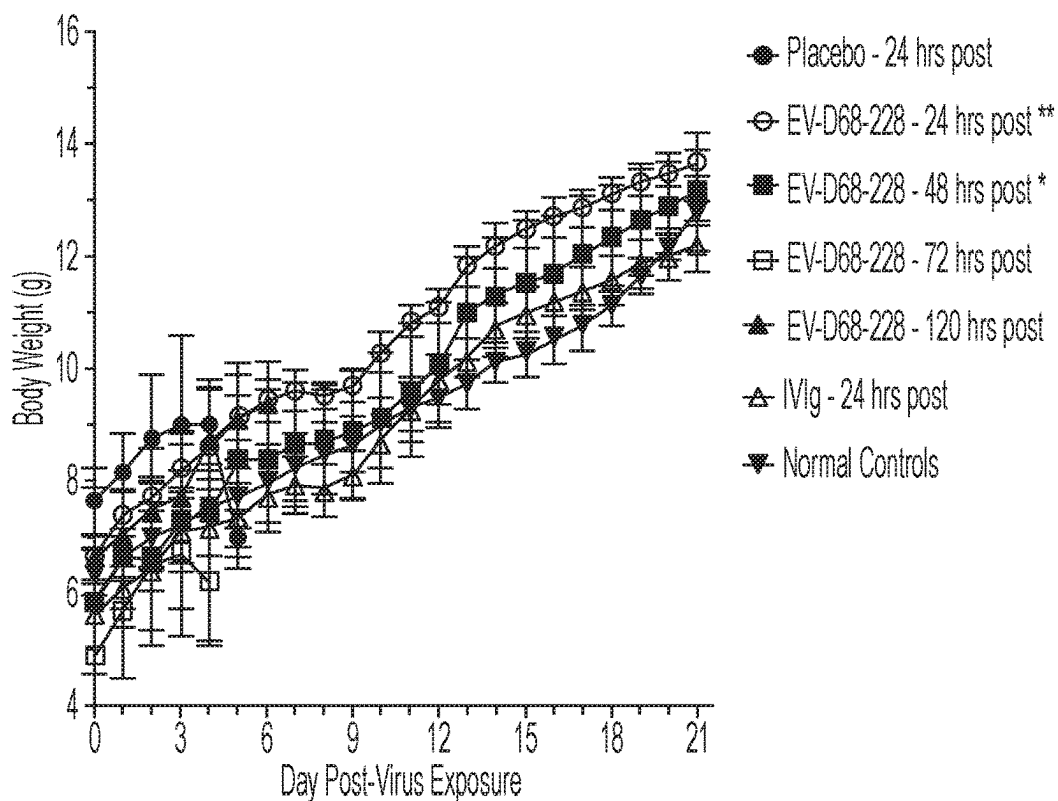


FIG. 29

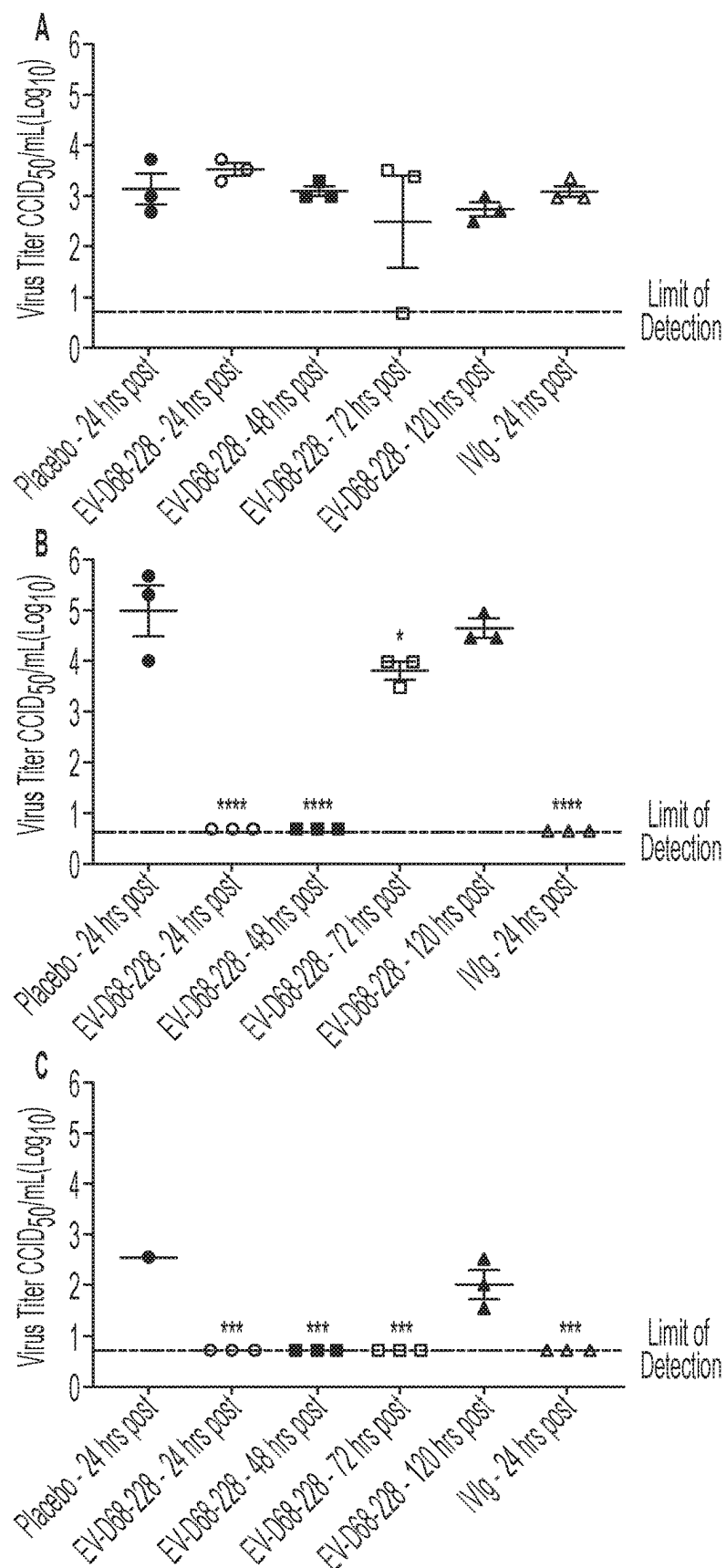


FIG. 30A-C

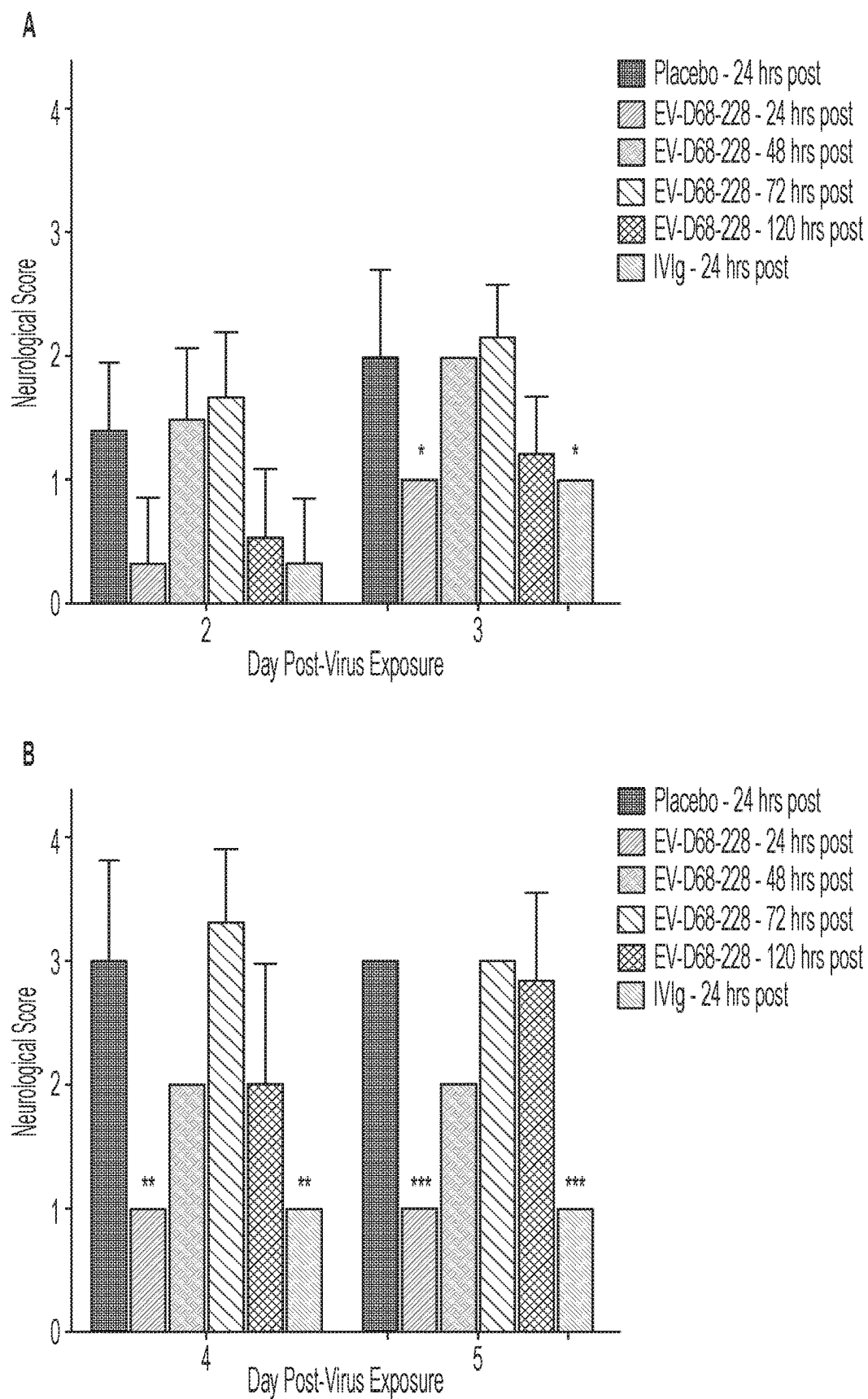


FIG. 31A-B

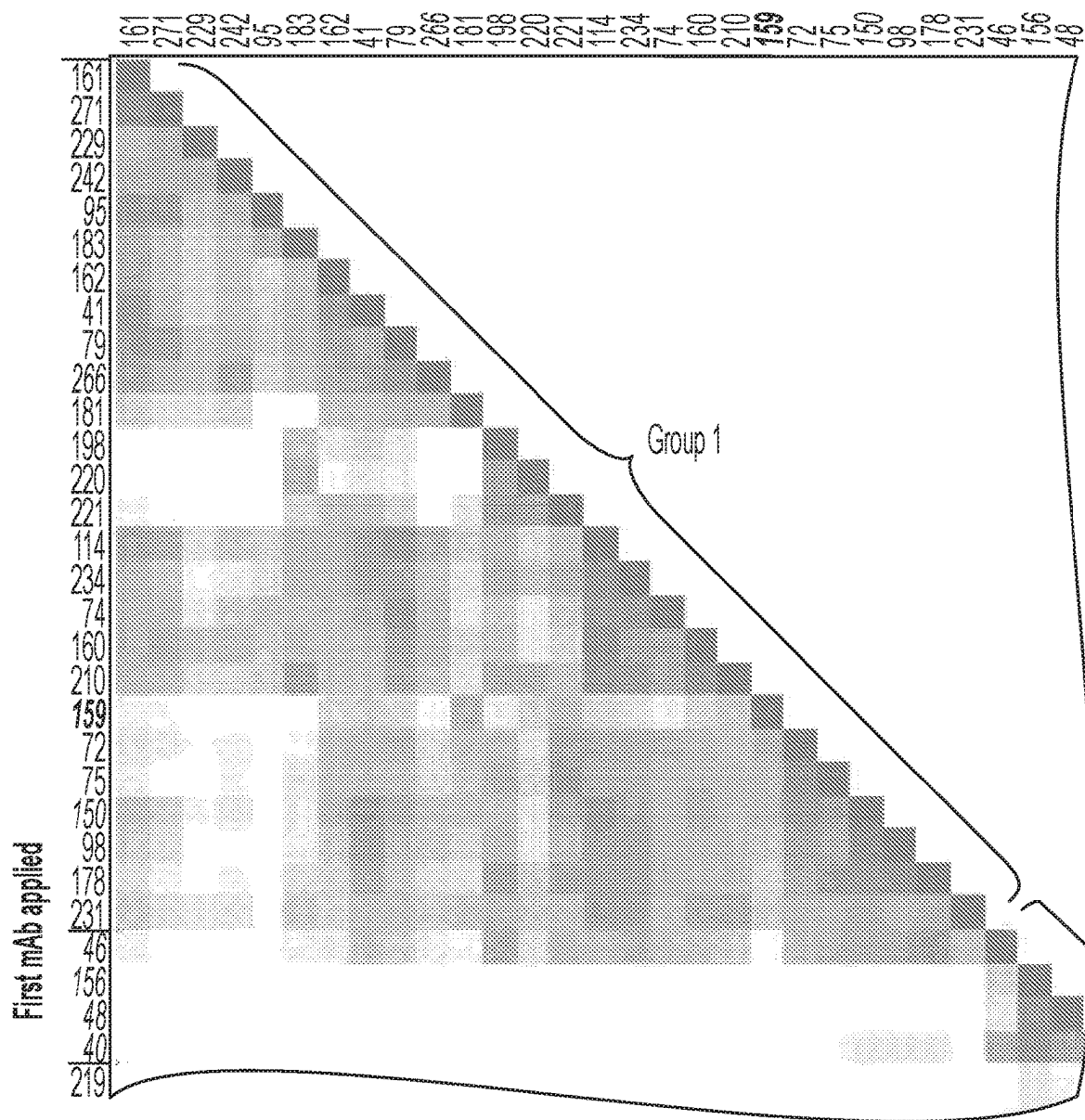


FIG. 32

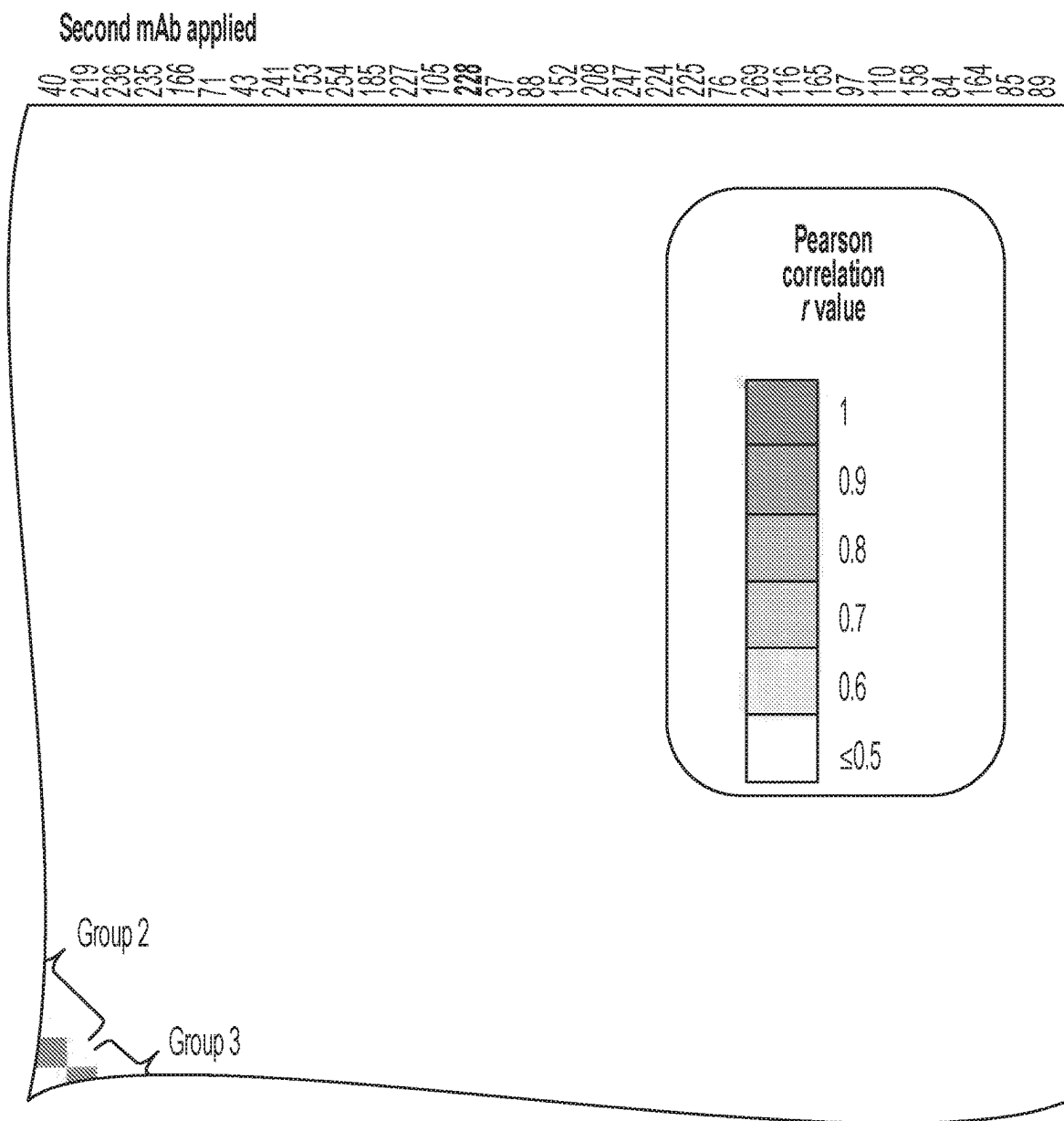


FIG. 32
CONTINUED

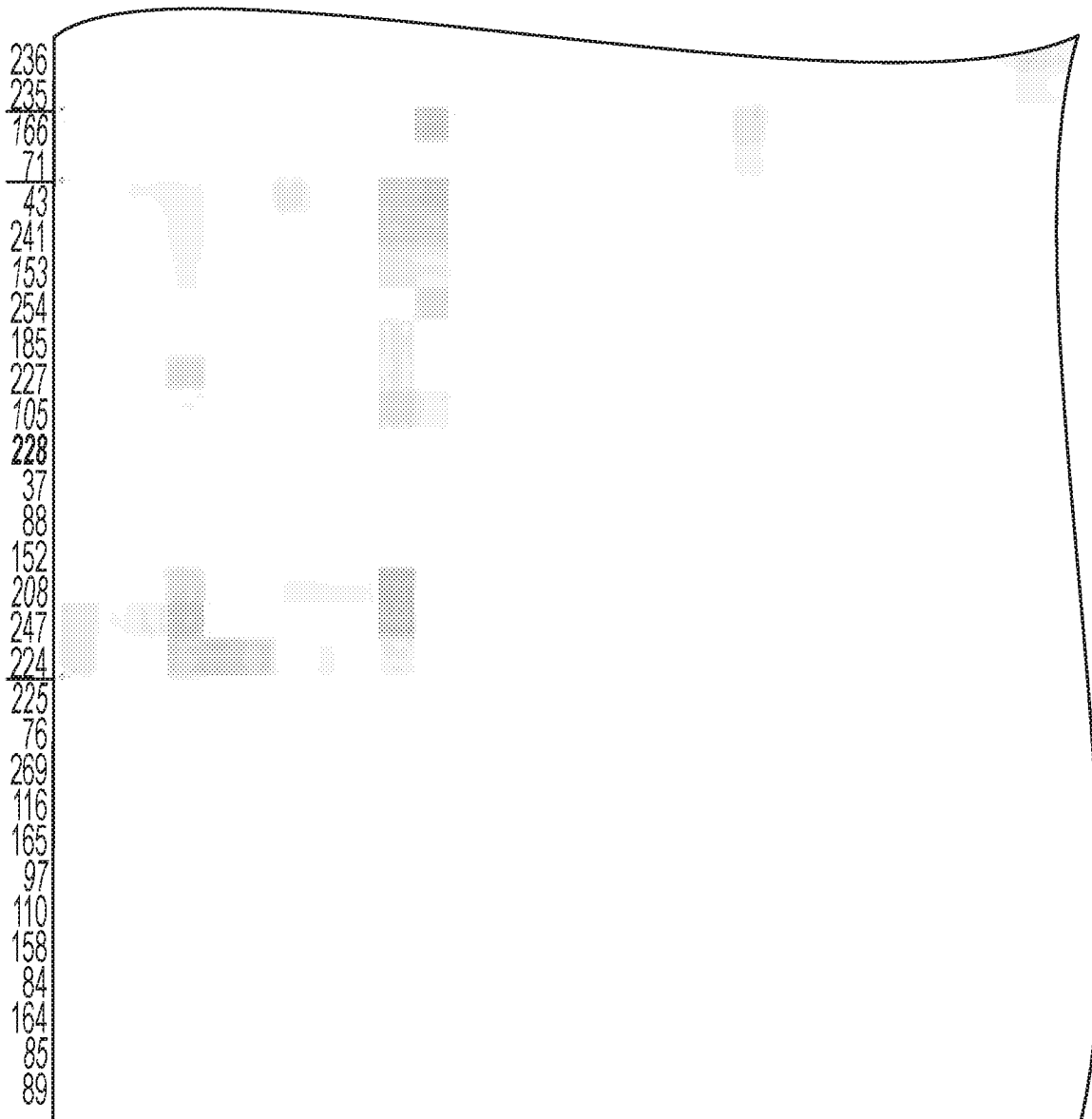


FIG. 32
CONTINUED

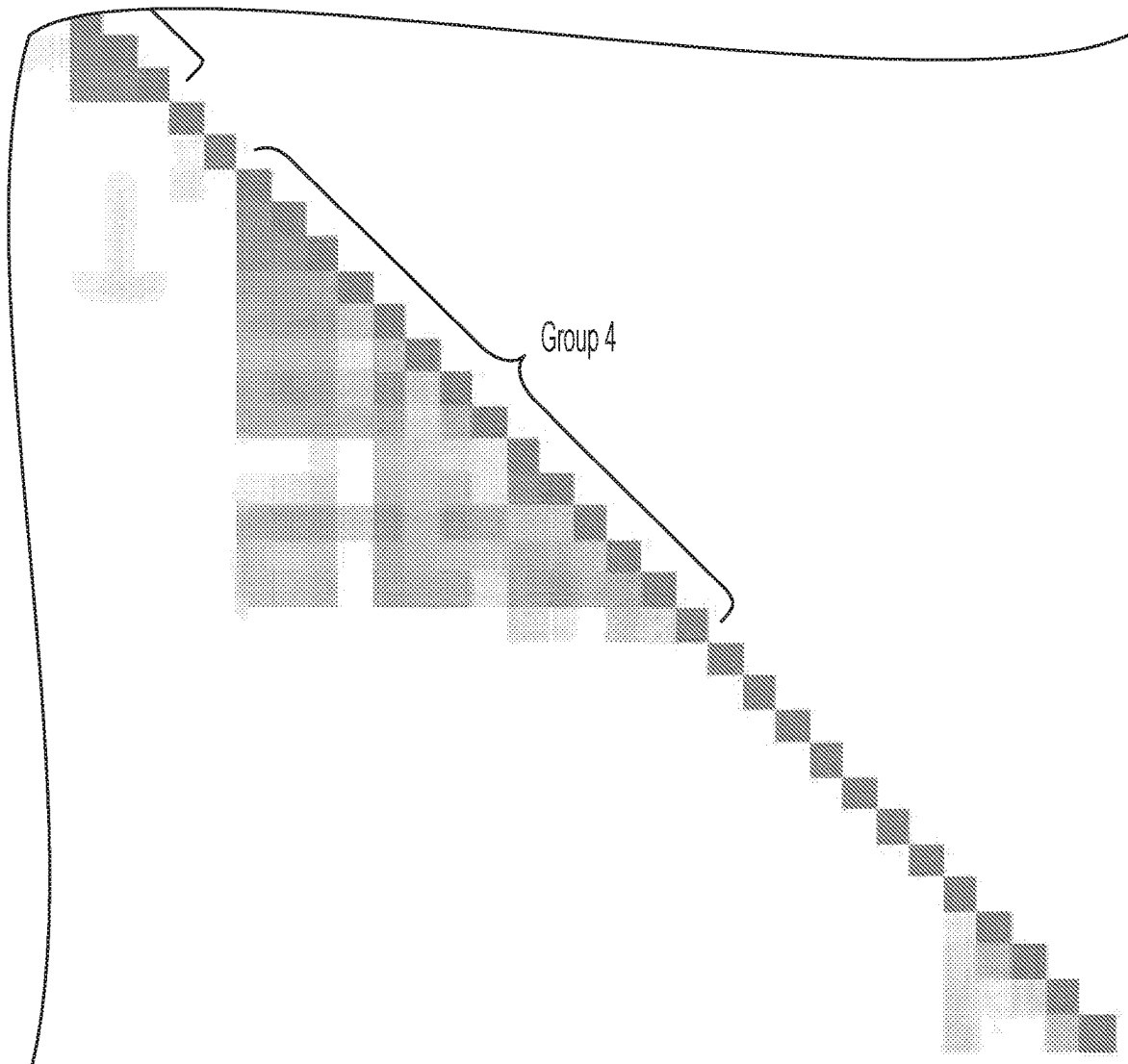


FIG. 32
CONTINUED

A Neutralization IC₅₀ in CCID₅₀ assay B Binding EC₅₀ in ELISA

		Clade and isolate tested									
Comp. group	mAb	B1		D	N/A	B1					
		IL/56	IL/56			KY/53	Fer.	MO/47	MO/49	IL/56	IL/52
1	159	+++	++	>	+	+++	>	+++	+++	+++	>
	150	+++	++	+	+	+++	+++	+++	+++	+++	>
	95	++	+	>	>	+++	+++	+++	+++	+++	+
	266	++	++	++	>	+++	+++	+	+++	+++	+
	41	+	+	>	>	+++	+++	+	+++	+++	>
	98	+	+	>	>	+	+	+	+	+	>
	231	+	+	>	>	>	>	>	>	>	>
	178	+	+	>	>	>	>	>	>	>	>
	162	+	+			+++	+++	+++	+++	+++	>
	161	>	>			+++	+++	+	+++	+++	>
	242	>	>			+	+	+	+	+	>
	72	>	>			+++	+++	+++	+++	+++	>
	74	>	>			+++	+++	+++	+++	+++	+
	75	>	>			+	+	+	+	+	>
	79	>	>			+++	+++	+++	+++	+++	+
	114	>	>			+	+	+	+	+	>
160	>	>			+	+	+	+	+	>	
181	>	>			+	+	+	+	+	>	
183	>	>			+	+	+	+	+	>	

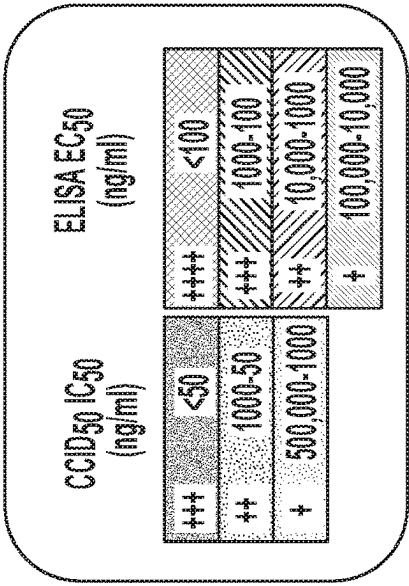


FIG. 33A-C

ELISA binding curves for two representative mAbs

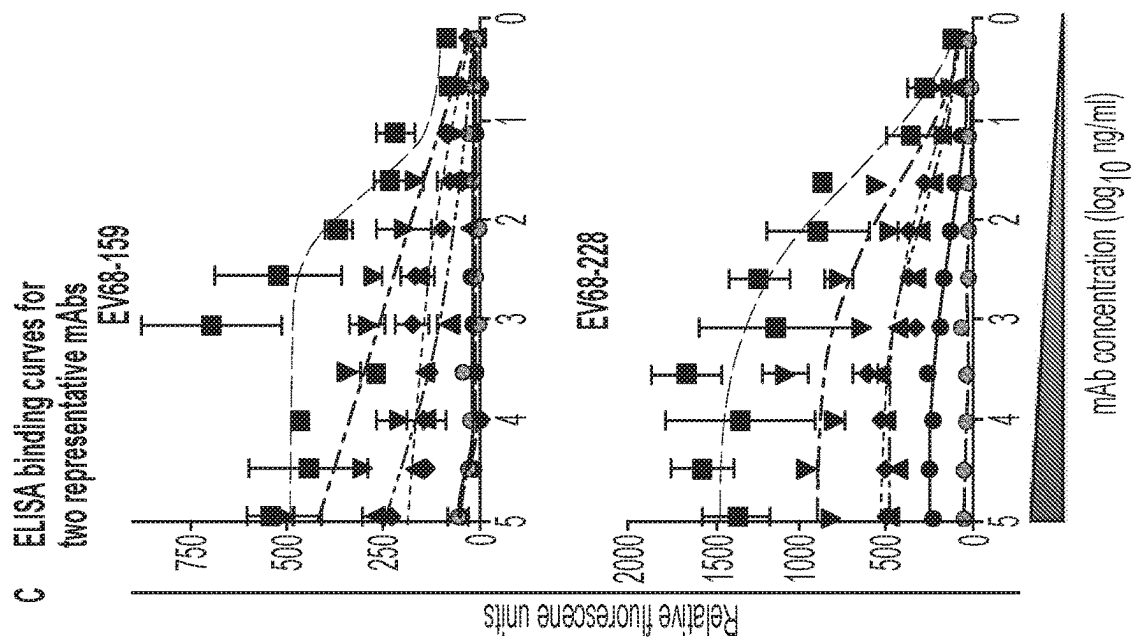


FIG. 33A-C
CONTINUED

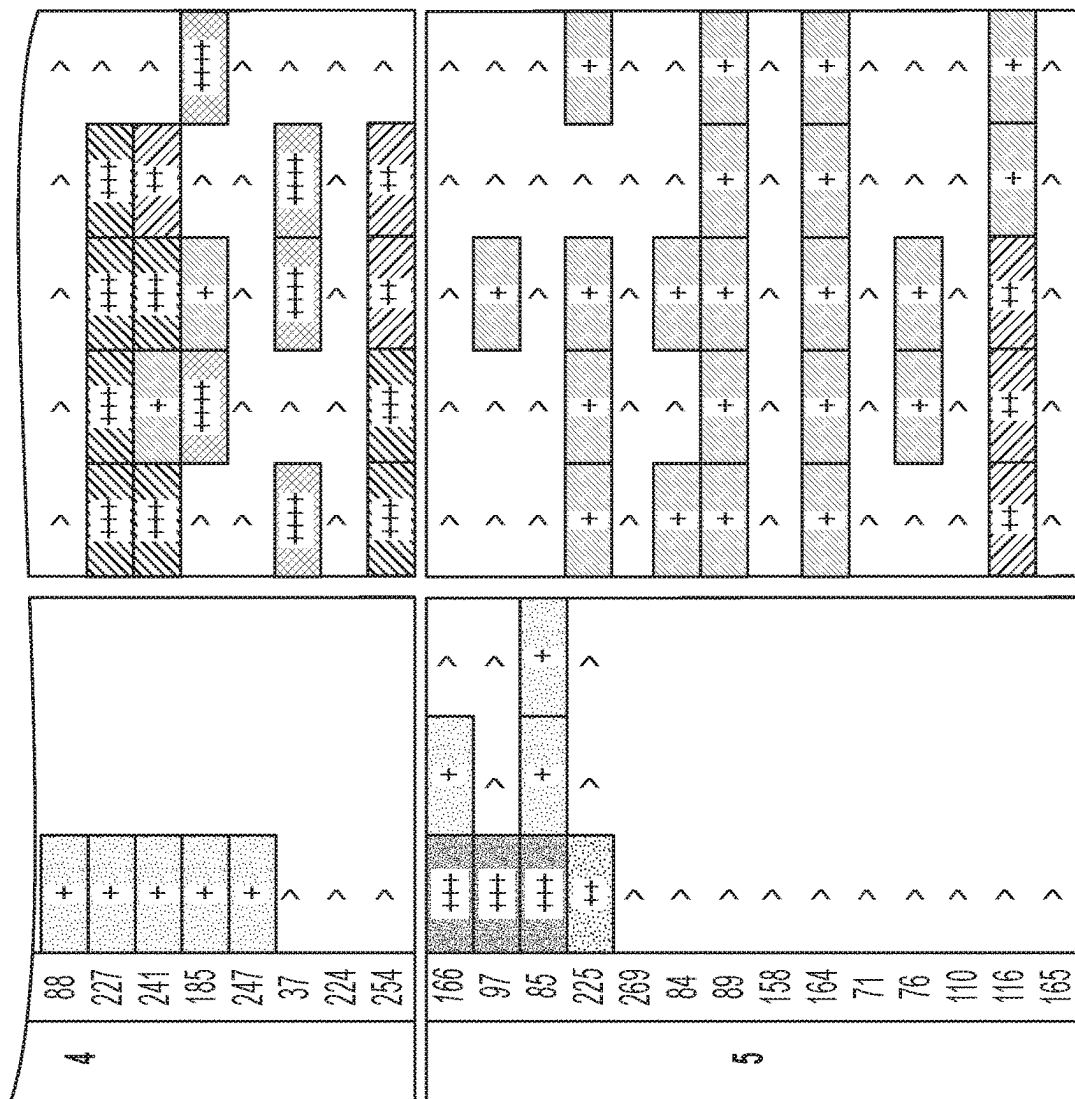
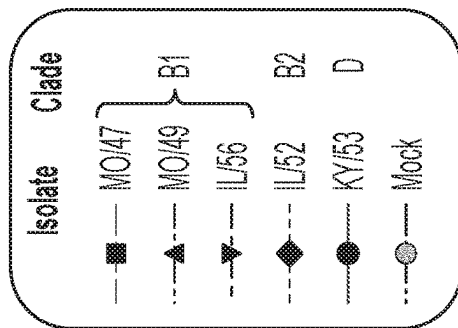
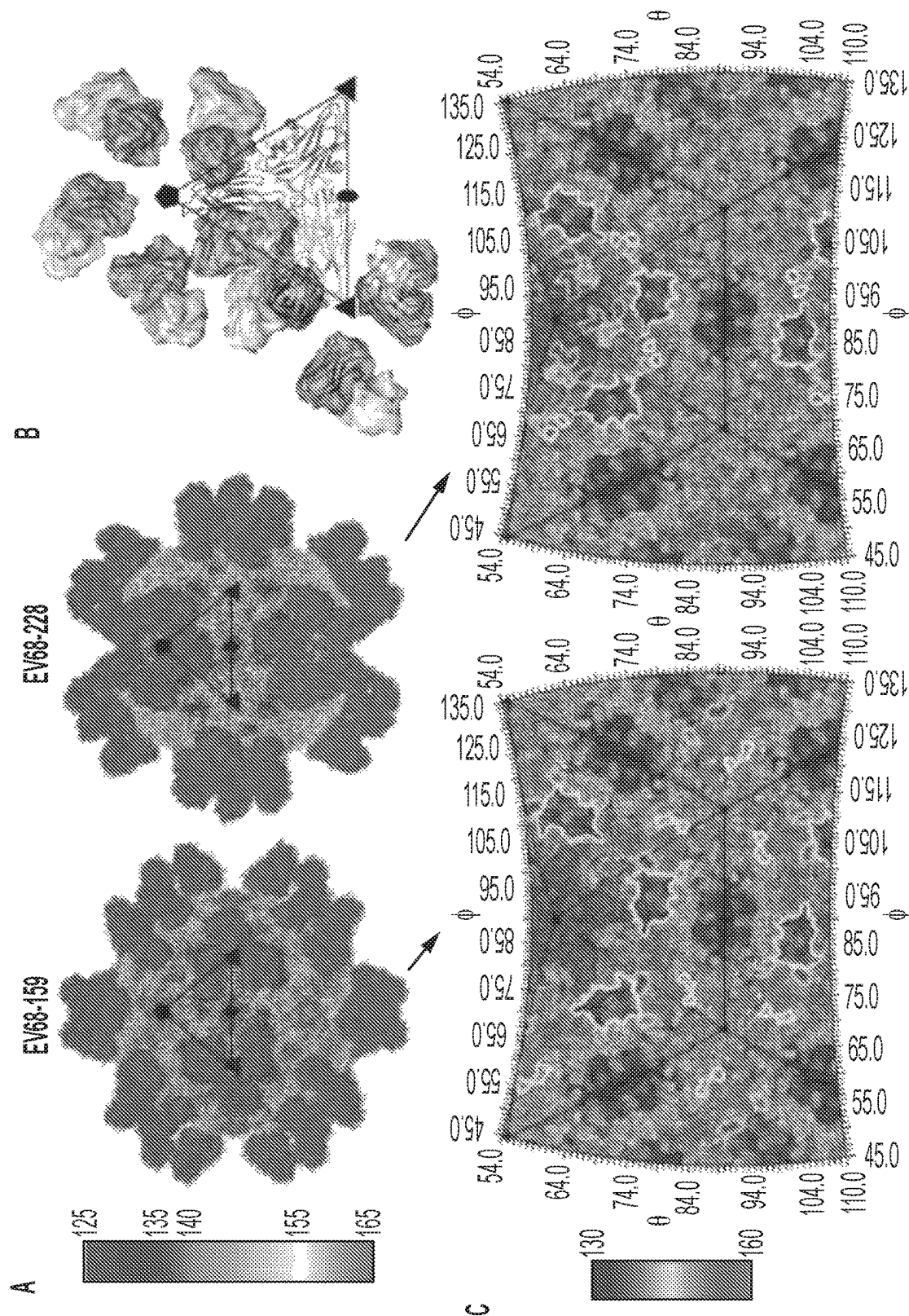


FIG. 33A-C
CONTINUED



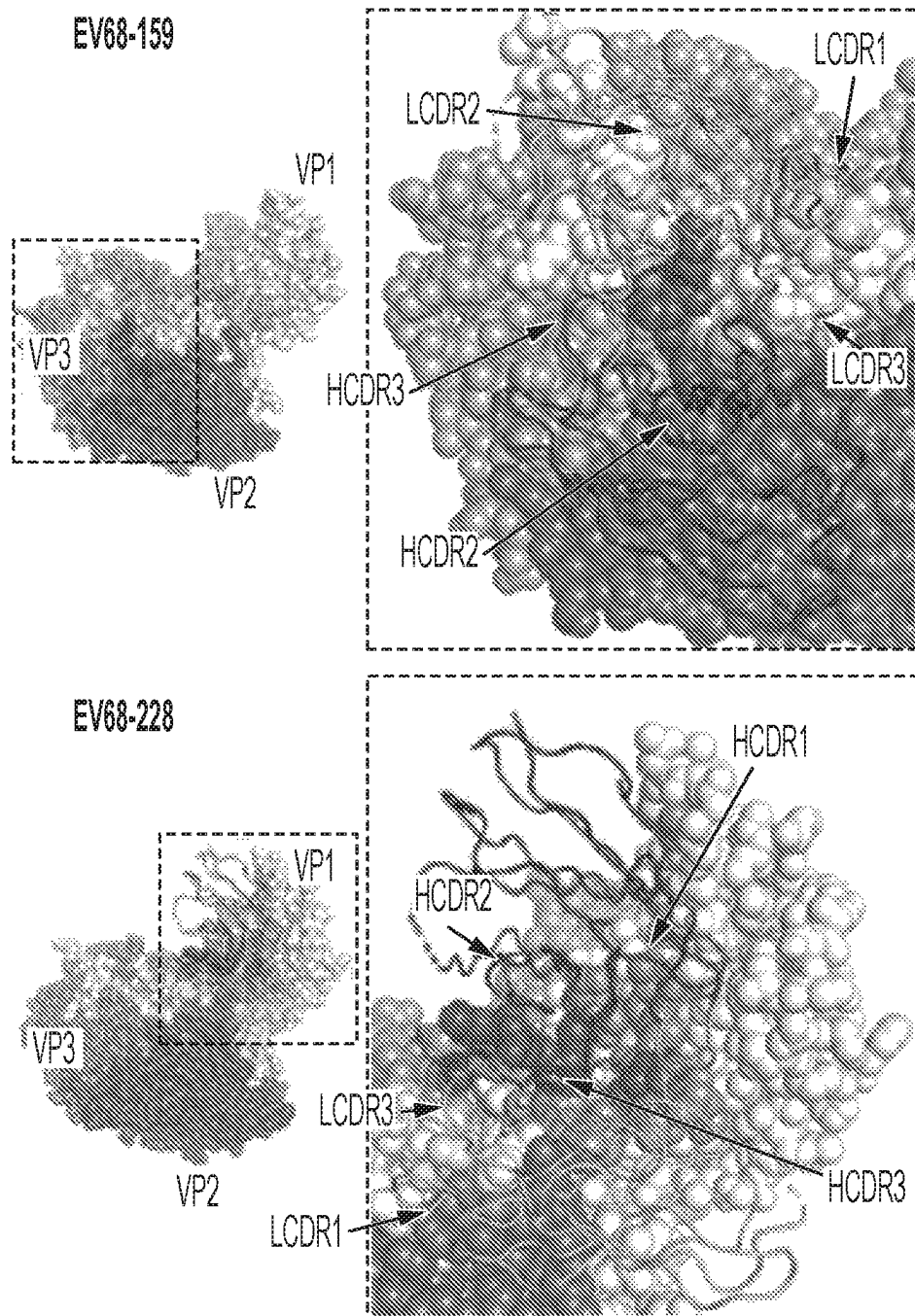


FIG. 35

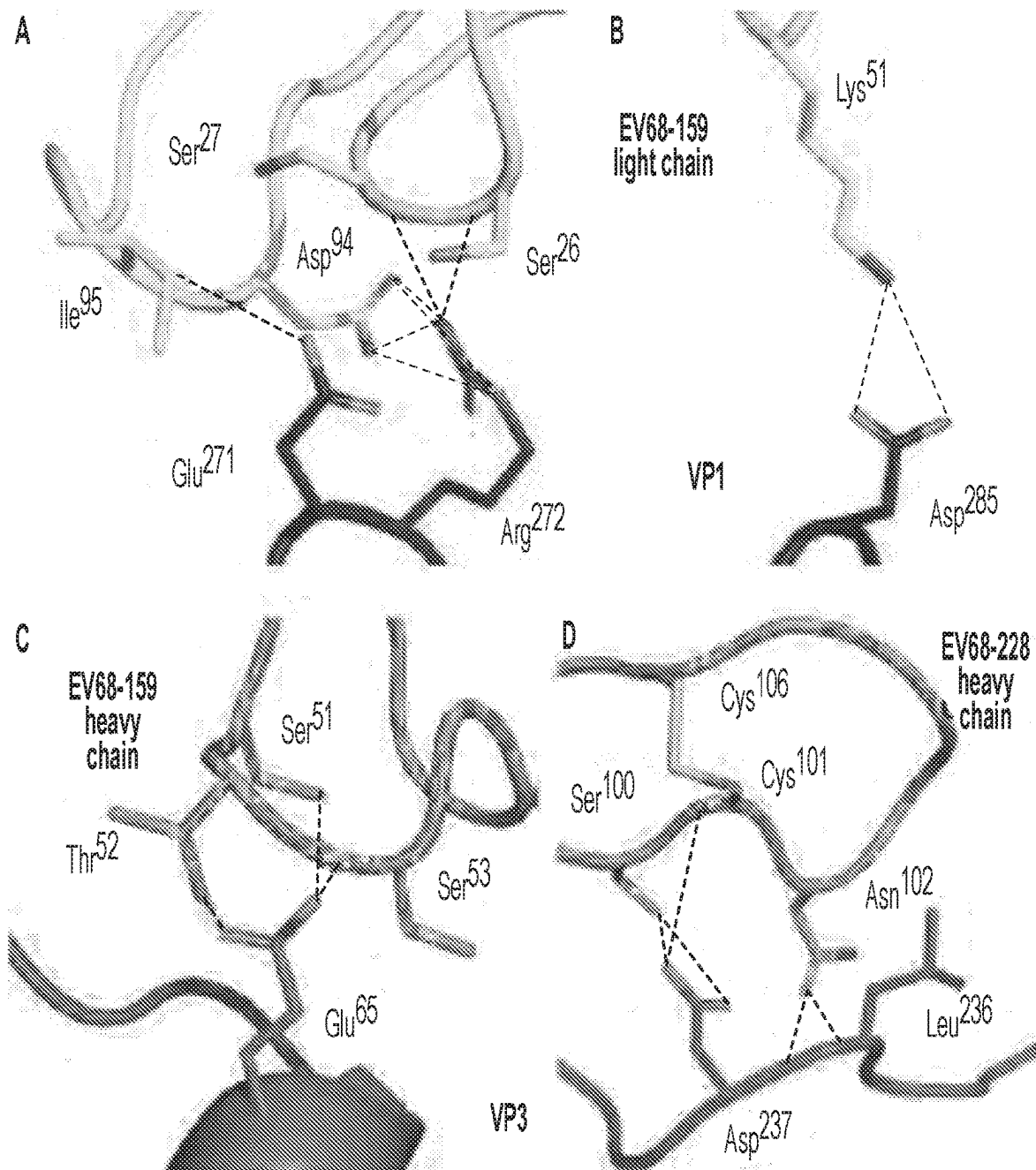


FIG. 36A-D

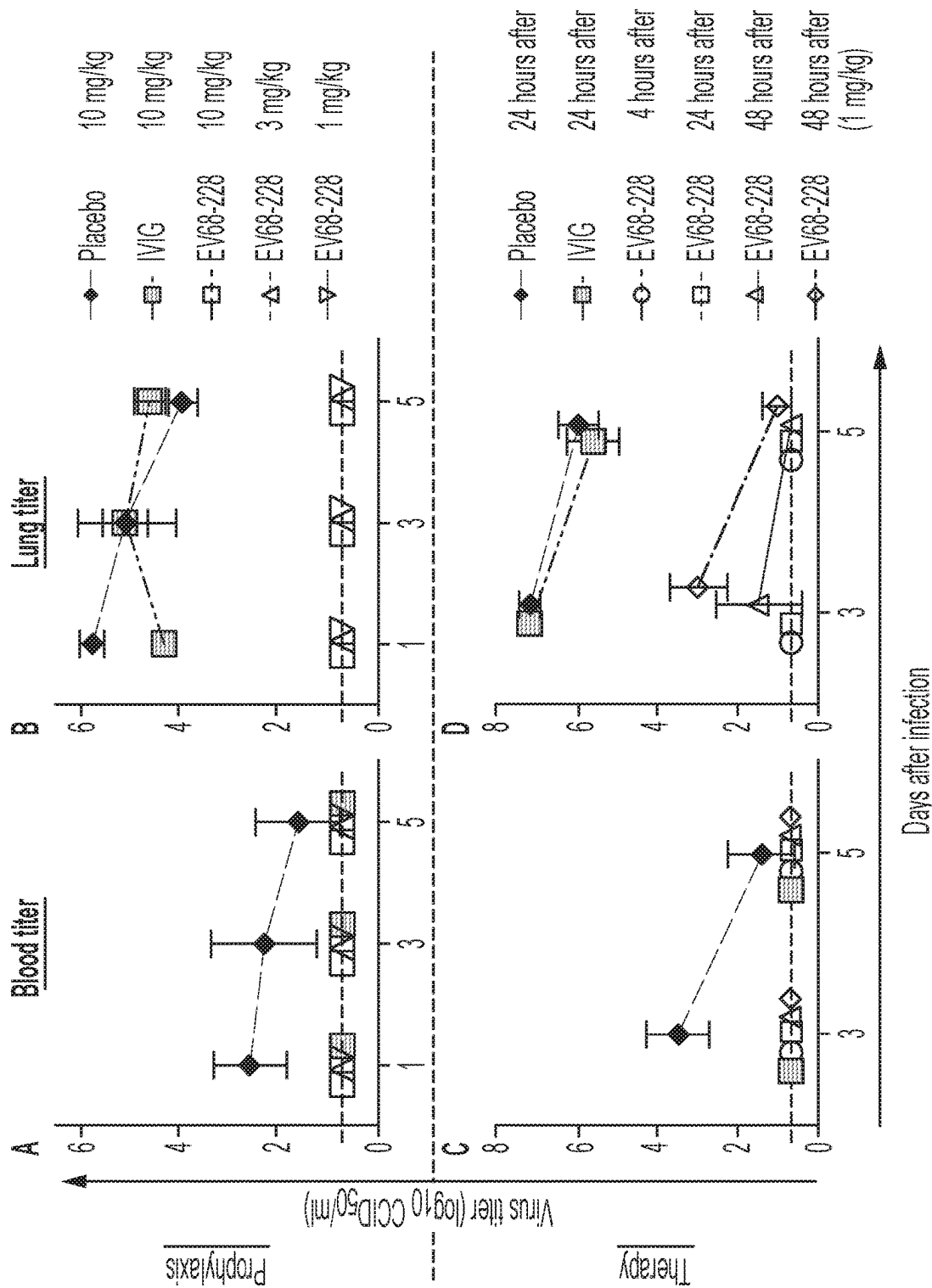


FIG. 37A-D

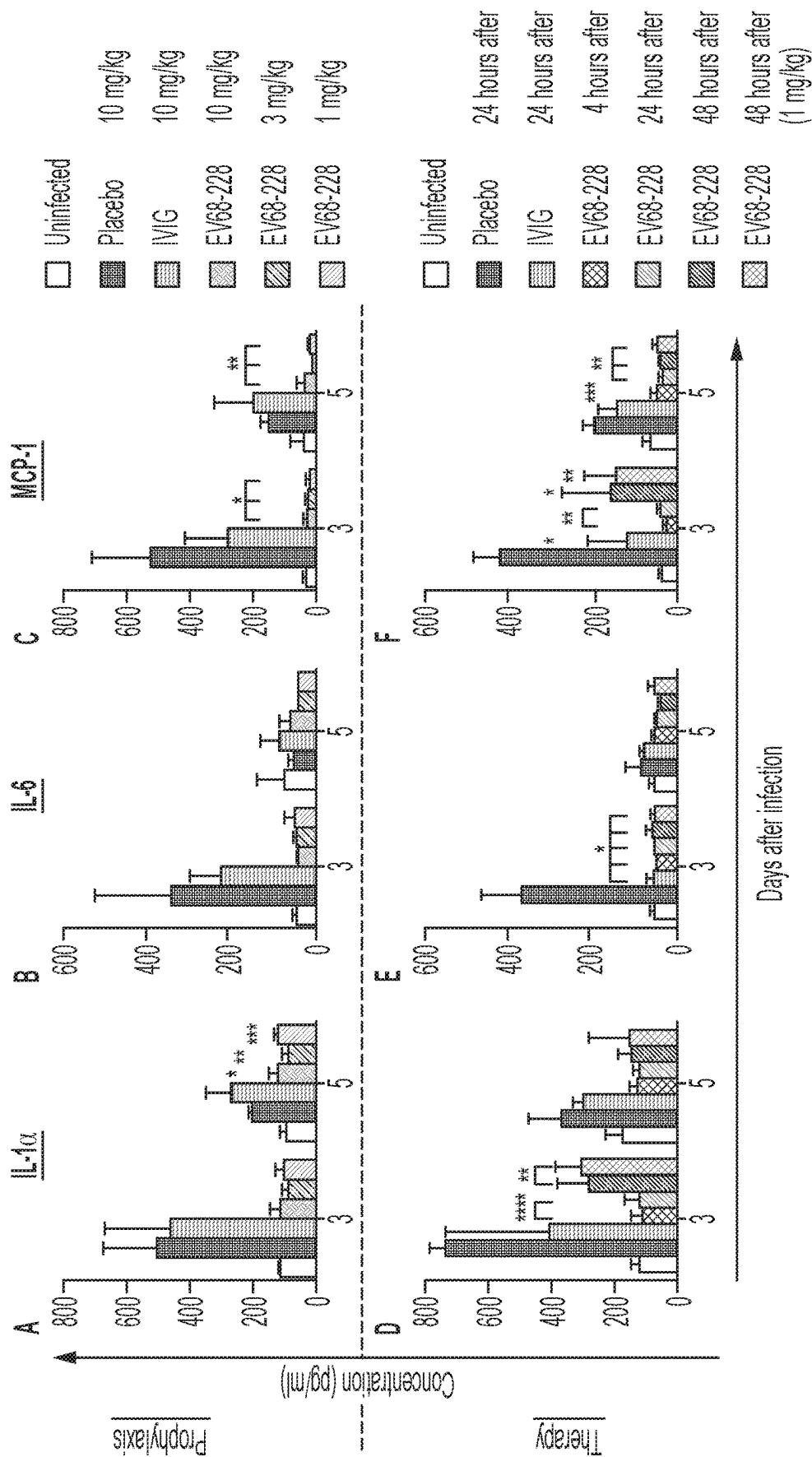


FIG. 38A-F

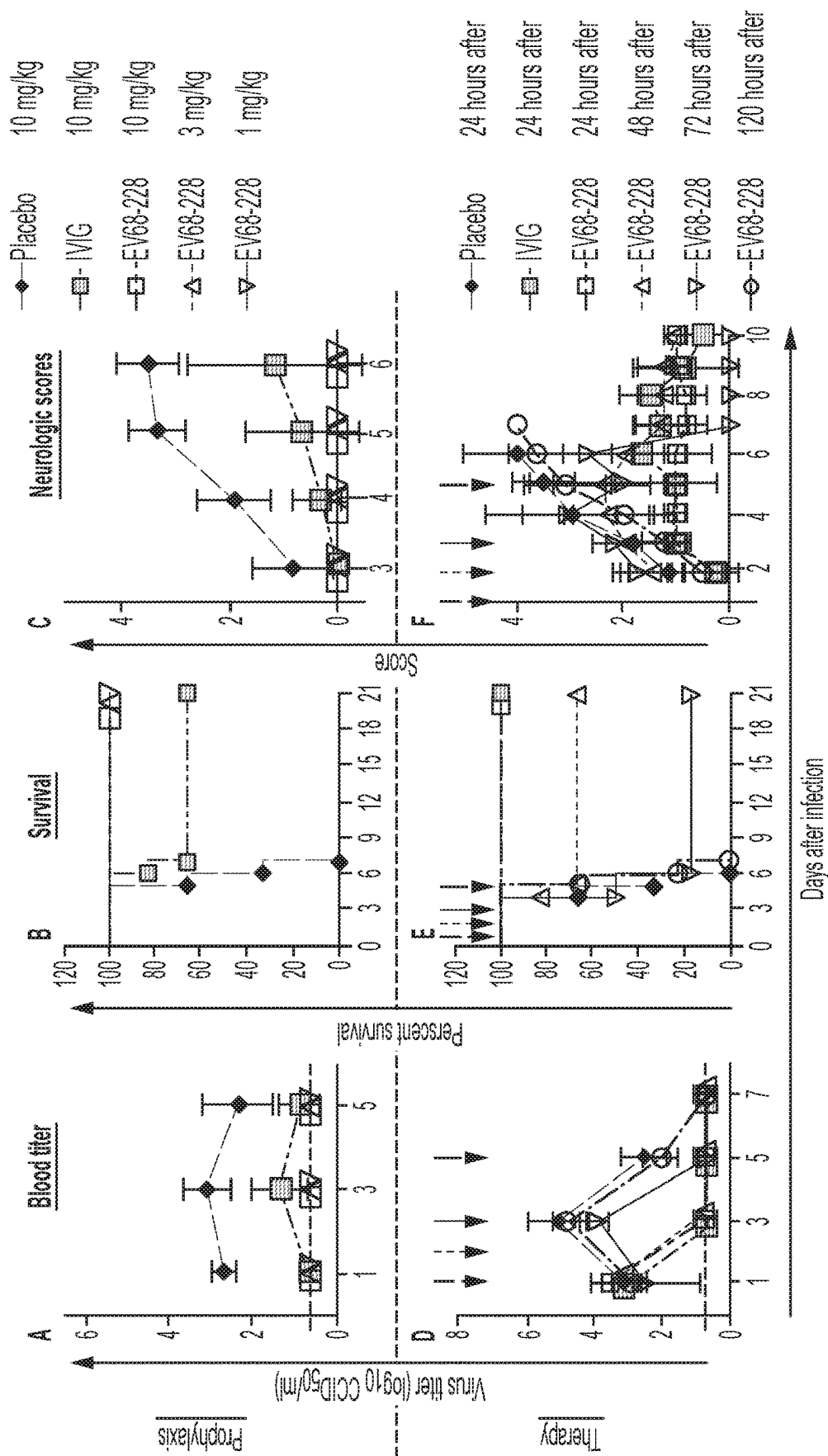


FIG. 39A-F

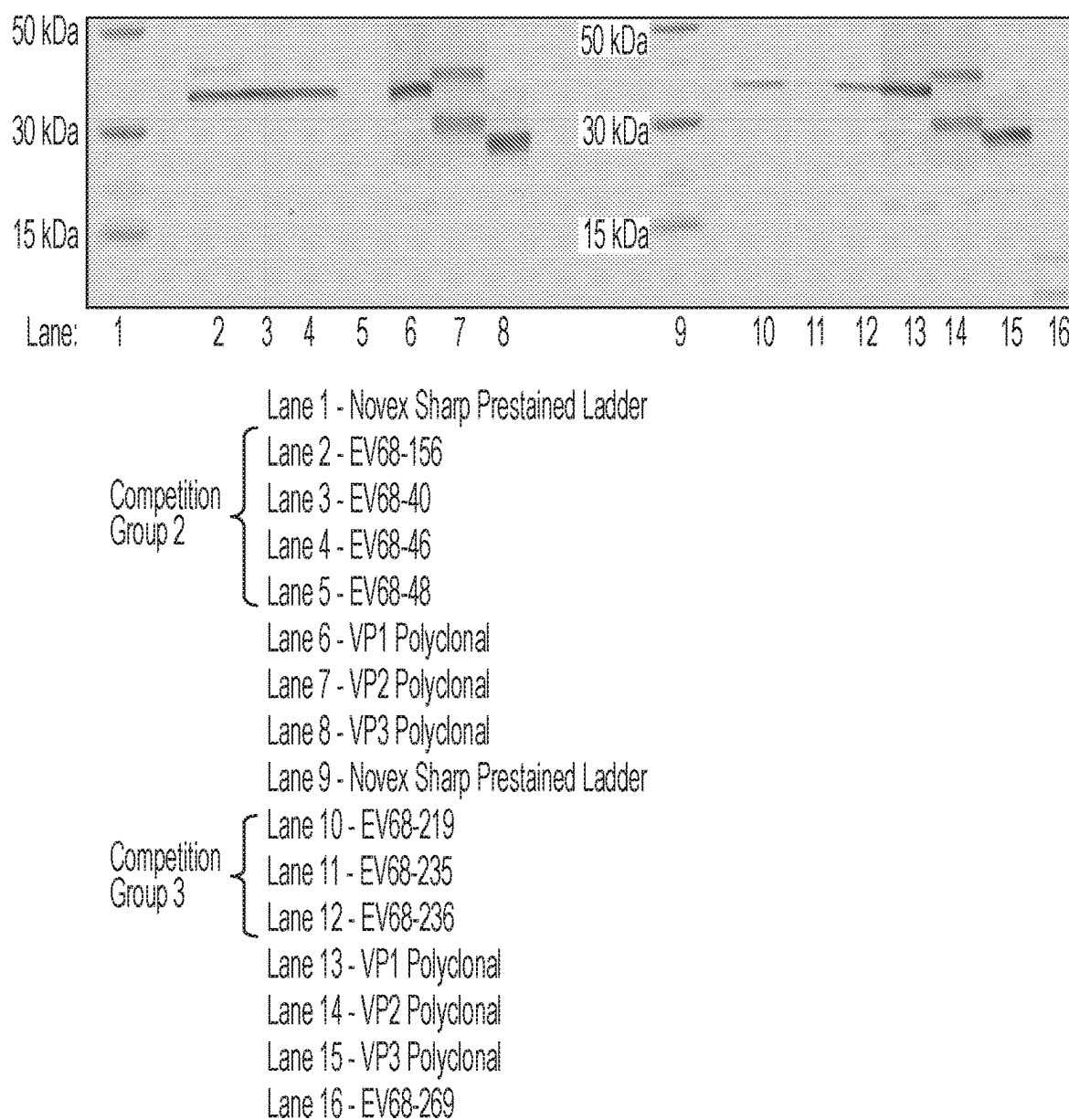


FIG. 40

Competition group	Neutralization IC ₅₀				Binding EC ₅₀					
	Clade and isolate tested									
	mAb	B1 IL/56	D KY/53	N/A Feron	B1 MO/47	B1 MO/49	B1 IL/56	B1 IL/52	D KY/53	
1	159	0.32	>	38,000	21	>	180	28	>	6
	150	16	28,000	2,800	12	56	36	15	>	1
	95	100	>	>	4,700	2,700	4,800	4,700	3,200	5
	266	250	70	>	57	14	>	4,200	30	5
	41	2,400	>	>	73	390	59	49	>	5
	98	2,400	>	>	13,000	68,000	52,000	>	>	5
	231	6,700	>	>	>	>	>	>	>	3
	178	16,000	>	>	>	>	>	>	>	1
	162	27,000	>	>	52	2,300	94	48	>	10
	161	>	>	>	1,700	>	50,000	2,000	>	8
	242	>	>	>	>	>	26,000	820	>	3
	72	>	>	>	160	150	290	48	>	1
	74	>	>	>	23	16	13	48	140	1
	75	>	>	>	>	>	>	>	>	1
	79	>	>	>	100	180	140	190	380	1
	114	>	>	>	>	>	>	>	>	1
	160	>	>	>	7,500	45,000	16,000	96,000	>	7
	181	>	>	>	1,400	18,000	2,500	50,000	>	1
	183	>	>	>	>	>	>	>	>	1

FIG. 41

	198	>		>	>	>	>	>	1	1	K
	210	>		>	>	>	>	>	1	1	L
	220	>		>	>	>	23,000	>	1	1	K
	221	>		>	>	>	>	>	3	1	L
	229	>		>	>	>	65	>	3	1	K
	234	>		>	5.5	6.5	>	>	3	1	K
	271	>		31,000	13,000	56,000	2,800	>	5	3	K
2	40	8.2	>	65,000	170	18,000	>	>	5	3	K
	48	11	5.2	20,000	940	11,000	46,000	>	5	3	K
	46	42	16,000	500	680	320	640	>	5	3	K
	156	370	>	40	51	36	31	>	3	1	L
											VP1
3	219	>		300	340	370	370	>	1	3	L
	235	>		940	1,100	310	10,000	>	3	3	L
	236	>		650	340	200	64,000	>	3	3	L
	153	1.6	>	>	>	>	>	>	2	1	L
	105	3.1	12	430	320	620	10,000	>	5	1	K
	228	3.7	5.2	82	78	54	540	>	3	1	K
	43	3.7	1,700	>	>	>	>	>	5	1	K
	208	790	4,200	>	>	>	>	>	1	1	K
	152	1,700	740	>	>	>	>	>	1	3	K

FIG. 41
CONTINUED

4	88	30,000	>	>	>	>	>	>	3	L
	227	32,000	>	>	740	480	400	>	3	K
	241	38,000	>	>	13,000	820	1,900	>	3	K
	185	38,000	>	>	11	28,000	>	9.2	1	K
	247	40,000	>	>	>	>	>	>	5	K
	37	>	>	>	17	14	13	>	5	L
	224	>	>	>	>	>	>	>	3	K
	254	>	>	>	420	980	2,000	5,500	5	K
5	166	5.5	48,000	>	>	>	>	>	1	L
	97	7.3	>	>	56,000	>	>	>	5	L
	85	14	3,000	>	>	>	>	>	3	K
	225	310	>	>	62,000	73,000	65,000	100,000	3	L
	269	>	>	>	>	>	>	>	5	L
	84	>	>	>	55,000	69,000	>	>	3	L
	89	>	>	>	40,000	55,000	27,000	43,000	3	L
	158	>	>	>	>	>	>	>	4	L
	164	>	>	>	30,000	30,000	32,000	26,000	11	K
	71	>	>	>	>	>	>	>	1	K
	76	>	>	>	13,000	16,000	>	>	1	L
	110	>	>	>	>	>	>	>	12	L
	116	>	>	>	1,200	4,200	1,700	17,000	1	L
	165	>	>	>	>	>	>	12,000	1	L

FIG. 41
CONTINUED

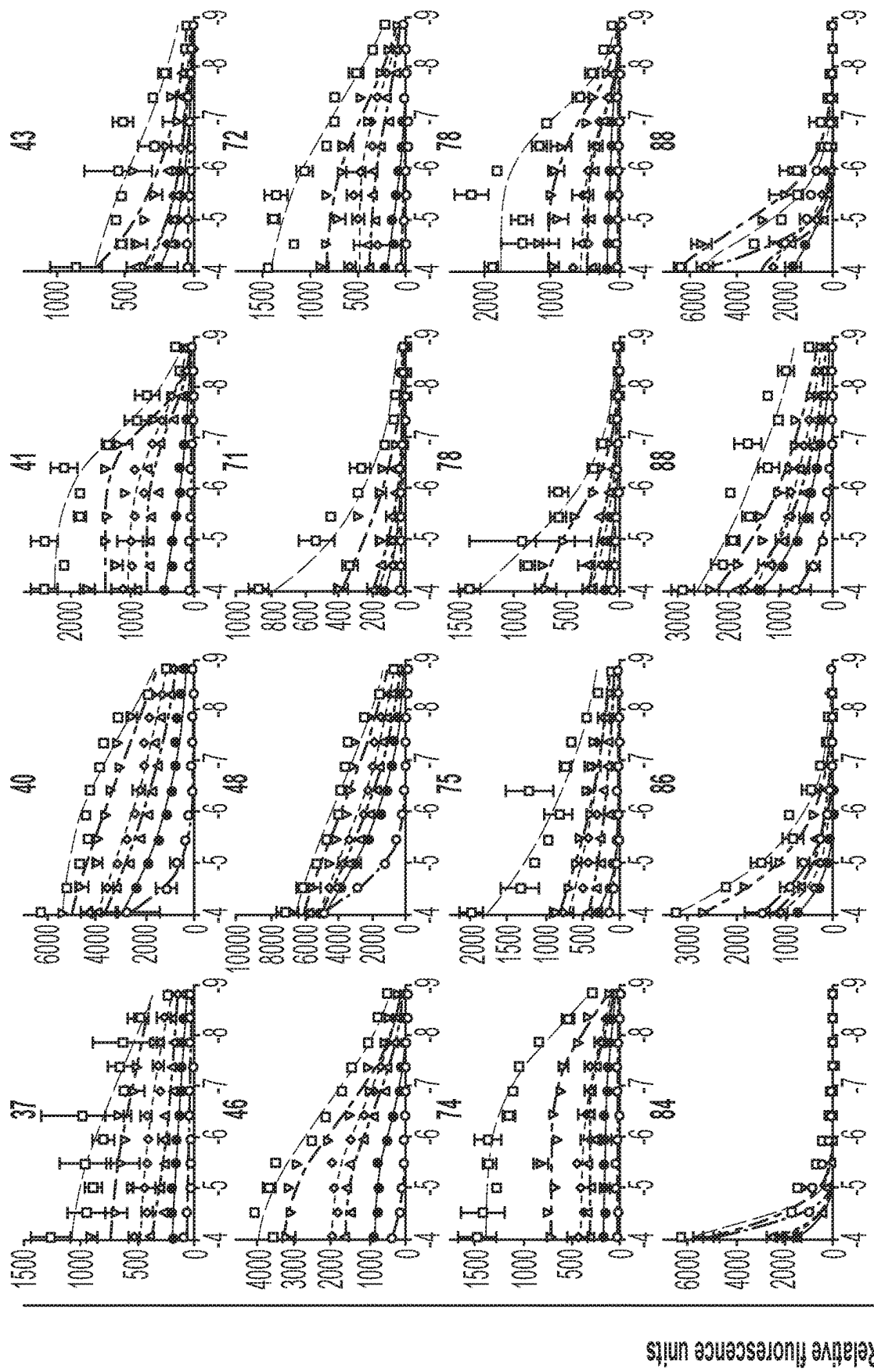


FIG. 42

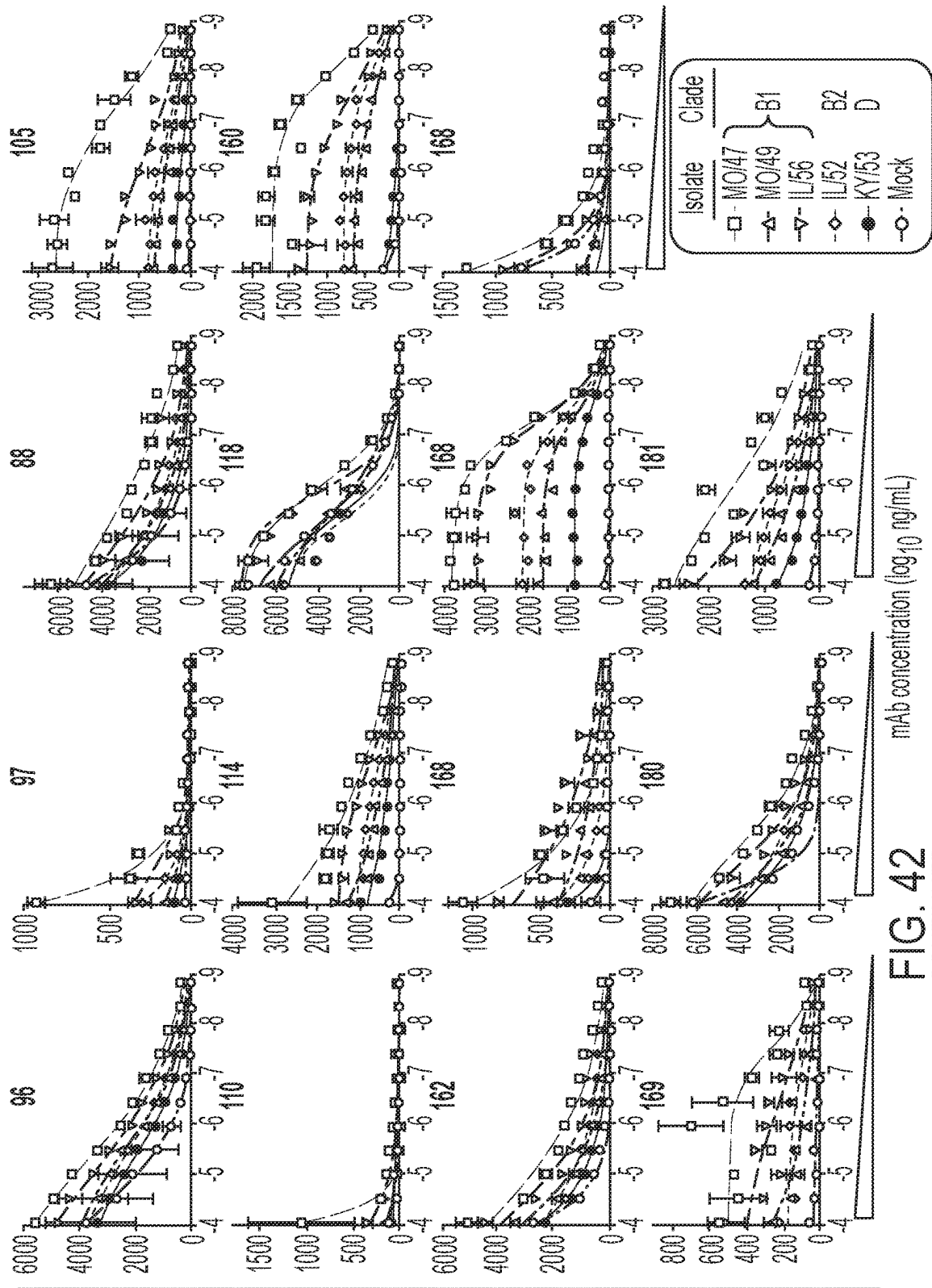


FIG. 42
CONTINUED

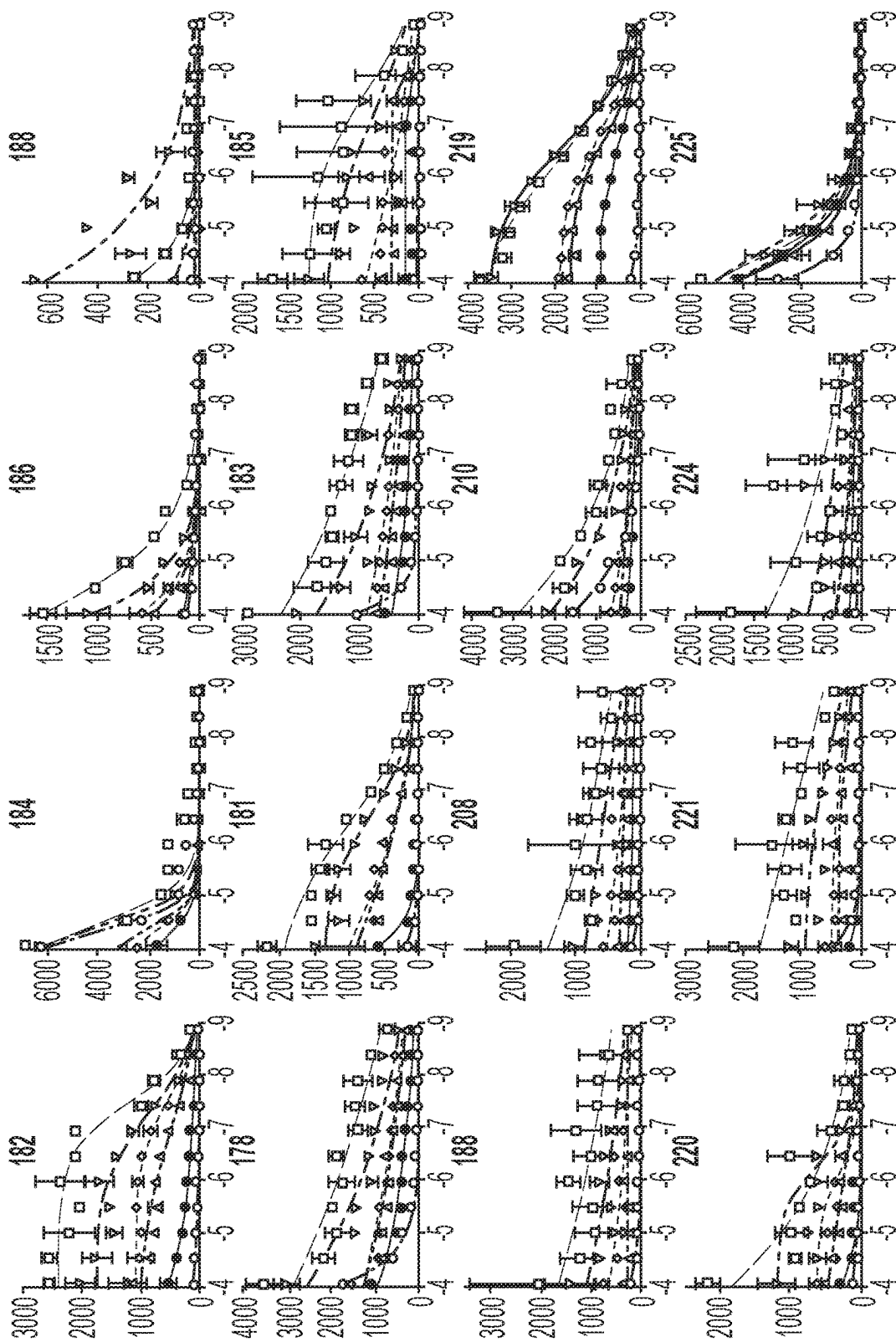
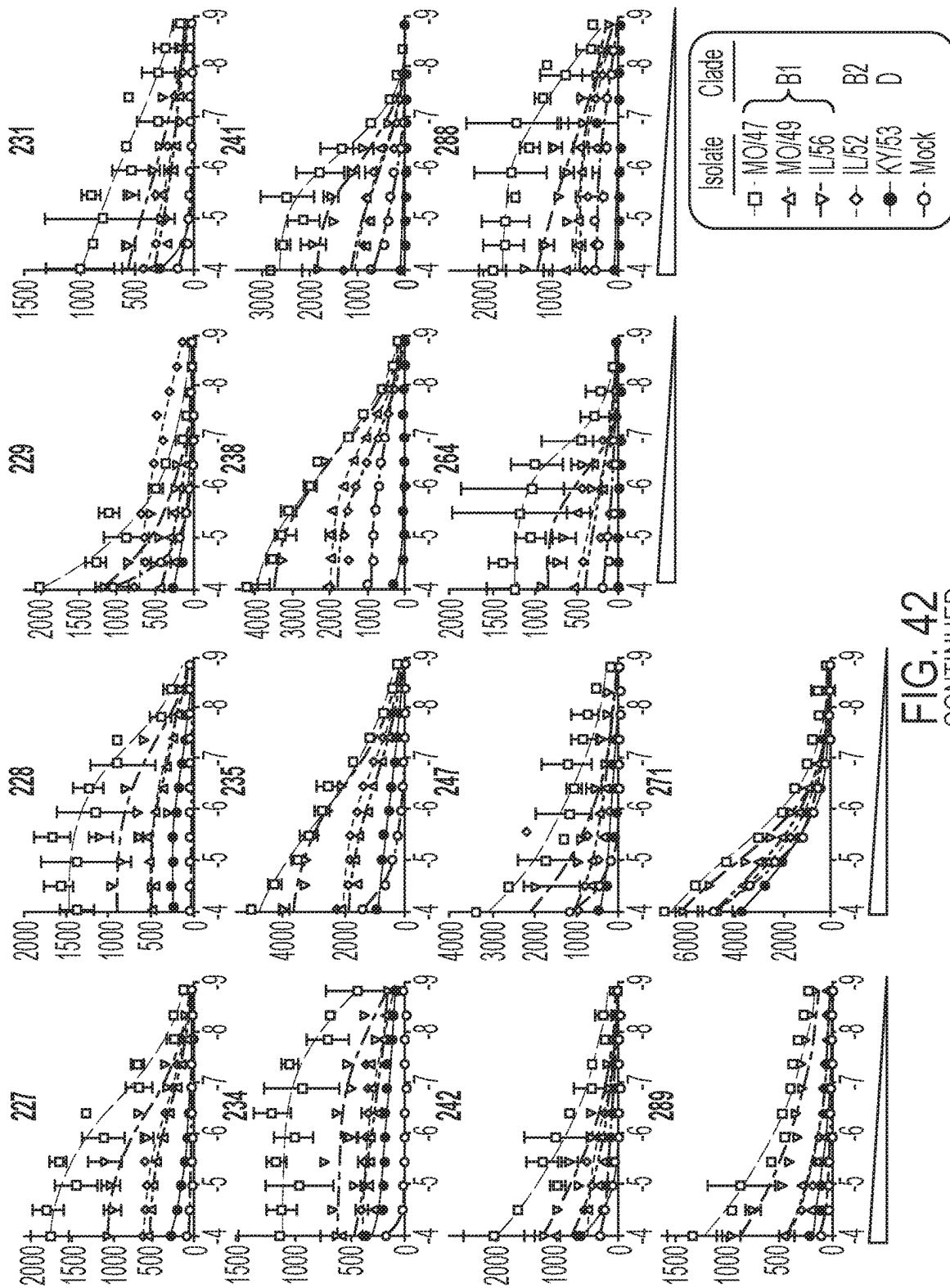


FIG. 42
CONTINUED

Relative fluorescence units



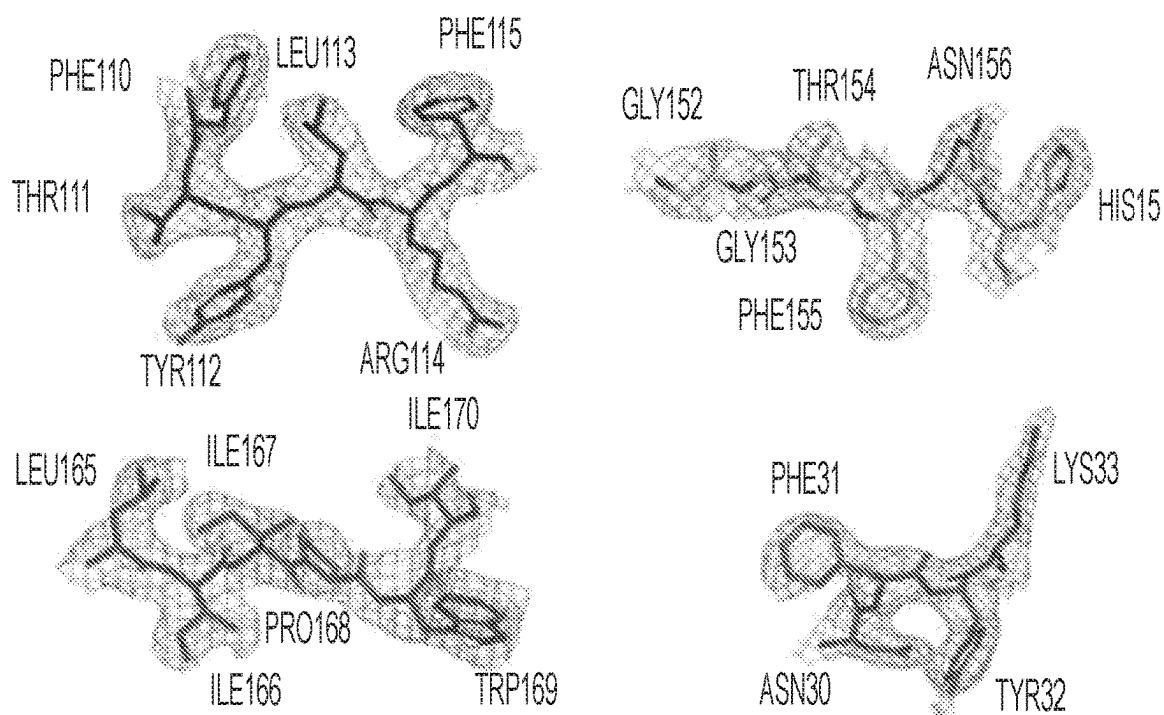


FIG. 43

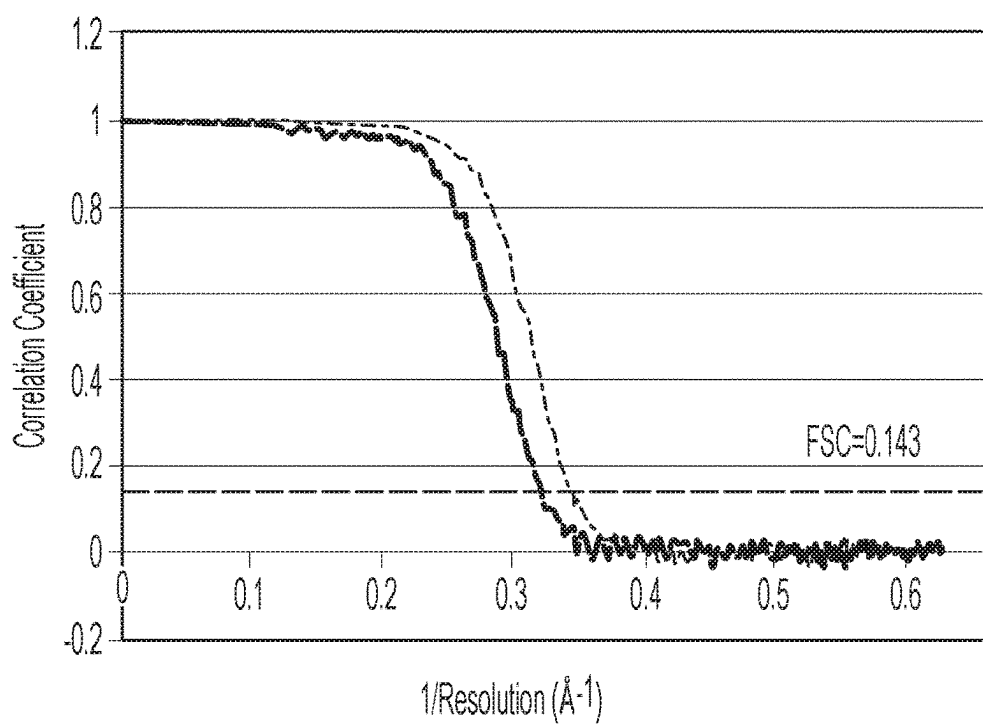


FIG. 44

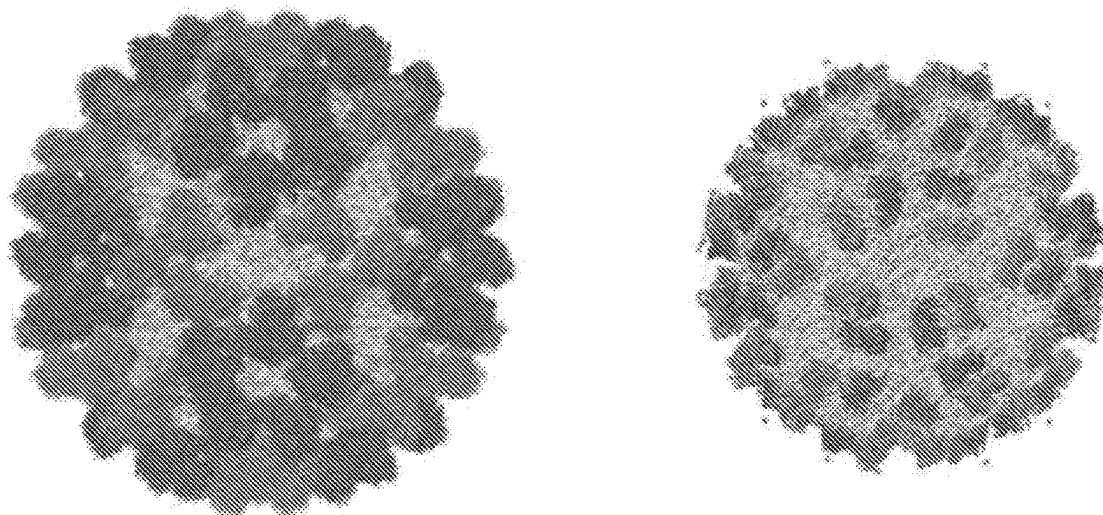


FIG. 45

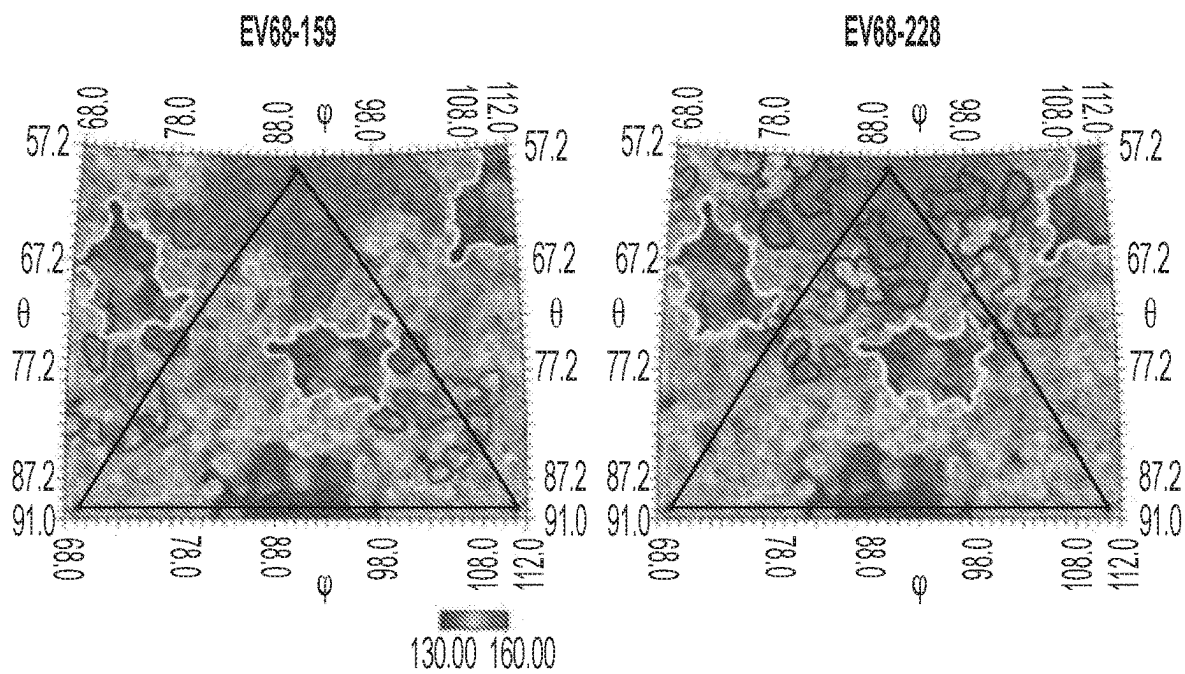


FIG. 46

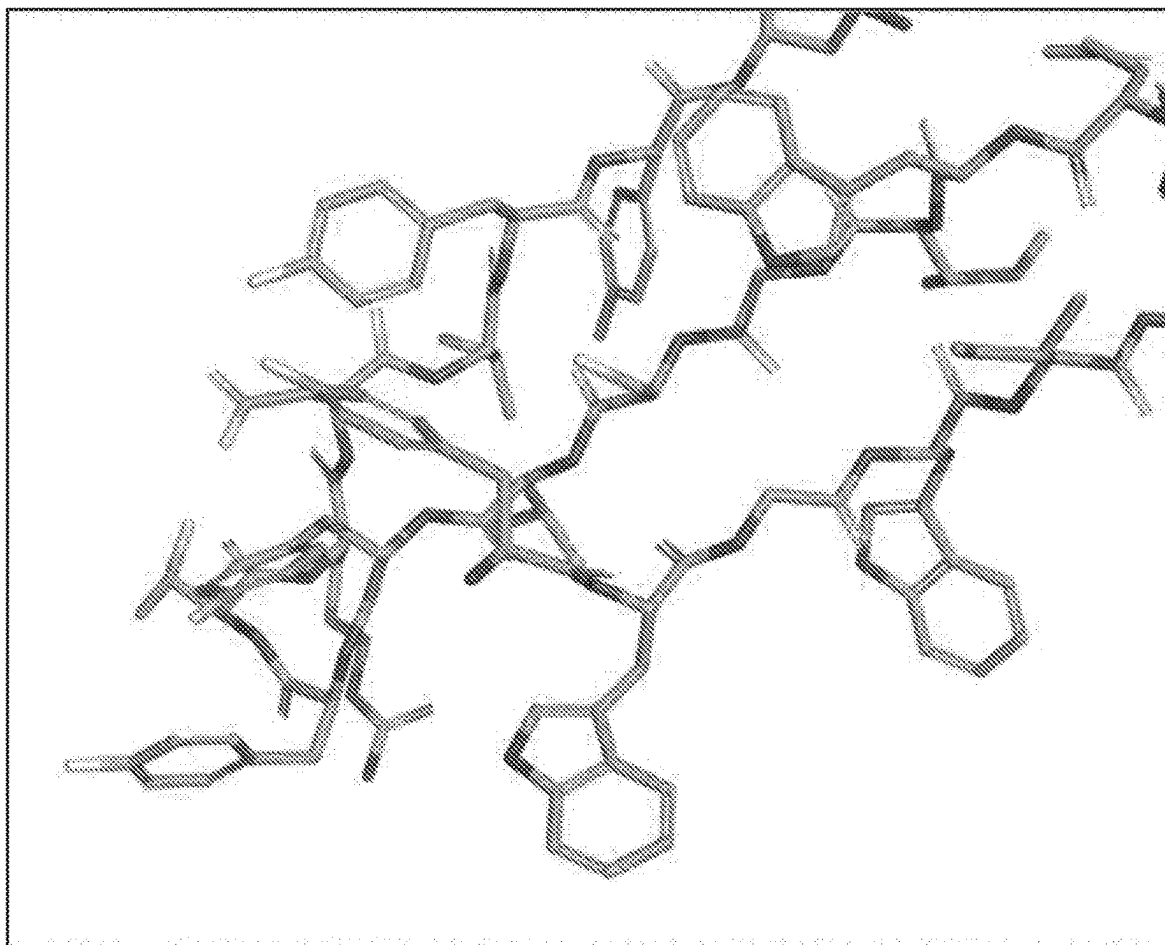


FIG. 47

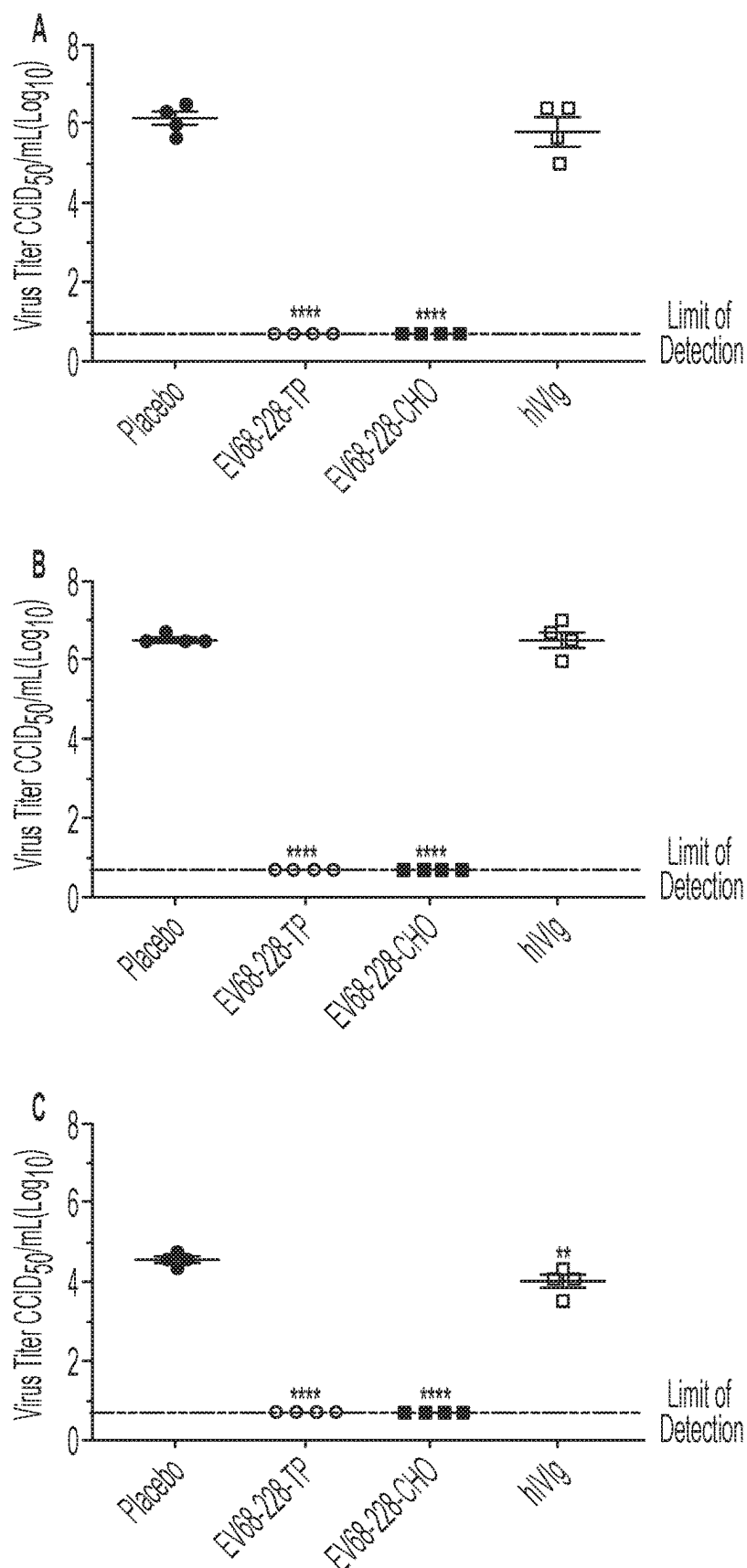


FIG. 48A-C

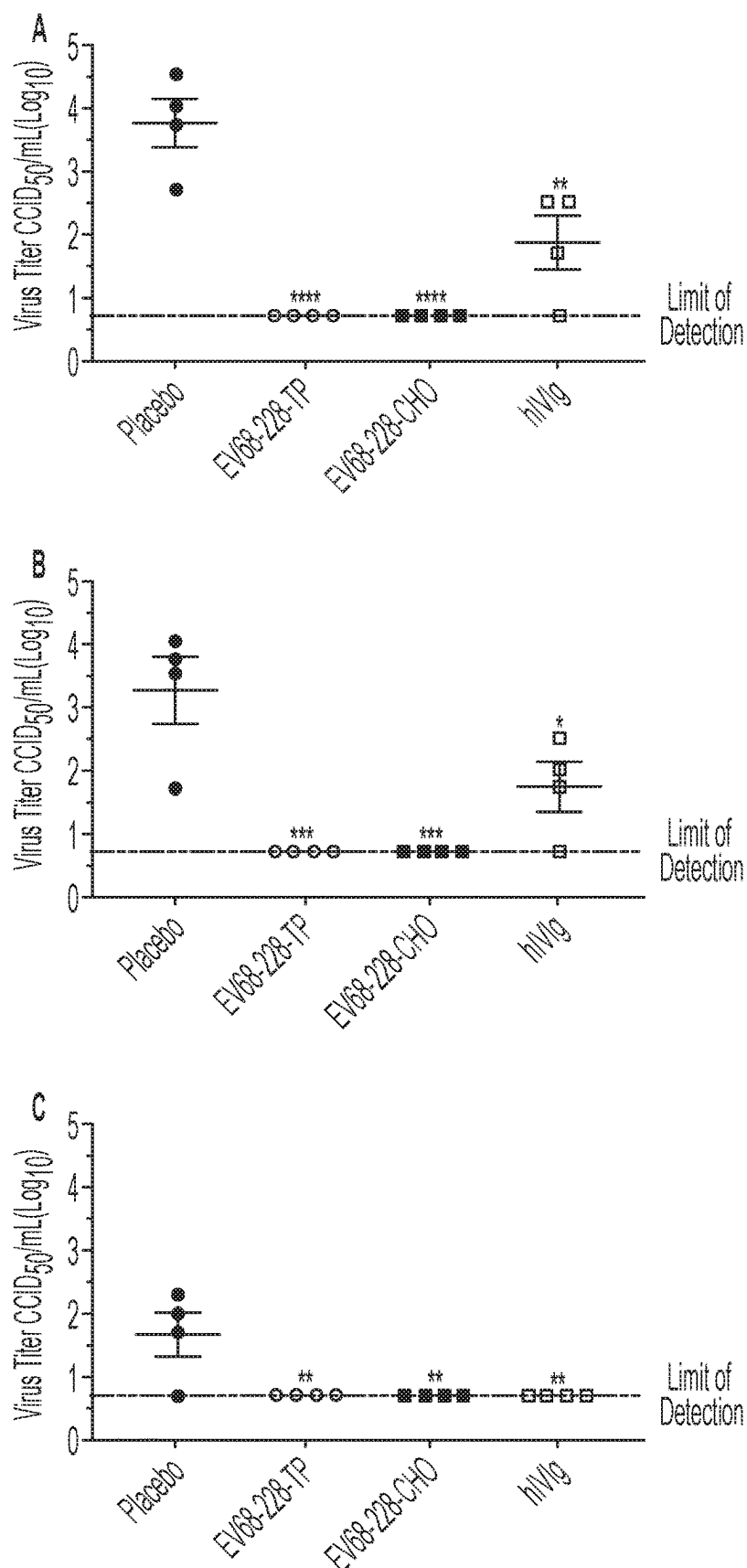
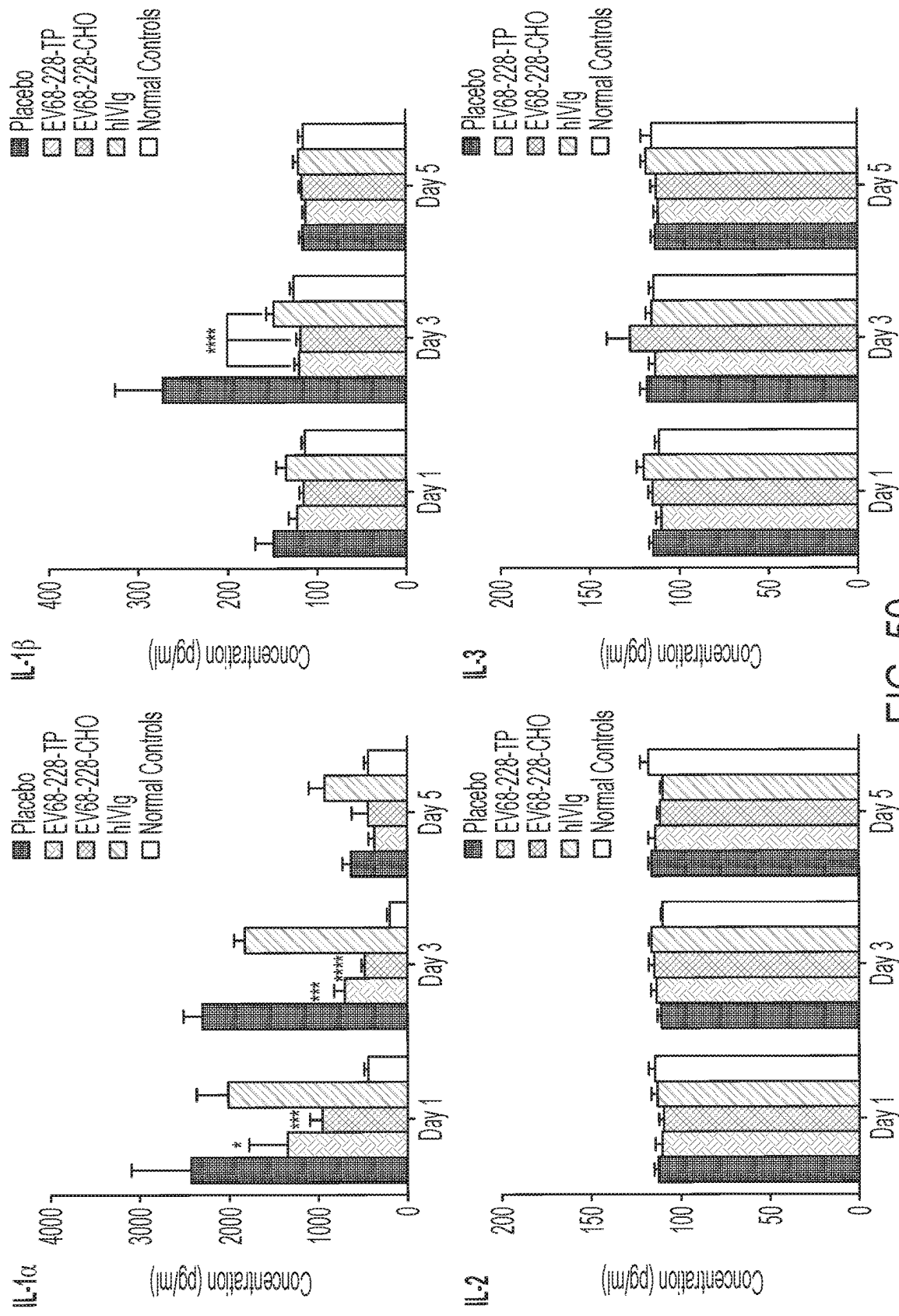


FIG. 49A-C



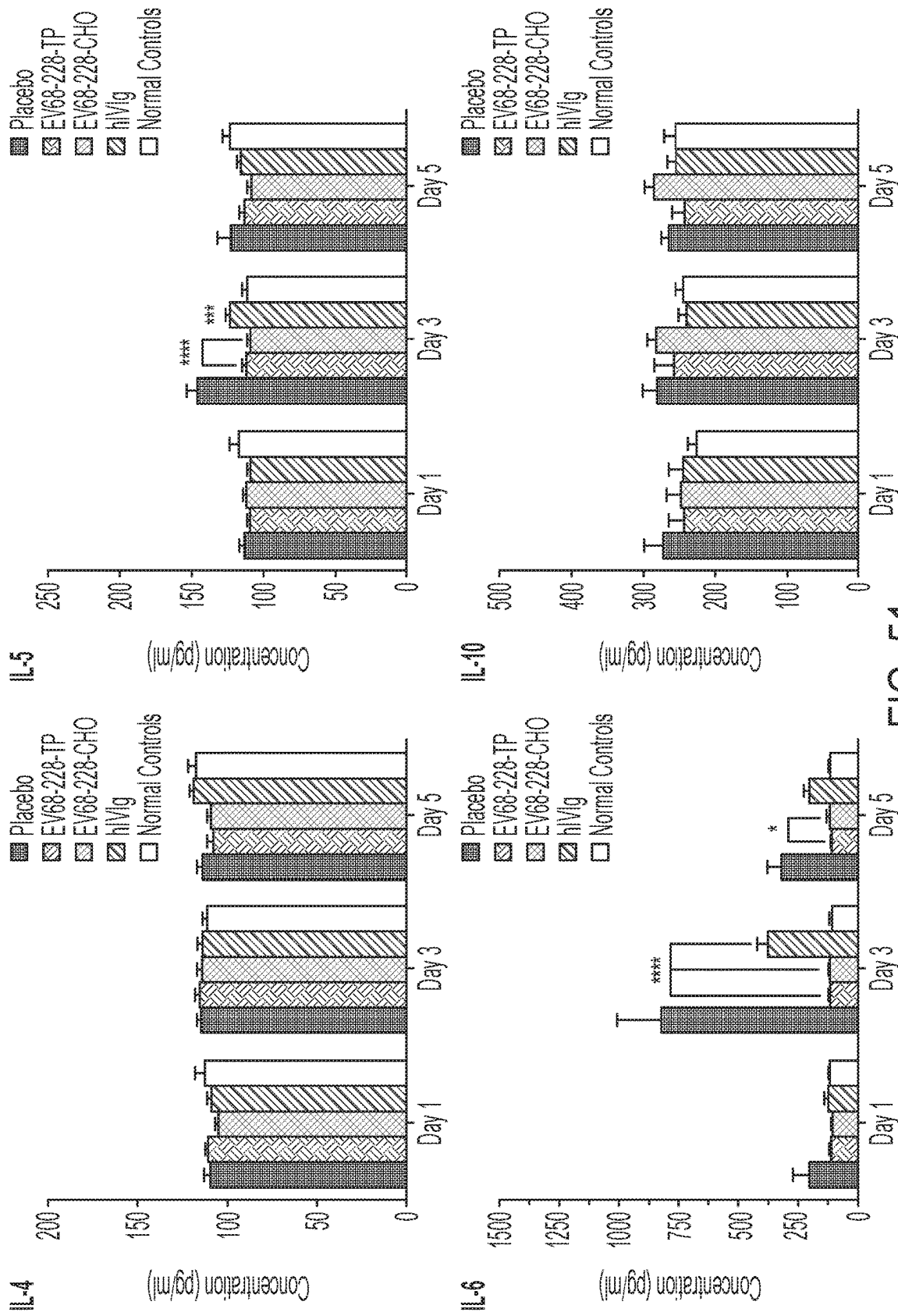
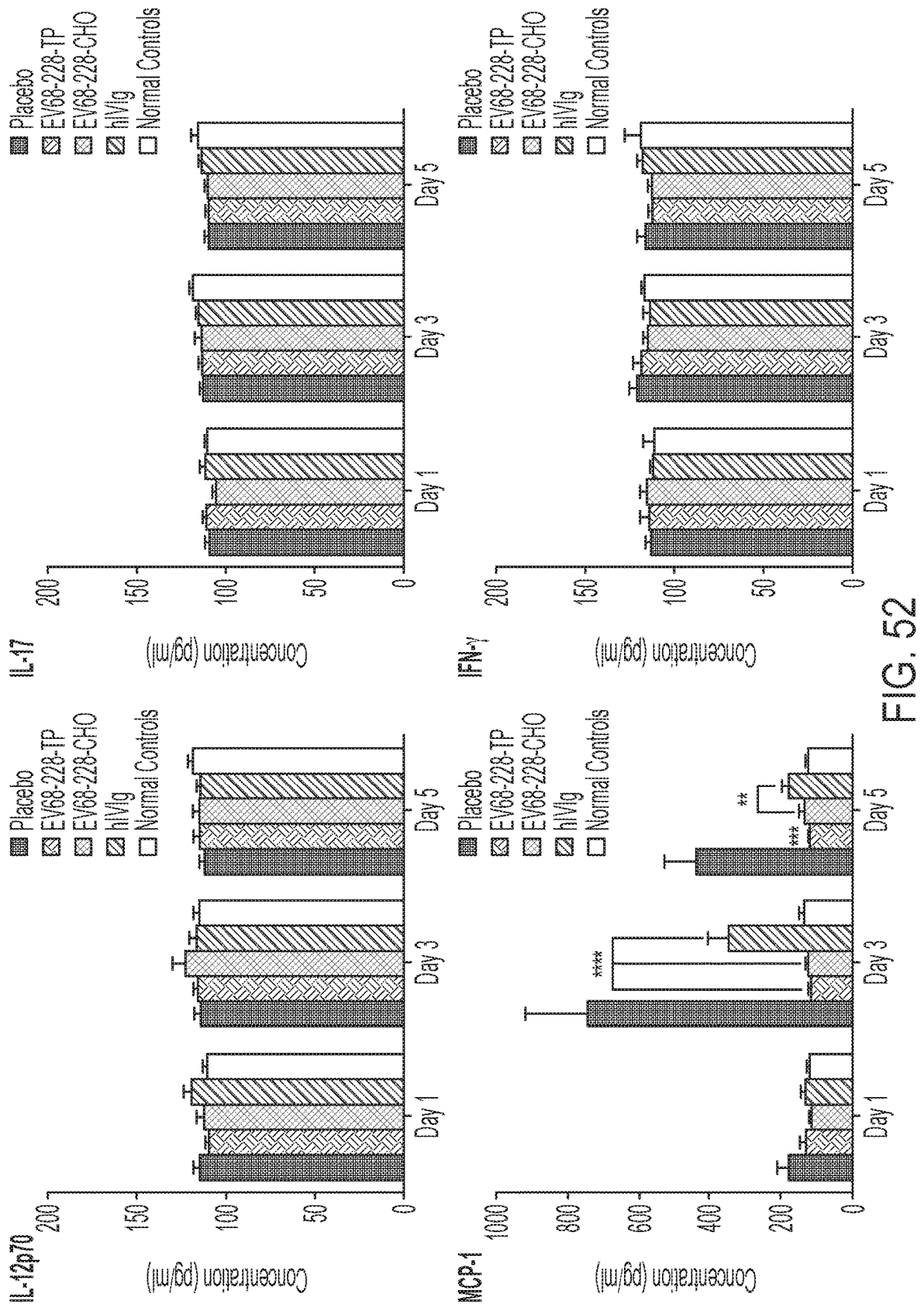


FIG. 51



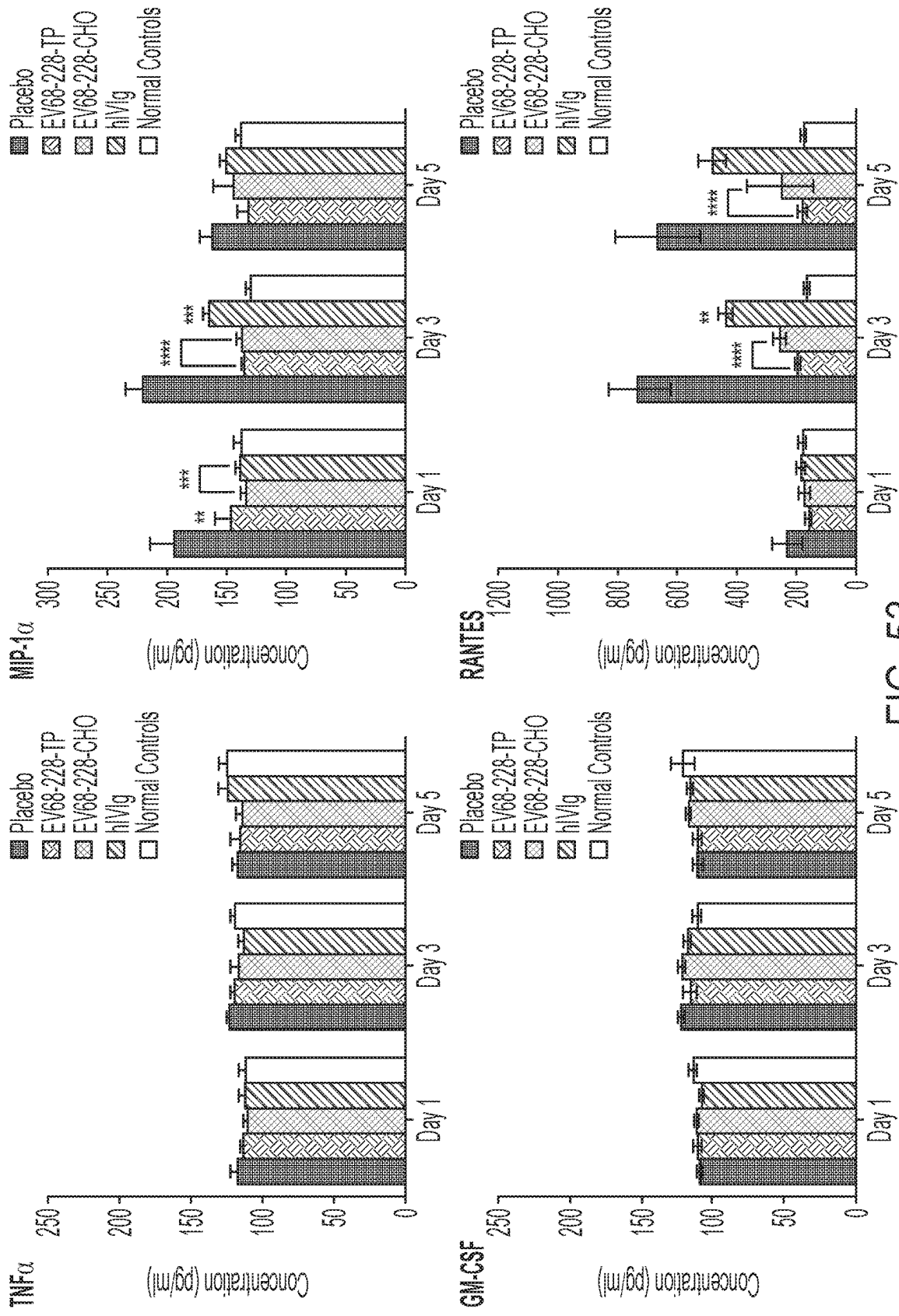


FIG. 53

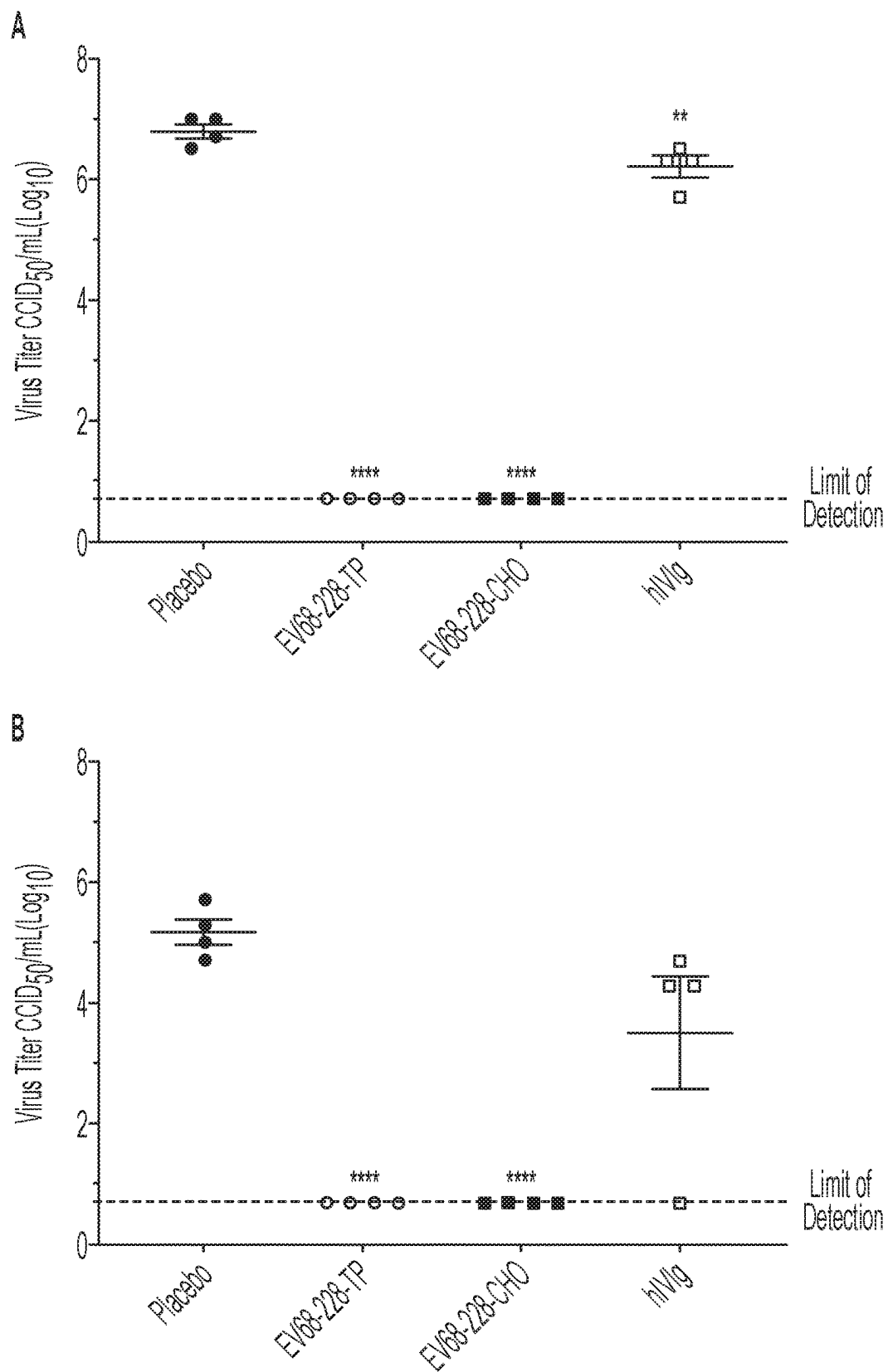


FIG. 54A-B

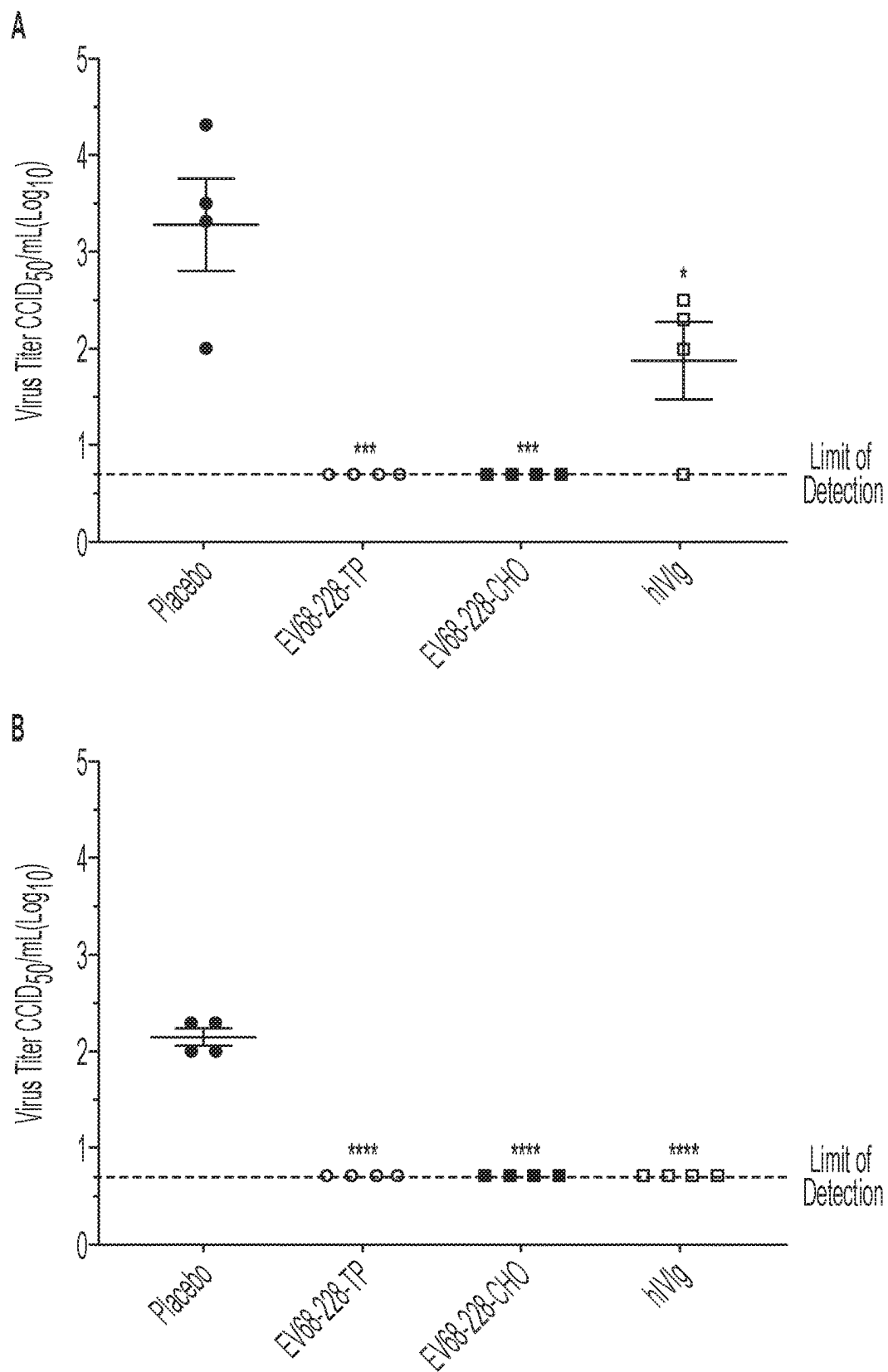
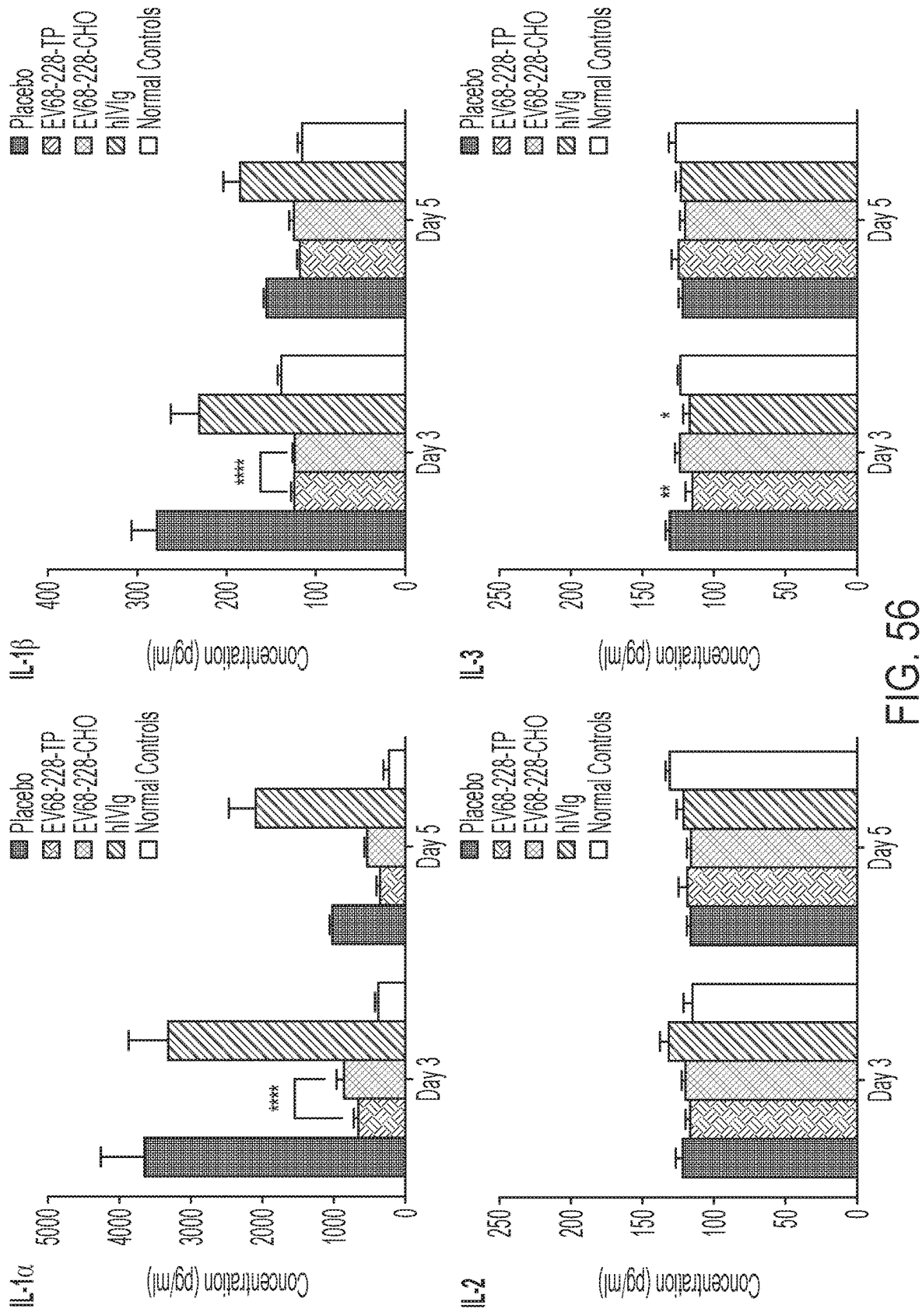
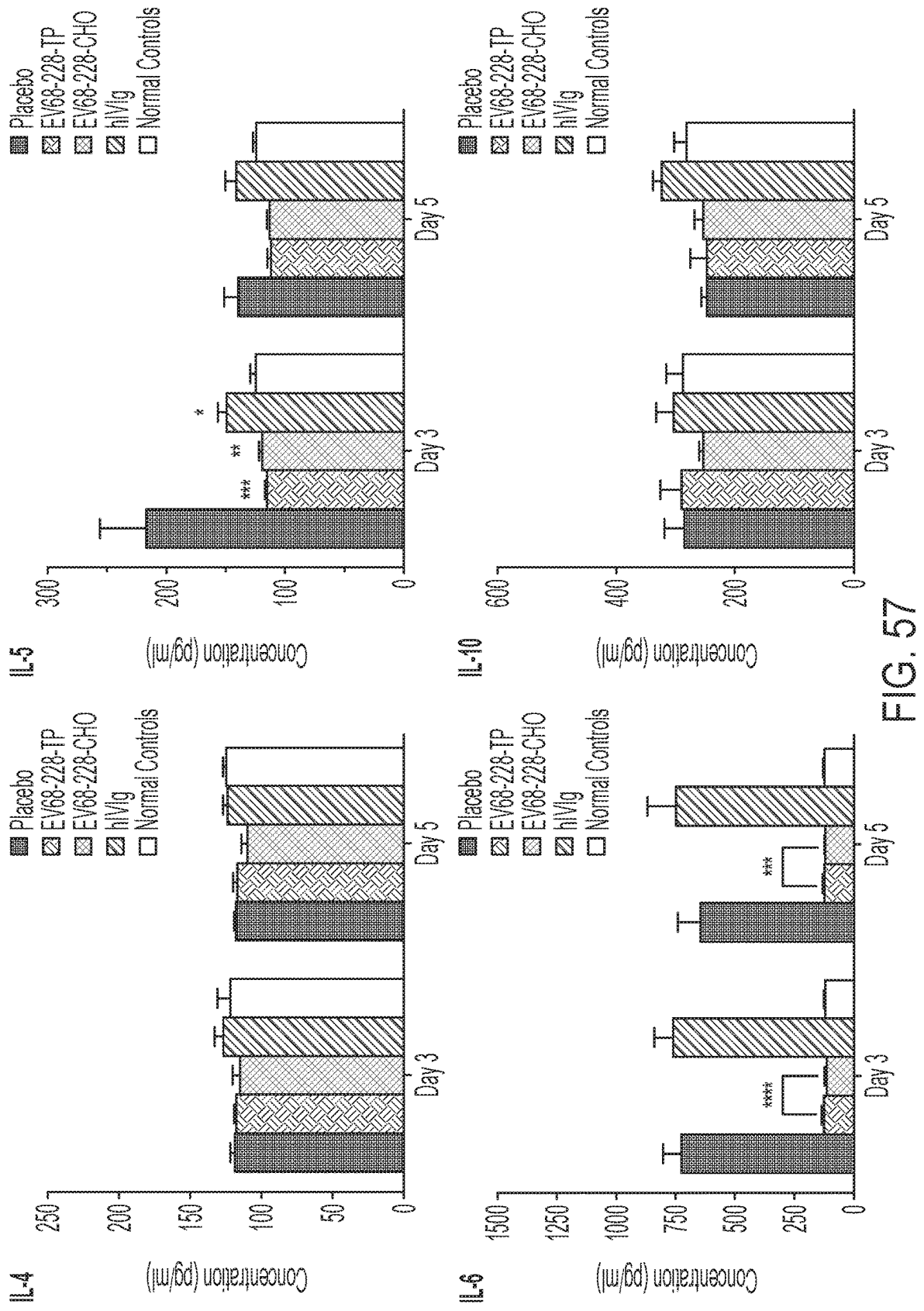
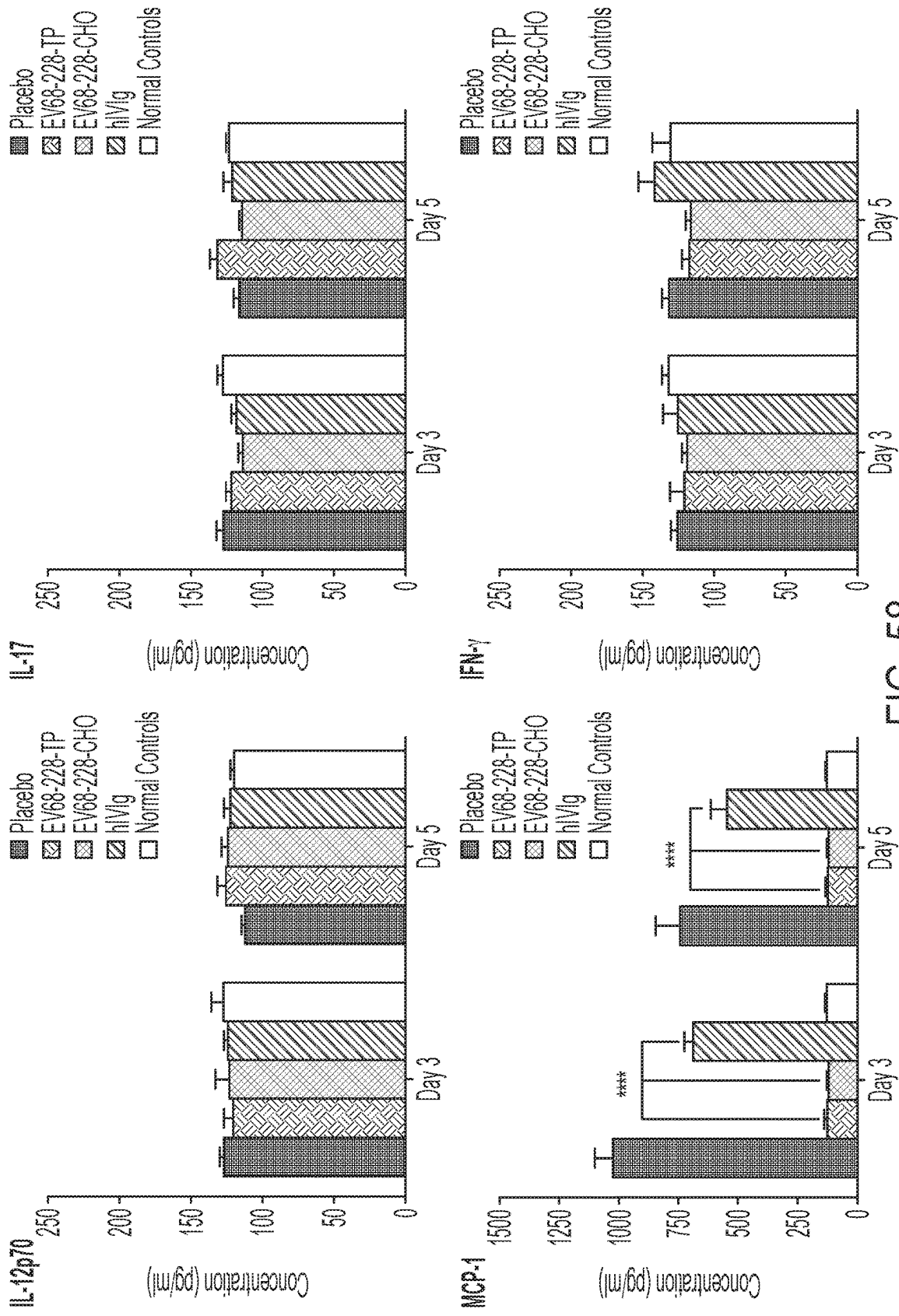


FIG. 55A-B







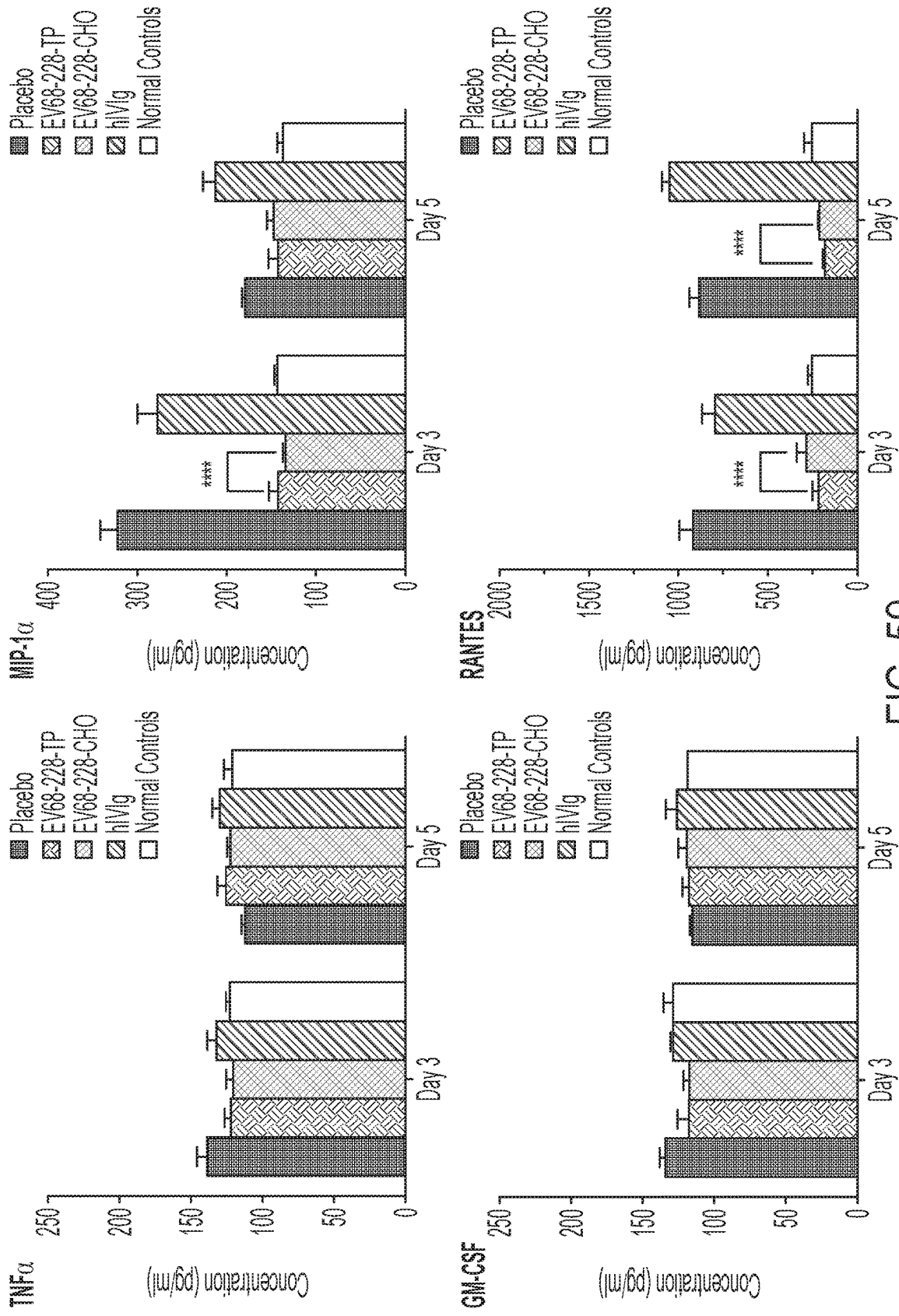


FIG. 59

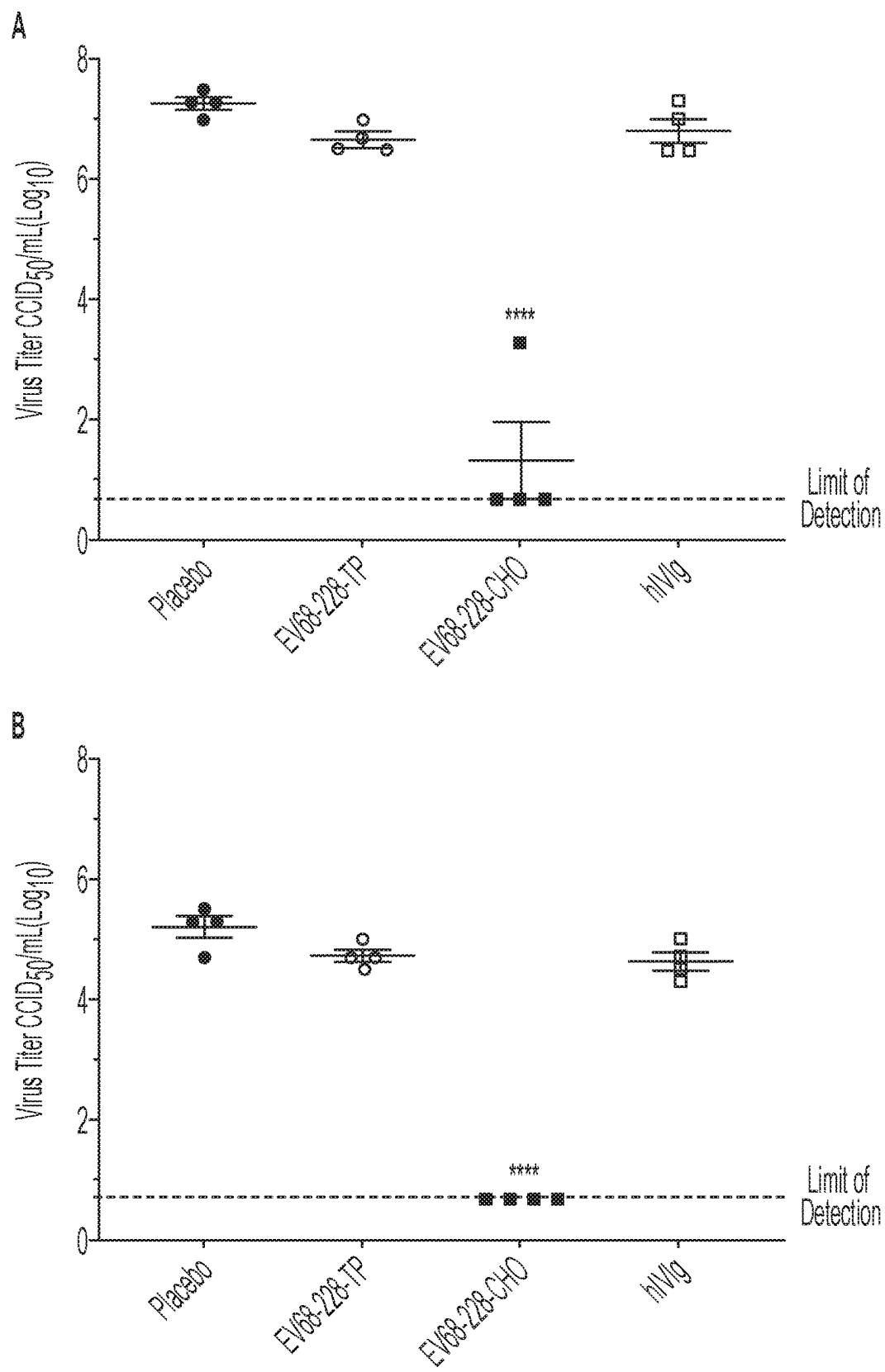


FIG. 60A-B

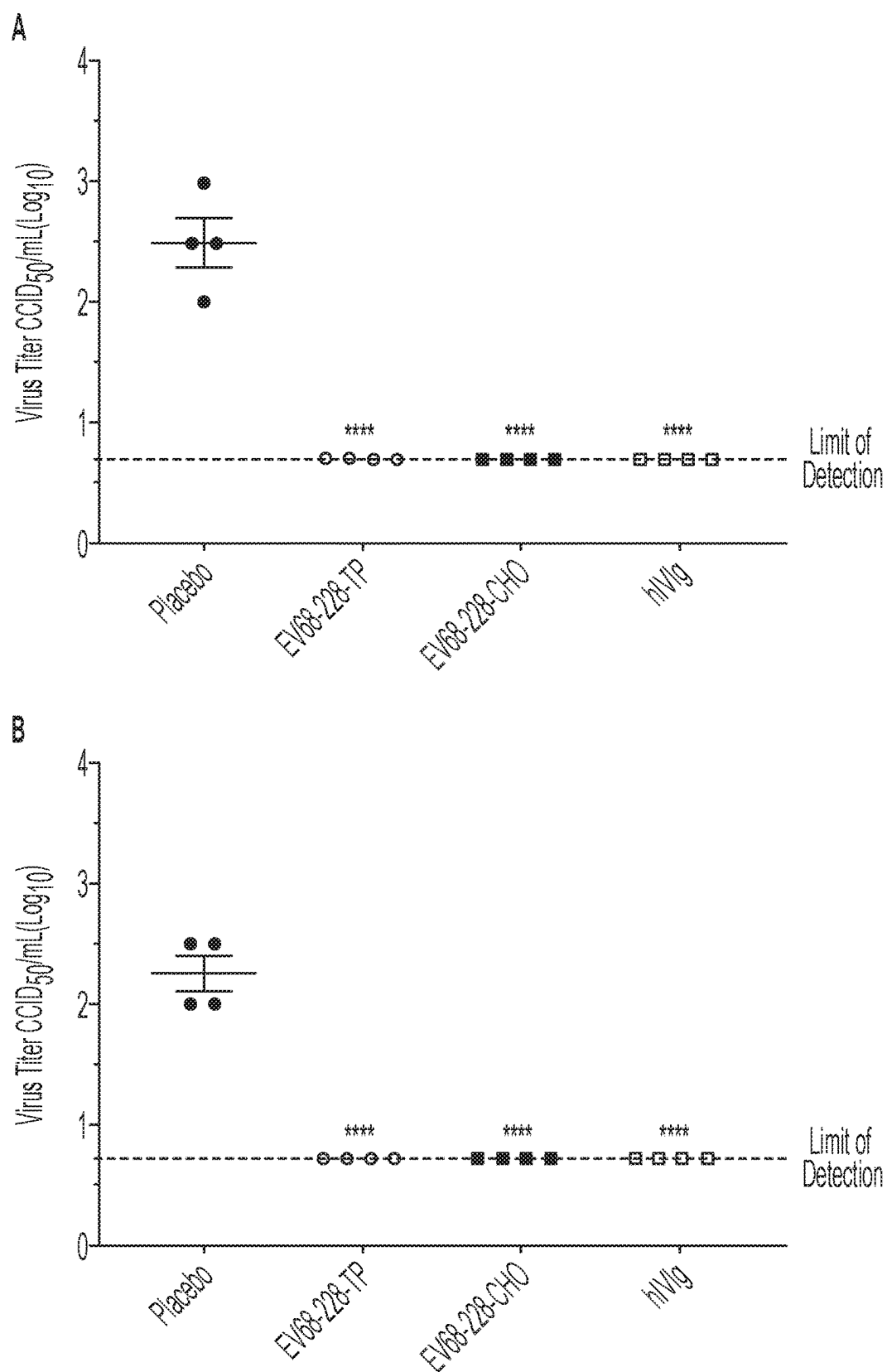
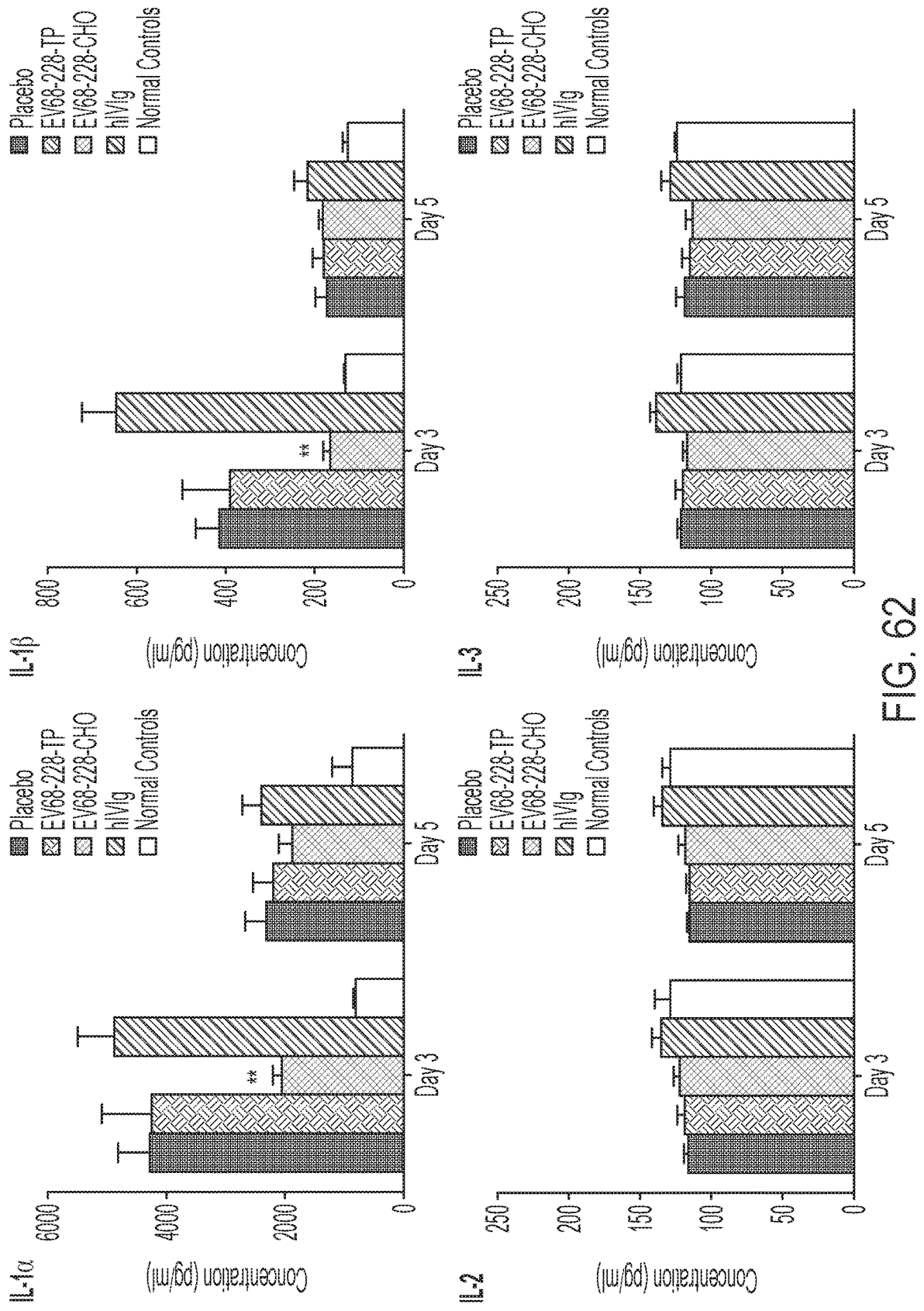
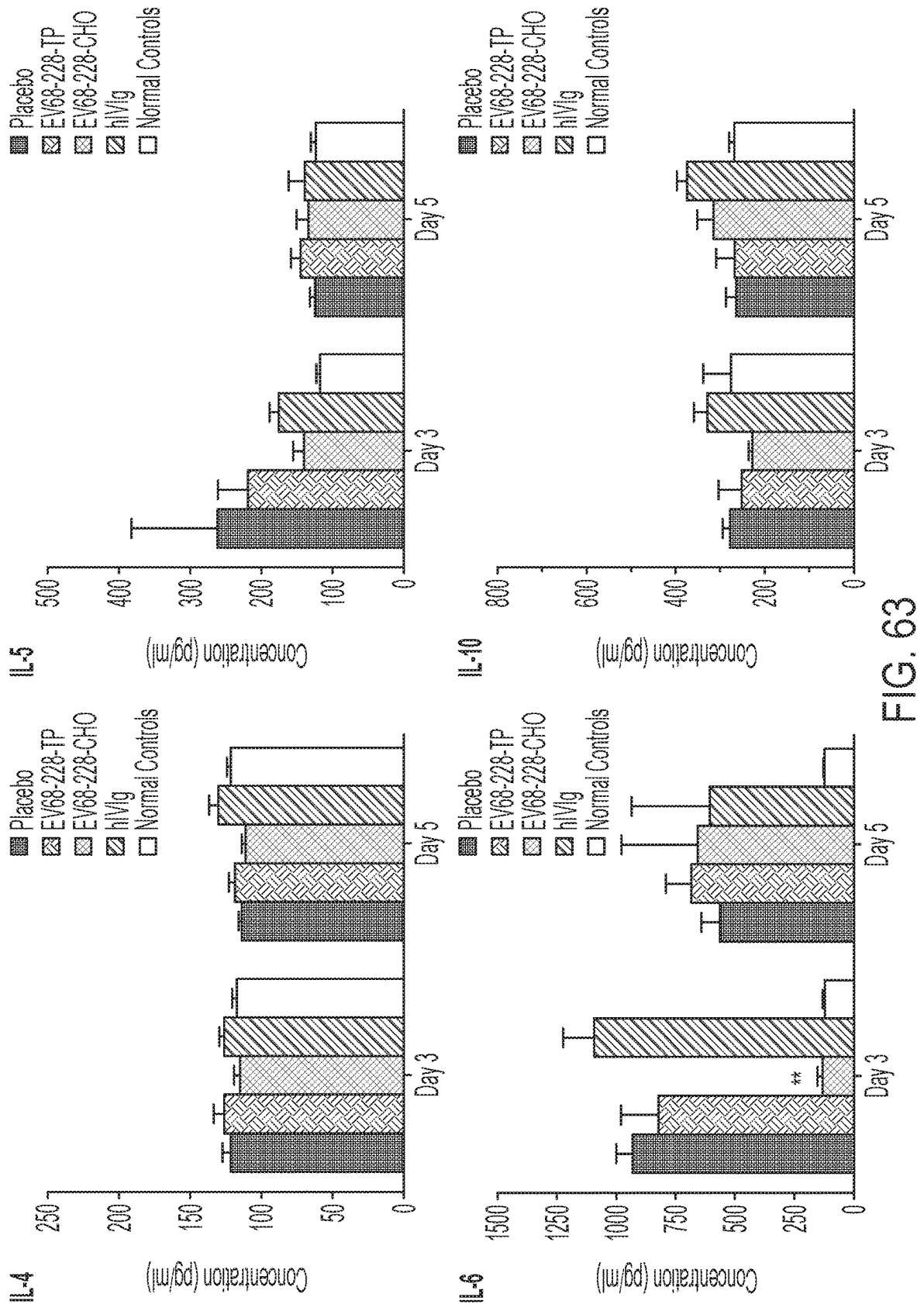
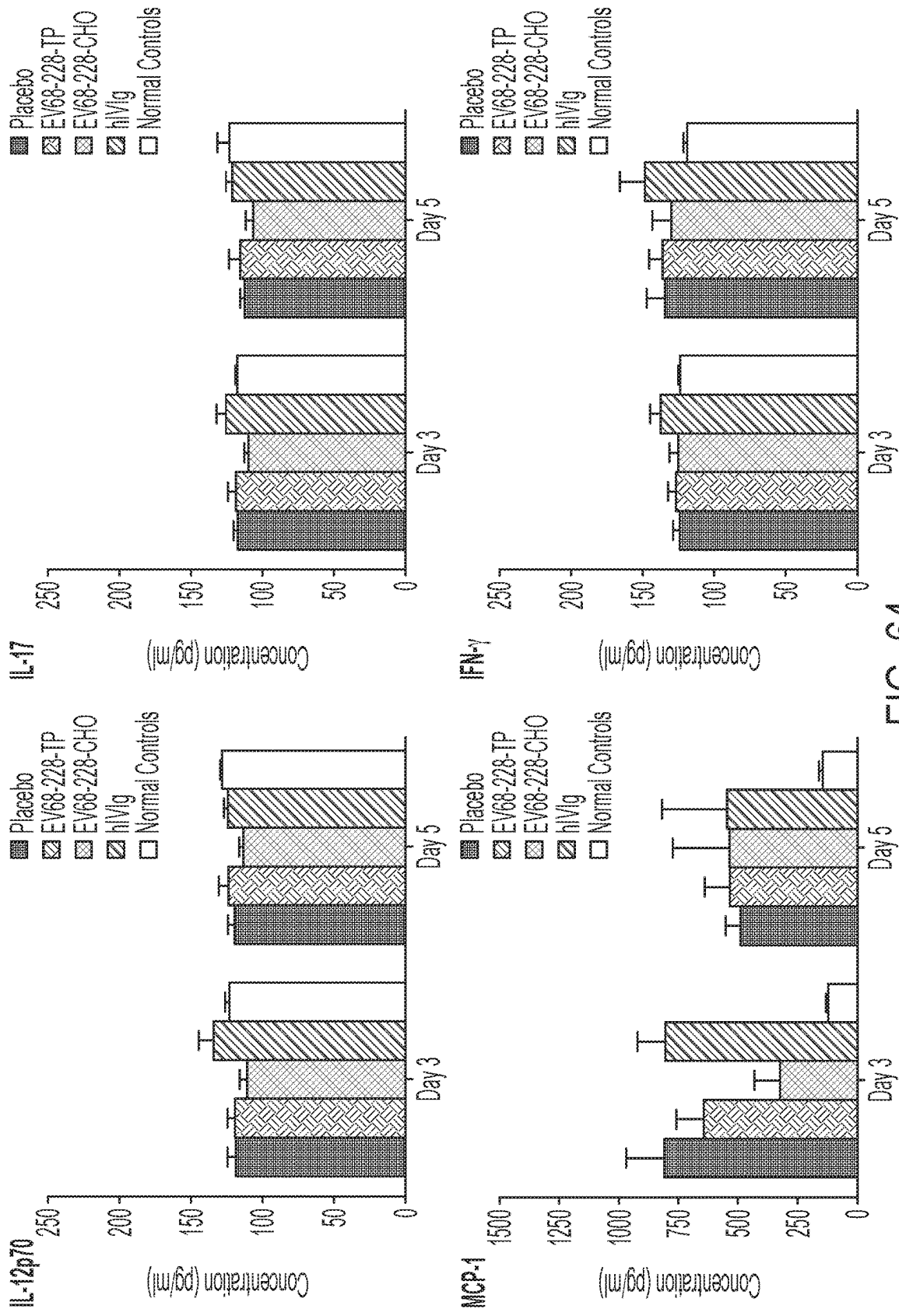
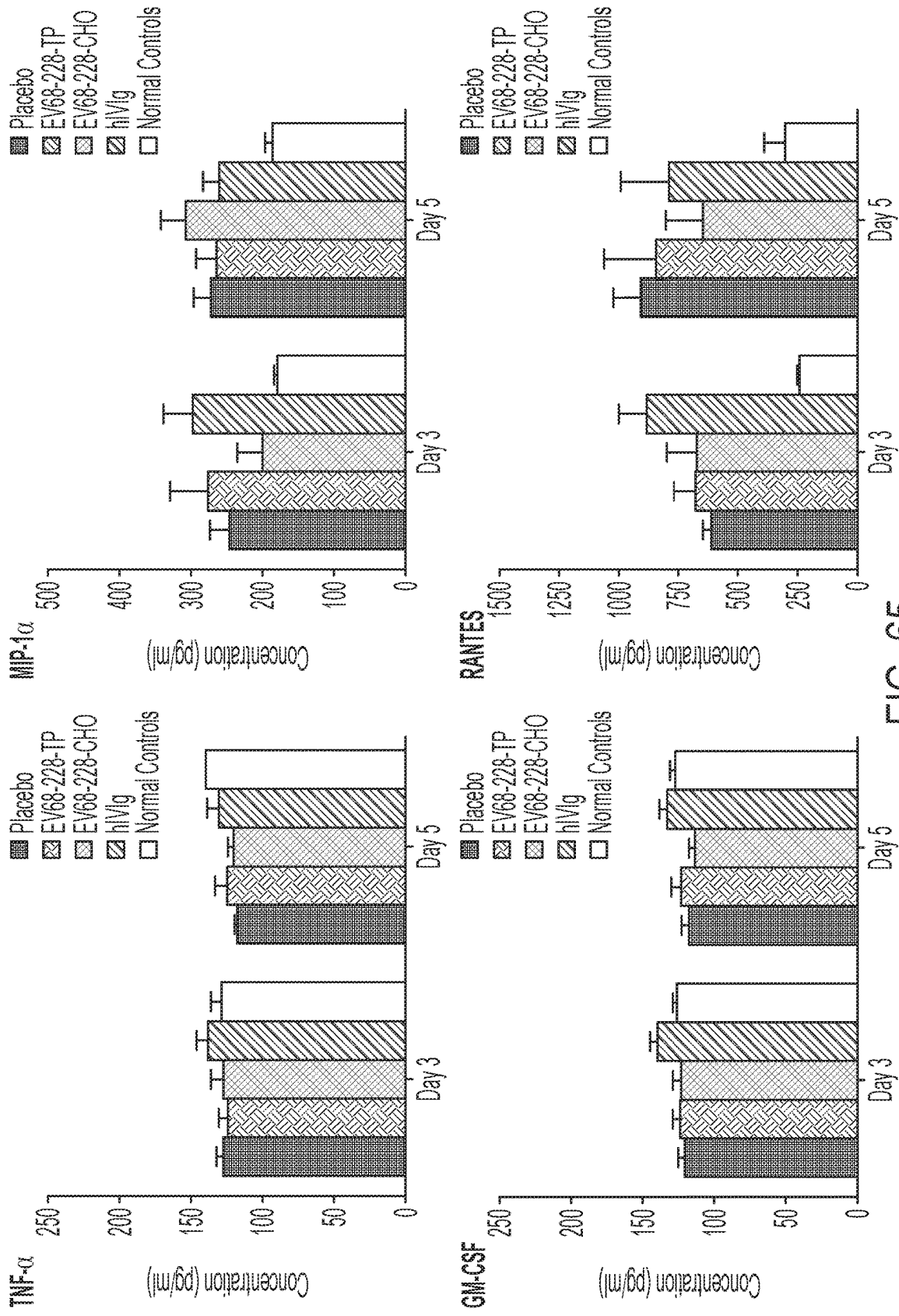


FIG. 61A-B









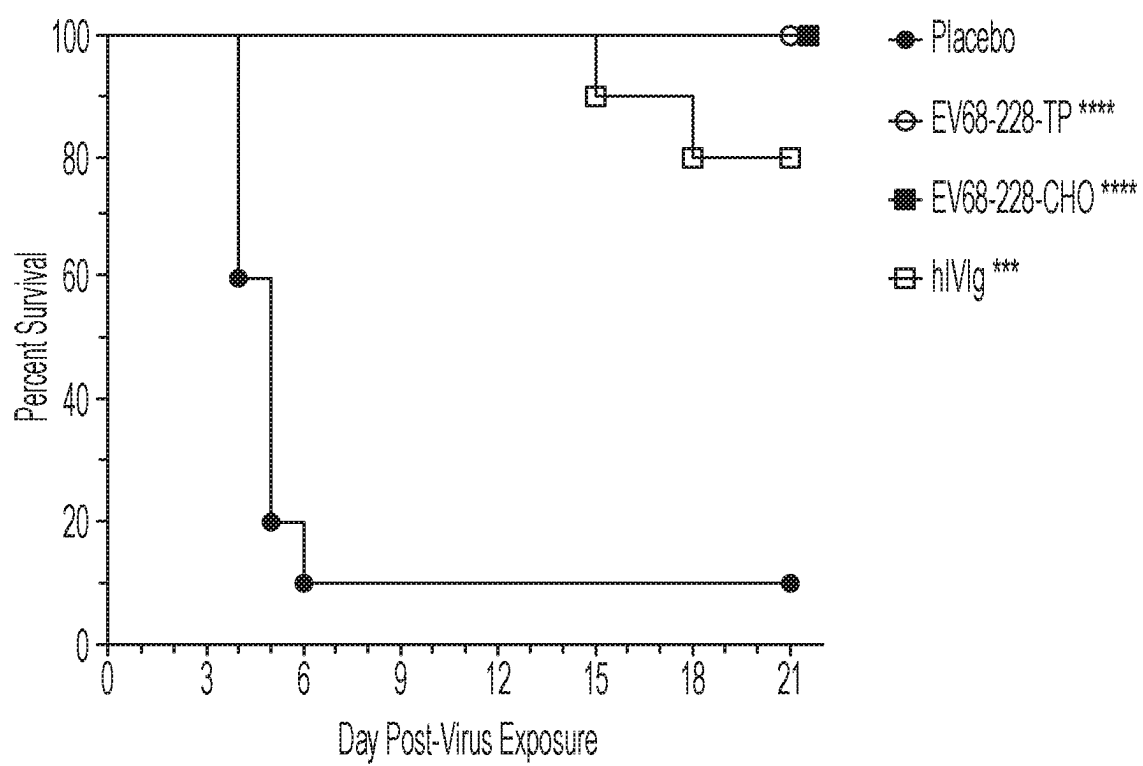


FIG. 66

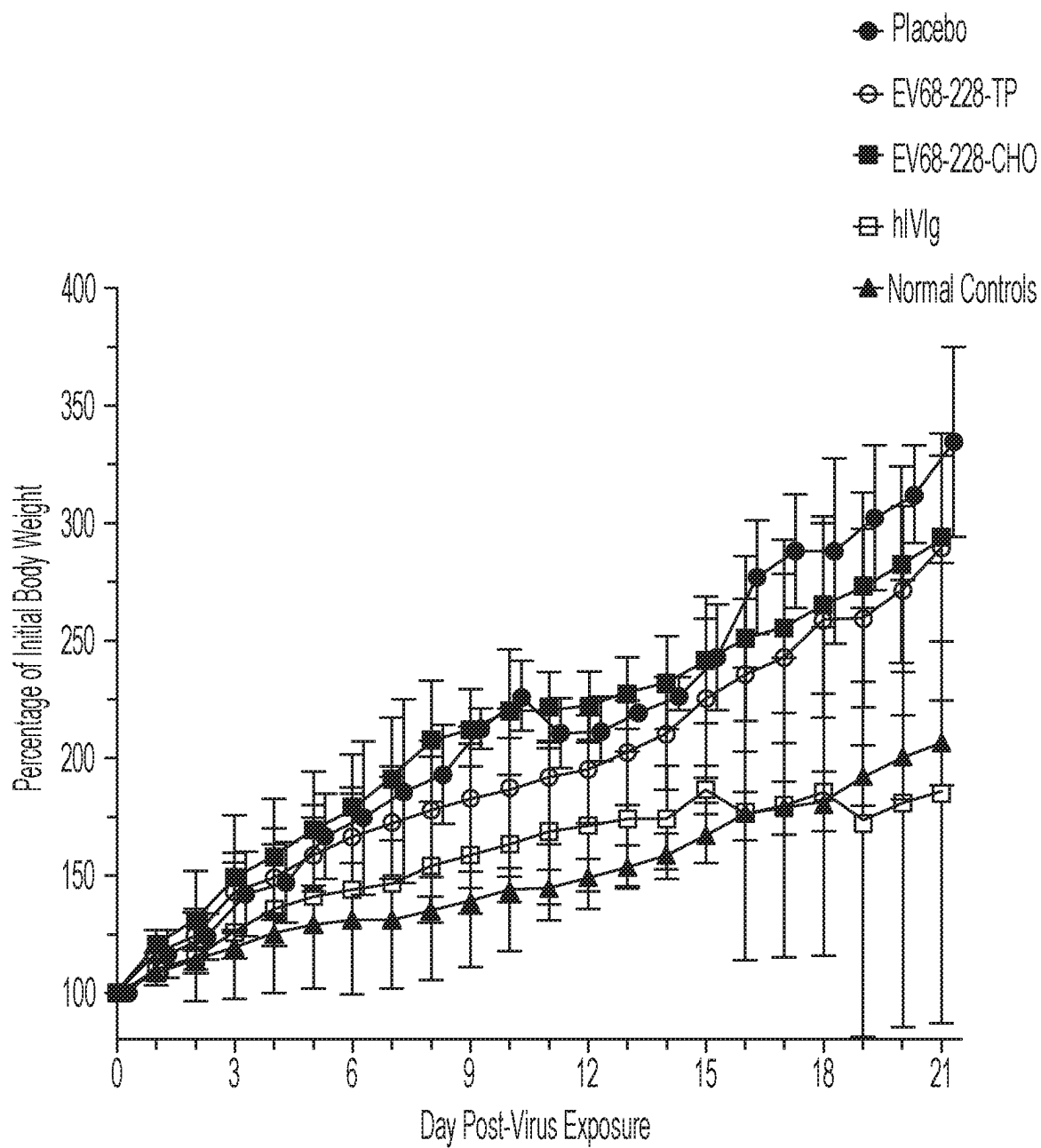


FIG. 67

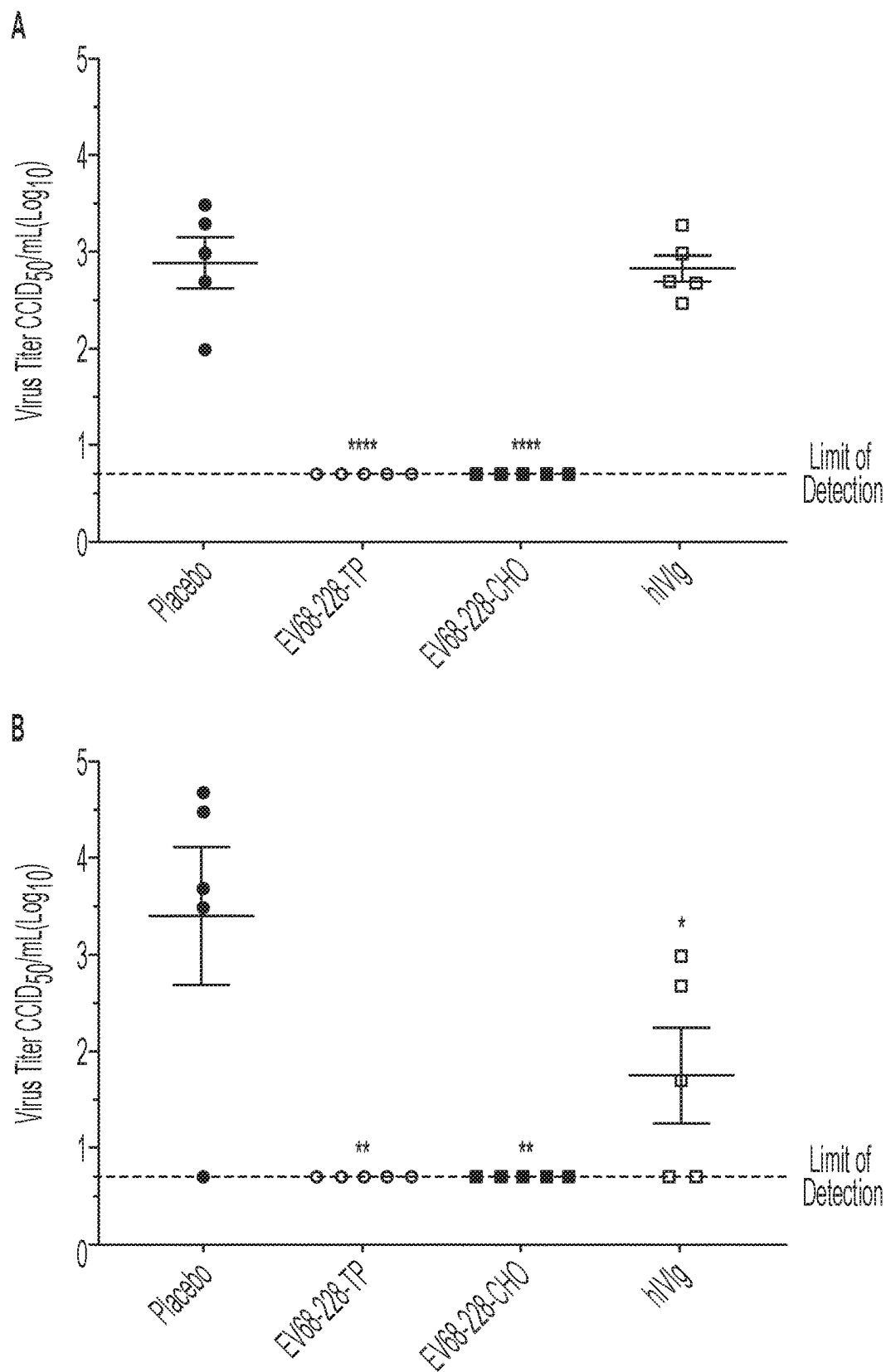


FIG. 68A-B

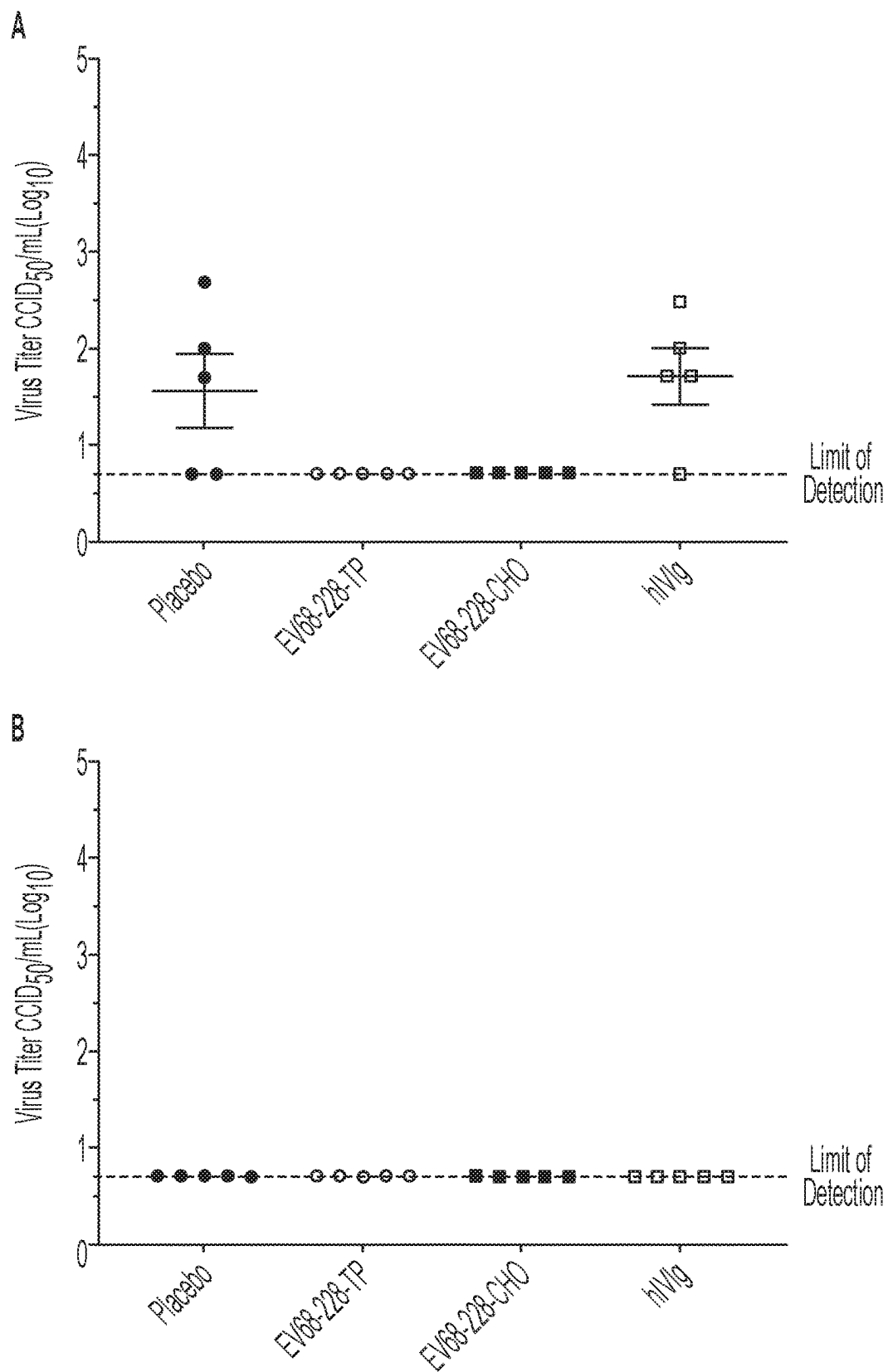


FIG. 69A-B

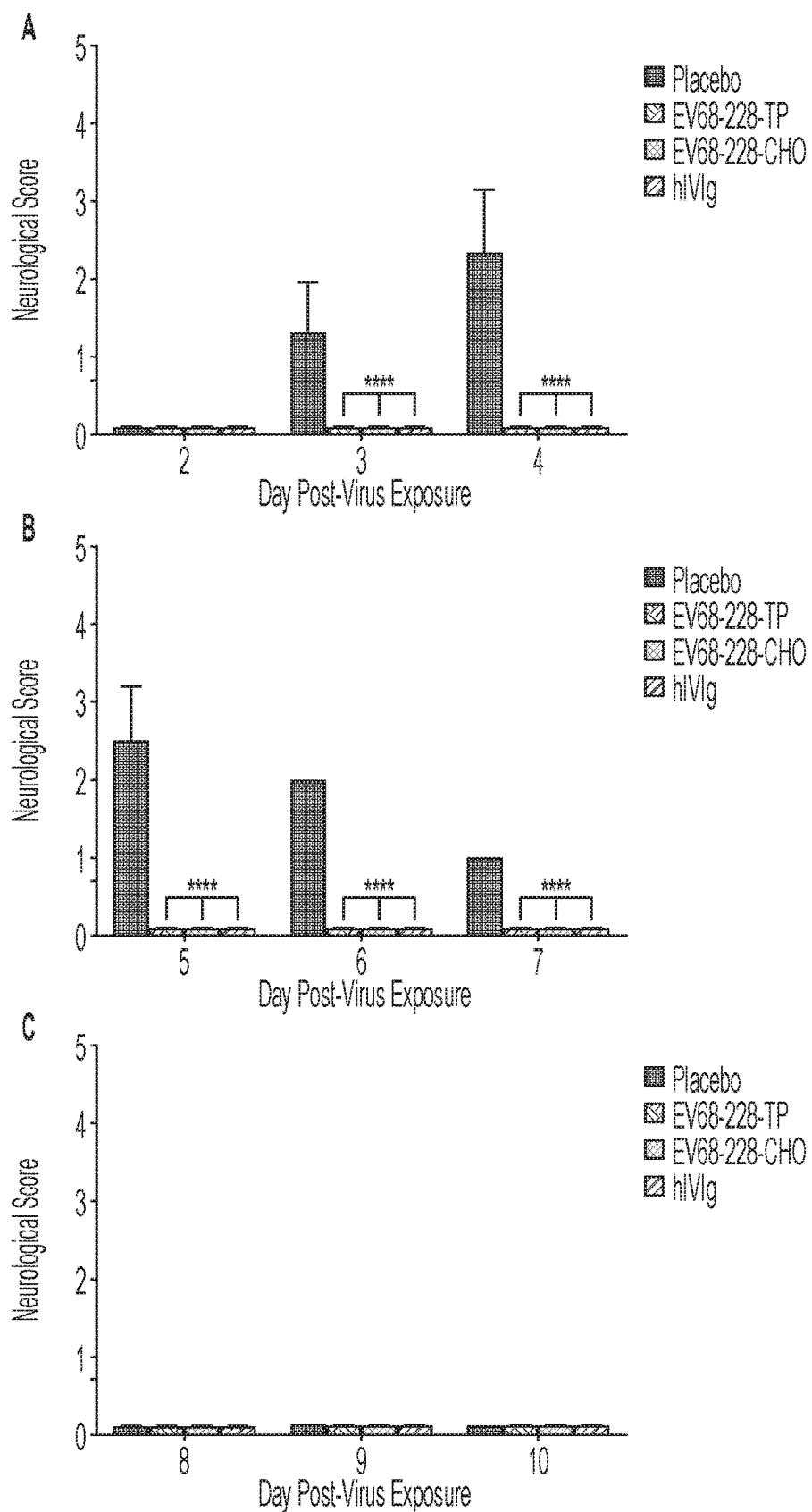


FIG. 70A-C

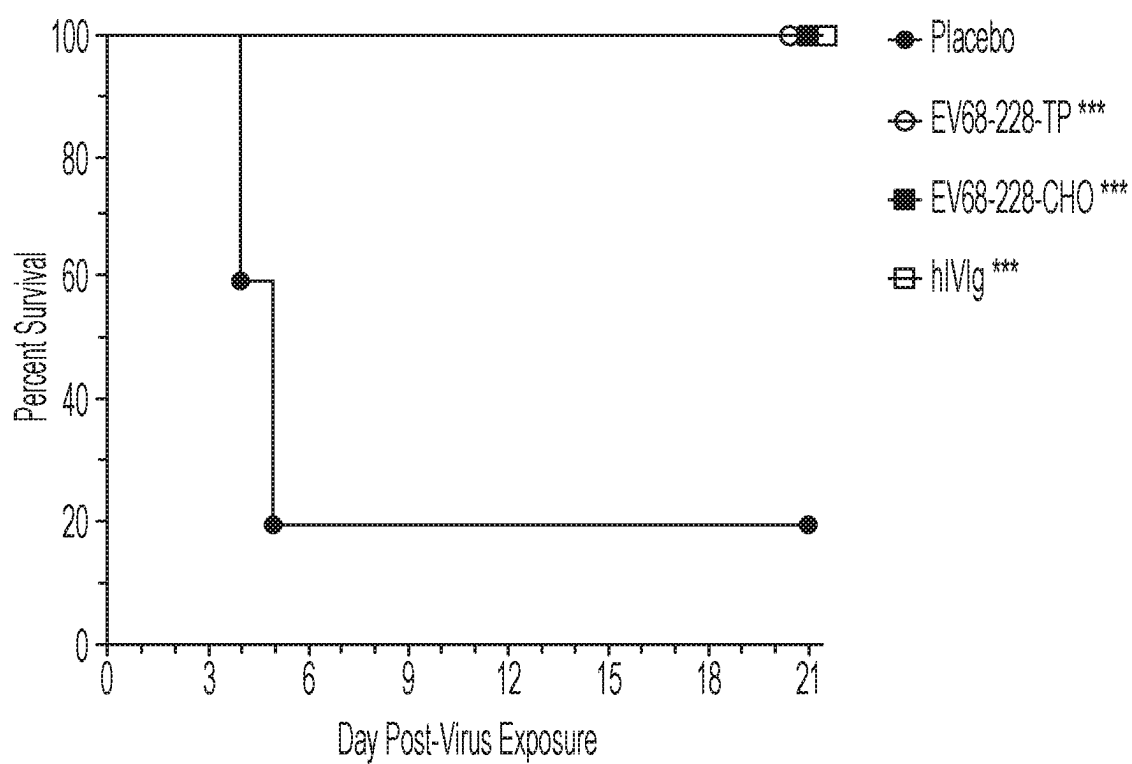


FIG. 71

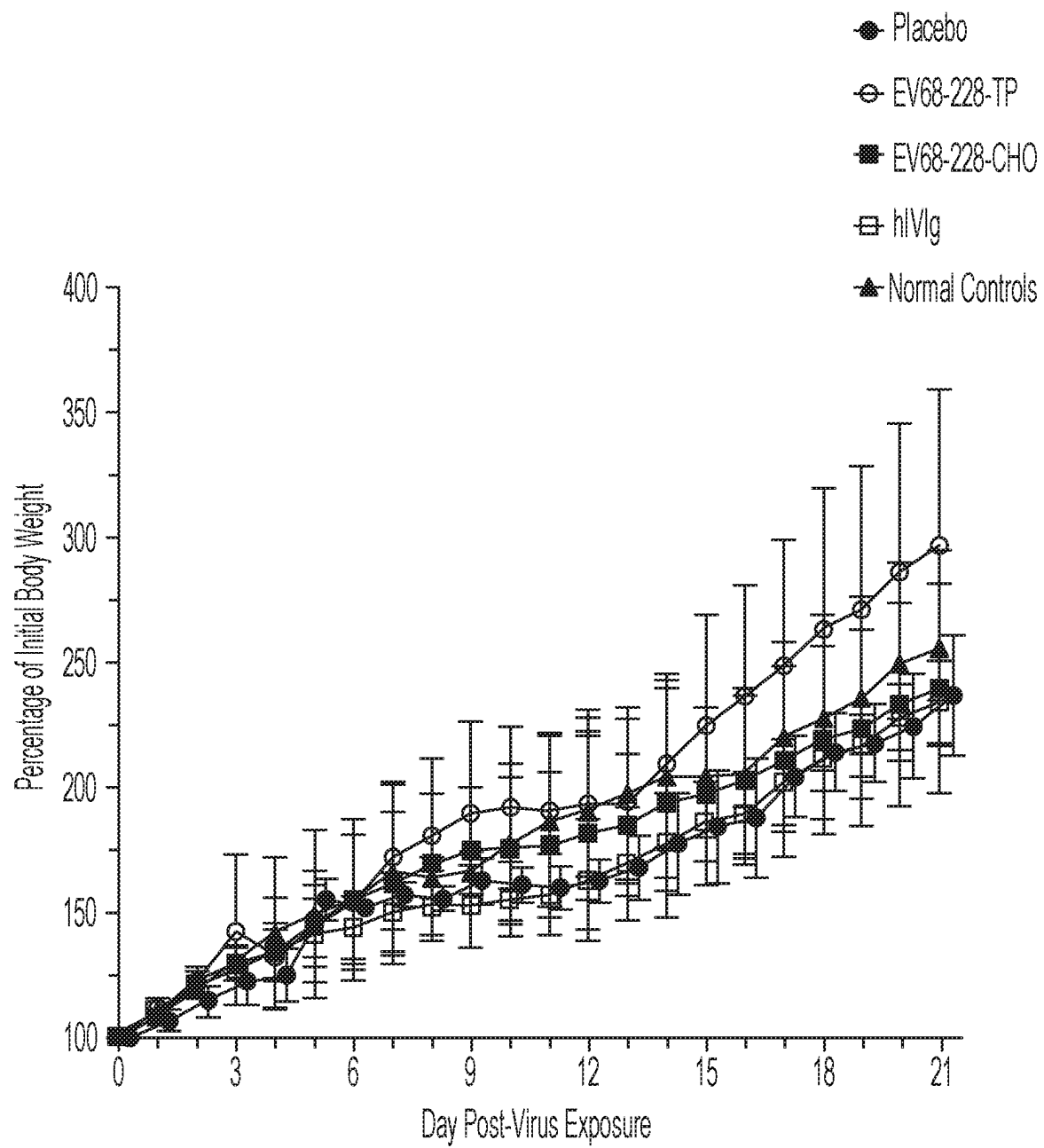


FIG. 72

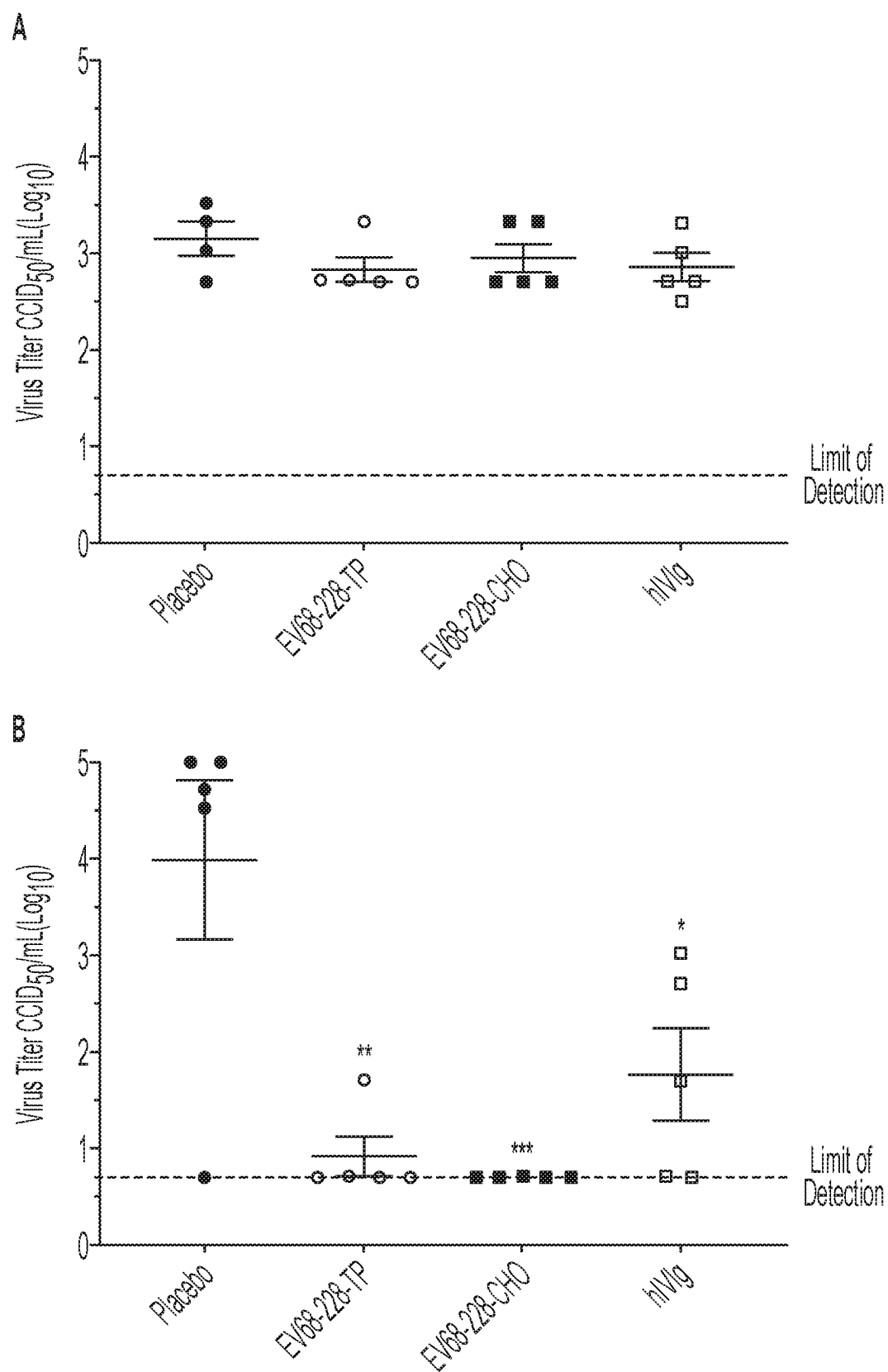
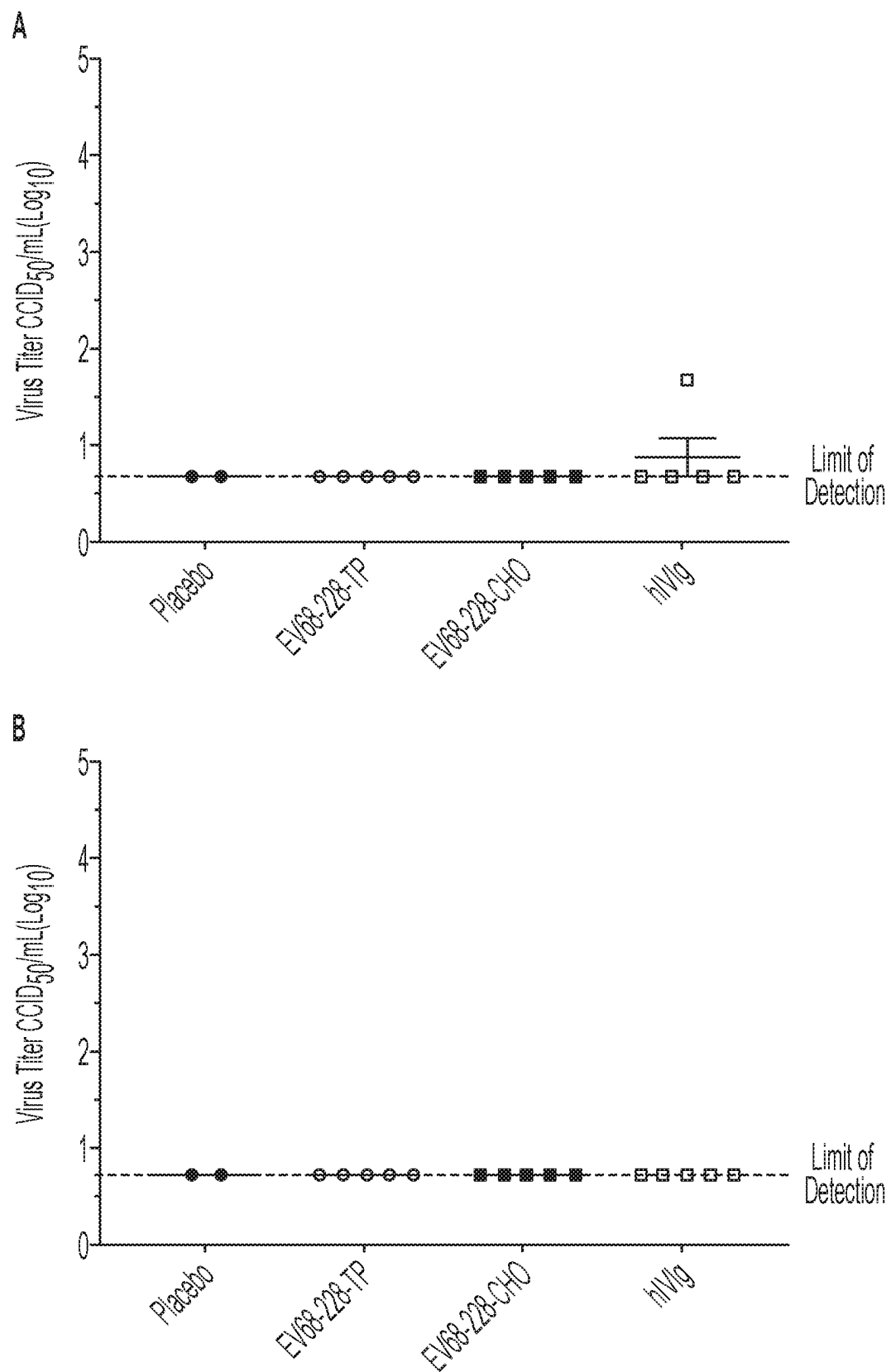


FIG. 73A-B



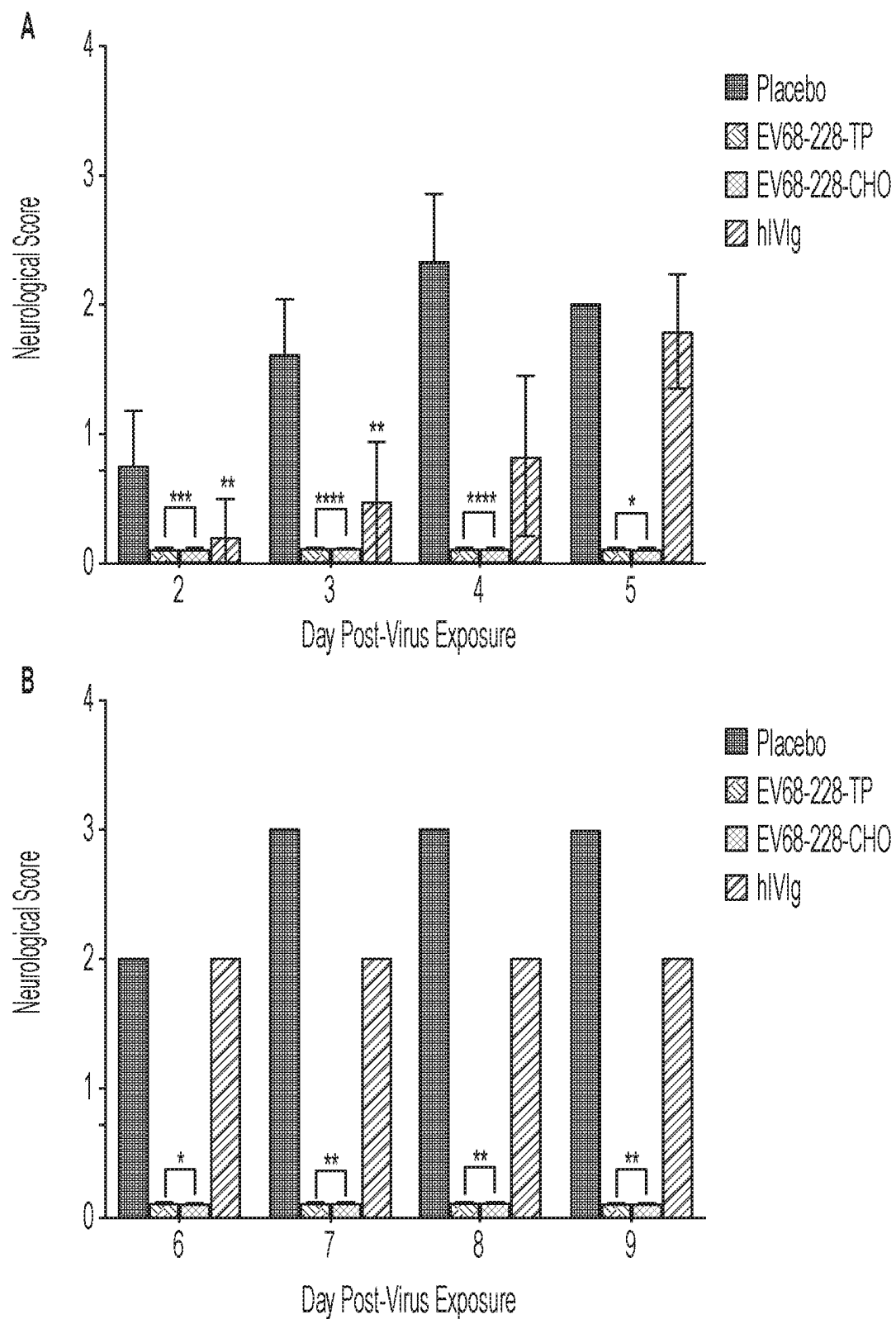


FIG. 75A-B

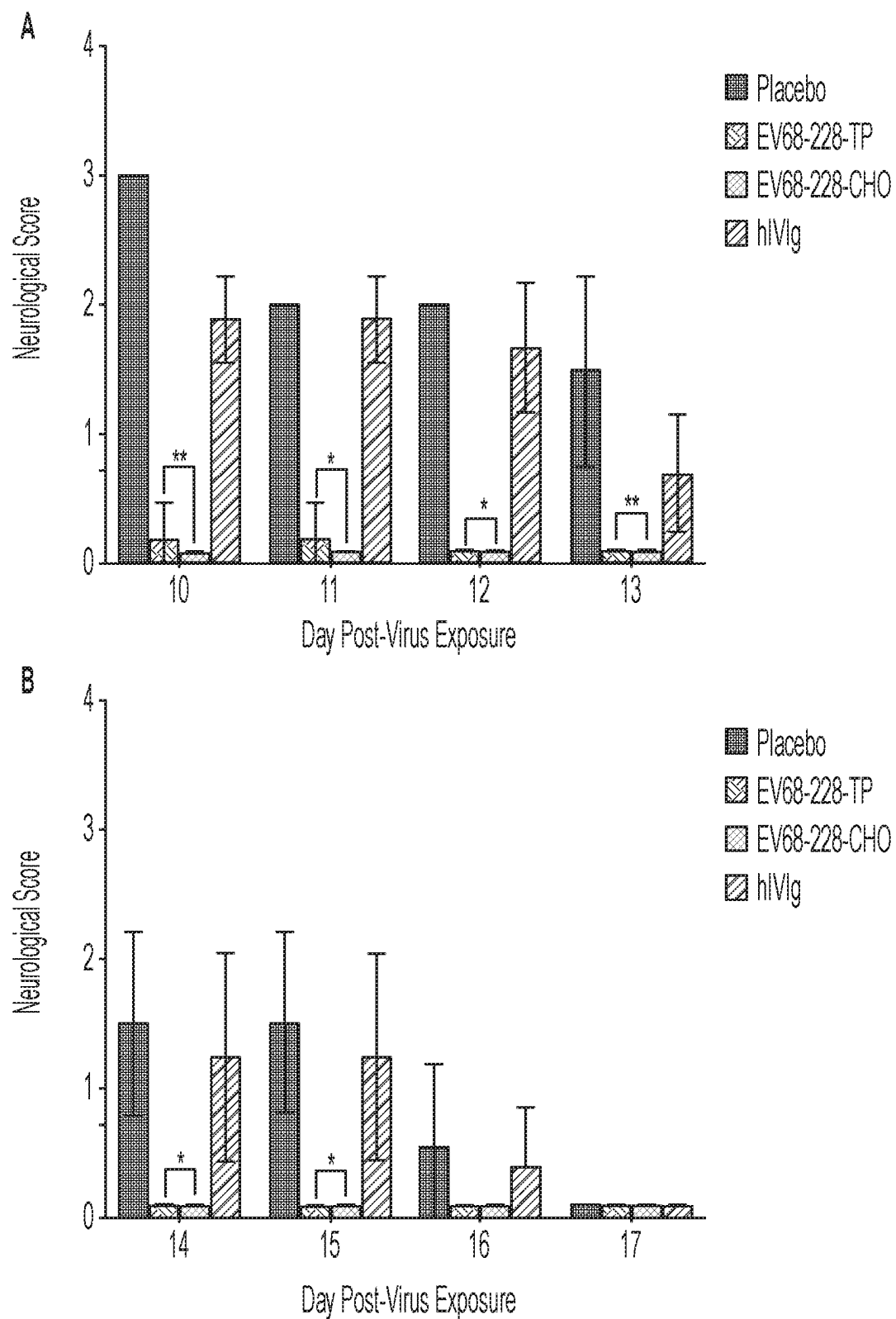


FIG. 76A-B

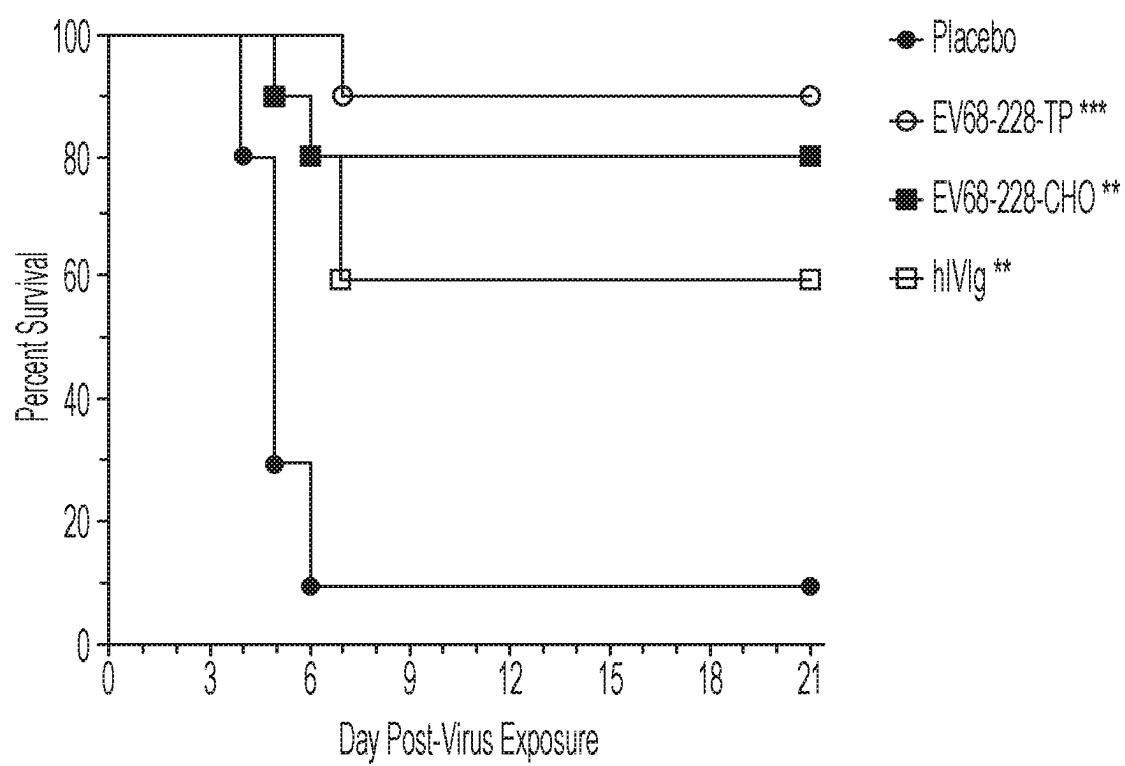


FIG. 77

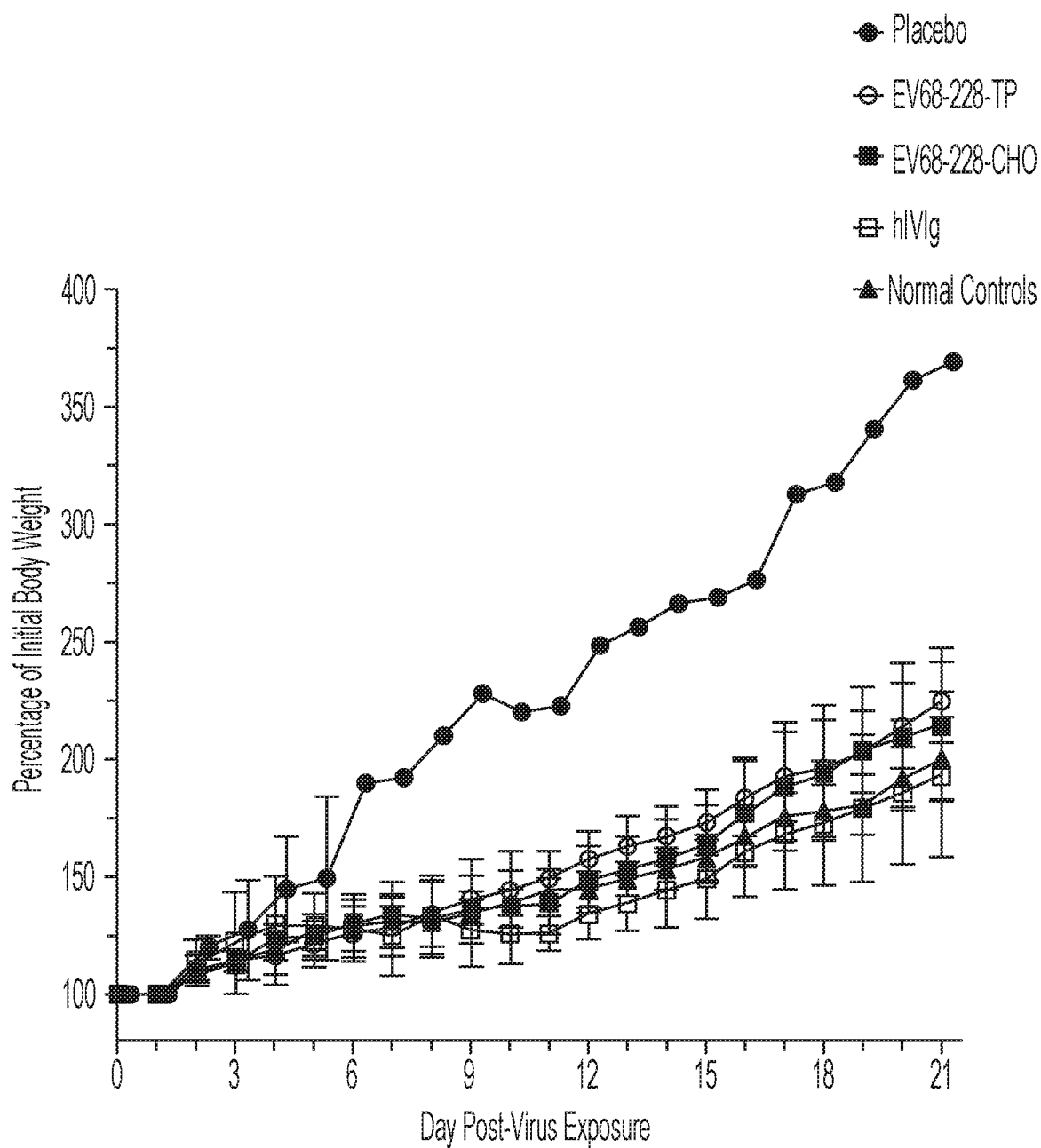


FIG. 78

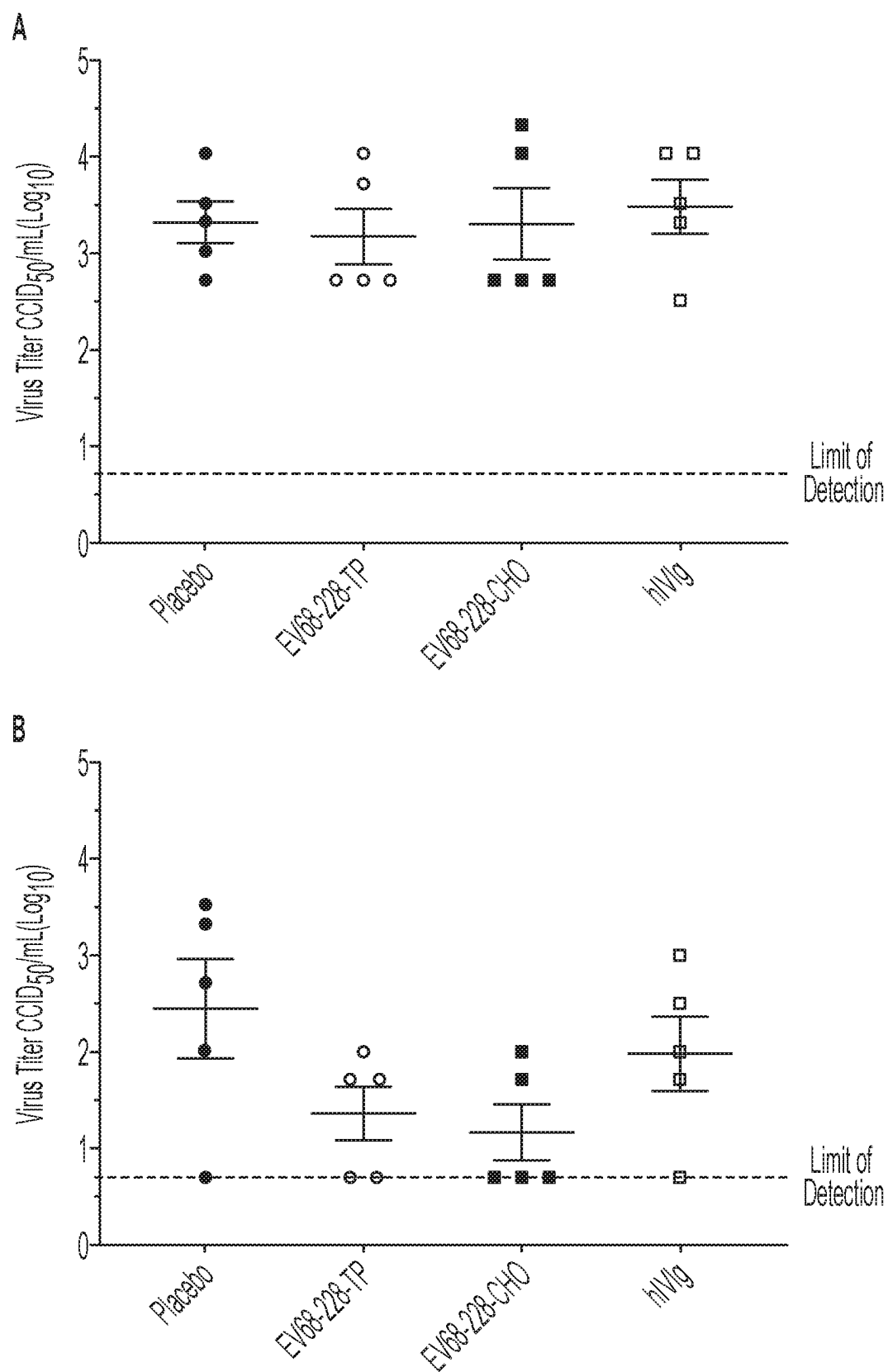


FIG. 79A-B

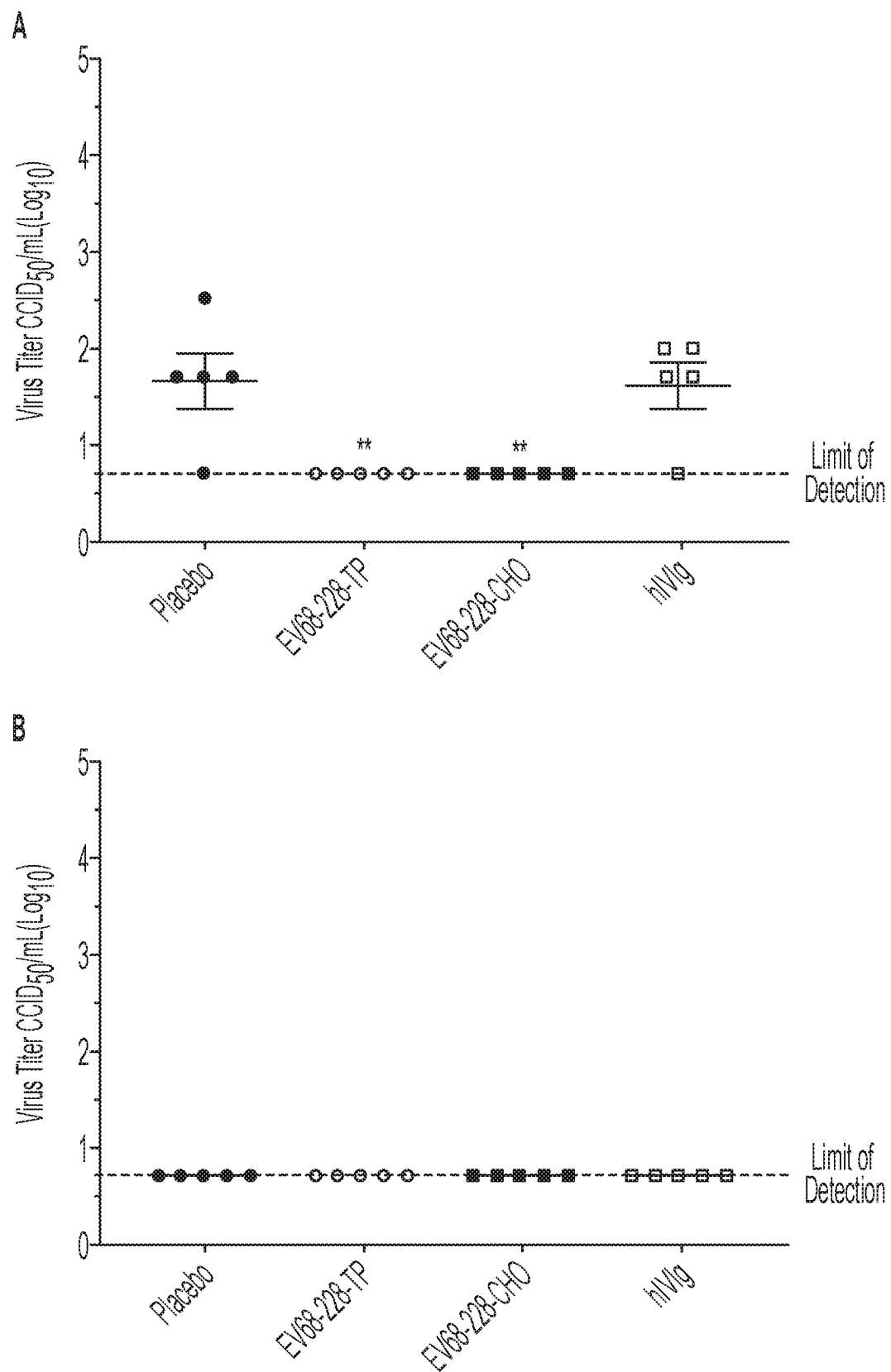


FIG. 80A-B

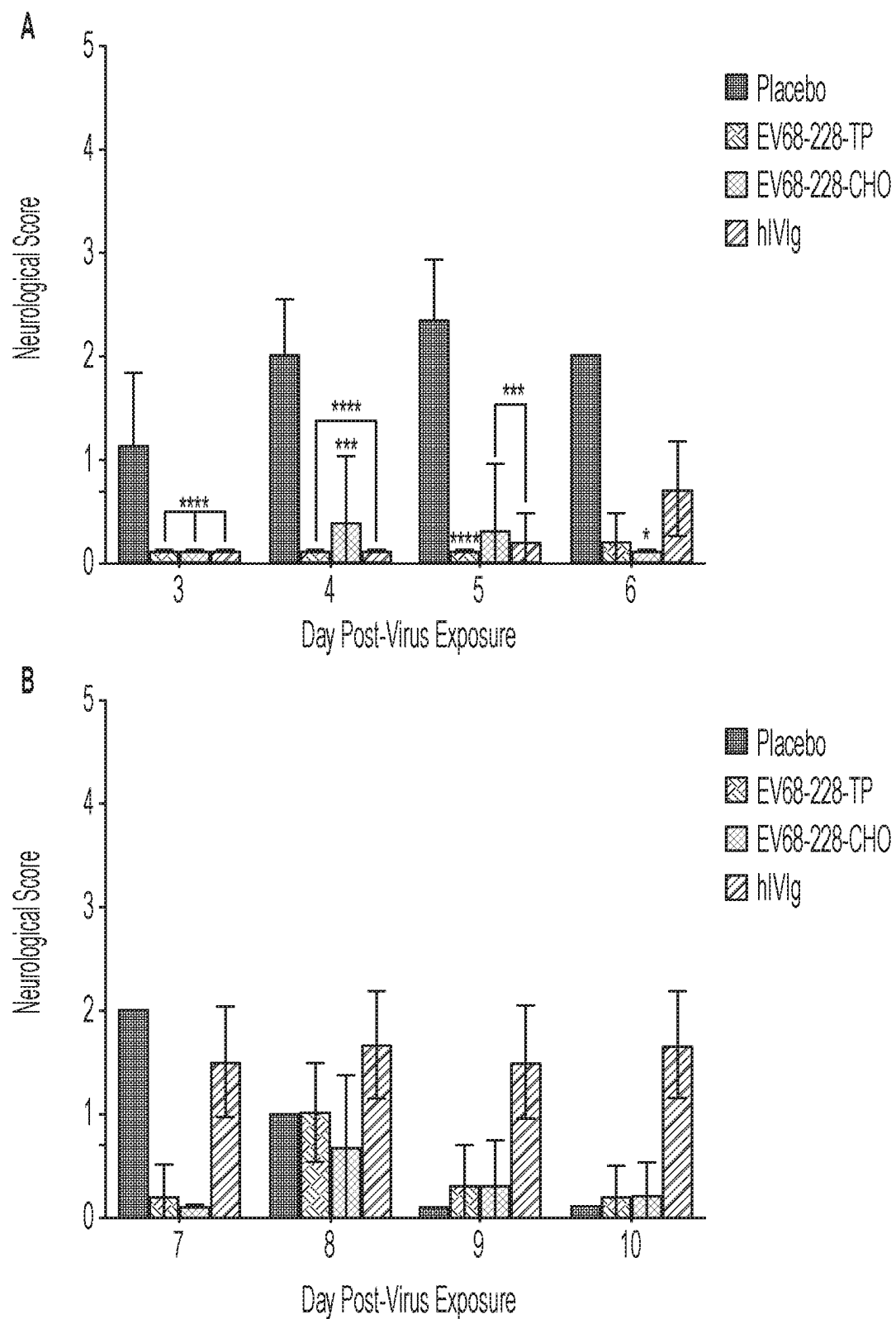


FIG. 81A-B

1

HUMAN MONOCLONAL ANTIBODIES TO ENTEROVIRUS D68

PRIORITY CLAIM

This application is a national phase application under 35 U.S.C. § 371 of International Application No. PCT/US2020/043415, filed Jul. 24, 2020, which claims benefit of priority to U.S. Provisional Application Ser. Nos. 62/878,955, 62/899,503 and 63/047,330, filed Jul. 26, 2019, Sep. 12, 2019 and Jul. 2, 2020, respectively, the entire contents of each application being incorporated by reference herein.

REFERENCE TO A SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Mar. 27, 2025, is named VBLTP0294US_ST25.txt and is 288,928 bytes in size.

BACKGROUND

1. Field of the Disclosure

The present disclosure relates generally to the fields of medicine, infectious disease, and immunology. More particularly, the disclosure relates to human antibodies binding to enterovirus D68 and methods of using such antibodies to diagnose and treat enterovirus D68 infections.

2. Background

Enterovirus-D68 (EV-D68) is a positive-sense, single-stranded RNA virus of the Enterovirus genus of the Picornaviridae family. Other common human pathogens within this genus include poliovirus, echovirus, coxsackievirus, rhinovirus, and other numbered enteroviruses such as enterovirus-A71 (Zell et al., 2017). While structurally and genetically similar, these viruses cause a wide variety of childhood diseases including neonatal sepsis, myocarditis, poliomyelitis, meningitis, respiratory tract infections, and hand, foot, and mouth disease. Much like the rhinoviruses, EV-D68 is primarily a respiratory pathogen that is acid-labile and replicates best at 33° C. (Oberste et al., 2004). These similarities are such that a strain of EV-D68 was even initially reported as rhinovirus 87 (Blomqvist et al., 2002).

From its initial identification in a California child with pneumonia in 1962 (Schieble et al., 1967) through the turn of the century, EV-D68 was detected only rarely (Khetsuriani et al., 2006). Since then, EV-D68 has been recognized increasingly as a pathogen of emerging importance due to its worldwide detection in outbreaks of primarily respiratory illness in children (Xiang and Wang, 2016). The largest ever known outbreak occurred in 2014 in the United States with 1,152 confirmed cases spanning all states except for Alaska (Oermann et al., 2015). This number likely grossly underestimates the actual number of cases of EV-D68 in 2014 because mild upper respiratory tract infection would not likely result in the specialized testing needed to detect this virus. Children with asthma experienced especially severe infections (Biggs et al., 2017), although a study of one institution found that EV-D68 was no more severe than non-EV-D68 enterovirus/rhinovirus infections in children with asthma (Overdahl et al., 2016). Over half of the hospitalized patients were admitted to intensive care units, with 80% receiving supplemental oxygen, 23% requiring

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non-invasive ventilation, 8% requiring intubation and mechanical ventilation, and 1% died (Midgley et al., 2015).

Concurrent with the 2014 outbreak, a small cluster of pediatric patients with acute onset flaccid paralysis and cranial nerve dysfunction was noted in Colorado (Messacar et al., 2015). This syndrome has been designated acute flaccid myelitis (AFM), defined as a poliomyelitis-like illness with typically asymmetric, flaccid limb weakness and myelitis of primarily gray matter seen on spinal cord imaging (Div. of Viral Diseases, CDC, 2018). Since the 2014 outbreak, many more cases and clusters of AFM have been associated with EV-D68. To date, 74 AFM cases with positive EV-D68 testing from any patient source have been identified across 6 continents (Messacar et al., 2018). While the Centers for Disease Control and Prevention (CDC) does not officially recognize EV-D68 as a proven cause of AFM (Nat'l Center for Immun Resp. Disease), many experts find the preponderance of evidence compelling enough to consider the relationship between the two to be causal (Messacar et al. 2018). Because of continued worldwide outbreaks, the World Health Organization Research and Development Blueprint now lists EV-D68 as a major public health risk (Ann. Rev. of Diseases, 2018).

As EV-D68 has emerged only recently as a priority pathogen, most initial studies focused on defining the epidemiology of the virus rather than characterizing the immune response. Therefore, the study of humoral immunity to EV-D68 is nascent. The role of serum antibodies in protection from other viruses of the Enterovirus genus is varied. For example, three doses of inactivated poliovirus vaccine approaches 100% induction of serum neutralizing antibodies and is 80-96% effective at preventing paralytic poliomyelitis; however, vaccination does not fully prevent enteric or nasopharyngeal poliovirus shedding (Vidor et al., 2018) due to its inability to induce nasal or duodenal IgA (Sutter et al., 2018). Studies of rhinovirus infection show that humoral immunity to specific serotypes of virus fails to protect reliably against homotypic virus reinfection within months (Howard et al., 2016). These differences in extent of protection associated with antibody responses likely are due to the differing sites of pathology for these viruses: secondary neuronal spread after initial enteral infection for polioviruses versus localized respiratory tract infection for rhinoviruses. EV-D68 infection can cause disease in the respiratory tract and is associated with disease in the central nervous system, so the role of antibodies in protection and disease pathogenesis is likely to be complex. A better understanding this role would aid in the development of vaccines and therapies for EV-D68 infections.

SUMMARY

Thus, in accordance with the present disclosure, there is provided a method of detecting enterovirus D68 (EV-D68) infection in a subject comprising (a) contacting a sample from said subject with an antibody or antibody fragment having clone-paired heavy and light chain CDR sequences from Tables 3 and 4, respectively; and (b) detecting EV-D68 in said sample by binding of said antibody or antibody fragment to a EV-D68 antigen in said sample. The sample may be a body fluid, such as blood, sputum, tears, saliva, mucous or serum, semen, cervical or vaginal secretions, amniotic fluid, placental tissues, urine, exudate, transudate, tissue scrapings, respiratory droplets or aerosol, feces, etc. Detection may comprise, for example, ELISA, RIA, lateral flow assay, Western blot, and the like. The method may

further comprise performing steps (a) and (b) a second time and determining a change in EV-D68 antigen levels as compared to the first assay.

The antibody or antibody fragment may be encoded by clone-paired variable sequences as set forth in Table 1, by light and heavy chain variable sequences having 70%, 80%, or 90% identity to clone-paired variable sequences as set forth in Table 1, or by light and heavy chain variable sequences having 95% identity to clone-paired sequences as set forth in Table 1. The antibody or antibody fragment may comprise light and heavy chain variable sequences according to clone-paired sequences from Table 2, may comprise light and heavy chain variable sequences having 70%, 80% or 90% identity to clone-paired sequences from Table 2, or may comprise light and heavy chain variable sequences having 95% identity to clone-paired sequences from Table 2. The antibody fragment may be a recombinant scFv (single chain fragment variable) antibody, Fab fragment, F(ab')₂ fragment, or Fv fragment.

In another aspect, the present disclosure provides a method of treating a subject infected with enterovirus D68 (EV-D68) or reducing the likelihood of infection of a subject at risk of contracting EV-D68, comprising delivering to said subject an antibody or antibody fragment having clone-paired heavy and light chain CDR sequences from Tables 3 and 4, respectively. The antibody or antibody fragment may be encoded by clone-paired variable sequences as set forth in Table 1, by light and heavy chain variable sequences having 70%, 80%, or 90% identity to clone-paired variable sequences as set forth in Table 1, or by light and heavy chain variable sequences having 95% identity to clone-paired sequences as set forth in Table 1. The antibody or antibody fragment may comprise light and heavy chain variable sequences according to clone-paired sequences from Table 2, may comprise light and heavy chain variable sequences having 70%, 80% or 90% identity to clone-paired sequences from Table 2, or may comprise light and heavy chain variable sequences having 95% identity to clone-paired sequences from Table 2. The antibody fragment may be a recombinant scFv (single chain fragment variable) antibody, Fab fragment, F(ab')₂ fragment, or Fv fragment. The antibody may be any isotype, including without limitation IgG, or a recombinant IgG antibody or antibody fragment comprising an Fc portion mutated to alter (eliminate or enhance) FcR interactions, to increase half-life and/or increase therapeutic efficacy, such as a LALA, N297, GASD/ALIE, YTE, DHS or LS mutation or glycan modified to alter (eliminate or enhance) FcR interactions such as enzymatic or chemical addition or removal of glycans or expression in a cell line engineered with a defined glycosylating pattern. The antibody may be a chimeric antibody or a bispecific antibody.

The antibody or antibody fragment may be administered prior to infection or after infection, e.g., such as at or less than about 7 days, about 5 days, about 3 days, about 2 days, or about 1 day following infection. Treating may treat lung infection and/or neurologic infection, or wherein treating reduces lung infection and/or neurologic infection. Delivering may comprise antibody or antibody fragment administration systemically, by aerosol delivery, etc., or genetic delivery with an RNA or DNA sequence or vector encoding the antibody or antibody fragment. The method may further comprise treating said subject with a second therapy, such as an anti-viral therapy or an anti-inflammatory therapy.

In yet another aspect, there is provided a monoclonal antibody, wherein the antibody or antibody fragment is characterized by clone-paired heavy and light chain CDR sequences from Tables 3 and 4, respectively. The antibody or

antibody fragment may be encoded by clone-paired variable sequences as set forth in Table 1, by light and heavy chain variable sequences having 70%, 80%, or 90% identity to clone-paired variable sequences as set forth in Table 1, or by light and heavy chain variable sequences having 95% identity to clone-paired sequences as set forth in Table 1. The antibody or antibody fragment may comprise light and heavy chain variable sequences according to clone-paired sequences from Table 2, may comprise light and heavy chain variable sequences having 70%, 80% or 90% identity to clone-paired sequences from Table 2, or may comprise light and heavy chain variable sequences having 95% identity to clone-paired sequences from Table 2. The antibody fragment may be a recombinant scFv (single chain fragment variable) antibody, Fab fragment, F(ab')₂ fragment, or Fv fragment. The antibody may be any isotype, including without limitation IgG, or a recombinant IgG antibody or antibody fragment comprising an Fc portion mutated to alter (eliminate or enhance) FcR interactions, to increase half-life and/or increase therapeutic efficacy, such as a LALA, N297, GASD/ALIE, YTE, DHS or LS mutation or glycan modified to alter (eliminate or enhance) FcR interactions such as enzymatic or chemical addition or removal of glycans or expression in a cell line engineered with a defined glycosylating pattern. The antibody may be a chimeric antibody or a bispecific antibody. The antibody or antibody fragment may further comprise a cell penetrating peptide and/or is an intrabody.

A hybridoma or engineered cell encoding an antibody or antibody fragment wherein the antibody or antibody fragment is characterized by clone-paired heavy and light chain CDR sequences from Tables 3 and 4, respectively. The hybridoma or engineered cell may encode an antibody or antibody fragment encoded by clone-paired variable sequences as set forth in Table 1, by light and heavy chain variable sequences having 70%, 80%, or 90% identity to clone-paired variable sequences as set forth in Table 1, or by light and heavy chain variable sequences having 95% identity to clone-paired sequences as set forth in Table 1. The hybridoma or engineered cell may contain an antibody or antibody fragment comprising light and heavy chain variable sequences according to clone-paired sequences from Table 2, comprising light and heavy chain variable sequences having 70%, 80% or 90% identity to clone-paired sequences from Table 2, or comprising light and heavy chain variable sequences having 95% identity to clone-paired sequences from Table 2. The antibody fragment may be a recombinant scFv (single chain fragment variable) antibody, Fab fragment, F(ab')₂ fragment, or Fv fragment. The antibody may be an IgG, or a recombinant IgG antibody or antibody fragment comprising an Fc portion mutated to alter (eliminate or enhance) FcR interactions, to increase half-life and/or increase therapeutic efficacy, such as a LALA, N297, GASD/ALIE, YTE, DHS or LS mutation or glycan modified to alter (eliminate or enhance) FcR interactions such as enzymatic or chemical addition or removal of glycans or expression in a cell line engineered with a defined glycosylating pattern. The antibody may be a chimeric antibody or a bispecific antibody. The antibody or antibody fragment further comprises a cell penetrating peptide and/or is an intrabody.

Also provided is a vaccine formulation comprising one or more antibodies or antibody fragments characterized by clone-paired heavy and light chain CDR sequences from Tables 3 and 4, respectively. The antibody or antibody fragment may be encoded by clone-paired variable sequences as set forth in Table 1, by light and heavy chain

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variable sequences having 70%, 80%, or 90% identity to clone-paired variable sequences as set forth in Table 1, or by light and heavy chain variable sequences having 95% identity to clone-paired sequences as set forth in Table 1. The antibody or antibody fragment may comprise light and heavy chain variable sequences according to clone-paired sequences from Table 2, may comprise light and heavy chain variable sequences having 70%, 80% or 90% identity to clone-paired sequences from Table 2, or may comprise light and heavy chain variable sequences having 95% identity to clone-paired sequences from Table 2. The antibody fragment may be a recombinant scFv (single chain fragment variable) antibody, Fab fragment, F(ab')₂ fragment, or Fv fragment. The antibody may be an IgG, or a recombinant IgG antibody or antibody fragment comprising an Fc portion mutated to alter (eliminate or enhance) FcR interactions, to increase half-life and/or increase therapeutic efficacy, such as a LALA, N297, GASD/ALIE, YTE, DHS or LS mutation or glycan modified to alter (eliminate or enhance) FcR interactions such as enzymatic or chemical addition or removal of glycans or expression in a cell line engineered with a defined glycosylating pattern. The antibody may be a chimeric antibody or a bispecific antibody. The antibody or antibody fragment further comprises a cell penetrating peptide and/or is an intrabody.

Still another embodiment comprises a vaccine formulation comprising one or more expression vectors encoding an antibody or antibody fragment characterized by clone-paired heavy and light chain CDR sequences from Tables 3 and 4, respectively. The antibody or antibody fragment may be encoded by clone-paired variable sequences as set forth in Table 1, by light and heavy chain variable sequences having 70%, 80%, or 90% identity to clone-paired variable sequences as set forth in Table 1, or by light and heavy chain variable sequences having 95% identity to clone-paired sequences as set forth in Table 1. The antibody or antibody fragment may comprise light and heavy chain variable sequences according to clone-paired sequences from Table 2, may comprise light and heavy chain variable sequences having 70%, 80% or 90% identity to clone-paired sequences from Table 2, or may comprise light and heavy chain variable sequences having 95% identity to clone-paired sequences from Table 2. The antibody fragment may be a recombinant scFv (single chain fragment variable) antibody, Fab fragment, F(ab')₂ fragment, or Fv fragment. The antibody may be an IgG, or a recombinant IgG antibody or antibody fragment comprising an Fc portion mutated to alter (eliminate or enhance) FcR interactions, to increase half-life and/or increase therapeutic efficacy, such as a LALA, N297, GASD/ALIE, YTE, DHS or LS mutation or glycan modified to alter (eliminate or enhance) FcR interactions such as enzymatic or chemical addition or removal of glycans or expression in a cell line engineered with a defined glycosylating pattern. The antibody may be a chimeric antibody or a bispecific antibody. The expression vector(s) is/are Sindbis virus or VEE vector(s). The vaccine formulation may be formulated for delivery by needle injection, jet injection, or electroporation. The vaccine formulation may further comprise one or more expression vectors encoding for a second antibody or antibody fragment, such as a distinct antibody or antibody fragment characterized by clone-paired heavy and light chain CDR sequences from Tables 3 and 4, respectively. The antibody or antibody fragment is encoded by clone-paired variable sequences as set forth in Table 1, by light and heavy chain variable sequences having 70%, 80%, or 90% identity to clone-paired variable sequences as set forth in Table 1, or by light and heavy chain variable

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sequences having 95% identity to clone-paired sequences as set forth in Table 1. The antibody or antibody fragment may comprise light and heavy chain variable sequences according to clone-paired sequences from Table 2, may comprise light and heavy chain variable sequences having 70%, 80% or 90% identity to clone-paired sequences from Table 2, or may comprise light and heavy chain variable sequences having 95% identity to clone-paired sequences from Table 2.

Still a further embodiment comprises a method of protecting the health of a placenta and/or fetus of a pregnant subject infected with or at risk of infection with enterovirus D68, comprising delivering to said subject an antibody or antibody fragment having clone-paired heavy and light chain CDR sequences from Tables 3 and 4, respectively. The antibody or antibody fragment may be encoded by clone-paired variable sequences as set forth in Table 1, by light and heavy chain variable sequences having 70%, 80%, or 90% identity to clone-paired variable sequences as set forth in Table 1, or by light and heavy chain variable sequences having 95% identity to clone-paired sequences as set forth in Table 1. The antibody or antibody fragment may comprise light and heavy chain variable sequences according to clone-paired sequences from Table 2, may comprise light and heavy chain variable sequences having 70%, 80% or 90% identity to clone-paired sequences from Table 2, or may comprise light and heavy chain variable sequences having 95% identity to clone-paired sequences from Table 2. The antibody fragment may be a recombinant scFv (single chain fragment variable) antibody, Fab fragment, F(ab')₂ fragment, or Fv fragment. The antibody may be an IgG, or a recombinant IgG antibody or antibody fragment comprising an Fc portion mutated to alter (eliminate or enhance) FcR interactions, to increase half-life and/or increase therapeutic efficacy, such as a LALA, N297, GASD/ALIE, YTE, DHS or LS mutation or glycan modified to alter (eliminate or enhance) FcR interactions such as enzymatic or chemical addition or removal of glycans or expression in a cell line engineered with a defined glycosylating pattern. The antibody may be a chimeric antibody or a bispecific antibody. The antibody or antibody fragment further comprises a cell penetrating peptide and/or is an intrabody.

The antibody or antibody fragment may be administered prior to infection or after infection, e.g., such as at or less than about 7 days, about 5 days, about 3 days, about 2 days, or about 1 day following infection. The subject may be a pregnant female, a sexually active female, or a female undergoing fertility treatments. Delivering may comprise antibody or antibody fragment administration, or genetic delivery with an RNA or DNA sequence or vector encoding the antibody or antibody fragment. The antibody or antibody fragment may increase the size of the placenta as compared to an untreated control. The antibody or antibody fragment may reduce viral load and/or pathology of the fetus as compared to an untreated control.

In still yet another embodiment, there is provided a method of determining the antigenic integrity, correct conformation and/or correct sequence of an enterovirus D68 antigen comprising (a) contacting a sample comprising said antigen with a first antibody or antibody fragment having clone-paired heavy and light chain CDR sequences from Tables 3 and 4, respectively; and (b) determining antigenic integrity, correct conformation and/or correct sequence of said antigen by detectable binding of said first antibody or antibody fragment to said antigen. The sample may comprise recombinantly produced antigen or may comprise a vaccine formulation or vaccine production batch. Detection

may comprise ELISA, RIA, western blot, a biosensor using surface plasmon resonance or biolayer interferometry, or flow cytometric staining.

The first antibody or antibody fragment is encoded by clone-paired variable sequences as set forth in Table 1, by light and heavy chain variable sequences having 70%, 80%, or 90% identity to clone-paired variable sequences as set forth in Table 1, or by light and heavy chain variable sequences having 95% identity to clone-paired sequences as set forth in Table 1. The first antibody or antibody fragment may comprise light and heavy chain variable sequences according to clone-paired sequences from Table 2, may comprise light and heavy chain variable sequences having 70%, 80% or 90% identity to clone-paired sequences from Table 2, or may comprise light and heavy chain variable sequences having 95% identity to clone-paired sequences from Table 2. The first antibody fragment may be a recombinant scFv (single chain fragment variable) antibody, Fab fragment, F(ab')₂ fragment, or Fv fragment. The method may further comprise performing steps (a) and (b) a second time to determine the antigenic stability of the antigen over time.

The method may further comprise (c) contacting a sample comprising said antigen with a second antibody or antibody fragment having clone-paired heavy and light chain CDR sequences from Tables 3 and 4, respectively; and (d) determining antigenic integrity of said antigen by detectable binding of said second antibody or antibody fragment to said antigen. The second antibody or antibody fragment is encoded by clone-paired variable sequences as set forth in Table 1, by light and heavy chain variable sequences having 70%, 80%, or 90% identity to clone-paired variable sequences as set forth in Table 1, or by light and heavy chain variable sequences having 95% identity to clone-paired sequences as set forth in Table 1. The second antibody or antibody fragment may comprise light and heavy chain variable sequences according to clone-paired sequences from Table 2, may comprise light and heavy chain variable sequences having 70%, 80% or 90% identity to clone-paired sequences from Table 2, or may comprise light and heavy chain variable sequences having 95% identity to clone-paired sequences from Table 2. The second antibody fragment may be a recombinant scFv (single chain fragment variable) antibody, Fab fragment, F(ab')₂ fragment, or Fv fragment. The method may comprise performing steps (c) and (d) a second time to determine the antigenic stability of the antigen over time.

Also provided is a human monoclonal antibody or antibody fragment, or hybridoma or engineered cell producing the same, wherein said antibody binds to EV-D68 across at least two viral clades.

An additional embodiment comprises a human monoclonal antibody or antibody fragment, or hybridoma or engineered cell producing the same, wherein said antibody binds to EV-D68 VP1, VP2 and VP3. The antibody may bind to EV-D68 VP1 DE loop, and/or to EV-D68 VP2 EE loop, and/or to EV-D68 VP3 N-terminal loop.

The use of the word "a" or "an" when used in conjunction with the term "comprising" in the claims and/or the specification may mean "one," but it is also consistent with the meaning of "one or more," "at least one," and "one or more than one." The word "about" means plus or minus 5% of the stated number.

It is contemplated that any method or composition described herein can be implemented with respect to any other method or composition described herein. Other objects, features and advantages of the present disclosure

will become apparent from the following detailed description. It should be understood, however, that the detailed description and the specific examples, while indicating specific embodiments of the disclosure, are given by way of illustration only, since various changes and modifications within the spirit and scope of the disclosure will become apparent to those skilled in the art from this detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

The following drawings form part of the present specification and are included to further demonstrate certain aspects of the present disclosure. The disclosure may be better understood by reference to one or more of these drawings in combination with the detailed description of specific embodiments presented herein.

FIG. 1. Antibody isolation. Sixty-four human monoclonal antibodies were isolated and screened using an indirect ELISA in which live virus isolates from the 2014 EV-D68 outbreak are directly coated onto an ELISA plate. Any hits are then fused with myeloma cells to make a hybridoma, which will theoretically continue dividing forever. After purifying antibodies, a number of in vitro characterization steps are performed to define their phenotypes.

FIG. 2. Antibody neutralization in vitro. In vitro neutralization is performed using the cell culture 50% infectious dose, or CCID₅₀ assay. Some antibodies are quite potent, some still very potent, and others weak, with about 1/3 of the mAbs showing some detectable neutralizing activity.

FIG. 3. Cross-reactive binding among clades. To compare across all mAbs, a heat map was generated using the half maximal effective concentration, or EC₅₀, from ELISAs. Some bind very well to all clades, while some have notable dropout, typically with the distant A2 clade.

FIG. 4. Competition reveals at least four main surface epitopes. Using Pearson correlation, the inventors generated relatedness values as the colored readout to cluster. This relatedness helps to sort the antibodies, but not actual interference percentage. The inventors have now described three to four phenotypes, which can be related to each other. This permits identification of candidates for clinical development.

FIG. 5. Clinical candidates. In addition to neutralization, further characterization revealed additional clinical candidates. mAb 159 does not cross-react in ELISA or CCID₅₀, even though it is incredibly potent and does not bind to a mock preparation of virus. mAb 105 is still quite potent, but it cross-reacts quite well.

FIGS. 6A-C. Expt. NIA-1849. Lung virus titers of EV-D68-infected AG129 mice treated with EV-D68-228. Treatment with EV-D68-228 significantly reduced lung virus titers on day 1 (FIG. 6A), day 3 (FIG. 6B), and day 5 (FIG. 6C) post-infection. No lung virus titers were detected in mice treated with 10, 3, or 1 mg/kg of EV-D68-228 at days 1, 3, or 5 post-infection. Treatment with IVIg was able to reduce lung virus titers only on day 1 post-infection. Guanidine significantly reduced lung virus titers on days 1 and 3 but not day 5 post-infection. **P<0.01, ****P<0.0001 compared to placebo-treated mice.

FIGS. 7A-C. Expt. NIA-1849. Blood virus titers of EV-D68-infected AG129 mice treated with EV-D68-228. Treatment with the EV-D68-228 mAb reduced blood virus titers at day 1 (FIG. 7A), day 3 (FIG. 7B), and day 5 (FIG. 7C) post-infection. Treatment with IVIg significantly reduced blood virus titers at days 1, 3, and 5 post-infection. Guanidine reduced blood virus titers on days 1 and 5 but not

day 3 post-infection. $^{**}P<0.01$, $^{***}P<0.001$, $^{****}P<0.0001$ compared to placebo-treated mice.

FIG. 8. Expt. NIA-1849. Lung concentrations of IL-1 α and IL-1 β from EV-D68-infected AG129 mice treated with EV-D68-228. Treatment with EV-D68-228 significantly reduced concentrations of IL-1 α and IL-1 β on day 3 post-infection compared to placebo-treated mice. Treatment with IVIg or guanidine did not significantly reduce lung concentrations of IL-1 α or IL-1 β . $^{****}P<0.0001$ compared to placebo-treated mice. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 9. Expt. NIA-1849. Lung concentrations of IL-6 and MCP-1 from EV-D68-infected AG129 mice treated with EV-D68-228. Treatment with EV-D68-228 significantly reduced concentrations of IL-6 and MCP-1 on day 3 post-infection compared to placebo-treated mice. Treatment with IVIg or guanidine significantly reduced lung concentrations of MCP-1 on day 3 post-infection. $^{*}P<0.05$, $^{**}P<0.01$, $^{***}P<0.001$, $^{****}P<0.0001$ compared to placebo-treated mice. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 10. Expt. NIA-1849. Lung concentrations of RANTES from EV-D68-infected AG129 mice treated with EV-D68-228. Treatment with EV-D68-228 significantly reduced concentrations of RANTES on days 3 and 5 post-infection compared to placebo-treated mice. Treatment with IVIg or guanidine did not significantly reduce lung concentrations of RANTES at any day post-infection. $^{**}P<0.01$, $^{***}P<0.001$, $^{****}P<0.0001$ compared to placebo-treated mice. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 11. Expt. NIA-1849. Lung concentrations of IL-2, IL-3, and IL-4 from EV-D68-infected AG129 mice treated with EV-D68-228. No significant changes in concentrations of IL-2, IL-3, or IL-4 were observed post-infection with EV-D68. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 12. Expt. NIA-1849. Lung concentrations of IL-5, IL-10, and IL-12p70 from EV-D68-infected AG129 mice treated with EV-D68-228. No significant changes in concentrations of IL-5, IL-10, or IL-12p70 were observed post-infection with EV-D68. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 13. Expt. NIA-1849. Lung concentrations of IL-17, IFN γ , and TNF α from EV-D68-infected AG129 mice treated with EV-D68-228. No significant changes in concentrations of IL-17, IFN γ , or TNF α were observed post-infection with EV-D68. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 14. Expt. NIA-1849. Lung concentrations of MIP-1 α and GM-CSF from EV-D68-infected AG129 mice treated with EV-D68-228. No significant changes in concentrations of MIP-1 α or GM-CSF were observed post-infection with EV-D68. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 15. Expt. NIA-1850. Survival of 10-day-old AG129 mice infected with EV-D68 and treated with EV-D68-228. (n=6 mice/group). Treatment with a dose of 10, 3 or 1 mg/kg of EV-D68-228 completely protected mice from mortality. A dose of 10 mg/kg of IVIg protected four of six mice from mortality. Guanidine at a dose of 100 mg/kg/day completely protected mice from mortality. All six of the placebo-treated mice succumbed to the infection $^{*}P<0.05$, $^{**}P<0.01$ compared to placebo-treated mice.

FIG. 16. Expt. NIA-1850. Mean body weights of EV-D68-infected 10-day-old AG129 mice treated with

EV-D68-228. Treatment with doses of 10, 3, or 1 mg/kg protected mice from infection-associated weight loss. Treatment with IVIg at a dose of 10 mg/kg also protected mice from weight loss. Mice treated with guanidine at a dose of 100 mg/kg/day were also protected from weight loss. $^{*}P<0.05$, $^{***}P<0.001$, $^{****}P<0.0001$ compared to placebo-treated mice.

FIGS. 17A-C. Expt. NIA-1850. Blood virus titers of 10-day-old AG129 mice infected with EV-D68 and treated with EV-D68-228. No blood virus titers were detected in mice treated with doses of 10, 3, or 1 mg/kg of EV-D68-228 day 1 (FIG. 17A), day 3 (FIG. 17B), or day 5 (FIG. 17C) post-infection. Treatment with a dose of 10 mg/kg of IVIg significantly reduced blood virus titers on days 1, 3, and 5 post-infection. Blood virus titers were also significantly reduced on days 1, 3, and 5 post-infection by treatment with guanidine at a dose of 100 mg/kg/day. $^{*}P<0.05$, $^{**}P<0.01$, $^{***}P<0.001$, $^{****}P<0.0001$ compared to placebo-treated mice.

FIG. 18. Expt. NIA-1850. Neurological scores of 10-day-old AG129 mice infected with EV-D68 and treated with EV-D68-228. Treatment with EV-D68-228 at doses of 10, 3, or 1 mg/kg prevented clinical signs of paralysis as measured by neurological scores. Treatment with a dose of 10 mg/kg of IVIg reduced neurological scores on days 3, 4, and 5 post-infection. No neurological scores were observed in the mice treated with 100 mg/kg/day of guanidine. $^{**}P<0.01$, $^{***}P<0.001$, $^{****}P<0.0001$ compared to placebo-treated mice. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIGS. 19A-B. Expt. NIA-1869. Lung virus titers of EV-D68-infected AG129 mice treated post-infection with EV-D68-228. Treatment with EV-D68-228 significantly reduced lung virus titers on day 3 (FIG. 19A) and day 5 (FIG. 19B) post-infection. No lung virus titers were detected in mice treated with 10 mg/kg of EV-D68-228 at within 24 hours post-infection. Treatment with EV-D68-228 at doses of 10 and 1 mg/kg at 48 hours post-infection significantly reduced lung virus titers at days 3 and 5 post-infection. Treatment with IVIg did not reduce lung virus titers when treated 24 hours post-infection. ($^{****}P<0.0001$ compared to placebo-treated mice).

FIGS. 20A-B. Expt. NIA-1869. Blood virus titers of EV-D68-infected AG129 mice treated post-infection with EV-D68-228. Treatment with the EV-D68-228 mAb reduced blood virus titers at day 3 (FIG. 20A) and day 5 (FIG. 20B) post-infection. Treatment with EV-D68-228 up to 48 hours post-infection reduced blood virus titers to below the limit of detection at both days 3 and 5 post-infection. Treatment with IVIg significantly reduced blood virus titers at days 3 and 5 post-infection. ($^{*}P<0.05$, $^{****}P<0.0001$ compared to placebo-treated mice).

FIG. 21. Expt. NIA-1869. Lung concentrations of IL-1 α and IL-1 β from EV-D68-infected AG129 mice treated post-infection with EV-D68-228. Treatment with EV-D68-228 significantly reduced concentrations of IL-1 α and IL-1 β on day 3 and day 5 post-infection compared to placebo-treated mice. Treatment with IVIg only significantly reduced lung concentrations of IL-1 α or IL-1 β on day 3 and day 5 post-infection. ($^{*}P<0.05$, $^{**}P<0.01$, $^{****}P<0.0001$ compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 22. Expt. NIA-1869. Lung concentrations of IL-5 and IL-6 from EV-D68-infected AG129 mice treated post-infection with EV-D68-228. Treatment with EV-D68-228 up to 48 hours post-infection significantly reduced concentra-

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tions of IL-5 and IL-6 on day 3 post-infection compared to placebo-treated mice. Treatment with IVIg 24 hours post-infection significantly reduced lung concentrations of IL-5 and IL-6 on day 3 post-infection. (**P<0.01, ***P<0.001, ****P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 23. Expt. NIA-1849. Lung concentrations of MCP-1 and RANTES from EV-D68-infected AG129 mice treated post-infection with EV-D68-228. Treatment with EV-D68-228 within 48 hours post-infection significantly reduced concentrations of MCP-1 and RANTES on days 3 and 5 post-infection compared to placebo-treated mice. Treatment with IVIg 24 hours post-infection significantly reduced lung concentrations of MCP-1 at day 3 post-infection and reduced concentrations of RANTES at days 3 and 5 post-infection. (**P<0.01, ***P<0.001, ****P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 24. Expt. NIA-1869. Lung concentrations of IL-2, IL-3, and IL-4 from EV-D68-infected AG129 mice treated post-infection with EV-D68-228. No significant changes in concentrations of IL-2, IL-3, or IL-4 were observed after infection with EV-D68. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 25. Expt. NIA-1869. Lung concentrations of IL-10, IL-12p70, and IL-17 from EV-D68-infected AG129 mice treated post-infection with EV-D68-228. No significant changes in concentrations of IL-5, IL-10, or IL-12p70 were observed after infection with EV-D68. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 26. Expt. NIA-1869. Lung concentrations of IFN γ and TNF α from EV-D68-infected AG129 mice treated post-infection with EV-D68-228. No significant changes in concentrations of IFN γ or TNF α were observed after infection with EV-D68. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 27. Expt. NIA-1869. Lung concentrations of MIP-1 α and GM-CSF from EV-D68-infected AG129 mice treated post-infection with EV-D68-228. No significant changes in concentrations of MIP-1 α or GM-CSF were observed after infection with EV-D68. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 28. Expt. NIA-1870. Survival of 10-day-old AG129 mice infected with EV-D68 and treated post-infection with EV-D68-228. (n=6 mice/group, n=7 mice/group treated 120 hours post-infection). Treatment with a dose of 10 mg/kg of EV-D68-228 completely protected mice from mortality at 24 hours post-infection. Treatment with EV-D68-228 at 48 hours post-infection protected four of six mice from mortality. Only one of six mice treated 72 hours post-infection with EV-D68-228 survived the infection. Despite none of the seven mice surviving in the group treated 120 hours post-infection with EV-D68-228, the survival curve was different than placebo-treated mice due to a delay in the day of death. A dose of 10 mg/kg of IVIg protected all six mice from mortality when treated 24 hours post-infection. All six of the placebo-treated mice succumbed to the infection (*P<0.05, **P<0.01, ***P<0.001 compared to placebo-treated mice).

FIG. 29. Expt. NIA-1870. Mean body weights of EV-D68-infected 10-day-old AG129 mice treated post-infection with EV-D68-228. Treatment with doses of 10 mg/kg of EV-D68-228 at 24 or 48 hours post-infection protected mice from infection-associated weight loss. Treatment with

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IVIg at a dose of 10 mg/kg 24 hours post-infection did not protect mice from weight loss. (*P<0.05, **P<0.01 compared to placebo-treated mice).

FIGS. 30A-C. Expt. NIA-1870. Blood virus titers of 10-day-old AG129 mice infected with EV-D68 and treated post-infection with EV-D68-228. Blood virus titers are shown at day 1 (FIG. 30A), day 3 (FIG. 30B), or day 5 (FIG. 30C) post-infection. Treatment with EV-D68-228 at a dose of 10 mg/kg significantly reduced blood virus titers on days 3 and 5 post-infection when administered 24, 48, or 72 hours after infection. IVIg at a dose of 10 mg/kg administered 24 hours post-infection also reduced blood virus titers at days 3 and 5 post-infection. (*P<0.05, ***P<0.001, ****P<0.0001 compared to placebo-treated mice).

FIGS. 31A-B. Expt. NIA-1870. Neurological scores of 10-day-old AG129 mice infected with EV-D68 and treated post-infection with EV-D68-228. Neurological scores are shown at days 2 and 3 post-infection (FIG. 31A) and days 4 and 5 post-infection (FIG. 31B). Treatment with EV-D68-228 at a dose of 10 mg/kg at 24 hours post-infection prevented clinical signs of paralysis on days 3, 4, and 5 as measured by neurological scores. Treatment with a dose of 10 mg/kg of IVIg 24 hours post-infection reduced neurological scores on days 3, 4, and 5 post-infection. (*P<0.05, **P<0.01, ***P<0.001, ****P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 32. Competition binding groups of mAbs from EV-D68-immune human subjects. Relatedness scores were generated from competition-binding ELISAs with a B1 clade EV-D68 isolate and used to cluster mAbs into four competition-binding groups designated 1 to 4. Clone numbers listed in red or blue are potentially neutralizing mAbs, with blue clone names indicating the two mAbs studied in detail in later figures.

FIGS. 33A-C. Neutralization potency and binding capacity of human mAbs. (FIG. 33A) MAb were ranked within competition-binding group (Comp. group, group 5 indicates the residual collection of singletons) by IC₅₀ value in a CCID₅₀ neutralization assay using a B1 clade isolate. The inventors also tested neutralization of a D clade and Fermon (Fer.) isolate for the 21 most potentially neutralizing mAbs. ">" denotes neutralization was not detected when tested in concentrations up to 50 μ g/mL. Blank cells indicate not tested. Clone numbers listed in red or blue are potentially neutralizing mAbs, with blue clone names indicating the two mAbs studied in detail in later figures. (FIG. 33B) Binding strength to live virus isolates or a mock virus preparation is denoted using EC₅₀ values generated by using (FIG. 33C) indirect ELISA with purified mAb dilutions to fit sigmoidal dose response curves. ">" indicates EC₅₀ value exceeds the maximum concentration tested of 100 μ g/mL, suggesting poor or no binding.

FIGS. 34A-C. Structural feature comparison between two immune complexes. (FIG. 34A) Radially colored cryo-EM maps of EV-D68:Fab EV68-159 (left) or EV-D68:Fab EV68-228 (right). Each map is projected down a two-fold axis of symmetry. The five-, three-, and two-fold axes of each asymmetric unit are depicted using a triangle outline labeled with one pentagon, two small triangles, and one oval, respectively. (FIG. 34B) Binding position comparison on an asymmetric unit. Viral proteins are colored in blue (VP1), green (VP2) and red (VP3). Fab molecules are colored in grey (EV68-159) or purple (EV68-228), and the heavy or light chains are shown in the same colors with dark or light intensities, respectively. (FIG. 34C) Footprints of EV68-159 Fab (left) or EV68-228 Fab (right). Radially

colored 2D projections of the viral surface were created with RIVEM. Virus surface residues facing any atoms from the Fab molecules within a distance of 4 Å are outlined in light blue (VP1), light green (VP2) and light red (VP3). The canyon region is outlined in yellow. Scale bars in (FIG. 34A) and (FIG. 34C) indicate radial distance measured in Å.

FIG. 35. Close-up view of the binding interfaces of EV68-159 and EV68-228. The viral capsid is shown as surface and the Fab is shown in a cartoon representation. VP1, VP2 and VP3 are colored in white, dark grey and silver, respectively. Heavy or light chains are colored in orange or yellow, respectively. Viral residues making interactions are colored based on the heavy and light chains, and the color intensities vary based on which of the VPs. The heavy and light chain complementarity-determining regions (HCDR and LCDR, respectively) involved in the binding interfaces are shown with arrows.

FIGS. 36A-D. Molecular detail of virion-Fab interactions. Representative interactions at the binding interface of EV-D68:Fab EV68-159 (FIGS. 36A-C) and EV-D68:Fab EV68-228 (FIG. 36D). Hydrogen bonds are colored in cyan and salt bridges are colored in magenta.

FIGS. 37A-D. MAb EV68-228 protects mice from EV-D68-induced respiratory disease, when used as either prophylaxis or therapy. Four-week-old AG129 strain mice (n=4 per time point) were inoculated with mouse-adapted B1 clade EV-D68 intranasally; antibody was administered intraperitoneally; and viral titers for indicated tissue were measured by a CCID₅₀ assay. (FIGS. 37A-B) Mice were inoculated with virus 24 hours after indicated dose of antibody, then viral titers were measured at indicated time points. (FIGS. 37C-D) Mice were inoculated with virus followed by 10 mg/kg (except where indicated) of antibody 4, 24, or 48 hours later, then viral titers were measured.

FIGS. 38A-F. MAb EV68-228 decreases lung inflammation in EV-D68 infected mice. Four-week-old AG129 mice (n=4 per time point) (FIGS. 38A-C) were inoculated with virus intranasally 24 hours after indicated dose of antibody or (FIGS. 38D-F) were inoculated with virus intranasally followed by 10 mg/kg (except where indicated) of antibody 4, 24, or 48 hours later, then cytokines were measured at indicated time points. Cytokines were quantified from lung homogenates using an ELISA. Values from the treatment groups were compared to the placebo group for each time point using a one-way ANOVA with Dunnett's T3 multiple comparisons test (*P<0.05, **P<0.01, ***P<0.001, ****P<0.0001). IL—interleukin; MCP—monocyte chemoattractant protein. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIGS. 39A-F. MAb EV68-228 protects mice from EV-D68-induced neurologic disease, when used as either prophylaxis or therapy. Ten-day-old mice were inoculated with mouse-adapted B1 clade EV-D68 intraperitoneally; antibody was administered intraperitoneally; and viral titers for indicated tissue were measured by a CCID₅₀ assay. (FIGS. 39A-C) Mice were inoculated with virus 24 hours after indicated dose of antibody, then (FIG. 39A) viral titers were measured (n=3 per time point), (FIG. 39B) survival was monitored, and (FIG. 39C) neurologic scores (n=6 per time point) were recorded at indicated time points. Higher scores indicate more severe motor impairment. (FIGS. 39D-F) Mice were inoculated with virus followed by a 10 mg/kg dose of antibody, then (FIG. 39D) viral titers were measured (n=3 per time point), (FIG. 39E) survival was monitored, and (FIG. 39F) neurologic scores (n=6 per time point, except n=9 for 120 hr post) were recorded. Colored vertical arrows indicate time of treatment.

FIG. 40. Western blot data. All mAbs were tested for the ability to stain B1 clade virus by western blot. All positive results are shown.

FIG. 41. Detailed characteristics of human mAbs. MABs were ranked within competition-binding group by IC₅₀ value in a CCID₅₀ neutralization assay using a B1 clade isolate. The inventors also tested neutralization of a D clade and Fermon (Fer.) isolate for the 21 most potently neutralizing mAbs. ">" denotes neutralization was not detected when tested at concentrations up to 50 µg/mL. Blank cells indicate not tested. Clone numbers listed with red or blue font are potently neutralizing mAbs, with blue clone names indicating the two mAbs studied in detail in later figures. Final IC₅₀ values for each mAb are the average of the logarithm of the EC₅₀ values from three separate experiments performed in duplicate. Binding strength to live virus isolates are denoted based on EC₅₀ values generated using indirect ELISA with purified mAb dilutions to fit sigmoidal dose response curves. ">" indicates the predicted EC₅₀ value exceeds the maximum concentration tested of 100 µg/mL, suggesting poor or no binding. Final EC₅₀ values for each mAb are the average of the logarithm of the EC₅₀ values from three separate experiments performed in duplicate. IgG subtypes were determined phenotypically. Light chain types are kappa (κ) or lambda (λ) and were determined genetically. WB indicates western blot reactivity, and all mAbs were tested, with blank rows indicating no reactivity.

FIG. 42. Indirect ELISA data for all mAbs. Binding strength to live virus isolates is denoted by EC₅₀ values generated by using indirect ELISA with purified mAb dilutions to fit sigmoidal dose response curves. Shown are representative data from one of three experiments, with error bars indicating the standard deviation of duplicate technical replicates. Final EC₅₀ values for each mAb are the average of the logarithm of the EC₅₀ values from all three experiments. The mock preparation of virus was from an RD cell monolayer that was not inoculated with virus but was prepared in the same manner as virus-infected cells.

FIG. 43. Representative densities from the EV-D68: Fab EV68-228 electron density map. Viral proteins are colored in blue (VP1), green (VP2), red (VP3) and magenta (VP4).

FIG. 44. Estimates of immune complex map resolutions. Map resolutions are estimated based on the gold-standard Fourier shell correlation (FSC) cutoff of 0.143. The final resolutions for EV68-159 (red curve) or EV68-228 (blue curve) complexes are 2.9 Å or 3.1 Å, respectively.

FIG. 45. Comparison of The Fab binding sites. EV68-159 and EV68-228 are colored in gold and blue, respectively, and the viral surface is colored in cyan. The left panel shows both variable domains and constant domains, whereas the right panel shows only the variable domains, demonstrating that the footprints of the two Fabs do not overlap.

FIG. 46. Roadmaps showing an enlarged view of the Fab footprints. The radially colored 2D projection of the viral surface was created with RIVEM. Virus surface residues facing any atoms from the Fab molecules within a distance of 4 Å are outlined in light blue (VP1), light green (VP2) and light red (VP3). The canyon region is outlined in yellow.

FIG. 47. Bulky side chains of the EV68-228 Fab heavy chain. This view shows an example of bulky side chains forming a hydrophobic interaction network to stabilize the EV68-228 Fab, which also is seen in the structure of the EV68-159 Fab.

FIGS. 48A-C. Expt. NIA-1930. Lung virus titers of EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Lung virus titers are shown on day 1 (FIG. 48A), day 3 (FIG. 48B), and day 5

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(FIG. 48C) post-infection. No lung virus titers were detected in mice treated with 10 mg/kg of EV68-228-TP or EV68-228-CHO at days 1, 3, or 5 post-infection. Treatment with 10 mg/kg of hIVlg only significantly reduced lung virus titers on day 5 post-infection. (**P<0.01, ****P<0.0001 compared to placebo-treated mice.)

FIGS. 49A-C. Expt. NIA-1930. Blood virus titers of EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Blood virus titers are shown at day 1 (FIG. 49A), day 3 (FIG. 49B), and day 5 (FIG. 49C) post-infection. No virus was detected in the blood of mice treated with 10 mg/kg of EV68-228-TP or EV68-228-CHO on days 1, 3, or 5 post-infection. Treatment with 10 mg/kg of hIVlg significantly reduced blood virus titers at days 1, 3, and 5 post-infection. (*P<0.05, **P<0.01, ***P<0.001, ****P<0.0001 compared to placebo-treated mice.)

FIG. 50. Expt. NIA-1930. Lung concentrations of IL-1 α , IL-1 β , IL-2, and IL-3 from EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced concentrations of IL-1 α on days 1, 3, and 5 post-infection. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO also reduced concentrations of IL-1 β on day 3 post-infection. Treatment with hIVlg only significantly reduced lung concentrations of IL-1 β on day 3 post-infection. No significant changes were observed in concentrations of IL-2 or IL-3 following infection. (*P<0.05, ***P<0.001, ****P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 51. Expt. NIA-1930. Lung concentrations of IL-4, IL-5, IL-6, and IL-10 from EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced concentrations of IL-5 on day 3 post-infection and IL-6 on days 3 and 5 post-infection. Treatment with 10 mg/kg of hIVlg significantly reduced lung concentrations of IL-5 and IL-6 on day 3 post-infection. No significant changes were observed in concentrations of IL-4 or IL-10 following infection. (*P<0.05, ***P<0.001, ****P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 52. Expt. NIA-1930. Lung concentrations of IL-12p70, IL-17, MCP-1, and IFN- γ from EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced concentrations of MCP-1 on days 3 and 5 post-infection. Treatment with 10 mg/kg of hIVlg significantly reduced lung concentrations of MCP-1 at day 3 and 5 post-infection. No significant changes were observed in concentrations of IL-12p70, IL-17, or IFN- γ following infection. (**P<0.01, ***P<0.001, ****P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 53. Expt. NIA-1930. Lung concentrations of TNF α , MIP-1 α , GM-CSF, and RANTES from EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced lung concentrations of MIP-1 α on days 1 and 3 post-infection and reduced concentrations of RANTES on days 3 and 5 post-infection. A 10 mg/kg dose of hIVlg significantly reduced concentrations of MIP-1 α on days 1 and 3 post-infection and also reduced concentrations of RANTES on day 3 post-

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infection. No significant changes in concentrations of TNF α or GM-CSF were observed after infection with EV-D68. (**P<0.01, ***P<0.001, P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIGS. 54A-B. Expt. NIA-1930. Lung virus titers of EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Lung virus titers are shown on day 3 (FIG. 54A) and day 5 (FIG. 54B) post-infection. No lung virus titers were detected in mice treated with 10 mg/kg of EV68-228-TP or EV68-228-CHO at days 3 or 5 post-infection. Treatment with 10 mg/kg of hIVlg only significantly reduced lung virus titers on day 3 post-infection. (**P<0.01, ***P<0.001, ****P<0.0001 compared to placebo-treated mice.)

FIGS. 55A-B. Expt. NIA-1930. Blood virus titers of EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Blood virus titers are shown at day 3 (FIG. 55A) and day 5 (FIG. 55B) post-infection. No virus was detected in the blood of mice treated with 10 mg/kg of EV68-228-TP or EV68-228-CHO on days 3 or 5 post-infection. Treatment with 10 mg/kg of hIVlg significantly reduced blood virus titers at days 3 and 5 post-infection. (*P<0.05, ***P<0.001, ****P<0.0001 compared to placebo-treated mice.)

FIG. 56. Expt. NIA-1930. Lung concentrations of IL-1 α , IL-1 β , IL-2, and IL-3 from EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced concentrations of IL-1 α and IL-1 β on day 3 post-infection. Concentrations of IL-3 were significantly reduced in mice treated with EV68-228-TP as well as mice treated with hIVlg on day 3 post-infection. No significant changes were observed in concentrations of IL-2 following infection. (*P<0.05, **P<0.01, ****P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 57. Expt. NIA-1930. Lung concentrations of IL-4, IL-5, IL-6, and IL-10 from EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced concentrations of IL-5 on day 3 post-infection and IL-6 on days 3 and 5 post-infection. Treatment with 10 mg/kg of hIVlg significantly reduced lung concentrations of IL-5 on day 3 post-infection. No significant changes were observed in concentrations of IL-4 or IL-10 following infection. (*P<0.05, **P<0.01, ***P<0.001, ****P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 58. Expt. NIA-1930. Lung concentrations of IL-12p70, IL-17, MCP-1, and IFN- γ from EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced concentrations of MCP-1 on days 3 and 5 post-infection. Treatment with 10 mg/kg of hIVlg significantly reduced lung concentrations of MCP-1 at day 3 and 5 post-infection. No significant changes were observed in concentrations of IL-12p70, IL-17, or IFN- γ following infection. (****P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 59. Expt. NIA-1930. Lung concentrations of TNF α , MIP-1 α , GM-CSF, and RANTES from EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-

228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced lung concentrations of MIP-1 α on day 3 post-infection and reduced concentrations of RANTES on days 3 and 5 post-infection. No significant changes in concentrations of TNF α or GM-CSF were observed after infection with EV-D68. (****P<0.0001 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIGS. 60A-B. Expt. NIA-1930. Lung virus titers of EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. Lung virus titers are shown on day 3 (FIG. 60A) and day 5 (FIG. 60B) post-infection. Only EV68-228-CHO significantly reduced lung virus titers on days 3 or 5 post-infection. (****P<0.0001 compared to placebo-treated mice).

FIGS. 61A-B. Expt. NIA-1930. Blood virus titers of EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. Blood virus titers are shown at day 3 (FIG. 61A) and day 5 (FIG. 61B) post-infection. No virus was detected in the blood of mice treated with 10 mg/kg of EV68-228-TP or EV68-228-CHO on days 3 or 5 post-infection. Treatment with 10 mg/kg of hIVIg significantly reduced blood virus titers at days 3 and 5 post-infection. (****P<0.0001 compared to placebo-treated mice).

FIG. 62. Expt. NIA-1930. Lung concentrations of IL-1 α , IL-1 β , IL-2, and IL-3 from EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. Only treatment with 10 mg/kg of EV68-228-CHO significantly reduced concentrations of IL-1 α and IL-1 β on day 3 post-infection. No significant changes were observed in concentrations of IL-2 or IL-3 following infection. (**P<0.01 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 63. Expt. NIA-1930. Lung concentrations of IL-4, IL-5, IL-6, and IL-10 from EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced concentrations of IL-6 on day 3 post-infection. No significant changes were observed in concentrations of IL-4, IL-5, or IL-10 following infection. (**P<0.01 compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 64. Expt. NIA-1930. Lung concentrations of IL-12p70, IL-17, MCP-1, and IFN- γ from EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. No significant changes were observed in concentrations of IL-12p70, IL-17, MCP-1, or IFN- γ following infection. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 65. Expt. NIA-1930. Lung concentrations of TNF α , MIP-1 α , GM-CSF, and RANTES from EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. No significant changes in concentrations of TNF α , MIP-1 α , GM-CSF, or RANTES were observed after infection with EV-D68. Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 66. Expt. NIA-1931. Survival of 10-day-old AG129 mice infected with EV-D68 and treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. (n=10 mice/group). Treatment with a dose of 10 mg/kg of EV68-228-TP or completely protected mice from mortality. All ten mice treated with 10 mg/kg of EV68-228-CHO were pro-

tected from mortality. A dose of 10 mg/kg of IVIg protected eight of ten mice from mortality. Nine of the ten of the placebo-treated mice succumbed to the infection (***P<0.001, ****P<0.0001 compared to placebo-treated mice).

FIG. 67. Expt. NIA-1931. Percentages of initial body weight of EV-D68-infected 10-day-old AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. No significant differences in weight loss were observed in mice treated 24 hours pre-infection with 10 mg/kg doses EV68-228-TP or EV68-228-CHO compared to placebo-treated mice. Treatment with a 10 mg/kg dose of hIVIg did not protect mice from weight loss.

FIGS. 68A-B. Expt. NIA-1931. Blood virus titers on days 1 and 3 post-infection of 10-day-old AG129 mice infected with EV-D68 and treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Blood virus titers are shown on day 1 (FIG. 68A) and day 3 (FIG. 68B) post-infection. Treatment 24 hours pre-infection with 10 mg/kg of EV68-228-TP significantly reduced virus titers on days 1 and 3 post-infection. Treatment 24 hours pre-infection with 10 mg/kg of EV68-228-CHO significantly reduced virus titers on days 1 and 3 post-infection. Treatment 24 hours pre-infection with a dose of 10 mg/kg of IVIg significantly reduced blood virus titers on day 3 post-infection. (*P<0.05, **P<0.01, ****P<0.0001 compared to placebo-treated mice).

FIGS. 69A-B. Expt. NIA-1931. Blood virus titers on days 5 and 7 post-infection of 10-day-old AG129 mice infected with EV-D68 and treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Blood virus titers are shown on day 5 (FIG. 69A) and day 7 (FIG. 69B) post-infection. Treatment 24 hours pre-infection with 10 mg/kg of EV68-228-TP or EV68-228-CHO reduced virus titers on day 5 post-infection. No virus was detected at day 7 post-infection.

FIGS. 70A-C. Expt. NIA-1931. Neurological scores of 10-day-old AG129 mice infected with EV-D68 and treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO on days 2-10 post-infection. Neurological scores are shown for days 2-4 (FIG. 70A), days 5-7 (FIG. 70B), and days 8-10 (FIG. 70C) post-infection. Treatment 24 hours pre-infection with 10 mg/kg of EV68-228-TP or EV68-228-CHO completely prevented clinical signs of paralysis as measured by neurological scores. No neurological scores were observed in mice treated with a dose of 10 mg/kg of IVIg 24 hours pre-infection. (****P<0.0001 compared to placebo-treated mice).

FIG. 71. Expt. NIA-1931. Survival of 10-day-old AG129 mice infected with EV-D68 and treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. (n=10 mice/group). Treatment with a dose of 10 mg/kg of EV68-228-TP or completely protected mice from mortality. All ten mice treated with 10 mg/kg of EV68-228-CHO were protected from mortality. A dose of 10 mg/kg of IVIg protected ten of ten mice from mortality. Eight of the ten of the placebo-treated mice succumbed to the infection (***P<0.001 compared to placebo-treated mice).

FIG. 72. Expt. NIA-1931. Percentages of initial body weight of EV-D68-infected 10-day-old AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. No significant differences in weight loss were observed in mice treated 24 hours post-infection with 10 mg/kg doses EV68-228-TP or EV68-228-CHO compared to placebo-treated mice. Treatment with a 10 mg/kg dose of hIVIg did not protect mice from weight loss.

FIGS. 73A-B. Expt. NIA-1931. Blood virus titers on days 1 and 3 post-infection of 10-day-old AG129 mice infected

with EV-D68 and treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Blood virus titers are shown on day 1 (FIG. 73A) and day 3 (FIG. 73B) post-infection. Treatment 24 hours post-infection with 10 mg/kg of EV68-228-TP significantly reduced virus titers on day 3 post-infection. Treatment 24 hours post-infection with 10 mg/kg of EV68-228-CHO significantly reduced virus titers on day 3 post-infection. Treatment 24 hours post-infection with a dose of 10 mg/kg of IVIg significantly reduced blood virus titers on day 3 post-infection. (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.0001$ compared to placebo-treated mice).

FIGS. 74A-B. Expt. NIA-1931. Blood virus titers on days 5 and 7 post-infection of 10-day-old AG129 mice infected with EV-D68 and treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Blood virus titers are shown on day 5 (FIG. 74A) and day 7 (FIG. 74B) post-infection. No virus was detected after treatment 24 hours post-infection with 10 mg/kg of EV68-228-TP or EV68-228-CHO on day 5 post-infection. No virus was detected in any mice at day 7 post-infection.

FIGS. 75A-B. Expt. NIA-1931. Neurological scores of 10-day-old AG129 mice infected with EV-D68 and treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO on days 2-9 post-infection. Neurological scores are shown for days 2-5 (FIG. 75A) and days 6-9 (FIG. 75B) post-infection. Treatment 24 hours post-infection with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced signs of paralysis as measured by neurological scores on days 2-9 post-infection. Treatment with hIVIg at a dose of 10 mg/kg 24 hours after infection only reduced neurological scores on days 2 and 3 post-infection. (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$ compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIGS. 76A-B. Expt. NIA-1931. Neurological scores of 10-day-old AG129 mice infected with EV-D68 and treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO on days 10-17 post-infection. Neurological scores are shown for days 10-13 (FIG. 76A) and days 14-17 (FIG. 76B) post-infection. Treatment 24 hours post-infection with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced signs of paralysis as measured by neurological scores on days 10-16 post-infection. Treatment with hIVIg at a dose of 10 mg/kg 24 hours after infection did not significantly reduce neurological scores. (* $P < 0.05$, ** $P < 0.01$ compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

FIG. 77. Expt. NIA-1931. Survival of 10-day-old AG129 mice infected with EV-D68 and treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. (n=10 mice/group). Treatment with a dose of 10 mg/kg of EV68-228-TP protected nine of ten mice from mortality. Eight of ten mice treated with 10 mg/kg of EV68-228-CHO were protected from mortality. A dose of 10 mg/kg of IVIg protected six of ten mice from mortality. Nine of the ten of the placebo-treated mice succumbed to the infection (** $P < 0.01$, *** $P < 0.001$ compared to placebo-treated mice).

FIG. 78. Expt. NIA-1931. Percentages of initial body weight of EV-D68-infected 10-day-old AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. No significant differences in weight loss were observed in mice treated 24 hours post-infection with 10 mg/kg doses EV68-228-TP or EV68-228-CHO compared to placebo-treated mice. Treatment with a 10 mg/kg dose of hIVIg did not protect mice from weight loss.

FIGS. 79A-B. Expt. NIA-1931. Blood virus titers on days 1 and 3 post-infection of 10-day-old AG129 mice infected with EV-D68 and treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. Blood virus titers are shown on day 1 (FIG. 79A) and day 3 (FIG. 79B) post-infection. Treatment 48 hours post-infection with 10 mg/kg of EV68-228-TP or EV68-228-CHO did not significantly reduce blood virus titers. Treatment 48 hours post-infection with a dose of 10 mg/kg of IVIg did not significantly reduce blood virus titers.

FIGS. 80A-B. Expt. NIA-1931. Blood virus titers on days 5 and 7 post-infection of 10-day-old AG129 mice infected with EV-D68 and treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. Blood virus titers are shown on day 5 (FIG. 80A) and day 7 (FIG. 80B) post-infection. No virus was detected after treatment 24 hours post-infection with 10 mg/kg of EV68-228-TP or EV68-228-CHO on day 5 post-infection. No virus was detected in any mice at day 7 post-infection.

FIGS. 81A-B. Expt. NIA-1931. Neurological scores of 10-day-old AG129 mice infected with EV-D68 and treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO on days 3-10 post-infection. Neurological scores are shown for days 3-6 (FIG. 81A) and days 7-10 (FIG. 81B) post-infection. Treatment 48 hours post-infection with 10 mg/kg of EV68-228-TP significantly reduced neurological scores on days 3-5 post-infection. A single administration of a 10 mg/kg dose of EV68-228-CHO 48 hours after infection significantly reduced signs of paralysis as measured by neurological scores on days 3-6 post-infection. Treatment with hIVIg at a dose of 10 mg/kg 48 hours after infection did not significantly reduce neurological scores post-infection. (* $P < 0.05$, *** $P < 0.001$, **** $P < 0.0001$ compared to placebo-treated mice). Designation of samples in graph corresponds left-to-right with vertical legend top-to-bottom.

DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

As discussed above, enterovirus D68 (EV-D68) can cause outbreaks of respiratory illness across the world, mostly in children. Clusters of acute flaccid myelitis, a poliomyelitis-like neuromuscular weakness syndrome, often occur concurrently with EV-D68 respiratory outbreaks. Seroepidemiologic studies have found that the serum of nearly everyone older than 2 to 5 years contains anti-EV-D68 neutralizing antibodies, suggesting that EV-D68 is a ubiquitous pathogen of childhood. However, knowledge of the viral epitopes against which the humoral immune response is directed is only inferred from previous studies of related viruses. Here, the inventors have generated 64 human monoclonal antibodies against EV-D68, some of which are potentially neutralizing. These are believed to be the first human mAbs against this pathogen and the inventors propose their use in detecting, treating and preventing EV-D68 infections. These and other aspects of the disclosure are described in detail below.

I. ENTEROVIRUS D68

Enterovirus 68 (EV68, EV-D68, HEV68) is a member of the Picornaviridae family, an enterovirus. First isolated in California in 1962 and once considered rare, it has been on a worldwide upswing in the 21st century. It is suspected of causing a polio-like disorder called acute flaccid myelitis.

EV68 is one of the more than one hundred types of enteroviruses, a group of ssRNA viruses containing the

polioviruses, coxsackieviruses, and echoviruses, and it is unenveloped. Unlike all other enteroviruses, EV68 displays acid lability and a lower optimum growth temperature, both characteristic features of the human rhinoviruses. It was previously called human rhinovirus 87 by some researchers. Children less than 5 years old and children with asthma appear to be most at risk for the illness, although illness in adults with asthma and immunosuppression have also been reported.

Since its discovery, EV68 had been described mostly sporadically in isolated cases. Six clusters (equal to or more than 10 cases) or outbreaks between 2005 and 2011 have been reported from the Philippines, Japan, the Netherlands, and the states of Georgia, Pennsylvania and Arizona in the United States. EV68 was found in 2 of 5 children during a 2012/13 cluster of polio-like disease in California. In August 2014, the virus caused clusters of respiratory disease in the United States. By mid-October 691 people in 46 states and the District of Columbia had come down with a respiratory illness caused by EV-D68 and five children died. In 2016, 29 cases were reported in Europe (5 in France and Scotland. 3 each in Sweden, Norway and Spain).

Since the year 2000, the original virus strains diversified and evolved a genetically distinct outbreak strain, clade B1. It is Clade B1, but not older strains, which has been associated with AFM and is neuropathic in animal models. Cases have been described to occur late in the enterovirus season (roughly the period of time between the spring equinox and autumn equinox), which is typically during August and September in the Northern Hemisphere.

EV68 almost exclusively causes respiratory illness, which varies from mild to severe, but can cause a range of symptoms, from none at all, to subtle flu-like symptoms, to debilitating respiratory illness and a suspected rare involvement in a syndrome with polio-like symptoms. Like all enteroviruses, it can cause variable rashes, abdominal pain and soft stools. Initial symptoms are similar to those for the common cold, including a runny nose, sore throat, cough, and fever. As the disease progresses, more serious symptoms may occur, including difficulty breathing as in pneumonia, reduced alertness, a reduction in urine production, and dehydration, and may lead to respiratory failure.

The degree of severity of symptoms experienced seems to depend on the demographic population in question. Experts estimate that the majority of the population has, in fact, been exposed to the enterovirus, but that no symptoms are exhibited in healthy adults. In contrast, EV-D68 is disproportionately debilitating in very young children, as well as the very weak. While several hundred people (472), mostly youth, have been exposed to the disease, less than a hundred of those patients have been diagnosed with severe symptoms (such as paralysis).

The virus has been suspected as one cause of acute flaccid myelitis a rare muscle weakness, usually due to polio, since two California children who tested positive for the virus had paralysis of one or more limbs reaching peak severity within 48 hours of onset. Recovery of motor function was poor at 6-month follow-up. As of October 2014, the CDC was investigating 10 cases of paralysis and/or cranial dysfunction in Colorado and other reports around the country, coinciding with the increase in enterovirus D68 activity. As of October 2014, it was believed that the actual number of cases might be 100 or more. As of 2018 the link of EV-D68 and the paralysis is strong, meeting six Bradford Hill criteria fully and two partially. The CDC issued a statement on Oct. 17, 2018 claiming "Right now, we know that poliovirus is not the cause of these AFM cases. CDC has tested every

stool specimen from the AFM patients, none of the specimens have tested positive for the poliovirus." In 2014, a real-time PCR test to speed up detection was developed by CDC.

There is no specific treatment and no vaccine, so the illness has to run its course; treatment is directed against symptoms (symptomatic treatment). Most people recover completely; however, some need to be hospitalized, and some have died as a result of the virus. Five EV68 paralysis cases were unsuccessfully treated with steroids, intravenous immunoglobulin and/or plasma exchange. The treatment had no apparent benefit as no recovery of motor function was seen. A 2015 study suggested the antiviral drug pleconaril may be useful for the treatment of EV-D68.

Since the virus is spread through saliva and phlegm as well as stool, washing hands is important. Sick people can attempt to decrease spreading the virus by basic sanitary measures, such as covering the nose and mouth when sneezing or coughing. Other measures including cleaning surfaces and toys. For hospitalized patients with EV-D68 infection, the CDC recommends transmission-based precautions, i.e., standard precautions, contact precautions, as is recommended for all enteroviruses, and to consider droplet precautions. According to the CDC in 2003, surfaces in healthcare settings should be cleaned with a hospital-grade disinfectant with an EPA label claim for any of several non-enveloped viruses (e.g., norovirus, poliovirus, rhinovirus).

A. Viral Epitopes

Measurements of serum antibody virus neutralization capacity reflect the activity of an entire polyclonal antibody repertoire. However, to fully understand the humoral response to a virus it is necessary to define the specific viral epitopes to which individual antibodies bind and determine whether antibody binding to specific epitopes protects against disease. Historically, four neutralizing immunopeptides (Nim) for viruses of the Enterovirus genus were identified through studies of murine monoclonal antibodies raised against rhinovirus-B14 (Rossmann et al., 1985). Studies of EV-D68 specific monoclonal antibody epitopes have not been performed to date. Determination of the crystal structure of EV-D68, however, suggested the likely location of the four Nims on the EV-D68 virion by observations of structural homology with rhinoviruses (Liu et al., 2015a). Studies comparing the amino acid sequences of the surface proteins of recent human isolates of EV-D68 from the U.S. (Zhang et al., 2015) or Japan (Imamura et al., 2014) to that of the Fermon reference virus strain suggest that mutations in the Nims and nearby flanking residues have occurred, especially in the BC and DE loops of capsid protein VP1 that are disordered in the crystal structure (Liu et al., 2015a). It is possible that these VP1 polymorphisms contribute to increased pathogenicity of the virus by eluding pre-existing humoral immunity. Murine monoclonal antibodies are available commercially (sources include GeneTex, ThermoFisher, and Sigma), but aside from listing the specific viral surface protein that was used as an immunogen, information is not available on the epitopes for these antibodies. Notably, some of these commercial antibodies were generated from immunogens of other Enterovirus genus viruses, suggesting the ability of heterotypic antibodies to bind to EV-D68. Further supporting this possibility, two murine mAbs cross-reacted to 40 different enterovirus species, although curiously EV-D68 was the only virus tested that they did not react to (Miao et al., 2009). Heterotypic molecular recognition of EV-D68 by antibodies induced by other types of enteroviruses might contribute to the observed

universal EV-D68 seroprevalence early in life. Understanding the fine specificity of human antibody epitopes on EV-D68 and the type specificity or breadth of such antibodies will require studies with cloned human monoclonal antibodies induced by natural infection.

During the 2014 EV-D68 outbreak in the U.S., nearly all viral isolates were of the newly emergent B1 clade, with fewer detections of virus from the closely related B2 or distantly related D clades (Tan et al., 2016). All but one of the subjects for this study were infected with B1 clade isolates (Table E). Since 2014, B3 clade viruses have dominated, and B1 clade viruses are no longer circulating (Dyrda et al., 2019); in 2018 all EV-D68 isolates sequenced by the U.S. Centers for Disease Control and Prevention were from the B3 clade (Kujawski et al., 2019). The inventors first measured the *in vitro* neutralization capability of each mAb in a 50% cell culture infectious dose (CCID₅₀) assay using a B1 clade EV-D68 isolate (FIG. 33A). Twenty-eight mAbs demonstrated neutralization with a half maximal inhibitory concentration (IC₅₀) below 50 µg/mL, with mAb EV68-159 exhibiting the strongest neutralization at an IC₅₀ value of 0.32 ng/mL (FIG. 41). The inventors further tested the 21 most potently neutralizing mAbs against a D clade isolate and found that 11 mAbs neutralized that virus, with 7 of those exhibiting at least a ten-fold decrease in potency by IC₅₀ value for the heterologous virus. The Fermon strain is an isolate from 1962 and is so distantly related to modern EV-D68 isolates that it does not fit into the clade classification scheme (Tan et al., 2016). Nine mAbs neutralized the Fermon laboratory reference strain, but less potently than they inhibited the contemporary B1 clade virus.

Recognizing that neutralization assays may underestimate cross-reactivity, the inventors used the same indirect ELISA approach described above to generate half-maximal effective concentrations (EC₅₀) of purified mAb for binding to representative EV-D68 isolates from the B1, B2, or D clades (FIGS. 33B-C, and FIG. 42). Of the mAbs with EC₅₀ values for binding of ≤1 µg/mL to B1 clade isolates, all bound to a B2 clade isolate, whereas about half also bound to a D clade isolate (FIG. 33B and FIG. 41). An additional class of mAbs was observed that bound weakly in general but cross-reacted to viruses from all clades tested.

To date, structural studies of antibody-EV-D68 interactions have been limited to murine mAbs (Zheng et al., 2019). The inventors selected two potently neutralizing human mAbs, the clade-specific mAb EV68-159 and the highly cross-reactive mAb EV68-228, to make immune complexes with antigen binding fragments (Fabs) and a B1 clade EV-D68 isolate for cryo-electron microscopy (cryo-EM) studies. The final density maps attained a resolution of 2.9 Å (EV68-159) or 3.1 Å (EV68-228) (FIG. 34A, FIG. 43, FIG. 44, and Table G). The structures revealed two distinct binding sites: EV68-159 attached around the three-fold axes of symmetry, whereas EV68-228 bound around the five-fold axes between depressions that form the canyon regions (FIGS. 34A-C, FIG. 45). Thus, for each Fab, a total of 60 copies bound to the virus particle. The Fab variable domains, which interacted with the viral surface, displayed strong densities similar to the viral capsid proteins, and an atomic model of each Fab was built together with the four viral capsid proteins. In contrast, the Fab constant domains, which are located further from the viral surface, displayed weaker densities and were excluded from atomic model building. The backbone of the polypeptide chains and the majority of amino acid side chains are well-ordered in the density maps, demonstrating the critical features of the binding interface between virus particle and Fab molecule.

B. Animal Models of Neurologic Disease

Measuring neutralization of viruses by antibodies *in vitro* is typically the most rapid and reproducible method to characterize the antiviral function of antibodies. However, this mode of testing does not measure effector functions mediated by the Fc portion of antibodies that operate only *in vivo*, such as complement activation or engagement with cell-surface Fc receptors. Consequently, the magnitude of *in vitro* neutralization capacity may not correlate perfectly with the levels of inhibition of viral replication or protection against disease observed *in vivo* (reviewed in (Lu et al., 2018)). Many of the Fc-mediated effector functions of antibodies require Fc receptor bearing cells of the innate or adaptive immune systems to mediate full protective effects. Small animal models of infection are needed for assessment of efficacy of humoral immunity in preclinical studies.

Investigators in Colorado (Hixon et al., 2017a; 2017b), Utah (Morrey et al., 2018), and China (Zhang et al., 2018) have reported murine models of EV-D68 since early 2017. In all three reported models, mice of no greater than 10 days old are inoculated with virus. Across these models, EV-D68 has been shown capable of causing flaccid limb paralysis and death when administered by intraperitoneal (Hixon et al., 2017a; 2017b; Morrey et al., 2018; Zhang et al., 2018), intramuscular (Hixon et al. 2017a; 2017b), or intracerebral routes (Hixon et al., 2017b). Notably, intranasal inoculation led to paralysis in only two of 73 outbred NIH Swiss Webster mice tested, with virus detected in the spinal cord (Hixon et al., 2017b), whereas intranasal inoculation of AG129 mice, which lack interferon type-I and -II receptors, caused paralysis in two of four mice with virus detected in muscle but not spinal cord (Morrey et al., 2018). However, all studies in which humoral immunity was evaluated in mice used non-physiologic routes of virus inoculation.

Viral antigen has been visualized in tissues by immunostaining in skeletal muscle (Morrey et al., 2018; Zhang et al., 2018) and spinal cord (Hixon et al., 2017b; Morrey et al., 2018). Involvement of both muscle and spinal cord suggests that EV-D68 may induce paralysis by two mechanisms: 1) direct pathologic effect on skeletal muscles and 2) loss of central motor neurons. Motor neurons in the spinal cord were infected, which is analogous to the pathogenesis of poliomyelitis. This pattern of murine neuron involvement in immunostaining studies also correlates with findings of gray matter change on spinal cord imaging in patients with AFM (Hixon et al., 2017b; Morrey et al., 2018). More sensitive real-time quantitative RT-PCR tests detected EV-D68 RNA chiefly in muscle and spinal cord, but also in brain, heart, lung, intestine, liver, spleen, kidney, and blood (Zhang et al., 2018). However, other than in spinal cord and muscle, the amounts of RNA detected were of questionable significance and may well reflect viral genome copies present in blood within these tissues rather than true tropism for these tissues.

Studies in these models of paralysis and lethality facilitated preliminary studies of the ability of antibodies to protect from neurologic disease. Heat-inactivated serum from immune mice passively transferred to naïve mice prevented neurologic disease, even when EV-D68 was inoculated directly by the intracerebral route (Hixon et al., 2017b). Also, therapeutic administration of human IVIG reduced motor impairment even up to 6 days after infection of mice (Hixon et al., 2017a), consistent with findings of high titers of EV-D68 specific antibodies in IVIG (Zhang et al., 2015). This observation provides hope that vaccines or human monoclonal antibodies could mediate a therapeutic effect after an individual patient is identified to have features of AFM temporally associated with EV-D68. Further, tar-

geted vaccination or immunoprophylaxis of populations such as infants and toddlers with waning maternal humoral immunity during a known outbreak of EV-D68 could prevent cases of AFM.

C. Animal Models of Respiratory Disease

The major limitations of current murine studies of EV-D68 infection are the use of non-physiologic routes of inoculation to induce paralysis and death and the lack of respiratory disease, which is the chief manifestation of EV-D68 infection in humans. For both cotton rats (*Sigmodon hispidus*) (Patel et al., 2016) and ferrets (*Mustela putorius furo*) (Zheng et al., 2017a) a single study of each has demonstrated that intranasal inoculation with EV-D68 results in replication of virus in the nose and lungs and induces increases in pro-inflammatory innate immune molecule transcripts in the lungs. Clinical symptoms consistent with upper respiratory tract infection (cough, nasal discharge, and dry nose) also developed in ~25% of inoculated ferrets. Neither study assessed for the presence of EV-D68 in muscle or central nervous system tissues nor noted apparent limb weakness or paralysis. Vaccination of cotton rats indicated that some levels of pre-existing humoral immunity did not protect fully against respiratory infection and may in fact be harmful (Patel et al., 2016), in findings that are described further below.

The theoretical advantage of cotton rats and ferrets as models for human respiratory viral infections is that their respiratory epithelia may more closely mimic that of humans, especially in terms of sialylation of glycans on epithelial surfaces, which are heavily α 2,6-linked in humans but not in mice (Gagneux et al., 2003). EV-D68 bound preferentially to α 2,6-linked sialic acids over α 2,3-linked sialic acids in an in vitro glycan array (Imamura et al., 2014). Staining of the respiratory tract with fluorescently labeled lectins indicated that cotton rats have only α 2,6-linked sialic acids in their lower respiratory tracts with a mix of α 2,6- and α 2,3-linked sialic acids in the upper airways (Blanco et al., 2018). Using similar lectin-based methods, ferrets were shown to have α 2,6-linked sialic acids on epithelial glycans of the upper respiratory tract (Leigh et al., 1995), with a mix of α 2,6- and α 2,3-linked sialic acids in the lower airways (Zheng et al., 2017a). However, more recent and sophisticated glycomic analyses of human (Walther et al., 2013), mouse (Bern et al., 2013), and ferret (Jia et al., 2014) respiratory tissues indicated that a more complex array of glycan modifications than simply density of α 2,6-linked sialylation likely determines the tropism of respiratory viruses for different animal species. Nonetheless, the success of initial studies in rats and ferrets at mimicking human EV-D68 respiratory disease are encouraging for their further development as models of human pathogenesis.

D. Vaccines

Experimental vaccine candidates for EV-D68 have not yet entered clinical development, however initial studies inoculating mice with either virus-like particle (VLP) vaccines made in yeast (Zhang et al., 2018a) or insect cells (Dai et al., 2018) or beta-propiolactone-inactivated EV-D68 (Zhang et al., 2018) have shown promise. The VLP vaccines induced antibody responses that protected mice from subsequent lethal intraperitoneal challenge. With each of these vaccine candidates, passive transfer of vaccine immune serum to naïve mice was sufficient for protection from lethal intraperitoneal challenge. A possible drawback of VLP based vaccines is the uncertainty of whether these synthetic constructs fully recapitulate all conformationally sensitive structures on the surface of the particles. Many potent human virus neutralizing antibodies for other viruses rec-

ognize complex quaternary antigenic sites with strict conformational constraints (Crowe, J E, 2017). The integrity of human antibody epitopes on VLPs cannot be assessed currently since these epitopes are unknown. Further tempering the promise of these murine vaccine studies are findings in cotton rats inoculated intramuscularly with either live or UV-inactivated EV-D68 and subsequently challenged intranasally with a homologous live virus, in which enhanced inflammation was seen in the lung compared to infection of naïve animals (Patel et al., 2016). Specifically, UV-inactivated virus vaccination did not limit viral replication in the lung or nose and skewed the resultant cytokine signature toward a Th2 phenotype, rather than a balanced Th1 and Th2 phenotype seen with immunity from live virus vaccination. Enhanced inflammation and disease have been noted with inactivated virus vaccine candidates for respiratory syncytial virus (Karron, R, 2018) and measles (Strebel et al., 2018). Further careful study of the immune correlates of protection or enhancement caused by EV-D68 vaccine candidates is necessary given these mixed findings.

II. MONOCLONAL ANTIBODIES AND PRODUCTION THEREOF

An “isolated antibody” is one that has been separated and/or recovered from a component of its natural environment. Contaminant components of its natural environment are materials that would interfere with diagnostic or therapeutic uses for the antibody, and may include enzymes, hormones, and other proteinaceous or non-proteinaceous solutes. In particular embodiments, the antibody is purified: (1) to greater than 95% by weight of antibody as determined by the Lowry method, and most particularly more than 99% by weight; (2) to a degree sufficient to obtain at least 15 residues of N-terminal or internal amino acid sequence by use of a spinning cup sequenator; or (3) to homogeneity by SDS-PAGE under reducing or non-reducing conditions using Coomassie blue or silver stain. Isolated antibody includes the antibody in situ within recombinant cells since at least one component of the antibody’s natural environment will not be present. Ordinarily, however, isolated antibody will be prepared by at least one purification step.

The basic four-chain antibody unit is a heterotetrameric glycoprotein composed of two identical light (L) chains and two identical heavy (H) chains. An IgM antibody consists of 5 basic heterotetramer units along with an additional polypeptide called J chain, and therefore contain 10 antigen binding sites, while secreted IgA antibodies can polymerize to form polyvalent assemblages comprising 2-5 of the basic 4-chain units along with J chain. In the case of IgGs, the 4-chain unit is generally about 150,000 daltons. Each L chain is linked to an H chain by one covalent disulfide bond, while the two H chains are linked to each other by one or more disulfide bonds depending on the H chain isotype. Each H and L chain also has regularly spaced intrachain disulfide bridges. Each H chain has at the N-terminus, a variable region (V_H) followed by three constant domains (C_H) for each of the alpha and gamma chains and four C_H domains for mu and isotopes. Each L chain has at the N-terminus, a variable region (V_L) followed by a constant domain (C_L) at its other end. The V_L is aligned with the V_H and the C_L is aligned with the first constant domain of the heavy chain (C_{H1}). Particular amino acid residues are believed to form an interface between the light chain and heavy chain variable regions. The pairing of a V_H and V_L together forms a single antigen-binding site. For the structure and properties of the different classes of antibodies, see,

e.g., Basic and Clinical Immunology, 8th edition, Daniel P. Stites, Abba I. Terr and Tristram G. Parslow (eds.), Appleton & Lange, Norwalk, Conn., 1994, page 71, and Chapter 6.

The L chain from any vertebrate species can be assigned to one of two clearly distinct types, called kappa and lambda based on the amino acid sequences of their constant domains (C_L). Depending on the amino acid sequence of the constant domain of their heavy chains (C_H), immunoglobulins can be assigned to different classes or isotypes. There are five classes of immunoglobulins: IgA, IgD, IgE, IgG, and IgM, having heavy chains designated alpha, delta, epsilon, gamma and mu, respectively. They gamma and alpha classes are further divided into subclasses on the basis of relatively minor differences in C_H sequence and function, humans express the following subclasses: IgG1, IgG2, IgG3, IgG4, IgA1, and IgA2.

The term "variable" refers to the fact that certain segments of the V domains differ extensively in sequence among antibodies. The V domain mediates antigen binding and defines specificity of a particular antibody for its particular antigen. However, the variability is not evenly distributed across the 110-amino acid span of the variable regions. Instead, the V regions consist of relatively invariant stretches called framework regions (FRs) of 15-30 amino acids separated by shorter regions of extreme variability called "hypervariable regions" that are each 9-12 amino acids long. The variable regions of native heavy and light chains each comprise four FRs, largely adopting a beta-sheet configuration, connected by three hypervariable regions, which form loops connecting, and in some cases forming part of, the beta-sheet structure. The hypervariable regions in each chain are held together in close proximity by the FRs and, with the hypervariable regions from the other chain, contribute to the formation of the antigen-binding site of antibodies (see Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991)). The constant domains are not involved directly in binding an antibody to an antigen, but exhibit various effector functions, such as participation of the antibody in antibody dependent cellular cytotoxicity (ADCC), antibody-dependent cellular phagocytosis (ADCP), antibody-dependent neutrophil phagocytosis (ADNP), and antibody-dependent complement deposition (ADCD).

The term "hypervariable region" when used herein refers to the amino acid residues of an antibody that are responsible for antigen binding. The hypervariable region generally comprises amino acid residues from a "complementarity determining region" or "CDR" (e.g., around about residues 24-34 (L1), 50-56 (L2) and 89-97 (L3) in the V_L , and around about 31-35 (H1), 50-65 (H2) and 95-102 (H3) in the V_H when numbered in accordance with the Kabat numbering system; Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991)); and/or those residues from a "hypervariable loop" (e.g., residues 24-34 (L1), 50-56 (L2) and 89-97 (L3) in the V_L , and 26-32 (H1), 52-56 (H2) and 95-101 (H3) in the V_H when numbered in accordance with the Chothia numbering system; Chothia and Lesk, J. Mol. Biol. 196:901-917 (1987)); and/or those residues from a "hypervariable loop"/CDR (e.g., residues 27-38 (L1), 56-65 (L2) and 105-120 (L3) in the V_L , and 27-38 (H1), 56-65 (H2) and 105-120 (H3) in the V_H when numbered in accordance with the IMGT numbering system; Lefranc, M. P. et al. Nucl. Acids Res. 27:209-212 (1999), Ruiz, M. et al. Nucl. Acids Res. 28:219-221 (2000)). Optionally the antibody has symmetrical insertions at one or more

of the following points 28, 36 (L1), 63, 74-75 (L2) and 123 (L3) in the V_L , and 28, 36 (H1), 63, 74-75 (H2) and 123 (H3) in the $V_{sub}H$ when numbered in accordance with AHO; Honneger, A. and Plunkthun, A. J. Mol. Biol. 309:657-670 (2001)).

By "germline nucleic acid residue" is meant the nucleic acid residue that naturally occurs in a germline gene encoding a constant or variable region. "Germline gene" is the DNA found in a germ cell (i.e., a cell destined to become an egg or in the sperm). A "germline mutation" refers to a heritable change in a particular DNA that has occurred in a germ cell or the zygote at the single-cell stage, and when transmitted to offspring, such a mutation is incorporated in every cell of the body. A germline mutation is in contrast to a somatic mutation which is acquired in a single body cell. In some cases, nucleotides in a germline DNA sequence encoding for a variable region are mutated (i.e., a somatic mutation) and replaced with a different nucleotide.

The term "monoclonal antibody" as used herein refers to an antibody obtained from a population of substantially homogeneous antibodies, i.e., the individual antibodies comprising the population are identical except for possible naturally occurring mutations that may be present in minor amounts. Monoclonal antibodies are highly specific, being directed against a single antigenic site. Furthermore, in contrast to polyclonal antibody preparations that include different antibodies directed against different determinants (epitopes), each monoclonal antibody is directed against a single determinant on the antigen. In addition to their specificity, the monoclonal antibodies are advantageous in that they may be synthesized uncontaminated by other antibodies. The modifier "monoclonal" is not to be construed as requiring production of the antibody by any particular method. For example, the monoclonal antibodies useful in the present disclosure may be prepared by the hybridoma methodology first described by Kohler et al., Nature, 256:495 (1975), or may be made using recombinant DNA methods in bacterial, eukaryotic animal or plant cells (see, e.g., U.S. Pat. No. 4,816,567) after single cell sorting of an antigen specific B cell, an antigen specific plasmablast responding to an infection or immunization, or capture of linked heavy and light chains from single cells in a bulk sorted antigen specific collection. The "monoclonal antibodies" may also be isolated from phage antibody libraries using the techniques described in Clackson et al., Nature, 352:624-628 (1991) and Marks et al., J. Mol. Biol., 222: 581-597 (1991), for example.

A. General Methods

It will be understood that monoclonal antibodies binding to EV-D68 will have several applications. These include the production of diagnostic kits for use in detecting and diagnosing EV-D68 infection, as well as for treating the same. In these contexts, one may link such antibodies to diagnostic or therapeutic agents, use them as capture agents or competitors in competitive assays, or use them individually without additional agents being attached thereto. The antibodies may be mutated or modified, as discussed further below. Methods for preparing and characterizing antibodies are well known in the art (see, e.g., Antibodies: A Laboratory Manual, Cold Spring Harbor Laboratory, 1988; U.S. Pat. No. 4,196,265).

The methods for generating monoclonal antibodies (MAbs) generally begin along the same lines as those for preparing polyclonal antibodies. The first step for both these methods is immunization of an appropriate host or identification of subjects who are immune due to prior natural infection or vaccination with a licensed or experimental vaccine. As is well known in the art, a given composition for

immunization may vary in its immunogenicity. It is often necessary therefore to boost the host immune system, as may be achieved by coupling a peptide or polypeptide immunogen to a carrier. Exemplary and preferred carriers are key-hole limpet hemocyanin (KLH) and bovine serum albumin (BSA). Other albumins such as ovalbumin, mouse serum albumin or rabbit serum albumin can also be used as carriers. Means for conjugating a polypeptide to a carrier protein are well known in the art and include glutaraldehyde, m-maleimidobenzoyl-N-hydroxysuccinimide ester, carbodiimide and bis-biazotized benzidine. As also is well known in the art, the immunogenicity of a particular immunogen composition can be enhanced by the use of non-specific stimulators of the immune response, known as adjuvants. Exemplary and preferred adjuvants in animals include complete Freund's adjuvant (a non-specific stimulator of the immune response containing killed *Mycobacterium tuberculosis*), incomplete Freund's adjuvants and aluminum hydroxide adjuvant and in humans include alum, CpG, MFP59 and combinations of immunostimulatory molecules ("Adjuvant Systems", such as AS01 or AS03). Additional experimental forms of inoculation to induce EV-D68-specific B cells is possible, including nanoparticle vaccines, or gene-encoded antigens delivered as DNA or RNA genes in a physical delivery system (such as lipid nanoparticle or on a gold biolistic bead), and delivered with needle, gene gun, transcutaneous electroporation device. The antigen gene also can be carried as encoded by a replication competent or defective viral vector such as adenovirus, adeno-associated virus, poxvirus, herpesvirus, or alphavirus replicon, or alternatively a virus like particle.

In the case of human antibodies against natural pathogens, a suitable approach is to identify subjects that have been exposed to the pathogens, such as those who have been diagnosed as having contracted the disease, or those who have been vaccinated to generate protective immunity against the pathogen or to test the safety or efficacy of an experimental vaccine. Circulating anti-pathogen antibodies can be detected, and antibody encoding or producing B cells from the antibody-positive subject may then be obtained.

The amount of immunogen composition used in the production of polyclonal antibodies varies upon the nature of the immunogen as well as the animal used for immunization. A variety of routes can be used to administer the immunogen (subcutaneous, intramuscular, intradermal, intravenous and intraperitoneal). The production of polyclonal antibodies may be monitored by sampling blood of the immunized animal at various points following immunization. A second, booster injection, also may be given. The process of boosting and titering is repeated until a suitable titer is achieved. When a desired level of immunogenicity is obtained, the immunized animal can be bled and the serum isolated and stored, and/or the animal can be used to generate MAbs.

Following immunization, somatic cells with the potential for producing antibodies, specifically B lymphocytes (B cells), are selected for use in the MAb generating protocol. These cells may be obtained from biopsied spleens, lymph nodes, tonsils or adenoids, bone marrow aspirates or biopsies, tissue biopsies from mucosal organs like lung or GI tract, or from circulating blood. The antibody-producing B lymphocytes from the immunized animal or immune human are then fused with cells of an immortal myeloma cell, generally one of the same species as the animal that was immunized or human or human/mouse chimeric cells. Myeloma cell lines suited for use in hybridoma-producing fusion procedures preferably are non-antibody-producing,

have high fusion efficiency, and enzyme deficiencies that render them incapable of growing in certain selective media which support the growth of only the desired fused cells (hybridomas). Any one of a number of myeloma cells may be used, as are known to those of skill in the art (Goding, pp. 65-66, 1986; Campbell, pp. 75-83, 1984). HMM2.5 cells or MFP-2 cells are particularly useful examples of such cells.

Methods for generating hybrids of antibody-producing spleen or lymph node cells and myeloma cells usually comprise mixing somatic cells with myeloma cells in a 2:1 proportion, though the proportion may vary from about 20:1 to about 1:1, respectively, in the presence of an agent or agents (chemical or electrical) that promote the fusion of cell membranes. In some cases, transformation of human B cells with Epstein Barr virus (EBV) as an initial step increases the size of the B cells, enhancing fusion with the relatively large-sized myeloma cells. Transformation efficiency by EBV is enhanced by using CpG and a Chk2 inhibitor drug in the transforming medium. Alternatively, human B cells can be activated by co-culture with transfected cell lines expressing CD40 Ligand (CD154) in medium containing additional soluble factors, such as IL-21 and human B cell Activating Factor (BAFF), a Type II member of the TNF superfamily. Fusion methods using Sendai virus have been described by Kohler and Milstein (1975; 1976), and those using polyethylene glycol (PEG), such as 37% (v/v) PEG, by Geffer et al. (1977). The use of electrically induced fusion methods also is appropriate (Goding, pp. 71-74, 1986) and there are processes for better efficiency (Yu et al., 2008). Fusion procedures usually produce viable hybrids at low frequencies, about 1×10^{-6} to 1×10^{-8} , but with optimized procedures one can achieve fusion efficiencies close to 1 in 200 (Yu et al., 2008). However, relatively low efficiency of fusion does not pose a problem, as the viable, fused hybrids are differentiated from the parental, unfused cells (particularly the unfused myeloma cells that would normally continue to divide indefinitely) by culturing in a selective medium. The selective medium is generally one that contains an agent that blocks the de novo synthesis of nucleotides in the tissue culture medium. Exemplary and preferred agents are aminopterin, methotrexate, and azaserine. Aminopterin and methotrexate block de novo synthesis of both purines and pyrimidines, whereas azaserine blocks only purine synthesis. Where aminopterin or methotrexate is used, the medium is supplemented with hypoxanthine and thymidine as a source of nucleotides (HAT medium). Where azaserine is used, the medium is supplemented with hypoxanthine. Ouabain is added if the B cell source is an EBV-transformed human B cell line, in order to eliminate EBV-transformed lines that have not fused to the myeloma.

The preferred selection medium is HAT or HAT with ouabain. Only cells capable of operating nucleotide salvage pathways are able to survive in HAT medium. The myeloma cells are defective in key enzymes of the salvage pathway, e.g., hypoxanthine phosphoribosyl transferase (HPRT), and they cannot survive. The B cells can operate this pathway, but they have a limited life span in culture and generally die within about two weeks. Therefore, the only cells that can survive in the selective media are those hybrids formed from myeloma and B cells. When the source of B cells used for fusion is a line of EBV-transformed B cells, as here, ouabain may also be used for drug selection of hybrids as EBV-transformed B cells are susceptible to drug killing, whereas the myeloma partner used is chosen to be ouabain resistant.

Culturing provides a population of hybridomas from which specific hybridomas are selected. Typically, selection

of hybridomas is performed by culturing the cells by single-clone dilution in microtiter plates, followed by testing the individual clonal supernatants (after about two to three weeks) for the desired reactivity. The assay should be sensitive, simple and rapid, such as radioimmunoassays, enzyme immunoassays, cytotoxicity assays, plaque assays dot immunobinding assays, and the like. The selected hybridomas are then serially diluted or single-cell sorted by flow cytometric sorting and cloned into individual antibody-producing cell lines, which clones can then be propagated indefinitely to provide mAbs. The cell lines may be exploited for MAb production in two basic ways. A sample of the hybridoma can be injected (often into the peritoneal cavity) into an animal (e.g., a mouse). Optionally, the animals are primed with a hydrocarbon, especially oils such as pristane (tetramethylpentadecane) prior to injection. When human hybridomas are used in this way, it is optimal to inject immunocompromised mice, such as SCID mice, to prevent tumor rejection. The injected animal develops tumors secreting the specific monoclonal antibody produced by the fused cell hybrid. The body fluids of the animal, such as serum or ascites fluid, can then be tapped to provide MAbs in high concentration. The individual cell lines could also be cultured in vitro, where the MAbs are naturally secreted into the culture medium from which they can be readily obtained in high concentrations. Alternatively, human hybridoma cells lines can be used in vitro to produce immunoglobulins in cell supernatant. The cell lines can be adapted for growth in serum-free medium to optimize the ability to recover human monoclonal immunoglobulins of high purity.

MAbs produced by either means may be further purified, if desired, using filtration, centrifugation and various chromatographic methods such as FPLC or affinity chromatography. Fragments of the monoclonal antibodies of the disclosure can be obtained from the purified monoclonal antibodies by methods which include digestion with enzymes, such as pepsin or papain, and/or by cleavage of disulfide bonds by chemical reduction. Alternatively, monoclonal antibody fragments encompassed by the present disclosure can be synthesized using an automated peptide synthesizer.

It also is contemplated that a molecular cloning approach may be used to generate monoclonal antibodies. Single B cells labelled with the antigen of interest can be sorted physically using paramagnetic bead selection or flow cytometric sorting, then RNA can be isolated from the single cells and antibody genes amplified by RT-PCR. Alternatively, antigen-specific bulk sorted populations of cells can be segregated into microvesicles and the matched heavy and light chain variable genes recovered from single cells using physical linkage of heavy and light chain amplicons, or common barcoding of heavy and light chain genes from a vesicle. Matched heavy and light chain genes form single cells also can be obtained from populations of antigen specific B cells by treating cells with cell-penetrating nanoparticles bearing RT-PCR primers and barcodes for marking transcripts with one barcode per cell. The antibody variable genes also can be isolated by RNA extraction of a hybridoma line and the antibody genes obtained by RT-PCR and cloned into an immunoglobulin expression vector. Alternatively, combinatorial immunoglobulin phagemid libraries are prepared from RNA isolated from the cell lines and phagemids expressing appropriate antibodies are selected by panning using viral antigens. The advantages of this approach over conventional hybridoma techniques are that approximately 10^4 times as many antibodies can be pro-

duced and screened in a single round, and that new specificities are generated by H and L chain combination which further increases the chance of finding appropriate antibodies.

Other U.S. patents, each incorporated herein by reference, that teach the production of antibodies useful in the present disclosure include U.S. Pat. No. 5,565,332, which describes the production of chimeric antibodies using a combinatorial approach; U.S. Pat. No. 4,816,567 which describes recombinant immunoglobulin preparations; and U.S. Pat. No. 4,867,973 which describes antibody-therapeutic agent conjugates.

B. Antibodies of the Present Disclosure

Antibodies according to the present disclosure may be defined, in the first instance, by their binding specificity. Those of skill in the art, by assessing the binding specificity/affinity of a given antibody using techniques well known to those of skill in the art, can determine whether such antibodies fall within the scope of the instant claims. For example, the epitope to which a given antibody bind may consist of a single contiguous sequence of 3 or more (e.g., 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20) amino acids located within the antigen molecule (e.g., a linear epitope in a domain). Alternatively, the epitope may consist of a plurality of non-contiguous amino acids (or amino acid sequences) located within the antigen molecule (e.g., a conformational epitope).

Various techniques known to persons of ordinary skill in the art can be used to determine whether an antibody "interacts with one or more amino acids" within a polypeptide or protein. Exemplary techniques include, for example, routine cross-blocking assays, such as that described in *Antibodies*, Harlow and Lane (Cold Spring Harbor Press, Cold Spring Harbor, N.Y.). Cross-blocking can be measured in various binding assays such as ELISA, biolayer interferometry, or surface plasmon resonance. Other methods include alanine scanning mutational analysis, peptide blot analysis (Reineke (2004) *Methods Mol. Biol.* 248: 443-63), peptide cleavage analysis, high-resolution electron microscopy techniques using single particle reconstruction, cryoEM, or tomography, crystallographic studies and NMR analysis. In addition, methods such as epitope excision, epitope extraction and chemical modification of antigens can be employed (Tomer (2000) *Prot. Sci.* 9: 487496). Another method that can be used to identify the amino acids within a polypeptide with which an antibody interacts is hydrogen/deuterium exchange detected by mass spectrometry. In general terms, the hydrogen/deuterium exchange method involves deuterium-labeling the protein of interest, followed by binding the antibody to the deuterium-labeled protein. Next, the protein/antibody complex is transferred to water and exchangeable protons within amino acids that are protected by the antibody complex undergo deuterium-to-hydrogen back-exchange at a slower rate than exchangeable protons within amino acids that are not part of the interface. As a result, amino acids that form part of the protein/antibody interface may retain deuterium and therefore exhibit relatively higher mass compared to amino acids not included in the interface. After dissociation of the antibody, the target protein is subjected to protease cleavage and mass spectrometry analysis, thereby revealing the deuterium-labeled residues which correspond to the specific amino acids with which the antibody interacts. See, e.g., Ehring (1999) *Analytical Biochemistry* 267: 252-259; Engen and Smith (2001) *Anal. Chem.* 73: 256A-265A. When the antibody neutralizes EV-D68, antibody escape mutant variant organisms can be isolated by propagating EV-D68 in vitro or

in animal models in the presence of high concentrations of the antibody. Sequence analysis of the EV-D68 gene encoding the antigen targeted by the antibody reveals the mutation (s) conferring antibody escape, indicating residues in the epitope or that affect the structure of the epitope allosterically.

The term “epitope” refers to a site on an antigen to which B and/or T cells respond. B-cell epitopes can be formed both from contiguous amino acids or noncontiguous amino acids juxtaposed by tertiary folding of a protein. Epitopes formed from contiguous amino acids are typically retained on exposure to denaturing solvents, whereas epitopes formed by tertiary folding are typically lost on treatment with denaturing solvents. An epitope typically includes at least 3, and more usually, at least 5 or 8-10 amino acids in a unique spatial conformation.

Modification-Assisted Profiling (MAP), also known as Antigen Structure-based Antibody Profiling (ASAP) is a method that categorizes large numbers of monoclonal antibodies (mAbs) directed against the same antigen according to the similarities of the binding profile of each antibody to chemically or enzymatically modified antigen surfaces (see US 2004/0101920, herein specifically incorporated by reference in its entirety). Each category may reflect a unique epitope either distinctly different from or partially overlapping with epitope represented by another category. This technology allows rapid filtering of genetically identical antibodies, such that characterization can be focused on genetically distinct antibodies. When applied to hybridoma screening, MAP may facilitate identification of rare hybridoma clones that produce mAbs having the desired characteristics. MAP may be used to sort the antibodies of the disclosure into groups of antibodies binding different epitopes.

The present disclosure includes antibodies that may bind to the same epitope, or a portion of the epitope. Likewise, the present disclosure also includes antibodies that compete for binding to a target or a fragment thereof with any of the specific exemplary antibodies described herein. One can easily determine whether an antibody binds to the same epitope as, or competes for binding with, a reference antibody by using routine methods known in the art. For example, to determine if a test antibody binds to the same epitope as a reference, the reference antibody is allowed to bind to target under saturating conditions. Next, the ability of a test antibody to bind to the target molecule is assessed. If the test antibody is able to bind to the target molecule following saturation binding with the reference antibody, it can be concluded that the test antibody binds to a different epitope than the reference antibody. On the other hand, if the test antibody is not able to bind to the target molecule following saturation binding with the reference antibody, then the test antibody may bind to the same epitope as the epitope bound by the reference antibody.

To determine if an antibody competes for binding with a reference anti-EV-D68 antibody, the above-described binding methodology is performed in two orientations: In a first orientation, the reference antibody is allowed to bind to the EV-D68 antigen under saturating conditions followed by assessment of binding of the test antibody to the EV-D68 antigen. In a second orientation, the test antibody is allowed to bind to the EV-D68 antigen molecule under saturating conditions followed by assessment of binding of the reference antibody to the EV-D68 antigen. If, in both orientations, only the first (saturating) antibody is capable of binding to EV-D68, then it is concluded that the test antibody and the reference antibody compete for binding to

EV-D68. As will be appreciated by a person of ordinary skill in the art, an antibody that competes for binding with a reference antibody may not necessarily bind to the identical epitope as the reference antibody but may sterically block binding of the reference antibody by binding an overlapping or adjacent epitope.

Two antibodies bind to the same or overlapping epitope if each competitively inhibits (blocks) binding of the other to the antigen. That is, a 1-, 5-, 10-, 20- or 100-fold excess of one antibody inhibits binding of the other by at least 50% but preferably 75%, 90% or even 99% as measured in a competitive binding assay (see, e.g., Junghans et al., *Cancer Res.* 1990 50:1495-1502). Alternatively, two antibodies have the same epitope if essentially all amino acid mutations in the antigen that reduce or eliminate binding of one antibody reduce or eliminate binding of the other. Two antibodies have overlapping epitopes if some amino acid mutations that reduce or eliminate binding of one antibody reduce or eliminate binding of the other.

Additional routine experimentation (e.g., peptide mutation and binding analyses) can then be carried out to confirm whether the observed lack of binding of the test antibody is in fact due to binding to the same epitope as the reference antibody or if steric blocking (or another phenomenon) is responsible for the lack of observed binding. Experiments of this sort can be performed using ELISA, RIA, surface plasmon resonance, flow cytometry or any other quantitative or qualitative antibody-binding assay available in the art. Structural studies with EM or crystallography also can demonstrate whether or not two antibodies that compete for binding recognize the same epitope.

In another aspect, there are provided monoclonal antibodies having clone-paired CDRs from the heavy and light chains as illustrated in Tables 3 and 4, respectively. Such antibodies may be produced by the clones discussed below in the Examples section using methods described herein.

In another aspect, the antibodies may be defined by their variable sequence, which include additional “framework” regions. These are provided in Tables 1 and 2 that encode or represent full variable regions. Furthermore, the antibodies sequences may vary from these sequences, optionally using methods discussed in greater detail below. For example, nucleic acid sequences may vary from those set out above in that (a) the variable regions may be segregated away from the constant domains of the light and heavy chains, (b) the nucleic acids may vary from those set out above while not affecting the residues encoded thereby, (c) the nucleic acids may vary from those set out above by a given percentage, e.g., 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% homology, (d) the nucleic acids may vary from those set out above by virtue of the ability to hybridize under high stringency conditions, as exemplified by low salt and/or high temperature conditions, such as provided by about 0.02 M to about 0.15 M NaCl at temperatures of about 50° C. to about 70° C., (e) the amino acids may vary from those set out above by a given percentage, e.g., 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% homology, or (f) the amino acids may vary from those set out above by permitting conservative substitutions (discussed below). Each of the foregoing applies to the nucleic acid sequences set forth as Table 1 and the amino acid sequences of Table 2.

When comparing polynucleotide and polypeptide sequences, two sequences are said to be “identical” if the sequence of nucleotides or amino acids in the two sequences is the same when aligned for maximum correspondence, as described below. Comparisons between two sequences are

typically performed by comparing the sequences over a comparison window to identify and compare local regions of sequence similarity. A "comparison window" as used herein, refers to a segment of at least about 20 contiguous positions, usually 30 to about 75, 40 to about 50, in which a sequence may be compared to a reference sequence of the same number of contiguous positions after the two sequences are optimally aligned.

Optimal alignment of sequences for comparison may be conducted using the Megalign program in the Lasergene suite of bioinformatics software (DNASTAR, Inc., Madison, Wis.), using default parameters. This program embodies several alignment schemes described in the following references: Dayhoff, M. O. (1978) A model of evolutionary change in proteins—Matrices for detecting distant relationships. In Dayhoff, M. O. (ed.) Atlas of Protein Sequence and Structure, National Biomedical Research Foundation, Washington D.C. Vol. 5, Suppl. 3, pp. 345-358; Hein J. (1990) Unified Approach to Alignment and Phylogeny pp. 626-645 Methods in Enzymology vol. 183, Academic Press, Inc., San Diego, Calif.; Higgins, D. G. and Sharp, P. M. (1989) CABIOS 5:151-153; Myers, E. W. and Muller W. (1988) CABIOS 4:11-17; Robinson, E. D. (1971) Comb. Theor 11:105; Santou, N. Nes, M. (1987) Mol. Biol. Evol. 4:406-425; Sneath, P. H. A. and Sokal, R. R. (1973) Numerical Taxonomy—the Principles and Practice of Numerical Taxonomy, Freeman Press, San Francisco, Calif.; Wilbur, W. J. and Lipman, D. J. (1983) Proc. Natl. Acad., Sci. USA 80:726-730.

Alternatively, optimal alignment of sequences for comparison may be conducted by the local identity algorithm of Smith and Waterman (1981) Add. APL. Math 2:482, by the identity alignment algorithm of Needleman and Wunsch (1970) J. Mol. Biol. 48:443, by the search for similarity methods of Pearson and Lipman (1988) Proc. Natl. Acad. Sci. USA 85: 2444, by computerized implementations of these algorithms (GAP, BESTFIT, BLAST, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group (GCG), 575 Science Dr., Madison, Wis.), or by inspection.

One particular example of algorithms that are suitable for determining percent sequence identity and sequence similarity are the BLAST and BLAST 2.0 algorithms, which are described in Altschul et al. (1977) Nucl. Acids Res. 25:3389-3402 and Altschul et al. (1990) J. Mol. Biol. 215:403-410, respectively. BLAST and BLAST 2.0 can be used, for example with the parameters described herein, to determine percent sequence identity for the polynucleotides and polypeptides of the disclosure. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information. The rearranged nature of an antibody sequence and the variable length of each gene requires multiple rounds of BLAST searches for a single antibody sequence. Also, manual assembly of different genes is difficult and error-prone. The sequence analysis tool IgBLAST (world-wide-web at ncbi.nlm.nih.gov/igblast/) identifies matches to the germline V, D and J genes, details at rearrangement junctions, the delineation of Ig V domain framework regions and complementarity determining regions. IgBLAST can analyze nucleotide or protein sequences and can process sequences in batches and allows searches against the germline gene databases and other sequence databases simultaneously to minimize the chance of missing possibly the best matching germline V gene.

In one illustrative example, cumulative scores can be calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0)

and N (penalty score for mismatching residues; always <0). Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T and X determine the sensitivity and speed of the alignment. The BLASTN program (for nucleotide sequences) uses as defaults a wordlength (W) of 11, and expectation (E) of 10, and the BLOSUM62 scoring matrix (see Henikoff and Henikoff (1989) Proc. Natl. Acad. Sci. USA 89:10915) alignments, (B) of 50, expectation (E) of 10, M=5, N=-4 and a comparison of both strands.

For amino acid sequences, a scoring matrix can be used to calculate the cumulative score. Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T and X determine the sensitivity and speed of the alignment.

In one approach, the "percentage of sequence identity" is determined by comparing two optimally aligned sequences over a window of comparison of at least 20 positions, wherein the portion of the polynucleotide or polypeptide sequence in the comparison window may comprise additions or deletions (i.e., gaps) of 20 percent or less, usually 5 to 15 percent, or 10 to 12 percent, as compared to the reference sequences (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid bases or amino acid residues occur in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the reference sequence (i.e., the window size) and multiplying the results by 100 to yield the percentage of sequence identity.

Yet another way of defining an antibody is as a "derivative" of any of the below-described antibodies and their antigen-binding fragments. The term "derivative" refers to an antibody or antigen-binding fragment thereof that immunospecifically binds to an antigen but which comprises, one, two, three, four, five or more amino acid substitutions, additions, deletions or modifications relative to a "parental" (or wild-type) molecule. Such amino acid substitutions or additions may introduce naturally occurring (i.e., DNA-encoded) or non-naturally occurring amino acid residues. The term "derivative" encompasses, for example, as variants having altered CH1, hinge, CH2, CH3 or CH4 regions, so as to form, for example antibodies, etc., having variant Fc regions that exhibit enhanced or impaired effector or binding characteristics. The term "derivative" additionally encompasses non-amino acid modifications, for example, amino acids that may be glycosylated (e.g., have altered mannose, 2-N-acetylglucosamine, galactose, fucose, glucose, sialic acid, 5-N-acetylneuraminic acid, 5-glycolneuraminic acid, etc. content), acetylated, pegylated, phosphorylated, amidated, derivatized by known protecting/blocking groups, proteolytic cleavage, linked to a cellular ligand or other protein, etc. In some embodiments, the altered carbohydrate modifications modulate one or more of the following: solubilization of the antibody, facilitation of subcellular transport and secretion of the antibody, promotion of antibody assembly, conformational integrity, and antibody-mediated effector function. In a specific embodiment, the altered

carbohydrate modifications enhance antibody mediated effector function relative to the antibody lacking the carbohydrate modification. Carbohydrate modifications that lead to altered antibody mediated effector function are well known in the art (for example, see Shields, R. L. et al. (2002) 5 "*Lack Of Fucose On Human IgG N-Linked Oligosaccharide Improves Binding To Human Fcγ RIII And Antibody-Dependent Cellular Toxicity*," J. Biol. Chem. 277(30): 26733-26740; Davies J. et al. (2001) "*Expression Of G_NTIII In A Recombinant Anti-CD20 CHO Production Cell Line: Expression Of Antibodies With Altered Glycoforms Leads To An Increase In ADCC Through Higher Affinity For FCγ RIII*," Biotechnology & Bioengineering 74(4): 288-294). Methods of altering carbohydrate contents are known to those skilled in the art, see, e.g., Wallick, S. C. et al. (1988) "*Glycosylation Of A VH Residue Of A Monoclonal Antibody Against Alpha (1 - - 6) Dextran Increases Its Affinity For Antigen*," J. Exp. Med. 168(3): 1099-1109; Tao, M. H. et al. (1989) "*Studies Of Aglycosylated Chimeric Mouse-Human IgG. Role Of Carbohydrate In The Structure And Effector Functions Mediated By The Human IgG Constant Region*," J. Immunol. 143(8): 2595-2601; Routledge, E. G. et al. (1995) "*The Effect Of Aglycosylation On The Immunogenicity Of A Humanized Therapeutic CD3 Monoclonal Antibody*," Transplantation 60(8):847-53; Elliott, S. et al. (2003) "*Enhancement Of Therapeutic Protein In Vivo Activities Through Glycoengineering*," Nature Biotechnol. 21:414-21; Shields, R. L. et al. (2002) "*Lack Of Fucose On Human IgG N-Linked Oligosaccharide Improves Binding To Human Fcγ RIII And Antibody-Dependent Cellular Toxicity*," J. Biol. Chem. 277(30): 26733-26740).

A derivative antibody or antibody fragment can be generated with an engineered sequence or glycosylation state to confer preferred levels of activity in antibody dependent cellular cytotoxicity (ADCC), antibody-dependent cellular phagocytosis (ADCP), antibody-dependent neutrophil phagocytosis (ADNP), or antibody-dependent complement deposition (ADCD) functions as measured by bead-based or cell-based assays or in vivo studies in animal models.

A derivative antibody or antibody fragment may be modified by chemical modifications using techniques known to those of skill in the art, including, but not limited to, specific chemical cleavage, acetylation, formulation, metabolic synthesis of tunicamycin, etc. In one embodiment, an antibody derivative will possess a similar or identical function as the parental antibody. In another embodiment, an antibody derivative will exhibit an altered activity relative to the parental antibody. For example, a derivative antibody (or fragment thereof) can bind to its epitope more tightly or be more resistant to proteolysis than the parental antibody. 50 C. Engineering of Antibody Sequences

In various embodiments, one may choose to engineer sequences of the identified antibodies for a variety of reasons, such as improved expression, improved cross-reactivity or diminished off-target binding. Modified antibodies may be made by any technique known to those of skill in the art, including expression through standard molecular biological techniques, or the chemical synthesis of polypeptides. Methods for recombinant expression are addressed elsewhere in this document. The following is a general discussion of relevant goals techniques for antibody engineering.

Hybridomas may be cultured, then cells lysed, and total RNA extracted. Random hexamers may be used with RT to generate cDNA copies of RNA, and then PCR performed using a multiplex mixture of PCR primers expected to amplify all human variable gene sequences. PCR product

can be cloned into pGEM-T Easy vector, then sequenced by automated DNA sequencing using standard vector primers. Assay of binding and neutralization may be performed using antibodies collected from hybridoma supernatants and purified by FPLC, using Protein G columns.

Recombinant full-length IgG antibodies can be generated by subcloning heavy and light chain Fv DNAs from the cloning vector into an IgG plasmid vector, transfected into 293 (e.g., Freestyle) cells or CHO cells, and antibodies can be collected and purified from the 293 or CHO cell supernatant. Other appropriate host cells systems include bacteria, such as *E. coli*, insect cells (S2, Sf9, Sf29, High Five), plant cells (e.g., tobacco, with or without engineering for human-like glycans), algae, or in a variety of non-human transgenic contexts, such as mice, rats, goats or cows.

Expression of nucleic acids encoding antibodies, both for the purpose of subsequent antibody purification, and for immunization of a host, is also contemplated. Antibody coding sequences can be RNA, such as native RNA or modified RNA. Modified RNA contemplates certain chemical modifications that confer increased stability and low immunogenicity to mRNAs, thereby facilitating expression of therapeutically important proteins. For instance, N1-methyl-pseudouridine (N1mΨ) outperforms several other nucleoside modifications and their combinations in terms of translation capacity. In addition to turning off the immune/eIF2α phosphorylation-dependent inhibition of translation, incorporated N1mΨ nucleotides dramatically alter the dynamics of the translation process by increasing ribosome pausing and density on the mRNA. Increased ribosome loading of modified mRNAs renders them more permissive for initiation by favoring either ribosome recycling on the same mRNA or de novo ribosome recruitment. Such modifications could be used to enhance antibody expression in vivo following inoculation with RNA. The RNA, whether native or modified, may be delivered as naked RNA or in a delivery vehicle, such as a lipid nanoparticle.

Alternatively, DNA encoding the antibody may be employed for the same purposes. The DNA is included in an expression cassette comprising a promoter active in the host cell for which it is designed. The expression cassette is advantageously included in a replicable vector, such as a conventional plasmid or minivector. Vectors include viral vectors, such as poxviruses, adenoviruses, herpesviruses, adeno-associated viruses, and lentiviruses are contemplated. Replicons encoding antibody genes such as alphavirus replicons based on VEE virus or Sindbis virus are also contemplated. Delivery of such vectors can be performed by needle through intramuscular, subcutaneous, or intradermal routes, or by transcutaneous electroporation when in vivo expression is desired.

The rapid availability of antibody produced in the same host cell and cell culture process as the final cGMP manufacturing process has the potential to reduce the duration of process development programs. Lonza has developed a generic method using pooled transfectants grown in CDACF medium, for the rapid production of small quantities (up to 50 g) of antibodies in CHO cells. Although slightly slower than a true transient system, the advantages include a higher product concentration and use of the same host and process as the production cell line. Example of growth and productivity of GS-CHO pools, expressing a model antibody, in a disposable bioreactor: in a disposable bag bioreactor culture (5 L working volume) operated in fed-batch mode, a harvest antibody concentration of 2 g/L was achieved within 9 weeks of transfection.

Antibody molecules will comprise fragments (such as F(ab'), F(ab')₂) that are produced, for example, by the proteolytic cleavage of the mAbs, or single-chain immunoglobulins producible, for example, via recombinant means. F(ab') antibody derivatives are monovalent, while F(ab')₂ antibody derivatives are bivalent. In one embodiment, such fragments can be combined with one another, or with other antibody fragments or receptor ligands to form "chimeric" binding molecules. Significantly, such chimeric molecules may contain substituents capable of binding to different epitopes of the same molecule.

In related embodiments, the antibody is a derivative of the disclosed antibodies, e.g., an antibody comprising the CDR sequences identical to those in the disclosed antibodies (e.g., a chimeric, or CDR-grafted antibody). Alternatively, one may wish to make modifications, such as introducing conservative changes into an antibody molecule. In making such changes, the hydropathic index of amino acids may be considered. The importance of the hydropathic amino acid index in conferring interactive biologic function on a protein is generally understood in the art (Kyte and Doolittle, 1982). It is accepted that the relative hydropathic character of the amino acid contributes to the secondary structure of the resultant protein, which in turn defines the interaction of the protein with other molecules, for example, enzymes, substrates, receptors, DNA, antibodies, antigens, and the like.

It also is understood in the art that the substitution of like amino acids can be made effectively on the basis of hydrophilicity. U.S. Pat. No. 4,554,101, incorporated herein by reference, states that the greatest local average hydrophilicity of a protein, as governed by the hydrophilicity of its adjacent amino acids, correlates with a biological property of the protein. As detailed in U.S. Pat. No. 4,554,101, the following hydrophilicity values have been assigned to amino acid residues: basic amino acids: arginine (+3.0), lysine (+3.0), and histidine (−0.5); acidic amino acids: aspartate (+3.0±1), glutamate (+3.0±1), asparagine (+0.2), and glutamine (+0.2); hydrophilic, nonionic amino acids: serine (+0.3), asparagine (+0.2), glutamine (+0.2), and threonine (−0.4), sulfur containing amino acids: cysteine (−1.0) and methionine (−1.3); hydrophobic, nonaromatic amino acids: valine (−1.5), leucine (−1.8), isoleucine (−1.8), proline (−0.5±1), alanine (−0.5), and glycine (0); hydrophobic, aromatic amino acids: tryptophan (−3.4), phenylalanine (−2.5), and tyrosine (−2.3).

It is understood that an amino acid can be substituted for another having a similar hydrophilicity and produce a biologically or immunologically modified protein. In such changes, the substitution of amino acids whose hydrophilicity values are within ±2 is preferred, those that are within ±1 are particularly preferred, and those within ±0.5 are even more particularly preferred.

As outlined above, amino acid substitutions generally are based on the relative similarity of the amino acid side-chain substituents, for example, their hydrophobicity, hydrophilicity, charge, size, and the like. Exemplary substitutions that take into consideration the various foregoing characteristics are well known to those of skill in the art and include: arginine and lysine; glutamate and aspartate; serine and threonine; glutamine and asparagine; and valine, leucine and isoleucine.

The present disclosure also contemplates isotype modification. By modifying the Fc region to have a different isotype, different functionalities can be achieved. For example, changing to IgG₁ can increase antibody dependent cell cytotoxicity, switching to class A can improve tissue distribution, and switching to class M can improve valency.

Alternatively or additionally, it may be useful to combine amino acid modifications with one or more further amino acid modifications that alter C1q binding and/or the complement dependent cytotoxicity (CDC) function of the Fc region of an IL-23p19 binding molecule. The binding polypeptide of particular interest may be one that binds to C1q and displays complement dependent cytotoxicity. Polypeptides with pre-existing C1q binding activity, optionally further having the ability to mediate CDC may be modified such that one or both of these activities are enhanced. Amino acid modifications that alter C1q and/or modify its complement dependent cytotoxicity function are described, for example, in WO/0042072, which is hereby incorporated by reference.

One can design an Fc region of an antibody with altered effector function, e.g., by modifying C1q binding and/or FcγR binding and thereby changing CDC activity and/or ADCC activity. "Effector functions" are responsible for activating or diminishing a biological activity (e.g., in a subject). Examples of effector functions include, but are not limited to: C1q binding; complement dependent cytotoxicity (CDC); Fc receptor binding; antibody-dependent cell-mediated cytotoxicity (ADCC); phagocytosis; down regulation of cell surface receptors (e.g., B cell receptor; BCR), etc. Such effector functions may require the Fc region to be combined with a binding domain (e.g., an antibody variable domain) and can be assessed using various assays (e.g., Fc binding assays, ADCC assays, CDC assays, etc.).

For example, one can generate a variant Fc region of an antibody with improved C1q binding and improved FcγRIII binding (e.g., having both improved ADCC activity and improved CDC activity). Alternatively, if it is desired that effector function be reduced or ablated, a variant Fc region can be engineered with reduced CDC activity and/or reduced ADCC activity. In other embodiments, only one of these activities may be increased, and, optionally, also the other activity reduced (e.g., to generate an Fc region variant with improved ADCC activity, but reduced CDC activity and vice versa).

FcRn binding. Fc mutations can also be introduced and engineered to alter their interaction with the neonatal Fc receptor (FcRn) and improve their pharmacokinetic properties. A collection of human Fc variants with improved binding to the FcRn have been described (Shields et al., (2001). High resolution mapping of the binding site on human IgG1 for FcγRI, FcγRII, FcγRIII, and FcRn and design of IgG1 variants with improved binding to the FcγR, (J. Biol. Chem. 276:6591-6604). A number of methods are known that can result in increased half-life (Kuo and Aveson, (2011)), including amino acid modifications may be generated through techniques including alanine scanning mutagenesis, random mutagenesis and screening to assess the binding to the neonatal Fc receptor (FcRn) and/or the in vivo behavior. Computational strategies followed by mutagenesis may also be used to select one of amino acid mutations to mutate.

The present disclosure therefore provides a variant of an antigen binding protein with optimized binding to FcRn. In a particular embodiment, the said variant of an antigen binding protein comprises at least one amino acid modification in the Fc region of said antigen binding protein, wherein said modification is selected from the group consisting of 226, 227, 228, 230, 231, 233, 234, 239, 241, 243, 246, 250, 252, 256, 259, 264, 265, 267, 269, 270, 276, 284, 285, 288, 289, 290, 291, 292, 294, 297, 298, 299, 301, 302, 303, 305, 307, 308, 309, 311, 315, 317, 320, 322, 325, 327, 330, 332, 334, 335, 338, 340, 342, 343, 345, 347, 350, 352,

354, 355, 356, 359, 360, 361, 362, 369, 370, 371, 375, 378, 380, 382, 384, 385, 386, 387, 389, 390, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 403, 404, 408, 411, 412, 414, 415, 416, 418, 419, 420, 421, 422, 424, 426, 428, 433, 434, 438, 439, 440, 443, 444, 445, 446 and 447 of the Fc region as compared to said parent polypeptide, wherein the numbering of the amino acids in the Fc region is that of the EU index in Kabat. In a further aspect of the disclosure the modifications are M252Y/S254T/T256E.

Additionally, various publications describe methods for obtaining physiologically active molecules whose half-lives are modified, see for example Kontermann (2009) either by introducing an FcRn-binding polypeptide into the molecules or by fusing the molecules with antibodies whose FcRn-binding affinities are preserved but affinities for other Fc receptors have been greatly reduced or fusing with FcRn binding domains of antibodies.

Derivatized antibodies may be used to alter the half-lives (e.g., serum half-lives) of parental antibodies in a mammal, particularly a human. Such alterations may result in a half-life of greater than 15 days, preferably greater than 20 days, greater than 25 days, greater than 30 days, greater than 35 days, greater than 40 days, greater than 45 days, greater than 2 months, greater than 3 months, greater than 4 months, or greater than 5 months. The increased half-lives of the antibodies of the present disclosure or fragments thereof in a mammal, preferably a human, results in a higher serum titer of said antibodies or antibody fragments in the mammal, and thus reduces the frequency of the administration of said antibodies or antibody fragments and/or reduces the concentration of said antibodies or antibody fragments to be administered. Antibodies or fragments thereof having increased in vivo half-lives can be generated by techniques known to those of skill in the art. For example, antibodies or fragments thereof with increased in vivo half-lives can be generated by modifying (e.g., substituting, deleting or adding) amino acid residues identified as involved in the interaction between the Fc domain and the FcRn receptor.

Beltramello et al. (2010) previously reported the modification of neutralizing mAbs, due to their tendency to enhance dengue virus infection, by generating in which leucine residues at positions 1.3 and 1.2 of CH2 domain (according to the IMGT unique numbering for C-domain) were substituted with alanine residues. This modification, also known as "LALA" mutation, abolishes antibody binding to FcγRI, FcγRII and FcγRIIIa, as described by Hessel et al. (2007). The variant and unmodified recombinant mAbs were compared for their capacity to neutralize and enhance infection by the four dengue virus serotypes. LALA variants retained the same neutralizing activity as unmodified mAb but were completely devoid of enhancing activity. LALA mutations of this nature are therefore contemplated in the context of the presently disclosed antibodies.

Altered Glycosylation. A particular embodiment of the present disclosure is an isolated monoclonal antibody, or antigen binding fragment thereof, containing a substantially homogeneous glycan without sialic acid, galactose, or fucose. The monoclonal antibody comprises a heavy chain variable region and a light chain variable region, both of which may be attached to heavy chain or light chain constant regions respectively. The aforementioned substantially homogeneous glycan may be covalently attached to the heavy chain constant region.

Another embodiment of the present disclosure comprises a mAb with a novel Fc glycosylation pattern. The isolated monoclonal antibody, or antigen binding fragment thereof, is present in a substantially homogenous composition repre-

sented by the GNGN or G1/G2 glycoform. Fc glycosylation plays a significant role in anti-viral and anti-cancer properties of therapeutic mAbs. The disclosure is in line with a recent study that shows increased anti-lentivirus cell-mediated viral inhibition of a fucose free anti-HIV mAb in vitro. This embodiment of the present disclosure with homogenous glycans lacking a core fucose, showed increased protection against specific viruses by a factor greater than two-fold. Elimination of core fucose dramatically improves the ADCC activity of mAbs mediated by natural killer (NK) cells but appears to have the opposite effect on the ADCC activity of polymorphonuclear cells (PMNs).

The isolated monoclonal antibody, or antigen binding fragment thereof, comprising a substantially homogenous composition represented by the GNGN or G1/G2 glycoform exhibits increased binding affinity for Fc gamma RI and Fc gamma RIII compared to the same antibody without the substantially homogeneous GNGN glycoform and with G0, G1F, G2F, GNF, GNGNF or GNGNFX containing glycoforms. In one embodiment of the present disclosure, the antibody dissociates from Fc gamma RI with a K_d of 1×10^{-8} M or less and from Fc gamma RIII with a K_d of 1×10^{-7} M or less.

Glycosylation of an Fc region is typically either N-linked or O-linked. N-linked refers to the attachment of the carbohydrate moiety to the side chain of an asparagine residue. O-linked glycosylation refers to the attachment of one of the sugars N-acetylgalactosamine, galactose, or xylose to a hydroxyamino acid, most commonly serine or threonine, although 5-hydroxyproline or 5-hydroxylysine may also be used. The recognition sequences for enzymatic attachment of the carbohydrate moiety to the asparagine side chain peptide sequences are asparagine-X-serine and asparagine-X-threonine, where X is any amino acid except proline. Thus, the presence of either of these peptide sequences in a polypeptide creates a potential glycosylation site.

The glycosylation pattern may be altered, for example, by deleting one or more glycosylation site(s) found in the polypeptide, and/or adding one or more glycosylation site(s) that are not present in the polypeptide. Addition of glycosylation sites to the Fc region of an antibody is conveniently accomplished by altering the amino acid sequence such that it contains one or more of the above-described tripeptide sequences (for N-linked glycosylation sites). An exemplary glycosylation variant has an amino acid substitution of residue Asn 297 of the heavy chain. The alteration may also be made by the addition of, or substitution by, one or more serine or threonine residues to the sequence of the original polypeptide (for O-linked glycosylation sites). Additionally, a change of Asn 297 to Ala can remove one of the glycosylation sites.

In certain embodiments, the antibody is expressed in cells that express beta (1,4)-N-acetylglucosaminyltransferase III (GnT III), such that GnT III adds GlcNAc to the IL-23p19 antibody. Methods for producing antibodies in such a fashion are provided in WO/9954342, WO/03011878, patent publication 20030003097A1, and Umana et al., *Nature Biotechnology*, 17:176-180, February 1999. Cell lines can be altered to enhance or reduce or eliminate certain post-translational modifications, such as glycosylation, using genome editing technology such as Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR). For example, CRISPR technology can be used to eliminate genes encoding glycosylating enzymes in 293 or CHO cells used to express recombinant monoclonal antibodies.

Elimination of monoclonal antibody protein sequence liabilities. It is possible to engineer the antibody variable

gene sequences obtained from human B cells to enhance their manufacturability and safety. Potential protein sequence liabilities can be identified by searching for sequence motifs associated with sites containing:

- 1) Unpaired Cys residues,
- 2) N-linked glycosylation,
- 3) Asn deamidation,
- 4) Asp isomerization,
- 5) SYE truncation,
- 6) Met oxidation,
- 7) Trp oxidation,
- 8) N-terminal glutamate,
- 9) Integrin binding,
- 10) CD11c/CD18 binding, or
- 11) Fragmentation

Such motifs can be eliminated by altering the synthetic gene for the cDNA encoding recombinant antibodies.

Protein engineering efforts in the field of development of therapeutic antibodies clearly reveal that certain sequences or residues are associated with solubility differences (Fernandez-Escamilla et al., *Nature Biotech.*, 22 (10), 1302-1306, 2004; Chennamsetty et al., *PNAS*, 106 (29), 11937-11942, 2009; Voynov et al., *Biocon. Chem.*, 21(2), 385-392, 2010) Evidence from solubility-altering mutations in the literature indicate that some hydrophilic residues such as aspartic acid, glutamic acid, and serine contribute significantly more favorably to protein solubility than other hydrophilic residues, such as asparagine, glutamine, threonine, lysine, and arginine.

Stability. Antibodies can be engineered for enhanced biophysical properties. One can use elevated temperature to unfold antibodies to determine relative stability, using average apparent melting temperatures. Differential Scanning Calorimetry (DSC) measures the heat capacity, C_p , of a molecule (the heat required to warm it, per degree) as a function of temperature. One can use DSC to study the thermal stability of antibodies. DSC data for mAbs is particularly interesting because it sometimes resolves the unfolding of individual domains within the mAb structure, producing up to three peaks in the thermogram (from unfolding of the Fab, C_H2 , and C_H3 domains). Typically unfolding of the Fab domain produces the strongest peak. The DSC profiles and relative stability of the Fc portion show characteristic differences for the human IgG₁, IgG₂, IgG₃, and IgG₄ subclasses (Garber and Demarest, *Biochem. Biophys. Res. Commun.* 355, 751-757, 2007). One also can determine average apparent melting temperature using circular dichroism (CD), performed with a CD spectrometer. Far-UV CD spectra will be measured for antibodies in the range of 200 to 260 nm at increments of 0.5 nm. The final spectra can be determined as averages of 20 accumulations. Residue ellipticity values can be calculated after background subtraction. Thermal unfolding of antibodies (0.1 mg/mL) can be monitored at 235 nm from 25-95° C. and a heating rate of 1° C./min One can use dynamic light scattering (DLS) to assess for propensity for aggregation. DLS is used to characterize size of various particles including proteins. If the system is not disperse in size, the mean effective diameter of the particles can be determined. This measurement depends on the size of the particle core, the size of surface structures, and particle concentration. Since DLS essentially measures fluctuations in scattered light intensity due to particles, the diffusion coefficient of the particles can be determined. DLS software in commercial DLA instruments displays the particle population at different diameters. Stability studies can be done conveniently using DLS. DLS measurements of a sample can show whether the particles

aggregate over time or with temperature variation by determining whether the hydrodynamic radius of the particle increases. If particles aggregate, one can see a larger population of particles with a larger radius. Stability depending on temperature can be analyzed by controlling the temperature in situ. Capillary electrophoresis (CE) techniques include proven methodologies for determining features of antibody stability. One can use an iCE approach to resolve antibody protein charge variants due to deamidation, C-terminal lysines, sialylation, oxidation, glycosylation, and any other change to the protein that can result in a change in pI of the protein. Each of the expressed antibody proteins can be evaluated by high throughput, free solution isoelectric focusing (IEF) in a capillary column (cIEF), using a Protein Simple Maurice instrument. Whole-column UV absorption detection can be performed every 30 seconds for real time monitoring of molecules focusing at the isoelectric points (pIs). This approach combines the high resolution of traditional gel IEF with the advantages of quantitation and automation found in column-based separations while eliminating the need for a mobilization step. The technique yields reproducible, quantitative analysis of identity, purity, and heterogeneity profiles for the expressed antibodies. The results identify charge heterogeneity and molecular sizing on the antibodies, with both absorbance and native fluorescence detection modes and with sensitivity of detection down to 0.7 µg/mL.

Solubility. One can determine the intrinsic solubility score of antibody sequences. The intrinsic solubility scores can be calculated using CamSol Intrinsic (Sormanni et al., *J Mol Biol* 427, 478-490, 2015). The amino acid sequences for residues 95-102 (Kabat numbering) in HCDR3 of each antibody fragment such as a scFv can be evaluated via the online program to calculate the solubility scores. One also can determine solubility using laboratory techniques. Various techniques exist, including addition of lyophilized protein to a solution until the solution becomes saturated and the solubility limit is reached, or concentration by ultrafiltration in a microconcentrator with a suitable molecular weight cut-off. The most straightforward method is induction of amorphous precipitation, which measures protein solubility using a method involving protein precipitation using ammonium sulfate (Trevino et al., *J Mol Biol*, 366: 449-460, 2007). Ammonium sulfate precipitation gives quick and accurate information on relative solubility values. Ammonium sulfate precipitation produces precipitated solutions with well-defined aqueous and solid phases and requires relatively small amounts of protein. Solubility measurements performed using induction of amorphous precipitation by ammonium sulfate also can be done easily at different pH values. Protein solubility is highly pH dependent, and pH is considered the most important extrinsic factor that affects solubility.

Autoreactivity. Generally, it is thought that autoreactive clones should be eliminated during ontogeny by negative selection, however it has become clear that many human naturally occurring antibodies with autoreactive properties persist in adult mature repertoires, and the autoreactivity may enhance the antiviral function of many antibodies to pathogens. It has been noted that HCDR3 loops in antibodies during early B cell development are often rich in positive charge and exhibit autoreactive patterns (Wardemann et al., *Science* 301, 1374-1377, 2003). One can test a given antibody for autoreactivity by assessing the level of binding to human origin cells in microscopy (using adherent HeLa or HEP-2 epithelial cells) and flow cytometric cell surface staining (using suspension Jurkat T cells and 293S human

embryonic kidney cells). Autoreactivity also can be surveyed using assessment of binding to tissues in tissue arrays.

Preferred residues ("Human Likeness"). B cell repertoire deep sequencing of human B cells from blood donors is being performed on a wide scale in many recent studies. Sequence information about a significant portion of the human antibody repertoire facilitates statistical assessment of antibody sequence features common in healthy humans. With knowledge about the antibody sequence features in a human recombined antibody variable gene reference database, the position specific degree of "Human Likeness" (HL) of an antibody sequence can be estimated. HL has been shown to be useful for the development of antibodies in clinical use, like therapeutic antibodies or antibodies as vaccines. The goal is to increase the human likeness of antibodies to reduce potential adverse effects and anti-antibody immune responses that will lead to significantly decreased efficacy of the antibody drug or can induce serious health implications. One can assess antibody characteristics of the combined antibody repertoire of three healthy human blood donors of about 400 million sequences in total and created a novel "relative Human Likeness" (rHL) score that focuses on the hypervariable region of the antibody. The rHL score allows one to easily distinguish between human (positive score) and non-human sequences (negative score). Antibodies can be engineered to eliminate residues that are not common in human repertoires.

D. Single Chain Antibodies

A single chain variable fragment (scFv) is a fusion of the variable regions of the heavy and light chains of immunoglobulins, linked together with a short (usually serine, glycine) linker. This chimeric molecule retains the specificity of the original immunoglobulin, despite removal of the constant regions and the introduction of a linker peptide. This modification usually leaves the specificity unaltered. These molecules were created historically to facilitate phage display where it is highly convenient to express the antigen binding domain as a single peptide. Alternatively, scFv can be created directly from subcloned heavy and light chains derived from a hybridoma or B cell. Single chain variable fragments lack the constant Fc region found in complete antibody molecules, and thus, the common binding sites (e.g., protein A/G) used to purify antibodies. These fragments can often be purified/immobilized using Protein L since Protein L interacts with the variable region of kappa light chains.

Flexible linkers generally are comprised of helix- and turn-promoting amino acid residues such as alanine, serine and glycine. However, other residues can function as well. Tang et al. (1996) used phage display as a means of rapidly selecting tailored linkers for single-chain antibodies (scFvs) from protein linker libraries. A random linker library was constructed in which the genes for the heavy and light chain variable domains were linked by a segment encoding an 18-amino acid polypeptide of variable composition. The scFv repertoire (approx. 5×10^6 different members) was displayed on filamentous phage and subjected to affinity selection with hapten. The population of selected variants exhibited significant increases in binding activity but retained considerable sequence diversity. Screening 1054 individual variants subsequently yielded a catalytically active scFv that was produced efficiently in soluble form. Sequence analysis revealed a conserved proline in the linker two residues after the V_H C terminus and an abundance of arginines and prolines at other positions as the only common features of the selected tethers.

The recombinant antibodies of the present disclosure may also involve sequences or moieties that permit dimerization or multimerization of the receptors. Such sequences include those derived from IgA, which permit formation of multimers in conjunction with the J-chain. Another multimerization domain is the Gal4 dimerization domain. In other embodiments, the chains may be modified with agents such as biotin/avidin, which permit the combination of two antibodies.

In a separate embodiment, a single-chain antibody can be created by joining receptor light and heavy chains using a non-peptide linker or chemical unit. Generally, the light and heavy chains will be produced in distinct cells, purified, and subsequently linked together in an appropriate fashion (i.e., the N-terminus of the heavy chain being attached to the C-terminus of the light chain via an appropriate chemical bridge).

Cross-linking reagents are used to form molecular bridges that tie functional groups of two different molecules, e.g., a stabilizing and coagulating agent. However, it is contemplated that dimers or multimers of the same analog or heteromeric complexes comprised of different analogs can be created. To link two different compounds in a step-wise manner, hetero-bifunctional cross-linkers can be used that eliminate unwanted homopolymer formation.

An exemplary hetero-bifunctional cross-linker contains two reactive groups: one reacting with primary amine group (e.g., N-hydroxy succinimide) and the other reacting with a thiol group (e.g., pyridyl disulfide, maleimides, halogens, etc.). Through the primary amine reactive group, the cross-linker may react with the lysine residue(s) of one protein (e.g., the selected antibody or fragment) and through the thiol reactive group, the cross-linker, already tied up to the first protein, reacts with the cysteine residue (free sulfhydryl group) of the other protein (e.g., the selective agent).

It is preferred that a cross-linker having reasonable stability in blood will be employed. Numerous types of disulfide-bond containing linkers are known that can be successfully employed to conjugate targeting and therapeutic/preventative agents. Linkers that contain a disulfide bond that is sterically hindered may prove to give greater stability in vivo, preventing release of the targeting peptide prior to reaching the site of action. These linkers are thus one group of linking agents.

Another cross-linking reagent is SMPT, which is a bifunctional cross-linker containing a disulfide bond that is "sterically hindered" by an adjacent benzene ring and methyl groups. It is believed that steric hindrance of the disulfide bond serves a function of protecting the bond from attack by thiolate anions such as glutathione which can be present in tissues and blood, and thereby help in preventing decoupling of the conjugate prior to the delivery of the attached agent to the target site.

The SMPT cross-linking reagent, as with many other known cross-linking reagents, lends the ability to cross-link functional groups such as the SH of cysteine or primary amines (e.g., the epsilon amino group of lysine). Another possible type of cross-linker includes the hetero-bifunctional photoreactive phenylazides containing a cleavable disulfide bond such as sulfosuccinimidyl-2-(p-azido salicylamido) ethyl-1,3'-dithiopropionate. The N-hydroxy-succinimidyl group reacts with primary amino groups and the phenylazide (upon photolysis) reacts non-selectively with any amino acid residue.

In addition to hindered cross-linkers, non-hindered linkers also can be employed in accordance herewith. Other useful cross-linkers, not considered to contain or generate a pro-

tected disulfide, include SATA, SPDP and 2-iminothiolane (Wawrzynczak & Thorpe, 1987). The use of such cross-linkers is well understood in the art. Another embodiment involves the use of flexible linkers.

U.S. Pat. No. 4,680,338, describes bifunctional linkers useful for producing conjugates of ligands with amine-containing polymers and/or proteins, especially for forming antibody conjugates with chelators, drugs, enzymes, detectable labels and the like. U.S. Pat. Nos. 5,141,648 and 5,563,250 disclose cleavable conjugates containing a labile bond that is cleavable under a variety of mild conditions. This linker is particularly useful in that the agent of interest may be bonded directly to the linker, with cleavage resulting in release of the active agent. Particular uses include adding a free amino or free sulphydryl group to a protein, such as an antibody, or a drug.

U.S. Pat. No. 5,856,456 provides peptide linkers for use in connecting polypeptide constituents to make fusion proteins, e.g., single chain antibodies. The linker is up to about 50 amino acids in length, contains at least one occurrence of a charged amino acid (preferably arginine or lysine) followed by a proline, and is characterized by greater stability and reduced aggregation. U.S. Pat. No. 5,880,270 discloses aminooxy-containing linkers useful in a variety of immunodiagnostic and separative techniques.

E. Multispecific Antibodies

In certain embodiments, antibodies of the present disclosure are bispecific or multispecific. Bispecific antibodies are antibodies that have binding specificities for at least two different epitopes. Exemplary bispecific antibodies may bind to two different epitopes of a single antigen. Other such antibodies may combine a first antigen binding site with a binding site for a second antigen. Alternatively, an antipathogen arm may be combined with an arm that binds to a triggering molecule on a leukocyte, such as a T-cell receptor molecule (e.g., CD3), or Fc receptors for IgG (FcγR), such as FcγRI (CD64), FcγRII (CD32) and Fc gamma RIII (CD16), so as to focus and localize cellular defense mechanisms to the infected cell. Bispecific antibodies may also be used to localize cytotoxic agents to infected cells. These antibodies possess a pathogen-binding arm and an arm that binds the cytotoxic agent (e.g., saporin, anti-interferon-α, *vinca* alkaloid, ricin A chain, methotrexate or radioactive isotope hapten). Bispecific antibodies can be prepared as full-length antibodies or antibody fragments (e.g., F(ab')₂ bispecific antibodies). WO 96/16673 describes a bispecific anti-ErbB2/anti-Fc gamma RIII antibody and U.S. Pat. No. 5,837,234 discloses a bispecific anti-ErbB2/anti-Fc gamma RI antibody. A bispecific anti-ErbB2/Fc alpha antibody is shown in WO98/02463. U.S. Pat. No. 5,821,337 teaches a bispecific anti-ErbB2/anti-CD3 antibody.

Methods for making bispecific antibodies are known in the art. Traditional production of full-length bispecific antibodies is based on the co-expression of two immunoglobulin heavy chain-light chain pairs, where the two chains have different specificities (Millstein et al., *Nature*, 305:537-539 (1983)). Because of the random assortment of immunoglobulin heavy and light chains, these hybridomas (quadromas) produce a potential mixture of ten different antibody molecules, of which only one has the correct bispecific structure. Purification of the correct molecule, which is usually done by affinity chromatography steps, is rather cumbersome, and the product yields are low. Similar procedures are disclosed in WO 93/08829, and in Traunecker et al., *EMBO J.*, 10:3655-3659 (1991).

According to a different approach, antibody variable regions with the desired binding specificities (antibody-

antigen combining sites) are fused to immunoglobulin constant domain sequences. Preferably, the fusion is with an Ig heavy chain constant domain, comprising at least part of the hinge, C_{H2}, and C_{H3} regions. It is preferred to have the first heavy-chain constant region (C_{H1}) containing the site necessary for light chain bonding, present in at least one of the fusions. DNAs encoding the immunoglobulin heavy chain fusions and, if desired, the immunoglobulin light chain, are inserted into separate expression vectors, and are co-transfected into a suitable host cell. This provides for greater flexibility in adjusting the mutual proportions of the three polypeptide fragments in embodiments when unequal ratios of the three polypeptide chains used in the construction provide the optimum yield of the desired bispecific antibody. It is, however, possible to insert the coding sequences for two or all three polypeptide chains into a single expression vector when the expression of at least two polypeptide chains in equal ratios results in high yields or when the ratios have no significant effect on the yield of the desired chain combination.

In a particular embodiment of this approach, the bispecific antibodies are composed of a hybrid immunoglobulin heavy chain with a first binding specificity in one arm, and a hybrid immunoglobulin heavy chain-light chain pair (providing a second binding specificity) in the other arm. It was found that this asymmetric structure facilitates the separation of the desired bispecific compound from unwanted immunoglobulin chain combinations, as the presence of an immunoglobulin light chain in only one half of the bispecific molecule provides for a facile way of separation. This approach is disclosed in WO 94/04690. For further details of generating bispecific antibodies see, for example, Suresh et al., *Methods in Enzymology*, 121:210 (1986).

According to another approach described in U.S. Pat. No. 5,731,168, the interface between a pair of antibody molecules can be engineered to maximize the percentage of heterodimers that are recovered from recombinant cell culture. The preferred interface comprises at least a part of the C_{H3} domain. In this method, one or more small amino acid side chains from the interface of the first antibody molecule are replaced with larger side chains (e.g., tyrosine or tryptophan). Compensatory "cavities" of identical or similar size to the large side chain(s) are created on the interface of the second antibody molecule by replacing large amino acid side chains with smaller ones (e.g., alanine or threonine). This provides a mechanism for increasing the yield of the heterodimer over other unwanted end-products such as homodimers.

Bispecific antibodies include cross-linked or "heteroconjugate" antibodies. For example, one of the antibodies in the heteroconjugate can be coupled to avidin, the other to biotin. Such antibodies have, for example, been proposed to target immune system cells to unwanted cells (U.S. Pat. No. 4,676,980), and for treatment of HIV infection (WO 91/00360, WO 92/200373, and EP 03089). Heteroconjugate antibodies may be made using any convenient cross-linking methods. Suitable cross-linking agents are well known in the art, and are disclosed in U.S. Pat. No. 4,676,980, along with a number of cross-linking techniques.

Techniques for generating bispecific antibodies from antibody fragments have also been described in the literature. For example, bispecific antibodies can be prepared using chemical linkage. Brennan et al., *Science*, 229: 81 (1985) describe a procedure wherein intact antibodies are proteolytically cleaved to generate F(ab')₂ fragments. These fragments are reduced in the presence of the dithiol complexing agent, sodium arsenite, to stabilize vicinal dithiols and

prevent intermolecular disulfide formation. The Fab' fragments generated are then converted to thionitrobenzoate (TNB) derivatives. One of the Fab'-TNB derivatives is then reconverted to the Fab'-thiol by reduction with mercaptoethylamine and is mixed with an equimolar amount of the other Fab'-TNB derivative to form the bispecific antibody. The bispecific antibodies produced can be used as agents for the selective immobilization of enzymes.

Techniques exist that facilitate the direct recovery of Fab'-SH fragments from *E. coli*, which can be chemically coupled to form bispecific antibodies. Shalaby et al., *J. Exp. Med.*, 175: 217-225 (1992) describe the production of a humanized bispecific antibody F(ab')₂ molecule. Each Fab' fragment was separately secreted from *E. coli* and subjected to directed chemical coupling in vitro to form the bispecific antibody. The bispecific antibody thus formed was able to bind to cells overexpressing the ErbB2 receptor and normal human T cells, as well as trigger the lytic activity of human cytotoxic lymphocytes against human breast tumor targets.

Various techniques for making and isolating bispecific antibody fragments directly from recombinant cell culture have also been described (Merchant et al., *Nat. Biotechnol.*, 16, 677-681 (1998) doi:10.1038/nbt0798-677pmid:9661204). For example, bispecific antibodies have been produced using leucine zippers (Kostelny et al., *J. Immunol.*, 148(5):1547-1553, 1992). The leucine zipper peptides from the Fos and Jun proteins were linked to the Fab' portions of two different antibodies by gene fusion. The antibody homodimers were reduced at the hinge region to form monomers and then re-oxidized to form the antibody heterodimers. This method can also be utilized for the production of antibody homodimers. The "diabody" technology described by Hollinger et al., *Proc. Natl. Acad. Sci. USA*, 90:6444-6448 (1993) has provided an alternative mechanism for making bispecific antibody fragments. The fragments comprise a V_H connected to a V_L by a linker that is too short to allow pairing between the two domains on the same chain. Accordingly, the V_H and V_L domains of one fragment are forced to pair with the complementary V_L and V_H domains of another fragment, thereby forming two antigen-binding sites. Another strategy for making bispecific antibody fragments by the use of single-chain Fv (sFv) dimers has also been reported. See Gruber et al., *J. Immunol.*, 152:5368 (1994).

In a particular embodiment, a bispecific or multispecific antibody may be formed as a DOCK-AND-LOCK™ (DNL™) complex (see, e.g., U.S. Pat. Nos. 7,521,056; 7,527,787; 7,534,866; 7,550,143 and 7,666,400, the Examples section of each of which is incorporated herein by reference.) Generally, the technique takes advantage of the specific and high-affinity binding interactions that occur between a dimerization and docking domain (DDD) sequence of the regulatory (R) subunits of cAMP-dependent protein kinase (PKA) and an anchor domain (AD) sequence derived from any of a variety of AKAP proteins (Baillie et al., *FEBS Letters*, 2005; 579: 3264; Wong and Scott, *Nat. Rev. Mol. Cell Biol.* 2004; 5: 959). The DDD and AD peptides may be attached to any protein, peptide or other molecule. Because the DDD sequences spontaneously dimerize and bind to the AD sequence, the technique allows the formation of complexes between any selected molecules that may be attached to DDD or AD sequences.

Antibodies with more than two valencies are contemplated. For example, trispecific antibodies can be prepared (Tutt et al., *J. Immunol.* 147: 60, 1991; Xu et al., *Science*, 358(6359):85-90, 2017). A multivalent antibody may be internalized (and/or catabolized) faster than a bivalent anti-

body by a cell expressing an antigen to which the antibodies bind. The antibodies of the present disclosure can be multivalent antibodies with three or more antigen binding sites (e.g., tetravalent antibodies), which can be readily produced by recombinant expression of nucleic acid encoding the polypeptide chains of the antibody. The multivalent antibody can comprise a dimerization domain and three or more antigen binding sites. The preferred dimerization domain comprises (or consists of) an Fc region or a hinge region. In this scenario, the antibody will comprise an Fc region and three or more antigen binding sites amino-terminal to the Fc region. The preferred multivalent antibody herein comprises (or consists of) three to about eight, but preferably four, antigen binding sites. The multivalent antibody comprises at least one polypeptide chain (and preferably two polypeptide chains), wherein the polypeptide chain(s) comprise two or more variable regions. For instance, the polypeptide chain(s) may comprise VD1-(X1)_n-VD2-(X2)_n-Fc, wherein VD1 is a first variable region, VD2 is a second variable region, Fc is one polypeptide chain of an Fc region, X1 and X2 represent an amino acid or polypeptide, and n is 0 or 1. For instance, the polypeptide chain(s) may comprise: VH-CH1-flexible linker-VH-CH1-Fc region chain; or VH-CH1-VH-CH1-Fc region chain. The multivalent antibody herein preferably further comprises at least two (and preferably four) light chain variable region polypeptides. The multivalent antibody herein may, for instance, comprise from about two to about eight light chain variable region polypeptides. The light chain variable region polypeptides contemplated here comprise a light chain variable region and, optionally, further comprise a C_L domain.

Charge modifications are particularly useful in the context of a multispecific antibody, where amino acid substitutions in Fab molecules result in reducing the mispairing of light chains with non-matching heavy chains (Bence-Jones-type side products), which can occur in the production of Fab-based bi-/multispecific antigen binding molecules with a VH/VL exchange in one (or more, in case of molecules comprising more than two antigen-binding Fab molecules) of their binding arms (see also PCT publication no. WO 2015/150447, particularly the examples therein, incorporated herein by reference in its entirety).

Accordingly, in particular embodiments, an antibody comprised in the therapeutic agent comprises

- (a) a first Fab molecule which specifically binds to a first antigen
- (b) a second Fab molecule which specifically binds to a second antigen, and wherein the variable domains VL and VH of the Fab light chain and the Fab heavy chain are replaced by each other,

wherein the first antigen is an activating T cell antigen and the second antigen is a target cell antigen, or the first antigen is a target cell antigen and the second antigen is an activating T cell antigen; and

wherein

- i) in the constant domain CL of the first Fab molecule under a) the amino acid at position 124 is substituted by a positively charged amino acid (numbering according to Kabat), and wherein in the constant domain CH1 of the first Fab molecule under a) the amino acid at position 147 or the amino acid at position 213 is substituted by a negatively charged amino acid (numbering according to Kabat EU index); or
- ii) in the constant domain CL of the second Fab molecule under b) the amino acid at position 124 is substituted by a positively charged amino acid (numbering according to Kabat), and wherein in the constant domain CH1 of

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the second Fab molecule under b) the amino acid at position 147 or the amino acid at position 213 is substituted by a negatively charged amino acid (numbering according to Kabat EU index).

The antibody may not comprise both modifications mentioned under i) and ii). The constant domains CL and CH1 of the second Fab molecule are not replaced by each other (i.e., remain unexchanged).

In another embodiment of the antibody, in the constant domain CL of the first Fab molecule under a) the amino acid at position 124 is substituted independently by lysine (K), arginine (R) or histidine (H) (numbering according to Kabat) (in one preferred embodiment independently by lysine (K) or arginine (R)), and in the constant domain CH1 of the first Fab molecule under a) the amino acid at position 147 or the amino acid at position 213 is substituted independently by glutamic acid (E), or aspartic acid (D) (numbering according to Kabat EU index).

In a further embodiment, in the constant domain CL of the first Fab molecule under a) the amino acid at position 124 is substituted independently by lysine (K), arginine (R) or histidine (H) (numbering according to Kabat), and in the constant domain CH1 of the first Fab molecule under a) the amino acid at position 147 is substituted independently by glutamic acid (E), or aspartic acid (D) (numbering according to Kabat EU index).

In a particular embodiment, in the constant domain CL of the first Fab molecule under a) the amino acid at position 124 is substituted independently by lysine (K), arginine (R) or histidine (H) (numbering according to Kabat) (in one preferred embodiment independently by lysine (K) or arginine (R)) and the amino acid at position 123 is substituted independently by lysine (K), arginine (R) or histidine (H) (numbering according to Kabat) (in one preferred embodiment independently by lysine (K) or arginine (R)), and in the constant domain CH1 of the first Fab molecule under a) the amino acid at position 147 is substituted independently by glutamic acid (E), or aspartic acid (D) (numbering according to Kabat EU index) and the amino acid at position 213 is substituted independently by glutamic acid (E), or aspartic acid (D) (numbering according to Kabat EU index).

In a more particular embodiment, in the constant domain CL of the first Fab molecule under a) the amino acid at position 124 is substituted by lysine (K) (numbering according to Kabat) and the amino acid at position 123 is substituted by lysine (K) or arginine (R) (numbering according to Kabat), and in the constant domain CH1 of the first Fab molecule under a) the amino acid at position 147 is substituted by glutamic acid (E) (numbering according to Kabat EU index) and the amino acid at position 213 is substituted by glutamic acid (E) (numbering according to Kabat EU index).

In an even more particular embodiment, in the constant domain CL of the first Fab molecule under a) the amino acid at position 124 is substituted by lysine (K) (numbering according to Kabat) and the amino acid at position 123 is substituted by arginine (R) (numbering according to Kabat), and in the constant domain CH1 of the first Fab molecule under a) the amino acid at position 147 is substituted by glutamic acid (E) (numbering according to Kabat EU index) and the amino acid at position 213 is substituted by glutamic acid (E) (numbering according to Kabat EU index).

F. Chimeric Antigen Receptors

Artificial T cell receptors (also known as chimeric T cell receptors, chimeric immunoreceptors, chimeric antigen receptors (CARs)) are engineered receptors, which graft an arbitrary specificity onto an immune effector cell. Typically,

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these receptors are used to graft the specificity of a monoclonal antibody onto a T cell, with transfer of their coding sequence facilitated by retroviral vectors. In this way, a large number of target-specific T cells can be generated for adoptive cell transfer. Phase I clinical studies of this approach show efficacy.

The most common form of these molecules are fusions of single-chain variable fragments (scFv) derived from monoclonal antibodies, fused to CD3-zeta transmembrane and endodomain. Such molecules result in the transmission of a zeta signal in response to recognition by the scFv of its target. An example of such a construct is 14g2a-Zeta, which is a fusion of a scFv derived from hybridoma 14g2a (which recognizes disialoganglioside GD2). When T cells express this molecule (usually achieved by oncoretroviral vector transduction), they recognize and kill target cells that express GD2 (e.g., neuroblastoma cells). To target malignant B cells, investigators have redirected the specificity of T cells using a chimeric immunoreceptor specific for the B-lineage molecule, CD19.

The variable portions of an immunoglobulin heavy and light chain are fused by a flexible linker to form a scFv. This scFv is preceded by a signal peptide to direct the nascent protein to the endoplasmic reticulum and subsequent surface expression (this is cleaved). A flexible spacer allows to the scFv to orient in different directions to enable antigen binding. The transmembrane domain is a typical hydrophobic alpha helix usually derived from the original molecule of the signaling endodomain which protrudes into the cell and transmits the desired signal.

Type I proteins are in fact two protein domains linked by a transmembrane alpha helix in between. The cell membrane lipid bilayer, through which the transmembrane domain passes, acts to isolate the inside portion (endodomain) from the external portion (ectodomain). It is not so surprising that attaching an ectodomain from one protein to an endodomain of another protein results in a molecule that combines the recognition of the former to the signal of the latter.

Ectodomain. A signal peptide directs the nascent protein into the endoplasmic reticulum. This is essential if the receptor is to be glycosylated and anchored in the cell membrane. Any eukaryotic signal peptide sequence usually works fine. Generally, the signal peptide natively attached to the amino-terminal most component is used (e.g., in a scFv with orientation light chain-linker-heavy chain, the native signal of the light-chain is used).

The antigen recognition domain is usually an scFv. There are however many alternatives. An antigen recognition domain from native T-cell receptor (TCR) alpha and beta single chains have been described, as have simple ectodomains (e.g., CD4 ectodomain to recognize HIV infected cells) and more exotic recognition components such as a linked cytokine (which leads to recognition of cells bearing the cytokine receptor). In fact, almost anything that binds a given target with high affinity can be used as an antigen recognition region.

A spacer region links the antigen binding domain to the transmembrane domain. It should be flexible enough to allow the antigen binding domain to orient in different directions to facilitate antigen recognition. The simplest form is the hinge region from IgG1. Alternatives include the CH₂CH₃ region of immunoglobulin and portions of CD3. For most scFv based constructs, the IgG1 hinge suffices. However, the best spacer often has to be determined empirically.

Transmembrane domain. The transmembrane domain is a hydrophobic alpha helix that spans the membrane. Gener-

ally, the transmembrane domain from the most membrane proximal component of the endodomain is used. Interestingly, using the CD3-zeta transmembrane domain may result in incorporation of the artificial TCR into the native TCR a factor that is dependent on the presence of the native CD3-zeta transmembrane charged aspartic acid residue. Different transmembrane domains result in different receptor stability. The CD28 transmembrane domain results in a brightly expressed, stable receptor.

Endodomain. This is the “business-end” of the receptor. After antigen recognition, receptors cluster and a signal is transmitted to the cell. The most commonly used endodomain component is CD3-zeta which contains 3 ITAMs. This transmits an activation signal to the T cell after antigen is bound. CD3-zeta may not provide a fully competent activation signal and additional co-stimulatory signaling is needed.

“First-generation” CARs typically had the intracellular domain from the CD3 ξ -chain, which is the primary transmitter of signals from endogenous TCRs. “Second-generation” CARs add intracellular signaling domains from various costimulatory protein receptors (e.g., CD28, 41BB, ICOS) to the cytoplasmic tail of the CAR to provide additional signals to the T cell. Preclinical studies have indicated that the second generation of CAR designs improves the antitumor activity of T cells. More recent, “third-generation” CARs combine multiple signaling domains, such as CD3z-CD28-41BB or CD3z-CD28-OX40, to further augment potency.

G. ADCs

Antibody Drug Conjugates or ADCs are a new class of highly potent biopharmaceutical drugs designed as a targeted therapy for the treatment of people with infectious disease. ADCs are complex molecules composed of an antibody (a whole mAb or an antibody fragment such as a single-chain variable fragment, or scFv) linked, via a stable chemical linker with labile bonds, to a biological active cytotoxic/anti-viral payload or drug. Antibody Drug Conjugates are examples of bioconjugates and immunoconjugates.

By combining the unique targeting capabilities of monoclonal antibodies with the cancer-killing ability of cytotoxic drugs, antibody-drug conjugates allow sensitive discrimination between healthy and diseased tissue. This means that, in contrast to traditional systemic approaches, antibody-drug conjugates target and attack the infected cell so that healthy cells are less severely affected.

In the development ADC-based anti-tumor therapies, an anticancer drug (e.g., a cell toxin or cytotoxin) is coupled to an antibody that specifically targets a certain cell marker (e.g., a protein that, ideally, is only to be found in or on infected cells). Antibodies track these proteins down in the body and attach themselves to the surface of cancer cells. The biochemical reaction between the antibody and the target protein (antigen) triggers a signal in the tumor cell, which then absorbs or internalizes the antibody together with the cytotoxin. After the ADC is internalized, the cytotoxic drug is released and kills the cell or impairs viral replication. Due to this targeting, ideally the drug has lower side effects and gives a wider therapeutic window than other agents.

A stable link between the antibody and cytotoxic/anti-viral agent is a crucial aspect of an ADC. Linkers are based on chemical motifs including disulfides, hydrazones or peptides (cleavable), or thioethers (non-cleavable) and control the distribution and delivery of the cytotoxic agent to the target cell. Cleavable and non-cleavable types of linkers have been proven to be safe in preclinical and clinical trials.

Brentuximab vedotin includes an enzyme-sensitive cleavable linker that delivers the potent and highly toxic anti-microtubule agent Monomethyl auristatin E or MMAE, a synthetic antineoplastic agent, to human specific CD30-positive malignant cells. Because of its high toxicity MMAE, which inhibits cell division by blocking the polymerization of tubulin, cannot be used as a single-agent chemotherapeutic drug. However, the combination of MMAE linked to an anti-CD30 monoclonal antibody (cAC10, a cell membrane protein of the tumor necrosis factor or TNF receptor) proved to be stable in extracellular fluid, cleavable by cathepsin and safe for therapy. Trastuzumab emtansine, the other approved ADC, is a combination of the microtubule-formation inhibitor mertansine (DM-1), a derivative of the Maytansine, and antibody trastuzumab (Herceptin®/Genentech/Roche) attached by a stable, non-cleavable linker.

The availability of better and more stable linkers has changed the function of the chemical bond. The type of linker, cleavable or non-cleavable, lends specific properties to the cytotoxic (anti-cancer) drug. For example, a non-cleavable linker keeps the drug within the cell. As a result, the entire antibody, linker and cytotoxic agent enter the targeted cancer cell where the antibody is degraded to the level of an amino acid. The resulting complex—amino acid, linker and cytotoxic agent—now becomes the active drug. In contrast, cleavable linkers are catalyzed by enzymes in the host cell where it releases the cytotoxic agent.

Another type of cleavable linker, currently in development, adds an extra molecule between the cytotoxic/antiviral drug and the cleavage site. This linker technology allows researchers to create ADCs with more flexibility without worrying about changing cleavage kinetics. Researchers are also developing a new method of peptide cleavage based on Edman degradation, a method of sequencing amino acids in a peptide. Future direction in the development of ADCs also include the development of site-specific conjugation (TDCs) to further improve stability and therapeutic index and a emitting immunoconjugates and antibody-conjugated nanoparticles.

H. BiTES

Bi-specific T-cell engagers (BiTEs) are a class of artificial bispecific monoclonal antibodies that are investigated for the use as anti-cancer drugs. They direct a host's immune system, more specifically the T cells' cytotoxic activity, against infected cells. BiTE is a registered trademark of Micromet AG.

BiTEs are fusion proteins consisting of two single-chain variable fragments (scFvs) of different antibodies, or amino acid sequences from four different genes, on a single peptide chain of about 55 kilodaltons. One of the scFvs binds to T cells via the CD3 receptor, and the other to an infected cell via a specific molecule.

Like other bispecific antibodies, and unlike ordinary monoclonal antibodies, BiTEs form a link between T cells and target cells. This causes T cells to exert cytotoxic/antiviral activity on infected cells by producing proteins like perforin and granzymes, independently of the presence of MHC I or co-stimulatory molecules. These proteins enter infected cells and initiate the cell's apoptosis. This action mimics physiological processes observed during T cell attacks against infected cells.

I. Intrabodies

In a particular embodiment, the antibody is a recombinant antibody that is suitable for action inside of a cell—such antibodies are known as “intrabodies.” These antibodies may interfere with target function by a variety of mecha-

nism, such as by altering intracellular protein trafficking, interfering with enzymatic function, and blocking protein-protein or protein-DNA interactions. In many ways, their structures mimic or parallel those of single chain and single domain antibodies, discussed above. Indeed, single-transcript/single-chain is an important feature that permits intracellular expression in a target cell, and also makes protein transit across cell membranes more feasible. However, additional features are required.

The two major issues impacting the implementation of intrabody therapeutic are delivery, including cell/tissue targeting, and stability. With respect to delivery, a variety of approaches have been employed, such as tissue-directed delivery, use of cell-type specific promoters, viral-based delivery and use of cell-permeability/membrane translocating peptides. With respect to the stability, the approach is generally to either screen by brute force, including methods that involve phage display and may include sequence maturation or development of consensus sequences, or more directed modifications such as insertion stabilizing sequences (e.g., Fc regions, chaperone protein sequences, leucine zippers) and disulfide replacement/modification.

An additional feature that intrabodies may require is a signal for intracellular targeting. Vectors that can target intrabodies (or other proteins) to subcellular regions such as the cytoplasm, nucleus, mitochondria and ER have been designed and are commercially available (Invitrogen Corp.; Persic et al., 1997).

By virtue of their ability to enter cells, intrabodies have additional uses that other types of antibodies may not achieve. In the case of the present antibodies, the ability to interact with the MUC1 cytoplasmic domain in a living cell may interfere with functions associated with the MUC1 CD, such as signaling functions (binding to other molecules) or oligomer formation. In particular, it is contemplated that such antibodies can be used to inhibit MUC1 dimer formation.

J. Purification

In certain embodiments, the antibodies of the present disclosure may be purified. The term "purified," as used herein, is intended to refer to a composition, isolatable from other components, wherein the protein is purified to any degree relative to its naturally-obtainable state. A purified protein therefore also refers to a protein, free from the environment in which it may naturally occur. Where the term "substantially purified" is used, this designation will refer to a composition in which the protein or peptide forms the major component of the composition, such as constituting about 50%, about 60%, about 70%, about 80%, about 90%, about 95% or more of the proteins in the composition.

Protein purification techniques are well known to those of skill in the art. These techniques involve, at one level, the crude fractionation of the cellular milieu to polypeptide and non-polypeptide fractions. Having separated the polypeptide from other proteins, the polypeptide of interest may be further purified using chromatographic and electrophoretic techniques to achieve partial or complete purification (or purification to homogeneity). Analytical methods particularly suited to the preparation of a pure peptide are ion-exchange chromatography, exclusion chromatography; polyacrylamide gel electrophoresis; isoelectric focusing. Other methods for protein purification include, precipitation with ammonium sulfate, PEG, antibodies and the like or by heat denaturation, followed by centrifugation; gel filtration, reverse phase, hydroxylapatite and affinity chromatography; and combinations of such and other techniques.

In purifying an antibody of the present disclosure, it may be desirable to express the polypeptide in a prokaryotic or eukaryotic expression system and extract the protein using denaturing conditions. The polypeptide may be purified from other cellular components using an affinity column, which binds to a tagged portion of the polypeptide. As is generally known in the art, it is believed that the order of conducting the various purification steps may be changed, or that certain steps may be omitted, and still result in a suitable method for the preparation of a substantially purified protein or peptide.

Commonly, complete antibodies are fractionated utilizing agents (i.e., protein A) that bind the Fc portion of the antibody. Alternatively, antigens may be used to simultaneously purify and select appropriate antibodies. Such methods often utilize the selection agent bound to a support, such as a column, filter or bead. The antibodies are bound to a support, contaminants removed (e.g., washed away), and the antibodies released by applying conditions (salt, heat, etc.).

Various methods for quantifying the degree of purification of the protein or peptide will be known to those of skill in the art in light of the present disclosure. These include, for example, determining the specific activity of an active fraction, or assessing the amount of polypeptides within a fraction by SDS/PAGE analysis. Another method for assessing the purity of a fraction is to calculate the specific activity of the fraction, to compare it to the specific activity of the initial extract, and to thus calculate the degree of purity. The actual units used to represent the amount of activity will, of course, be dependent upon the particular assay technique chosen to follow the purification and whether or not the expressed protein or peptide exhibits a detectable activity.

It is known that the migration of a polypeptide can vary, sometimes significantly, with different conditions of SDS/PAGE (Capaldi et al., 1977). It will therefore be appreciated that under differing electrophoresis conditions, the apparent molecular weights of purified or partially purified expression products may vary.

III. ACTIVE/PASSIVE IMMUNIZATION AND TREATMENT/PREVENTION OF ENTEROVIRUS D68 INFECTION

A. Formulation and Administration

The present disclosure provides pharmaceutical compositions comprising anti-EV-D68 antibodies and antigens for generating the same. Such compositions comprise a prophylactically or therapeutically effective amount of an antibody or a fragment thereof, or a peptide immunogen, and a pharmaceutically acceptable carrier. In a specific embodiment, the term "pharmaceutically acceptable" means approved by a regulatory agency of the Federal or a state government or listed in the U.S. Pharmacopeia or other generally recognized pharmacopeia for use in animals, and more particularly in humans. The term "carrier" refers to a diluent, excipient, or vehicle with which the therapeutic is administered. Such pharmaceutical carriers can be sterile liquids, such as water and oils, including those of petroleum, animal, vegetable or synthetic origin, such as peanut oil, soybean oil, mineral oil, sesame oil and the like. Water is a particular carrier when the pharmaceutical composition is administered intravenously. Saline solutions and aqueous dextrose and glycerol solutions can also be employed as liquid carriers, particularly for injectable solutions. Other suitable pharmaceutical excipients include starch, glucose, lactose, sucrose, gelatin, malt, rice, flour, chalk, silica gel,

sodium stearate, glycerol monostearate, talc, sodium chloride, dried skim milk, glycerol, propylene, glycol, water, ethanol and the like.

The composition, if desired, can also contain minor amounts of wetting or emulsifying agents, or pH buffering agents. These compositions can take the form of solutions, suspensions, emulsion, tablets, pills, capsules, powders, sustained-release formulations and the like. Oral formulations can include standard carriers such as pharmaceutical grades of mannitol, lactose, starch, magnesium stearate, sodium saccharine, cellulose, magnesium carbonate, etc. Examples of suitable pharmaceutical agents are described in "Remington's Pharmaceutical Sciences." Such compositions will contain a prophylactically or therapeutically effective amount of the antibody or fragment thereof, preferably in purified form, together with a suitable amount of carrier so as to provide the form for proper administration to the patient. The formulation should suit the mode of administration, which can be oral, intravenous, intraarterial, intrabuccal, intranasal, nebulized, bronchial inhalation, intrarectal, vaginal, topical or delivered by mechanical ventilation.

Active vaccines are also envisioned where antibodies like those disclosed are produced in vivo in a subject at risk of EV-D68 infection. Such vaccines can be formulated for parenteral administration, e.g., formulated for injection via the intradermal, intravenous, intramuscular, subcutaneous, aerosol or even intraperitoneal routes. Administration by intradermal and intramuscular routes are contemplated. The vaccine could alternatively be administered by a topical route directly to the mucosa, for example by nasal drops, inhalation, by nebulizer, or via intrarectal or vaginal delivery. Pharmaceutically acceptable salts include the acid salts and those which are formed with inorganic acids such as, for example, hydrochloric or phosphoric acids, or such organic acids as acetic, oxalic, tartaric, mandelic, and the like. Salts formed with the free carboxyl groups may also be derived from inorganic bases such as, for example, sodium, potassium, ammonium, calcium, or ferric hydroxides, and such organic bases as isopropylamine, trimethylamine, 2-ethylamino ethanol, histidine, procaine, and the like.

Passive transfer of antibodies, known as artificially acquired passive immunity, generally will involve the use of intravenous or intramuscular injections. The forms of antibody can be human or animal blood plasma or serum, as pooled human immunoglobulin for intravenous (IVIG) or intramuscular (IG) use, as high-titer human IVIG or IG from immunized or from donors recovering from disease, and as monoclonal antibodies (MAb). Such immunity generally lasts for only a short period of time, and there is also a potential risk for hypersensitivity reactions, and serum sickness, especially from gamma globulin of non-human origin. However, passive immunity provides immediate protection. The antibodies will be formulated in a carrier suitable for injection, i.e., sterile and syringeable.

Generally, the ingredients of compositions of the disclosure are supplied either separately or mixed together in unit dosage form, for example, as a dry lyophilized powder or water-free concentrate in a hermetically sealed container such as an ampoule or sachette indicating the quantity of active agent. Where the composition is to be administered by infusion, it can be dispensed with an infusion bottle containing sterile pharmaceutical grade water or saline. Where the composition is administered by injection, an ampoule of sterile water for injection or saline can be provided so that the ingredients may be mixed prior to administration.

The compositions of the disclosure can be formulated as neutral or salt forms. Pharmaceutically acceptable salts include those formed with anions such as those derived from hydrochloric, phosphoric, acetic, oxalic, tartaric acids, etc., and those formed with cations such as those derived from sodium, potassium, ammonium, calcium, ferric hydroxides, isopropylamine, triethylamine, 2-ethylamino ethanol, histidine, procaine, etc.

2. ADCC

Antibody-dependent cell-mediated cytotoxicity (ADCC) is an immune mechanism leading to the lysis of antibody-coated target cells by immune effector cells. The target cells are cells to which antibodies or fragments thereof comprising an Fc region specifically bind, generally via the protein part that is N-terminal to the Fc region. By "antibody having increased/reduced antibody dependent cell-mediated cytotoxicity (ADCC)" is meant an antibody having increased/reduced ADCC as determined by any suitable method known to those of ordinary skill in the art.

As used herein, the term "increased/reduced ADCC" is defined as either an increase/reduction in the number of target cells that are lysed in a given time, at a given concentration of antibody in the medium surrounding the target cells, by the mechanism of ADCC defined above, and/or a reduction/increase in the concentration of antibody, in the medium surrounding the target cells, required to achieve the lysis of a given number of target cells in a given time, by the mechanism of ADCC. The increase/reduction in ADCC is relative to the ADCC mediated by the same antibody produced by the same type of host cells, using the same standard production, purification, formulation and storage methods (which are known to those skilled in the art), but that has not been engineered. For example the increase in ADCC mediated by an antibody produced by host cells engineered to have an altered pattern of glycosylation (e.g., to express the glycosyltransferase, GnTIII, or other glycosyltransferases) by the methods described herein, is relative to the ADCC mediated by the same antibody produced by the same type of non-engineered host cells.

3. CDC

Complement-dependent cytotoxicity (CDC) is a function of the complement system. It is the processes in the immune system that kill pathogens by damaging their membranes without the involvement of antibodies or cells of the immune system. There are three main processes. All three insert one or more membrane attack complexes (MAC) into the pathogen which cause lethal colloid-osmotic swelling, i.e., CDC. It is one of the mechanisms by which antibodies or antibody fragments have an anti-viral effect.

IV. ANTIBODY CONJUGATES

Antibodies of the present disclosure may be linked to at least one agent to form an antibody conjugate. In order to increase the efficacy of antibody molecules as diagnostic or therapeutic agents, it is conventional to link or covalently bind or complex at least one desired molecule or moiety. Such a molecule or moiety may be, but is not limited to, at least one effector or reporter molecule. Effector molecules comprise molecules having a desired activity, e.g., cytotoxic activity. Non-limiting examples of effector molecules which have been attached to antibodies include toxins, anti-tumor agents, therapeutic enzymes, radionuclides, antiviral agents, chelating agents, cytokines, growth factors, and oligo- or polynucleotides. By contrast, a reporter molecule is defined as any moiety which may be detected using an assay. Non-limiting examples of reporter molecules which have

been conjugated to antibodies include enzymes, radiolabels, haptens, fluorescent labels, phosphorescent molecules, chemiluminescent molecules, chromophores, photoaffinity molecules, colored particles or ligands, such as biotin.

Antibody conjugates are generally preferred for use as diagnostic agents. Antibody diagnostics generally fall within two classes, those for use in *in vitro* diagnostics, such as in a variety of immunoassays, and those for use in *in vivo* diagnostic protocols, generally known as "antibody-directed imaging." Many appropriate imaging agents are known in the art, as are methods for their attachment to antibodies (see, for e.g., U.S. Pat. Nos. 5,021,236, 4,938,948, and 4,472,509). The imaging moieties used can be paramagnetic ions, radioactive isotopes, fluorochromes, NMR-detectable substances, and X-ray imaging agents.

In the case of paramagnetic ions, one might mention by way of example ions such as chromium (III), manganese (II), iron (III), iron (II), cobalt (II), nickel (II), copper (II), neodymium (III), samarium (III), ytterbium (III), gadolinium (III), vanadium (II), terbium (III), dysprosium (III), holmium (III) and/or erbium (III), with gadolinium being particularly preferred. Ions useful in other contexts, such as X-ray imaging, include but are not limited to lanthanum (III), gold (III), lead (II), and especially bismuth (III).

In the case of radioactive isotopes for therapeutic and/or diagnostic application, one might mention astatine²¹¹, ¹⁴carbon, ⁵¹chromium, ³⁶chlorine, ⁵⁷cobalt, ⁵⁸cobalt, copper⁶⁷, ¹⁵²Eu, gallium⁶⁷, ³hydrogen, iodine¹²³, iodine¹²⁵, iodine¹³¹, indium¹¹¹, ⁵⁹iron, ³²phosphorus, rhenium¹⁸⁶, rhenium¹⁸⁸, ⁷⁵selenium, ³⁵sulphur, technetium^{99m} and/or yttrium⁹⁰. ¹²⁵I is often being preferred for use in certain embodiments, and technetium^{99m} and/or indium¹¹¹ are also often preferred due to their low energy and suitability for long range detection. Radioactively labeled monoclonal antibodies of the present disclosure may be produced according to well-known methods in the art. For instance, monoclonal antibodies can be iodinated by contact with sodium and/or potassium iodide and a chemical oxidizing agent such as sodium hypochlorite, or an enzymatic oxidizing agent, such as lactoperoxidase. Monoclonal antibodies according to the disclosure may be labeled with technetium^{99m} by ligand exchange process, for example, by reducing pertechnetate with stannous solution, chelating the reduced technetium onto a Sephadex column and applying the antibody to this column. Alternatively, direct labeling techniques may be used, e.g., by incubating pertechnetate, a reducing agent such as SnCl_2 , a buffer solution such as sodium-potassium phthalate solution, and the antibody. Intermediary functional groups which are often used to bind radioisotopes which exist as metallic ions to antibody are diethylenetriaminepentaacetic acid (DTPA) or ethylene diaminetetraacetic acid (EDTA).

Among the fluorescent labels contemplated for use as conjugates include Alexa 350, Alexa 430, AMCA, BODIPY 630/650, BODIPY 650/665, BODIPY-FL, BODIPY-R6G, BODIPY-TMR, BODIPY-TRX, Cascade Blue, Cy3, Cy5, 6-FAM, Fluorescein Isothiocyanate, HEX, 6-JOE, Oregon Green 488, Oregon Green 500, Oregon Green 514, Pacific Blue, REG, Rhodamine Green, Rhodamine Red, Renografin, ROX, TAMRA, TET, Tetramethylrhodamine, and/or Texas Red.

Additional types of antibodies contemplated in the present disclosure are those intended primarily for use *in vitro*, where the antibody is linked to a secondary binding ligand and/or to an enzyme (an enzyme tag) that will generate a colored product upon contact with a chromogenic substrate. Examples of suitable enzymes include urease, alkaline phosphatase, (horseradish) hydrogen peroxidase or glucose oxi-

dase. Preferred secondary binding ligands are biotin and avidin and streptavidin compounds. The use of such labels is well known to those of skill in the art and are described, for example, in U.S. Pat. Nos. 3,817,837, 3,850,752, 3,939,350, 3,996,345, 4,277,437, 4,275,149 and 4,366,241.

Yet another known method of site-specific attachment of molecules to antibodies comprises the reaction of antibodies with hapten-based affinity labels. Essentially, hapten-based affinity labels react with amino acids in the antigen binding site, thereby destroying this site and blocking specific antigen reaction. However, this may not be advantageous since it results in loss of antigen binding by the antibody conjugate.

Molecules containing azido groups may also be used to form covalent bonds to proteins through reactive nitrene intermediates that are generated by low intensity ultraviolet light (Potter and Haley, 1983). In particular, 2- and 8-azido analogues of purine nucleotides have been used as site-directed photoprobes to identify nucleotide binding proteins in crude cell extracts (Owens & Haley, 1987; Atherton et al., 1985). The 2- and 8-azido nucleotides have also been used to map nucleotide binding domains of purified proteins (Khatoun et al., 1989; King et al., 1989; Dholakia et al., 1989) and may be used as antibody binding agents.

Several methods are known in the art for the attachment or conjugation of an antibody to its conjugate moiety. Some attachment methods involve the use of a metal chelate complex employing, for example, an organic chelating agent such as diethylenetriaminepentaacetic acid anhydride (DTPA); ethylenetriaminetetraacetic acid; N-chloro-p-toluenesulfonamide; and/or tetrachloro-3 α -6 α -diphenylglycyl-3 attached to the antibody (U.S. Pat. Nos. 4,472,509 and 4,938,948). Monoclonal antibodies may also be reacted with an enzyme in the presence of a coupling agent such as glutaraldehyde or periodate. Conjugates with fluorescein markers are prepared in the presence of these coupling agents or by reaction with an isothiocyanate. In U.S. Pat. No. 4,938,948, imaging of breast tumors is achieved using monoclonal antibodies and the detectable imaging moieties are bound to the antibody using linkers such as methyl-p-hydroxybenzimidate or N-succinimidyl-3-(4-hydroxyphenyl)propionate.

In other embodiments, derivatization of immunoglobulins by selectively introducing sulfhydryl groups in the Fc region of an immunoglobulin, using reaction conditions that do not alter the antibody combining site are contemplated. Antibody conjugates produced according to this methodology are disclosed to exhibit improved longevity, specificity and sensitivity (U.S. Pat. No. 5,196,066, incorporated herein by reference). Site-specific attachment of effector or reporter molecules, wherein the reporter or effector molecule is conjugated to a carbohydrate residue in the Fc region have also been disclosed in the literature (O'Shannessy et al., 1987). This approach has been reported to produce diagnostically and therapeutically promising antibodies which are currently in clinical evaluation.

V. IMMUNODETECTION METHODS

In still further embodiments, the present disclosure concerns immunodetection methods for binding, purifying, removing, quantifying and otherwise generally detecting EV-D68 and its associated antigens. While such methods can be applied in a traditional sense, another use will be in quality control and monitoring of vaccine and other virus stocks, where antibodies according to the present disclosure can be used to assess the amount or integrity (i.e., long term

stability) of antigens in viruses. Alternatively, the methods may be used to screen various antibodies for appropriate/desired reactivity profiles.

Other immunodetection methods include specific assays for determining the presence of EV-D68 in a subject. A wide variety of assay formats are contemplated, but specifically those that would be used to detect EV-D68 in a fluid obtained from a subject, such as saliva, blood, plasma, sputum, semen, urine, respiratory droplets or aerosol. In particular, semen has been demonstrated as a viable sample for detecting EV-D68 (Purpura et al., 2016; Mansuy et al., 2016; Barzon et al., 2016; Gornet et al., 2016; Duffy et al., 2009; CDC, 2016; Halfon et al., 2010; Elder et al. 2005). The assays may be advantageously formatted for non-healthcare (home) use, including lateral flow assays (see below) analogous to home pregnancy tests. These assays may be packaged in the form of a kit with appropriate reagents and instructions to permit use by the subject of a family member.

Some immunodetection methods include enzyme linked immunosorbent assay (ELISA), radioimmunoassay (RIA), immunoradiometric assay, fluoroimmunoassay, chemiluminescent assay, bioluminescent assay, and Western blot to mention a few. In particular, a competitive assay for the detection and quantitation of EV-D68 antibodies directed to specific parasite epitopes in samples also is provided. The steps of various useful immunodetection methods have been described in the scientific literature, such as, e.g., Doolittle and Ben-Zeev (1999), Gulbis and Galand (1993), De Jager et al. (1993), and Nakamura et al. (1987). In general, the immunobinding methods include obtaining a sample suspected of containing EV-D68, and contacting the sample with a first antibody in accordance with the present disclosure, as the case may be, under conditions effective to allow the formation of immunocomplexes.

These methods include methods for purifying EV-D68 or related antigens from a sample. The antibody will preferably be linked to a solid support, such as in the form of a column matrix, and the sample suspected of containing the EV-D68 or antigenic component will be applied to the immobilized antibody. The unwanted components will be washed from the column, leaving the EV-D68 antigen immunocomplexed to the immobilized antibody, which is then collected by removing the organism or antigen from the column.

The immunobinding methods also include methods for detecting and quantifying the amount of EV-D68 or related components in a sample and the detection and quantification of any immune complexes formed during the binding process. Here, one would obtain a sample suspected of containing EV-D68 or its antigens and contact the sample with an antibody that binds EV-D68 or components thereof, followed by detecting and quantifying the amount of immune complexes formed under the specific conditions. In terms of antigen detection, the biological sample analyzed may be any sample that is suspected of containing EV-D68 or EV-D68 antigen, such as a tissue section or specimen, a homogenized tissue extract, a biological fluid, including blood and serum, or a secretion, such as feces or urine.

Contacting the chosen biological sample with the antibody under effective conditions and for a period of time sufficient to allow the formation of immune complexes (primary immune complexes) is generally a matter of simply adding the antibody composition to the sample and incubating the mixture for a period of time long enough for the antibodies to form immune complexes with, i.e., to bind to EV-D68 or antigens present. After this time, the sample-antibody composition, such as a tissue section, ELISA plate,

dot blot or Western blot, will generally be washed to remove any non-specifically bound antibody species, allowing only those antibodies specifically bound within the primary immune complexes to be detected.

In general, the detection of immunocomplex formation is well known in the art and may be achieved through the application of numerous approaches. These methods are generally based upon the detection of a label or marker, such as any of those radioactive, fluorescent, biological and enzymatic tags. Patents concerning the use of such labels include U.S. Pat. Nos. 3,817,837, 3,850,752, 3,939,350, 3,996,345, 4,277,437, 4,275,149 and 4,366,241. Of course, one may find additional advantages through the use of a secondary binding ligand such as a second antibody and/or a biotin/avidin ligand binding arrangement, as is known in the art.

The antibody employed in the detection may itself be linked to a detectable label, wherein one would then simply detect this label, thereby allowing the amount of the primary immune complexes in the composition to be determined. Alternatively, the first antibody that becomes bound within the primary immune complexes may be detected by means of a second binding ligand that has binding affinity for the antibody. In these cases, the second binding ligand may be linked to a detectable label. The second binding ligand is itself often an antibody, which may thus be termed a "secondary" antibody. The primary immune complexes are contacted with the labeled, secondary binding ligand, or antibody, under effective conditions and for a period of time sufficient to allow the formation of secondary immune complexes. The secondary immune complexes are then generally washed to remove any non-specifically bound labeled secondary antibodies or ligands, and the remaining label in the secondary immune complexes is then detected.

Further methods include the detection of primary immune complexes by a two-step approach. A second binding ligand, such as an antibody that has binding affinity for the antibody, is used to form secondary immune complexes, as described above. After washing, the secondary immune complexes are contacted with a third binding ligand or antibody that has binding affinity for the second antibody, again under effective conditions and for a period of time sufficient to allow the formation of immune complexes (tertiary immune complexes). The third ligand or antibody is linked to a detectable label, allowing detection of the tertiary immune complexes thus formed. This system may provide for signal amplification if this is desired.

One method of immunodetection uses two different antibodies. A first biotinylated antibody is used to detect the target antigen, and a second antibody is then used to detect the biotin attached to the complexed biotin. In that method, the sample to be tested is first incubated in a solution containing the first step antibody. If the target antigen is present, some of the antibody binds to the antigen to form a biotinylated antibody/antigen complex. The antibody/antigen complex is then amplified by incubation in successive solutions of streptavidin (or avidin), biotinylated DNA, and/or complementary biotinylated DNA, with each step adding additional biotin sites to the antibody/antigen complex. The amplification steps are repeated until a suitable level of amplification is achieved, at which point the sample is incubated in a solution containing the second step antibody against biotin. This second step antibody is labeled, as for example with an enzyme that can be used to detect the presence of the antibody/antigen complex by histoenzymol-

ogy using a chromogen substrate. With suitable amplification, a conjugate can be produced which is macroscopically visible.

Another known method of immunodetection takes advantage of the immuno-PCR (Polymerase Chain Reaction) methodology. The PCR method is similar to the Cantor method up to the incubation with biotinylated DNA, however, instead of using multiple rounds of streptavidin and biotinylated DNA incubation, the DNA/biotin/streptavidin/antibody complex is washed out with a low pH or high salt buffer that releases the antibody. The resulting wash solution is then used to carry out a PCR reaction with suitable primers with appropriate controls. At least in theory, the enormous amplification capability and specificity of PCR can be utilized to detect a single antigen molecule.

A. ELISAs

Immunoassays, in their most simple and direct sense, are binding assays. Certain preferred immunoassays are the various types of enzyme linked immunosorbent assays (ELISAs) and radioimmunoassays (RIA) known in the art. Immunohistochemical detection using tissue sections is also particularly useful. However, it will be readily appreciated that detection is not limited to such techniques, and western blotting, dot blotting, FACS analyses, and the like may also be used.

In one exemplary ELISA, the antibodies of the disclosure are immobilized onto a selected surface exhibiting protein affinity, such as a well in a polystyrene microtiter plate. Then, a test composition suspected of containing the EV-D68 or EV-D68 antigen is added to the wells. After binding and washing to remove non-specifically bound immune complexes, the bound antigen may be detected. Detection may be achieved by the addition of another anti-EV-D68 antibody that is linked to a detectable label. This type of ELISA is a simple "sandwich ELISA." Detection may also be achieved by the addition of a second anti-EV-D68 antibody, followed by the addition of a third antibody that has binding affinity for the second antibody, with the third antibody being linked to a detectable label.

In another exemplary ELISA, the samples suspected of containing the EV-D68 or EV-D68 antigen are immobilized onto the well surface and then contacted with the anti-EV-D68 antibodies of the disclosure. After binding and washing to remove non-specifically bound immune complexes, the bound anti-EV-D68 antibodies are detected. Where the initial anti-EV-D68 antibodies are linked to a detectable label, the immune complexes may be detected directly. Again, the immune complexes may be detected using a second antibody that has binding affinity for the first anti-EV-D68 antibody, with the second antibody being linked to a detectable label.

Irrespective of the format employed, ELISAs have certain features in common, such as coating, incubating and binding, washing to remove non-specifically bound species, and detecting the bound immune complexes. These are described below.

In coating a plate with either antigen or antibody, one will generally incubate the wells of the plate with a solution of the antigen or antibody, either overnight or for a specified period of hours. The wells of the plate will then be washed to remove incompletely adsorbed material. Any remaining available surfaces of the wells are then "coated" with a nonspecific protein that is antigenically neutral with regard to the test antisera. These include bovine serum albumin (BSA), casein or solutions of milk powder. The coating allows for blocking of nonspecific adsorption sites on the

immobilizing surface and thus reduces the background caused by nonspecific binding of antisera onto the surface.

In ELISAs, it is probably more customary to use a secondary or tertiary detection means rather than a direct procedure. Thus, after binding of a protein or antibody to the well, coating with a non-reactive material to reduce background, and washing to remove unbound material, the immobilizing surface is contacted with the biological sample to be tested under conditions effective to allow immune complex (antigen/antibody) formation. Detection of the immune complex then requires a labeled secondary binding ligand or antibody, and a secondary binding ligand or antibody in conjunction with a labeled tertiary antibody or a third binding ligand.

"Under conditions effective to allow immune complex (antigen/antibody) formation" means that the conditions preferably include diluting the antigens and/or antibodies with solutions such as BSA, bovine gamma globulin (BGG) or phosphate buffered saline (PBS)/Tween. These added agents also tend to assist in the reduction of nonspecific background.

The "suitable" conditions also mean that the incubation is at a temperature or for a period of time sufficient to allow effective binding. Incubation steps are typically from about 1 to 2 to 4 hours or so, at temperatures preferably on the order of 25° C. to 27° C., or may be overnight at about 4° C. or so.

Following all incubation steps in an ELISA, the contacted surface is washed so as to remove non-complexed material. A preferred washing procedure includes washing with a solution such as PBS/Tween, or borate buffer. Following the formation of specific immune complexes between the test sample and the originally bound material, and subsequent washing, the occurrence of even minute amounts of immune complexes may be determined.

To provide a detecting means, the second or third antibody will have an associated label to allow detection. Preferably, this will be an enzyme that will generate color development upon incubating with an appropriate chromogenic substrate. Thus, for example, one will desire to contact or incubate the first and second immune complex with a urease, glucose oxidase, alkaline phosphatase or hydrogen peroxidase-conjugated antibody for a period of time and under conditions that favor the development of further immune complex formation (e.g., incubation for 2 hours at room temperature in a PBS-containing solution such as PBS-Tween).

After incubation with the labeled antibody, and subsequent to washing to remove unbound material, the amount of label is quantified, e.g., by incubation with a chromogenic substrate such as urea, or bromocresol purple, or 2,2'-azino-di-(3-ethyl-benzthiazoline-6-sulfonic acid (ABTS), or H₂O₂, in the case of peroxidase as the enzyme label. Quantification is then achieved by measuring the degree of color generated, e.g., using a visible spectra spectrophotometer.

In another embodiment, the present disclosure contemplates the use of competitive formats. This is particularly useful in the detection of EV-D68 antibodies in sample. In competition-based assays, an unknown amount of analyte or antibody is determined by its ability to displace a known amount of labeled antibody or analyte. Thus, the quantifiable loss of a signal is an indication of the amount of unknown antibody or analyte in a sample.

Here, the inventor proposes the use of labeled EV-D68 monoclonal antibodies to determine the amount of EV-D68 antibodies in a sample. The basic format would include

contacting a known amount of EV-D68 monoclonal antibody (linked to a detectable label) with EV-D68 antigen or particle. The EV-D68 antigen or organism is preferably attached to a support. After binding of the labeled monoclonal antibody to the support, the sample is added and incubated under conditions permitting any unlabeled antibody in the sample to compete with, and hence displace, the labeled monoclonal antibody. By measuring either the lost label or the label remaining (and subtracting that from the original amount of bound label), one can determine how much non-labeled antibody is bound to the support, and thus how much antibody was present in the sample.

B. Western Blot

The Western blot (alternatively, protein immunoblot) is an analytical technique used to detect specific proteins in a given sample of tissue homogenate or extract. It uses gel electrophoresis to separate native or denatured proteins by the length of the polypeptide (denaturing conditions) or by the 3-D structure of the protein (native/non-denaturing conditions). The proteins are then transferred to a membrane (typically nitrocellulose or PVDF), where they are probed (detected) using antibodies specific to the target protein.

Samples may be taken from whole tissue or from cell culture. In most cases, solid tissues are first broken down mechanically using a blender (for larger sample volumes), using a homogenizer (smaller volumes), or by sonication. Cells may also be broken open by one of the above mechanical methods. However, it should be noted that bacteria, virus or environmental samples can be the source of protein and thus Western blotting is not restricted to cellular studies only. Assorted detergents, salts, and buffers may be employed to encourage lysis of cells and to solubilize proteins. Protease and phosphatase inhibitors are often added to prevent the digestion of the sample by its own enzymes. Tissue preparation is often done at cold temperatures to avoid protein denaturing.

The proteins of the sample are separated using gel electrophoresis. Separation of proteins may be by isoelectric point (pI), molecular weight, electric charge, or a combination of these factors. The nature of the separation depends on the treatment of the sample and the nature of the gel. This is a very useful way to determine a protein. It is also possible to use a two-dimensional (2-D) gel which spreads the proteins from a single sample out in two dimensions. Proteins are separated according to isoelectric point (pH at which they have neutral net charge) in the first dimension, and according to their molecular weight in the second dimension.

In order to make the proteins accessible to antibody detection, they are moved from within the gel onto a membrane made of nitrocellulose or polyvinylidene difluoride (PVDF). The membrane is placed on top of the gel, and a stack of filter papers placed on top of that. The entire stack is placed in a buffer solution which moves up the paper by capillary action, bringing the proteins with it. Another method for transferring the proteins is called electroblotting and uses an electric current to pull proteins from the gel into the PVDF or nitrocellulose membrane. The proteins move from within the gel onto the membrane while maintaining the organization they had within the gel. As a result of this blotting process, the proteins are exposed on a thin surface layer for detection (see below). Both varieties of membrane are chosen for their non-specific protein binding properties (i.e., binds all proteins equally well). Protein binding is based upon hydrophobic interactions, as well as charged interactions between the membrane and protein. Nitrocellulose membranes are cheaper than PVDF but are far more

fragile and do not stand up well to repeated probings. The uniformity and overall effectiveness of transfer of protein from the gel to the membrane can be checked by staining the membrane with Coomassie Brilliant Blue or Ponceau S dyes. Once transferred, proteins are detected using labeled primary antibodies, or unlabeled primary antibodies followed by indirect detection using labeled protein A or secondary labeled antibodies binding to the Fc region of the primary antibodies.

C. Lateral Flow Assays

Lateral flow assays, also known as lateral flow immunochromatographic assays, are simple devices intended to detect the presence (or absence) of a target analyte in sample (matrix) without the need for specialized and costly equipment, though many laboratory-based applications exist that are supported by reading equipment. Typically, these tests are used as low resources medical diagnostics, either for home testing, point of care testing, or laboratory use. A widely spread and well-known application is the home pregnancy test.

The technology is based on a series of capillary beds, such as pieces of porous paper or sintered polymer. Each of these elements has the capacity to transport fluid (e.g., urine) spontaneously. The first element (the sample pad) acts as a sponge and holds an excess of sample fluid. Once soaked, the fluid migrates to the second element (conjugate pad) in which the manufacturer has stored the so-called conjugate, a dried format of bio-active particles (see below) in a salt-sugar matrix that contains everything to guarantee an optimized chemical reaction between the target molecule (e.g., an antigen) and its chemical partner (e.g., antibody) that has been immobilized on the particle's surface. While the sample fluid dissolves the salt-sugar matrix, it also dissolves the particles and in one combined transport action the sample and conjugate mix while flowing through the porous structure. In this way, the analyte binds to the particles while migrating further through the third capillary bed. This material has one or more areas (often called stripes) where a third molecule has been immobilized by the manufacturer. By the time the sample-conjugate mix reaches these strips, analyte has been bound on the particle and the third 'capture' molecule binds the complex. After a while, when more and more fluid has passed the stripes, particles accumulate and the stripe-area changes color. Typically there are at least two stripes: one (the control) that captures any particle and thereby shows that reaction conditions and technology worked fine, the second contains a specific capture molecule and only captures those particles onto which an analyte molecule has been immobilized. After passing these reaction zones, the fluid enters the final porous material—the wick—that simply acts as a waste container. Lateral Flow Tests can operate as either competitive or sandwich assays. Lateral flow assays are disclosed in U.S. Pat. No. 6,485,982.

D. Immunohistochemistry

The antibodies of the present disclosure may also be used in conjunction with both fresh-frozen and/or formalin-fixed, paraffin-embedded tissue blocks prepared for study by immunohistochemistry (IHC). The method of preparing tissue blocks from these particulate specimens has been successfully used in previous IHC studies of various prognostic factors and is well known to those of skill in the art (Brown et al., 1990; Abbondanzo et al., 1990; Allred et al., 1990).

Briefly, frozen-sections may be prepared by rehydrating 50 ng of frozen "pulverized" tissue at room temperature in phosphate buffered saline (PBS) in small plastic capsules;

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pelletting the particles by centrifugation; resuspending them in a viscous embedding medium (OCT); inverting the capsule and/or pelletting again by centrifugation; snap-freezing in -70°C . isopentane; cutting the plastic capsule and/or removing the frozen cylinder of tissue; securing the tissue cylinder on a cryostat microtome chuck; and/or cutting 25-50 serial sections from the capsule. Alternatively, whole frozen tissue samples may be used for serial section cuttings.

Permanent-sections may be prepared by a similar method involving rehydration of the 50 mg sample in a plastic microfuge tube; pelletting; resuspending in 10% formalin for 4 hours fixation; washing/pelletting; resuspending in warm 2.5% agar; pelletting; cooling in ice water to harden the agar; removing the tissue/agar block from the tube; infiltrating and/or embedding the block in paraffin; and/or cutting up to 50 serial permanent sections. Again, whole tissue samples may be substituted.

E. Immunodetection Kits

In still further embodiments, the present disclosure concerns immunodetection kits for use with the immunodetection methods described above. As the antibodies may be used to detect EV-D68 or EV-D68 antigens, the antibodies may be included in the kit. The immunodetection kits will thus comprise, in suitable container means, a first antibody that binds to EV-D68 or EV-D68 antigen, and optionally an immunodetection reagent.

In certain embodiments, the EV-D68 antibody may be pre-bound to a solid support, such as a column matrix and/or well of a microtiter plate. The immunodetection reagents of the kit may take any one of a variety of forms, including those detectable labels that are associated with or linked to the given antibody. Detectable labels that are associated with or attached to a secondary binding ligand are also contemplated. Exemplary secondary ligands are those secondary antibodies that have binding affinity for the first antibody.

Further suitable immunodetection reagents for use in the present kits include the two-component reagent that comprises a secondary antibody that has binding affinity for the first antibody, along with a third antibody that has binding affinity for the second antibody, the third antibody being linked to a detectable label. As noted above, a number of exemplary labels are known in the art and all such labels may be employed in connection with the present disclosure.

The kits may further comprise a suitably aliquoted composition of the EV-D68 or EV-D68 antigens, whether labeled or unlabeled, as may be used to prepare a standard curve for a detection assay. The kits may contain antibody-label conjugates either in fully conjugated form, in the form of intermediates, or as separate moieties to be conjugated by the user of the kit. The components of the kits may be packaged either in aqueous media or in lyophilized form.

The container means of the kits will generally include at least one vial, test tube, flask, bottle, syringe or other container means, into which the antibody may be placed, or preferably, suitably aliquoted. The kits of the present disclosure will also typically include a means for containing the antibody, antigen, and any other reagent containers in close confinement for commercial sale. Such containers may include injection or blow-molded plastic containers into which the desired vials are retained.

F. Vaccine and Antigen Quality Control Assays

The present disclosure also contemplates the use of antibodies and antibody fragments as described herein for use in assessing the antigenic integrity of a viral antigen in a sample. Biological medicinal products like vaccines differ from chemical drugs in that they cannot normally be characterized molecularly; antibodies are large molecules of

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significant complexity and have the capacity to vary widely from preparation to preparation. They are also administered to healthy individuals, including children at the start of their lives, and thus a strong emphasis must be placed on their quality to ensure, to the greatest extent possible, that they are efficacious in preventing or treating life-threatening disease, without themselves causing harm.

The increasing globalization in the production and distribution of vaccines has opened new possibilities to better manage public health concerns but has also raised questions about the equivalence and interchangeability of vaccines procured across a variety of sources. International standardization of starting materials, of production and quality control testing, and the setting of high expectations for regulatory oversight on the way these products are manufactured and used, have thus been the cornerstone for continued success. But it remains a field in constant change, and continuous technical advances in the field offer a promise of developing potent new weapons against the oldest public health threats, as well as new ones—malaria, pandemic influenza, and HIV, to name a few—but also put a great pressure on manufacturers, regulatory authorities, and the wider medical community to ensure that products continue to meet the highest standards of quality attainable.

Thus, one may obtain an antigen or vaccine from any source or at any point during a manufacturing process. The quality control processes may therefore begin with preparing a sample for an immunoassay that identifies binding of an antibody or fragment disclosed herein to a viral antigen. Such immunoassays are disclosed elsewhere in this document, and any of these may be used to assess the structural/antigenic integrity of the antigen. Standards for finding the sample to contain acceptable amounts of antigenically correct and intact antigen may be established by regulatory agencies.

Another important embodiment where antigen integrity is assessed is in determining shelf-life and storage stability. Most medicines, including vaccines, can deteriorate over time. Therefore, it is critical to determine whether, over time, the degree to which an antigen, such as in a vaccine, degrades or destabilizes such that it is no longer antigenic and/or capable of generating an immune response when administered to a subject. Again, standards for finding the sample to contain acceptable amounts of antigenically intact antigen may be established by regulatory agencies.

In certain embodiments, viral antigens may contain more than one protective epitope. In these cases, it may prove useful to employ assays that look at the binding of more than one antibody, such as 2, 3, 4, 5 or even more antibodies. These antibodies bind to closely related epitopes, such that they are adjacent or even overlap each other. On the other hand, they may represent distinct epitopes from disparate parts of the antigen. By examining the integrity of multiple epitopes, a more complete picture of the antigen's overall integrity, and hence ability to generate a protective immune response, may be determined.

Antibodies and fragments thereof as described in the present disclosure may also be used in a kit for monitoring the efficacy of vaccination procedures by detecting the presence of protective EV-D68 antibodies. Antibodies, antibody fragment, or variants and derivatives thereof, as described in the present disclosure may also be used in a kit for monitoring vaccine manufacture with the desired immunogenicity.

VI. EXAMPLES

The following examples are included to demonstrate preferred embodiments. It should be appreciated by those of

skill in the art that the techniques disclosed in the examples that follow represent techniques discovered by the inventor to function well in the practice of embodiments, and thus can be considered to constitute preferred modes for its practice. However, those of skill in the art should, in light of the present disclosure, appreciate that many changes can be made in the specific embodiments which are disclosed and still obtain a like or similar result without departing from the spirit and scope of the disclosure.

Example 1—Materials and Methods

Animals. Four-week-old male and female AG129 mice from a specific-pathogen-free colony maintained at the Utah Science Technology and Research (USTAR) building at Utah State University. The mice were bred and maintained on irradiated Teklad Rodent Diet (Harlan Teklad) and autoclaved tap water at the USTAR building of Utah State University.

Antibodies and Compound. The monoclonal antibody (mAb) EV-D68-228 was provided by James Crowe at Vanderbilt University Medical Center. EV-D68-228 was provided in solution at a concentration of 1.134 mg/ml and was diluted in sterile saline to doses of 10, 3, and 1 mg/kg for treatment. rRSV-90 was provided in solution at a concentration of 5 mg/ml and was used as a negative control antibody at a dose of 10 mg/kg. Intravenous immunoglobulin (IVIg, Carimune, CSL Behring, King of Prussia, PA) was purchased from a local pharmacy and was used as a comparator to the EV-D68-228 mAb. Guanidine HCl (guanidine) was obtained from Sigma-Aldrich (St. Louis, MO) and served as a positive control.

Virus. Enterovirus D68 was obtained from BEI Resources, NIAID, NIH: Enterovirus D68, US/MO/14-18949, NR-49130. The virus was serially passaged 30 times in the lungs of 4-week-old AG129 mice and then plaque-purified three times in Rhabdomyosarcoma (RD) cells obtained from the American Type Culture Collection (Manassas, VA). The resulting virus stock was amplified twice in RD cells to create a working stock. The virus used for infection was designated EV-D68 MP30 PP.

Experiment design. A total of 78 mice were randomized into 6 groups of 12 mice each with a group of 6 mice used for normal controls as shown in Table I. Mice were treated via intraperitoneal (IP) administration of EV-D68-228 mAb,

IVIg, or placebo mAb 24 hours prior to infection. Mice were infected via intranasal (IN) instillation of $1 \times 10^{4.5}$ CCID₅₀ of EV-D68 MP30 PP in a 90 μ l volume of MEM. Treatment with guanidine started 4 hours post-infection and continued twice daily for 5 days. Mice were weighed prior to treatment and daily thereafter. Four mice from each treatment group were euthanized on days 1, 3, and 5 post-infection for evaluation of lung virus titers, blood virus titers, and lung cytokine concentrations.

Lung Cytokine/Chemokine Evaluations. Each sample of lung homogenate was tested for cytokines and chemokines using a chemiluminescent ELISA-based assay according to the manufacturer's instructions (Quansys Biosciences Q-Plex™ Array, Logan, UT). The Quansys multiplex ELISA is a quantitative test in which 16 distinct capture antibodies have been applied to each well of a 96-well plate in a defined array. Each sample supernatant was tested at for the following: IL-1 α , IL-1 β , IL-2, IL-3, IL-4, IL-5, IL-6, IL-10, IL-12p70, IL-17, MCP-1, IFN- γ , TNF α , MIP-1 α , GM-CSF, and RANTES. Definitions of abbreviations are: IL—interleukin; MCP—monocyte chemoattractant protein; IFN—interferon; TNF—tumor necrosis factor, MIP—macrophage inflammatory protein; GM-CSF—granulocyte/macrophage colony stimulating factor; and RANTES—regulated upon activation, normal T cell expressed and secreted.

Statistical analysis. All figures and statistical analyses were completed using Prism 8.0.2. (GraphPad Software Inc.). For each day post-infection, lung and blood virus titers from treated groups were compared to lung and blood titers from placebo-treated mice using a one-way analysis of variance (ANOVA). For each cytokine/chemokine, the concentrations from treated mice were compared to placebo-treated mice using a two-way ANOVA.

Ethics regulation of laboratory animals. This study was conducted in accordance with the approval of the Institutional Animal Care and Use Committee of Utah State University dated Mar. 2, 2019 (expires Mar. 1, 2022). The work was done in the AAALAC-accredited Laboratory Animal Research Center of Utah State University. The U. S. Government (National Institutes of Health) approval was renewed Mar. 9, 2018 (PHS Assurance No. D16-00468 [A3801-01]) in accordance to the National Institutes of Health Guide for the Care and Use of Laboratory Animals (Revision; 2011).

TABLE I

Experimental Design to Text Efficacy of EV-D68-228 for treatment of an EV-D68 respiratory infection in mice							
Number/ Cage	Group No.	Infected	Compound	Dosage	Route	Treatment Schedule	Observations
12	1	Yes	rRSV-90 (Placebo)	10 mg/kg	IP	Once, 24 hours pre-infection	4 mice per group euthanized at days 1, 3, and 5 post-infection for lung virus titers, blood virus titers, and lung cytokines.
12	3	Yes	EV-D68-228	10 mg/kg			
12	5	Yes	EV-D68-228	3 mg/kg			
12	7	Yes	EV-D68-228	1 mg/kg			
12	9	Yes	IVIg (Carimune NF)	10 mg/kg			
12	11	Yes	Guanidine	100 mg/kg/day		b.i.d. x 5 beginning 4 hours post-infection	
6	2	No	Normal Controls	—	—	—	2 mice per group euthanized at days 1, 3, and 5 post-infection for lung cytokines.

The objective of this study was to determine the efficacy of treatment with EV-D68-228 for an Enterovirus D68 (EV-D68) respiratory infection in four-week-old AG129 mice. This study determined the efficacy of an EV-D68-228 mAb for treatment of an EV-D68 respiratory infection in four-week-old AG129 mice.

FIGS. 6A-C show lung virus titers for EV-D68-infected AG129 mice treated with EV-D68-228. No lung virus titers were detected at days 1, 3, or 5 post-infection in mice treated with doses of 10, 3, or 1 mg/kg of EV-D68-228. Treatment with a dose of 10 mg/kg of IVIg significantly reduced lung virus titers on day 1 post-infection, but not on days 3 or 5 post-infection. Guanidine treatment at a dose of 100 mg/kg/day significantly reduced lung virus titers on days 1 and 3 post-infection, but not on day 5 post-infection.

FIGS. 7A-C show blood virus titers for EV-D68-infected AG129 mice treated with EV-D68-228. No blood virus titers were detected at days 1, 3, or 5 post-infection in mice treated with doses of 10, 3, or 1 mg/kg of EV-D68-228. Treatment with IVIg at a dose of 10 mg/kg also reduced blood virus titers at days 1, 3, and 5 post-infection. Guanidine significantly reduced blood virus titers on days 1 and 5 but not day 3 post-infection.

FIG. 8 shows lung concentrations of IL-1 α and IL-1 β from EV-D68-infected AG129 mice treated with EV-D68-228. Treatment with EV-D68-228 significantly reduced concentrations of IL-1 α and IL-1 β on day 3 post-infection. Treatment with IVIg or guanidine did not significantly reduce lung concentrations of IL-1 α or IL-1 β .

Lung concentrations of IL-6 and MCP-1 from EV-D68-infected AG129 mice treated with EV-D68-228 are shown in FIG. 9. Treatment with EV-D68-228 significantly reduced concentrations of IL-6 and MCP-1 on day 3 post-infection. Treatment with IVIg or guanidine significantly reduced lung concentrations of MCP-1 but not IL-6 on day 3 post-infection.

Lung concentrations of RANTES from EV-D68-infected AG129 mice treated with EV-D68-228 are shown in FIG. 10. Treatment with EV-D68-228 significantly reduced concentrations of RANTES on days 3 and 5 post-infection compared to placebo-treated mice. Treatment with IVIg or guanidine did not significantly reduce lung concentrations of RANTES at days 1, 3, or 5 post-infection.

FIG. 11 shows lung concentrations of IL-2, IL-3, and IL-4 from EV-D68-infected AG129 mice treated with EV-D68-228. No significant changes in concentrations of IL-2, IL-3, or IL-4 were observed post-infection with EV-D68.

FIG. 12 shows lung concentrations of IL-5, IL-10, and IL-12p70 from EV-D68-infected AG129 mice treated with EV-D68-228. No significant changes in concentrations of IL-5, IL-10, or IL-12p70 were observed post-infection with EV-D68.

Lung concentrations of IL-17, IFN γ , and TNF α from EV-D68-infected AG129 mice treated with EV-D68-228 are shown in FIG. 13. No significant changes in concentrations of IL-17, IFN γ , or TNF α were observed post-infection with EV-D68.

Lung concentrations of MIP-1 α and GM-CSF from EV-D68-infected AG129 mice treated with EV-D68-228 are shown in FIG. 14. No significant changes in concentrations of MIP-1 α or GM-CSF were observed post-infection with EV-D68.

Example 3—Discussion

This study determined the efficacy of EV-D68-228, a mAb against EV-D68, for treatment of a respiratory infection caused by EV-D68 in four-week-old AG129 mice.

Doses of 10, 3, and 1 mg/kg of EV-D68-228 were highly effective at reducing lung virus titers, blood virus titers, and pro-inflammatory cytokines caused by EV-D68 infection. Lung concentrations of IL-1 α , IL-1 β , IL-6, MCP-1, and RANTES were significantly reduced by treatment with EV-D68-228.

EV-D68-228 was highly effective at preventing infection with EV-D68.

Example 4—Materials and Methods

Animals. Ten-day-old AG129 mice from a specific-pathogen-free colony maintained at the Utah Science Technology and Research (USTAR) building at Utah State University. The mice were bred and maintained on irradiated Teklad Rodent Diet (Harlan Teklad) and autoclaved tap water at the USTAR building of Utah State University.

Antibodies and Compound. The monoclonal antibody (mAb) EV-D68-228 was provided by James Crowe at Vanderbilt University Medical Center. EV-D68-228 was provided in solution at a concentration of 1.134 mg/ml and was diluted in sterile saline to doses of 10, 3, and 1 mg/kg for treatment. rRSV-90 was provided in solution at a concentration of 5 mg/ml and was used as a negative control antibody at a dose of 10 mg/kg. Intravenous immunoglobulin (IVIg, Carimune, CSL Behring, King of Prussia, PA) was purchased from a local pharmacy and was used as a comparator to the EV-D68-228 mAb. Guanidine HCl (guanidine) was obtained from Sigma-Aldrich (St. Louis, MO) and served as a positive control.

Virus. Enterovirus D68 was obtained from BEI Resources, NIAID, NIH: Enterovirus D68, US/MO/14-18949, NR-49130. The virus was serially passaged 30 times in the lungs of 4-week-old AG129 mice and then plaque-purified three times in Rhabdomyosarcoma (RD) cells obtained from the American Type Culture Collection (Manassas, VA). The resulting virus stock was amplified twice in RD cells to create a working stock. The virus used for infection was designated EV-D68 MP30 PP.

Experiment design. A total of 42 mice were randomized into 7 groups of 6 mice each as shown in Table II. Mice were treated via intraperitoneal (IP) administration of EV-D68-228 mAb, IVIg, or placebo mAb 24 hours prior to infection. Mice were infected via intraperitoneal (IP) administration of $1 \times 10^{6.5}$ CCID₅₀ of EV-D68 MP30 PP in a 100 μ l volume of MEM. Treatment with guanidine started 4 hours post-infection and continued twice daily for 5 days. Mice were weighed prior to treatment and daily thereafter. All mice were observed daily for morbidity, mortality, and neurological scores through day 21. Neurological scores (NS) were recorded as follows: NS0—no observable paralysis, NS1—abnormal splay of hindlimb but normal or slightly slower gait, NS2—hindlimb partially collapsed and foot drags during use for forward motion, NS3—rigid paralysis of hindlimb and hindlimb is not used for forward motion, NS4—rigid paralysis in hindlimbs and no forward motion. Any animals observed with a score of NS4 were humanely euthanized.

Statistical analysis. Kaplan-Meier survival curves were generated Prism 8.0.2. (GraphPad Software Inc.). Survival curves were compared using the Log-rank (Mantel-Cox) test followed by a Gehan-Breslow-Wilcoxon test. For each day post-infection, blood virus titers from treated groups were compared to lung and blood titers from placebo-treated mice using a one-way analysis of variance (ANOVA). Mean body weights were compared using a one-way ANOVA. Neuro-

logical scores were compared using a Kruskal-Wallis test followed by a Dunn's multiple comparisons test.

Ethics regulation of laboratory animals. This study was conducted in accordance with the approval of the Institutional Animal Care and Use Committee of Utah State University dated Mar. 2, 2019 (expires Mar. 1, 2022). The work was done in the AAALAC-accredited Laboratory Animal Research Center of Utah State University. The U. S. Government (National Institutes of Health) approval was renewed Mar. 9, 2018 (PHS Assurance No. D16-00468 [A3801-01]) in accordance to the National Institutes of Health Guide for the Care and Use of Laboratory Animals (Revision; 2011).

Neurological scores in 10-day-old AG129 mice infected with EV-D68 and treated with EV-D68-228, IVIg, or guanidine are shown in FIG. 18. No neurological scores were observed in mice treated with doses of 10, 3, or 1 mg/kg of the EV-D68-228 Ab. Neurological scores were observed in mice treated with IVIg at a dose of 10 mg/kg on days 4, 5, and 6 post-infection. However, neurological scores in mice treated with IVIg were significantly reduced compared to placebo-treated mice. No neurological scores were observed in the mice treated with guanidine at a dose of 100 mg/kg/day.

TABLE II

Experimental Design to Test Efficacy of EV-D68-228 for treatment of an EV-D68 neurological infection in mice							
Number/ Cage	Group No.	Infected	Compound	Dosage	Route	Treatment Schedule	Observations
6	1	Yes	rRSV-90 (Placebo)	10 mg/kg	IP	Once, 24 hours pre-infection	Mice observed daily for survival, body weights, and neurological scores.
6	3	Yes	EV-D68-228	10 mg/kg			Blood collected from 3 mice per group on days 1, 3, and 5 post- infection for blood virus titers. Observed for normal weight gain.
6	5	Yes	EV-D68-228	3 mg/kg			
6	7	Yes	EV-D68-228	1 mg/kg			
6	9	Yes	IVIg (Carimune NF)	10 mg/kg			
6	11	Yes	Guanidine	100 mg/kg/day		b.i.d. x 5 beginning 4 hours post-infection	
6	2	No	Normal Controls	—	—	—	

Example 5—Results

The objective of this study was to determine the efficacy of treatment with EV-D68-228 for an Enterovirus D68 (EV-D68) neurological infection in 10-day-old AG129 mice. This study determined the efficacy of an EV-D68-228 mAb for treatment of an EV-D68 neurological infection in 10-day-old AG129 mice.

FIG. 15 shows Kaplan-Meier survival curves for 10-day-old AG129 mice infected with EV-D68 and treated with EV-D68-228, IVIg, or guanidine. Doses of 10, 3, or 1 mg/kg of EV-D68-228 provided 100% protection from mortality in each group of six mice infected with EV-D68. Two of six mice that were treated with IVIg at a dose of 10 mg/kg did not survive the infection. Treatment with guanidine at a dose of 100 mg/kg/day protected all six mice from mortality.

FIG. 16 shows mean body weights for 10-day-old AG129 mice infected with EV-D68 and treated with EV-D68-228, IVIg, or guanidine. Doses of 10, 3, or 1 mg/kg protected mice from infection-associated weight loss. A dose of 10 mg/kg of IVIg also protected mice from weight loss. In addition, guanidine at a dose of 100 mg/kg/day protected mice from weight loss.

FIGS. 17A-C show blood virus titers for EV-D68-infected AG129 mice treated with EV-D68-228. No blood virus titers were detected at days 1, 3, or 5 post-infection in mice treated with doses of 10, 3, or 1 mg/kg of EV-D68-228. Treatment with IVIg at a dose of 10 mg/kg also reduced blood virus titers at days 1, 3, and 5 post-infection. Guanidine treatment at a dose of 100 mg/kg/day significantly reduced blood virus titers on days 1, 3, and 5 post-infection.

Example 6—Discussion

This study determined the efficacy of EV-D68-228, a mAb against EV-D68, for treatment of a neurological infection caused by EV-D68 in 10-day-old AG129 mice.

Doses of 10, 3, and 1 mg/kg of EV-D68-228 provided protection from mortality and weight loss in EV-D68-infected mice. In addition, no blood virus titers were detected at days 1, 3, or 5 post-infection in mice treated with EV-D68-228. No paralysis, as measured by neurological scores, was observed in mice treated with doses of 10, 3, or 1 mg/kg of EV-D68-228.

EV-D68-228 appeared to provide protection from EV-D68 infection compared to commercial IVIg as indicated by survival, blood virus titers, and neurological scores observed in mice treated with IVIg at a dose of 10 mg/kg.

EV-D68-228 provided equivalent protection from mortality and paralysis when compared to treatment with the positive control, guanidine. However, while blood virus titers were still detected in guanidine-treated mice on day 3 post-infection, no virus was detected in the blood of mice treated with EV-D68-228 at doses of 10, 3, or 1 mg/kg at days 1, 3, or 5 post-infection.

Example 7—Efficacy of EV-D68-228 on Respiratory Infection in Mice

Materials and Methods

Animals. Four-week-old male and female AG129 mice from a specific-pathogen-free colony maintained at the Utah Science Technology and Research (USTAR) building at Utah State University. The mice were bred and maintained

on irradiated Teklad Rodent Diet (Harlan Teklad) and autoclaved tap water at the USTAR building of Utah State University.

Antibodies and Compound. The monoclonal antibody (mAb) EV-D68-228 was provided by James Crowe at Vanderbilt University Medical Center. EV-D68-228 was provided in solution at a concentration of 1.134 mg/ml and was diluted in sterile saline to doses of 10 or 1 mg/kg for treatment. rRSV-90 was provided in solution at a concentration of 5 mg/ml and was used as a negative control antibody at a dose of 10 mg/kg. Intravenous immunoglobulin (IVIg, Carimune, CSL Behring, King of Prussia, PA) was purchased from a local pharmacy and was used as a comparator to the EV-D68-228 mAb.

Virus. Enterovirus D68 was obtained from BEI Resources, NIAID, NIH: Enterovirus D68, US/MO/14-18949, NR-49130. The virus was serially passaged 30 times in the lungs of 4-week-old AG129 mice and then plaque-purified three times in Rhabdomyosarcoma (RD) cells obtained from the American Type Culture Collection (Manassas, VA). The resulting virus stock was amplified twice in RD cells to create a working stock. The virus used for infection was designated EV-D68 MP30 PP.

Experiment design. A total of 52 mice were randomized into 6 groups of 8 mice each with a group of 4 mice used for normal controls as shown in Table C. Mice were treated via

IL-10, IL-12p70, IL-17, MCP-1, IFN- γ , TNF α , MIP-1 α , GM-CSF, and RANTES. Definition of abbreviations are: IL—interleukin; MCP—monocyte chemoattractant protein; IFN—interferon; TNF—tumor necrosis factor; MIP—macrophage inflammatory 3 protein; GM-CSF—granulocyte/macrophage colony stimulating factor; and RANTES—regulated upon activation, normal T cell expressed and secreted.

Statistical analysis. All figures and statistical analyses were completed using Prism 8.2.0. (GraphPad Software Inc.). For each day post-infection, lung and blood virus titers from treated groups were compared to lung and blood titers from placebo-treated mice using a one-way analysis of variance (ANOVA). For each cytokine/chemokine, the concentrations from treated mice were compared to placebo-treated mice using a two-way ANOVA.

Ethics regulation of laboratory animals. This study was conducted in accordance with the approval of the Institutional Animal Care and Use Committee of Utah State University. The work was done in the AAALAC-accredited Laboratory Animal Research Center of Utah State University. The U.S. Government (National Institutes of Health) approval was renewed (PHS Assurance No. D16-00468 [A3801-01]) in accordance to the National Institutes of Health Guide for the Care and Use of Laboratory Animals (Revision; 2011).

TABLE C

Expt. NIA-1869. Experimental Design - Therapeutic Efficacy of EV-D68-228 for treatment of an EV-D68 respiratory infection in mice.								
Number/ Cage	Group No.	Infected	Compound	Dosage	Route	Treatment Schedule	Observations	
12	1	Yes	rRSV-90 (Placebo)	10 mg/kg	IP	Once, 24 hours post-infection	4 mice per group euthanized at days 3 and 5	
12	3	Yes	EV-D68-228	10 mg/kg		Once, 4 hours post-infection		
12	5	Yes	EV-D68-228	10 mg/kg		Once, 24 hours post-infection	lung virus titers, blood virus titers, and lung cytokines.	
12	7	Yes	EV-D68-228	10 mg/kg		Once, 48 hours post-infection		
12	9	Yes	EV-D68-228	1 mg/kg		Once, 24 hours post-infection		
12	11	Yes	IVIg (Carimune NF)	10 mg/kg				
4	2	No	Normal Controls	—	—	—	2 mice per group euthanized at days 3 and 5 post-infection for lung cytokines.	

intraperitoneal (IP) administration of EV-D68-228 mAb, IVIg, or placebo mAb at 4, 24, or 48 hours post-infection. Mice were infected via intranasal (IN) instillation of 1 $\times$ 104.5 CCID50 of EV-D68 MP30 PP in a 90 μ l volume of MEM. Mice were weighed prior to treatment and daily thereafter. Four mice from each treatment group were euthanized on days 3 and 5 post-infection for evaluation of lung virus titers, blood virus titers, and lung cytokine concentrations.

Lung Cytokine/Chemokine Evaluations. Each sample of lung homogenate was tested for cytokines and chemokines using a chemiluminescent ELISA-based assay according to the manufacturer's instructions (Quansys Biosciences Q-Plex™ Array, Logan, UT). The Quansys multiplex ELISA is a quantitative test in which 16 distinct capture antibodies have been applied to each well of a 96-well plate in a defined array. Each sample supernatant was tested at the following: IL-1 α , IL-1 β , IL-2, IL-3, IL-4, IL-5, IL-6,

Results and Discussion

This study determined the therapeutic efficacy of an EV-D68-228 mAb for treatment of an EV-D68 respiratory infection in four-week-old AG129 mice.

FIG. 19 shows lung virus titers for EV-D68-infected AG129 mice treated post-infection with EV-D68-228. No lung virus titers were detected at days 3 or 5 post-infection in mice treated with doses of 10 mg/kg of EV-D68-228 when treatment started at 4 or 24 hours after infection. In mice treated 48 hours post-infection, a dose of 10 mg/kg of EV-D68-228 significantly reduced lung virus titers on days 3 and 5 post-infection. Even a dose of 1 mg/kg given 48 hours post-infection significantly reduced lung virus titers on days 3 and 5 post-infection. Treatment with a dose of 10 mg/kg of IVIg given 24 hours post-infection did not significantly reduce lung virus titers on days 3 or 5 post-infection.

FIG. 20 shows blood virus titers for EV-D68-infected AG129 mice treated post-infection with EV-D68-228. No blood virus titers were detected at days 3 or 5 post-infection in mice treated with doses of 10 mg/kg of EV-D68-228 at 4, 24, or 48 hours after infection. A lower dose of 1 mg/kg of EV-D68-228 given 48 hours post-infection reduced blood virus titers below the limit of detection. Treatment with IVIg at a dose of 10 mg/kg also reduced blood virus titers at days 3, and 5 post-infection.

FIG. 21 shows lung concentrations of IL-1 α and IL-1 β from EV-D68-infected AG129 mice treated post-infection with EV-D68-228. Therapeutic treatment with EV-D68-228 significantly reduced concentrations of IL-1 α and IL-1 β on days 3 and 5 post-infection. Treatment with IVIg significantly reduced lung concentrations of IL-1 α or IL-1 β , but only on day 3 post-infection.

Lung concentrations of IL-5 and IL-6 from EV-D68-infected AG129 mice treated post-infection with EV-D68-228 are shown in FIG. 224. Therapeutic treatment with EV-D68-228 significantly reduced concentrations of IL-5 and IL-6 on day 3 post-infection. Treatment with IVIg 24 hours post-infection significantly reduced lung concentrations of IL-5 and IL-6 on day 3 post-infection.

Lung concentrations of MCP-1 and RANTES from EV-D68-infected AG129 mice treated post-infection with EV-D68-228 are shown in FIG. 23. Therapeutic treatment with EV-D68-228 significantly reduced lung concentrations of MCP-1 and RANTES on days 3 and 5 post-infection compared to placebo-treated mice. Treatment with IVIg significantly reduced lung concentrations of MCP-1 on day 3 post-infection and reduced concentrations of RANTES on days 3 and 5 post-infection.

FIG. 24 shows lung concentrations of IL-2, IL-3, and IL-4 from EV-D68-infected AG129 mice treated post-infection with EV-D68-228. No significant changes in concentrations of IL-2, IL-3, or IL-4 were observed post-infection with EV-D68.

FIG. 25 shows lung concentrations of IL-10, IL-12p70, and IL-17 from EV-D68-infected AG129 mice treated post-infection with EV-D68-228. No significant changes in concentrations of IL-10, IL-12p70, or IL-17 were observed after infection with EV-D68.

Lung concentrations of IFN γ and TNF α from EV-D68-infected AG129 mice treated post-infection with EV-D68-228 are shown in FIG. 26. No significant changes in concentrations of IFN γ or TNF α were observed after infection with EV-D68.

Lung concentrations of MIP-1 α and GM-CSF from EV-D68-infected AG129 mice treated post-infection with EV-D68-228 are shown in FIG. 27. No significant changes in concentrations of MIP-1 α or GM-CSF were observed following infection with EV-D68.

Thus, this study determined the therapeutic efficacy of EV-D68-228, a mAb against EV-D68, for treatment of a respiratory infection caused by EV-D68 in four-week-old AG129 mice. Doses of 10 and 1 mg/kg of EV-D68-228 were highly effective at reducing lung virus titers, blood virus titers, and pro-inflammatory cytokines caused by EV-D68 infection when administered within 48 hours post-infection. Lung concentrations of IL-1 α , IL-1 β , IL-5, IL-6, MCP-1, and RANTES were significantly reduced by treatment with EV-D68-228. As EV-D68-228 was highly effective as a therapeutic treatment for an EV-D68 respiratory infection in AG129 mice, plethysmography could be used in future studies to determine the impact of EV-D68-228 treatment on lung function.

Example 8—Efficacy of EV-D68-228 on Respiratory Infection in Mice

Materials and Methods

The objective of this study was to determine the therapeutic efficacy of treatment with EV-D68-228 for an Enterovirus D68 (EV-D68) neurological infection in 10-day-old AG129 mice.

Animals. Ten-day-old AG129 mice from a specific-pathogen-free colony maintained at the Utah Science Technology and Research (USTAR) building at Utah State University. The mice were bred and maintained on irradiated Teklad Rodent Diet (Harlan Teklad) and autoclaved tap water at the USTAR building of Utah State University.

Antibodies and Compound. The monoclonal antibody (mAb) EV-D68-228 was provided by James Crowe at Vanderbilt University Medical Center. EV-D68-228 was provided in solution at a concentration of 1.134 mg/ml and was diluted in sterile saline to doses of 10 mg/kg for treatment. rRSV-90 was provided in solution at a concentration of 5 mg/ml and was used as a negative control antibody at a dose of 10 mg/kg. Intravenous immunoglobulin (IVIg, Carimune, CSL Behring, King of Prussia, PA) was purchased from a local pharmacy and was used as a comparator to the EV-D68-228 mAb.

Virus. Enterovirus D68 was obtained from BEI Resources, NIAID, NIH: Enterovirus D68, US/MO/14-18949, NR-49130. The virus was serially passaged 30 times in the lungs of 4-week-old AG129 mice and then plaque-purified three times in Rhabdomyosarcoma (RD) cells obtained from the American Type Culture Collection (Manassas, VA). The resulting virus stock was amplified twice in RD cells to create a working stock. The virus used for infection was designated EV-D68 MP30 PP.

Experiment design. A total of 42 mice were randomized into 6 groups of 6 treated-mice each with one group of 9 mice to account for deaths prior to treatment each as shown in Table D. One group of 3 mice was used as normal controls. Mice were infected via intraperitoneal (IP) administration of $1 \times 10^{6.5}$ CCID₅₀ of EV-D68 MP30 PP in a 100 μ l volume of MEM. Mice were treated via intraperitoneal (IP) administration of EV-D68-228 mAb at 24, 48, 72, or 120 hours post-infection. Treatment with IVIg or placebo Ab was given once 24 hours post-infection. Mice were weighed prior to treatment and daily thereafter. All mice were observed daily for morbidity, mortality, and neurological scores through day 21. Neurological scores (NS) were recorded as follows: NS0—no observable paralysis, NS1—abnormal splay of hindlimb but normal or slightly slower gait, NS2—hindlimb partially collapsed and foot drags during use for forward motion, NS3—rigid paralysis of hindlimb and hindlimb is not used for forward motion, NS4—rigid paralysis in hindlimbs and no forward motion. Any animals observed with a score of NS4 were humanely euthanized.

Statistical analysis. Kaplan-Meier survival curves were generated Prism 8.2.0. (GraphPad Software Inc.). Survival curves were compared using the Log-rank (Mantel-Cox) test followed by a Gehan-Breslow-Wilcoxon test. For each day post-infection, blood virus titers from treated groups were compared to lung and blood titers from placebo-treated mice using a one-way analysis of variance (ANOVA). Mean body weights were compared using a one-way ANOVA. Neurological scores were compared using a Kruskal-Wallis test followed by a Dunn's multiple comparisons test.

Ethics regulation of laboratory animals. This study was conducted in accordance with the approval of the Institutional Animal Care and Use Committee of Utah State University. The work was done in the AAALAC-accredited Laboratory Animal Research Center of Utah State University. The U. S. Government (National Institutes of Health) approval was renewed (PHS Assurance No. D16-00468 [A3801-01]) in accordance to the National Institutes of Health Guide for the Care and Use of Laboratory Animals (Revision; 2011).

TABLE D

Expt. NIA-1870. Experimental Design - Therapeutic efficacy of EV-D68-228 for treatment of an EV-D68 neurological infection in mice.							
Number/ Cage	Group No.	Infected	Compound	Dosage	Route	Treatment Schedule	Observations
6	1	Yes	rRSV-90 (Placebo)	10 mg/kg	IP	Once, 24 hours post-infection	Mice observed daily for survival, body weights, and neurological scores. Blood collected from 3 mice per group on days 1, 3, 5, and 7 post- infection for blood virus titers.
6	3	Yes	EV-D68-228	10 mg/kg		Once, 24 hours post-infection	
6	5	Yes	EV-D68-228			Once, 48 hours post-infection	
6	7	Yes	EV-D68-228			Once, 72 hours post-infection	
9	9	Yes	EV-D68-228)			Once, 120 hours post-infection	
6	11	Yes	IVIg (Carimune NF)	10 mg/kg		Once, 24 hours post-infections	Observed for normal weight gain.
3	2	No	Normal Controls	—	—	—	

Results and Discussion

This study determined the efficacy of an EV-D68-228 mAb for therapeutic treatment of an EV-D68 neurological infection in 10-day-old AG129 mice.

FIG. 28 shows Kaplan-Meier survival curves for 10-day-old AG129 mice infected with EV-D68 and treated post-infection with EV-D68-228. With a single administration of EV-D68-228 at a dose of 10 mg/kg, a significant survival benefit was observed in mice treated at 24, 48, and 120 hours post-infection. The difference in survival benefit in the mice treated 120 hours post-infection was due to an increase in the day of death as all seven mice that were treated still succumbed to the infection. A dose of 10 mg/kg of IVIg completely protected mice from mortality when administered 24 hours post-infection.

FIG. 29 shows mean body weights for 10-day-old AG129 mice infected with EV-D68 and treated post-infection with EV-D68-228. A single dose of 10 mg/kg of EV-D68-228 protected mice from infection-associated weight loss when administered 24 or 48 hours after infection. A single administration of IVIg administered 24 hours post-infection did not protect mice from weight loss.

FIGS. 30A-C shows blood virus titers for EV-D68-infected AG129 mice treated post-infection with EV-D68-228. A single administration of 10 mg/kg of EV-D68-228 at 24, 48, or 72 hours post-infection significantly reduced blood virus titers at days 3 and 5 post-infection. Treatment with IVIg at a dose of 10 mg/kg 24 hours after infection reduced blood virus titers at days 3 and 5 post-infection. No virus titers were detected in any of the surviving mice at day seven post-infection.

Neurological scores in 10-day-old AG129 mice infected with EV-D68 and treated post-infection with EV-D68-228 are shown in FIGS. 31A-B. Neurological scores on days 3,

4, and 5 were significantly reduced in mice treated 24 hours post-infection with EV-D68-228 at a dose of 10 mg/kg. IVIg at a dose of 10 mg/kg given 24 hours post-infection reduced neurological scores at days 3, 4, and 5 post-infection. No comparisons in neurological score were completed after day 5 post-infection due to the death of all the placebo-treated mice.

Thus, this study determined the therapeutic efficacy of EV-D68-228, a mAb against EV-D68, for treatment of a neurological infection caused by EV-D68 in 10-day-old

AG129 mice. When administered within 48 hours post-infection, a single dose of 10 mg/kg of EV-D68-228 protected mice from mortality and weight loss. A significant reduction in blood virus titers was observed in mice treated within 72 hours post-infection. A reduction in paralysis as measured by neurological scores was observed in mice treated with 10 mg/kg of EV-D68 at 24 hours post-infection. As EV-D68-228 could be used as a prophylactic therapy during an outbreak of EV-D68, determining the efficacy of EV-D68-228 when administered at various time points prior to infection would be beneficial.

Example 9—Materials and Methods

Study design. The inventors designed this study to try to identify any antibodies that humans can make in response to EV-D68 infection. Therefore, they used live virus isolates in an indirect ELISA screen to identify B cells secreting EV-D68-binding antibodies, and then electrofused those B cells with myeloma cells to create monoclonal antibody secreting hybridomas. The inventors then characterized the neutralization and binding properties of these individual mAbs in vitro using CCID₅₀, ELISA, and cryo-EM based techniques. They pursued in vivo experiments to generate pre-clinical data supporting the development of mAb EV68-228 as a prophylactic and/or therapeutic agent in humans. For this purpose, they studied the effectiveness of mAb EV68-228 at protecting mice from EV-D68 infection as compared to human IVIG, which is widely used to treat humans with AFM based on theoretical benefit, but this IVIG treatment so far has not been proven to be effective. An advantage of the AG129 murine model of infection is that the inventors could measure the effect of antibody treatment in both respiratory and neurologic models of infection.

Cell lines. RD cells (human, female origin) were obtained from the American Type Culture Collection (ATCC CCL-136). RD cells were cultured in 5% CO₂ at 37° C. in Dulbecco's Modified Eagle Medium (DMEM) (ThermoFisher Scientific) supplemented with 10% heat inactivated fetal bovine serum (HI-FBS; HyClone), 1 mM sodium pyruvate and 1% penicillin-streptomycin-amphotericin B (ThermoFisher Scientific). For structural studies, RD cells were cultured in DMEM (Sigma-Aldrich) supplemented with 10% HI-FBS (Sigma-Aldrich) and nonessential amino acids (NEAA, Life Technologies). The ExpiCHO (hamster, female origin) cell line was purchased from ThermoFisher Scientific and cultured according to the manufacturer's protocol. The HMM2.5 line is a non-secreting mouse-human heteromyeloma cell line (sex information is not available) that was generated by fusing a murine myeloma cell line with a human myeloma cell line (Posner et al., 1987). This cell line was cultured as described previously (Yu et al., 2008). All cell lines were tested on a monthly basis for *Mycoplasma* and found to be negative in all cases.

Viruses. See Table K for a list of the virus isolates used in this study. EV-D68 isolates were propagated for two generations in RD cell monolayer cultures for use in enzyme linked immunosorbent assays (ELISAs) described below. RD cell monolayers were inoculated with a given virus isolate and monitored until 70 to 90% cell death was observed. This cell culture flask then was frozen to -80° C., thawed, and the contents scraped and collected into a 50 mL conical tube. This preparation was sonicated three times for 20 s in an inverted cup sonicator at maximum power settings (Fisherbrand), vortexed for 30 s, and then sonicated two more times for 20 s. Cell debris was pelleted, and the virus-containing supernatant was spun over a 30% sucrose in PBS (w/v) cushion at 10° C. for four hrs at 100,000×g. Supernatant was discarded, and the pellet allowed to soak in 0.01% (w/v) bovine serum albumin (BSA) in NTE buffer (20 mM Tris, 120 mM NaCl, 1 mM EDTA pH 8.0) overnight at 4° C. The resuspended pellet then was clarified further by centrifugation at 10,000×g for 10 min, before storage of virus aliquots at -80° C. until ready for use.

For structural studies the US/MO/14-18947 isolate was used. Virus was passaged in RD cells and stored at -80° C. before large scale propagation. RD cells were grown to 80% confluency and were infected with EV-D68 at a multiplicity of infection of 0.01. Two days post-infection, the cells were collected together with the supernatant and spun down. The cell pellets were collected and after multiple freeze/thaw cycles spun down to remove cell debris. All supernatants were combined and pelleted at 210,000×g for 2 hours. The pellets were incubated and resuspended in 250 mM HEPES (pH=7.5), 250 mM NaCl buffer, then supplemented with final concentration 5 mM MgCl₂, 0.01 mg/mL DNase (Sigma-Aldrich), 0.8 mg/mL trypsin, 15 mM EDTA and 1% (w/v) n-lauryl-sarcosine. The sample was then pelleted at 210,000×g for 2 hours, resuspended, and loaded onto a potassium tartrate gradient (10 to 40%, w/v) for the last round of ultracentrifugation at 160,000×g for 2 hours. The purified virus sample, which was observed as a blue band in the middle of the tube, was extracted and buffered exchanged into 20 mM Tris, 120 mM NaCl, 1 mM EDTA (pH=8.0) buffer to remove potassium tartrate.

Detection of virus load by CCID₅₀ assay. Titration of virus stocks or virus in murine blood or lung samples was performed by CCID₅₀ assay in RD cell culture monolayers. Briefly, increasing 10-fold dilutions of the samples were applied to RD cell monolayers in triplicate wells (50 µL) of a 96 well plate, incubated for five days in 5% CO₂ at 33° C.,

and then fixed with 1% paraformaldehyde and stained with crystal violet. Wells with any cytopathic effect were scored as positive for virus, and titers were determined using a formula based on the Spearman-Kaerber equation (Ramakrishnan, 2016); the limit of detection was 136 CCID₅₀/mL.

Virus neutralization assay. Virus neutralization assays were performed in a CCID₅₀ format using the indicated viruses, essentially as described previously for poliovirus (Weldon et al., 2016). Virus was incubated with increasing concentrations of mAb in duplicate for one hr at 33° C., then each suspension was added to a monolayer of RD cells in technical quadruplicate wells (50 µL) of a 96-well plate. After five days incubation in 5% CO₂ at 33° C., cells were fixed with 1% paraformaldehyde and stained with crystal violet. Wells with any cytopathic effect were scored as positive for virus, and half maximal inhibitory concentrations (IC₅₀) were determined using a formula based on the Spearman-Kaerber equation (Ramakrishnan, 2016); the limits of detection were 57 µg/mL to 4.8 pg/mL.

Mouse models. Ten-day old (neurologic model) or four-week old (respiratory model) male and female AG129 mice (deficient in receptors for interferon α/β and γ) were obtained from a specific-pathogen-free colony maintained at the Utah Science Technology and Research (USTAR) building at Utah State University. The mice were bred and maintained on irradiated Teklad Rodent Diet (Harlan Teklad) and autoclaved tap water at the US TAR building of Utah State University. This study was conducted in accordance with the approval of the Institutional Animal Care and Use Committee of Utah State University dated Mar. 2, 2019 (expires Mar. 1, 2022). The work was done in the AAALAC-accredited Laboratory Animal Research Center of Utah State University. The U. S. Government (National Institutes of Health) approval was renewed Mar. 9, 2018 (PHS Assurance No. D16-00468[A3801-01]) in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals (Revision; 2011).

Antibody and control treatments were diluted in PBS and administered by intraperitoneal injection at indicated time points before or after EV-D68 inoculation. Guanidine HCl (Sigma-Aldrich) served as a positive control for treatment (Hurst et al., 2019), started 4 hours post-infection and continued twice daily for 5 days. A suspension of mouse adapted EV-D68 was administered by intraperitoneal injection (neurologic model; 10^{6.5} CCID₅₀ in a 100 µL volume of MEM) or intranasal instillation (respiratory model; 10^{4.5} CCID₅₀ in a 90 µL volume of MEM). Mice were weighed prior to treatment and daily thereafter. Mice were euthanized humanely at indicated time points post-infection for measurement of lung virus titers, blood virus titers, or lung cytokine concentrations, as indicated. For the neurologic model, all mice were observed daily for morbidity, mortality, and neurological scores through day 21. Neurological scores (NS) were recorded as follows: NS0—no observable paralysis, NS1—abnormal splay of hindlimb but normal or slightly slower gait, NS2—hindlimb partially collapsed and foot drags during use for forward motion, NS3—rigid paralysis of hindlimb and hindlimb is not used for forward motion, NS4—rigid paralysis in hindlimbs and no forward motion. Any animals observed with a score of NS4 were euthanized humanely.

Lung cytokine/chemokine evaluations. Each sample of lung homogenate was tested for cytokines and chemokines using quantitative chemiluminescent ELISA-based assays according to the manufacturer's instructions (Quansys Biosciences Q-Plex™ Array, Logan, UT). The Quansys multi-

plex ELISA is a quantitative test in which 16 distinct capture antibodies are applied to each well of a 96-well plate in a defined array.

Generation of human monoclonal antibodies (mAbs). Subjects were identified from the Childhood Onset of Asthma (COAST) birth cohort (Lemanske, 2002) who had laboratory documented EV-D68 upper respiratory tract infections (Bochkov et al., 2014). After written informed consent was obtained, peripheral blood was collected and stored at room temperature until peripheral blood mononuclear cells (PBMCs) could be purified using SepMate tubes (Stemcell Technologies), per the manufacturer's protocol, and then cryopreserved in 10% (v/v) dimethyl sulfoxide (DMSO) in fetal bovine serum (FBS) and stored in the vapor phase of liquid nitrogen. Lymphoblastoid cell lines (LCLs) were generated as described previously (Smith and Crowe, 2015) from memory B cells within the PBMCs by mixing with Epstein-Barr virus, cell cycle checkpoint kinase 2 (chk2) inhibitor (Sigma-Aldrich), CpG (Sigma-Aldrich), and cyclosporin A (Sigma-Aldrich) in Medium A (STEM-CELL Technologies). One week later, LCLs were counted and then expanded on a feeder layer of γ -irradiated, human PBMCs from an unrelated donor. In one more week, LCL supernatants were screened for the presence of EV-D68-reactive IgG by indirect ELISA using live EV-D68 virus as the antigen, comprising cell culture grown EV-D68 virus generated from a 2014 clinical isolate. LCLs from wells containing virus-reactive antibodies were fused to HMM2.5 myeloma cells by electrofusion, as previously described (Yu et al., 2008). After the fusion reaction, hybridoma lines were cultured in a selection medium containing HAT media supplements (Sigma-Aldrich) and ouabain (Sigma-Aldrich) in 384-well plates before screening of supernatants for antibody production. Two weeks later, supernatants from the resulting hybridoma cell lines were screened by indirect ELISA with live virus as antigen, and cell lines from wells with EV-D68-reactive antibodies were expanded in culture and then cloned by single-cell flow cytometric sorting into 384-well cell culture plates. These cloned cells were expanded in Medium E until about 50% confluent in 12-well tissue culture treated plates (Corning) and their supernatants screened for virus binding by ELISA. Wells with the highest signal in ELISA were selected as the mAb-producing hybridoma cell lines for further use.

mAb isotype and gene sequence analysis. The isotype and subclass of secreted antibodies were determined using mouse anti-human IgG1, IgG2, IgG3 or IgG4 antibodies conjugated with horseradish peroxidase (Southern Biotech). Antibody heavy- and light-chain variable region genes were sequenced from RNA obtained from hybridoma lines that had been cloned biologically by flow cytometric sorting. Total RNA was extracted using the RNeasy Mini kit (Qiagen). A modified 5' RACE (Rapid Amplification of cDNA Ends) approach was used (Turchaninova et al., 2016). Briefly, 5 μ L total RNA was mixed with cDNA synthesis primer mix (10 μ M each) and incubated for 2 min at 70° C. and then decrease the incubation temperature to 42° C. to anneal the synthesis primers (1 to 3 min). After incubation, a mixture containing 5 $\times$ first-strand buffer (Clontech), DTT (20 mM), 5' template switch oligo (10 μ M), dNTP solution (10 mM each) and 10 $\times$ SMARTScribe Reverse Transcriptase (Clontech) was added to the primer-annealed total RNA reaction and incubated for 60 min at 42° C. The first-strand synthesis reaction was purified using the Ampure Size Select Magnetic Bead Kit at a ratio of 1.8 $\times$ (Beckman Coulter). Following, a single PCR amplification reaction containing 5 μ L first-strand cDNA, 2 $\times$ Q5 High Fidelity Mastermix

(NEB), dNTP (10 mM each), forward universal primer (10 μ M) and reverse primer mix (0.2 μ M each in heavy-chain mix, 0.2 μ M each in light-chain mix) were subjected to thermal cycling with the following conditions: initial denaturation for 90 s followed by 30 cycles of denaturation at 98° C. for 10 s, annealing at 60° C. for 20 s, and extension at 72° C. for 40 s, followed by a final extension step at 72° C. for 4 min. All primer sequences used in this protocol were described previously ((Turchaninova et al., 2016). The first PCR reaction was purified using the AMPure Size Select Magnetic Bead Kit at a ratio of 0.6 $\times$ (Beckman Coulter) Amplicon libraries were then prepared according to the Multiplex SMRT Sequencing protocol (Pacific Biosciences) and sequenced on a Sequel platform instrument (Pacific Biosciences). Raw sequencing data was demultiplexed and circular consensus sequences (CCS) were determined using the SMRT Analysis tool suite (Pacific Biosciences). The identities of gene segments, complementarity-determining regions (CDRs), and mutations from germline genes were determined by alignment using the ImMunoGeneTics database (Giudicelli and Lefranc, 2011).

Antibody production and purification. For hybridoma-derived mAb, hybridoma cells were grown to exhaustion in Hybridoma SFM (1 $\times$) serum free medium (Gibco). For recombinant mAb production, cDNA encoding the genes of heavy and light chains were synthesized and cloned into a DNA plasmid expression vector encoding a full-length IgG1 protein (McLean et al., 2000) and transformed into *E. coli* cells. mAb proteins were produced after transient transfection of ExpiCHO cells following the manufacturer's protocol. The resulting secreted IgGs were purified from filtered culture supernatants by fast protein liquid chromatography (FPLC) on an ÄKTA instrument using a Protein G column (GE Healthcare Life Sciences). Purified mAbs were buffer exchanged into PBS, filtered using sterile 0.45- μ m pore size filter devices (Millipore), concentrated, and stored in aliquots at -80° C. until use. An aliquot of each mAb also was biotinylated directly in 96-well format using the EZ-Link NHS-PEG₄-biotin kit (ThermoFisher Scientific) with a 20-fold molar excess of biotin to mAb, followed by buffer exchange back to PBS using a desalting plate (Zeba, 7 kDa cutoff). Hybridoma-derived mAbs were used in in vitro experiments, and recombinant mAbs were used in in vivo experiments. Pooled human immunoglobulin was purchased as intravenous immunoglobulin (IVIG, Carimune, CSL Behring, King of Prussia, PA). RSV90 is a recombinant human IgG1 mAb produced in the inventors' laboratory that was used as a negative control, placebo mAb in mouse experiments. Polyclonal anti-VP1, -VP2, and -VP3 antibodies used in western blot were purchased from Genetex.

Fab fragment production. Fab fragments were generated and purified via Pierce Fab Preparation Kit (ThermoFisher Scientific). The Immobilized Papain vial spin column and Zeba Spin Desalting Column were equilibrated with digestion buffer (35 mg cysteine-HCl per 10 mL of supplied Fab Digestion Buffer, pH ~7.0) before use. The NAb Protein A Plus Spin Column was equilibrated with PBS buffer before use. The original IgG samples were passed through the Zeba Spin Desalting Column, and 0.5 mL of the prepared IgG samples were applied on the Immobilized Papain vial and incubated at 37° C. for 5 hours Fab digestion. Then the final Fab fragments were buffer exchanged to PBS and stored at 4° C.

EV-D68-specific ELISA. Wells of medium binding, black fluorescent immunoassay microtiter plates (Greiner Bio-One) were coated with virus stocks diluted in 100 mM bicarbonate/carbonate buffer, pH 9.6 and incubated at 4° C.

overnight. Plates were blocked with 2% BSA in Dulbecco's phosphate-buffered saline (DPBS) containing 0.05% Tween-20 (DPBS-T) for 1 hr. For mAb screening assays, hybridoma culture supernatants were added to the wells and incubated for 2 hr at ambient temperature. The bound antibodies were detected using Fc-specific goat anti-human IgG conjugated with HRP (Southern Biotech) and QuantaBlu fluorogenic peroxidase substrate (ThermoFisher Scientific). After 20 min, 100 mM glycine (pH 10.5) was added to quench the reaction, and the emission was measured at 420 nm after excitement at 325 nm using a Synergy H1 microplate reader (Biotek). For dose-response and cross-reactivity assays, serial dilutions of purified mAbs were applied to the wells in duplicate technical replicates and mAb binding was detected as above; the experiments were performed at least three times. For the competition ELISA, microtiter plates were first coated with virus, and then a purified mAb was added at 100 µg/mL and allowed to incubate at 33° C. for 3 hr. Then, a biotinylated mAb was spiked into this mixture at a final concentration of 5 µg/mL and allowed to incubate at ambient temperature for 1.5 hr. After a wash and 30 min of incubation with avidin-peroxidase (ThermoFisher Scientific), mAb binding was detected as above.

Western blot. B1 clade virus preparation was mixed with denaturing and reducing loading buffer, boiled at 100° C. for 5 min, and then run on an SDS-PAGE gel along with Novex sharp pre-stained protein standard (ThermoFisher Scientific). Protein was transferred to a membrane, blocked in blocking buffer (Li-Cor), and then cut into strips so that individual lanes could be stained with purified mAb in blocking buffer. An IRDye 800CW-conjugated goat anti-human secondary antibody (Li-Cor) was used to detect mAb binding. Strips were reassembled to visualize molecular weight and imaged on an Odyssey CLx near infrared imager (Li-Cor).

Cryo-EM sample preparation and data collection. For both EV-D68:Fab EV68-159 and EV-D68:Fab EV68-228 complexes, purified EV-D68 viruses and Fabs were mixed at a molar ratio of 1:200. After incubating at room temperature for 45 to 60 min, 3.5 µL of virus-Fab mixture sample were added to a glow-discharged 400 mesh lacey carbon film copper grid (Ted Pella Inc.). Grids were plunge frozen (Cryoplunge 3 system, Gatan) in liquid ethane after being blotted for 3.5 s in 75 to 80% humidity. Cryo-EM datasets were collected on a 300 kV Titan Krios Microscope (Thermo Fisher Scientific). For the EV-D68:Fab EV68-228 dataset, movies were collected using the program Legion (Subway et al., 2005) with a K3 Direct Detection Camera (Gatan) at a magnification of 64,000×, resulting in a super resolution pixel size of 0.662 Å, with a defocus range from 0.7 to 2 µm. A total electron dose of 44.2 electrons/Å² over 2.6 seconds of exposure was recorded over 50 frames. The EV-D68:Fab EV68-159 dataset was acquired with a K2 Summit direct electron detector (Gatan) at a nominal magnification of 81,000×, resulting in a super resolution pixel size of 0.874 Å, a defocus range from 0.7 to 3.5 µm. A total electron dose of 31.4 electrons/Å² over 12 seconds of exposure was split into 60 frames. Overall, 462 movies and 732 movies were acquired for the EV-D68:Fab EV68-228 and EV-D68:Fab EV68-159 datasets, respectively.

Image processing. For both datasets, motion correction was performed on the raw movie frames via MotionCor2 (Zheng et al., 2017b) as implemented in Appion (Lander et al., 2009) during data collection. The contrast transfer function (CTF) was estimated on the aligned frames with CTFFIND4 (Rohou and Grigorieff, 2015). Particle picking templates were generated using the Appion Manual Picker (Lander et al., 2009) and templates for auto picking were

obtained through 2-dimensional (2D) classification in XMIPP (Sorzano et al., 2004). These templates were then used for auto-picking in FindEM (Roseman, 2004) and particles were extracted using RELION. These particles were then subjected to multiple rounds of 2D and 3D classifications in RELION (Scheres, 2012). This resulted in 20,194 and 30,554 particles for the EV-D68:Fab EV68-228 and EV-D68:Fab EV68-159 datasets which were selected for final 3D icosahedral reconstructions using the program JSPR following the gold-standard refinement method (Guo and Jiang, 2014). The final resolutions for both maps were estimated based on a gold-standard Fourier shell correlation cutoff of 0.143 (Scheres and Chen, 2012). Map sharpening was done in RELION (Scheres, 2012) post-processing. Data collection parameters and related items are summarized in Table G.

Model building, refinement and analysis. The same methods were used for the atomic structures of both EV-D68:Fab EV68-159 and EV-D68:Fab EV68-228. The X-ray crystallography model of the EV-D68 Fermon strain (PDB code: 4WM8) was selected as a starting reference for model building and was manually fitted into the density maps using the program Chimera (Pettersen et al., 2004). Using the initial fitting as a basis, the models were rebuilt in Coot (Emsley et al., 2010) and refined using real-space-refinement in PHENIX (Adams et al., 2010) to correct for outliers and poorly fitted rotamers. Chimera (Pettersen et al., 2004), Coot (Emsley et al., 2010) and CCP4i2-PISA (Potterton et al., 2018) were used to determine the binding interface residues. The final atomic models were validated in MolProbity (Chen et al., 2012). Refinement statistics are described in Table G.

Selection of neutralization escape mutant virus. A clade B1 EV-D68 isolate was passaged under selection with increasing amounts of purified mAb in RD cells. After incubating mAb and virus for 1 hr at 33° C., this mixture was added to a cell monolayer for 2 hr at 33° C. The monolayer then was rinsed thrice, and mAb containing medium was added back. This culture was incubated at 33° C. until at least 70% cytopathic effect was observed (cells lifted off of plate), at which point the cells and supernatant together were collected and frozen to -80° C. This sample was thawed and sonicated in the same microfuge tube in an inverted cup sonicator at maximum power 3×20 s, vortexed for 30 s at maximum power, and sonicated again 2×20 s. Cellular debris was clarified for 10 min at 10,000×g. Then the virus-containing supernatant was mixed 1:1 with fresh medium containing mAb at higher concentration. Over three passages, mAb concentration was increased from 5 to 50 to 500 ng/mL. Viral RNA was harvested using TRI Reagent and Direct-zol RNA MiniPrep kit (Zymo Research). In triplicate, the inventors generated cDNA templates, from which a 3,080 bp amplicon covering the P1 region of the viral genome was generated with the PrimeScript One Step RT-PCR Kit Ver. 2 (Takara) and primers 5'-CCTCCGGCCCCCTGAAT (Fwd; SEQ ID NO: 641) and 5'-CCATTGAATCCCTGGGCCTT (Rev; SEQ ID NO: 642). They used a Pacific Biosciences (PacBio) next generation sequencing platform to generate sequences of each of the three replicates. 2,000 reads of each sequencing run were used to quantitate the percentage of reads in which each mutation was observed. Mutations were determined as compared to a wild-type consensus sequence of all of the reads from the negative control selection mAb selection.

Quantification and statistical analysis. Technical and biological replicates are indicated in the methods and figure legends. Statistical analyses were performed using Prism v8 (GraphPad).

Competition-binding assay. ELISA fluorescence values were normalized to a percentage of maximal binding determined from a control well without an irrelevant prior competing mAb added. The Pearson correlation of each biotinylated mAb to each other biotinylated mAb was calculated using the median inhibition percentage from three different experiments using the corn method of the Pandas Python package (McKinney, 2010). Hierarchical clustering was then performed on these Pearson correlations using the clustermap method of the Seaborn Python package. The clustering information was exported in newick format and imported into Interactive Tree of Life v4 (Letunic and Bork, 2019), which was used to display the hierarchically-clustered heatmap before importation into Excel (Microsoft) for final display.

Antibody ELISA binding experiments. EC_{50} values for mAb binding were determined after log transformation of antibody concentration using four-parameter sigmoidal dose-response nonlinear regression analysis constrained to a bottom value of zero and top value less than the maximal fluorescent value of the mAb with the highest saturation fluorescence value.

Virus assays. MAb IC_{50} values were calculated using a formula based on the Spearman-Kärber equation (Ramkrishnan, 2016). Viral titers in murine plasma and lungs were compared using a one-way analysis of variance (ANOVA) and Dunnett's multiple comparisons test, with a single pooled variance. A value of $p < 0.05$ was considered significant.

Lung cytokine/chemokine evaluations. For each cytokine/chemokine, the concentrations from treated mice were compared to placebo-treated mice using a Brown-Forsythe one-way ANOVA test and Dunnett's T3 multiple comparisons test, with individual variances computed for each comparison. This analysis was chosen because the inventors did not assume equal standard deviations for each measurement.

In vivo protection studies. Survival curves were generated using the Kaplan-Meier method and curves compared using the log rank test (Mantel-Cox). Neurologic scores were compared using a chi-square test.

Example 10—Results

After obtaining written informed consent, twelve subjects who had previous documented EV-D68 respiratory tract infections during the 2014 outbreak in the U.S. donated blood, from which the inventors isolated peripheral blood mononuclear cells (PBMCs). The subjects were 12 to 15 years old when infected and 16 to 18 years old at time of blood collection. Each subject had a history of EV-D68-associated respiratory disease, and none had symptoms of AFM (Table E). The collected PBMCs were transformed in vitro by inoculation with Epstein-Barr virus to generate memory B cell derived lymphoblastoid cell lines (LCLs), which secrete antibodies. LCL culture supernatants were used in an indirect ELISA to screen for the presence of EV-D68-reactive IgGs. The inventors selected cultures with antibodies that bound to laboratory-grown live virus preparations of EV-D68 generated from 2014 clinical isolates but that did not bind to a similarly prepared uninfected cell supernatant. After electrofusion of LCLs secreting EV-D68-specific antibodies with a non-secreting myeloma cell line,

the resulting hybridoma cells were single-cell sorted to generate clonal hybridomas secreting fully human mAbs (Table F).

The inventors sought to determine how many major antigenic sites on the virus surface are bound by human mAbs made in response to natural infection. To identify groups of antibodies that recognized similar epitopes, they determined whether the mAbs could compete with the binding of each of the other mAbs to live virus in an indirect ELISA. For competition-binding experiments, virus was coated directly onto an ELISA plate and then incubated with high concentrations of one unlabeled mAb. Next, mAbs labeled by biotinylation were added at a lower concentration, and the ability of the second mAb to bind the virus in the presence of the first mAb was determined. The inventors then used a Pearson correlation with the inhibition data to determine the relatedness of the antibody binding patterns to each other and identified four main competition-binding groups (FIG. 32), which they termed groups 1 to 4. They used each mAb to stain a western blot of EV-D68 preparations and found that nearly all mAbs in competition-binding groups 2 and 3 bound to linear epitopes in the VP1 protein, whereas only a single other mAb bound to any protein in the virus preparation (FIG. 40).

During the 2014 EV-D68 outbreak in the U.S., nearly all viral isolates were of the newly emergent B1 clade, with fewer detections of virus from the closely related B2 or distantly related D clades (Tan et al., 2016). All but one of the subjects for this study were infected with B1 clade isolates (Table E). Since 2014, B3 clade viruses have dominated, and B1 clade viruses are no longer circulating (Dyrdak et al., 2019); in 2018 all EV-D68 isolates sequenced by the U.S. Centers for Disease Control and Prevention were from the B3 clade (Kujawski et al., 2019). The inventors first measured the in vitro neutralization capability of each mAb in a 50% cell culture infectious dose ($CCID_{50}$) assay using a B1 clade EV-D68 isolate (FIG. 33A). Twenty-eight mAbs demonstrated neutralization with a half maximal inhibitory concentration (IC_{50}) below 50 $\mu\text{g/mL}$, with mAb EV68-159 exhibiting the strongest neutralization at an IC_{50} value of 0.32 ng/mL (FIG. 41). The inventors further tested the 21 most potently neutralizing mAbs against a D clade isolate and found that 11 mAbs neutralized that virus, with 7 of those exhibiting at least a ten-fold decrease in potency by IC_{50} value for the heterologous virus. The Fermon strain is an isolate from 1962 and is so distantly related to modern EV-D68 isolates that it does not fit into the clade classification scheme (Tan et al., 2016). Nine mAbs neutralized the Fermon laboratory reference strain, but less potently than they inhibited the contemporary B1 clade virus.

Recognizing that neutralization assays may underestimate cross-reactivity, the inventors used the same indirect ELISA approach described above to generate half-maximal effective concentrations (EC_{50}) of purified mAb for binding to representative EV-D68 isolates from the B1, B2, or D clades (FIGS. 33B-C, and FIG. 42). Of the mAbs with EC_{50} values for binding of $\leq 1 \mu\text{g/mL}$ to B1 clade isolates, all bound to a B2 clade isolate, whereas about half also bound to a D clade isolate (FIG. 33B and FIG. 41). An additional class of mAbs was observed that bound weakly in general but cross-reacted to viruses from all clades tested.

To date, structural studies of antibody-EV-D68 interactions have been limited to murine mAbs (Zheng et al., 2019). The inventors selected two potently neutralizing human mAbs, the clade-specific mAb EV68-159 and the highly cross-reactive mAb EV68-228, to make immune complexes with antigen binding fragments (Fabs) and a B1 clade

EV-D68 isolate for cryo-electron microscopy (cryo-EM) studies. The final density maps attained a resolution of 2.9 Å (EV68-159) or 3.1 Å (EV68-228) (FIG. 34A, FIG. 43, FIG. 44, and Table G). The structures revealed two distinct binding sites: EV68-159 attached around the three-fold axes of symmetry, whereas EV68-228 bound around the five-fold axes between depressions that form the canyon regions (FIGS. 34A-C, FIG. 45). Thus, for each Fab, a total of 60 copies bound to the virus particle. The Fab variable domains, which interacted with the viral surface, displayed strong densities similar to the viral capsid proteins, and an atomic model of each Fab was built together with the four viral capsid proteins. In contrast, the Fab constant domains, which are located further from the viral surface, displayed weaker densities and were excluded from atomic model building. The backbone of the polypeptide chains and the majority of amino acid side chains are well-ordered in the density maps, demonstrating the critical features of the binding interface between virus particle and Fab molecule.

For both models, the viral surface residues that were facing and within a 4 Å distance from the Fab were identified as the footprint (FIG. 34C, FIG. 46, and Table H). The footprints show that both Fab molecules sit within one protomer. In the EV-D68:Fab EV68-159 complex, each Fab masked a viral surface area around 990 Å². At the binding interface (FIG. 35), essential interactions were found between the EV68-159 light chain and three residues on the C-terminus of VP1: Glu271 and Arg272 (FIG. 36A) and Asp285 (FIG. 36B). Residues Glu271 and Arg272 formed hydrogen bonds with CDR3 and CDR1. Arg272 and Asp285 formed salt bridges with CDR3 and CDR2 residues, respectively. The heavy chain of EV68-159 contributed 77% of the masked surface areas. A series of hydrogen bonds was found between the heavy chain complementarity-determining region 2 (CDR2) and CDR3 and the VP3 N-terminal loop before the B-β strand (βB) (FIG. 36C).

In the EV-D68:Fab EV68-228 complex, each Fab masked approximately 1,170 Å² of the viral capsid surface. Similar to EV68-159, the heavy chain of the EV68-228 Fab dominated the interaction with the viral capsid by masking around 84% of the surface area. The binding interface (FIG. 35) was stabilized mainly by hydrogen bonds formed between the heavy chain CDRs and the VP1βB as well as the VP3 C-terminus (FIG. 36D). The light chain CDR1 interacted with the VP2 EF loop. In addition, hydrogen bonds formed between the heavy chain framework region (FR) 3 and the VP1 DE loop. Furthermore, a salt bridge formed between the light chain CDR3 and the VP1 C-terminus. Overall, the EV68-228 Fab bound the viral surface around the five-fold axes and recognized the classical picornavirus neutralizing immunogenic sites (Nlms) NIm-IB (VP1 DE loop) and NIm-II (VP2 EF loop) (Rossmann et al., 1985).

Bulky side chains were found at the interface for both Fabs and act to stabilize the structures through hydrophobic interaction networks (FIG. 47). Furthermore, disulfide bonds also were detected around CDR1 and CDR3 in heavy and light chains. Another pair of cysteines, Cys101 and Cys106, were found within the CDR3 of the EV68-228 heavy chain and were at the correct distance and orientation to form a disulfide bond (FIG. 36D). Specifically, when the contour levels were reduced, the densities of the two cysteine side chains connected. As described above for the EV-D68:Fab EV68-228 complex, hydrogen bonds were observed between the heavy chain CDR3 and the VP3 C-terminal residues adjacent to the canyon involving Cys101, forming both a hydrogen bond and a disulfide bond. These cysteine residues play critical roles in stabilizing both Fab structure and the virus-Fab binding interface.

The inventors next sought to determine if the potentially neutralizing and highly cross-reactive human mAb EV68-228 could prevent or treat infection and disease in small animal models of EV-D68 infection. They tested for this antiviral activity in vivo using two different established models of infection causing either respiratory or AFM-like neurologic disease in AG129 strain mice that are deficient in receptors for interferon α/β and γ (Evans et al., 2019; Hurst et al., 2019). First, the inventors tested whether antibodies could reduce viremia and lung virus replication in the respiratory model of infection. MAb EV68-228 administered systemically as prophylaxis a day before virus inoculation provided sterilizing immunity in the blood (FIG. 37A) and lungs (FIG. 37B) at each of the concentrations tested, whereas human IVIG only sterilized the blood. Induction of pro-inflammatory cytokine secretion was inhibited in the lungs of EV68-228 treated mice (FIGS. 38A-C). When used as treatment given at increasing times after virus inoculation, again all treatments were highly effective at sterilizing the blood (FIG. 37C), but only EV68-228 had efficacy in the lungs (FIG. 37D). The inventors similarly observed reduced pro-inflammatory cytokine levels in the lungs of EV68-228-treated mice (FIGS. 38D-F).

Next the inventors assessed the effect of passive transfer of antibodies in a neurologic model of infection that mimics AFM disease. EV68-228 prophylaxis provided sterilizing immunity of the blood (FIG. 39A) and complete protection from death (FIG. 39B) or development of any neurologic disease (FIG. 39C), whereas IVIG immunity protected only partially. Given therapeutically, EV68-228 treatment sterilized the blood within 24 hours of administration (FIG. 39D) at each of the time points given. EV68-228 improved survival (FIG. 39E) and neurologic disease when given as late as 48 hours after infection; when given at 72 hours post-infection, the mouse that survived improved clinically (FIG. 39F and Table I).

TABLE E

Characteristics of subjects who provided peripheral blood mononuclear cells

Subject	Age at donation ¹	Gender ²	Ethnicity ³	Race ⁴	GenBank accession numbers for virus isolate	Virus Clade	Illness date	Donation date	Symptoms ⁵	Severity ⁶
1	18	F	Unk.	W	KX255352 KX255415	B1 B1	Aug. 11, 2014	Dec. 27, 2017	R, H	Mod.
2	17	F	NHL	B & W	KX255378	B1	Sep. 5, 2014	Aug. 7, 2017	NR	NR
3	17	M	NHL	W	KX255367	B1	Sep. 11, 2014	Aug. 1, 2017	NR	Mild
4	17	M	NHL	W	KX255385 KX433163	B1	Sep. 11, 2014	Jul. 19, 2017	C, R, H, SP	Mild

TABLE E-continued

Characteristics of subjects who provided peripheral blood mononuclear cells										
Subject	Age at donation ¹	Gender ²	Ethnicity ³	Race ⁴	GenBank accession numbers for virus isolate	Virus Clade	Illness date	Donation date	Symptoms ⁵	Severity ⁶
5	16	M	NHL	W	KX255382	B1	Aug. 3, 2014	Jul. 12, 2017	NR	NR
6	17	F	NHL	W	KX255364	B1	Aug. 13, 2014	Jul. 5, 2017	C, R	Mod.
					KX255373	B1				
7	17	M	NHL	W	KX433162	A1	Aug. 7, 2012	Jun. 21, 2017	C, R, H, W	Severe
8	17	M	NHL	W	KX255407	B1	Sep. 23, 2014	Jun. 19, 2017	C, R, H, W	Mild
9	17	M	NHL	W	KX255379	B1	Sep. 26, 2014	Jun. 8, 2017	C, R	Mild
10	18	F	NHL	W	KX255411	B1	Sep. 4, 2014	Jun. 6, 2017	C	Mild
11	18	F	NHL	W	KX255390	B1	Jul. 18, 2014	May 30, 2017	C, H, R, W	Mod.
					KX255362	B1				
					KX255403	B1				
12	18	M	NHL	W	KX255389	B1	Sep. 26, 2014	Jan. 22, 2018	C, R, W	Mod.

¹Age in years;²F: female, M: male;³Unk.: unknown, NHL: not Hispanic or Latino;⁴B: black, W: white;⁵C: cough, H: hoarse, R: rhinitis, SP: sinus pain, W: wheeze, NR: not recorded;⁶Mod.: moderate, NR: not recorded.

TABLE F

Sequence characteristics of human mAbs								
Clone	Heavy chain				Light chain			
	V gene	J gene	D gene	CDR1	V gene	J gene	CDR3	
EV68-37	V3-30*04	J4*02	D6-13*01	ARPTLPYSNNWYAPEY (SEQ ID NO: 259)	LV3-10*01	LJ2*01	SSTDSSGNPVL (SEQ ID NO: 451)	
EV68-40	V3-23*04	J4*02	D6-6*01	AKVVRIA AVLYYFDY (SEQ ID NO: 262)	KV3-11*01	KJ5*01	QQRSSWPIT (SEQ ID NO: 454)	
EV68-41	V3-23*04	J3*02	D1-1*01	ARVKSTTGT TALVF DI (SEQ ID NO: 265)	LV8-61*01	LJ3*02	GLYMGSGIWI (SEQ ID NO: 457)	
EV68-43	V3-30*03	J6*02	D4*17*01	AKDKHGFDFYYGVDV (SEQ ID NO: 268)	KV1-12*01	KJ1*01	QQADSFPR T (SEQ ID NO: 460)	
EV68-46	V3-30*03	J4*02	D1-26*01	ARRRPGSFPGLCDY (SEQ ID NO: 271)	KV3-5*03	KJ2*01	QQHNSYSYT (SEQ ID NO: 463)	
EV68-48	V1-69*06	J3*01	D5-12*01	ARMYSGHDGVDV (SEQ ID NO: 274)	KV3-11*01	KJ1*01	QQRSTWPPGM (SEQ ID NO: 466)	
EV68-71	V3-39*01	J3*02	D4-17*01	ARHLTHLYGDYVTPDALDI (SEQ ID NO: 277)	KV3-20*01	KJ1*01	QQYSNSRLT (SEQ ID NO: 469)	
EV68-72	V3-48*02	J5*02	D3-3*01	ARAHGRIVNSGVVISRFDP (SEQ ID NO: 280)	LV2-14*01	LJ2*01	SSYTTSNTLVV (SEQ ID NO: 472)	
EV68-74	V1-2*02	J4*02	D2-2*01	ARMGCRSDRCYSTNYNFDQ (SEQ ID NO: 283)	LV3-21*02	LJ2*01	QVWD SGIDVV (SEQ ID NO: 475)	
EV68-75	V3-21*01	J4*02	D6-6*01	ARERGHSTSSSYFDS (SEQ ID NO: 286)	LV3-21*01	LJ3*01	RVWSDSDTHRV (SEQ ID NO: 478)	
EV68-76	V3-15*01	J4*02	D3-22*01	STGPYYDTSGYPQPF DY (SEQ ID NO: 289)	LV3-1*01	LJ2*01	QAWDSSSTVV (SEQ ID NO: 481)	
EV68-79	V3-21*01	J6*02	D3-9*01	ARDRPIMEGEGDEL TGYVYQYYAMDV ND (SEQ ID NO: 643)		ND	ND	
EV68-84	V1-2*02	J4*02	D5-24*01	ARAGRNGYDY (SEQ ID NO: 295)	LV3-25*03	LJ2*01	QSGDSSGTYLV (SEQ ID NO: 487)	
EV68-85	V3-23*01	J4*02	D4-11*01	TVPWGNYN DYVSDY (SEQ ID NO: 298)	KV3-11*01	KJ5*01	HQHSTWPRGT (SEQ ID NO: 490)	

TABLE F-continued

Sequence characteristics of human mAbs							
Clone	Heavy chain				Light chain		
	V gene	J gene	D gene	CDR1	V gene	J gene	CDR3
EV68-88	V3-30-3*01J4*02	D6-13*01	ARHFLPYSSSWYQGFNY (SEQ ID NO: 301)		LV6-57*01	LJ2*01	QSYDNSNRAVV (SEQ ID NO: 493)
EV68-89	V3-33*01	J4*02	D4-17*01 ARGVPYGDLTGLVY (SEQ ID NO: 304)		LV6-57*01	LJ3*02	QSYDNSDRV (SEQ ID NO: 496)
EV68-95	V2-26*01	J4*01	D2-15*01 ARLLVAGTFLPSHYFDY (SEQ ID NO: 307)		LV3-21*01	LJ2*01	QVWDSSRNHPV (SEQ ID NO: 499)
EV68-97	V3-48*01	J4*02	D2-15*01 IRQVGADFSGRGFDY (SEQ ID NO: 310)		LV3-1*01	LJ3*02	QAWDSSSTAV (SEQ ID NO: 502)
EV68-98	V3-48*01	J3*02	D1-14*01 ATARHITNDGFDI (SEQ ID NO: 314)		KV1-9*01	KJ3*01	QQLNSHPRMFT (SEQ ID NO: 505)
EV68-105	V4-38-2*01J4*02	D2-21*02	ARGPGHCYGDCCYAYYFDQ (SEQ ID NO: 316)		KV1-12*01	KJ3*01	QQANSFPFFT (SEQ ID NO: 508)
EV68-110	V1-69*06	J4*02	D2-8*01 ARSLPYCTNDVCSNQNTFDY (SEQ ID NO: 316)		LV3-25*03	LJ2*01	QSADSSGTYVV (SEQ ID NO: 511)
EV68-114	V3-23*01	J4*02	D4-17*01 VRRFPMTTTSFDS (SEQ ID NO: 325)		LV2-11*01	LJ2*01	GAYAGFNAL (SEQ ID NO: 517)
EV68-116	V3-7*03	J4*01	D2-2*01 VREGVRRVVVRSTGYFDF (SEQ ID NO: 328)		LV3-16*01	LJ3*02	YSTDSSGYQRA (SEQ ID NO: 520)
EV68-150	V4-31*03	J5*02	D2-21*02 ARHVVTASGWDFP (SEQ ID NO: 331)		KV3-11*01	KJ2*01	QQRSRWPPPYT (SEQ ID NO: 523)
EV68-152	V3-30-3*01J6*02	D3-22*01	ARVTADYYESSGKVF (SEQ ID NO: 337)		KV1-5*03	KJ1*01	QQYTFSWT (SEQ ID NO: 529)
EV68-153	ND	ND	ND		LV1-47*01	LJ2*01	AAWDDRLSGVV (SEQ ID NO: 644)
EV68-156	V4-59*01	J1*01	D2-8*02 ARGSMPhi (SEQ ID NO: 346)		LV1-47*01	LJ1*01	AAWDDSLKAPV (SEQ ID NO: 538)
EV68-158	V3-23*01	J3*02	D3-22*01 AKDSHSMIVVDHAFDI (SEQ ID NO: 352)		LV3-21*02	LJ1*01	QVWDSYNVHYV (SEQ ID NO: 544)
EV68-159	V3-21*01	J4*02	D1-14*01 AREEGFRAYNLY (SEQ ID NO: 355)		LV1-47*01	LJ2*01	AAWDDILSGVV (SEQ ID NO: 547)
EV68-160	V3-30*04	J4*02	D2-8*01 ARDWDRLVRSVAGY (SEQ ID NO: 358)		LV2-11*01	LJ3*02	CSYAGTYTWV (SEQ ID NO: 550)
EV68-161	V4-61*01	J6*02	D3-3*01 ERRLRILSIERNYYAMDV (SEQ ID NO: 361)		KV3-20*01	KJ1*01	QQYGPSPWT (SEQ ID NO: 553)
EV68-162	V3-30-3*01J6*02	D2-15*01	ARDHVPPKDCSDGNCHSDYGMVDV (SEQ ID NO: 364)		KV1-5*03	KJ2*01	QQHNSYSYT (SEQ ID NO: 556)
EV68-164	V3-48*02	J6*02	D3-10*01 ARVYTMRLRGASMDV (SEQ ID NO: 370)		KV3-11*01	KJ3*01	QLRITWPPIFT (SEQ ID NO: 562)
EV68-165	V1-2*02	J4*02	D2-2*01 ARVKCSSANCYGNFDY (SEQ ID NO: 373)		LV2-8*01	LJ2*01	SSYAGSNNLV (SEQ ID NO: 565)
EV68-166	V5-51*01	J5*01	D5-12*01 ARQTTQNSGYDRWFDS (SEQ ID NO: 376)		LV1-44*01	LJ3*02	AAWDDSLNGWV (SEQ ID NO: 568)
EV68-178	ND	ND	ND		KV1-5*03	KJ1*01	QQYNSYPLT (SEQ ID NO: 645)
EV68-183	V3-23*01	J4*02	D3-3*01 AISVPLLRFLFLEWQHPFDF (SEQ ID NO: 379)		KV3-15*01	KJ1*01	HQYINWPPWT (SEQ ID NO: 571)
EV63-183	V3-21*01	J4*02	D4-11*01 VRPTMTTNTFDS (SEQ ID NO: 382)		LV2-11*01	LJ3*02	CSHAGSHTWV (SEQ ID NO: 574)
EV68-185	V3-30*03	J3*02	D3-22*01 PKVIPHPYDSSGDDAFDI (SEQ ID NO: 385)		KV3-15*03	KJ5*01	QQYSKLPIT (SEQ ID NO: 577)

TABLE F-continued

Sequence characteristics of human mAbs									
Clone	Heavy chain				Light chain				
	V gene	J gene	D gene	CDR1	V gene	J gene	CDR3		
EV68-208	V1-69*01	J4*03	D3-10*01	ARFISTASYVPGTFEDV (SEQ ID NO: 391)	KV3-11*01	KJ2*02	QQRSDWPPGT (SEQ ID NO: 583)		
EV68-210	V3-21*01	J4*02	D4-17*01	ARMVRNTVTAFDY (SEQ ID NO: 394)	LV2-23*02	LJ3*02	CSYGGNNSWM (SEQ ID NO: 586)		
EV68-219	V1-24*01	J4*02	D2-21*01	ATWGVVEVVNGRRDYFDS (SEQ ID NO: 397)	LV3-25*03	LJ2*01	QSADNTRITV (SEQ ID NO: 589)		
EV68-220	V3-48*03	J3*01	D3-22*01	ARDVRDCSALTCPRRGDAFDF (SEQ ID NO: 400)	KV2-30*01	KJ3*01	MQGTHWPRT (SEQ ID NO: 592)		
EV68-221	V3-21*01	J4*02	D2-15*01	VKVGGSKHQYYFDY (SEQ ID NO: 403)	LV1-44*01	LJ2*01	AAWDDSLNGVV (SEQ ID NO: 595)		
EV68-224	V1-18*01	J4*02	D2-2*02	ARERCSTSTCYSRYADY (SEQ ID NO: 406)	KV1-39*01	KJ1*01	QQSYRSPT (SEQ ID NO: 598)		
EV68-225	V3-48*03	J3*02	D3-9*01	MREGLTYDSTI (SEQ ID NO: 409)	LV4-69*01	LJ3*02	QTWGTGFRV (SEQ ID NO: 601)		
EV68-227	V3-9*01	J3*02	D3-16*01	AKDDYEGAGFDI (SEQ ID NO: 412)	KV3-11*01	KJ5*01	QQRSNWPIT (SEQ ID NO: 604)		
EV68-228	V4-38-2*02	J4*02	D2-15*01	VRHEGSCNDGSCYGSFVDN (SEQ ID NO: 415)	KV1-12*01	KJ4*01	QQADSFIT (SEQ ID NO: 607)		
EV68-229	ND	ND	ND	ND	KV3-11*01	KJ4*01	QQRSNWPPLT (SEQ ID NO: 646)		
EV68-231	V3-23*01	J2*01	D2-21*02	ARGGTFHNWYFDL (SEQ ID NO: 418)	KV3-15*01	KJ1*01	QQYKNWPRT (SEQ ID NO: 610)		
EV68-234	V3-23*01	J6*03	D5-12*01	AKGTITYSYYMAV (SEQ ID NO: 421)	KV3-20*01	KJ4*01	QQYGTSTI (SEQ ID NO: 613)		
EV68-235	V1-24*01	J4-02	D2-21*01	ATWGIEVVNGRDEFFDS (SEQ ID NO: 424)	LV3-25*03	LJ2*01	QSADTRITV (SEQ ID NO: 616)		
EV68-236	V1-24*01	J4-02	D2-21*01	ATWGVAVVSGRRDYFDS (SEQ ID NO: 427)	LV3-25*03	LJ2*01	QTADIKYTV (SEQ ID NO: 619)		
EV68-241	V4-30-4*01	J5*02	D3-3*01	ARAYAYEFWSGYPNWFDP (SEQ ID NO: 430)	KV1-39*01	KJ1*01	QQSYSPWPT (SEQ ID NO: 622)		
EV68-242	ND	ND	ND	ND	KV1-5*03	KJ1*01	QQYNSLPWT (SEQ ID NO: 647)		
EV68-247	V3-30*01	J4*02	D2-2*01	ARGLGYCSGTGGCTPFEY (SEQ ID NO: 433)	KV1-39*01	KJ4*01	QQSDSAPPT (SEQ ID NO: 625)		
EV68-254	V4-34*02	J5*02	D2-2*01	VRVPRRGFECSFGCDD- TACRYGHTWFDP (SEQ ID NO: 436)	KV1-39*01	KJ1*01	QQSYLTPTT (SEQ ID NO: 628)		
EV68-266	V3-11*01	J2*01	D2-8*02	AGSKVGYTTGRRNWYFDL (SEQ ID NO: 442)	LV2-8*01	LJ2*01	SSYAGNNLV (SEQ ID NO: 634)		
EV68-269	V1-46*01	J3*02	D2-2*02	ARDLVVVVPVMSRRAPDI (SEQ ID NO: 445)	LV3-21*02	LJ2*01	QVWDSTTDHGV (SEQ ID NO: 637)		
EV68-271	V1-2*06	J4*02	D3-16*02	ARDYRDDYMWGSYRPLDY (SEQ ID NO: 448)	KV3-11*01	KJ4*01	QQRSNGLT (SEQ ID NO: 640)		

Sequence characteristics as identified by the International ImMunoGeneTics (IMGT) Information System, world-wide-web at imgt.org. When IMGT could not rule out multiple genes, the first call is listed.
 ND: not determined.

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TABLE G

Cryo-EM data acquisition parameters and refinement statistics		
	EV-D68: Fab EV68-159	EV-D68: Fab EV68-228
Cryo-EM data acquisition and processing		
Magnification	81,000X	64,000X
Camera	K2 Summit direct electron detector (Gatan)	K3 Direct Detection Camera (Gatan)
Voltage (kV)	300	300
Pixel size (Å)	0.874	0.662
Defocus range (µm)	0.7-3. 5	0.7-2.0
Total electron dose (electrons/Å ²)	31.4	44.2
Particles picked	42,078	27,390
Particles used	30,554	20,194
Map resolution (FSC threshold = 0.143)	2.9	3.1
Model building and refinement		
Model building reference (PDB code)	4WM8	4WM8
MolProbity score	1.93	1.96
Clash score	9.05	8.7
Rotamer outliers (%)	0.46	0.00
	R.m.s. deviations	
Bond length (Å)	0.005	0.005
Bond angles (°)	0.691	0.689
	Ramachandran plot (%)	
Favored	93.08	91.70
Allowed	6.92	8.20
Outliers	0.00	0.10

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TABLE H

Structural contact amino acid residues of EV-D68 and respective Fabs		
Viral amino acid	Fab amino acid	Potential interactions
EV68-159		
VP1: GLU271	L: ILE95	Hydrogen bond
VP1: ARG272	L: SER26	Hydrogen bond
	L: SER27	Hydrogen bond
	L: ASP94	Salt bridge
VP1: ASP285	L: LYS51	Salt bridge
VP3: GLU59	H: ARG102	Hydrogen bond
VP3: SER60	H: GLY100	Hydrogen bond
VP3: MET64	H: TYR56	Hydrogen bond
VP3: GLU65	H: SER51	Hydrogen bond
	H: THR52	Hydrogen bond
	H: SER53	Hydrogen bond
EV68-228		
VP1: LYS71	H: SER30	Hydrogen bond
VP1: ARG72	H: TYR-27	Hydrogen bond
VP1: SER73	H: ASN31	Hydrogen bond
VP1: GLY129	H: SER73	Hydrogen bond
VP1: LYS268	L: ASP92	Salt bridge
VP2: ASN136	L: SER30	Hydrogen bond
VP3: GLY234	H: ASN102	Hydrogen bond
VP3: LEU236	H: ASN102	Hydrogen bond
VP3: ASP237	H: TYR34	Hydrogen bond
	H: SER100	Hydrogen bond
	H: CYS101	Hydrogen bond
	H: ASN102	Hydrogen bond
VP3: HIS238	H: TYR53	Hydrogen bond
	H: TYR54	Hydrogen bond
VP3: GLU243	H: TYR33	Hydrogen bond

30 Heavy and light chains are labeled as H and L, respectively. Notably, only direct interactions that are clearly observed in the electron density maps at high resolution are listed. This differs from contact residues highlighted in the roadmaps (FIG. 34C and FIG. 46), which use a 4 Å cutoff for displaying the overall footprint at maximum hydrogen bonding distance.

Table I – Neurologic scores of individual mice treated with antibody after EV-68 inoculation

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Day	Placebo - 24 hrs post						EV-D68-228 - 24 hrs post					
2	1	2	1	1	2	0	0	1	1	0	0	0
3	2	2	2	1	3	1	1	1	1	1	1	1
4	3	3	4	2	4	2	1	1	1	1	1	1
5	4	4		3		3	1	1	1	1	1	1
6				4		4	1	1	1	1	1	1
7							1	1	0	1	1	1
8							1	1	0	1	1	1
9							1	1	1	1	1	1
10							1	1	1	1	1	1

Day	EV-D68-228 - 48 hrs post						EV-D68-228 - 72 hrs post					
2	1	2	1	1	2	2	1	2	2	1	2	2
3	2	2	2	2	2	2	2	3	2	2	2	2
4	2	2	2	2	2	4	4	0	3	3	4	4
5	2	2	2	2	4			0	3	3		
6	2	2	2	2				0	4	4		
7	1	2	1	1				0				
8	1	2	1	1				0				
9	1	2	1	1				0				
10	1	1	1	1				0				

Day	EV-D68-228 - 120 hrs post										IVlg - 24 hrs post				
2	1	0	1	0	1	1	1	0	0	0	0	1	0	0	1
3	1	1	1	2	1	1	1	2	1	1	1	1	1	1	1
4	1	1	2	3	2	3	2	2	3	1	1	1	1	1	1
5	2	2	3	3	3	4	3	4	4	1	1	1	1	1	1
6	3	4	4	3	4		4			2	2	1	1	2	2
7	4			4						1	2	2	1	1	1
8										1	2	2	1	2	2
9										0	2	2	0	1	1
10										0	1	1	0	0	0

Each column is the data from an individual mouse within the indicated treatment group corresponding to the graph in FIG. 39F. Day represents the number of days after EV-D68 inoculation the score was determined. Mice were sacrificed or already dead upon a score of 4, so no further scores are listed in days after a 4 was recorded. Underlined scores indicate instances of a mouse improving clinically.

Notably, the competition-binding studies used full-length IgG molecules, so the competition seen is functional as would occur in human tissues and does not necessarily indicate that there are only four structurally distinct epitopes on the viral surface. The lack of western blot reactivity of EV68-159 and EV68-228 correlates with the findings in the structural

TABLE J

Neutralization escape amino acid mutants						
Amino acid			Average percent of reads ¹			Contact residue ²
Position	WT ³	Mutant	RSV-90	EV68-159	EV68-228	EV68-159 EV68-228
VP2 135	HIS	TYR	0	13	0	Adjacent
VP2 139	ALA	THR	20	2	3	Adjacent
VP3 42	SER	ASN	7	15	3	
VP1 71	LYS	MET	12	11	33	Contact
VP1 126	ALA	VAL	1	3	0	
VP1 157	LYS	GLU	6	17	4	
VP1 270	ARG	LYS	1	0	1	Adjacent

¹Each of the mAbs listed was used as selection for three passages of EV-D68 in RD cell culture, with RSV-90 being the same negative control mAb used as placebo treatment for in vivo experiments. For each treatment, the structural genes were sequenced using next generation sequencing in 3 technical replicates with 2000 sequences analyzed per replicate. Numbers listed are the average of these replicates.
²Contact residues as determined by cryo-EM structures from this manuscript, which are virus amino acids located within 4 Å of Fab amino acids. Adjacent refers to amino acids that are immediately adjacent to contact residues.
³WT: wild-type. Represents the consensus sequence of the RSV-90 selected virus sequences.

TABLE K

EV-68 isolates used						
Strain name	Abbreviation	Clade	Source	Source ID	GenBank Accession	Pubmed ID
US/MO/14-18947	MO/47	B1	BEI	NR-49129	KM851225	25414503
US/MO/14-18949	MO/49	B1	BEI	NR-49130	KM851227	25414503
US/IL/14-18956	IL/56	B1	BEI	NR-49133	MK268345	25414503
US/IL/14-18952	IL/52	B2	BEI	NR-49131	KM851230	25414503
US/KY/14-18953	KY/53	D	BEI	NR-49132	KM851231	25414503
Fermon	Fer.	1962 reference strain	ATCC	VR-1826	AY426531	15302951
Mouse adapted US/MO/14-18949		B1	Hurst and Tarbet, USU	NA	MH708882	30521834

ATCC: American Type Culture Collection; BEI: Biodefense and Emerging Infections Research Resources Repository, NIAID, NIH; USU: Utah State University; NA: not applicable.

Example 11—Discussion

These studies reveal diverse features of the human B cell response to EV-D68 infection. The inventors attempted to be unbiased in their approach to isolating these mAbs, using live virus isolates as the screening antigen. The diversity in antibody phenotypes that they recovered may be a result of this strategy, as the inventors observed a broad range of cross-reactivity among clades of EV-D68, both with binding and neutralization. Interestingly, strong binding to live virus particles did not necessarily predict high neutralizing potency. Of the competition-binding groups observed, only groups 2 and 3 exhibited uniformity in phenotype. MABs in both groups were cross-reactive, but group 2 mAbs neutralized virus whereas group 3 mAbs did not. Nearly all of the group 2 and 3 mAbs bound to VP1 in western blot (FIGS. 40 and 41), suggesting that they bind to linear epitopes.

studies that show both epitopes span all three major viral surface proteins. The conformation-dependent nature of the epitopes of these two potently neutralizing mAbs is notable because recent diagnostic advances using peptide microarray (Mishra et al., 2019) and phage library (Schubert et al., 2019) technologies scanned for antibodies in the cerebrospinal fluid of patients with AFM that recognize linear epitopes. Detection of antibodies recognizing linear epitopes currently can be used in valuable diagnostic tools; however, these studies reveal these tests are at best only partially informative about the quality of antibody response these patients make in response to enterovirus infection. The structures also suggest the molecular basis for antibody-mediated neutralization. By contacting all three structural proteins within a protomer, both mAbs appear capable of inhibiting dynamic structural transitions necessary for infection, which are poorly understood.

The disulfide bond in the CDR3 of EV68-228 heavy chain is a structural moiety the inventors have now observed in broadly neutralizing antiviral human mAbs for a number of viruses, including both hepatitis C (Flyak et al., 2018) and influenza A virus (Bangaru et al., 2019). The intervening four to five amino acids between cysteines forms a smaller structured loop at the most distal tip of the full CDR3 loop, stabilizing the CDR3 in a preconfigured state optimal for binding the viral antigen. For EV68-228 specifically, the Cys101 also directly interacts with VP3 via a hydrogen bond, so the cysteine participates in both CDR3 loop stabilization and interaction with target.

The three VP1 residues that interact with EV68-159 light chain (Glu271, Arg272 and Asp285) and Glu59 on the N-terminal loop of VP3, which interacts with EV68-159 heavy chain, are adjacent to the sialic acid receptor binding site (Liu et al., 2015), suggesting that the EV68-159 Fab may block virus from binding sialic acid receptors. In particular, these three VP1 residues are located on a 22-amino acid VP1 C-terminal peptide that is bound by antibodies found in the cerebrospinal fluid of patients with AFM (Mishra et al., 2019; Schubert et al., 2019). Furthermore, the interaction of the EV68-159 Fab heavy chain with the VP3 N-terminal loop may prevent the virus from uncoating, since the N-termini of the four VPs contribute to capsid stability (Filman et al., 1989). EV68-228 may prevent the virus from uncoating by binding VP1 β B, inhibiting the externalization of the N-terminus of VP1 that is required for entry. In addition, the antibody footprint includes residues on the C-terminus of VP3, which is not part of a classical NIm. These residues are adjacent to the canyon receptor binding site, suggesting that mAb EV68-228 also may block virus binding to receptors.

Finally, at a time when poliovirus types 2 and 3 have been eradicated, AFM is on the rise, and the role of EV-D68 in causing epidemics of this paralytic disease is increasingly evident. Given how well prophylaxis with human mAb EV68-228 works in vivo, these data suggest that an effective EV-D68 vaccine might prevent AFM disease. Indeed, recent studies indicate that virus-like particle (Zhang et al., 2018a; Dai et al., 2018) and inactivated EV-D68 (Patel et al., 2016) vaccine candidates are immunogenic and protective against infection in mice. However, a study of cotton rats vaccinated with inactivated EV-D68 suggested that they may have suffered worse respiratory disease upon subsequent EV-D68 infection (Messacar et al., 2016). While this finding could suggest the possibility of antibody-dependent enhancement (ADE) of EV-D68 infection, in mice the inventors did not observe ADE caused by polyclonal or monoclonal antibodies, within the range of antibody concentrations they tested. Also, the prospect of using mAb EV68-228 as a therapy early during EV-D68 infection is appealing, especially since this antibody potentially neutralizes a diverse set of viral isolates without obvious autoreactive binding to human cell materials (FIG. 33C). Even though IVIG protected mice from AFM-like disease due to EV-D68 in prior in vivo studies (Hixon et al., 2017a), so far IVIG has not been shown to confer benefit for humans with AFM (Messacar et al., 2016). However, IVIG is a complex mixture of polyclonal antibodies with only a small fraction that recognize EV-D68. MAb prophylaxis or therapy for EV-D68 associated AFM is more promising than IVIG because of the high specificity, high potency, and lower antibody dose that can be used. It is possible, however, that a cocktail of mAbs directed at multiple epitopes may be more protective than mAb monotherapy. A mAb cocktail theoretically would provide a higher barrier to emergence of mAb resistant virus, but the

inventors did not observe resistance in vivo (FIGS. 37A-D and FIGS. 39A-F). Even under conditions optimized for selecting EV68-228 resistant viruses in vitro, the inventors could only identify virus genomes with mutations of unclear significance (Table J). In the absence of a reverse genetics system for making recombinant viruses with these mutations, they were unable to verify specifically if these mutations caused escape from neutralization. Therefore, they find emergence of resistance during potential therapeutic use unlikely. These experiments also provide hope for therapeutic efficacy in patients with severe respiratory disease due to EV-D68, which is the clinical syndrome that brought the 2014 EV-D68 outbreak to the attention of public health authorities prior to recognition of the association with AFM (Midgley et al., 2014). Overall, the studies presented here show that natural EV-D68 infection of humans induces B cells encoding broad and potentially neutralizing antibodies that can prevent or treat infection and disease in both the respiratory tract and the nervous system.

Example 12

The objective of this study was to determine the efficacy of treatment with tobacco-produced EV68-228 for an Enterovirus D68 (EV-D68) respiratory infection in four-week-old AG129 mice.

Materials and Methods

Animals. Four-week-old male and female AG129 mice from a specific-pathogen-free colony maintained at the Utah Science Technology and Research (USTAR) building at Utah State University. The mice were bred and maintained on irradiated Teklad Rodent Diet (Harlan Teklad) and autoclaved tap water.

Antibodies and Compound. The monoclonal antibody (mAb) EV68-228 produced in tobacco plants (EV68-228-TP) as well as Chinese hamster ovary (CHO) cells (EV68-228-CHO) was provided by the inventors. Both the tobacco-produced and CHO antibodies were provided as solutions and were dosed at a concentration of 10 mg/kg. A tobacco-produced anti-HIV (Anti-HIV-TP) antibody was used as a negative control antibody at a dose of 10 mg/kg. Intravenous immunoglobulin (HIVIg, Carimune, CSL Behring, King of Prussia, PA) was purchased from a local pharmacy and was used as a positive control.

Virus. Enterovirus D68 was obtained from BEI Resources, NIAID, NIH: Enterovirus D68, US/MO/14-18949, NR-49130. The virus was serially passaged 30 times in the lungs of 4-week-old AG129 mice and then plaque-purified three times in Rhabdomyosarcoma (RD) cells obtained from the American Type Culture Collection (Manassas, VA). The resulting virus stock was amplified twice in RD cells to create a working stock. The virus used for infection was designated EV-D68 MP30 PP.

Experiment design. A total of 118 mice were randomized into 4 groups of 12 mice, 8 groups of 8, and an additional group of 6 mice used for normal controls as shown in Table 1. Mice were treated via intraperitoneal (IP) administration of EV68-228 mAb, HIVIg, or placebo mAb at 24 hours pre-infection or, 24- or 48-hours post-infection. Mice were infected via intranasal (IN) instillation of $1 \times 10^{4.5}$ CCID₅₀ of EV-D68 MP30 PP in a 90 μ l volume of MEM. Mice were weighed prior to treatment and daily thereafter. Four mice from each treatment group were euthanized on days 1, 3 and 5 post-infection for evaluation of lung virus titers, blood virus titers, and lung cytokine concentrations. For the mice

treated 24- and 48-hours post-infection, samples were only collected on days 3 and 5 post-infection.

Lung Cytokine/Chemokine Evaluations. Each sample of lung homogenate was tested for cytokines and chemokines using a chemiluminescent ELISA-based assay according to the manufacturer's instructions (Quansys Biosciences Q-Plex™ Array, Logan, UT). The Quansys multiplex ELISA is a quantitative test in which 16 distinct capture antibodies have been applied to each well of a 96-well plate in a defined array. Each sample supernatant was tested at for the following: IL-1 α , IL-1 β , IL-2, IL-3, IL-4, IL-5, IL-6, IL-10, IL-12p70, IL-17, MCP-1, IFN- γ , TNF α , MIP-1 α , GM-CSF, and RANTES. Definition of abbreviations are: IL—interleukin; MCP—monocyte chemoattractant protein; IFN—interferon; TNF—tumor necrosis factor; MIP—macrophage inflammatory protein; GM-CSF—granulocyte/macrophage colony stimulating factor; and RANTES—regulated upon activation, normal T cell expressed and secreted.

Statistical analysis. All figures and statistical analyses were completed using Prism 8.4.2. (GraphPad Software Inc.). For each day post-infection, lung and blood virus titers from treated groups were compared to lung and blood titers from placebo-treated mice using a one-way analysis of variance (ANOVA). For each cytokine/chemokine, the concentrations from treated mice were compared to placebo-treated mice using a two-way ANOVA.

Ethics regulation of laboratory animals. This study was conducted in accordance with the approval of the Institutional Animal Care and Use Committee of Utah State University dated Mar. 2, 2019 (expires Mar. 1, 2022). The work performed done in the AAALAC-accredited Laboratory Animal Research Center of Utah State University. The U.S. Government (National Institutes of Health) approval was renewed Mar. 9, 2018 (PHS Assurance No. D16-00468 [A3801-01]) in accordance to the National Institutes of Health Guide for the Care and Use of Laboratory Animals (Revision; 2011).

Results and Discussion

This study determined the efficacy of a tobacco-produced EV68-228 mAb for treatment of an EV-D68 respiratory infection in four-week-old AG129 mice.

FIGS. 48A-C show lung virus titers for EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. No lung virus titers were detected at days 1, 3, or 5 post-infection in mice treated with doses of 10 mg/kg of EV68-228-TP or EV68-228-CHO. Treatment with a dose of 10 mg/kg of hIVlg only significantly reduced lung virus titers on day 5 post-infection.

FIGS. 49A-C show blood virus titers for EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. No blood virus titers were detected at days 1, 3, or 5 post-infection in mice treated with doses of 10 mg/kg of EV68-228-TP or EV68-228-CHO. Treatment with hIVlg at a dose of 10 mg/kg also reduced blood virus titers at days 1, 3, and 5 post-infection.

FIG. 50 shows lung concentrations of IL-1 α , IL-1 β , IL-2, and IL-3 from EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Significant reductions in lung concentration of IL-1 α were observed on days 1 and 3 post-infection in mice treated with EV68-228-TP as well as mice treated with EV68-228-CHO. Both the tobacco-produced and the CHO-produced treatments reduced concentrations of IL-1 β at day 3 post-infection. Treatment with hIVlg did not significantly affect lung cytokine concentrations of IL-1 α or IL-1 β . No significant changes in lung concentrations of IL-2 or IL-3 were observed post-infection with EV-D68.

Lung concentrations of IL-4, IL-5, IL-6, and IL-10 from EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO are shown in FIG. 51. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced concentrations of IL-5 on day 3 post-infection and reduced concentrations of IL-6 on days 3 and 5 post-infection. Treatment with hIVlg significantly reduced lung concentrations of IL-5 and IL-6 on day

TABLE L

Expt. NIA-1930/Experimental Design: Efficacy of EV68-228-TP for treatment of an EV-D68 respiratory infection in mice							
No./Cage	Group No.	Infected	Treatment	Dose	Treatment Schedule	Route	Observations
12	1	Yes	Anti-HIV-TP*	10 mg/kg	Once, 24 hours pre-infection	IP	4 mice/group sacrificed at days 1, 3, and 5 post-infection for: lung virus titers, and lung cytokines.
12	2	Yes	EV68-228-TP				
12	3	Yes	EV68-228-CHO				
12	4	Yes	hIVlg				
8	5	Yes	Anti-HIV-TP*	10 mg/kg	Once, 24 hours post-infection	IP	4 mice/group sacrificed at days 3 and 5 post-infection for: lung virus titers, and lung cytokines.
8	6	Yes	EV68-228-TP				
8	7	Yes	EV68-228-CHO				
8	8	Yes	hIVlg				
8	9	Yes	Anti-HIV-TP*	10 mg/kg	Once, 48 hours post-infection	IP	2 mice/group sacrificed at days 1, 3, and 5 for normal cytokine controls
8	10	Yes	EV68-228-TP				
8	11	Yes	EV68-228-CHO				
8	12	Yes	hIVlg				
6	13	No	Normal Controls	—	—	—	

3 post-infection. No significant changes in lung concentrations of IL-4 or IL-10 were observed after infection with EV-D68.

FIG. 52 shows lung concentrations of IL-12p70, IL-17, MCP-1, and IFN γ from EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced lung concentrations of MCP-1 days 3 and 5 post-infection. Treatment with hIVlg significantly reduced lung concentrations of MCP-1 on days 3 and 5 post-infection. No significant changes in concentrations of IL-12p70, IL-17, or IFN γ were observed after infection with EV-D68.

FIG. 53 shows lung concentrations of TNF α , MIP-1 α , GM-CSF, and RANTES from EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO reduced lung virus concentrations of MIP-1 α on days 1 and 3 post-infection and reduced concentrations of RANTES on days 3 and 5 post-infection. A 10 mg/kg dose of hIVlg reduced lung concentrations of MIP-1 α on days 1 and 3 post-infection and reduced concentrations of RANTES on day 3 post-infection. No significant changes in concentrations of TNF α or GM-CSF were observed after infection with EV-D68.

FIGS. 54A-B show lung virus titers for EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. No lung virus titers were detected at days 3 or 5 post-infection in mice treated with doses of 10 mg/kg of EV68-228-TP or EV68-228-CHO. Treatment with a dose of 10 mg/kg of hIVlg only significantly reduced lung virus titers on day 3 post-infection.

FIGS. 55A-B show blood virus titers for EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. No blood virus titers were detected at days 3 or 5 post-infection in mice treated with doses of 10 mg/kg of EV68-228-TP or EV68-228-CHO. Treatment with hIVlg at a dose of 10 mg/kg also reduced blood virus titers at days 3 and 5 post-infection.

FIG. 56 shows lung concentrations of IL-1 α , IL-1 β , IL-2, and IL-3 from EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Significant reductions in lung concentrations of IL-1 α and IL-1 β were observed on day 3 post-infection in mice treated with EV68-228-TP as well as mice treated with EV68-228-CHO. Treatment with hIVlg did not significantly affect lung cytokine concentrations of IL-1 α or IL-1 β . Concentrations of IL-3 were significantly reduced in mice treated with EV68-228-TP as well as mice treated with hIVlg on day 3 post-infection. No significant changes in lung concentrations of IL-2 were observed post-infection with EV-D68.

Lung concentrations of IL-4, IL-5, IL-6, and IL-10 from EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO are shown in FIG. 57. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced concentrations of IL-5 on day 3 post-infection and reduced concentrations of IL-6 on days 3 and 5 post-infection. Treatment with hIVlg significantly reduced lung concentrations of IL-5 on day 3 post-infection. No significant changes in concentrations of IL-4 or IL-10 in the lung tissue were observed after infection with EV-D68.

FIG. 58 shows lung concentrations of IL-12p70, IL-17, MCP-1, and IFN γ from EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO significantly reduced lung concentrations

of MCP-1 days 3 and 5 post-infection. Treatment with hIVlg significantly reduced lung concentrations of MCP-1 on days 3 and 5 post-infection. No significant changes in concentrations of IL-12p70, IL-17, or IFN γ were observed after infection with EV-D68.

FIG. 59 shows lung concentrations of TNF α , MIP-1 α , GM-CSF, and RANTES from EV-D68-infected AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Treatment with 10 mg/kg of EV68-228-TP or EV68-228-CHO reduced lung virus concentrations of MIP-1 α on day 3 post-infection and reduced concentrations of RANTES on days 3 and 5 post-infection. A 10 mg/kg dose of hIVlg did not significantly reduce lung concentrations of MIP-1 α or RANTES post-infection. No significant changes in concentrations of TNF α or GM-CSF were observed after infection with EV-D68.

FIGS. 60A-B show lung virus titers for EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. Only treatment with the 10 mg/kg dose of EV68-228-CHO reduced lung virus titers when treatment was given 48 hours after infection. Neither treatment with 10 mg/kg of EV68-228-TP or hIVlg significantly reduced lung virus titers when treatment was administered 48 hours after infection.

FIGS. 61A-B show blood virus titers for EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. No blood virus titers were detected at days 3 or 5 post-infection in mice treated with doses of 10 mg/kg of EV68-228-TP or EV68-228-CHO. Treatment with hIVlg at a dose of 10 mg/kg also reduced blood virus titers at days 3 and 5 post-infection.

FIG. 62 shows lung concentrations of IL-1 α , IL-1 β , IL-2, and IL-3 from EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. Significant reductions in lung concentrations of IL-1 α and IL-1 β were observed on day 3 post-infection in mice treated with EV68-228-CHO but not in the mice treated with EV68-228-TP. Treatment with hIVlg did not significantly affect lung cytokine concentrations of IL-1 α or IL-1 β . No significant changes in lung concentrations of IL-2 or IL-3 were observed post-infection with EV-D68.

Lung concentrations of IL-4, IL-5, IL-6, and IL-10 from EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO are shown in FIG. 63. Treatment with 10 mg/kg of EV68-228-CHO significantly reduced concentrations of IL-6 on day 3 post-infection. Treatment with EV68-228-TP or hIVlg did not significantly reduce lung concentrations of IL-6. No significant changes in concentrations of IL-4, IL-5, or IL-10 in the lung tissue were observed after infection with EV-D68.

FIG. 64 shows lung concentrations of IL-12p70, IL-17, MCP-1, and IFN γ from EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. When treatment was administered 48 hours after infection, no significant changes were observed in lung concentrations of IL-12p70, IL-17, MCP-1, or IFN γ in mice infected with EV-D68.

FIG. 65 shows lung concentrations of TNF α , MIP-1 α , GM-CSF, and RANTES from EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. When treatment was administered 48 hours after infection, no significant changes were observed in lung concentrations of TNF α , MIP-1 α , GM-CSF, or RANTES in mice infected with EV-D68.

Conclusions

This study determined the efficacy of a tobacco-produced EV68-228 mAb against EV-D68 for treatment of a respira-

tory infection caused by EV-D68 in four-week-old AG129 mice. The CHO-produced EV68-228 Ab was used as a comparator.

When treatment was administered either 24 hours pre-infection or 24 hours post-infection, the tobacco-produced Ab produced similar reductions in lung virus titers, blood virus titers, and lung cytokine concentrations when compared to the CHO-produced Ab.

When administered 48 hours post-infection, slight differences in the efficacy of the Abs were observed. The CHO-produced Ab was able to significantly reduce lung virus titers on days 3 and 5 post-infection while the tobacco-produced Ab was not able to reduce lung virus titers at either day 3 or 5 post-infection. Both the tobacco-produced and CHO-produced Abs were able to reduce blood virus titers on days 3 and 5 post-infection. In addition, the CHO-produced Ab significantly reduced lung concentrations of IL-1 α , IL-1 β , and IL-6 while the tobacco-produced Ab did not produce similar reductions in cytokine concentrations.

These differences in lung virus titers and lung cytokine concentrations were only observed when treatment was administered 48 hours after infection and the variability may be due to the limits of Ab treatment post-infection. However, in the previous study (NIA-1869) evaluating post-treatment of an EV-D68 respiratory infection with the CHO-produced antibody, a dose of 1 mg/kg was able to reduce lung concentrations of IL-1 α , IL-1 β , IL-5, MCP-1 and RANTES. Additional studies would be valuable to determine if there are significant differences between the tobacco-produced and the CHO-produced antibody.

Example 13

The objective of this study was to determine the efficacy of treatment with tobacco-produced EV68-228 antibody for an Enterovirus D68 (EV-D68) neurological infection in 10-day-old AG129 mice. Antibody produced in Chinese hamster ovary cells was used as a comparator for efficacy.

Materials and Methods

Animals. Ten-day-old AG129 mice from a specific-pathogen-free colony maintained at the Utah Science Technology and Research (USTAR) building at Utah State University. The mice were bred and maintained on irradiated Teklad Rodent Diet (Harlan Teklad) and autoclaved tap water at the USTAR building of Utah State University.

Antibodies and Compound. The monoclonal antibody (mAb) EV68-228 produced in tobacco plants (EV68-228-TP) as well as Chinese hamster ovary (CHO) cells (EV68-228-CHO) was provided by the inventors at Vanderbilt University Medical Center. Both the tobacco-produced and CHO antibodies were provided as solutions and were dosed at a concentration of 10 mg/kg. A tobacco-produced anti-HIV (Anti-HIV-TP) antibody was used as a negative control antibody at a dose of 10 mg/kg. Intravenous immunoglobulin

(IVIg, Carimune, CSL Behring, King of Prussia, PA) was purchased from a local pharmacy and was used as a positive control.

Virus. Enterovirus D68 was obtained from BEI Resources, NIAID, NIH: Enterovirus D68, US/MO/14-18949, NR-49130. The virus was serially passaged 30 times in the lungs of 4-week-old AG129 mice and then plaque-purified three times in Rhabdomyosarcoma (RD) cells obtained from the American Type Culture Collection (Manassas, VA). The resulting virus stock was amplified twice in RD cells to create a working stock. The virus used for infection was designated EV-D68 MP30 PP.

Experiment design. A total of 125 mice were randomized into 12 groups of 10 mice each as shown in Table 1 with an additional 5 mice used as normal controls for weight gain. Mice were treated via intraperitoneal (IP) administration of EV68-228-TP or EV68-228-CHO, IVIg, or placebo 24 hours before, 24- or 48-hours post-infection. Mice were infected via intraperitoneal (IP) administration of $1 \times 10^{6.5}$ CCID₅₀ of EV-D68 MP30 PP in a 0.1 mL volume of MEM. Mice were weighed prior to treatment and daily thereafter. Blood was collected from groups of 5 mice on days 1, 3, 5, and 7 post-infection for evaluation of blood virus titers. All mice were observed daily for morbidity, mortality, and neurological scores through day 21. Neurological scores (NS) were recorded as follows: NS0—no observable paralysis, NS1—abnormal splay of hindlimb but normal or slightly slower gait, NS2—hindlimb partially collapsed and foot drags during use for forward motion, NS3—rigid paralysis of hindlimb and hindlimb is not used for forward motion, NS4—rigid paralysis in hindlimbs and no forward motion. Any animals observed with a score of NS4 were humanely euthanized.

Statistical analysis. Kaplan-Meier survival curves were generated Prism 8.4.2. (GraphPad Software Inc.). Survival curves were compared using the Log-rank (Mantel-Cox) test followed by a Gehan-Breslow-Wilcoxon test. For each day post-infection, blood virus titers from treated groups were compared to lung and blood titers from placebo-treated mice using a one-way analysis of variance (ANOVA). Mean body weights were compared using a one-way ANOVA. Neurological scores were compared using a Kruskal-Wallis test followed by a Dunn's multiple comparisons test.

Ethics regulation of laboratory animals. This study was conducted in accordance with the approval of the Institutional Animal Care and Use Committee of Utah State University dated Mar. 2, 2019 (expires Mar. 1, 2022). The work was done in the AAALAC-accredited Laboratory Animal Research Center of Utah State University. The U. S. Government (National Institutes of Health) approval was renewed Mar. 9, 2018 (PHS Assurance No. D16-00468 [A3801-01]) in accordance to the National Institutes of Health Guide for the Care and Use of Laboratory Animals (Revision; 2011).

TABLE M

Expt. NIA-1931/Experimental Design: Efficacy of EV68-228-TP for treatment of an EV-D68 neurological infection in mice							
No./Cage	Group No.	Infected	Compound	Dosage	Route	Treatment Schedule	Observations
10	1	Yes	Anti-HIV-TP (Placebo)	10 mg/kg	IP	Once, 24 hours pre-infection	Mice observed daily for survival, body weights, and
10	2	Yes	EV68-228-TP				

TABLE M-continued

Expt. NIA-1931/Experimental Design: Efficacy of EV68-228-TP for treatment of an EV-D68 neurological infection in mice							
No./ Cage	Group No.	Infected	Compound	Dosage	Route	Treatment Schedule	Observations
10	3	Yes	EV68-228-CHO				neurological scores.
10	4	Yes	hIVIg				Blood collected
10	5	Yes	Anti-HIV-TP (Placebo)	10 mg/kg	IP	Once, 24 hours post- infection	from 5 mice per group on days 1, 3, 5, and 7 post- infection for blood virus titers.
10	6	Yes	EV68-228-TP				
10	7	Yes	EV68-228-CHO				
10	8	Yes	hIVIg				
10	9	Yes	Anti-HIV-TP (Placebo)	10 mg/kg	IP	Once, 48 hours post- infection	
10	10	Yes	EV68-228-TP				
10	11	Yes	EV68-228-CHO				
10	12	Yes	hIVIg				
5	13	No	Normal Controls	—	—	—	Observed for normal weight gain

Results and Discussion

This study determined the efficacy of a tobacco-produced EV68-228 mAb for treatment of an EV-D68 neurological infection in 10-day-old AG129 mice.

FIG. 66 shows Kaplan-Meier survival curves for 10-day-old AG129 mice infected with EV-D68 and treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. Both the EV68-228-TP or EV68-228-CHO treatment 24 hours prior to infection provided complete protection from mortality. A single 10 mg/kg dose of hIVIG protected eight of ten mice (80%) from mortality. Nine of the ten placebo-treated mice succumbed to the infection.

FIG. 67 shows percentages of initial body weights for 10-day-old AG129 mice infected with EV-D68 and treated 24 hours prior to infection with EV68-228-TP or EV68-228-CHO. No significant protection from weight loss was observed in mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO since the one surviving mouse in the placebo-treated group gained weight after recovering from infection.

FIG. 68A-B show blood virus titers on days 1 and 3 post-infection for EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. No blood virus titers were detected at days 1 or 3 post-infection in mice treated with a dose of 10 mg/kg of EV68-228-TP or EV68-228-CHO. Treatment with IVIg at a dose of 10 mg/kg also reduced blood virus titers at day 3 post-infection.

FIGS. 69A-B show blood virus titers on days 5 and 7 post-infection for EV-D68-infected AG129 mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO. No blood virus titers were detected at day 5 post-infection in mice treated with a dose of 10 mg/kg of EV68-228-TP or EV68-228-CHO. Treatment with IVIg at a dose of 10 mg/kg did not significantly reduce blood virus titers at days 5 post-infection. No virus was detected in the blood of any of the treatment groups at day 7 post-infection.

Neurological scores on days 2-10 post-infection in 10-day-old AG129 mice infected with EV-D68 and treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO are shown in FIGS. 70A-B. No neurological scores were observed in mice treated with a dose of 10 mg/kg of EV68-228-TP or EV68-228-CHO at any day post-infection. In addition, no neurological scores were observed in mice treated with IVIg at a dose of 10 mg/kg at any time

post-infection. No significant reductions in neurological scores were observed after day 7 post-infection because the only surviving placebo-treated mouse did not display any signs of paralysis.

Kaplan-Meier survival curves of 10-day-old AG129 mice infected with EV-D68 and treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO are shown in FIG. 71. Treatment with a 10 mg/kg dose of EV68-228-TP or EV68-228-CHO 24 hours after infection completely protected mice from mortality. A single dose of hIVIG given 24 hours after infection also protected all ten mice (100%) from mortality. Eight of the ten placebo-treated mice succumbed to the infection.

FIG. 72 shows percentages of initial body weights for 10-day-old AG129 mice infected with EV-D68 and treated 24 hours after infection with EV68-228-TP or EV68-228-CHO. No significant protection from weight loss was observed in mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO since two surviving mice in the placebo-treated group gained weight infection.

Blood virus titers on days 1 and 3 post-infection from 10-day-old mice infected with EV-D68 and treated 24 hours after infection with EV68-228-TP or EV68-228-CHO are shown in FIGS. 73A-B. No significant reduction in blood virus titers were observed on day 1 post-infection since the treatment was administered 24 hours after infection. On day 3 post-infection, treatment with EV68-228-TP and EV68-228-CHO significantly reduced blood virus titers. Treatment with hIVIG 24 hours after infection significantly reduced blood virus titers on day 3 post-infection.

Blood virus titers on days 5 and 7 post-infection from 10-day-old mice infected with EV-D68 and treated 24 hours after infection with EV68-228-TP or EV68-228-CHO are shown in FIGS. 74A-B. No virus was detected in the blood of mice treated 24 hours after infection with EV68-228-TP or EV68-228-CHO, however, only two placebo-treated mice were still alive for collection of blood and virus was not detected in the blood of either animal. Due to the lack of virus titers in placebo-treated animals, no statistical significance could be determined for any of the treatment groups. No virus was detected in the blood of any of the groups at day 7 post-infection.

FIGS. 75A-B show neurological scores on days 2-9 post-infection in 10-day-old AG129 mice treated 24 hours post-infection with EV68-228-TP or EV68-228-CHO. Treatment with EV68-228-TP or EV68-228-CHO 24 hours

after infection significantly reduced signs of paralysis as indicated by reduced neurological scores at days 2-9 post-infection. A single 10 mg/kg dose of hIVIg 24 hours after infection reduced neurological scores at days 2 and 3 post-infection but did not improve the outcome for the remainder of the study.

Neurological scores on days 10-17 post-infection in 10-day-old AG129 mice treated 24 hours after infection with EV68-228-TP or EV68-228-CHO are shown in FIGS. 76A-B. Both EV68-228-TP and EV68-228-CHO significantly reduced neurological scores on days 10-15 post-infection. The administration of hIVIg did not significantly reduce neurological scores at days 10-17 post-infection.

FIG. 77 shows Kaplan-Meier survival curves for 10-day-old AG129 mice infected with EV-D68 and treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. Treatment with EV68-228-TP at a dose of 10 mg/kg protected nine of ten mice (90%) from mortality. Eight of ten mice (80%) survived the infection with treated 48 hours after infection with EV68-228-CHO. A single 10 mg/kg dose of hIVIg protected six of ten mice (60%) from mortality. One of the ten placebo-treated mice survived the infection.

FIG. 78 shows percentages of initial body weights for 10-day-old AG129 mice infected with EV-D68 and treated 48 hours after infection with EV68-228-TP or EV68-228-CHO. No significant protection from weight loss was observed in mice treated 24 hours pre-infection with EV68-228-TP or EV68-228-CHO as the one surviving mouse in the placebo-treated group gained weight following infection.

FIGS. 79A-B show blood virus titers on days 1 and 3 post-infection for EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. On day 1 post-infection, none of the treatment groups were significantly different since no treatments had occurred at that point. At day 3 post-infection, blood virus titers were lower in mice treated with 10 mg/kg of EV68-228-TP or EV68-228-CHO although the differences were not statistically significant. This is likely due to treatment being administered just 24 hours prior to the blood collection on day 3. Treatment with IVIg at a dose of 10 mg/kg did not significantly reduce blood virus titers at day 1 or day 3 post-infection.

FIGS. 80A-B show blood virus titers on days 5 and 7 post-infection for EV-D68-infected AG129 mice treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO. No blood virus titers were detected at day 5 post-infection in mice treated with a dose of 10 mg/kg of EV68-228-TP or EV68-228-CHO. Treatment with IVIg at a dose of 10 mg/kg did not significantly reduce blood virus titers at days 5 post-infection. No virus was detected in the blood of any of the treatment groups at day 7 post-infection.

Neurological scores on days 3-10 post-infection in 10-day-old AG129 mice infected with EV-D68 and treated 48 hours post-infection with EV68-228-TP or EV68-228-CHO are shown in FIGS. 81A-B. Treatment with 10 mg/kg of EV68-228-TP significantly reduced neurological scores at days 3-5 post-infection. Treatment with 10 mg/kg of EV68-228-CHO significantly reduced neurological scores on days 3-6 post-infection. Neurological scores were significantly reduced in mice treated with 10 mg/kg of hIVIg on days 3-5 post-infection. No significant reduction in neurological scores were observed in any of the treatment groups on days 7-10 post-infection.

Conclusions

This study determined the efficacy of a tobacco-produced EV68-228 mAb against EV-D68 neurological infection 10-day-old AG129 mice.

The tobacco-produced mAb provided similar protection from mortality compared to the CHO-produced mAb when treatment was administered 24 hours prior to infection as well as 24 or 48 hours after infection.

In addition, similar reductions in blood virus titers were observed in mice treated with both EV68-228-TP and EV68-228-CHO mAbs in 10-day-old AG129 mice infected with EV-D68.

Similar protection from paralysis as indicated by reductions in neurological scores was observed in mice treated with EV68-228-TP as well as EV68-228-CHO.

In summary, the tobacco-produced EV68-228 provided similar protection from mortality, reductions in blood virus titers, and protection from paralysis when compared to the CHO-produced EV68-228.

TABLE 1

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence Region	
EV-D68-37	1	heavy	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCCTGAGACTCTCATGTGACGCTCTGGATTCATCTTCAGTCGCTACGCTCTGCACTGGGTCCGCCAGGCTCCAGGCAAGGGCTGGACTGGGTGGCAGTTATATCATATGATGCAAGAAATTCATATTACACAGACTCCGCTGAAGGGCCGATTACCATCTCCAGAGACAATTCAAAAACACGCTGTTTCTGCAGATGAACAGTCTGAGAGCTGACGACACGCTGTCTATTACTGTGCGAGACCGACTTTGCCCTACAGCAACAACGTACGCGCCTGAATACTGGGGCCAGGGAACCCCTGGTCACGCTCTCCTCA	
	2	light	TCCTATGAGCTGACACAGCCACCTCGGTGTCAGTGTCCCAGGACAAACGGCCAGGATCACCTGCTCTGGAGATGCATTACCAAAAAATATGCTTCTTGGTACCAGCAGAAGTCAGGCCAGGCCCTGTGCTGGTCATCTATGAAGACCAAAACGACCCTCCGGGATCCCTGAGAGATTCTCTGGCTCCAGCTCAGGACAAATGGCCACCTTGACTATCAGTGGGGCCAGGTGGAAGATGAAAGTGACTACTACTGTTCTCTCGACAGACAGCAGTGGTAATCCTGTGCTATTTCGGCGGAGGACCAAGTTGACCGTCCTA	
EV-D68-	3	heavy	GAGGTGCAGCTGGTAGAGTCTGGGGGAGACTGGTACAGCCTGGGGGTCCCTGAGACTCTCCTGTGACGCTCTGGAT	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable	Sequence Region
40				TCTCCTTTAGCAGCTATGCCATGGCCTGGGTCCGCCAGGC TCCAGGGAAGGGGCTGCAGTGGGTCTCATCTATTAGTGGT AACGGTAATGGGAGATCCCTATGCAGATTCTCTGAAGGGCC GGTTCACCACTCCAAAGACCTTTCCAAGTATACCTGTAT CTGCAAATGAACAATCTGAGACCCGAGGACACGGCCATAT ATTACTGTGCGAAAGTTGTCCGTATAGCAGCTGTTTGTAT TACTTTGACTATTGGGGCCCGGGAACCCAGGTTACCGTCT CCTCA
	4	light		GAAATTGTGTTGACACAGTCTCCAGCCACCTGTCTTTGTC TCCAGGGGAAAGAGCCACCTCTCCTGCAGGGCCAGTCA GAGTGTTAGCACCTACTTAGCCTGGTACCAACAAAGCCTG GCCAGGCTCCCAGGCTCCTCATCTATGAAGCATCCACCAG GGCCACTGGCATCCCAGGCAGGTTCAAGTGGCAGTGGGTCT GGGACAGACTTCACTCTCATCATCAGCAGCCTAGAGCCTG AAGATTTTGAGTTTATCACTGTGACGAGCGCAGCTCCTGG CCGATCACCTTCGGCCAAGGGACACGACTGGAGATTGAA
EV- D68- 41	5	heavy		GACGTGCAGCTGGTGGAGTCTGGGGGGGCTTGGTACAG CCTGGGGGGTCCCTGAGACTCTCCTGTGCTGCCTCTGGAT TCACTTTTAGCAACTATGCCATGACCTGGGTCCGCCAGGCT CTAGGGAAGGGGCTGGAGTGGGTCTCTTCTATTAGTGGTA GTGGTGGCCTCACATATTTCCGCACACTCCGTGAAGGGCCG GCTCACCATCTCCAGAGACAACCTCCAAGAATACCTCTATC TGCAATGAGCAGCCTGAGAGCCGAAGACACGGCCGTATA TTACTGTGCGAGAGTGAAAGTACAACCTGGAACGACGGCG TTAGTTTTGATATCTGGGGCCAAGGACAATGGTCACCGT CTCTTCG
	6	light		CAGACTGTGGTGACCCAGGAGCCATCGTTCTCAGTGTCCC CTGGAGGGACAGTCACTCACTTGTGGCTTGAGTTCTGG CTCAGTCTCTAGTAGTTACTACCCAGTGGTACCAGCAGA CCCCAGGCCAGGCTCCACGCACGCTCATCTACAGTATAAA CAGACGTTCTTCTGGGGTCCCTGATCGCTTCTCTGGCTTCA TCCTTGGGAACAAAGCTGCCCTCACCATCAGGGGGGCCCA GGCAGATGATGAATCTGATTATTACTGTGGGCTGTATATGG GTAGTGGCATTGGATCTTCGGCGGAGGGACCAAGCTGAC CGTCTTA
EV- D68- 43	7	heavy		CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAG CCTGGGAGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTCATTAACTATGGCATGCACTGGGTCCGCCAGGCT CCAGGCAAGGGGCTGGAGTGGGTGGCAGTTATATCAATG ATGGAAGTTATAACTACGATGCAGACTCCGTGAAGGGCCG ATTCACCATCTCCAGAGACAATCCAAGAACAAGGTTTATC TACAAATGAACAGCCTGAGACCTGAGGACACGGCTGTGTA TTCTGTGCGAAAGACAAACACGGTGACTTCGACTACTACG GTGTGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCT CCCCA
	8	light		GACGTCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATC TATAGGAGACAGAGTCACCATCACTTGTCTGGGCGGGTCAG GGAATTAGCAGCTGGTTAGCCTGGTATCAGCAGAAACCAG GGAAAGCCCTAAGCTCCTGATCTATGCTGCATCCAATTG CAAAGTGGGGTCCCATCAGGTTTACGGCAGTGGATCTG GGACAGATTCTACTCTCACCATCAGCAGCCTGCAGCCTGA AGATTTTGGAATTACTATTGTCAACAGGCTGACAGTTTCC CTCGGACGTTTCGGCCAAGGGACCAAGGTGGAAATCAA
EV- D68- 46	9	heavy		CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAG CCTGGGAGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTCAGTAGTTATGGCATACACTGGGTCCGCCAGGC TCCAGGCAAGGGGCTGGAGTGGGTGGCAGTTATATCCTAT GATGGAAGTGATAACACCTATGCACCTTTTGTGAACGGCC GATTCAACATCTCCAGAGACAATCCAAGAACAAGCTGTAT CTGCAAATGAACAGCCTGAGAGCTGACGACACGGCTGTGT ATTACTGTGCGAGGCGTCGGCCTGGGAGCTTCCAGGACT TTGCGACTACTGGGGCCAGGGAGCCTGGTACCGTCTCC TCA
	10	light		GACATCCAGATGACCCAGTCGCCTTCCACTATGTCTGCTTC GTAGGAGACAGAGTCACCATCACTTGCCTGGGCGAGTCAG AGTATTAGTAAGTGGTTGGCCTGGTATCAGCAGAAGCCAG GGAAAGCCCTAAACTCCTGATCTATAAGGCGTCTACCTTA CAAACCTGGGGTCCCATCAAGGTTACGGCAGTGGATCTG GGACAGAATTCACTCTCACCATCAACAGCCTGCAGCCTGAT GATTTTGCAACTTATTACTGCCAACAGCATAATAGTTATTCTG TACACTTTTGGCCAGGGGACCAAGGTGGAGATCAAG

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence Region	
EV-D68-48	11	heavy	CAGGTGCAGCTGGTGCAGTCTGGGGCTGAGGTGAAGAAG CCTGGGTCTCGGTGAAGGTCTCCTGCAAGGCTTCTGGAG GCAGCTTCAGCAGACTTACTATCATCTGGGTGCGCAGGC CCCTGGACAAGGGCTTGAGTGGATGGGAGGGCACATCCC TATCTTTGGAACAACAACTACGCACTGAAGTTCAGGGCA GAGTCACGATTACCGCGGACAAAACACGAGCACAGCCTA CATGGAGCTGAGCAGCCTGAGATCTGAGGACACGGCCATA TATTACTGTGCGAGAATGTATAGTGGCCATGACGGCGTTGA TGTCTGGGGCCAAGGGACACTGGTCACCGTCTCTTCA	
	12	light	GAAATTGTGTTGACACAGTCTCCAGCCACCCGTCTTTGTCT TCCAGGGGAAAGAGCCACCCCTCTCCTGCAAGGGCAGTCA GAGTGTAGGAGCTACTTAGCCTGGTACCAACACAAACCT GGCCAGGCTCCCAGGCTCCTCATCTATGATGCATCCAACA GGGCCAAGGGCATCCAGCCAGGTTCACTGGCAGTGGGT CTGGGACAGACTTCACTCTCACCATCAGCAGCCTAGAGCC TGAAGATTTTGCAGTTTATTACTGTGTCAGCAGCGTAGCACCT GGCCTCCGGGAATGTTTCGGCCAAGGGACAGGGTGGAAA TCAAA	
EV-D68-71	13	heavy	CAGCTGCAGCTGCAGGAGTCGGGGCCAGGACTGGTGAAG CATTCGGAGACCCTGTCCCTCACCTGCAGTGTCTCTGGTG GCTCCATCAGCAGTGGTTTTTACTACTGGGGCTGGATCCG CCAGCCCCAGGGAAGGGCTAGAGTGGATTGGACTAT CTATGATAGTGGGAGAACCCTATGACAACCCGTCCCTCAAG AGTCGAGTCACCATATCCGACAGACGCTCAAGAAGCAGT TCTCACTGACACTGAGGTCTGCGACCGCCGCGACACGG CTGTGTATTCTGTGCGAGACACCTTACCACCTCTACGGT GACTACGTCACCCCTGATGCTTTAGATATCTGGGGCCAAG GGACAATGGTCACCGTCTCTTCA	
	14	light	GAAATTGTGTTGACGAGTCTCCAGGCACCCGTCTCTTGTCT TCCAGGGGAAAGAGCCACCCCTCTCTTGCAGGGCCAGTCAG AGCGTTAGTAGCAGCTTCTTAGCCTGGTACCAGCAGAAAC CTGGCCAGGCTCCCAGGCTCCTCATCTATGTTGCGTCCAG CAGGGCCACCGGCATCCAGACAGGTTAGAGGCAGTGG GTCTGGGACAGACTTCACTCTCACCATCAGCAGACTGGAG CCTGAAGATTTTGCAGTATATTACTGTGAGCAGTATAGTAA CTCAGCTCTGACGTTTCGGCCAAGGGACCAAGTGGAAATC AAA	
EV-D68-72	15	heavy	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTACAG CCTGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTCACTACCTATAGCATGAACGGGTCCGCCAGGCT CCAGGGAAGGGGCTGGAGTGGATTTATACATTAGTAGTG GTAGTAGTAACATATACTACGACAGCTCTGTGAAGGGCCG ATTCAACATCTCCAGAGACAATGCCAAGAACCTCACTGAATC TGCAAATGAGCAGCCTGAGAGACGAGGACACGGCTGTGTA TTACTGTGCGAGAGCCACGGACGTATAGTGAATTCTGGA GTGGTTATTAGTAGGTTTCGACCCCTGGGGCCAGGGAATCC TGGTCACCGTCTCTCTCA	
	16	light	CAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTC CTGGACAGTCGATCACCATCTCCTGCACTGGAACAGCAG TGACGTTGGTGGTTATACTATGTCTCCTGGTACCAACTGC ACCCAGGCAAAGCCCCAACTCATGATTTTGGAGTCACT TATCGGCCCTCAGGGGTTTCTAATCGCTTCTCTGGCTCCAA GTCTGGCAACACGGCCTCCCTGACCATCTCTGGGCTCCAG GCTGAGGACGAGGCTGATTATTTCTGCAGCTCCTATACAAC CAGCAACACTCTCGTGGTGTTCGGCGGAGGGACCAAGCTG ACCGTCCTA	
EV-D68-74	17	heavy	CAGGTGCAGCTGGTGCAGTCTGGGGCTGAGGTGAAGAAG CCTGGGGCTCAGTGAAGGTCTCCTGTAGGACTTCTGGAT ACACCTTCACGGCTTACTATATGCACTGGGTGCGACAGGC CCCCGGACAAGGGCTGAGTGGATGGGAAGGATCAACCC GAGCAGTGGTGGCGCACAGTATGCACAGAAGTTTCAGGGC AGGGTCACCATGACCAGGGACACGTCCATCAGCACAACCT ACATGACCCCTGAGCGGGCTGACATCTGACGACACGGCCGT GTTTTACTGTGCGAGAATGGTTGTCTGATGACCGGTGC TATTCGACCAACTACAACTTTGACCAGTGGGGCCAGGGAA CCCTGGTCACCGTCTCTCTCA	
	18	light	TCCTATGTGCTGACTCAGCCACCCCTCGGTGTGAGTGGCCC CAGGACAGACGGCCAGGATTCCTGTGGGGGAAACAACAT TGGAACATAAAGTGTTCAGTGGTACGACAGAAGCCAGGC CAGGCCCTGTGCTGGTCTCTCTAATGACAGCGACCGGC CCTCAGGGATCCCTGAGCGATTCTCTGGCTCCAAGTCTGG GAACACGGCCACCCCTGACCATCAGCAGGGTCGAAGCCGG	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence	Region
			GGATGAGGCCGACTATTATTGTCAGGTGTGGGATAGTGGT ATTGATGTCGTTTTTCGGCGGAGGGACCAAGCTGACCGTCC TA	
EV- D68- 75	19	heavy	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCCTGGTCAAG CCTGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACATCAGTCCCTATGGCATGAAGTGGGTCCGCCAGGC TCCAGGGAAGGGGCTGGAGTGGGTCTCATTCATTAGTAGT AGTAGTCGTACACATATTACGCAGACTCAGTGAAGGGCC GTTTCACCATCTCCAGAGACAACGCCAAGAATTCACTGTCT CTGCAAATGAACAGCCTGAGAGCCGAGGACACGGCTGTGT ATTACTGTGCGAGAGAGAGGGGCCATAGCACCTCGTCCTC ATACTTTGACTCCTGGGGCCAGGGAACCTGGTCACCGTC TCCTCA	
	20	light	TCCTATGTGCTGACTCAGCCACCCTCAGTGTCTATGGCCC CAGGAAAGACGGCCAGGATTACCTGTGGGGAAACAACAT TGGAACTAAAACGTGAGCTGGTACCAGCAGAAGCCAGGC CAGGCCCTGTGCTGGTCATGTATTATGATAGTGACCGGC CCTCAGGGATCCCTGAGCGATTCTCTGGCTCCAACCTCTGG GAACACGGCCACCCTGACCATCAACAGGGTCGAGGCCGG GGATGAGGCCGACTATTACTGTGCGGTGTGGGATAGTGAT ACTGATCATCGAGTGTTTCGGCGGGGGACCAAGCTGACC GTCCTA	
EV- D68- 76	21	heavy	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTAAAG CCTGGGGGGTCCCTAGACTCTCCTGTGCAGCCTCTGGAT TCACTTTCAGTAACGCCTGGATTAGCTGGGTCCGCCAGGC TCCAGGGAAGGGGCTGGAGTGGGTGGCCGTATTCAAACC AAAACGTATGGAGGGACAACAGACTACGCTGCACCCGTGA AAGGCAGATTCAACATCTCAAGAGATGATTCAAAAAACAG TTGTATCTGCAAATGAACAGCCTGAAAACCGAGGACACAG CCTTGATATTATTGTAGCACAGGACCGTATTACTATGATACTA GTGGTTACCCCCAACCCCTTGACTACTGGGGCCAGGGAAC CCTGGTCACCGTCTCCTCA	
	22	light	TCCTATGAGCTGACTCAGCCACCCTCAGTGTCCGTGTCCC CAGGACAGACAGCCAGCATCACCTGCTCTGGAGATAAATT GGGAGATAAATATGCTTGCTGGTATCAGCAGAAGCCAGGC CAGTCCCCTGTGCTGGTCATCTATCAAGATACCAAGCGGC CCTCAGGGATCCCTGAGCGATTCTCTGGCTCCAACCTCTGG GAACACAGCCACTCTGACCATCAGCGGGACCCAGGCTATG GATGAGGCTGACTATTACTGTGAGCGTGGGACAGCAGCA CTGTGGTATTCGGCGGAGGGACCAAGCTGACCGTCTCTA	
EV- D68- 80	23	heavy	GAGGTACAGATGGTGGAGTCTGGGGGAGGCCTGGTCAAG CCGGGGGGGTCCCTCAGACTCTCCTGTTAGTCTCTGGAT TCGACTTCAGTAGATACACCATGAAGTGGTTCGCCAGGCT CCAGGGGAGGGGCTGAAGTGGGTCTCGTCCATTAGTAGTA CTAGTCTTTACACATTTCTATGCGGACTCAGTGAAGGGCCGA TTCTCCATCTCCAGAGACAACGCCAGGGTTCCCTGTCTCT ACAAATGAGCAGCCTGAGACCCGAAGATACGGCTGTCTAT TATTGTGCGAGAGTCGTTGGTCCCGCCGAGTTAGATTACT GGGGCCAGGGAGTGTGGTCACCGTCTCCTCA	
	24	light	GTCACCCAGTTGACCCAGTCTCCATCCCTCCTGTCTGCTTC TGTGCGGGACAGAGTCACCATCACTTGCCGGGCAAGTCAG GACATTGGAGTTGACTTAGGTTGGTTTCAGCAGAGACCTG GGAAAGCCCCCTAACTCCTGATCTATGGTGCCTCCAGGTT GCAGAGTGGGGTCCCATCACGCTTCAGCGGGCGTGGATC TGGCACATTTTTCTACTCTCACCATCAGCAGCCTGCAGCCTG AAGATTTTACAACCTACTTCTGTCTTCAAGATTATAATTACC CTTGACGCTCGGCCAGGGGACCACGGTGGGAGTCAAA	
EV- D68- 84	25	heavy	CAGGTGCACCTGGTGCAGTCTGGGTCTGCGGTGAGGAAG CCTGGGGCCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGAT ACACCTTCACCGACTACTATATTCACTGGGTGCGACAGGC CCCTGGACAAGGCCCTTGAAGTGGATGGGATGGATCAACCT AAAACGTGGTGGCTCAAATTATACACAGAGGTTTCAGGCCAG GGTCACCATGACCTGGGACACGTCATCAGTACAGCCCTAC ATGGAGTTGAGCAGGCTGAGATCTGACGACACGGCCGTGT ATTATTGTGCGAGGGCGGGCAGAAATGGCTACGACTACTG GGGCCAGGGAACCCCTGGTCACCGTCTCCTCA	
	26	light	TCCTATGAGCTGACACAGCCACCCTCGGTGTCTAGTGTCCC CAGGACAGACGGCCAGGATCACCTGCTCTGCAGATGCATT GCCAAGCAATATGCTTATTGGTACCAACAGAAGCCAGGC CAGGCCCTGTGTTGATGATATATCAAGACTGAGAGGC CCTCAGGGATCCCTGAGCGATTCTCTGGGTCCAGTCAAG	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable	Sequence Region
				GACAACAGTCACGTTGACCATCAGTGGAGTCCAGGCAGAA GACGAGGCTGACTATTACTGTCAATCAGGAGACAGCAGTG GTACTTATCTAGTTTTTCGGCGGAGGGACCAAGCTGACCGT CCTA
EV- D68- 85	27	heavy		GAGGTACAGCTCTTGGAGTCTGGGGAGGCTTGGTACAGC CTGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATT CAAATTAGAAACTATGCCATGACCTGGGTCCGCCAGGCT CCAGGGAAGGGGCTGGAGTGGGTCTCAACTATTACTAGTG GTGGTAGTACAGAGTACGACAGCCCGTGAAGGGCCGGT TCATCATCTCCAGAGACAATTCCAAGAACAGTTATATTG CAAATGAACAGCCTGAGAGCCGACGACACGGCCGTATATT ACTGTACAGTGCCGTGGGGTAACTACAATGACTACGTGTCT GACTACTGGGGCCAGGGAACCTGGTCCCGTCTCTCA 28 light GAAGTTGTATTGACACAGTCTCCAGCCACCTGTCTTTGTCTCCAGGGCAAAGAGCCACCTCTCCTGCAGGGCCAGTCAGCGTGTGGCAACTCCTTAGCCTGGTACCAACAAAACCTGGCCAGGCTCCCAGCCTCCTCATCTATGATGCTTCCAAGAGGGCCACTGGCATCCCAGCCAGGTTCACTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCATCAGCCTAGAGTCTGAAGATTTTGCAAGTTATTACTGTCAACACATAGCACCTGGCCTCGGGGGACCTTCGGCCAAGGGACACGACTGGAGATTAA A
EV- D68- 88	29	heavy		CAGGTGCACCTGGTTGAGTCTGGGGAGGCGTGGTCCAG CCTGGGAGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCATCTTCAGTAGATATCCTATGCACTGGGTCCGCCAGGCT CCAGGCAAGGGTCTGGAGTGGGTGGCACTTATATCATATG ATGGAACAATAAAATACTACGACAGCTCCGTGAAGGGCCG ATTCAACATCTCCAGAGACAATTCCAAGAACACGCTGTTTC TCCAATGAACAGCCTGAGAGCTGAGGACACGGCTGTCTA TTACTGTGCGAGACATTTCTCCCATATAGCAGTAGTTGGT ACCAGGGCTTTAACTACTGGGCCAGGGAATCCTGGTCAC CGTCTCCTCA 30 light AATTTTATGCTGACTCAGCCCCACTCTGTGTGGAGTCTCCGGGGAAGACGGTAACCATCTCCTGCACCCGACGAGTGGCAGCATTGGCCCAACTATGTGCACTGGTACCAGCAGCGCCGGGCAGTTCCCCACCCCTATAATCTTTGAAGATAGTCAAAGACCGTCTGGGTCCCTGATCGGTTCTCTGGTCCATC GACAGCTCCTCCAATTCTGCCTCCCTCACCATCTCTGGACT GAGGACTGACGACGAGGCTGACTACTACTGTGAGTCTTAT GATAACAGCAATCGGGCTGTTGTATTCGGCGGAGGACCA AGCTGACCGTCTCA
EV- D68- 89	31	heavy		CAGGTGCAGCTGGTGGAGTCTGGGGAGGCGTGGTCCAG CCTGGGAGGTCCCTGAGACTCTCCTGTGAAGCGTCTGGAT TCCTCTTCAGTCGCTATGGCATGCACTGGGTCCGCCAGGC TCCAGGCAAGGGCTGGACTGGGTGGCAGTTATATCGTAT GATGAAAATAAGAAATATATGCAGACTCTGTGAAGGGCCG ATTCAACATCTCCAGAGACAATTCAAAGAACAGTTGTATC TGCAAGTGAACAGCCTGAGAGTCCAGGACACGGCTGTTTA TTACTGTGCGAGAGGTGTCCCGTACGGTGACACCCCTTACA GGGCTTGTCTACTGGGCCAGGGAACCTGGTCAACCGTCT CCTCA 32 light AATTTTATGCTGACTCAGCCGCACTCTGTGTGGAGTCTCCGGGGAAGACGGTAACCATCTCCTGCACCCGACGAGTGGCACCATTGGCCAGCACTATGTGCACTGGTACCAGCAGCGCCGGGCAGTGCCCCCACCCTGTAATCTATGAGGATAACCAAAGACCTCTGGGTCCCTGATCGGTTCTCTGGCTCCATCGACAGCTCCTCCAACCTCTGCCTCCCTCACCATCTCTGGAATGAACTGAGGACGAGGCTGACTACTACTGTGAGTCTTATGATAACAGCGATCGGGTGTTCGGCGGAGGACCAAGCTGACCGTCTCT
EV- D68- 95	33	heavy		CAGGTCACCTTGAAGGAGTCTGGTCTGTGCTGGTGAAAC CCACAGAGACCTCAGCTGACCTGCACCGTCTCTGGGT CTCACTCAGGAACGCTAGAATGGGAGTGAGCTGGATCCGT CAGCCCCCAGGGAAGGCCCTGGAGTGGCTTGACACATTT TTTCGAATGACGAAAAATCTACAACACATCTCTGAAGAGC AGACTCTCCATCTCAAGGACACCTCCAAAAGCCAGGTGG TCCTTACCATGACAGCATGGACCCCTTGGACACAGCCAC ATATTTCTGTGCACGGCTACTGGTGGCTGGTACTTTCTCTC CCTCTCACTACTTTGACTACTGGGGCCAGGAATCTGGTC ACCGTCTCTCA

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable	Sequence Region
	34	light	TCCTATGTGCTGACTCAGCCACCCTCAGTGTCA GTGACCC CAGGAAAGACGGCCAGGATTACCTGTGGGGGAAACAACAT TGGACTTAAAGTGTCTTCTGTAC CAGGAGAGGCCAGAC CAGGCCCTGTGGTGGTCATCTATTATGATAGCGCCCGGC CCTCAGGGATCCCTGAGCGAATCTCTGGCTCCAAGTCTGG GAACACGGCCACCCTGACCATCACCAGGTCGAAGCCGG GGATGAGGCCGACTATTTTGT CAGGTGTGGGATAGTAGT CGTAATCATCCGGTCTTCGGCGGAGGACCAAAGTACCG TCCTC	
EV- D68- 97	35	heavy	GAGCTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTACAG CCTGGGGGCTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTCAGTACCTATAGCATGAATTGGGTCCGCCAGGCT CCAGGGAAGGGCTGGAGTGGGTTTCATACATTAGTAGTA GTAGTAGTACCATACAGTACGCAGACTCTGTGAAGGGCCG ATTCAACATCTCCAGAGACAATGCCAAGAACTCACTGTATC TGCAATGAATAGCCTGAGAGCCGAAGACACGGCCGTGTA TTATTGTACGAGACAGGTGCGGGCGGATTTCAGTGGCCGC GGCTTTGACTACTGGGGCCAGGGAACCTGTCTACCGTCT CCTCA	
	36	light	TCCTATGAGCTGACTCAGCCACCCTCAGTGTCCGTGTCCC CAGGACAGACAGCCACCATCACCCTGCTTTGGAGATAAATT GGGGGATAAATATGCTTGCTGGTATCAGCAGAAGCCAGGC CAGTCCCCTGTGTGGTCATCTACCAAGATAGCAAGCGGC CCTCAGGGATCCCTGAGCGATTCTCTGGCTCCAAGTCTGG GAACACAGCCACTCTGACCATCAGCGGGACCCAGGCAATG GATGAGGCTGACTATTACT-GT- CAGGCGTGGGACAGCAGCACTGCAGTGTTCGGCGGAGGG ACCAAGCTGACCGTCCTC	
EV- D68- 98	37	heavy	GAGGTACAAC TAGTGGAGTCAGGGGAGGCTTGGTACAG CCTGGGGGCTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCAAGTTTTCCGTCTATGCCTTGAGTTGGGTCCGCCAGGCT CCAGGGAAGGGCTGGAGTGGATTTCATATATTAGTAGTA GTGGTTCTACCATATATTATTAGACTCTGTGAAGGGCCGA TTCACCATCTCCAGAGACAATGTCGGGAAC TCACTGTTTGT GCAAAATGAACAGCCTGAGAGCCGAGGACACGGGTATTTAT TACTGTGCGACAGCCCGCCACATCACC AATGATGTTTTGA TATTTGGGGCCAAGGACAATGGTCATCGTCTCTTCA GACATCCAGTTGACCCAGTCTCCATCCTTCTGTCTGCATC TGTAGGAGACAGAGTCACCATCACTTGCCGGGCCAGTCAG GGCATTAGTAGGTTTTTAGCCTGGTATCAGCAGAACCAGG GAAAGCCCCTAAGCTCCTGATCTATTCTGCGTCCACTTTAC AAAGGGGGTCCCATCAAGGTT CAGCGGCAGTGGATTTGG GACAGATTTCACTCTCACAATCAGTAGCCTTCAGCCTGAAG ATTTTGCAACTTATTACTGTCAACAAC TTAATAGTCACCCCC GAATGTTCACTTTCGGCCCTGGGACCACAGTGGATATCAA G	
EV- D68- 105	39	heavy	CAGGTGCAGCTGCAGGAGTCGGGCC CAGGACTGGTGAAG CCTTCGGAGACCCTGTCCCTCACCTGCCTGTGTCTGGTT ACTTAATCAGCAATGGTTACTACTGGGCTGGATCCGGCA GCCCCCGGGAAGGGCTGGAGTGGATTGGGAGTATCTA TCATACTAGAAGCCTACTACAACCCGTCCCTCAAGAGTC GAGTCAGCATCTCAGTAGACAGTCCAAGAACCGGTTCTC CCTGAGGCTGAGGCTGTGACCGCCGAGACACGGCCTTT TATTACTGTGCGAGAGGCC CAGGCCACTGTTATGGTGATG ACGACTGTACGCGTACTACTTTGACCACTGGGGCCAGGG AACCCCGGTCACCGTCTCCCA	
	40	light	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTTCATC TGTAGGAGACAGAGTCACCATCACTTGTCGGGCGAGTCAG GGTATTAGCAACTGGTTAGCCTGGTATCAGCAGAACCAG GGAAAGCTCCTAAACTCCTGATCTATGATGCCTCCAGTTG CAAAGTGGGGTCCCATCAAGGTT CAGCGGCAGTGGATCTG GGACAGATTCTACTCTCACCATCAACAGCCTGCAGCCTGAA GATTTTGCAACTTACTATTGTCAACAGGCTAACAGTTTCCCT TTCACTTTCGGCCCTGGGACCAAAGTGGATATCAAA	
EV- D68- 110	41	heavy	CAGGTGCAGCTGGTGCAGTCTGGGGCTGAGGTGAAGAAG CCTGGGTCTCGGTGAAGGTCTCCTGCAAGGCTTCTGGAG GCACCTTCAGCAGGTTTGCTATCAGTCTGGTGCAGACAGGC CCCTGGACAAGGGCTTCAGTGGATGGGAGGGATCCTCCCT ATCTTTGGTACAGCAAACTACGCACAGAAGTTC CAGGGCA GAGTCACGATTACCGCGGACACATCCACGAGCACAGCCTA CATGGAGCTGAGCAGCTGAGATCTGAGGACACGGCCGT	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable	Sequence Region
				GTATTACTGTGCGAGATCCCTCCCTATTGTACTAATGATG TATGCTCAAACCAGAACACATTGACTACTGGGGCCAGGG AACCCCTGGTCACCGTCTCCTCA
	42	light		TCCTATGAGCTGACACAGCCACCCTCGGTGTCA GTGTCCC CAGGACAGGCGGCCAGGATCACCTGCTCTGGAGATGCATT GCCTAAGCAATATGCTTATTTGTAC CAGCAGAAGCCAGGC CAGGCCCTGTGTTGGTGATATATGAAGACAATAAGAGGC CCTCAGGGATCCCTGAGCGATTCTCTGGCTCCAGCTCAGG GACACAGTCACGTTGACCATCAGTGGAGTCCAGGCAGAA GACGAGGCTGACTATTACTGTCAATCAGCAGACAGCAGTG GTACTTATGTGGTATTCTGGCGGAGGGACCAAGCTGACCGT CCTA
EV- D68- 111	43	heavy		CAGGTGCAGCTGCAGGAGTCGGGCCAGGACTGGTGAAG CCTTCACAGACCCTGTCCCTCACCTGCAGTGTCTCTGGTG GCTCCATCAGCAGTGGTGATTACTACTGGAGTTGGATCCG CCAGCCCCCAGGGAAGGGCTGGAGTGGAATTGGGTACAT CTATTACAGTGGGAGCACCTACTACAACCCGTCCCTCAAGA GTCGAGTTACCATATCAGTAGACAGTCCAAGAACCAGTTC TCCCTGAAGCTGAGCTCTGTGACTGCCGAGACACGGCCG TGTATTACTGTGCCAGCCGCTACGGTGACCCGATAGGGGA CAACTGGTTCGACCCCTGGGGCCAGGGAACCTGGTCAC CGTCTCCTCA
	44	light		TCCTATGTGCTGACTCAGCCACCCTCAGTGTCA GTGACCC CAGGAAAGACCGGCCAGGATTACCTGTGGGGGAAACAACAT TGGACTTAAAAGTGCTTTCTGGTACCAGGAGAGGCCAGAC CAGGCCCTGTGGTGGTCATCTATTATGATAGCCTCCGCGC CCTCAGGGATCCCTGAGCGAATCTCTGGCTCCAAGTCTGG GAACACGGCCACCCTGACCATCACCAGGGTCGAAGCCGG GGATGAGGCCGACTATTTTTGTG CAGGTGTGGATAGTAGT CGTAATCATCCGGTCTTCGGCGGAGGGACCAACTGACCG TCCTC
EV- D68- 114	45	heavy		GAGGTGCAACTGTTGGAGTCGGGGGAGGCTTGGTGCAG CCGGGGGGGTCCCTGAGACTCTCCTGCGCAGCCTCCGGA TTCAGGTTTAGCTTCTATGGCATGACCTGGGTCCGCCAGG CTCCAGGGAAGGGCTGGAGTGGGTCTCAAGTATTAGTGG TACTGGTGCTACCAGAACTGCGCAGACTCCGTGAAGGGC CGGTTCAACCATCTCCAGAGACAACCTCAAGAACACGCTGT ACCTGCAAAATGGACAGCCTGAGAGTCGACGACACGGCCGT TTTTTATTGTGTGAGACGGTTCCCGATGACCACGGTGACAA GCTTTGACTCTTGGGGCCAGGGAACCTGGTCAACGCTCTC CTCA
	46	light		CAGTCTGCCCTGACTCAGCCTCGCTCAGTGTCCGGGTCTC CTGGACAGTCAGTCACCATCTCCTGCACTGGAACCAAGCAG TGATGTTGGTGGTTATAACTTTGTCTCTGGTACCAACAAC ACCCGGGCAAAGCCCCCAAACCTCATGATTTTTGATGTCACT GGGCGGCCCTCAGGGGTCCCTGATCGCTTCTCTGGCTCCA AGTCTGGCAACACGGCCTCCCTGACCATCGCTGGGCTCCA GGCTGAAGATGAGGCTGATTATTATTGTGGCGCATATGCG GGCTTTAACGCTCTTTTCGGCGGAGGGACCAACTGACCG TCCTA
EV- D68- 116	47	heavy		GAGGTGCAGCTGGTGCAGTCGGGGGAGGCTTGGTCCGG CCGGGGGGGTCCGTGAGACTCTCCTGTGTAGCCTCTGGAT TCCCTTCAATATGTTTTGGATGGGTGGGTCCGCCAGACT CCAGGGAAGGGACTGGAGTGGGTGGCCAAATATAAAACAG GATGGCAGTGAGAAATACTATGTCAGCTCCGTGAAGGGCC GATTCGCCATCTCCAGAGACAATGCCAAGAACTCTCTCTTT CTTCAAATGGACAGCCTGAGTGTGCGGGGACACGGCCATCT ATTATTGTGTGAGAGAGGGGTGCGAAGGGTCGTGTAAG TAGTACCGGTTACTTCGACTTTTGGGGCCAGGGACAGCTG GTCACCGTCTCCTCA
	48	light		TCCTATGAGCTGACACAGCCACCCTCGATGTCA GTGTCCC CAGGACAAACGGCCAGGATCACCTGCTCTGGAGATGCAGT GCCAATAAAATATGTTTATTTGGTACCAACAGAGGTCAAGCC AGGCCCTGTATTAGTCACTATGAAGACGACAGACGACC CTCCGGGATCTCTGAGAGATTCTCTGGCTCCAGTTCAGGG ACAACGGCCACCTTGACTATCACTGGGGCCAGGTGGAGG ATGAAGGTGACTACTATTGCTATTCAACAGACAGTAGTGGT TATCAGAGAGCGTTTCGGCGGGGGACACGCTGACCGTC CTA

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence Region	
EV-D68-150	49	heavy	CGGGTGCAGCTGCAGGAGTCGGCCCCAGGACTGGTGAGG CCTTCAGAGACCCTGTCCCTCACCTGCAGTGTCTCTGGTG GCCCCATCAGCAATGGTCCTATTACTGGAGCTGGATCCG CCAGCACCCAGGGAAGGGCCTGGAGTGGATTGGATTTCATC TTTTACAGTGGGAGCACCAACTACAACCCGTCCTCCGGG GGCGCGTAACCATGGCAGTGGACACGTCTAAGAACCAGTT CTCCCTGAGGCTGAACCTCTGTGACTGCCGCGGACACGGCC GTTTATTACTGTGCGAGACATGTGGTACTGCGTCGGGGT GGTTCGACCCCTGGGGCCAGGAACCTGGTCACCGTCT CCTCA	
	50	light	GAAATTGTGTTGACACAGTCTCCAGCCACCCTGTCTTTGTG TCCAGGGGAAAGAGCCACCCTCTCCTGCAGGGCCAGTCA GAGTGTGGCACCAGCTTAGCCTGGTACCAACAGAAACCT GGCCAGGCTCCAGGGTCCTCATCTATGATGCATTCAAGA GGGCCACTGGCATCCAGCCAGGTTCACTGGCAGTGGGT CTGGGACAGAGTTCACCTCTCACCATCAGCAGCCTCGAGCC TGAAGATTTGCAAGTTTATTACTGTCAGCAGCGTAGCAGGT GGCCTCCCCGTACACTTTTGGCCAGGGGACCAAGCTGGA GATCAAA	
EV-D68-151	51	heavy	CGGGTGCAGCTGCAGGAGTCGGCCCCAGGACTGGTGAGG CCTTCAGAGACCCTGTCCCTCACCTGCAGTGTCTCTGGTG GCCCCATCAGCAATGGTCCTATTACTGGAGCTGGATCCG CCAGCACCCAGGGAAGGGCCTGGAGTGGATTGGATTTCATC TTTTACAGTGGGAGCACCAACTACAACCCGTCCTCCGGG GGCGCGTAACCATGGCAGTGGACACGTCTAAGAACCAGTT CTCCCTGAGGCTGAACCTCTGTGACTGCCGCGGACACGGCC GTTTATTACTGTGCGAGACATGTGGTACTGCGTCGGGGT GGTTCGACCCCTGGGGCCAGGAACCTGGTCACCGTCT CCTCA	
	52	light	GAAATTGTGTTGACACAGTCTCCAGCCACCCTGTCTTTGTG TCCAGGGGAAAGAGCCACCCTCTCCTGCAGGGCCAGTCA GAGTGTGGCACCAGCTTAGCCTGGTACCAACAGAAACCT GGCCAGGCTCCAGGGTCCTCATCTATGATGCATTCAAGA GGGCCACTGGCATCCAGCCAGGTTCACTGGCAGTGGGT CTGGGACAGAGTTCACCTCTCACCATCAGCAGCCTCGAGCC TGAAGATTTGCAAGTTTATTACTGTCAGCAGCGTAGCAGGT GGCCTCCCCGTACACTTTTGGCCAGGGGACCAAGCTGGA GATCAAA	
EV-D68-152	53	heavy	CAGGTGCATCTGGTGGAGTCTGGGGGAGGCGTGGTCCAG CCTGGGAGGTCCCTGAGACTCTCCTGTGCAGACTCTGGAG TCACCTTCAGTGACAATGCTTTGTACTGGGTCCGCCAGGCT CCAGGCAAGGGGCTGGAGTGGGTGCGAGTTATCTCATATG ATGGAAGCAGTAGATACTACGCAGACTCCGTGAGGGGCCG GTTCACCATATCCAGAGACAATCCAAGGACACGCTGTATC TGCAATGAACAGACTGAGAGCTGAGGACACGGCTATTTAT TACTGTGCGAGAGTCACAGCGGATTACTATGAGAGTAGTG GCAAGGTGTTTTGGGGCCAGGGAGCCCTGGTCGTCGCTC CTCA	
	54	light	GACATCCAGATGACCCAGTCTCCTTCCACCCTGTCTGCATC TGTGGGGGACAGAGTCTCCATCACTTGCCGGGCCAGTCAG AGTGTTAGGAGCTGGTTGGCCTGGTATCAGCACAAACCAG GGAAAGCCCTAAACTCCTGATCTATAAGGCGTCTAGTTTA GAAAGTGGGGTCCCATCAAGGTTCAAGCGCAGTGGATCTG GGACAGAATTCACTCTCACCATCAGCAGCCTGCAGGCTGA TGATTTTGCAACTTATTACTGCCAACAGTATCAGACTTTTTT CTGGACGTTTCGGCCAAGGGACCAAGGTGAAGTCAAA	
EV-D68-154	55	heavy	CAGGTGCAGCTGCAGGAGTCGGGGCCAGCACTGGTGAAG CCTTCGGAGACCCTGTCCCTCACCTGCAGTGTCTCTGGTG GCTCCATCAGTGATCACTACTGGAGCTGGATCCGGCAGCC CCCAGGGAAGGGACTGGAGTGGATTGGTACATCTATACC AGTGGGACCACTACAACCCCTCCCTCAAGAGTCGAG TCACCATATCAGTAGACACATCCAAGAAGCAGTTCTCCCTG AATCTGAGGTCTGTGACCGCCGACACACGGCCGTGTATT ACTGTGCGAGAAGCTAGAAACGGTGATCCGTTTCTACTAC TACCACTACATGGAGCTCTGGGGCAAAGGGACACGGTGA TCGTCTCATCA	
	56	light	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCAC CCCTGGAGAGCCGGCCTCCATCTCCTGCAGGTCAGTCAG AGCCTCCTGCAGAGTGATGGGTACAGCTATTTGGATTGGT ACCTGCAGAAGCCAGGGCAGTCTCCACAGCTCCTGATCTA TTTGGGTTCTAATCGGGCCTCCGGGGTCCCTGACAGGTTT AGTGTCAATTGGATCAGGCACATATTTTACACTGAAAATCAG	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence	Region
			CAGAGTGGAGGCTGAGGATGTTGGCGTTTATTTCTGCATG CAAGCTCTACAACTCCGTGGACGTTTCGGCCAAGGGACCA AGGTGGAATCAAA	
EV- D68- 155	57	heavy	CAGGTGCAGCTGCAGGAGTCGGGCCAGGACTGGTGAAG CCTTCGGAGACCCTGTCCCTCACCTGCAGTGTCTCGTGGCG GCTCCATCGGTGATTACCACTGGAAGTGGATCCGGCAGCC CGCCGGGAAGGGCTGGAGTGGATTGGCGGTATACATAG CAGTGGGAACACTGACTACAACCCCTCCCTCAAGAGTCGA GTCACCATGTCTAGTAGACACGTCCAAGAACCAGTTCTCCCT GAAACTGAGGTCTGTGACCGCCGCGGACACGGCCGTGTAT TACTGTGCGAGGCAGAAATGTTTTGATATCTGGGGCCAAGG GACAAATGGTCACCGTCTCTTCA	
	58	light	GACATCGTGATGACCCAGTCTCCAGACTCCCTGGCTCTGT CTCTGGGCGAGAGGGCCACCATCACTGCAAGTCCAGCCA GAGTGTTCATTCAGCTCCAACAATAAGAACTACTTAGCTTG GTACCAGCAGAAACCAGGACAGCCTCCTAAGTTGCTCATTT ACTGGGCATCTACCCGGGAATCCGGGGTCCCTGACCGATT CAGTGGCAGCGGGTCTGAGACAGATTTCACTCTCACCATC AGCAGCCTGCAGGCTGAAGATGTGCAGTTTATTTCTGTCA GCAATTTTATACTACTCCGCTCACTTTCGGCGGAGGGACCA AGGTGGAGATCAAA	
EV- D68- 156	59	heavy	CAGGTACAGATGCAGGAGTCGGGCCAGGGCTGGTGAAG GCTTCGGAGACCCTGTCCCTCACCTGCAGTGTCTCTGGTA TCTCCATCAATAACTACTATTGGAGTTGGTTCCGCCAGCCC CCCGGGAAGGGCTGGAGTGGATCGGATATGTCTATTCTA CTGGGAGTTCAGTACAATCCCTCCCTCGAGCGTCGAGC CACCATGTCTAGTAGACACGTCCAACAACAACCTCTCCCTGA GGCTGACGTCTGTGACCACTGCGGACACGGCCGTCTACTA CTGTGCGCGGGGGAGTATGCCGCATATCTGGGGCCAGGG CCTCCTGGTCACCGTCTCTCTCA	
	60	light	CAGTCTGTGCTGACTCAGCCACCCTCAGCGTCCGGGACCC CCGGACAGAGGGTCCACATCTCCTGTTCTGGTAGCACCTC CAACATCGAGACTAATTATGTATACTGGTACCAGCAGGTTC CAGGAACCGCCCCAAGCCCTCGTCTATAGGAATGATCA GGCCCCCTCGGGGGTCCCTGACCGATTCTCTGGCTCCAAG TCTGGCACCTCGGCCCTCCCTGGTTCAGTGGGCTCCGGA CCGAGGATGAGGCTGCTTATTATTGTGCAGCTTGGGATGA CAGTCTGAAAGCTCCGGTCTTCGGAGCTGGGACCAAGGTC GCCGTCTC	
EV- D68- 157	61	heavy	GAGGTGCAGCTGGTGGAGTCTGGGGAGGCTTGATAAAG CCGGGGGGGTCCCTTAGACTCTCCTGTGCAGCCTCTGGAA TCACTTTCAGTAACGCCTGGATGAGCTGGGTCCGCCAGGC TCCAGGGAAGGGGCTGGAATGGGTGGCCGTATTGAAAGC AAAATTGACGGTGGGACAATAGACTACGCTACACCCGTGA AAGGCAGATTACCATCTCAAGAGATGATTCAAAAAACAG CTGTACCTGCAAAATGAACAGCCTGAAAAACCGAGGACACAG CCGTCTATTACTGTACCAAGACAGGGCTACTATGATAGA AGTGGTTATTGGGTGCTCGGGAACCACTTTGACTACTGGG GCCAGGGAATCTGTGTCACCGTCTCCTCA	
	62	light	CAGTCTGTGCTGACGACGCGCCCTCAGTGTCTGGGGCC CCAGGGCTGAGGGTCACCATCTCCTGTACTGGGAGCAGCA CCAACATCGGGGAGGTTATGATGTACTGGTACCAGCA CCTTCAGGAACAGCCCCAACTCCTCATCTATGGTAACA GCAATCGGCCCTCAGGGGTCCCTGACCGATTCTCTGGCTC CAAGTCTGGCACCTCAGCCTCCCTGGCCATCACTGGGCTC CAGGCTGACGATGCGGCTGATTATTACTGCCAGTCTATG ACAGAAAGCCTGAGTACTTATGTCTTCGGAACCTGGGACCAA GGTCACCGTCTCA	
EV- D68- 158	63	heavy	GAGGTGCAGCTGTTGGAGTCTGGGGAGGCTTGGTACAG CCTGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTTAGCAGCTATGCCATGAACCTGGTCCGCCAGGC TCCAGGGAAGGGCTGGAGTGGGTCTCAGGTATTAGTGGT ACTACAGGTAGCACATACTACGACAGCTCCGTGAAGGGCC GGTTCACCATCTCCAGAGACAATTCCAAGAACACGGTGCAT CTGCAAAATGAACAGCCTGAGAGCCGAGGACACGGCCGTAT ATTACTGTGCGAAAGATTCTCACTCCATGATAGTAGTTGAT CATGCTTTTGATATCTGGGGCCAAGGGACAATGGTCACCG TCTCTTCA	
	64	light	TCCTATGTGCTGACTCAGCCACCCTCGGTGTCTAGTGGCCC CAGGACAGACGGCCAGGATTACCTGTGGGGGAAACAATAT TGGAACTAAAGTGTGCACTGGTACCAGCAGAGGCCAGGC	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence	Region
			CAGGCCCTGTGCTGGTCTCTATGATGATAGCGACCGGC CCTCAGGGATCCCTGAGCGATTCTCTGGCTCCAACCTCTGG GAACACGGCCACCTTGACCATCAGCAGGGTCGAAGCCGG GGATGAGGCCGACTATTACTGTCAAGGTGTGGATAGTTAT AATGTTCAATTATGTCTTCGGAACAGGGACCAAGGTCACCGT CCTA	
EV- D68- 159	65	heavy	GAGGTGCAGCTGGTGGAGTCTGGGGGGGCTGGTCAAG CCGGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTCAGTACTTATATCATGACCTGGGTCCGCCAGGCC CCAGGGAGGGGGCTGGAGTGGGTCTCATCCATTAGTACCA GTAGTGTTTACACATTCTATGCAGATTCAGTGAAGGGCCGG TTCACCATCTCCAGAGACAACGCCAAGAATTCAAGTGTATCT GCAGATGAACAGCCTGAGAGCCGACGACACGGCTGTTTAT TACTGTGCGAGGGAAGAAGGGTTTCGAGCTTATAACCTATA CTGGGGCCAGGGAACCTGGTCACTGTCTCCTCA	
	66	light	CAGTCTGTGCTGACTCAGCCACCGTCAGCGTCTGGGACCC CCGGGCAGAGGGTCACCATCTCTTGTCTGGAAGCAGCTC CAACATCGAATAACAATTATGTTTACTGGTACCAGAAATTCCC AGGAACGGCCCCCAAACTCCTCATCTATAAAAAAATCAGC GGCCCTCAGGGGTCCCTGACCGATTCTCTGGCTCCAAGTC TGGCACCCTCAGCCTCCCTGGCCATTAGTGGGCTCCGGTCC GAGGATGAGGGTGATTATTACTGTGCAGCATGGGATGACA TCCTGAGTGGTGTGGTTTTTCGGCGGGGGACCAAGCTGAC CGTCCTC	
EV- D68- 160	67	heavy	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAG CCTGGGAGGTCCAGAACTCTCCTGTGCAGCCTCTGGAT TCACGTTCAAGTAGGTTTGGTATGCAGCTGGGTCCGCCAGGC TCCAGGCAAGGGGCTGGAGTGGGTGGCAGTTATTTCTGTTT GATGGAAGTAATAGATACTACGCAGACTCCGTGAAGGGGC GATTCAACATCACAGAGACAATTCCAAGAACACATTGTAT CTGCAATGAACAACCTGAGACCTGAGGACACGGCTGTAT ATTACTGCGCGAGAGATTGGGATAGGCTGGTTCGTTCGGC GGTTGGCTACTGGGGCCAGGGAACCTGGTCAGCGTCTC CTCA	
	68	light	CAGTCTGCCCTGACTCAGCCTCGCTCAGTGTCCGGTCTC CTGGACAGTCAGTCACCATCTCCTGCAGTGGAAACAGTAAT GATGTTGGTGGTTATAAATTTGTCTCCTGGTATCAGCAACA CCCAGGCAAGCCCCCAAACTAATGATTTTGTATGTCATTA GGCGGCCCTCAGGGGTCCCTGGTCGCTTCTCTGGCTCCAA GTCTGGCGACACGGCTCCCTTATCATCTCTGGACTCCAG GCTGAGGATGAGGCTGATTATTACTGCTGCTCATATGCAG GCACCTACACCTGGGTATTTCGGCGAGGGACCACACTGAC CGTCCTA	
EV- D68- 161	69	heavy	CAGGTGCACCTGCAGGAGTCGGGGCCACGACTGGTGAAG CCTTCGGAGACCCTGTCCCTCACCTGCAGTGTCTCTGGTG GCTCCGTGAGCACTGCCACTTACTACTGGAGCTGGATCCG GCAGTCCCAGGGAGGGGACTGGAGTGGATAGGATATATC TATTCAGTGGTAACACCAACTACAACCCCTCCCTTAAGAG TCGAGTCAACATTTCTTTAGACACGCCAACCAACGAGCTCT CCCTGACGTTGACCTCTGTGACCGCTGCGGACACGGCCAT TTATTATTGTGAGAGCGCTTACGTATTCTGAGTATTGAGA GGAATACTACGCTATGGACGCTCTGGGGCCAAGGGACCCC GGTACCGTCTCCTCA	
	70	light	GAAGTTGTGTTGACGCAGACTCCAGGCACCCCTGTCTTTGT CTCCGGGGGAAGGAGCACCCCTCTCCTGCAGGGCCAGTC AGAGGGTTGTCAACAACACTTAGCCTGGTACCAGCAGAG AGCTGGCCAGGCTCCAGGCTCCTCATTTTGGTGCATCC AACAGGGCCACTGGCATCCAGACAGGTTCAAGTGGCAGTG GGTCTGGGACAGACTTCACTCTCACCATCAGGAAGCTGGA GCCCCAAGATTTTGAGTGATTACTGTCAACAATATGGTA GCCCGTGGACGTTTCGGCCACGGGACCAAGGTGGAATGA AA	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence Region	
EV-D68-162	71	heavy	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAG CCTGGGAGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTCAGTAGTTATGCTATGCATTGGGTCCGCCAGGCT CCAGGCAGGGGCTGGAGTGGGTGGCAGTTATATCATATG ATGCAAGCAAGAAATACCACGCAGACTCCGTGAAGGGCCG ATTCACAATCTCCAGAGACAGTTCGAAGAACACGCTGTTTC TGCAATGAATAGCCTGAAACCTGAGGACACGGCTGTGTA TTACTGTGCGAGAGATCATGTCCCCCAAGGATTGCAGT GATGGTAATTGCCACTCGGACTACGGTATGGACGTCTGGG GCCAAGGGACCACGGTCACCGTCTCCTCA	
	72	light	GACATCCAGATGACCCAGTCGCTTCCACTATGTCTGCTTC TGTAGGAGACAGAGTCACCATCACTTGCCGGGCCAGTCAG AGTATTAGTAAGTGGTTGGCTTGGTATCAGCAGAAGCCAG GGAAAGCCCTAAACTCCTGATCTATAAGGCGCTACCTTA CAAACCTGGGGTCCCATCAAGGTTACGCGGCAGTGGATCTG GGACAGAATTCACCTCTCACCATCAACAGCCTGCAGCCTGAT GATTTTGCAACTTATTACTGCCAACAGCATAATAGTTATTTCG TACACTTTTGGCCAGGGGACCAGGTGGAGATCAAG	
EV-D68-163	73	heavy	CAGGTCCAGCTTGTGCAGTCTGGGGCTGAGGTGAAGAAGC CTGGGGCCTCAGTGAAGATTCTTGCAGGGCTTCTGGATA CTCCTTCACTAACTTTGCTGTGCATTGGGTGCGCCAGGCC CCCGGACAAAGACTTGAGTGGATGGGATGGATCAACCCGTG GCAATAGAAACACAAAGTATTACACAACCTTTCAGGGCAGA GTCACCATTACCAGGGACACATCCGCGAACACAGCCTACA TGGAACTGAGCAGCCTGAGATCTCAAGACACGGCTGTGTA TTACTGTGCGAGACTTCCGATAGCAGCAGCTGGCAGGGGC TGGTTGCGACCCCTGGGGCAGGGAACCCCTGGTCACCGTCT CCTCC	
	74	light	GAAATTGTGTTGACGCAGTCTCCAGGCACCCGTGCTTTGTCT TCCAGGGGAAAGAGCCACCCCTCTCCTGCAGGGCCAGTCA GAGTGTATATCAGCACCTACTTAGCCTGGTACCAGCAGAGA CCTGGCCAGGCTCCAGGGTCTCATCTATGATGTATCCA CCAGGGCCACTGGCATCCAGACAGGTTTCAGTGGCAGTG GGTCTGGGACAGACTTCACTCTCACCATCAGCAGACTGGA GCCTGAAGATTTTGAGTCTATTTCTGTCAACAGTATGGTA GTTCAACCGGCGACGTTTCGGCCAAAGGACCAAGGTGGAAAT CAAA	
EV-D68-164	75	heavy	GAAGTGCAGCTGGTGGAGTCTGGGGGAGACTTGGTACAG CCTGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTCAATACTTATGGCATGAACGGGTCCGCCAGGCT CCAGGGAAGGGACTGGAGTGGGTTTCATACATTAGTAGTG CCACCACTACCTTCTACTACGCAGACTCTGTGAGGGGCCG ATTCACCATCTCCAGAGACAATGCCAAGAATTCATATTTCT GCACATGAAGAGCCTGAGAGACGAAGACACGGCAGTTTAT TACTGTGCGAGAGTCTATATGCTTCGCGGAGCGAGTAT GGACGTCTGGGGCCACGGGACCACGGTCACCGTCTCCTC A	
	76	light	GAAATTGTGTTGACACAGTCTCCAGCCACCCGTGCTTTGTCT TCCAGGGGAAAGAGCCACCCCTCTCCTGCAGGGCCAGTCA GAGTGTGGCACCTACCTGGCCTGGTACCAACAGAAACCT GGCCAGGCTCCAGGCTCCTCATCTATGATTAGCCAAACA GGGCCACTGGCATCCAGCCCGGTTTCAGTGGCAGTGGGT CTGGGACAGACTTCACTCTCACCATCAGCAGCCTAGAGCC TGAAGATTTTGAGTTTATTACTGTCAGCTGCGCATCACCT GGCCTCCTATATTCATTTTCGGCCCTGGGACCAAAGTGGA TGTCAA	
EV-D68-165	77	heavy	CAGGTGCACCTGGTGCAGTCTGGGGCTGAGGTGAAGAAG TCTGGGGCCTCAGTGAAGTCTCCTGCAAGACTTTTGGAT ACACCTTCACCGCCTACTATATGCACTGGGTGCGACAGGC CCCTGGACAGGGGCTGAGTGGATGGGATGGATCAACCC TATCAGTGGTGGCACAACTATGCACAGAAGTTTCAGGGC AGGGTCTCCATGACCAGGGACACGTCCATCAGCACAGCCT ACATGGGCTGAGCAGGCTGAGACCTGACGACACGGCCG TCTATTACTGTGCGAGAGTGAAGTGTAGTAGTCCAACTGC TATGGGAACTTTGACTACTGGGGCCAGGGTACCCTGGTCA CCGTCTCCTCA	
	78	light	CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTC CTGGAGAGTCAGTCACCATCTCCTGCACTGGAACACAGCAG TGACGTTGGTGGTTATAACTATGTCTCCTGGTACCAACAAC ACCCAGGCAAAGCCCCCAAACATCATGATTATGAGGTCAGT AAGCGGCCTCAGGGGTCCCTGATCGCTTCTCTGGCTCCA AGTCTGGCAACACGGCCTCCCTGACCGTCTCTGGGCTCCA	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence	Region
			GGCTGAGGATGAGGCTGATTATTACTGCAGCTCATATGCA GGCAGCAACAATTGGTATTTCGGCGGAGGGACCAAGCTGA CCGTCTCA	
EV- D68- 166	79	heavy	GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAG CCCGGTGAGTCTCTGAAGATCTCCTGTAGGGTCTCTGGAT ACAGGTTTACCAACTACCGGATCGGCTGGGTGCGCCAGAT GCCCGGAAAGGCTTGGAGTGGATGGGATCATCTATCC CGGTGGCTCTGATACCAGATACAGTCCGTCCCTCCAAGGC CAGGTCAACCATGTCAGTGCACAAGTCCATCAGCACCGCCT ACCTGATGTGGAGCAGCCTGAAGGCCTCGGACACCGCCAT GTATTACTGTGCTCGACAGACCACTCAAATAGTGGCTACG ATAGATGGTTTGACTCCTGGGGCCAGGGAACCCACGTCAC CGTCTCTCA	
	80	light	CAGTCTGTACTGACTCAGCCACCCTCAGCGTCTGGGACCC CCGGGCAGAGGGTCACCATCTCTTGTCTGGAAGCACCTC CAGCATCGGAAGTAATATTGTAATTTGGTACCAACACCTCC CAGGAACGGCCCCCAAACCTCATCTATTAATAATCAG CGGCCCTCAGGGGTCCCTGACCGATTCTCTGGCTCCAAGT CTGGCACCTCAGCCTCCCTGGCCATCAGTGGGCTCCAGTC TGAGGATGAGGCTGACTATTACTGTGCAGCATGGGATGAC AGCCTGAATGGTTGGGTGTTTCGGCGGAGGGACCAAGCTG ACCGTCTG	
EV- D68- 181	81	heavy	GAGGTGCAGGTGTTGGAGTCTGGGGGAGGCTTGGTACAG CCTGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAA TCACCTTTAGCAGGCATATGAGCTGGGTCCGCCAGGC TCCAGGAAGGGGCTGGAGTGGGTCTCAGCTATTAGTGG GAGTGGTGGTAGCACATATCATGCAGACTCCGTGAAGGCG CGCTTCACCATCTCCAGAGACAGTTCCAGAGCACGCTGT ATCTGCAATGAACAGCCTGAGAGCCGAGGACACGGCCGT TTATTACTGTGCGATCTCGTGCCATTATTACGATTTTGGTA GTGGTTTCAACACCTTTTGACTTCTGGGGCCAGGGAACC CTGGTCACCGTCTCTCA	
	82	light	GAAATAGTGATGACGAGTCTCCAGCCTCCCTGTCTGTGT CTCCAGGGGAAAGAGTCACCTCTCCTGCAGGGCCAGTCA GAGTGTGGCAGCACCTTAGCCTGGTACCAGCACAAACCT GGCCAGGCTCCAGGCTCCTCATCTCTGGTGCATCCACCA GGGCCACTGGTGTCCAGCCAGGTTCACTGGCAGTGGGT CTGGGACAGAGTTCACTCTCACCATCAGCAGTCTGCAGTC TGAAGATTTGCAAGTTTACTACTGTCAACAGTATATTAAGT GCCTCCGTGGACGTTTCGGCCAAGGGACCAAGGTGGAAT CAAA	
EV- D68- 183	83	heavy	GAGGTGCGGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAA CCTGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTCAATACATATTCATGAGCTGGGTCCGCCAGGCT CCAGGGAAGGGCTGGAGTGGGTGCGCTCCATTAGTAGTA CCGGAAGTTACATATACAATGCAGACTCACTGAAGGGCCG ATTACCATCTCCAGAGACAACGCCAAGAACTCCCTGTTTC TGCAATGAACAGCCTGAGAGTCAAGACACGGCTGTGTA TTACTGTGTGAGATTACCATGACTACAGTGAATACTTTG ACTCATGGGGCCAGGGAACCTGGTCACCGTCTCTCTCA	
	84	light	CAGTCTGCCCTGACTCAGCCTCGCTCAGTGTCCGGGTCTC CTGGACAGTCAGTCACCATCTCCTGCACTGGAACACAGCAG TGATGTTGGTGCTTATAGCTATGTCTCCTGGTACCAACAAC ACCCAGGCAAGCCCCCAAACCTCATGATTATGATGTCTAC AGGCGGCCCTCAGGGGTCCCTGGTCTCTCTGGCTCCA AGTCTGGCAACACGGCTCCCTGACCGTCTCTGGGCTCCA GGCTGAGGATGAGGCTGATTACTACTGTCTGCTCACATGCA GGCAGTCACACCTGGGTGTTTCGGCGGAGGGACCAAGGTG ACCGTCTCA	
EV- D68- 185	85	heavy	CAGCTGCAGGTGGTGGCGTCTGGGGGAGGCGTGGTCCAG CCTGGGAGGTCCCTGAGACTCTCCTGTAAAGCCTCTGGAT TCACGTTCAACCAATTATGGCATGCACTGGGTTCCGCCAGGC GCCAGGCAAGGGGCTGGAGTGGGTGGCTTTTATATCATAC GATGGAGGTAATAAATTTTATGCAGACTCTGTGAAGGGCCG ATTACCATCTCCAGAGACAATTCAGGAACACGGTTTATC TGCAATGAACAGCCTGAGAGTGGCGGACACGGCTATGTA TTACTGTCCGAAGGTCATCCCCACCGTATTATGATAGTA GTGGTGATGATGCTTTTGATATCTGGGGCCAAGGACAAAT GGTCGCCATTTCTCA	
	86	light	GAAATAGTGATGACGAGTCTCCAGCCACCCTGTCTGTGT CTCCAGGGGAAAGAGCCACCCTCTCCTGTGGGCCAGTCA	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence	Region
			GAGTATTAGCCGCACTTAGCCTGGTATCAGCAAAACCTG GCCAGGCTCCCCGACTCCTCATCTATGGTGCATCCACCAG GGCCACTGGTATCCCCGCCAAGTTCAGTGGCAGTGGGTCT GGGACAGACTTCACTCTCACCCTCAGCAGCCTGCAGTCTG AAGACCTGCAGTTTATTACTGTCAGCAGTATAGTAAGTTG CCTATCACCTTCGGCCAAGGGACACGACTGGAGATTAA	
EV- D68- 200	87	heavy	GAGGTGCAGCTGGTGGAGTCTGGGGAGGCTTGGTACAG CCGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCCCCTTTCAGTAGTTATAGCATGAGCTGGGTCCGCCAGGC TCCAGGGAAGGGCTGGAGTGGGTCTCATACATAAGTGGT AGTGGTGGTGACATATACTACGCAGACTCTGTGAAGGGCC GATTACCATCTCCAGGGACAATGCCAGGAACCTCACTGTCT CTGCAATGAACAGCCTGAGAGCCGACGACACGGCTGTGT ATTACTGTGCGAGAGGGCTGGTAGCAAACTGGTACAAG GTACTTTGACTACTGGGGCCAGGGAACCTGGTCACCGTC TCCTCA	
	88	light	GAAATTGTGTTGACACAGTCTCCAGCCACCTGTCTTTGTC TCCAGGGGAAAGAGCCACCTCTCCTGCAGGGCCAGTCA GAGTGTTAGGAGTTACTTAGCCTGGTACCAACAGAGACCT GGCCAGGCTCCAGGCTCCTCATCTACGATGCATCCAACA GGGCCACTGGCATCCAGTCAGGTTCACTGGCAGTGGGT CTGGGACAGACTTCACTCTCACCATCAGCAGCCTAGAGCC TGAAGATTTGCAGTTTATTACTGTCAGCAGCGTAGCTACT GGCTCCGTTCACTTTTCGGCGAGGGACCAAGGTGGAGAT CAAA	
EV- D68- 208	89	heavy	CAGGTGCAGCTGGTGCAGTCTGGGGCTGAGGTGAAGAAG CCGGGGTCTCGGTGAAGGTCTCCTGCAAGGCTTCTGGAG GCACCTTCAGGAGGTTTGCTATCAGCTGGGTGCGACAGGC CCCTGGACAAGGGCTTGAGTGGATGGGAGGGATCATCCCT ATCCTAGGTAGAGGAAAGTACGCACAGAAGTTCCAGGGCA GAGTCAGGATTACCGCGGACGAATCCACGAGCACAGCCTA CATGGAGCTGAGCAGCCTGAGATCTGAAGACACGGCCGTG TATTACTGTGCGAGATTTATTTTCGACAGCCTCCTATGTTCC GGGGACCTTCGAGGACGTCTGGGGCCAAGGGACCACGGT CACCGTCTCCTCA	
	90	light	GACATTGTGTTGACACAGTCTCCAGCCACCTGTCTTTGTC TCCAGGGGAAAGAGCCACCTCTCCTGCAGGGCCAGTCA GAGTGTTAGCAGCTACTTAGCCTGGTACCAACAGAAACCT GGCCAGGCTCCAGGCTCCTCATCTATGATGCATCCAACA GGGCCGCTGGCATCCAGCCAGGTTCACTGGCAGTGGGT CTGAGACAGACTTCACTCTCACCATCAGCAGCCTAGAGCC TGAAGATGTGCGGTTTATTACTGTCAGCAGCGTAGCGACT GGCTCCGGGACTTTTGCCAGGGGACCAACGTGGAGA TCAA	
EV- D68- 210	91	heavy	GAGGTGCAGCTGGTGGAGTCTGGGGAGGCCTGGTCAAG CCGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTCAGGAACATAACATCAACTGGGTCCGCCAGGCT CCAGGGAAGGGGCTGGAGTGGGTCTCATCCATTAGTAGTA CTGGTAGTTACATACACTACGCAGATTTAGTGAAGGGCCGA TTCACCATCTCCAGAGACAACGCCAAGAAGTCACTGTATCT GCAAAATGAACAGCCTGAGAGTCGAGGACACGGCTGTATAT TACTGTGCGCAATGGTTAGGAATACGGTGACTGCCTTTG ACTACTGGGGCCAGGGAACCTGGTCTCCGTCTCCTCA	
	92	light	CAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTC CTGGCCAGTCGATCACCATCTCCTGCCTGGAACAGCAG TGATGTTGGTGGCTATAACTTTGTCTCCTGGTACCAACAAC AGCCAGGCAGAGCCCCAACTCCTTATTATGAAGTCATT AAGCGGCCCTCAGGGGTTTCTGATCGCTTCTCTGGCTCCA AGTCTGGCGACACGGCTCCCTGACAATCTCTGGGCTCCA GGCTGAGGACGAGGCTGATTATTACTGTCTCATATGGG GGTAACAACCTCTGGATGTTTCGGCGAGGGACCATGCTGA CCGTCCTA	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence Region	
EV-D68-219	93	heavy	CAGGTCCAGGTGGTACAGTCTGGGGCTGAGATGAAGAAGC CTGGGGCCTCAGTGAAGGTCTCCTGCAAGGTTTCCGGATA CAGGCTCATTTGATTTACCCTTGCACTGGGTGCGACAGGCT CCTGGAAAAGGGCTTGAGTGGATGGGACTTTTGTATCCTG AAAAGGCTGAAGCCATCTACTCACAGAAATTCAGGACAAG GTCACCATAAGCGAGGACACATCTATCGACACAGCCTACAT GGAACCTGAACAGCCTGCGCTCTGAAGACACGGCCGTCTAC TACTGTGCAACTTGGGGAGTGAAGTGGTGAATGGGAGAA GGGACTACTTTGACTCCTGGGGCCAGGGAACCTGGTCAC CGTCTCCTCA	
	94	light	TCCTATGAGCTGACACAGCCACCCTCGGTGTGAGTGTCCC CAGGACAGACGGCCAGGATCAGATGCTATGCAGATGTGTT GTCAAACCAATATACTTACTGGTATCAACAGAAGCCAGGCC AGGCCCCGTGTGTTGGTGATATATAAGACACTGAGAGGCC CTCAGGGATCCCTGAGCGATTTGCTGGCTCCAGCTCAGGG ACAACAGTCACCTTAGTCATCAATGGAGTCCGGGCAGAGG ACGAGGCTTACTATTACTGTCAATCAGCCGACAAACACCAGA ATTACGGTTTTCGGCGAGGGACCAAGCTGTCCGTCTCA	
EV-D68-220	95	heavy	GGGGTGCAGCTGGTGGAGTCTGGGGGAGGATTGATACAG CCTGGAGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTCAGTAGTTTGAATGAAGTGGGTCCGCCAGGCT CCAGGGAAGGGGCTGGAGTGGGTTTCATACATTAGTACTA GTGGTAGTACCATATACTACGCACTCTGTCAAGGGCCG ATTCACCATCTCCAGAGACAACGCCAGGAACCTCACTGTCTC TGCAATGAACAGCCTGAGAGCCAGGACACGGCTGTTTA TTACTGTGCGAGAGAGCTGAGGGATTGTAGTGCTCTTACG TGCCCCGAAGGGGAGATGCTTGTGATTTCTGGGGCCGTG GGACAAGGGTCACCGTCTCTTCA	
	96	light	GATGTTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCA CCTTGACAGTCGGCCTCCATCTCCTGCAGGTCTAGTCAA AGCCTCGTATACAGTGATGGAACACCTTCTGAATTGGTT TCAGCAGAGGGCAGGCCAATCTCCAAGGCGCTAATTTAT AAGGTTTCTAACCGGGACTCTGGGGTCCAGACAGATTCA GCGGCAGTGGGTGAGGCACTGATTTCACTGAAAATCAG CAGGGTGGAGGCTGAGGATGTTGGGGTTATTACTGCATG CAAGGTACACACTGGCCTCGCACTTTCGGCCCTGGGACCA AAGTGGATATCAA	
EV-D68-221	97	heavy	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTGGTCAAG CCGGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCAGTTTCAGTGTCTATCCCATGAAGTGGGTCCGCCAGGCT CCAGGGAAGGGGCTGGAGTGGGTCTCATCCATAAGTAGTA GTAGTCGTTACATATCCTACGCACTCACTGAGGGGTGCG AATCACCATCTCCAGAGACAACGCCAAGAACTCACTGTATC TGCAATGAACAGCCTGAGAGTCGAGGACACGGCTGTGTA TTACTGTGTGAAGTCGGGGGTAGCAACACCAATACTACT TTGACTACTGGGGCCAGGATCCCTGGTCAACGCTCTCCTC A	
	98	light	CAGTCTGTGCTGACTCAGCCACCCTCAACGTCTGGGATCC CCGGGCAGACGGTCACCATCTCTTGTCTGGAAGCAGGTCC CAACATCGGAAGTTATATGTTAACTGGTACGAGCAACTCC CAGGAACGGCCCCAACTCGTCATCTTAAATAATAATCAG CGTCCCTCAGGGGTCCCTGACCGATCTCTGGCTCCAAGT CTGGCACCTCAGCCTCCCTGGCCATCAGTGGGCTCCAGTC TGAGGGTGAAGCTGATTATTACTGTGCAGCATGGGATGAC AGCCTGAATGGTGTGGTTTTCGGCGGAGGGACCAAGCTGA CCGTCCTCA	
EV-D68-224	99	heavy	CAGGTTTCAGTTGGTGCAGTCTGGAGCTGAGGTGAAGAGGC CTGGGGCCTCAGTGAAGGTCTTCTGCAAGGCTTCTGGTTA CACCTTTACGAATTATGACATCATCTGGGTGCGACAGGCCC CTGGACAAGGGCTTGAGTGGGTGGGTGGATCAGCACTTA CAATGGTAACACAACTATGAACAGAACCCTCCAGGGCAGA GTCACCATGACCACAGACACATCCAGGACAGCCTACA TGGAGCTGAGGAGCCTGAGATCTGACGACACGGCCGTTTA CTATTGTGCGAGAGAGCGTTGTAGTACTAGTACCTGCTATA GTCGTTATGCTGACTACTGGGGCCAGGGAACCTGGTCAC CGTCTCCTCA	
	100	light	GACATCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATC TGTGGGAGACAGAGTCACCATCACTTGCCGGGCAGAGTCAG AGCATTAATATCTATTGAAATTGGTATCAGCAAAAACCGGG GAAAGCCCCTAAGGTCCTGATCTATGTGTCATCCAGTTTGC AAAGTGGGGTCCCATCAAGGTTCCGTGGCAGTGGGTCTGG GACAGATTTTCATCCTCACCATCAGCAGTCTGCACCTGAAG	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence Region	
			ACTTTGCAACCTACTACTGTCAACAGAGTTACAGGTCCCCT CGGACGTTTCGGCCAAGGGACCGAGGTGGAATCAAA	
EV- D68- 225	101	heavy	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGCACAG CCTGGAGAGTCCATGAGACTCTCCTGTGTAGCCTCTGGAT TCACCTTAAGTCGTTATGAAGTGAAGTGGTCCGCCAGGC TCCAGGGAAGGGCTAGAGTGGCTTTCATACATTAGCAGT GGTGGTCTCTCCATATACTACGCAGACTCTGTGAAGGGCC GATTCACGATCTCCAGAACAGCGCCGAGAACTCACTGGA ACTACAAATGTCCACCCTGAGGACCGAGGACACGGCTGTT TATTATTGTATGAGAGAGGGTCTTACTTATTATGATAGTACT ATTTGGGGCCAGGGAACCTGGTCGCCGTCTCCTCA	
	102	light	CAGAATGTGCTGACTCAATCGCCCTCTGCCTCTGCCTCCCT GGGAGCCTCGGTCAAACCTCACCTGCACCTGAAACAGTGGG CACAGCAGATACGCCATCGCATGGCATCAACATCAGCCAC AGAGGGGCCCTCGGTTCTGTATGAAGATTAATAGTGATGG CAGACACATCAGGGGGGACGGCATCTCTGATCGCTTCTCA GGCTCCGCCCTTGGGGCTGAGCGTCATCTACCATCTCCA GCCTCCAGCCTGAGGATGAGGCTGACTATTATTGTCTAGAC CTGGGGCACTGGCTTTCGGGTGTTTCGGCGAGGGACCAA ACTGACCGTCCTA	
EV- D68- 227	103	heavy	GAAGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTACAG CCTGGCAGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTTGATGAATATGCCATGCACTGGGTCCGGCAAGTT CCAGGGAAGGGCCTGGAGTGGGTCTCAGTATTAGTTGGA ATGGTGGTAGCAAGGCTATGCGGACTCTGTGAAGGGCCG ATTACCATCTCCAGAGATAACGCCAGGTATCCCTGTCTC TGCAATGAACAGTCTGAGAAGTGAAGACACGGCCTTATAT TACTGCGCAAAAGATGATTACGAGGGGGCTGGTTTGTATAT CTGGGGCCAAGGGACAGTGGTCACCGTCTCTTCA	
	104	light	GAAATTGTGTTGACACAGTCTCCAGCCACCCTGTCTTTGTC TCCAGGGGAAAGAGCCACCCTCTCCTGCAGGGCCAGTCA GAGTGTAGCAGCTACTTAGGCTGGTACCAACAGAAATCTG GCCAGGCTCCCAGGCTCCTCATCTATGGTGCATCCAGCAG GGCCACTGGCATCCAGCCAGGTTCACTGGCAGTGGGTCT GGGACAGACTTCACTCTCACCATCAGCAGCCTAGAGCCTG AAGATTTTGCAATTTATTACTGTCTAGCAGCGTAGCAACTGG CCTATCACCTTCGGCCAAGGGACACGACTGGAGATTAAA	
EV- D68- 228	105	heavy	CAGGTGCAGCTGCAGAGTCCGGGCCAGGACTGGTGAAG CCTTCGGAGACCCTGTCCCTCACCTGCACTGTCTCTGGCT ATTTAATTAGCAATGGTTACTACTGGGGCTGGATCCGGCAG TCCCCCGGAAGGGCTGGAGTGGATTGGGAGTATCTATT ATACTAGGACACCTACTACAAGTGGTCCCTCAAGAGTCGA ATCACCATATCAGTGGACACGTCCAAGAAACAGTTCTCCCT GAAAGTTGATTCTGTGACCGCCGACAGACCGGCCGTCTAT TACTGTGTGAGACATGAGGGTCTTGCAATGATGGAAGCT GTTACGGCTCGTTCGTTGACAACTGGGGCCAGGGAACCT GGTCACCGTCTCCTCA	
	106	light	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATC TGTAGGAGACAGAGTGACCCTCACTGTGCGGCGAGTCAG GATATTAGCAGCTGGTAGCCTGGTATCAGCAGAAACCAG GGAAAGCCCTAAGCTCCTGATCTATGCTGCATCCAGTTTG CAAAGTGGGGTCCCATCAAGGTTACGCGCAGTGGATCTG GGACACATTCACTCTCACCATCAGCAGCCTGCAGCCTGA AGATTCGCAACTTACTTTTGTCAACAGGCTGACAGTTTCAT CACTTTCGGCGAGGGACCAAGGTGGAGATCAAA	
EV- D68- 231	107	heavy	GAGGTGCAACTGTTGGAGTCTGGGGGAGGCTTGGTACAG CCGGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGATT TCACCTTCAGCAGCTACACCATGGCCTGGGTCCGCCAGGC TCCAGGGAAGGGCTTGAATGGGTCTCATCTATTAGTGGT GATGGTGTAGCACAAAAACGCAGACTCCGTGAAGGGCC GATTCCTCGTCTCCAGAGACAATTCCAAGAACACACTTTT CTGCAACTGAACAGTCTGAGAGCCGAGGACACGGCCTTTT ATTACTGTGCGAGGGGGGACCTTCCATAACTGGTACTT CGATCTCTGGGCGGTGGCGTCTGGTCACTGTCTCCTCA	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence	Region
	108	light	GAAATAGTGATGACGCAGTCTCCAGCCACCCGTGTCTGTGTCTCCAGGGGAAACAGCCACCCCTCTCCTGCAGGGCCAGTCA GAGTATTGGCGACAACCTTAGCCTGGTATCAGCAGAAACCTGGCCAGGCTCCAGGCTCCTCATCTCTGGTGCATCCACAA GGCCCACTGATTTCAGCCAGGTTCCGTGGCAGTGGGTC TGGGACAGAGTTCACTCTCACCATCAGCAGCTGCAGTCT GAAGACTTTGCAGTTTATTACTGTGTCAGCAGTATAAAACTG GCCTCGGACGTTCCGGCCAGGGACCAAGGTGGAAGTCAG A	
EV-D68-234	109	heavy	GAGGTGCAGCTGTTGGAGTCTGGGGGAGGCTTGGTACAG CCGGGGGGGTCCTGAGACTCTCCTGTGCAGCCTCTGGAT TCACCTTTAGCACCTATGCCATGAGCTGGGTCCGCCAGGC TCCAGGGAAGGGCCGGAGTGGGTCTCAGGTATTAGTGG TAGTGGTGGTAGCACAACCTACGCAGACTCTGCGAAGGGC CGGTTACCATCTCCAGAGACAATCCAGAACACCGCTGTA TCTGCAAAATGAACAGCCTGAGAGCCGAGGACACGGCCGTA TATTATTGTGCGAAGGGACATTACTTACTCTACTACTAC ATGGCCGTCTGGGGCAAAGGGACCACGGTCACCGTCTCCT CA	
	110	light	GAAATTGTGTTGACGCAGTCTCCAGGCACCCGTGCTTTGTCTCCAGGGGAAAGAGGCACCCCTCTCCTGCAGGGCCAGTCA GAGTGTTAGGAGCAGCTACTTAGCCTGGTACCAGCAGAGA CCTGGCCAGGCTCCAGGCTCCTCATCTATGGTGCATCCA GCAGGGCCACTGGCATCCAGACAGGTTCACTGGCAGTG GGTCTGGGACAGACTTCACGCTCACCATCGGCAGACTGGA GCCTGAAGATTTTGAGTGTATTACTGTGTCAGCAGTATGGTA CCTCAATAACTTTCCGGCGAGGGACCAAGGTGGAGATCAA A	
EV-D68-235	111	heavy	CAGGTCCAGGTGGTACAATCTGGGGCTGAGATGAGGAAGC CTGGGGCCTCAGTGAGGGTCTCCTGCAAGGTTTCCGGATA CAGGCTCACTGATTTACCCTTGCACTGGGTGCGACAGGCT CCTGGAAAAGGGCTTGAATGGATTGGATTGTTGATCTTGA AAAGCGCGAAATCATCTACGCACAGAAATTTAGGGCAAA GTCACCATAACCGAGGACACATCTGCAGACACCGCTACA TGGAACTGAACAGCCTGCGATCTGAAGACACGGCCGTCTA CTACTGTGCAACTTGGGGAATTGAGGTGGTGAATGGGAGG GACGAATCTTTGACTCTGGGGCCAGGGAACCCGTGTCT CCGTCTCCTCA	
	112	light	TCCTATGAGCTGACACAGCCACCCTCGGTGTGAGTGTCCC CAGGACAGACGGCCAGGATCAGATGCTATGCAGATGTGTT GTCAAAGCAATATACTTACTGGTATCAGCAGAAGCCAGGCC AGGCCCTGTGTTGGTGATATATAAGACACTGAGAGGCC CTCAGGGATCCCTGAGCGATTTGCTGGTCCAGCTCAGGG ACAACAGTCACCTTGATCATCAATGGAGTCCGGACAGAGG ACGAGGCGTACTATTACTGTCAATCAGCCGACACCAGAATT ACGGTTTTTCGGCGGAGGGACCAAGCTGTCCGTCTCA	
EV-D68-236	113	heavy	CAGGTCCAGGTGGTCCAGTCTGGGGCTGAGATGAAGAAG CCTGGGGCCTCAGTGAAGGTCTCCTGCAAGGTTTCCGGAT ATAGTCTCAGTGATTTACCCTTGCACTGGGTGCGACAGGCT CCTGGAAAAGGGCTTGAGTGGATGGGACTTTTGTATCTTAT AAACGGTGAATCATCTACGCACAGACATTCAGGGCAAA GTCACCATAAGCGAGGACACATCGATAGACACAGCCTACA TGGAACTCAACAGCCTGCGATCTGAAGACACGGCCGTGTA CTATTGTGCAACTTGGGGAGTTGCGGTGGTGAAGTGGGAGA AGGGACTACTTTGACTCTGGGGCCAGGGAACCCGTGTCA CCGTCTCCTCA	
	114	light	TCCTATGAGCTGACACAGCCACCCTCGGTGTGAGTGTCCC CAGGACAGACGGCCAGGATCAGATGCTATGCAGCTGTATT GTCAAACCAATATACTTACTGGTATCAACAGAAGCCAGGCC AGGCCCTGTGTTGGTGATATATAAGACACTGAGAGGCC CTCAGGGATCCCTGAGCGATTTGCTGGTCCAGCTCAGGG ACAACAGTCACCTTGATCATCAATGGAGTCCGGACAGAGG ACGAGGCTTACTATTACTGTCAACAGCCGACACCAATAT ACGGTTTTTCGGCGGAGGGACCAAGCTGTCCGTCTCA	
EV-D68-241	115	heavy	CAGGTGCAGCTGCAGAGTCCGGGCCAGGACTGGTGAAG CCTTCACAGACCCTGTCCCTCACCTGCAGTGTCTCTGGTG GCTCCATCACCAGTGGTGATTACTACTGGAATTGGATCCGC CAGCCCCCAGGGAAGGGCCTGGAGTGGATTGGGTACATC TATCAGAGTGGGACCACCTACTACAACCCGTCCCTCAAGA GTCGAGTTACCATATCAGTAGACACGTCCTAAGAACAGGTT TCCCTGAAGTTGTCTCTGTGACTGCCGACAGACACGGCCG	

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable	Sequence Region
				TGTATTTTGTGCCAGAGCCTACGCTTATGAATTTTGGAGC GGTTACCCCTAACTGGTTCGACCCCTGGGGCCTGGGAACCC TGGTCACCGTCTCATCA
	116	light		GACATCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATC TGTGGAGACAGAGTCACCATCACTTGCCGGGCAAGTCAG CGCATTAGTACCTATGTAAATTGGTATCAGGTGAAAGCAGG GACAGCCCCTAAGGTCCTGATCTATGCTGCGTCCAGTTTG CAAAC TGGGGTCCCATCAAGGTTCACTGGCAGTGGATCTG GGACAGATTCTACTCTCACCATTGTCTAGTCTACAACTGAA GATTTTACAACCTACTTCTGTCAACAGAGTTACAGTCCCCC GTGGACGTTTCGGCCAAGGACCAAGGTGGAATCAAA
EV- D68- 247	117	heavy		CAGGTGCAGTTGGTGGAGTCTGGGGGAGGCTTGGTCCAG CCTGGTAGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGGT TCATCTTCAATAGATATGCCATGCACCTGGGTCCGCCAGGCT CCAGGCAAGGGGCTCGAGTGGGTGGCTCTTATATCATATG ATGGAATTAATAAATATTACGCAGACTCCGTGAAGGGCCGA TTCTCCATCTCCAGAGACAATTCCAAGAGTACGCTGTATCT GCAAA TGAACAGCCTCAGAGCTGAGGACACGGCTATCTTT TACTGTGCGAGAGGACTAGGATATTGTAGTGGTACCGGTG GTAGCTGTACACCTTTGAATATTGGGGCCAGGGAATCCT GGTCACCGTCTCCTCA
	118	light		GACATCCAGATGACCCAGTCTCCACCCTCCCTGTCTGCAT CTGTTGGAGACAGAGTCACCATCACTTGCCGGGCAAGTCA GAGCATTAAGAAATATTTAAATTGGTATCAGCAGAAACCAG GGAATGCCCTAAGCTCCTCATCTATGGCGCATCCAATTG CAAAC TGGGGTCCCATCAAGGTTCACTGGCAGTGGATCTG GGACAGATTTCGCTCTCTCCATCAGCAGTCTGGAAC TGA GATTTTGAACCTACTACTGTCAACAGAGTGACAGTGGCCC TCCCACTTTTCGGCGAGGACCAAGGTGGAGTTCAAA
EV- D68- 254	119	heavy		CAGGTGCAGCTGCAACAGCGGGGCGCAGGGCTGTTGAAG CCCTCGGAGACCCTGTCCCTTACCTGCGAAATCTATGGTG CATCCCTCAATGATTACGACTGGACCTGGATCCGCCAGCC CCCAGGGAAGAGGCTGGAGTGGATTGGGGTCATCAATCGT CGTGACACTGTTGACTACAACCCGTCCTCAAGAGTCGGG TCACCCTCTCACTTGACACGTCCAAGAACCAACTTTCCCTG AGTCTGAGTTCTGTGACCGCGCGGACACGGCTGTTTATT ACTGTGTGAGAGTCCACCTCGGGGCTTTGAAGGGTCTTT CGGATTTTGTGATGATATGCTGCGCTACGGGCATACC TGGTTTCGACCCCTGGGGCCAGGGAACCTGGTCACCGTCT CCTCA
	120	light		GACATTCAGATGACCCAGTCTCCTTCCCTCCCTGTCTGCATC TGTGGCGACAGAGTCACCATCACTTGCCGGGCAAGTCAG AGTATTAGAGATTATTTAAATTGGTATCAACAAAGACCAGG GAAAGCCCC TAAAGTCTGATCTTTGCTGGTTCCCGTTTGG AAAGTGGGGTCCCATCGAGTTTAGAGGCCGTGGATCTGG GACAGAATTCACTCTCACCATCAGCGATCTGCAACCTGAG GATTTTGAACCTTACTACTGTCAACAGAGTTACCTTACACCT CCGACATTCGGCCAAGGGACCAACCGTCGATATCAAA
EV- D68- 260	121	heavy		CAGGTCACCTTGAAGGAGTCTGGTCTGTGCTGGTGAAAC CCACAGAGACCCTCAGCTGACCTGCACCGTCTCTGGATT CTCACTCCGCAATGGTAGAATGGGTGTGAGCTGGATCCGT CAGCCCCCAGGGAAGGCCCTGGAGTGGCTTGACACATTT TTGCGAGTGACGAAAATCTTACAGTACATCTCAGAGGACC AGGCTCTCCATCTCAGGGACACCTCCAAAAGCCAAGTGG TCCTTAGCATGACCGACATGGACCCCTGTGGACACAGCCAC ATATTACTGTGCGCGGATTTGAAGTTTGGGACAATGAGGG CCGCATACTACTTTGACTACTGGGGCCAGGGAGCCCTGGT CCCCGTCTCCTCA
	122	light		TCCTATGTGCTGACTCAGCCACCCTCAGTGTCACTGGGCC CAGGAAAGACGGCCAGGATTACCTGTGGGGGAATCAACAT TGGAAATTAAGACTGTACTGGTACCAGCAGAAAGCCAGGC CAGGCCCTATGTTGGTCACTATTATGATAGCAGCCGGC CCTTAGGGATCCCTGAGCGATTCTCTGGCTCCAAGTCTGG GAACACGGCCACCCCTGACCATCAGCAGGTCGAAGCCGG GGATGAGGCCGACTATTACTGTCAGGTGTGGGATAGTAGT AGTGATCATGTTGTATTTCGGCGGAGGGACCAAGCTGACCG CCCTA
EV- D68- 266	123	heavy		CAGGTGCACCTTGGTGGAGTCTGGGGGAGGCTTGGTCAAG CCTGGAGGGTCCCTGAGACTCTCCTGTGCAGTTTCTGGAT TCACCTTCAGTGACTACTACATGAGCTGGATCCGCCAGGC TCCGGGGAAGGACTGGAGTGGCTTTCATACATTAGTAGT

TABLE 1-continued

NUCLEOTIDE SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable	Sequence Region
				AGTGGTAGTACCATATACTACGCAGACTCTATGAAGGGCC GATTACCATCTCCAGGGACAACGCCAGGAACCTCACTCTAT CTGCAATGAACAGCCTGAGAGTCGAGGACACGGCCTTT ATTACTGTGCGGGTCAAAGGTGGCTATACTACTGGTCG AAGGAAGTGGTACTTCGATCTCTGGGGCGTGGCACCCCTG GTCACTGTCTCCTCA
	124	light		CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTC CTGGACAGTCAGTCACCATCTCCTGCACTGGAACAGCAG TGACGTGGTGCTTATAACTATGTCTCTGGTACCAACAGC ACCCAGGCAAAGCCCCAAACTCATGATTATGAGGTCACT AAGCGGCCCTCAGGGGTCCCTGATCGCTTCTCTGGCTCCA AGTCTGGCAACACGGCCTCTCTGACCGTCTCTGGGTCCA GGCTGAGGATGAGGCTGATTATTACTGCAGCTCATATGCA GGCAACAACAATTAGTCTTCGGCGGAGGGACCAAGCTGA CCGTCTCA
EV- D68- 269	125	heavy		CAGGTGCAGTTGCTGCAGTCTGGGTCTGAGGTGAGGAAAC CTGGGGCCTCAGTGAACATTCAGTGAAGGCATCTGGATT CACTTTCACCGACTTCTATTTACACTGGGTGCGACAGGCCC CTGGACAAGGGCTTGAGTGGATGGGGATAATCAACCTGA AACCCTGTGAGACAACCTACTCAGAGAAGTTTCAGGGCAGA GTCACCATGACCAGGGACACGTCCACGAGTGTAGTGAATC TGGAGTGAGGAGCCTGAGATCTGAGGACACGGCCATATA TTACTGTGCGAGAGATCTCGTTGTCTAGTCCCCGTGAA TGTCTCGGCGTGCCCTTGACATTTGGGGCCAAGGATTAT GGTCACAGTCTCCTCA
	126	light		TCCTATGTGCTGACTCAGCCACCTCGGTGTGAGTGGCCC CAGGACAGACGGCCAGGATTCCTCTGGGGGCAACAACAT TGAACGTAAAAGTGTCCACTGGTACCAGCAGAGGCCAGGC CAGGCCCTGTGTTGGTCTCTATGATGATACTGTCCGGC CCTCAGGTATCCCTGAGCGATTCTCTGGCTCCAACCTCCG GAGCACGGCCACCTGACCATCAGCAGGGTCCGAGCCGG GGATGAGGGCGACTATTATTGTGAGGTGTGGACAGCACC ACTGACCATGGGGTCTTCGGCGGAGGGACCAAGCTGACC GTCCTA
EV- D68- 271	127	heavy		CGGGTGCAGCTGGTGCAGTCTGGGGCTGAGGTGAAGAAG CCTGGGGCCTCAGTGAAGTCTCCTGTGAGACTTCTGGAT ACATTTTCAGCGCTACTACATCTATTGGGTGCGACAGGCC CCTGGACAAGGACTTGAGTGGATGGGACGGATGAACGCTA AGAGTGGAGGCGCAAACTGCACAGCAGTTTCAGGGCAG ACTCACCATGACCAGGGACATGTCCGTGAGCAGCCTAC ATGGAAGTGAAGGCTGCGATCGGACGACACGGCCGTC TATTATTGTGCGAGAGACTATAGGGATGACTACATGTGGG GAGTTATCGGCCTTTAGACTACTGGGGCCAGGGAACCTG GTCACCGTCTCCTCA
	128	light		GAAGTTGTGTTGACACAGTCTCCAGCCACCTGTCTTTGTC TCCAGGGGAAAGAGCCTCCCTCTCCTGACGGCCAGTCAG AGTGTAGCGGCTACTTAGCCTGGTACCACACAACCTG GCCAGGCTCCCAGGCTCCTCATCTATGATGCATCCAACAG GGCCGCTGGCATCCAGCCAGGTTCCGTGGCAGTGGGT TGGGACAGACTTCACTCTCACCATCAGCAGCTGGAGCCT GAAGATTTTGCAGTTTATTTCTGTGACGCGTAGCAACGG GCTCACTTTCGGCGGAGGACCAAGGTCGAGATCAAA

TABLE 2

PROTEIN SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable	Sequence
EV- D68- 37	129	heavy		QVQLVESGGGVVQPGRLRLSCAASGFIFSRVALHWVRQ APGKGLDWAVISYDARNYYTDSVKGRFTISRDNKNTL FLQMNSLRADDVAVYCARPTLPYSNNWYAPEYWGQGT LTVSS
	130	light		SYELTQPPSVSVSPGQTARITCSGDALPKKYASWYQKSG QAPVLVIYEDTKRPSGIPERFSGSSSGTMATLTISGAQVED ESDYICSTDSGNPVLPGGKLTIVL
EV- D68-	131	heavy		EVQLVESGGDLVQPGGSLRLSCAASGFSSSYAMAWVRQ APGKGLQWVSSISGNGNGRSYADSLKGRFTTSKDLSTYTL

TABLE 2-continued

PROTEIN SEQUENCES FOR ANTIBODY VARIABLE REGIONS			
Clone	SEQ ID NO:	Chain	Variable Sequence
40			YLQMNLRPEDTAIYYCAKVRIAALVYFDYWGPGTQVT VSS
	132	light	EIVLTQSPATLSLSPGERATLSCRASQSVSTYLAWYQQKP GQAPRLLIYEASTRATGIPARFSGSGSGTDFTLISSLEPED FAVYHCQQRSSWPIITFGQGTREIE
EV- D68- 41	133	heavy	DVQLVESGGGLVQPGGSLRLSCAASGFTFSNYAMTWVRQ ALGKGLEWVSSISGSGGLTYFAHSVKGRLTISRDNKNTLY LQMSSLRAEDTAVYYCARVKSTTGTALVFDIWGQGTMTV VSS
	134	light	QTVVTQEPSFSVSPGGTVTLTCGLSSGVSSTSYPSWYQ QTPGQAPRTLIYSINRRSSGVPDRFSGFILGNKAALTIRGAQ ADDESYYCYGLYMGSIGIIFGGGTKLTVL
EV- D68- 43	135	heavy	QVQLVESGGGVVQPGRLRLSCAASGFTFINYGMHWVRQ APGKGLEWVAVISNDGSYNDADSVKGRFTISRDNKNTLY YLQMNSLRPEDTAVYFCADKHGDFDYGVVDVWGQGTTV TVSP
	136	light	DVQMTQSPSSVSASIGDRVTITCRAGQGISWLAHYQQKP GKAPKLLIYAASNLQSGVPSRFSGSGSGTDFTLTISLQPE DFGTYCQQADSFPRTFGQGTKEIK
EV- D68- 46	137	heavy	QVQLVESGGGVVQPGRLRLSCAASGFTFSYGIHWVRQ APGKGLEWVAVISYDSDNTYAPFVNGRFTISRDNKNTLY YLQMNSLRADDTAVYYCARRRPGSFPGLCDYWGQALVT VSS
	138	light	DIQMTQSPSTMSASVGDRTITCRASQISKWLAWYQQKP GKAPKLLIYKASTLQTVPSRFSGSGSGTEFTLTINSLQPD DFATYYCQQHNSYSYTFGQGTKEIK
EV- D68- 48	139	heavy	QVQLVQSGAEVKKPGSSVKVCKASGGSFRLTIHWVRQA PGQGLEWMGGHIPIFGTTNYALKFQGRVTITADKTTSTAYM ELSSLRSEDTAIYYCARMYSGHDGVDVWGQGLTVTVSS
	140	light	EIVLTQSPATLSLSPGERATLSCRASQSVRSYLAWYQHKP GQAPRLLIYDASNRAKGI PARFSGSGSGTDFTLTISLEPED FAVYYCQQRSTWPPGMFGQGTREIK
EV- D68- 71	141	heavy	QLQLQESGPGLVKHSETLSLTCTVSGGSISSGFYYWGWR QPPGKLEWIGTIYDSGRTYDNP SLKSRVTISADTSKKQFS LTLRSATAADTAVYFCARHLTHLYGDYVTPDALDIWGQGT MTVTVSS
	142	light	EIVLTQSPGTLSLSPGERATLSCRASQSVSSFLAWYQQK PGQAPRLLIYGASSRATGIPDRFRSGSGSGTDFTLTISRLEP EDFAVYYCQQYSNSRLTFGQGTKEIK
EV- D68- 72	143	heavy	EVQLVESGGGLVQPGGSLRLSCAASGFTFTTYSMNWVRQ APGKGLEWISYISSGSSNIYADSVKGRFTISRDNKNSLNL QMSLRDEDTAVYYCARAHGRIVNSGVVISRFDPWGQGIL VTVSS
	144	light	QSALTQPASVSGSPGQSITISCTGTSSDVGGYNYVSWYQL HPGKAPKLMI FEVITYRPSGVSNRFSGSKSGNTASLTISGLQ AEDEADYFCSSYTTNTLVVFGGGTKLTVL
EV- D68- 74	145	heavy	QVQLVQSGAEVKKPGASVKVSCRTSGYFTTAYYMHWVRQ APGQGPFWMGRIINPSSGGAQYAKFQGRVTMTDRDTSIST TYMTLSGLTSDDTAVFYCARMGCRSDRCYSTNYNFDQWG QGTLLTVSS
	146	light	SYVLTQPPSVSVAPGQTARIPCGGNNIGTKSVHWYQQKPG QAPVLVVSNDSDRPSGIPERFSGSKSGNTATLTISRVEAGD EADYYCQVWDSGIDVVFGGGTKLTVL
EV- D68- 75	147	heavy	EVQLVESGGGLVKPGGSLRLSCAASGFTISPYGMNWVRQ APGKGLEWVSFISSSRYTYADSVKGRFTISRDNKNSLS LQMNSLRADDTAVYYCARERGHSTSSSYFDSWGQGLTVT VSS
	148	light	SYVLTQPPSVSLAPGKTARITCGGNNIGTKTVSWYQQKPG QAPVLVMYYDSDRPSGIPERFSGNSGNTATLTINRVEAG DEADYYCRVWDSDDHVRVFGGGTKLTVL

TABLE 2-continued

PROTEIN SEQUENCES FOR ANTIBODY VARIABLE REGIONS			
Clone	SEQ ID NO:	Chain	Variable Sequence
EV-D68-76	149	heavy	EVQLVESGGGLVQPGGSLRLSCAASGFTFSNAWISWVRQ APGKGLEWVGRIQTKTDGGTTDYAAPVKGRFTISRDDSKN TLYLQMNSLKTEDTALYYCSTGPPYYDTSGYPQPFQYWG QGTLVTVSS
	150	light	SYELTQPPSVSVSPGQTASITCSGDKLGDKYACWYQQKP GQSPVLVIYQDTRPSGIPERFSGNSGNTATLTISGTQAM DEADYYCQAWDSSTVFGGGTKLTVL
EV-D68-80	151	heavy	EVQMVESGGGLVQPGGSLRLSCSVSGFDPSRYTMNWFR QAPGEGLEWVSSISSTSLYTFYADSVKGRFISRDNAGGS LSLQMSLRPEDTAVYYCARVVGPAELDYWGQGLVTVS S
	152	light	VTQLTQSPSSLSASVGDRTTITCRASQDIGVDLGFQQR GKAPKLLIYGASRLQSGVPSRFSRSGSGTFFTLTISSLQPE DFTTYFCLQDYNYPWTLGQGTIVGVK
EV-D68-84	153	heavy	QVHLVQSGSAVRKPGASVKVSCASGYTFTDYYIHVVRQ APGQGLEWGMWINPKTGGSNYTRFQARVTMTWDTISIT AYMELSRRLSDDTAVYYCARAGRNGDYWGQGLVTVSS
	154	light	SYELTQPPSVSVSPGQTARITCSADALPKQYAYWYQQKPG QAPVLMYQDTERPSGIPERFSGSSSGTTVTLTISGVQAED EADYYCQSGDSSGTYLVFGGGTKLTVL
EV-D68-85	155	heavy	EVQLLESGGGLVQPGGSLRLSCAASGFKFRNYAMTWVRQ APGKGLEWVSTITSGGSTYADAVKGRFISRDNKNTLYL QMNSLRADDTAVYYCTVPWGNVNDYVSDYWGQGLVTV SS
	156	light	EVVLTQSPATLSLSPGQRATLSCRASQVRGNSLAWYQQK PGQAPSLTIYDASKRATGIPARFSGSGSGTDFTLTIIISLE SE DFAVYYCHQHSWPRGTFGQGTLEIK
EV-D68-88	157	heavy	QVHLVESGGGVVQPGRLRLSCAASGFIFSRYPMHVVRQ APGKGLEWVALISYDGNKYYADSVKGRFTISRDNKNTL FLQMNSLRAEDTAVYYCARHFLPYSSSWYQGFNYWGQGI LTVVSS
	158	light	NFMLTQPHSVSESPGKTVTISCTRSSGIATNYVQWYQQR PGSSPTPIIFEDSQRPSPGVDRFSGSIDSSNSASLTISGLR TDDEADYYCQSYDNSRAVFGGGTKLTVL
EV-D68-89	159	heavy	QVQLVESGGGVVQPGRLRLSCEASGFLFSRYGMHWVR QAPGKGLDWAVISYDGNKYYADSVKGRFTISRDNKNT LYLQVNSLRVEDTAVYYCARGVPYGDTLTGLVYWGQGLV TVSS
	160	light	NFMLTQPHSVSESPGKTVTISCTRSSGIATNYVQWYQQR PGSAPTTVIYEDNQRPSPGVDRFSGSIDSSNSASLTISGL KTEDEADYYCQSYDNSDRVFGGGTKLTVL
EV-D68-95	161	heavy	QVTLKESGPVLVKPTETLTCTVSGFSLRNARMGVSWIR QPPGKALEWLAHIFSNDEKSYNTSLKSRLSISKDTSKSQV LTMTSMDPLDTATYFCARLLVAGTFLPSHYFDYWGQGLV TVSS
	162	light	SYVLTQPPSVSVTPGKTARITCGNNIGLKSVMFYQERPD QAPVVVIYYDSARPSGIPERISGKSGNTATLTITRVEAGDE ADYFCQVWDSRNHPVFGGGTKLTVL
EV-D68-97	163	heavy	ELQLVESGGGLVQPGGSLRLSCAASGFTFSTYSMNWVRQ APGKGLEWVSYISSSSSTIQYADSVKGRFTISRDNKNSLY LQMNSLRAEDTAVYYCTRQVGADFGSGRFDYWGQGLTLL TVSS
	164	light	SYELTQPPSVSVSPGQTATITCFGDKLGDKYACWYQQKPG QSPVLVIYQDSKRPSPGIPERFSGKSGNTATLTISGTQAMD EADYYCQAWDSSTAVFGGGTKLTVL
EV-D68-98	165	heavy	EVQLVESGGGLVQPGGSLRLSCAASGFKFSVYALSWVRQ APGKGLEWISYISSSGSTIYSDSVKGRFTISRDNVGNLFLV QMNSLRAEDTGIYYCATARHITNDGFDIWGQGTMIIVSS
	166	light	DIQLTQSPSFLSASVGDRTTITCRASQGISRFLAWYQQKPG KAPKLLIYASLTQRGVPSPRFSGSGFGTDFTLTIISSLQPEDF ATYYCQQLNSHPRMFTFGPGTTVDIK

TABLE 2-continued

PROTEIN SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence	
EV-D68-105	167	heavy	QVQLQESGPGLVKPSSETLSLTCAVSGYLI SNGYWGWIRQ PPGKGLEWIGSIYHTRSTYYNPSLKSRSVISVDT SKNRFSL	
			RLRSVTAADTAFYYCARGPGHCYGD DDCYAYYFDQWGQ GTPVTVSP	
	168	light	DIQMTQSPSSVSSSVGDRVTITCRASQGISNWLAWYQQNP GKAPKLLIYDASSLQSGVPSRFSGSGSGTDFTLTINSLQPE	
			DFATYYCQANSFPFTFGPGTKVDIK	
EV-D68-110	169	heavy	QVQLVQSGAEVKKPGSSVKVSCASGGTFSRFAISWVRQ APGQGLQWMMGILPIFGTANYAQKFQGRVTITADTSTSTA	
	170	light	YMESSLRSED TAVYYCARSLPYCTNDVCSNQNTFDYWG QGTLTVSS SYELTQPPSVSVSPGQAARITCSGDALPKQYAYWYQQKP GQAPVLVIYEDNKRPSGIPERFSGSSSGTTVTLTISGVQAE DEADYYCQADSSGTYYVFGGGTKLTVL	
EV-D68-111	171	heavy	QVQLQESGPGLVKPSQTLSTCTVSGGSISSGDYYWSWIR QPPGKLEWIGIYIYSGSTYYNPSLKSRTISVDT SKNQFS	
	172	light	LKLSSVTAADTAVYYCASRYGDP IGDNWFDPWGQGT LVT SS SYVLTQPPSVSVTPGKTARITCGNNIGLKS VFYQERPD QAPVVVIYD SARPSGIPERISGSKSGNTATLTITRVEAGDE ADYFCQVWDSRNHPVFGGGTKLTVL	
EV-D68-114	173	heavy	EVQLLESGGGLVQPGGSLRLSCAASGFRFSFYGMTWVRQ APGKGLEWVSSISGTGATRCADSVKGRFTISRDN SKNTL	
	174	light	YLQMDSLRVDDTAVFYCVRFPMTT VTSFDSWGQGT LVT VSS QSALTQPRSVSGSPGQSVTISCTGTSSDVGGYNFVSWYQ QHPGKAPKLMIFDVTGRPSGVDRFSGSKSGNTASLT IAG LQAEDEADYYCGAYAGFNALFGGGTKLTVL	
EV-D68-116	175	heavy	EVQLVQSGGGLVRPGGSLRLSCVAGSPFNMFWMGWVR QTPGKGLEWVANIKQDGSEKYYVDSVKGRFAISRDN AKNS	
	176	light	LFLQMDLSLVGDTAIYYCVREGVRRVVRSTGYFDW GQ GQLTVSS SYELTQPPSMVSVSPGQTARITCSGDAVPIKYVYWYQQRSG QAPVLVIYEDDRPSGISERFSGSSSGTTATLTITGAQVED EGDYCYSTDSSGYQRAFGGGTTLTVL	
EV-D68-150	177	heavy	RVQLQESAPGLVRPSETLSLTCSVSGGPISNGPYWSWIR QHPGKLEWIGIFIFYSGSTNYNPSLRGRVTMAVDTSKNQF	
	178	light	SLRLNSVTAADTAVYYCARHVVTASGWFPDWPWGQGT LVT SS EIVLTQSPATLSLSPGERATLSCRASQSVGTDLAWYQQKP GQAPRVLIYDAFKRATGIPARFSGSGSGTEFTLTIS SLEPED FAVYYCQQRSRWPPPYTFGQGT KLEIK	
EV-D68-151	179	heavy	RVQLQESAPGLVRPSETLSLTCSVSGGPISNGPYWSWIR QHPGKLEWIGIFIFYSGSTNYNPSLRGRVTMAVDTSKNQF	
	180	light	SLRLNSVTAADTAVYYCARHVVTASGWFPDWPWGQGT LVT SS EIVLTQSPATLSLSPGERATLSCRASQSVGTDLAWYQQKP GQAPRVLIYDAFKRATGIPARFSGSGSGTEFTLTIS SLEPED FAVYYCQQRSRWPPPYTFGQGT KLEIK	
EV-D68-152	181	heavy	QVHLVESGGGVVQPGSLRLSCADSGVTFS DNALYWVRQ APGKGLEWVAVISYDGSSRYADSVRGRFTISRDN SKDTL	
	182	light	YLQMNRLRAEDTAIYYCARVTADYYESSGKVFWGQ GALVV VSS DIQMTQSPSTLSASVGRV SITCRASQSVRSWLAWYQH KP GKAPKLLIYKASSLESVPSRFSGSGSGTEFTLTIS SLQAD DFATYYCQYQTFSWTFGQGT TVEVK	
EV-D68-154	183	heavy	QVQLQESGPALVKPSETLSLTCTVSGGSI SDHYWSWIRQP PGKLEWIGIYITSGTTNYP SLKSRVTISVDT SKKQFSLNL	
	184	light	RSVTAADTAVYYCARSL ETVIRFYYYHYMDVWGKT TVIVS S DIVMTQSPSLSLPVTGPGEPAISCRSSQSL LQSDGYSYLDWY LQKPGQSPQLLIYLGSNRASGVDRFSGVIGSGTYFTLKISR VEAEDVGVIYFCMQALQTPWTFGQGT KVEIK	
EV-D68-155	185	heavy	QVQLQESGPGLVKPSSETLSLTCTVAGGSIGDYHWNWIRQP AGKGLEWIGRIHSSGNTDYNPSLKSRTVMSVDT SKNQFSL	
			KLRSVTAADTAVYYCARQNVFDIWGQGT MVTVSS	

TABLE 2-continued

PROTEIN SEQUENCES FOR ANTIBODY VARIABLE REGIONS			
Clone	SEQ ID NO:	Chain	Variable Sequence
	186	light	DIVMTQSPDSLALSLGERATINCKSSQSVLFSSNNKNYLAW YQQKPGQPPKLLIYWASTRESGVDPDRFSGSGSETDFTLTIS SLQAEDVAVYFCQQFYTTPLTFGGGTKVEIK
EV- D68- 156	187	heavy	QVQMQESGPGLVKASETSLTCSVSGISINNYYSWFRQP PGKGLEWIGYVYSTGSSKYNPSLERRATMSVDTSNNNFSL RLTSVTTADTAVYYCARGSMPHIWGQGLLVTVSS
	188	light	QSVLTQPPSASGTPGQRVITISCSGSTSNIETNYVWYQQV PGTAPKPLVYRNDQRPSGVDPDRFSGSKSGTSASLVISGLR TEDEAAYYCAAWDDSLKAPVFGAGTKVAVL
EV- D68- 157	189	heavy	EVQLVESGGGLIKPGGSLRLSCAASGITFSNAWMSWVRQA PGKGLEWVGRIESKIDGGTIDYATPVKGRFTISRDDSKNTL YLQMNSLKTEDTAVYYCTDQGYDRSGYVWVGNHFDY WGQGLLVTVSS
	190	light	QSVLTQPPSVSGAPGLRVITISCTGSSTNIGAGYDVHWYQH LPGTAPKLLIYGNSNRPSGVDPDRFSGSKSGTSASLAITGLQ ADDAADYYCQSYDRSLSTYVFGTGKTVL
EV- D68- 158	191	heavy	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMNWRQ APGKGLEWVSGISGTTGTYADSVKGRFTISRDNKNTV HLQMNSLRAEDTAVYYCAKDSHSMIVVDHAFDIWQGTM VTSS
	192	light	SYVLTQPPSVSVAPGQTARITCGGNNIGTKSVHWYQQRPG QAPVLVVYDDSDRPSGIPERFSGSGSNGTATLTISRVEAGD EADYYCQVWDSYNVHYVFGTGKTVL
EV- D68- 159	193	heavy	EVQLVESGGGLVKPGGSLRLSCAASGFTFSTYIMTWVRQA PGRGLEWVSSISTSSVYTFYADSLKGRFTISRDNKNSVYL QMNSLRADDTAVYYCAREEGFRAYNLYWGQGLTVTVSS
	194	light	QSVLTQPPSASGTPGQRVITISCSGSSSNI EYNYVWYQKF PGTAPKLLIYKNNQRPSGVDPDRFSGSKSGTSASLAISGLRS EDEGDYYCAAWDDILSGVVFGGGTKLTVL
EV- D68- 160	195	heavy	QVQLVESGGGVVQPGRSQKLSCAASGFTFSRFGMHVVR QAPGKLEWVAVISFDGSNRYADSVKGRFTITRDNSKNT LYLQMNNLRPEDTAVYYCARDWDLVRSVAVGYWGQGLTV SVSS
	196	light	QSALTQPRSVSGSPGQSVTISCTGTSDNVGGYNFVSWYQ QHPGKAPKLMIFDVIRRPSPGPRFSGSKSGDASLIISGL QAEDEADYYCCSYAGTYTWVFGAGTTLTVL
EV- D68- 161	197	heavy	QVHLQESGPRLVKPSSETLSLTCTVSGGSVSTATYYWSWIR QSPGRGLEWIGYIYSSGNTNYPNLSKRVITSLDTPNNQLS LTLTSVTAADTAIYYCERRLRILSIERNYYAMDVWGQGPV TVSS
	198	light	EVVLTQTPGTLSLSPGEGATLSCRASQRVVNNYLAWYQQ RAGQAPRLLI FGASNRATGIPDRFSGSGSGTDFTLTIRKLE PEDFAVYYCQQYGSPTFGHGTVKEMK
EV- D68- 162	199	heavy	QVQLVESGGGVVQPGRSRLSCAASGFTFSSYAMHWVR QAPGRGLEWVAVISYDASKKYHADS VKGRFTISRDSKNT LFLQMNSLKPEDTAVYYCARDHVPKDCSDGNCHSDYGM DVWGQGTTVTVSS
	200	light	DIQMTQSPSTMSASVGDVITICRASQSI SKWLAWYQQKP GKAPKLLIYKASTLQTGTGVPSPRFSGSGSGTEFTLTINSLQPD DFATYYCQQHNSYSYTFGQGTKVEIK
EV- D68- 163	201	heavy	QVQLVQSGAEVKKPGASVKISCKASGYSTNFVAVHWVRQ APGQRLEWMGWINPGNRNTKYSHNFQGRVTITRDTISANT AYMELSSLSRQDTAVYYCARLPFAAAGRGWDPWGQGLT VTSS
	202	light	EIVLTQSPGTLSLSPGERATLSCRASQSVISTYLAWYQQRP GQAPRVLIYDVSTRATGIPDRFSGSGSGTDFTLTISRLEPE DFAVYFCHQYGSPPATFGQGTKVEIK
EV- D68- 164	203	heavy	EVQLVESGGDLVQPGGSLRLSCAASGFTFNITYGMNWRQ APGKGLEWVSYISSATTTFFYADSVRGRFTISRDNKNSLF LHMKSLRDEDTAVYYCARVYTMRLGASMDVWGHGTTVT SS
	204	light	EIVLTQSPATLSLSPGERATLSCRASQSVGTYLAWYQQKP GQAPRLLIYDSANRATGIPARFSGSGSGTDFTLTISSLEPED FAVYYCQLRITWPPIFTFGPGTKVDVK

TABLE 2-continued

PROTEIN SEQUENCES FOR ANTIBODY VARIABLE REGIONS				
Clone	SEQ ID NO:	Chain	Variable Sequence	
EV-D68-165	205	heavy	QVHLVQSGAEVKKSGASVKVCKTFGYTFTAYYMHWVRQ APGQGPPEWMGWINPISGGTNYAQKFQGRVSMTRDTSIST AYMGLSRLRPDDTAVYYCARVKCSSANCYGNFDYWGQG TLVTVSS	
	206	light	QSALTQPPSASGSPGESVTISCTGTSSDVGGYNYVSWYQ QHPGKAPKLMIEVSKRPSGVPDRFSGSKSGNTASLTVSG LQAEDEADYYCCSSYAGSNLNVFGGGTKLTVL	
EV-D68-166	207	heavy	EVQLVQSGAEVKKPGESLKISCKGSGYRFTNYRIGWVRQM PGKGLEWMGIIYPGGSDTRYSPSLQGQVMTMSVDKSISTAY LMWSSLKASDTAMYCARQTTQNSGYDRWFDSWGQGTHT VTVSS	
	208	light	QSVLTQPPSASGTPGQRTVISCSTSSIGSNIVNWWYQHLP GTAPKLLIYINNQRPSGVPDRFSGSKSGTSASLAISGLQSE DEADYYCAAWDDSLNGWVFGGGTKLTVL	
EV-D68-181	209	heavy	EVQVLESGGGLVQPGGSLRLSCAASGITFSRHTMSWVRQ APGKGLEWWSAISGGGTYHADSVKGRFTISRDSKSTL YLQMNSLRADTAVYYCAISVPLLRFLWFQHPDFWGGQ TLVTVSS	
	210	light	EIVMTQSPASLSVSPGERVTLSCRASQSVGSTLAWYQHQP GQAPRLLISGASTRATGVPARFSGSGSGTEFTLTISLQSE DFAVYYCHQYINWPPWTFGQGTKEIK	
EV-D68-183	211	heavy	EVRLVESGGGLVKPGGSLRLSCAASGFTFTNTYSMSWVRQ APGKGLEWVASISSTGSYIYNADSLKGRFTISRDNKNSLF LQMNSLRVEDTAVYYCVRFMTMTVTNFDSWGQGLTVTS S	
	212	light	QSALTQPRSVSGSPGQSVTISCTGTSSDVGAYSIVSWYQ QHPGKAPKLMIDVYRRPSGVPGRFSGSKSGNTASLTVSG LQAEDEADYYCCSHAGSHTWVFGGGTKVTVL	
EV-D68-185	213	heavy	QLQVVASGGGVVQPGSLRLSCKASGFTFTNYGMHWVR QAPGKGLEWVAFISYDGGNKFYADSVKGRFTISRDNRNRT VYLQMNSLRVADTAMYYCPKVIHPHYDSSGDDAFDIWQ GTMVAISS	
	214	light	EIVMTQSPATLSVSPGERATLSCWASQISRNLAWYQQKP GQAPRLLIYGASTRATGIPAKFSGSGSGTDFTLIVSSLOSE DLAVYYCQQYSKLPITFGQGTREIK	
EV-D68-200	215	heavy	EVQLVESGGGLVQPGGSLRLSCAASGFPSSYSMSWVRQ APGKGLEWVSYISGGGDIYYADSVKGRFTISRDNARNSL SLQMNSLRADDTAVYYCARGLVATTGTRYFDYWGQGLTV TVSS	
	216	light	EIVLTQSPATLSLSPGERATLSCRASQSVSYLAWYQQRP GQAPRLLIYDASNRAAGIPVRFSGSGSGTDFTLTISLLEPED FAVYYCQQRSYWPPPTFGGGTKVEIK	
EV-D68-208	217	heavy	QVQLVQSGAEVKKPGSSVKVCKASGGTFRFAISWVRQ APGQGLEWMGGIIPILGRGKYAQKFQGRVRIADESTSTAY MELSSLRSEDVAVYYCARFISTASYVPGTFEDVWGQGTTV TVSS	
	218	light	DIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKP GQAPRLLIYDASNRAAGIPARFSGSGSETDFTLTISLLEPED VAVYYCQQRSDWPPGTFGQGTNVEIK	
EV-D68-210	219	heavy	EVQLVESGGGLVKPGGSLRLSCAASGFTFRNINWVRQ APGKGLEWVSSISSTGSYIHYADLVKGRFTISRDNKNSLY LQMNSLRVEDTAVYYCARMVRNTVTFDYWGQGLTVSVS S	
	220	light	QSALTQPASVSGSPGQSITISCTGTSSDVGGYNFVSWYQQ QPGRAPKLLIYEVIKRPSGVSDRFSGSKSGDTASLTISGLQ AEDEADYYCCSYGGNNSWMFGGGTMLTVL	
EV-D68-219	221	heavy	QVQVQSGAEMKKPGASVKVCKVSGYRLIDLPLHWVRQ APGKGLEWMGLFDPEKAEAIYSQKFQDKVTISEDTSIDTAY MELNSLRSEDVAVYYCATWGEVNVNRRDYFDSWGQGT LVTVSS	
	222	light	SYELTQPPSVSVSPGQTARITCYADVLNQQYTYWYQQKPG QAPVLVIYKDTERPSPGIPERFAGSSSGTTLVLVINGVRAED EAYYYCQASDNTRITVFGGGTKLSVL	

TABLE 2-continued

PROTEIN SEQUENCES FOR ANTIBODY VARIABLE REGIONS			
Clone	SEQ ID NO:	Chain	Variable Sequence
EV-D68-220	223	heavy	GVQLVESGGGLIQPGGSLRLSCAASGFTFSSFEMNWVRQ APGKGLEWWSYISTSGSTIYYADSVKGRFTISRDNARNSLS LQMNSLRAEDTAVYYCARDVRDCSALTCPRRGDAFDWFG RGTRVTVSS
	224	light	DVVMTQSPPLSLPVTLGQSASISCRSSQLVYSYGNTYLNW FQQRPGQSPRRLIYKVSNRDSGVPDRFSGSGSGTDFTLKI SRVEAEDVGVYYCMQGTHTWPTFGPGTKVDIK
EV-D68-221	225	heavy	EVQLVESGGGLVKPGGSLRLSCAASGFSFVYPMNWVRQ APGKGLEWVSSISSSSRYISYADSLRGRITISRDNANKSLYL QMNSLRVEDTAVYYCVKVGGSKHQYFDYWGGSLVTVS S
	226	light	QSVLTQPPSTSGIPGQTVTISCSGSRNIGSYTVNWYEQLP GTAPKLVIFFNNQRPSPGVPDRFSGSKSGTSASLAISGLQSE GEADYYCAAWDDSLNGVVFGGGKLTVL
EV-D68-224	227	heavy	QVQLVQSGAEVKRPGASVKVFCASGYTFTNYDIIWVRQA PGQGLEWVGWISTYNGNTNIEQNLRGRVTMTDTSTSTA YMERLSLRSDDTAVYYCARERCSTSTCYSRYADYWGQGT LVTVSS
	228	light	DIQMTQSPSSLASASVGDRVTITCRASQSIINIYNWYQQKPG KAPKVLIIYAASSLQSGVPSRFRGSGSGTDFILTISLQPEDF ATYYCQQSYRSRPTFGQGTVEIK
EV-D68-225	229	heavy	EVQLVESGGGLAQPGEMLRSCVASGFTLSRYEVNWVRQ APGKGLEWLSYISSGGPSIYYADSVKGRFTISRNSAENSLE LQMSTLRTEDTAVYYCMREGLTYDSTIWGQGLTVAVSS
	230	light	QNVLTQSPSASASLGASVKLTCTLNSGHSRYAIAWHQHQP QRGPRFLMKINSDDRHIRGDSIDRFSGSASGAERHLTISS LQPEDEADYYCQTWGTGFRVFGGKLTVL
EV-D68-227	231	heavy	EVQLVESGGGLVQPGSLRLSCAASGFTFDEYAMHWVRQ VPGKGLEWVSGISWNGGSKGYADSVKGRFTISRDNARYS LSLQMNSLRTEDTALYYCAKDDYEGAGFDIWGQGTVVTVS S
	232	light	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLGWYQQKS GQAPRLLIYGASSRATGIPARFSGSGSGTDFTLTISLQSE FAIYYCQQRSNWPIITFGQGTREIK
EV-D68-228	233	heavy	QVQLQESGPGLVKPSSETLSLTCTVSGYLIISNGYYGWIRQ SPGKGLEWIGSIYYTRDTYYNWSLKRITISVDTSKKQFSLK LYSVTAADTAVYYCVRHEGSCNDGSCYGSFVDNWGQGT LVTVSS
	234	light	DIQMTQSPSSVSASVGDRVTLTCRASQDISSWLAWYQQK PGKAPKLLIIYAASSLQSGVPSRFRGSGSGTHFTLTISLQ EDFATYFCQQADSFITFGGKTKVEIK
EV-D68-231	235	heavy	EVQLLESGGGLVQPGGSLRLSCAASDFTFSSYTMWVRQ APGKGLEWVSSISGDGVSTKNADSVKGRFSVSRDNSKNTL FLQLNSLRAEDTAFYYCARGGTFHNWYFDLWGRGVLTV SS
	236	light	EIVMTQSPATLSVSPGETATLSCRASQSIGDNLAWYQQKP GQAPRLLIYGASTRATDFPARFRGSGSGTEFTLTISLQSE DFAVYYCQQYKNWPRTFGRGKVEVR
EV-D68-234	237	heavy	EVQLLESGGGLVQPGGSLRLSCAASGFTFSTYAMSWVRQ APGKGPEWVSGISGSGGSTNYADSAKGRFTISRDNKNTL YLQMNSLRAEDTAVYYCAKGTITYSYMMAVWGKGTVT VSS
	238	light	EIVLTQSPGTLSLSPGERGTLSKRASQSVRSYLLAWYQQ PGQAPRLLIYGASSRATGIPDRFSGSGSGTDFTLTIGRLEP EDFAVYYCQQYGTSTITFGGKTKVEIK
EV-D68-235	239	heavy	QVQVQSGAEMRKPGASVRVSKVSGYRLTDLPLHWVR QAPGKGLEWIGFVDLEKREIYYAQKFGKVTITEDTSADTAY MELNSLRSEDVAVYYCATWGIEVNVNRRDEFFDSWGQGT LVS
	240	light	SYELTQPPSVSVSPGQTARITCYADVLSKQYTYWYQQKPG QAPVLVIYKTERPSGIPERFAGSSSGTDTVTLIINGVTEDE AAYYCQSADTRITVFGGKTKLSVL
EV-D68-236	241	heavy	QVQVQSGAEMKPGASVKVSKVSGYSLDPLPLHWVRQ APGKGLEWMLFDPIINGEIIYAQTFQKVTISETSIDTAYM ELNSLRSEDVAVYYCATWGVAVVSGRRDYFDSWGQGT LVS

TABLE 2-continued

PROTEIN SEQUENCES FOR ANTIBODY VARIABLE REGIONS			
Clone	SEQ ID NO:	Chain	Variable Sequence
	242	light	SYELTQPPSVSVSPGQTARITCYAAVLNSQYTYWYQQKPG QAPVLVIYKDTERPSPGIPERFAGSSSGTTVTLIINGV RTEDE AYYYCQTADTKYTVFGGGTKLSVL
EV- D68- 241	243	heavy	QVQLQESGPGPLVKPSQTLSTCTVSGGSITSGDYWNWIR QPPGKLEWIGYIYHSGTTYNPSLKSRTTISVDTSKNRFS LKLSSVTAADTAVYFCARAYAYEFWSGYPNWFDPWGLGT LVTVSS
	244	light	DIQMTQSPSSLASVGDRTTITCRASQRISTVYNNWYQVKA GTAPKVLIIYAASSLQGTGVPSRFSGSGSGTDFLTITVSLQPE DFTTYFCQQSYSPPWTFGGGTKVEIK
EV- D68- 247	245	heavy	QVQLVESGGGLVQPGRSLRLSCAASGFIENRYAMHWVRQ APGKGLEWVALISYDGINKYYADSVKGRFSISRDNSTLY LQMNSLRADTAIFYCARGLYCSGTGSGCTPFYWGQGI LVTVSS
	246	light	DIQMTQSPSSLASVGDRTTITCRASQSIKKYLNWYQQKP GNAPKLLIYGASNLQGTGVPSRFSGSGSGTDFALSISLETE DFATYYCQSDSAPPTFGGGTKVEFK
EV- D68- 254	247	heavy	QVQLQQRGAGLLKPSETLSLTCIYGASLNDYDWTWIRQP PGKRLEWIGVINRRDVTVDYNPSLKSRTLSLDTSKNQLSLS LSSVTAADTAVYICVRVPRRGFEFGFCDDTACRYGHT WFDPWGQGTLLVTVSS
	248	light	DIQMTQSPSSLASVGDRTTITCRASQSIIRDYLNWYQQRP GKAPKVLIFAGSRLESGVPSRFRGRGSGTEFTLTISDLQPE DFATYYCQSYLTPPTFGGTTVDIK
EV- D68- 260	249	heavy	QVTLKESGPFVLVKPTETLTCTVSGFSLRNGRMGVSWIR QPPGKALEWLAHIFASDEKSYSTQRTLSISRDTSKSQVV LSMTDMDPVDATATYCARILKFGTMRAAYYFDYWQGAL VPVSS
	250	light	SYVLTQPPSVSVAPGKTARITCGGINIGIRTVHWYQQKPGQ APMLVIYYDSRPLGIPERFSGSKSGNTATLTISRVEAGDE ADYYCQVWDSSSDHVVFGGGTKLTAL
EV- D68- 266	251	heavy	QVHLVESGGGLVKPGGSLRLSCAVSGFTFSDYYMSWIRQ APGKGLEWLSYISSSGSTIYADSMKGRFTISRDNARNLSY LQMNSLRVEDTAFYYCAGSKVGYTTGRRNWYFDLWGRG TLVTVSS
	252	light	QSALTQPPSASGSPGQSVTISCTGTSSDVGAYNYVSWYQ QHPGKAPKLMIEVTKRPSGVDRFSGSKSGNTASLTVSG LQAEDEADYYCSSYAGNNNLVFGGGTKLTVL
EV- D68- 269	253	heavy	QVQLLQSGSEVRKPGASVNIHCKASGFTFTDFYLHWVRQA PGQGLEWMGIINPETGETTYSQKFGGRVTMRDTSSTSVVN LEVRSLRSEDTAIYYCARDLVVVVPVEMSRRAFDIWGQGIM VTVSS
	254	light	SYVLTQPPSVSVAPGQTARIPCGGNNIERKSVHWYQQRP GQAPVLVYDDTVRPSGIPERFSGNSGSTATLTISRVGAG DEGDYYCQVWDSTTDHGVFGGGTKLTVL
EV- D68- 271	255	heavy	RVQLVQSGAEVKKPGASVKVSCQTSGYIFSAYYIYWVRQA PGQGLEWMGRMNAKSGGANTAQPFQGRITMTRDMSVS TAYMELSLRLSDDTAVYYCARDYRDDYMWGSYRPLDYW GGGTLVTVSS
	256	light	EVVLTQSPATLSLSPGERASLSCRASQSVSGYLAWYQHKP GQAPRLLIYDASNRAAGIPARFRGSGSGTDFLTISLSLEPE DFAVYFCQQRNGLTFGGGTKVEIK

TABLE 3

CDR HEAVY CHAIN SEQUENCES			
Clone	CDRH1 (SEQ ID NO:)	CDRH2 (SEQ ID NO:)	CDRH3 (SEQ ID NO:)
EV- D68-37	GFIFSRYA (257)	ISYDARNS (258)	ARPTLPYSNNWYAPEY (259)
EV- D68-40	GFSFSSYA (260)	ISGNNGNR (261)	AKVVRIAAVLYYFDY (262)

TABLE 3-continued

CDR HEAVY CHAIN SEQUENCES			
Clone	CDRH1 (SEQ ID NO:)	CDRH2 (SEQ ID NO:)	CDRH3 (SEQ ID NO:)
EV- D68-41	GFTFSNYA (263)	ISGSGGLT (264)	ARVKSTGTALVFDI (265)
EV- D68-43	GFTFINYG (266)	ISNDGSYN (267)	AKDKHGDYDYGVDV (268)
EV- D68-46	GFTFSSYG (269)	ISYDGSND (270)	ARRRPGSFPGLCDY (271)
EV- D68-48	GGSF SRLT (272)	HIPIFGTT (273)	ARMYSGHDGV DV (274)
EV- D68-71	GGSISSGFYY (275)	IYDSGRT (276)	ARHLTHLYGDYVTPDALDI (277)
EV- D68-72	GFTFTTYS (278)	ISSGSSNI (279)	ARAHGRIVNSGVVISRFDP (280)
EV- D68-74	GYTFTAYY (281)	INPSSGGA (282)	ARMGCRSDRCYSTNYNFDQ (283)
EV- D68-75	GFTISPYG (284)	ISSSSRYT (285)	ARERGHSTSSSYFDS (286)
EV- D68-76	GFTFSNAW (287)	IQTKIDGGIT (288)	STGPYYDTSGYPQPFDY (289)
EV- D68-80	GFDFSRYT (290)	ISSTSLYT (291)	ARVVGPAELDY (292)
EV- D68-84	GYTFTDYY (293)	INPKTGGG (294)	ARAGRNGYDY (295)
EV- D68-85	GFKFRNYA (296)	ITSGGST (297)	TVPWGNYN DYVSDY (298)
EV- D68-88	GFIFSRYP (299)	ISYDGNKK (300)	ARHFLPYSSSWYQGFNY (301)
EV- D68-89	GFLFSRYG (302)	ISYDGNKK (303)	ARGVPYGD TLTGLVY (304)
EV- D68-95	GFSLRNARMG (305)	IFSNDEK (306)	ARLLVAGTFLPSHYFDY (307)
EV- D68-97	GFTFSTYS (308)	ISSSSSTI (309)	TRQVGADFSGRGFDY (310)
EV- D68-98	GFKFSVYA (311)	ISSSGSTI (312)	ATARHITNDGF DI (313)
EV- D68- 105	GYLISNGYY (314)	IYHTRST (315)	ARGPGHCYGD DDCYAYYFDQ (316)
EV- D68- 110	GGTFSRFA (317)	ILPIFGTA (318)	ARSLPYCTNDVCSNQNTFDY (319)
EV- D68- 111	GGSISSGDYY (320)	IYYSGST (321)	ASRYGDP IGDNWFDP (322)
EV- D68- 114	GFRFSFYG (323)	ISGTGATR (324)	VRRFPMTT VTSFDS (325)
EV- D68- 116	GPPFNMFV (326)	IKQDGSEK (327)	VREGVRRVVVRSTGYFDF (328)
EV- D68- 150	GGPISNGPYY (329)	IFYSGST (330)	ARHVVTASGW FDP (331)

TABLE 3-continued

CDR HEAVY CHAIN SEQUENCES			
Clone	CDRH1 (SEQ ID NO:)	CDRH2 (SEQ ID NO:)	CDRH3 (SEQ ID NO:)
EV- D68- 151	GGPISNGPYY (332)	IFYSGST (333)	ARHVVTASGWFD P (334)
EV- D68- 152	GVTFSDNA (335)	ISYDGSSR (336)	ARVTADYIYESSGKVF (337)
EV- D68- 154	GGGISDHY (338)	IYTS GTT (339)	ARSLETVIRFYHYMDV (340)
EV- D68- 155	GGSIGDYH (341)	IHSSGNT (342)	ARQNVFDI (343)
EV- D68- 156	GISINNY (344)	VYSTGSS (345)	ARGSM PHI (346)
EV- D68- 157	GITFSNAW (347)	IESKIDGGTI (348)	TTDQGYDRSGYWVGNHFDY (349)
EV- D68- 158	GFTFSSYA (350)	ISGTTGST (351)	AKDSHSMIVVDHAFDI (352)
EV- D68- 159	GFTFSTYI (353)	ISTSSVYT (354)	AREEGFRAYNLY (355)
EV- D68- 160	GFTFSRFG (356)	ISFDGSNR (357)	ARDWDR LVRSAVGY (358)
EV- D68- 161	GGSVSTATYY (359)	IYSSGNT (360)	ERRLRILSIERNYYAMD V (361)
EV- D68- 162	GFTFSSYA (362)	ISYDASKK (363)	ARDHVPPKDCSDGNCHSDYGM DV (364)
EV- D68- 163	GYSFTNFA (365)	INPGNRNT (366)	ARLP IAAAGRGWFDP (367)
EV- D68- 164	GFTFNTYG (368)	ISSATTTF (369)	ARVYTMLRGASMDV (370)
EV- D68- 165	GYTFTAYY (371)	INPISGGT (372)	ARVKCSSANCYGNFDY (373)
EV- D68- 166	GYRFTNYR (374)	IYPGSDT (375)	ARQTTQNSGYDRWFDS (376)
EV- D68- 181	GITFSRHT (377)	ISGSGGST (378)	AISVPLLR FLEWFQHPFDF (379)
EV- D68- 183	GFTFNTYS (380)	ISSIGSYI (381)	VRFTMTTVTNFDS (382)
EV- D68- 185	GFTFTNYG (383)	ISYDGGNK (384)	PKVIPHPYDSSGDDAFDI (385)

TABLE 3-continued

CDR HEAVY CHAIN SEQUENCES			
Clone	CDRH1 (SEQ ID NO:)	CDRH2 (SEQ ID NO:)	CDRH3 (SEQ ID NO:)
EV- D68- 200	GFPFSSYS (386)	ISGSGGDI (387)	ARGLVATTGTRYFDY (388)
EV- D68- 208	GGTFRRFA (389)	IIPILGRG (390)	ARFISTASYVPGTFEDV (391)
EV- D68- 210	GFTFRNYN (392)	ISSIGSYI (393)	ARMVRNTVTAFDY (394)
EV- D68- 219	GYRLIDL (395)	FDPEKAEA (396)	ATWGVVNVNGRDYFDS (397)
EV- D68- 220	GFTFSSFE (398)	ISTSGSTI (399)	ARDVRDCSALTCPRRGDAFDF (400)
EV- D68- 221	GFSFSVYP (401)	ISSSSRYI (402)	VKVGGSKHQYYFDY (403)
EV- D68- 224	GYFTNYD (404)	ISTYNGNT (405)	ARERCSTSTCYSRYADY (406)
EV- D68- 225	GFTLSRYE (407)	ISSGGPSI (408)	MREGLTYDSTI (409)
EV- D68- 227	GFTFDEYA (410)	ISWNGGSK (411)	AKDDYEGAGFDI (412)
EV- D68- 228	GYLISNGYY (413)	IYYTRDT (414)	VRHEGSCNDGSCYGSFVDN (415)
EV- D68- 231	DFTFSSYT (416)	ISGDGVST (417)	ARGGT FHNWYFDL (418)
EV- D68- 234	GFTFSTYA (419)	ISGSGGST (420)	AKGTITYSYYMAV (421)
EV- D68- 235	GYRLTDLP (422)	VDLEKREI (423)	ATWGIEVVNGRDEFFDS (424)
EV- D68- 236	GYSLSDL (425)	FDPINGEI (426)	ATWGVAVVSGRRDYFDS (427)
EV- D68- 241	GGSIITSGDYY (428)	IYHSGTT (429)	ARAYAYEFWSGYPNWFDP (430)
EV- D68- 247	GFIFNRYA (431)	ISYDGINK (432)	ARGLGYCSGTGGSCPTPEY (433)
EV- D68- 254	GASLNDYD (434)	INRRDTV (435)	VRVPRRGFEFSFGFCDDTACRYGHTWFDP (436)
EV- D68- 260	GFSLENGRMG (437)	IFASDEK (438)	ARILKFMTMRAAYYFDY (439)
EV- D68- 266	GFTFSDYY (440)	ISSSGSTI (441)	AGSKVGYTTGRRNWYFDL (442)

TABLE 3-continued

CDR HEAVY CHAIN SEQUENCES			
Clone	CDRH1 (SEQ ID NO:)	CDRH2 (SEQ ID NO:)	CDRH3 (SEQ ID NO:)
EV-D68-269	GFTFTDFY (443)	INPETGET (444)	ARDLVVVVPVEMSRRAFDI (445)
EV-D68-271	GYIFSAYY (446)	MNAKSGGA (447)	ARDYRDDYMWGSYRPLDY (448)

TABLE 4

CDR LIGHT CHAIN SEQUENCES			
Clone	CDRL1 (SEQ ID NO:)	CDRL2 (SEQ ID NO:)	CDRL3 (SEQ ID NO:)
EV-D68-37	ALPKKY (449)	EDT (450)	SSTDSSGNPVL (451)
EV-D68-40	QSVSTY (452)	EAS (453)	QQRSSWPIT (454)
EV-D68-41	SGSVSSSY (455)	SIN (456)	GLYMGSGIWI (457)
EV-D68-43	QGISSW (458)	AAS (459)	QQADSFPR (460)
EV-D68-46	QSISKW (461)	KAS (462)	QQHNSYSY (463)
EV-D68-48	QSVRSY (464)	DAS (465)	QQRSTWPPGM (466)
EV-D68-71	QSVSSSF (467)	GAS (468)	QQYSNSRLT (469)
EV-D68-72	SSDVGGYNY (470)	EVT (471)	SSYTSTNLVV (472)
EV-D68-74	NIGTKS (473)	NDS (474)	QVWDSGIDVV (475)
EV-D68-75	NIGTKT (476)	YDS (477)	RVWDSSTDHRV (478)
EV-D68-76	KLGDYK (479)	QDT (480)	QAWDSSSTVV (481)
EV-D68-80	QDIGVD (482)	GAS (483)	LQDYNYPWT (484)
EV-D68-84	ALPKQY (485)	QDT (486)	QSGDSSGTYLV (487)
EV-D68-85	QRVGNS (488)	DAS (489)	HQHSTWPRGT (490)
EV-D68-88	SGSIATNY (491)	EDS (492)	QSYDNSRAVV (493)
EV-D68-89	SGTIASNY (494)	EDN (495)	QSYDNSDRV (496)
EV-D68-95	NIGLKS (497)	YDS (498)	QVWDSRNHPV (499)
EV-D68-97	KLGDYK (500)	QDS (501)	QAWDSSSTAV (502)
EV-D68-98	QGISRF (503)	SAS (504)	QQLNSHPRMFT (505)

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TABLE 4-continued

CDR LIGHT CHAIN SEQUENCES			
Clone	CDRL1 (SEQ ID NO:)	CDRL2 (SEQ ID NO:)	CDRL3 (SEQ ID NO:)
EV-D68-105	QGISNW (506)	DAS (507)	QQANSFPPT (508)
EV-D68-110	ALPKQY (509)	EDN (510)	QSADSSGTYVV (511)
EV-D68-111	NIGLKS (512)	YDS (513)	QVWDSRNHPV (514)
EV-D68-114	SSDVGGYNY (515)	DVT (516)	GAYAGFNAL (517)
EV-D68-116	AVPIKY (518)	EDD (519)	YSTDSSGYQRA (520)
EV-D68-150	QSVGTD (521)	DAF (522)	QQRSRWPPPYT (523)
EV-D68-151	QSVGTD (524)	DAF (525)	QQRSRWPPPYT (526)
EV-D68-152	QSVRSW (527)	KAS (528)	QQYQTFSTW (529)
EV-D68-154	QSLQSDGYSY (530)	LGS (531)	MQALQTPWT (532)
EV-D68-155	QSVLFSSNNKNY (533)	WAS (534)	QQFYTTPLT (535)
EV-D68-156	TSNIETNY (536)	RND (537)	AAWDDSLKAPV (538)
EV-D68-157	STNIGAGYD (539)	GNS (540)	QSYDRSLSTYV (541)
EV-D68-158	NIGTKS (542)	DDS (543)	QVWDSYNVHYV (544)
EV-D68-159	SSNIEYNY (545)	KNN (546)	AAWDDILSGVV (547)
EV-D68-160	SNDVGGYNY (548)	DVI (549)	CSYAGTYTWV (550)
EV-D68-161	QRVVNNY (551)	GAS (552)	QQYGSPWT (553)
EV-D68-162	QSISKW (554)	KAS (555)	QQHNSYSY (556)
EV-D68-163	QSVISTY (557)	DVS (558)	HQYGSSPAT (559)

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TABLE 4-continued

CDR LIGHT CHAIN SEQUENCES			
Clone	CDRL1 (SEQ ID NO:)	CDRL2 (SEQ ID NO:)	CDRL3 (SEQ ID NO:)
EV-D68-164	QSVGTY (560)	DSA (561)	QLRITVVPPIFT (562)
EV-D68-165	SSDVGGINY (563)	EVS (564)	SSYAGSNNLV (565)
EV-D68-166	TSSIGSNI (566)	INN (567)	AAWDDSLNGWV (568)
EV-D68-181	QSVGST (569)	GAS (570)	HQYINWPPWT (571)
EV-D68-183	SSDVGAYS (572)	DVY (573)	CSHAGSHTWV (574)
EV-D68-185	QISIRN (575)	GAS (576)	QQYSKLPIT (577)
EV-D68-200	QSVRSY (578)	DAS (579)	QQRSYWPFFT (580)
EV-D68-208	QSVSSY (581)	DAS (582)	QQRSDWPPGT (583)
EV-D68-210	SSDVGGYNF (584)	EVI (585)	CSYGGNNSWM (586)
EV-D68-219	VLSNQY (587)	KDT (588)	QSADNTRITV (589)
EV-D68-220	QSLVYSDGNTY (590)	KVS (591)	MQGTHWPRT (592)
EV-D68-221	RSNIGSYT (593)	NNN (594)	AAWDDSLNGVV (595)
EV-D68-224	QSINIY (596)	AAS (597)	QQSYRSPT (598)
EV-D68-225	SGHSRYA (599)	INSDGRH (600)	QWTGTGFRV (601)
EV-D68-227	QSVSSY (602)	GAS (603)	QQRSNWPIT (604)
EV-D68-228	QDISSW (605)	AAS (606)	QQADSFIT (607)
EV-D68-231	QSIGDN (608)	GAS (609)	QQYKNWPRT (610)
EV-D68-234	QSVRSSY (611)	GAS (612)	QQYGTSTIT (613)
EV-D68-235	VLSKQY (614)	KDT (615)	QSADTRITV (616)
EV-D68-236	VLSNQY (617)	KDT (618)	QTADTKYTV (619)
EV-D68-241	QRISTY (620)	AAS (621)	QQSYSPPT (622)
EV-D68-247	QSIKKY (623)	GAS (624)	QQSDSAPPT (625)
EV-D68-254	QSIRDY (626)	AGS (627)	QQSYLTPPT (628)
EV-D68-260	NIGIRT (629)	YDS (630)	QVWDSSSDHVV (631)
EV-D68-266	SSDVGAYNY (632)	EVT (633)	SSYAGNNLV (634)

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TABLE 4-continued

CDR LIGHT CHAIN SEQUENCES			
Clone	CDRL1 (SEQ ID NO:)	CDRL2 (SEQ ID NO:)	CDRL3 (SEQ ID NO:)
EV-D68-269	NIERKS (635)	DDT (636)	QVWDSTTDHGV (637)
EV-D68-271	QSVSGY (638)	DAS (639)	QQRSNGLT (640)

All of the compositions and methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the compositions and methods of this disclosure have been described in terms of preferred embodiments, it will be apparent to those of skill in the art that variations may be applied to the compositions and methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit and scope of the disclosure. More specifically, it will be apparent that certain agents which are both chemically and physiologically related may be substituted for the agents described herein while the same or similar results would be achieved. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the disclosure as defined by the appended claims.

VII. REFERENCES

The following references, to the extent that they provide exemplary procedural or other details supplementary to those set forth herein, are specifically incorporated herein by reference.

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U.S. Pat. No. 3,850,752
U.S. Pat. No. 3,939,350
U.S. Pat. No. 3,996,345
U.S. Pat. No. 4,196,265
U.S. Pat. No. 4,275,149
U.S. Pat. No. 4,277,437
U.S. Pat. No. 4,366,241
U.S. Pat. No. 4,472,509
U.S. Pat. No. 4,554,101
U.S. Pat. No. 4,680,338
U.S. Pat. No. 4,816,567
U.S. Pat. No. 4,867,973
U.S. Pat. No. 4,938,948
U.S. Pat. No. 5,021,236
U.S. Pat. No. 5,141,648
U.S. Pat. No. 5,196,066
U.S. Pat. No. 5,563,250
U.S. Pat. No. 5,565,332
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SEQUENCE LISTING

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tctaatecgt tctctggctc caagtctggc aacacggcct ccctgacat ctctgggctc      240
caggctgagg acgaggctga ttatttctgc agctcctata caaccagcaa cactctcgtg      300
gtgttcggcg gagggaccaa gctgaccgtc cta                                           333

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<210> SEQ ID NO 17
<211> LENGTH: 378
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 17

caggtgcagc tgggtcagtc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc      60
tcctgtagga cttctggata caccttcacg gcctactata tgactgggt gcgacaggcc      120

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cccggacaag ggcctgagtg gatgggaagg atcaaccga gcagtgggtg cgcacagtat 180
gcacagaagt ttcagggcag ggtcaccatg accagggaca cgtccatcag cacaacctac 240
atgacctga gggggtgac atctgacgac acggcgtgt tttactgtgc gagaatgggt 300
tgtcgtagtg accgggtgcta ttcgaccaac tacaactttg accagtgggg ccaggggaacc 360
ctggtcaccg tctcctca 378

```

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<210> SEQ ID NO 18
<211> LENGTH: 321
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 18

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tcctatgtgc tgactcagcc accctcgggtg tcagtggccc caggacagac ggccaggatt 60
ccctgtgggg gaaacaacat tggaactaaa agtggttact ggtaccagca gaagccaggc 120
caggcccctg tgctgggtcgt ctctaatac agcgaccggc cctcagggat ccctgagcga 180
ttctctggct ccaagtctgg gaacacggcc accctgacca tcagcagggt cgaagccggg 240
gatgaggccg actattattg tcaggtgtgg gatagtggta ttgatgtcgt tttcggcgga 300
gggaccaagc tgaccgtcct a 321

```

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<210> SEQ ID NO 19
<211> LENGTH: 366
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 19

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gagggtgcagc tgggtggagtc tgggggaggc ctgggtcaagc ctgggggggc cctgagactc 60
tcctgtgcag cctctggatt cactatcagt ccctatggca tgaactgggt ccgccaggct 120
ccaggggaagg ggctggagtg ggtctcattc attagtagta gtagtcgtta cacatattac 180
gcagaactcag tgaagggccg tttaccatc tccagagaca acgccaagaa ttcactgtct 240
ctgcaaatga acagcctgag agccgaggac acggctgtgt attactgtgc gagagagagg 300
ggccatagca cctcgtcctc atactttgac tcttggggcc agggaaacct ggtcaccgtc 360
tcctca 366

```

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<210> SEQ ID NO 20
<211> LENGTH: 324
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 20

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tcctatgtgc tgactcagcc accctcagtg tcattggccc caggaaagac ggccaggatt 60
acctgtgggg gaaacaacat tggaactaaa actgtgagct ggtaccagca gaagccaggc 120
caggcccctg tgctgggtcat gtattatgat agtgaccggc cctcagggat ccctgagcga 180
ttctctggct ccaactctgg gaacacggcc accctgacca tcaacagggt cgaggccggg 240
gatgaggccg actattactg tcgggtgtgg gatagtata ctgatcatcg agtggtcggc 300
ggggggacca agctgaccgt ccta 324

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<210> SEQ ID NO 21
<211> LENGTH: 381
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 21
gaggtgcagc tgggtggagtc tgggggaggc ttggtaaagc ctggggggtc ccttagactc      60
tcctgtgcag cctctggatt cactttcagt aacgcctgga ttagctgggt ccgccaggct      120
ccagggaagg ggctggagtg ggttggccgt attcaaacca aaactgatgg agggacaaca      180
gactacgctg caccogtgaa aggcagattc accatctcaa gagatgattc aaaaaacacg      240
ttgtatctgc aaatgaacag cctgaaaacc gaggacacag ccttgattta ttgtagcaca      300
ggaccgtatt actatgatac tagtggttac cccaaccct ttgactactg gggccaggga      360
accctggtca ccgtctctc a                                     381

```

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<210> SEQ ID NO 22
<211> LENGTH: 318
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 22
tcctatgagc tgactcagcc accctcagtg tccgtgtccc caggacagac agccagcatc      60
acctgctctg gagataaatt gggagataaa tatgcttgct ggtatcagca gaagccaggc      120
cagtcccctg tgctggctcat ctatcaagat accaagcggc cctcagggat ccctgagcga      180
ttctctggct ccaactctgg gaacacagcc actctgacca tcagcgggac ccaggctatg      240
gatgaggctg actattactg tcaggcgtgg gacagcagca ctgtggtatt cggcggaggg      300
accaagctga ccgtocta                                     318

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<210> SEQ ID NO 23
<211> LENGTH: 354
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 23
gaggtacaga tgggtggagtc tgggggaggc ctggtcaagc cggggggggtc cctcagactc      60
tcctgttcag tctctggatt cgacttcagt agatacacca tgaactgggt ccgccaggct      120
ccaggggagg ggctgaagtg ggtctcgtcc attagtagta ctagtcttta cacattctat      180
gcggaactcag tgaagggcgg attctccatc tccagagaca acgcccaggg ttccctgtct      240
ctacaaatga gcagcctgag acccgaagat acggctgtct attattgtgc gagagtcgtt      300
ggtcccgccg agttagatta ctggggccag ggagtgtggg tcaccgtctc ctca          354

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<210> SEQ ID NO 24
<211> LENGTH: 321
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 24
gtcaccagtc tgaccagtc tccatctcc ctgtctgctt ctgtcgggga cagagtcacc      60

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atcacttgcc gggcaagtca ggacattgga gttgacttag gttggtttca gcagagacct	120
gggaaagccc ctaaaactcct gatctatggt gcctccaggt tgcagagtgg ggtcccatca	180
cgcttcacgc ggcgtggatc tggcacattt ttcactctca ccatcagcag cctgcagcct	240
gaagatttta caacttactt ctgtcttcaa gattataatt acccttggac gctcggccag	300
gggaccacgg tgggagtc aa	321

<210> SEQ ID NO 25
 <211> LENGTH: 351
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 25

caggtgcacc tgggtcagtc tgggtctgcg gtgaggaagc ctggggcctc agtgaaggtc	60
tcctgcaagg cttctggata caccttcacc gactactata ttcactgggt gcgacaggcc	120
cctggacaag gccttgagtg gatgggatgg atcaacccta aaactgggtg ctcaaattat	180
acacagaggt ttcaggccag ggtcaccatg acctgggaca cgtccatcag tacagcctac	240
atggagttga gcaggtgag atctgacgac acggccgtgt attattgtgc gagggcgggc	300
agaaatggct acgactactg gggccaggga accctggtea ccgtctctc a	351

<210> SEQ ID NO 26
 <211> LENGTH: 324
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 26

tcctatgagc tgacacagcc accctcggtg tcagtgtccc caggacagac ggccaggatc	60
acctgctctg cagatgcatt gccaaagcaa tatgcttatt ggtaccaaca gaagccaggc	120
caggcccctg tgttgatgat atatcaagac actgagaggg cctcagggat cctgagcga	180
ttctctgggt ccagctcagg gacaacagtc acgttgacca tcagtggagt ccaggcagaa	240
gacgaggctg actattactg tcaatcagga gacagcagtg gtacttatct agttttcggc	300
ggagggacca agctgaccgt ccta	324

<210> SEQ ID NO 27
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 27

gaggtacagc tcttgagtc tgggggaggc ttggtacagc ctggggggtc cctgagactc	60
tcctgtgcag cctctggatt caaattdaga aactatgcca tgacctgggt ccgccaggct	120
ccagggaagg ggctggagtg ggtctcaact attactagtg gtggtagtag agagtagcga	180
gacgccgtga agggccgggt catcatctcc agagacaatt ccaagaacac gttatatttg	240
caaatgaaca gcctgagagc cgacgacacg gccgtatatt actgtacagt gccgtggggt	300
aactacaatg actacgtgtc tgactactgg ggccaggga ccctggtecc cgtctctca	360

<210> SEQ ID NO 28
 <211> LENGTH: 324

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 28

gaagttgtat tgacacagtc tccagccacc ctgtctttgt ctcaggggca aagagccacc      60
ctctcctgca gggccagtcg gctgtttggc aactccttag cctggtacca acaaaaacct      120
ggccaggctc ccagcctcct catctatgat gcttccaaga gggccactgg catcccagcc      180
aggttcagtg gcagtgggtc tgggacagac ttcactctca ccatcatcag cctagagtct      240
gaagattttg cagtttatta ctgtcaccaa catagcacct ggctcggggg gaccttcggc      300
caagggacac gactggagat taaa                                           324

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<210> SEQ ID NO 29
<211> LENGTH: 372
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 29

cagggtgcacc tgggtgagtc tgggggaggc gtggtccagc ctgggaggtc cctgagactc      60
tcctgtgcag cctctggatt catcttcagt agataccta tgcactgggt ccgccaggct      120
ccaggcaagg gtctggagtg ggtggcactt atatcatatg atggaacaa taaatactac      180
gcagactccg tgaagggcgc attcaccatc tccagagaca attccaagaa cacgctgttt      240
ctccaaatga acagcctgag agctgaggac acggctgtct attactgtgc gagacatttc      300
ctcccatata gcagtagttg gtaccagggc tttaactact ggggccaggg aatcctggtc      360
accgtctcct ca                                           372

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<210> SEQ ID NO 30
<211> LENGTH: 336
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 30

aatatttatgc tgactcagcc ccactctgtg tcggagtctc cggggaagac ggtaaccatc      60
tcctgcaccc gcagcagtg ggcagcttgc accaactatg tgcagtggta ccagcagcgc      120
ccgggcagtt cccccacccc tataatcttt gaagatagtc aaagaccgtc tggggtcctt      180
gatcggttct ctggtcccat cgacagctcc tccaattctg cctccctcac catctctgga      240
ctgaggagctg acgacgaggc tgactactac tgtcagtcct atgataacag caatcgggct      300
gttgtattcg gcggaggggc caagctgacc gtccta                                           336

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<210> SEQ ID NO 31
<211> LENGTH: 366
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 31

cagggtgcagc tgggtgagtc tgggggaggc gtggtccagc ctgggaggtc cctgagactc      60
tcctgtgaag cgtctggatt cctcttcagt cgctatggca tgcactgggt ccgccaggct      120

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ccaggcaagg ggctggactg ggtggcagtt atatcgatg atggaaataa gaaatattat	180
gcagactctg tgaagggcgc attcaccatc tccagagaca attcaaagaa cacgttgat	240
ctgcaagtga acagcctgag agtcgaggac acggctgttt attactgtgc gagaggtgtc	300
ccgtacgggtg acacccttac agggctgtgc tactggggcc agggaaacct ggtaaccgtc	360
tcctca	366

<210> SEQ ID NO 32
 <211> LENGTH: 330
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 32

aattttatgc tgactcagcc gcactctgtg tcggagtctc cggggaagac ggtaaccatc	60
tcctgcaccc gcagcagtg caccattgcc agcaactatg tgcagtggta ccagcagcgc	120
ccgggcagtg cccccaccac tgtaatctat gaggataacc aaagaccctc tggggtcct	180
gatcggttct ctggtccat cgacagctcc tccaactctg cctccctcac catctctgga	240
ctgaagactg aggacgaggc tgactactac tgtcagctct atgataacag cgatcggtg	300
ttcgcgaggag ggaccaagct gaccgtcctt	330

<210> SEQ ID NO 33
 <211> LENGTH: 375
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 33

caggtcacct tgaaggagtc tggctctgtg ctggtgaaac ccacagagac cctcacgctg	60
acctgcaccc tctctgggtt ctcactcagg aacgctagaa tgggagtgag ctggatccgt	120
cagccccag ggaaggccct ggagtggctt gcacacattt tttcgaatga cgaaaaatcc	180
tacaacacat ctctgaagag cagactctcc atctccaagg acacctccaa aagccagggtg	240
gtccttacca tgaccagcat ggaccctttg gacacagcca catatttctg tgcacggcta	300
ctggtggctg gtactttct cccctctcac tactttgact actggggcca aggaatctg	360
gtcacgtct cctca	375

<210> SEQ ID NO 34
 <211> LENGTH: 324
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 34

tcctatgtgc tgactcagcc accctcagtg tcagtgacct caggaaagac ggccaggatt	60
acctgtgggg gaaacaacat tggacttaaa agtgtcttct ggtaccagga gagggcagac	120
caggcccctg tgggtgctat ctattatgat agcgcccgcc cctcagggat ccctgagcga	180
atctctggct ccaagtctgg gaacacggcc accctgacca tcaccagggt cgaagccggg	240
gatgaggcgg actatttttg tcagggtgtg gatagtagtc gtaatcatcc ggtcttcggc	300
ggagggacca aactgaccgt cctc	324

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<210> SEQ ID NO 35
<211> LENGTH: 366
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 35
gagctgcagc tgggtggagtc tgggggaggc ttggtacagc ctgggggggc cctgagactc      60
tcctgtgcag cctctggatt caccttcagt acctatagca tgaattgggt ccgccaggct      120
ccagggaagg ggctggagtg ggtttcatac attagtagta gtagtagtac catacagtac      180
gcagactctg tgaagggccg attcaccatc tccagagaca atgccaagaa ctcaactgtat      240
ctgcaaatga atagcctgag agccgaagac acggccgtgt attattgtac gagacaggtc      300
ggggcggatt tcagtggccg cggccttgac tactggggcc agggaacctt gctcaccgtc      360
tcctca                                           366

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<210> SEQ ID NO 36
<211> LENGTH: 318
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 36
tcctatgagc tgactcagcc accctcagtg tccgtgtccc caggacagac agccaccatc      60
acctgctttg gagataaatt gggggataaa tatgcttgct ggtatcagca gaagccaggc      120
cagtcccctg tgttggtcat ctaccaagat agcaagcggc cctcagggat cctgagcga      180
ttctctggct ccaagtctgg gaacacagcc actctgacca tcagcgggac ccaggcaatg      240
gatgaggctg actattactg tcaggcgtgg gacagcagca ctgcagtgtt cggcggaggg      300
accaagctga cgtctctc                                           318

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<210> SEQ ID NO 37
<211> LENGTH: 360
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 37
gaggtacaac tagtgagtc agggggaggc ttggtacagc ctgggggggc cctgagactc      60
tcctgtgcag cctctggatt caagtittcc gtctatgcct tgagttgggt ccgccaggct      120
ccagggaagg ggctggagtg gatttcatat attagtagta gtggttctac catatattat      180
tcagactctg tgaagggccg attcaccatc tccagagaca atgtcgggaa ctcaactgtt      240
gtgcaaatga acagcctgag agccgaggac acgggtatattt attactgtgc gacagcccgc      300
cacatcacca atgatggttt tgatatgttg ggccaaggga caatgggtcat cgtctcttca      360

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<210> SEQ ID NO 38
<211> LENGTH: 327
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 38
gacatccagt tgaccagtc tccatccttc ctgtctgcat ctgtaggaga cagagtcacc      60

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atcacttgcc gggccagtc gggcattagt aggttttttag cctgggtatca gcagaaacca	120
gggaaagccc ctaagctcct gatctattct gcgtccactt tacaaagggg ggtcccatca	180
aggttcagcg gcagtggatt tgggacagat ttcactctca caatcagtag ccttcagcct	240
gaagattttg caacttatta ctgtcaacaa cttaatagtc acccccgaaat gtteactttc	300
ggccctggga ccacagtggga tatcaag	327

<210> SEQ ID NO 39
 <211> LENGTH: 381
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 39

caggtgcagc tgcaggagtc gggcccagga ctggtgaagc cttcggagac cctgtccctc	60
acctgcgctg tgtctgggta cttaatcagc aatggttact actggggctg gatccggcag	120
ccccccggga aggggctgga gtggattggg agtatctatc atactagaag cacctactac	180
aaccctgccc tcaagagtcg agtcagcatc tcagtagaca cgtccaagaa ccggttctcc	240
ctgaggtctga ggtctgtgac cgccgcagac acggcctttt attactgtgc gagaggccca	300
ggccactgtt atggtgatga cgactgctac gcgtactact ttgaccagtg gggccaggga	360
accccggtca ccgtctcccc a	381

<210> SEQ ID NO 40
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 40

gacatccaga tgacccagtc tccatcttcc tgggtctcat ctgtaggaga cagagtcacc	60
atcacttgtc gggcgagtc ggggtattagc aactgggttag cctgggtatca gcagaacca	120
gggaaagctc ctaaaactcct gatctatgat gcctccagtt tgcaaagtgg ggtcccatca	180
aggttcagcg gcagtggatc tgggacagat ttcactctca ccatcaacag cctgcagcct	240
gaagattttg caacttacta ttgtcaacag gctaacagtt tccctttcac ttccggccct	300
gggaccaaag tggatatcaa a	321

<210> SEQ ID NO 41
 <211> LENGTH: 381
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 41

caggtgcagc tgggtcagtc tggggctgag gtgaagaagc ctgggtcctc ggtgaaggtc	60
tcctgcaagg cttctggagg caccttcagc aggtttgcta tcagctgggt gcgacaggcc	120
cctggacaag ggcttcagtg gatgggaggg atcctcccta tctttggtac agcaaaactac	180
gcacagaagt tccagggcag agtcacgatt accgcggaca catccacgag cacagcctac	240
atggagctga gcagcctgag atctgaggac acggcctgtg attactgtgc gagatccctc	300
ccctattgta ctaatgatgt atgctcaaac cagaacacat ttgactactg gggccaggga	360
accctggtca ccgtctcctc a	381

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<210> SEQ ID NO 42
<211> LENGTH: 324
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 42
tcctatgagc tgacacagcc accctcgggtg tcagtgtccc caggacaggc ggccaggatc      60
acctgctctg gagatgcatt gcctaagcaa tatgcttatt ggtaccagca gaagccaggc      120
caggccccctg tgttggtgat atatgaagac aataagaggc cctcagggat ccctgagcga      180
ttctctggct ccagctcagg gacaacagtc acgttgacca tcagtggagt ccaggcagaa      240
gacgaggctg actattactg tcaatcagca gacagcagtg gtacttatgt ggtattcggc      300
ggagggacca agctgaccgt ccta                                           324

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<210> SEQ ID NO 43
<211> LENGTH: 369
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 43
cagggtgcagc tgcaggagtc gggcccagga ctggtgaagc cttcacagac cctgtccctc      60
acctgcactg tctctggtgg ctccatcagc agtggtgatt actactggag ttggatccgc      120
cagccccccag ggaagggcct ggagtggatt gggtagatct attacagtgg gagcacctac      180
tacaaccctg ccctcaagag tcgagttacc atatcagtag acacgtccaa gaaccagttc      240
tccctgaagc tgagctctgt gactgccgca gacacggccg tgtattactg tgccagccgc      300
tacggtgacc cgatagggga caactgggtc gacccctggg gccagggaac cctggtcacc      360
gtctcctca                                           369

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<210> SEQ ID NO 44
<211> LENGTH: 324
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 44
tcctatgtgc tgactcagcc accctcagtg tcagtgacct caggaaagac ggccaggatt      60
acctgtgggg gaaacaacat tggacttaaa agtgtcttct ggtaccagga gaggccagac      120
caggccccctg tgggtggtcat ctattatgat agcgcccgcc cctcagggat ccctgagcga      180
atctctggct ccaagtctgg gaacacggcc accctgacca tcaccagggt cgaagccggg      240
gatgagggcg actatttttg tcagggtggt gatagtagtc gtaatcatcc ggtcttcggc      300
ggagggacca aactgaccgt cctc                                           324

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<210> SEQ ID NO 45
<211> LENGTH: 363
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 45

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gaggtgcaac tgttgagtc ggggggaggc ttggtgcagc cgggggggtc cctgagactc	60
tcctgcgcag cctccgatt caggtttagc ttctatggca tgacctgggt ccgccaggct	120
ccagggaagg ggctggagtg ggtctcaagt attagtggta ctggtgctac cagaaactgc	180
gcagactccg tgaagggccg gttcaccatc tccagagaca actccaagaa cacgctgtac	240
ctgcaaatgg acagcctgag agtcgacgac acggccgttt tttattgtgt gagacggttc	300
ccgatgacca cggtgacaag ctttgactct tggggccagg gaacctgggt caccgtctcc	360
tca	363

<210> SEQ ID NO 46
 <211> LENGTH: 327
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 46

cagttgccc tgactcagcc tcgctcagtg tccgggtctc ctggacagtc agtcaccatc	60
tcctgcactg gaaccagcag tgatgttggg ggttataact ttgtctcctg gtaccaacaa	120
cacccgggca aagcccccaa actcatgatt tttgatgtca ctgggcggcc ctcaggggtc	180
cctgatcgct tctctggctc caagtctggc aacacggcct ccctgacat cgctgggctc	240
caggctgaag atgaggctga ttattattgt ggcgcataat cgggctttaa cgctcttttc	300
ggcggaggga ccaaactgac cgtccta	327

<210> SEQ ID NO 47
 <211> LENGTH: 375
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 47

gaggtgcagc ttggtcagtc ggggggaggc ttggtccggc cgggggggtc cgtgagactc	60
tcctgtgtag cctctggatt ccccttcaat atgttttggg tgggtgggt ccgccagact	120
ccagggaagg gactggagtg ggtggccaat ataaacagg atggcagtga gaaatactat	180
gtcgactccg tgaagggccg attcgccatc tccagagaca atgccaagaa ctctctcttt	240
cttcaaatgg acagcctgag tgcgggggac acggccatct attattgtgt cagagagggg	300
gtgcgaaggg tcgctgtacg tagtaccggt tacttcgact tttggggcca gggacagctg	360
gtcaccgtct cctca	375

<210> SEQ ID NO 48
 <211> LENGTH: 324
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 48

tcctatgagc tgacacagcc accctcgatg tcagtgtccc caggacaaac ggccaggatc	60
acctgctctg gagatgcagt gccaaataaaa tatgtttatt ggtaccaaca gaggtcaggc	120
caggcccctg tattagtcat ctatgaagac gacagacgac cctccgggat ctctgagaga	180
ttctctggct ccagttcagg gacaacggcc accttgacta tcaactggggc ccaggtggag	240
gatgaaggtg actactattg ctattcaaca gacagtagtg gttatcagag agcgttcggc	300

-continued

ggggggacca cgctgaccgt ccta 324

<210> SEQ ID NO 49
 <211> LENGTH: 363
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 49

cggtgacgc tgcaggagtc ggccccagga ctggtgaggc cttcagagac cctgtccctc 60
 acctgcagtg tctctggtgg ccccatcagc aatggtcctt attactggag ctggatccgc 120
 cagcaccag ggaagggcct ggagtggatt ggattcatct tttacagtgg gagcaccaac 180
 tacaaccctg cctccgggg gcgcgtaacc atggcagtg acacgtctaa gaaccagttc 240
 tccctgaggc tgaactctgt gactgccgcg gacacggccg tttattactg tgcgagacat 300
 gtggtgactg cgtcggggtg gttcgacccc tggggccagg gaaccctggt caccgtctcc 360
 tca 363

<210> SEQ ID NO 50
 <211> LENGTH: 327
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 50

gaaattgtgt tgacacagtc tccagccacc ctgtctttgt ctccaggga aagagccacc 60
 ctctcctgca gggccagtc gagtggtgg accgacttag cctggtacca acagaaacct 120
 ggccaggctc ccagggtcct catctatgat gcattcaaga gggccactgg catcccagcc 180
 aggttcagtg gcagtggtgc tgggacagag ttcactctca ccatcagcag cctcgagcct 240
 gaagattttg cagtttatta ctgtcagcag cgtagcaggt ggccctcccc gtacactttt 300
 ggccagggga ccaagctgga gatcaaa 327

<210> SEQ ID NO 51
 <211> LENGTH: 363
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 51

cggtgacgc tgcaggagtc ggccccagga ctggtgaggc cttcagagac cctgtccctc 60
 acctgcagtg tctctggtgg ccccatcagc aatggtcctt attactggag ctggatccgc 120
 cagcaccag ggaagggcct ggagtggatt ggattcatct tttacagtgg gagcaccaac 180
 tacaaccctg cctccgggg gcgcgtaacc atggcagtg acacgtctaa gaaccagttc 240
 tccctgaggc tgaactctgt gactgccgcg gacacggccg tttattactg tgcgagacat 300
 gtggtgactg cgtcggggtg gttcgacccc tggggccagg gaaccctggt caccgtctcc 360
 tca 363

<210> SEQ ID NO 52
 <211> LENGTH: 327
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 52

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gaaattgtgt tgacacagtc tccagccacc ctgtctttgt ctccagggga aagagccacc      60
ctctcctgca gggccagtc gagtggtggc accgacttag cctggtacca acagaaacct      120
ggccaggctc ccagggtcct catctatgat gcattcaaga gggccactgg catcccagcc      180
aggttcagtg gcagtggtgc tgggacagag ttcactctca ccatcagcag cctcgagcct      240
gaagattttg cagtttatta ctgtcagcag cgtagcaggt ggctccccc gtacactttt      300
ggccagggga ccaagctgga gatcaaa                                     327

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<210> SEQ ID NO 53

<211> LENGTH: 366

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 53

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caggtgcacg tgggtgagtc tgggggaggc gtggtccagc ctgggaggtc cctgagactc      60
tcctgtgcag actctggagt caccttcagt gacaatgctt tgtactgggt ccgccaggct      120
ccaggcaagg ggctggagtg ggtcgcagtt atctcatatg atggaagcag tagatactac      180
gcagactccg tgagggggcgc gttcaccata tccagagaca attccaagga cagcgtgtat      240
ctgcaaatga acagactgag agctgaggac acggctatatt attactgtgc gagagtcaca      300
gcggattact atgagagtag tggcaagggt ttttggggcc agggagccct ggtcgtcgtc      360
tcctca                                     366

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<210> SEQ ID NO 54

<211> LENGTH: 321

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 54

```

gacatccaga tgacccagtc tccttccacc ctgtctgcat ctgtggggga cagagtctcc      60
atcacttgcc gggccagtc gagtggtagg agctggttgg cctggtatca gcacaaacca      120
gggaaagccc ctaaacctct gatctataag gcgtctagtt tagaaagtgg ggtcccatca      180
aggttcagcg gcagtggtgc tgggacagaa ttcactctca ccatcagcag cctgcaggct      240
gatgatattg caacttatta ctgccaacag tatcagactt tttcctggac gttcggccaa      300
gggaccacgg tggaagtcaa a                                     321

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<210> SEQ ID NO 55

<211> LENGTH: 372

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 55

```

caggtgcagc tgcaggagtc gggcccagca ctggtgaagc ctcgggagac cctgtccctc      60
acctgcactg tctctggtgg ctccatcagt gatcactact ggagctggat ccggcagccc      120
ccagggaagg gactggagtg gattggctac atctatacca gtgggaccac caactacaac      180
ccctccctca agagtcgagt caccatatca gtagacacat ccaagaagca gttctcctg      240

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-continued

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aatctgaggt ctgtgaccgc cgcagacacg gccgtgtatt actgtgcgag aagtctagaa    300
acgggtgatcc gtttctacta ctaccactac atggacgtct ggggcaaagg gaccacgggtg    360
atcgtctcat ca                                                              372

```

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<210> SEQ ID NO 56
<211> LENGTH: 336
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 56
gatattgtga tgactcagtc tccactctcc ctgcccgta cccctggaga gccggcctcc    60
atctcctgca ggtctagta gagcctctcg cagagtgatg ggtacagcta tttggattgg    120
tacctgcaga agccagggca gtctccacag ctectgatct atttgggttc taatcggggc    180
tccgggggtcc ctgacagggt cagtgtcatt ggatcaggca catattttac actgaaaatc    240
agcagagtgg aggctgagga tgttggcggt tattttctgca tgcaagctct acaaactccg    300
tggaagtctc gccaaaggac caaggtggaa atcaaaa                                336

```

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<210> SEQ ID NO 57
<211> LENGTH: 342
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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```

<400> SEQUENCE: 57
cagggtgcagc tgcaggagtc gggcccagga ctggtgaagc cttcggagac cctgtccctc    60
acctgcactg tcgctggcgg ctccatcggt gattaccact ggaactggat ccggcagccc    120
gccgggaagg ggctggagtg gattgggcgt atacatagca gtgggaacac tgactacaac    180
ccctccctca agagtcgagt caccatgtca gtagacacgt ccaagaacca gttctccctg    240
aaactgaggt ctgtgaccgc cgcggacacg gccgtgtatt actgtgcgag gcaaaatgtt    300
tttgatatct ggggccaagg gacaatggtc accgtctctt ca                        342

```

```

<210> SEQ ID NO 58
<211> LENGTH: 339
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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```

<400> SEQUENCE: 58
gacatcgtga tgaccagtc tccagactcc ctggctctgt ctctgggcga gagggccacc    60
atcaactgca agtccagcca gagtgtttta ttcagctcca acaataagaa ctacttagct    120
tggtaccagc agaaaccagg acagcctcct aagtgtctca tttactgggc atctaccggg    180
gaatccgggg tccctgaccg attcagtggc agcgggtctg agacagattt cactctcacc    240
atcagcagcc tgcaggctga agatgtggca gtttatttct gtcagcaatt ttatactact    300
ccgctcactt tcggcggagg gaccaaggtg gagatcaaa                                339

```

```

<210> SEQ ID NO 59
<211> LENGTH: 342
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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-continued

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 59

caggtacaga tgcaggagtc gggcccaggg ctggtgaagg cttcggagac cctgtccctc	60
acctgcagtg tctctggtat ctccatcaat aactactatt ggagttgggt ccgccagccc	120
cccgggaagg gcctggagtg gatcggatat gtctattcta ctgggagttc caagtacaat	180
ccctccctcg agcgtcgagc caccatgtca gtagacacgt ccaacaacaa cttctccctg	240
aggctgacgt ctgtgaccac tgcggacacg gccgtctact actgtgcgcg ggggagtatg	300
ccgcatactc ggggccaggg cctcctggtc accgtctcct ca	342

<210> SEQ ID NO 60

<211> LENGTH: 330

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 60

cagttctgtc tgactcagcc accctcagcg tccgggaccc ccggacagag ggtcaccatc	60
tcctgttctg gtagcacctc caacatcgag actaattatg tatactggta ccagcaggtc	120
ccaggaacgg cccccaagcc cctcgtctat aggaatgac agcgcacctc gggggtcctc	180
gaccgattct ctggctccaa gtctggcacc tcggcctccc tggtcacag tgggtcccg	240
accgaggatg aggtgctta ttattgtgca gcttgggatg acagtctgaa agtcctggtc	300
ttcggagctg ggaccaaggt cgcgcctctc	330

<210> SEQ ID NO 61

<211> LENGTH: 390

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 61

gaggtgcagc tgggtggagtc tgggggaggc ttgataaagc cgggggggtc ccttagactc	60
tcctgtgcag cctctggaat cactttcagt aacgcctgga tgagctgggt ccgccaggct	120
ccaggggaagg ggctggaatg ggttggccgt attgaaagca aaattgacgg tgggacaata	180
gactacgcta caccctgaa aggcagattc accatctcaa gagatgattc aaaaaacacg	240
ctgtacctgc aaatgaacag cctgaaaacc gaggacacag ccgtctatta ctgtaccaca	300
gaccagggct actatgatag aagtggttat tgggtcgtcg ggaaccactt tgactactgg	360
ggccagggaa tcctgggtcac cgtctcctca	390

<210> SEQ ID NO 62

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 62

cagttctgtc tgacgcagcc gccctcagtg tctggggccc cagggtgag ggtcaccatc	60
tcctgtactg ggagcagcac caacatcggg gcaggttatg atgtacctg gtaccagcac	120
cttcaggaa cagccccaa actcctcatc tatggttaaca gcaatcggcc ctcaggggtc	180
cctgaccgat tctctggctc caagtctggc acctcagcct ccctggccat cactgggtc	240

-continued

caggctgacg atgcggctga ttattactgc cagtcctatg acagaagcct gagtacttat 300
gtcttcggaa ctgggaccaa ggtcaccgtc cta 333

<210> SEQ ID NO 63
<211> LENGTH: 369
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 63
gaggtgcagc tgttgagtc tgggggaggc ttggtacagc ctggggggtc cctgagactc 60
tcctgtgcag cctctggatt caccttagc agctatgcca tgaactgggt ccgccaggct 120
ccagggaagg ggctggagtg ggtctcaggt attagtggta ctacaggtag cacatactac 180
gcagactccg tgaagggccg gttcaccatc tccagagaca attccaagaa cacggtgcat 240
ctgcaaatga acagcctgag agccgaggac acggccgtat attactgtgc gaaagattct 300
cactccatga tagtagttga tcatgctttt gatatctggg gccaaaggac aatggtcacc 360
gtctcttca 369

<210> SEQ ID NO 64
<211> LENGTH: 324
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 64
tcctatgtgc tgactcagcc accctcgggt tcaagtggccc caggacagac ggccaggatt 60
acctgtgggg gaaacaatat tggaactaaa agtgtgcact ggtaccagca gaggccaggc 120
caggccccctg tgctggctgt ctatgatgat agcgaccggc cctcagggat cctgagcga 180
ttctctggct ccaactctgg gaacacggcc accctgacca tcagcagggt cgaagccggg 240
gatgaggccg actattactg tcagggtgtg gatagttata atgttcatta tgtcttcgga 300
acagggacca aggtcaccgt ccta 324

<210> SEQ ID NO 65
<211> LENGTH: 357
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 65
gaggtgcagc tgggtgagtc tggggggggc ctggtaagc cggggggggtc cctgagactc 60
tcctgtgcag cctctggatt caccttcagt acttatatca tgacctgggt ccgccaggcc 120
ccagggaggg ggctggagtg ggtctcatcc attagtacca gtagtggtta cacattctat 180
gcagattcac tgaagggccg gttcaccatc tccagagaca acgccaagaa ttcagtgtat 240
ctgcagatga acagcctgag agccgacgac acggctgttt attactgtgc gagggaagaa 300
gggtttcgag cttataacct atactggggc cagggaaccc tggtaactgt ctctca 357

<210> SEQ ID NO 66
<211> LENGTH: 330
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 66

cagtctgtgc tgactcagcc accgtcagcg tctgggaccc cgggcagag ggtcaccatc	60
tcttggtctg gaagcagctc caacatcgaa tacaattatg tttactggta ccagaaatc	120
ccaggaacgg cccccaaact cctcatctat aaaaataatc agcggccctc aggggtccct	180
gaccgattct ctggctccaa gtctggcacc tcagcctccc tggccattag tgggtccgg	240
tccgaggatg aggggtgatta ttactgtgca gcatgggatg acatcctgag tgggtgtggt	300
ttcggcgggg ggaccaagct gaccgtcctc	330

<210> SEQ ID NO 67

<211> LENGTH: 363

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 67

cagggtgcagc tgggtgagtc tgggggaggc gtggtccagc ctgggaggtc ccagaaactc	60
tcctgtgcag cctctggatt cacgttcagt aggtttggtg tgcactgggt ccgccaggct	120
ccaggaacgg ggctggagtg ggtggcagtt atttcgtttg atggaagtaa tagatactac	180
gcagactccg tgaaggggagc attcaccatc accagagaca attccaagaa cacattgtat	240
ctgcaaatga acaacctgag acctgaggac acggctgtat attactgcgc gagagattgg	300
gataggctgg ttcgttcggc ggttggtctac tggggccagg gaacctggt cagcgtctcc	360
tca	363

<210> SEQ ID NO 68

<211> LENGTH: 330

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 68

cagtctgccc tgactcagcc tcgctcagtg tccgggtctc ctggacagtc agtcaccatc	60
tcctgcactg gaaccagtaa tgatgttggg ggttataact ttgtctcctg gtatcagcaa	120
caccaggaag aagcccccaa actaatgatt tttgatgtca ttaggcggcc ctgaggggtc	180
cctggctcgt tctctggctc caagtctggc gacacggcct cccttatcat ctctggactc	240
caggctgagg atgaggctga ttattactgc tgctcatatg caggcaccta cacctgggta	300
ttcggcgcag ggaccacact gaccgtccta	330

<210> SEQ ID NO 69

<211> LENGTH: 378

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 69

cagggtgcacc tgcaggagtc gggccacga ctgggtgaagc cttcggagac cctgtccctc	60
acctgcactg tctctgggtg ctcgctcagc actgccactt actactggag ctggatccgg	120
cagtccccag ggaggggact ggagtggata ggatatactt attccagtgg taacaccaac	180
tacaaccctt cccttaagag tcgagtcacc atttctttag acacgccccaa caaccagctc	240

-continued

tccttgacgt tgacctctgt gaccgctgcg gacacggcca tttattattg tgagaggcgc	300
ttacgtattc tgagtattga gaggaactac tacgctatgg acgtctgggg ccaagggacc	360
cgggtcaccg tctcctca	378

<210> SEQ ID NO 70
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 70

gaagttgtgt tgacgcagac tccaggcacc ctgtctttgt ctccggggga aggagccacc	60
ctctcctgca gggccagtc gagggttgtc aacaactact tagcctggta ccagcagaga	120
gctggccagg ctcccaggct cctcattttt ggtgcatcca acagggccac tggcatcca	180
gacaggttca gtggcagtg gctcgggaca gacttcactc tcaccatcag gaagctggag	240
cccgaagatt ttgcagtga ttactgtcaa caatatggta gcccgtaggac gttcggccac	300
gggaccaagg tggaaatgaa a	321

<210> SEQ ID NO 71
 <211> LENGTH: 390
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 71

cagggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgagactc	60
tcctgtgcag cctctggatt caccttcagt agttatgcta tgcattgggt ccgccaggct	120
ccaggcaggg ggctggagtg ggtggcagtt atatcatatg atgcaagcaa gaaataccac	180
gcagactccg tgaagggccg attcacaatc tccagagaca gttccaagaa cagctgttt	240
ctgcaaatga atagcctgaa acctgaggac acggctgtgt attactgtgc gagagatcat	300
gtccccccca aggattgcag tgatggtaat tgccactcgg actacggtat ggacgtctgg	360
ggccaaggga ccacggtcac cgtctcctca	390

<210> SEQ ID NO 72
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 72

gacatccaga tgacccagtc gccttccact atgtctgctt ctgtaggaga cagagtcacc	60
atcacttgcc gggccagtc gagtattagt aagtggttgg cctgggtatca gcagaagcca	120
gggaaagccc ctaaactcct gatctataag gcgtctacct taaaaactgg ggtcccatca	180
aggttcagcg gcagtggatc tgggacagaa ttcactctca ccatcaacag cctgcagcct	240
gatgatatttg caacttatta ctgccaacag cataatagtt attcgtacac ttttggccag	300
gggaccaagg tggagatcaa g	321

<210> SEQ ID NO 73
 <211> LENGTH: 366
 <212> TYPE: DNA

-continued

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 73

caggtcacgc ttgtgcagtc tggggctgag gtgaagaagc ctggggcctc agtgaagatt	60
tcttgcaagg cttctggata ctcttctact aactttgctg tgcattgggt gcgccaggcc	120
cccgacaaaa gacttgagtg gatgggatgg atcaaccctg gcaatagaaa cacaaagtat	180
tcacacaact ttcagggcag agtcaccatt accagggaca catccgcgaa cacagcctac	240
atggaactga gcagcctgag atctcaagac acggctgtgt attactgtgc gagacttccg	300
atagcagcag ctggcagggg ctgggttcgac cctgggggcc agggaacctt ggtcaccgtc	360
tcctcc	366

<210> SEQ ID NO 74

<211> LENGTH: 324

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 74

gaaattgtgt tgacgcagtc tccaggcacc ctgtctttgt ctccagggga aagagccacc	60
ctctcctgca gggccagtcag gagtggtatc agcacctact tagcctggta ccagcagaga	120
cctggccagg ctcccagggt cctcatctat gatgtatcca ccagggccac tggcatccca	180
gacaggttca gtggcagtggt gtctgggaca gacttacttc tcaccatcag cagactggag	240
cctgaagatt ttgcagctca tttctgtcac cagtatggtg gttcacccgc gacgttcggc	300
caagggacca aggtggaat caaa	324

<210> SEQ ID NO 75

<211> LENGTH: 363

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 75

gaagtgcagc tgggtgagtc tgggggagac ttggtacagc ctggggggtc cctgagactc	60
tcctgtgcag cctctggatt caccttcaat acttatggca tgaactgggt ccgccagggt	120
ccagggaagg gactggagtg ggtttctatac attagtagtg ccaccactac cttctactac	180
gcagactctg tgagggggccg attcaccatc tccagagaca atgccaagaa ttcactatct	240
ctgcacatga agagcctgag agacgaagac acggcagttt attactgtgc gagagtctat	300
actatgcttc gcggagcgag tatggacgtc tggggccacg ggaccacggg caccgtctcc	360
tca	363

<210> SEQ ID NO 76

<211> LENGTH: 327

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 76

gaaattgtgt tgacacagtc tccagccacc ctgtctttgt ctccagggga aagagccacc	60
ctctcctgca gggccagtcag gagtggtggc acctacctgg cctggtacca acagaaacct	120

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ggccaggctc ccaggctect catctatgat tcagccaaca gggccactgg catcccagcc 180
cggttcagtg gcagtgggtc tgggacagac ttcactctca ccatcagcag cctagagcct 240
gaagattttg cagtttatta ctgtcagctg cgcacacact ggctcctat attcactttc 300
ggccctggga ccaaagtgga tgtcaaa 327

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<210> SEQ ID NO 77
<211> LENGTH: 369
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 77

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caggtgcacc tgggtcagtc tggggctgag gtgaagaagt ctggggcctc agtgaaggtc 60
tcctgcaaga cttttggata caccttcacc gcctactata tgcactgggt gcgacaggcc 120
cctggacagg ggctgagtg gatgggatgg atcaacccta tcagtggtag cacaactat 180
gcacagaagt ttcagggcag ggtctccatg accagggaca cgtccatcag cacagcctac 240
atgggctga gcaggctgag acctgacgac acggccgtct attactgtgc gagagtgaag 300
tgtagtagtg ccaactgcta tgggaacttt gactactggg gccagggtac cctggtcacc 360
gtctcctca 369

```

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<210> SEQ ID NO 78
<211> LENGTH: 330
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 78

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cagttgccc tgactcagcc tccctccgcg tccgggtctc ctggagagtc agtcaccatc 60
tcctgcaactg gaaccagcag tgacgttggt ggttataact atgtctcctg gtaccaacaa 120
caccaggcca aagcccccaa actcatgatt tatgaggtea gtaagcggcc ctgagggtc 180
cctgatcgct tctctggctc caagtctggc aacacggcct ccctgaccgt ctctgggctc 240
caggctgagg atgaggctga ttattactgc agctcatatg caggcagcaa caatttgta 300
ttcggcggag ggaccaagct gaccgtccta 330

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<210> SEQ ID NO 79
<211> LENGTH: 369
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 79

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gaggtgcagc tgggtcagtc tggagcagag gtgaaaaagc ccggtgagtc tctgaagatc 60
tcctgtaagg gttctggata caggtttacc aactaccgga tcggctgggt gcgccagatg 120
cccggaag gcctggagtg gatggggatc atctatcccg gtggctctga taccagatac 180
agtccgtccc tccaaggcca ggteaccatg tcagtcgaca agtccatcag caccgcctac 240
ctgatgtgga gcagcctgaa ggcctcggac accgccatgt attactgtgc tcgacagacc 300
actcaaaata gtggctacga tagatggttt gactcctggg gccagggaac ccagtcacc 360
gtctcctca 369

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<210> SEQ ID NO 80
<211> LENGTH: 330
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 80

cagtctgtac tgactcagcc accctcagcg tctgggaccc cgggcagag ggtcaccatc      60
tcttggtctg gaagcacctc cagcatcgga agtaatatg taaattggta ccaacacctc      120
ccaggaacgg cccccaaact cctcatctat attaataatc agcggccctc aggggtccct      180
gaccgattct ctgggtccaa gtctggcacc tcagcctccc tggccatcag tgggtccag      240
tctgaggatg aggctgacta ttactgtgca gcatgggatg acagcctgaa tggttgggtg      300
ttcggcggag ggaccaagct gaccgtcctg                                     330

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<210> SEQ ID NO 81
<211> LENGTH: 378
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 81

gaggtgcagg tggtggagtc tgggggaggc ttggtacagc ctggggggtc cctgagactc      60
tcctgtgcag cctctggaat cacctttagc aggcatacta tgagctgggt ccgccaggct      120
ccaggaaggg ggctggagtg ggtctcagct attagtggga gtggtggtag cacatatcat      180
gcagactccg tgaagggcgc cttcaccatc tccagagaca gttccaagag cacgtgtgat      240
ctgcaaatga acagcctgag agccgaggac acggccggtt attactgtgc gatctccgtg      300
ccattattac gatttttggg gtggtttcaa cacccttttg acttctgggg ccaggggaacc      360
ctggtcaccg tctcctca                                     378

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<210> SEQ ID NO 82
<211> LENGTH: 324
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 82

gaaatagtga tgacgcagtc tccagcctcc ctgtctgtgt ctccagggga aagagtcacc      60
ctctcctgca gggccagtcg gactgttggc agcaccttag cctggtagca gcacaaacct      120
ggccaggctc ccaggctcct catctctggt gcatccacca gggccactgg tgtcccagcc      180
aggttcagtg gcagtgggtc tgggacagag ttcactctca ccatcagcag tctgcagtct      240
gaagattttg cagtttacta ctgtcaccag tatattaact ggctccgtg gacgttcggc      300
caagggacca aggtggaaat caaa                                     324

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<210> SEQ ID NO 83
<211> LENGTH: 360
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 83

gaggtgcggc tgggtggagtc tgggggaggc ctgggtcaaac ctggggggtc cctgagactc      60

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tcctgtgcag cctctggatt caccttcaat acatattcca tgagctgggt cgcagggt	120
ccaggaagg ggctggagt ggctgcctcc attagtagta ccggaagta catatacaat	180
gcagactcac tgaagggcgc attcaccatc tccagagaca acgccaagaa ctccctgttt	240
ctgcaaatga acagcctgag agtcgaagac acggctgtgt attactgtgt gagattcacc	300
atgactacag tgactaactt tgactcatgg ggccaggga ccctgggtcac cgtctcctca	360

<210> SEQ ID NO 84
 <211> LENGTH: 330
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 84

cagtctgccc tgactcagcc tcgctcagtg tccgggtctc ctggacagtc agtcaccatc	60
tcctgcactg gaaccagcag tgatgttggg gcttatagct atgtctcctg gtaccaacaa	120
caccaggaag aagcccccaa actcatgatt tatgatgtct acaggcgccc ctccaggggc	180
cctggctcgt tctctggctc caagtctggc aacacggcct ccctgaccgt ctctgggctc	240
caggctgagg atgaggctga ttactactgc tgctcacatg caggcagtc cacctgggtg	300
ttcggcggag ggaccaaggt gaccgtccta	330

<210> SEQ ID NO 85
 <211> LENGTH: 378
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 85

cagctgcagg tgggtggcgc tgggggaggc gtggtccagc ctgggaggtc cctgagactc	60
tcctgtaaag cctctggatt caggttcacc aattatggca tgcaactggg tcgccaggcg	120
ccaggcaagg ggctggagt gggtggcttt atatcatacg atggaggtaa taaattttat	180
gcagactctg tgaagggcgc attcaccatc tccagagaca attccaggaa caccgtttat	240
ctgcaaatga acagcctgag agtggcggac acggctatgt attactgtcc gaaggtcac	300
ccccaccgt attatgatag tagtggtgat gatgcttttg atatctgggg ccaagggaca	360
atggtcgcca tttcttca	378

<210> SEQ ID NO 86
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 86

gaaatagtga tgacgcagtc tccagccacc ctgtctgtgt ctccaggga aagagccacc	60
ctctcctgct gggccagtc gagtattagc cgcaacttag cctgggtatc gcaaaaacct	120
ggccaggctc cccgactcct catctatggt gcatccacca gggccactgg tatccccgcc	180
aagttcagtg gcagtgggtc tgggacagac ttcactctca ccgtcagcag cctgcagtct	240
gaagaccttg cagtttatta ctgtcagcag tatagtaagt tgccatcac cttcgcccaa	300
gggacacgac tggagattaa a	321

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<210> SEQ ID NO 87
<211> LENGTH: 366
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 87

gaggtgcagc tggaggagtc tgggggaggc ttggtacagc cggggggggtc cctgagactc      60
tcctgtgcag cctctggatt ccccttcagt agttatagca tgagctgggt ccgccaggct      120
ccagggaagg ggctggagtg ggtctcatatc ataagtggta gtggtgggtga catatactac      180
gcagactctg tgaagggccg attcaccatc tccagggaca atgccaggaa ctcactgtct      240
ctgcaaatga acagcctgag agccgacgac acggctgtgt attactgtgc gagagggtg      300
gtagcaacaa ctggtacaag gtactttgac tactggggcc agggaaccct ggtcaccgtc      360
tcctca                                           366

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<210> SEQ ID NO 88
<211> LENGTH: 324
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 88

gaaattgtgt tgacacagtc tccagccacc ctgtctttgt ctccaggga aagagccacc      60
ctctcctgca gggccagtca gagtgttagg agttacttag cctggtacca acagagacct      120
ggccaggctc ccaggctcct catctacgat gcatccaaca gggccactgg catcccagtc      180
aggttcagtg gcagtgggtc tgggacagac ttcactctca ccatcagcag cctagagcct      240
gaagattttg cagtttatta ctgtcagcag cgtagctact ggctccgtt cactttcggc      300
ggagggaaca aggtggagat caaa                                           324

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<210> SEQ ID NO 89
<211> LENGTH: 372
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 89

caggtgcagc tgggtgcagtc tggggctgag gtgaagaagc cggggtcctc ggtgaaggtc      60
tcctgcaagg cttctggagg caccttcagg aggtttgcta tcagctgggt gcgacaggcc      120
cctggacaag ggcttgagtg gatgggaggg atcatcccta tcctaggtag aggaaagtac      180
gcacagaagt tccagggcag agtcaggatt accgcggacg aatccacgag cacagcctac      240
atggagctga gcagcctgag atctgaagac acggccgtgt attactgtgc gagatattatt      300
tcgacagcct cctatgttcc ggggaccttc gaggacgtct ggggccaagg gaccacggtc      360
accgtctcct ca                                           372

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<210> SEQ ID NO 90
<211> LENGTH: 324
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 90

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gacattgtgt tgacacagtc tccagccacc ctgtctttgt ctccagggga aagagccacc	60
ctctcctgca gggccagtc gagtgtagc agctacttag cctggtacca acagaaacct	120
ggccaggctc ccaggctcct catctatgat gcatccaaca gggccgctgg catcccagcc	180
aggttcagtg gcagtggtc tgagacagac ttcactctca ccatcagcag cctagagcct	240
gaagatgttg cggtttatta ctgtcagcag cgtacgcact ggctccggg gacttttggc	300
caggggacca acgtggagat caaa	324

<210> SEQ ID NO 91
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 91

gaggtgcagc tgggtgagtc tgggggaggc ctggtcaagc cgggggggtc cctgagactc	60
tcctgtgcag cctctggatt caccttcagg aactataaca tcaactgggt ccgccaggct	120
ccagggaagg ggctggagtg ggtctcatcc attagtagta ctggtagtta catacactac	180
gcagatttag tgaagggcgc attcaccatc tccagagaca acgccaagaa ctactgtat	240
ctgcaaatga acagcctgag agtcgaggac acggctgtat attactgtgc gcgaatggtt	300
aggaatacgg tgactgcctt tgactactgg ggccagggaa ccctggtctc cgtctcctca	360

<210> SEQ ID NO 92
 <211> LENGTH: 330
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 92

cagtctgccc tgactcagcc tgcctccgtg tctgggtctc ctggccagtc gatcaccatc	60
tcctgcactg gaaccagcag tgatgttggc ggctataact ttgtctcctg gtaccaacaa	120
cagccaggca gagcccccaa actccttatt tatgaagtca ttaagcggcc ctccaggggtt	180
tctgatcgct tctctggctc caagtctggc gacacggcct ccctgacaat ctctgggctc	240
caggctgagg acgaggctga ttattactgc tgctcatatg ggggtaacaa ctcttgatg	300
ttcggcggag ggaccatgct gaccgtccta	330

<210> SEQ ID NO 93
 <211> LENGTH: 372
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 93

caggtccagg tggtagagtc tggggctgag atgaagaagc ctggggcctc agtgaaggtc	60
tcctgcaagg ttccggata caggctcatt gatttaccct tgcactgggt gcgacaggct	120
cctggaaaag ggcttgagtg gatgggactt tttgatcctg aaaaggctga agccatctac	180
tcacagaaat tccaggacaa ggtcaccata agcgaggaca catctatcga cacagcctac	240
atggaactga acagcctgcg ctctgaagac acggccgtct actactgtgc aacttgggga	300
gttaggtggt tgaatgggag aagggactac tttgactcct ggggccaggg aaccctggtc	360

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accgtctctct ca 372

<210> SEQ ID NO 94
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 94

tcctatgagc tgacacagcc accctcggtg tcagtgtccc caggacagac ggccaggatc	60
acatgctatg cagatgtgtt gtcaaaccac tatacttact ggtatcaaca gaagccaggc	120
caggcccctg tgttggtgat atataaagac actgagaggc cctcagggat cctgagcga	180
tttgcctggt ccagctcagg gacaacagtc accctagtca tcaatggagt ccgggcagag	240
gacgaggctt actattactg tcaatcagcc gacaacacca gaattacggt ttccggcgga	300
gggaccaagc tgcctgtct a	321

<210> SEQ ID NO 95
 <211> LENGTH: 384
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 95

ggggtgcagc tgggtgagtc tgggggagga ttgatacagc ctggagggtc cctgagactc	60
tcctgtgcag cctctggatt caccttcagt agttttgaaa tgaactgggt ccgccaggct	120
ccagggaagg ggctggagtg ggtttcatat attagtacta gtggtagtac catatactac	180
gcagactctg tcaagggcgc attcaccatc tccagagaca acgccaggaa ctactgtct	240
ctgcaaatga acagcctgag agccgaggac acggctgttt attactgtgc gagagacgtg	300
agggattgta gtgctcttac gtgccccga aggggagatg cttttgattt ctggggccgt	360
gggacaaggg tcacgtcttc ttca	384

<210> SEQ ID NO 96
 <211> LENGTH: 336
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 96

gatgttgta tgactcagtc tccactctcc ctgcccgtca cccttgaca gtcggcctcc	60
atctcctgca ggtctagtca aagcctcgta tacagtgatg gaaacaccta cttgaattgg	120
tttcagcaga ggccaggcca atctccaagg cgcctaattt ataaggtttc taaccgggac	180
tctgggggtc cagacagatt cagcggcagt gggtcaggca ctgatttcac actgaaaatc	240
agcagggtgg aggctgagga tgttggggtt tattactgca tgcaaggtag aactggcct	300
cgcactttcg gccctgggac caaagtggat atcaaa	336

<210> SEQ ID NO 97
 <211> LENGTH: 363
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 97

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gaggtgcagc tgggtggagtc tgggggaggc ctgggtcaagc cgggggggtc cctgagactc	60
tcctgtgcag cctctggatt cagtttcagt gtctatccca tgaactgggt ccgccaggct	120
ccaggaaggg ggctggagtg ggtctcatcc ataagtagta gtagtcgtta catatcctac	180
gcagactcac tgaggggtcg aatcaccatc tccagagaca acgccaagaa ctcaactgtat	240
ctgcaaatga acagcctgag agtcgaggac acggctgtgt attactgtgt gaaagtcggg	300
ggtagcaaac accaatacta ctttgactac tggggccagg gatccctggt caccgtctcc	360
tca	363

<210> SEQ ID NO 98
 <211> LENGTH: 330
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 98

cagtctgtgc tgactcagcc accctcaacg tctgggatcc cggggcagac ggtcaccatc	60
tcttgttctg gaagcaggtc caacatcgga agttatactg ttaactggta cgagcaactc	120
ccaggaacgg cccccaaact cgtcatcttt aataataatc agcgtccctc aggggtccct	180
gaccgattct ctgggtccaa gtctggcacc tcagcctccc tggccatcag tgggtccag	240
tctgaggggtg aagctgatta ttactgtgca gcatgggatg acagcctgaa tgggtgtggt	300
ttcggcggag ggaccaagct gaccgtccta	330

<210> SEQ ID NO 99
 <211> LENGTH: 372
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 99

caggttcagt tgggtcagtc tggagctgag gtgaagaggc ctggggcctc agtgaaggtc	60
ttctgcgaagg cttctgggta cacccttacg aattatgaca tcatctgggt gcgacaggcc	120
cctggacaag ggcttgagtg ggtgggctgg atcagcactt acaatggtaa cacaaactat	180
gaacagaacc tccagggcag agtcaccatg accacagaca catccacgag cacagcctac	240
atggagctga ggagcctgag atctgacgac acggccgttt actattgtgc gagagagcgt	300
tgtagtacta gtacctgcta tagtcgttat gctgactact ggggccaggg aaccctggtc	360
accgtctcct ca	372

<210> SEQ ID NO 100
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 100

gacatccaga tgaccagtc tccatcctcc ctgtctgcat ctgtgggaga cagagtcacc	60
atcacttgcc gggcaagtca gacattaat atctatttga attggtatca gcaaaaaccg	120
gggaaagccc ctaaggctct gatctatgct gcatccagtt tgcaaaagtgg ggtcccatca	180
aggttcctgt gcagtgggtc tgggacagat ttcacctcca ccatcagcag tctgcaacct	240

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gaagactttg caacctacta ctgtcaacag agttacaggt cccctcggac gtteggccaa	300
gggaccgagg tggaaatcaa a	321

<210> SEQ ID NO 101
 <211> LENGTH: 357
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 101

gaggtgcagc tgggtggagtc tgggggaggc ttggcacagc ctggagagtc catgagactc	60
tcctgtgtag cctctggatt caccttaagt cgttatgaag tgaactgggt ccgccaggct	120
ccaggaagg ggctagagtg gctttcatac attagcagtg gtggtccttc catatactac	180
gcagactctg tgaagggccg attcacgac tccagaaaca gcgccagaa ctcaactggaa	240
ctacaaatgt ccacctgag gaccgaggac acggctgttt attattgtat gagagagggt	300
cttacttatt atgatagtag tatttggggc cagggaaccc tggtcgccgt ctctca	357

<210> SEQ ID NO 102
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 102

cagaatgtgc tgactcaatc gccctctgcc tctgcctccc tgggagcctc ggtcaaacac	60
acctgcactc tgaacagtgg gcacagcaga tacgccatcg catggcatca acatcagcca	120
cagaggggccc ctccgttctc gatgaagatt aatagtgtat gcagacacat caggggggac	180
ggcatctctg atcgtctctc aggtccgcc tctggggctg agcgtcatct caccatctcc	240
agcctccagc ctgaggatga ggctgactat tattgtcaga cctggggcac tggctttcgg	300
gtgttcggcg gagggaccaa actgaccgtc cta	333

<210> SEQ ID NO 103
 <211> LENGTH: 357
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 103

gaagtgcagc tgggtggagtc tgggggaggc ttggtacagc ctggcaggtc cctgagactc	60
tcctgtgcag cctctggatt cacctttgat gaatatgcca tgcaactgggt ccggcaagtt	120
ccaggaagg gcctggagtg ggtctcaggt attagtgtga atggtggttag caaaggctat	180
gcggactctg tgaagggccg attcaccatc tccagagata acgccaggta ttccctgtct	240
ctgcaaatga acagtctgag aactgaggac acggccttat attactgcgc aaaagatgat	300
tacgaggggg ctggttttga tatctggggc caaggagacag tggtcaccgt ctcttca	357

<210> SEQ ID NO 104
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 104

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gaaattgtgt tgacacagtc tccagccacc ctgtctttgt ctccagggga aagagccacc	60
ctctcctgca gggccagtc gagtgtagc agctacttag gctggtacca acagaaatct	120
ggccaggctc ccaggctcct catctatggt gcatccagca gggccactgg catcccagcc	180
aggttcagtg gcagtggttc tgggacagac ttcactctca ccatcagcag cctagagcct	240
gaagattttg caatttatta ctgtcagcag cgtagcaact ggcctatcac ctteggccaa	300
gggacacgac tggagattaa a	321

<210> SEQ ID NO 105
 <211> LENGTH: 378
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 105

cagggtgcagc tgcaggagtc gggcccagga ctggtgaagc ctteggagac cctgtccctc	60
acctgcactg tctctggeta ttttaattagc aatgggtact actggggctg gatccggcag	120
tccccggga aggggctgga gtggattggg agtatctatt atactaggga cacctactac	180
aactgggtccc tcaagagtcg aatcaccata tcagtggaca cgtccaagaa acagttctcc	240
ctgaagttgt attctgtgac cgccgcagac acggccgtct attactgtgt gagacatgag	300
ggttcttgca atgatggaag ctgttacggc tcgttcgttg acaactgggg ccaggggaacc	360
ctggtcaccg tctctca	378

<210> SEQ ID NO 106
 <211> LENGTH: 318
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 106

gacatccaga tgacccagtc tccatcttcc gtgtctgcat ctgtaggaga cagagtgacc	60
ctcacttgtc gggcgagtc ggatattagc agctggtag cctgggtatca gcagaaacca	120
gggaaagccc ctaagctcct gatctatgct gcatccagtt tgcaaagtgg ggtcccatca	180
aggttcagcg gcagtggtgc tgggacacat ttcactctca ccatcagcag cctgcagcct	240
gaagatttcg caacttactt ttgtcaacag gctgacagtt tcatcacttt cggcggaggg	300
accaaggtgg agatcaaa	318

<210> SEQ ID NO 107
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 107

gaggtgcaac tggtggagtc tgggggaggc ttggtacagc cgggggggtc cctgagactc	60
tcctgtgcag cctctgattt caccttcagc agctacacca tggcctgggt ccgccaggct	120
ccaggaagg ggcttgaatg ggtctcatct attagtgggt atggtgttag cacaaaaaac	180
gcagactccg tgaagggcgc attctccgtc tccagagaca attccaagaa cacacttttt	240
ctgcaactga acagtctgag agccgaggac acggcctttt attactgtgc gagggggggg	300

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accttcata actggtactt cgatctctgg ggccgtggcg tcttggtcac tgtctcctca 360

<210> SEQ ID NO 108
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 108

gaaatagtga tgacgcagtc tccagccacc ctgtctgtgt ctccagggga aacagccacc 60
 ctctcctgca gggccagtc gagtattggc gacaacttag cctgggtatca gcagaaacct 120
 ggccaggctc ccaggctcct catctctggt gcattccaca gggccactga tttccagcc 180
 aggttcctgt gcagtggtgc tgggacagag ttcactctca ccatcagcag cctgcagtct 240
 gaagactttg cagtttatta ctgtcagcag tataaaaact ggctcggac gttcggccga 300
 gggaccaagg tggaagtcag a 321

<210> SEQ ID NO 109
 <211> LENGTH: 363
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 109

gaggtgcagc tgttgagtc tgggggagc ttggtacagc cgggggggct cctgagactc 60
 tcctgtgcag cctctggatt cacctttagc acctatgcca tgagctgggt ccgccaggct 120
 ccagggaagg ggccggagtg ggtctcaggt attagtggta gtggtggtag cacaactac 180
 gcagactctg cgaaggccgc gttcaccatc tccagagaca attccaagaa cacgctgtat 240
 ctgcaaatga acagcctgag agccgaggac acggccgtat attattgtgc gaaagggacc 300
 attacttact cctactacta catggccgct tggggcaaag ggaccacggt caccgtctcc 360
 tca 363

<210> SEQ ID NO 110
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 110

gaaattgtgt tgacgcagtc tccaggcacc ctgtctttgt ctccagggga aagaggcacc 60
 ctctcctgca gggccagtc gagtgtagg agcagctact tagcctggta ccagcagaga 120
 cctggccagg ctcccaggct cctcatctat ggtgcatcca gcagggccac tggcatccca 180
 gacaggttca gtggcagtg gtctgggaca gacttcacgc tcaccatcgg cagactggag 240
 cctgaagatt ttgcagtgt ttactgtcag cagtatggta cctcaataac tttcggcgga 300
 gggaccaagg tgagatcaa a 321

<210> SEQ ID NO 111
 <211> LENGTH: 372
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 111

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caggtcacagg tggtaacaac tggggctgag atgaggaagc ctggggcctc agtgagggtc	60
tcctgcaagg ttcccgata caggetcact gatttaccct tgcactgggt gcgacaggct	120
cctggaaaag ggcttgaatg gattggattt gttgatcttg aaaagcgca aatcatctac	180
gcacagaaat ttcagggaac agtcaccata accgaggaca catctgcaga caccgcctac	240
atggaactga acagcctgag atctgaagac acggccgtct actactgtgc aacttgggga	300
attgaggtgg tgaatgggag ggacgaattc ttgactcct ggggccaggg aaccctggtc	360
tccgtctcct ca	372

<210> SEQ ID NO 112
 <211> LENGTH: 318
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 112

tcctatgagc tgacacagcc accctcggtg tcagtgtccc caggacagac ggccaggatc	60
acatgctatg cagatgtgtt gtcaaagcaa tatacttact ggtatcagca gaagccaggc	120
caggccctg tgttggtgat atataagac actgagaggc cctcagggat ccctgagcga	180
tttctggct ccagctcagg gacaacagtc accctgatca tcaatggagt ccggacagag	240
gacgaggcgt actattactg tcaatcagcc gacaccagaa ttacggtttt cgccggaggg	300
accaagctgt ccgtccta	318

<210> SEQ ID NO 113
 <211> LENGTH: 372
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 113

caggtcacagg tggtcacagc tggggctgag atgaagaagc ctggggcctc agtgaagggtc	60
tcctgcaagg ttcccgata tagtctcagt gatttaccct tgcactgggt gcgacaggct	120
cctggaaaag ggcttgagtg gatgggactt ttgatccta taaacgggtga aatcatctac	180
gcacagacat tccagggaac agtcaccata agcgaggaca catcgataga cacagcctac	240
atggaactca acagcctgag atctgaagac acggccgtgt actattgtgc aacttgggga	300
gttgccgtgg tgagtgggag aagggaactac ttgactcct ggggccaggg aaccctggtc	360
accgtctcct ca	372

<210> SEQ ID NO 114
 <211> LENGTH: 318
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 114

tcctatgagc tgacacagcc accctcggtg tcagtgtccc caggacagac ggccaggatc	60
acatgctatg cagctgtatt gtcaaaccac tatacttact ggtatcaaca gaagccaggc	120
caggccctg ttttggtgat atataagac actgagaggc cctcagggat ccctgagcga	180
tttctggct ccagctcagg gacaacagtc accctgatca tcaatggagt ccggacagag	240

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gacgaggctt actattactg tcaaacagcc gacaccaa atacggtttt cggcggaggg	300
accaagctgt ccgtccta	318

<210> SEQ ID NO 115
 <211> LENGTH: 378
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 115

cagggtgcagc tgcaggagtc gggcccagga ctggtgaagc cttcacagac cctgtccctc	60
acctgcactg tctctgggtg ctccatcacc agtggtgatt actactggaa ttggatccgc	120
cagcccccag ggaagggcct ggagtggatt gggtagatct atcacagtgg gaccacctac	180
tacaaccctg cctcaagag tcgagttacc atatcagtag acacgtccaa gaacaggttc	240
tccctgaagt tgtcctctgt gactgccgca gacacggccg tgtatttttg tgccagagcc	300
tacgcttatg aattttggag cggttaccct aactgggtcg acccctgggg cctgggaacc	360
ctggtcaccg tctcatca	378

<210> SEQ ID NO 116
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 116

gacatccaga tgacccagtc tccatcctcc ctgtctgcat ctgttgaga cagagtcacc	60
atcacttgcc gggcaagtca gcgcattagt acctatgtaa attggtatca ggtgaaagca	120
gggacagccc ctaaggctct gatctatgct gcgtccagtt tgcaaaactgg ggtcccatca	180
aggttcagtg gcagtggtac tgggacagat ttcactctca ccattgtcag tctacaacct	240
gaagatttta caacctactt ctgtcaacag agttacagtc ccccgtaggac gttcggccaa	300
gggaccaagg tggaaatcaa a	321

<210> SEQ ID NO 117
 <211> LENGTH: 378
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 117

cagggtgcagt tgggtggagtc tgggggaggc ttggtccagc ctggtaggtc cctgagactc	60
tcctgtgcag cctctgggtt catcttcaat agatatgcca tgcactgggt ccgccaggct	120
ccaggcaagg ggctcgagtg ggtggctctt atatcatatg atggaattaa taaatattac	180
gcagactccg tgaagggccg attctccatc tccagagaca attccaagag tacgctgtat	240
ctgcaaatga acagcctcag agctgaggac acggctatct tttactgtgc gagaggacta	300
ggatattgta gtggtaccgg tggtagctgt acaccctttg aatattgggg ccagggaatc	360
ctggtcaccg tctcctca	378

<210> SEQ ID NO 118
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence

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<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 118

gacatccaga tgacccagtc tccaccctcc ctgtctgcat ctgttggaga cagagtcacc	60
atcacttgcc gggcaagtca gaggattaag aaatatataa attggtatca gcagaaacca	120
gggaatgccc ctaagctcct catctatggc gcacccaatt tgcaaaactgg ggtcccatca	180
aggttcagtg gcagtggatc tgggacagat ttgcgtctct ccatcagcag tctggaaact	240
gaagattttg caacttacta ctgtcaacag agtgacagtg cccctcccac ttgcggcgga	300
gggaccaagg tggagttcaa a	321

<210> SEQ ID NO 119

<211> LENGTH: 405

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 119

cagggtgcagc tgcaacagcg gggcgagggg ctgttgaagc cctcggagac cctgtccctt	60
acctgcgaaa tctatggtgc atccctcaat gattacgact ggacctggat ccgccagccc	120
ccagggaaga ggctggagtg gattggggtc atcaatcgtc gtgacactgt tgactacaac	180
ccgtccctca agagtcgggt caccctctca ctgacacgt ccaagaacca actttccctg	240
agtctgagtt ctgtgaccgc cgcggacacg gctgtttatt actgtgtgag agtcccacgt	300
cggggctttg aagggtcttt cggattttgt gatgatactg cctgccgcta cgggcatacc	360
tggttcgacc cctggggcca ggaaccctg gtcaccgtct cctca	405

<210> SEQ ID NO 120

<211> LENGTH: 321

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 120

gacattcaga tgacccagtc tccttcctcc ctgtctgcat ctgttggcga cagagtcacc	60
atcacttgcc gggcaagtca gaggattaga gattatttaa attggtatca acaaagacca	120
gggaaagccc ctaaagtctc gatctttgct ggttcccgtt tggaaagtgg ggtcccatcg	180
aggtttagag gccgtggatc tgggacagaa ttcactctca ccatcagcga tctgcaacct	240
gaggattttg caacttacta ctgtcaacag agttacctta cacctccgac attcggccaa	300
gggaccacgg tcgatataca a	321

<210> SEQ ID NO 121

<211> LENGTH: 375

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 121

caggtcacct tgaaggagtc tggctctgtg ctggtgaaac ccacagagac cctcacgctg	60
acctgcaccg tctctggatt ctactccgc aatggtagaa tgggtgtgag ctggatccgt	120
cagcccccag ggaaggccct ggagtggtct gcacacattt ttgcgagtga cgaaaaatct	180

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tacagtacat ctcagaggac caggtctctcc atctccaggg acacctccaa aagccaagtg	240
gtccttagca tgaccgacat ggaccctgtg gacacagcca catattactg tgcgcggatt	300
ttgaagtttg ggacaatgag ggccgcatac tactttgact actggggcca gggagccctg	360
gtccccgtct cctca	375

<210> SEQ ID NO 122
 <211> LENGTH: 324
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 122

tcctatgtgc tgactcagcc accctcagtg tcagtggccc caggaaagac ggccaggatt	60
acctgtgggg gaatcaacat tggaattaga actgtacact ggtaccagca gaagccaggc	120
caggccccta tgttggtcat ctattatgat agcgaccggc ccttagggat cctgagcga	180
ttctctggct ccaagtctgg gaacacggcc accctgacca tcagcagggt cgaagccggg	240
gatgaggccg actattactg tcagggtgtg gatagtagta gtgatcatgt tgtattcggc	300
ggagggacca agctgacgc ccta	324

<210> SEQ ID NO 123
 <211> LENGTH: 375
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 123

caggtgcact tgggtgagtc tgggggaggc ttggtcaagc ctggagggtc cctgagactc	60
tcctgtgcag tttctggatt caccttcagt gactactaca tgagctggat ccgccaggct	120
ccggggaagg gactggagtg gctttcatac attagtagta gtggtagtag catatactac	180
gcagactcta tgaaggccg attcaccatc tccagggaca acgccaggaa ctactctat	240
ctgcaaatga acagcctgag agtcgaggac acggcctttt attactgtgc ggggtcaaag	300
gttggtata ctactggtc aaggaactgg tacttcgac tctggggccg tggcaccctg	360
gtcactgtct cctca	375

<210> SEQ ID NO 124
 <211> LENGTH: 330
 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 124

cagtctgccc tgactcagcc tccctccgcg tccgggtctc ctggacagtc agtcaccatc	60
tcctgcactg gaaccagcag tgacgttggg gcttataact atgtctcctg gtaccaacag	120
caccagggca aagccccaa actcatgatt tatgaggtca ctaagcggcc ctcaggggtc	180
cctgatcgct tctctggctc caagtctggc aacacggcct ctctgaccgt ctctgggctc	240
caggctgagg atgaggctga ttattactgc agctcatatg caggcaacaa caatttagtc	300
ttcggcggag ggaccaagct gaccgtccta	330

<210> SEQ ID NO 125
 <211> LENGTH: 378

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 125

cagggtgcagt tgcctgcagtc tgggtctgag gtgaggaaac ctggggcctc agtgaacatt	60
cactgtaagg catctggatt cactttcacc gacttctatt tacactgggt gcgacaggcc	120
cctggacaag ggcttgagtg gatggggata atcaaccctg aaaccgggta gacaacctac	180
tcacagaagt ttcagggcag agtcaccatg accagggaca cgtccacgag tgtagtgaat	240
ctggaagtga ggagcctgag atctgaggac acggccatat attactgtgc gagagatctc	300
gttgctcgtag tccccgtga aatgtctcgg cgtgcctttg acatttgggg ccaagggtt	360
atggtcacag tctctca	378

<210> SEQ ID NO 126
<211> LENGTH: 324
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 126

tcctatgtgc tgactcagcc accctcgggtg tcagtggccc caggacagac ggccaggatt	60
ccctgtgggg gcaacaacat tgaacgtaaa agtgtccact ggtaccagca gaggccaggc	120
caggccccctg tggtggctgt ctatgatgat actgtccggc cctcaggtat ccctgagcga	180
ttctctggct ccaactcggg gagcacggcc accctgacca tcagcagggt cggagccggg	240
gatgagggcg actattattg tcagggtgtg gacagcacca ctgaccatgg ggtcttcggc	300
ggagggacca agctgaccgt ccta	324

<210> SEQ ID NO 127
<211> LENGTH: 375
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 127

cgggtgcagc tgggtgcagtc tgggtctgag gtgaagaagc ctggggcctc agtgaaggtc	60
tcctgtcaga cttctggata cattttcagc gcctactaca tctattgggt gcgacaggcc	120
cctggacaag gacttgagtg gatgggacgg atgaacgcta agagtggagg cgcaaacact	180
gcacagcagt ttcagggcag actcaccatg accagggaca tgtccgtcag cacagcctac	240
atggaactga gcaggctgcg atcggacgac acggccgtct attattgtgc gagagactat	300
agggatgact acatgtgggg gagttatcgg cctttagact actggggcca gggaaacctg	360
gtcacctgtct cctca	375

<210> SEQ ID NO 128
<211> LENGTH: 318
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 128

gaagttgtgt tgacacagtc tccagccacc ctgtctttgt ctccaggga aagagcctcc	60
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ctctcctgca gggccagtcg gagtggttagc ggctacttag cctgggtacca acacaaacct 120
ggccaggctc ccaggctcct catctatgat gcatccaaca gggccgctgg catcccagcc 180
aggttcctg gacgtgggtc tgggacagac ttcactctca ccatcagcag cctggagcct 240
gaagattttg cagtttattt ctgtcagcag cgtagcaacg ggctcacttt cggcggaggg 300
accaaggtcg agatcaaa 318

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<210> SEQ ID NO 129
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 129

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Gln Val Gln Leu Val Glu Ser Gly Gly Val Val Gln Pro Gly Arg
1      5      10     15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ile Phe Ser Arg Tyr
20     25     30
Ala Leu His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Asp Trp Val
35     40     45
Ala Val Ile Ser Tyr Asp Ala Arg Asn Ser Tyr Tyr Thr Asp Ser Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Phe
65     70     75     80
Leu Gln Met Asn Ser Leu Arg Ala Asp Asp Thr Ala Val Tyr Tyr Cys
85     90     95
Ala Arg Pro Thr Leu Pro Tyr Ser Asn Asn Trp Tyr Ala Pro Glu Tyr
100    105    110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115    120

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<210> SEQ ID NO 130
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 130

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Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1      5      10     15
Thr Ala Arg Ile Thr Cys Ser Gly Asp Ala Leu Pro Lys Lys Tyr Ala
20     25     30
Ser Trp Tyr Gln Gln Lys Ser Gly Gln Ala Pro Val Leu Val Ile Tyr
35     40     45
Glu Asp Thr Lys Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
50     55     60
Ser Ser Gly Thr Met Ala Thr Leu Thr Ile Ser Gly Ala Gln Val Glu
65     70     75     80
Asp Glu Ser Asp Tyr Tyr Cys Ser Ser Thr Asp Ser Ser Gly Asn Pro
85     90     95
Val Leu Phe Gly Gly Thr Lys Leu Thr Val Leu
100    105

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<210> SEQ ID NO 131
<211> LENGTH: 122
<212> TYPE: PRT

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 131

Glu Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr
20           25           30
Ala Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Gln Trp Val
35           40           45
Ser Ser Ile Ser Gly Asn Gly Asn Gly Arg Ser Tyr Ala Asp Ser Leu
50           55           60
Lys Gly Arg Phe Thr Thr Ser Lys Asp Leu Ser Lys Tyr Thr Leu Tyr
65           70           75           80
Leu Gln Met Asn Asn Leu Arg Pro Glu Asp Thr Ala Ile Tyr Tyr Cys
85           90           95
Ala Lys Val Val Arg Ile Ala Ala Val Leu Tyr Tyr Phe Asp Tyr Trp
100          105          110
Gly Pro Gly Thr Gln Val Thr Val Ser Ser
115          120

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<210> SEQ ID NO 132
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 132

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1           5           10           15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Thr Tyr
20           25           30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35           40           45
Tyr Glu Ala Ser Thr Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly
50           55           60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Ile Ile Ser Ser Leu Glu Pro
65           70           75           80
Glu Asp Phe Ala Val Tyr His Cys Gln Gln Arg Ser Ser Trp Pro Ile
85           90           95
Thr Phe Gly Gln Gly Thr Arg Leu Glu Ile Glu
100          105

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<210> SEQ ID NO 133
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 133

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr
20           25           30
Ala Met Thr Trp Val Arg Gln Ala Leu Gly Lys Gly Leu Glu Trp Val
35           40           45

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Ser Ser Ile Ser Gly Ser Gly Gly Leu Thr Tyr Phe Ala His Ser Val
50 55 60

Lys Gly Arg Leu Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Ser Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Val Lys Ser Thr Thr Gly Thr Thr Ala Leu Val Phe Asp Ile
100 105 110

Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 134
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 134

Gln Thr Val Val Thr Gln Glu Pro Ser Phe Ser Val Ser Pro Gly Gly
1 5 10 15

Thr Val Thr Leu Thr Cys Gly Leu Ser Ser Gly Ser Val Ser Ser
20 25 30

Tyr Tyr Pro Ser Trp Tyr Gln Gln Thr Pro Gly Gln Ala Pro Arg Thr
35 40 45

Leu Ile Tyr Ser Ile Asn Arg Arg Ser Ser Gly Val Pro Asp Arg Phe
50 55 60

Ser Gly Phe Ile Leu Gly Asn Lys Ala Ala Leu Thr Ile Arg Gly Ala
65 70 75 80

Gln Ala Asp Asp Glu Ser Asp Tyr Tyr Cys Gly Leu Tyr Met Gly Ser
85 90 95

Gly Ile Trp Ile Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 135
<211> LENGTH: 122
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 135

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ile Asn Tyr
20 25 30

Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Val Ile Ser Asn Asp Gly Ser Tyr Asn Tyr Asp Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Lys Val Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Phe Cys
85 90 95

Ala Lys Asp Lys His Gly Asp Phe Asp Tyr Tyr Gly Val Asp Val Trp
100 105 110

Gly Gln Gly Thr Thr Val Thr Val Ser Pro

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115	120
<210> SEQ ID NO 136	
<211> LENGTH: 107	
<212> TYPE: PRT	
<213> ORGANISM: Artificial sequence	
<220> FEATURE:	
<223> OTHER INFORMATION: Synthetic polypeptide	
<400> SEQUENCE: 136	
Asp Val Gln Met Thr Gln Ser Pro Ser Ser Val Ser Ala Ser Ile Gly	
1 5 10 15	
Asp Arg Val Thr Ile Thr Cys Arg Ala Gly Gln Gly Ile Ser Ser Trp	
20 25 30	
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile	
35 40 45	
Tyr Ala Ala Ser Asn Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly	
50 55 60	
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro	
65 70 75 80	
Glu Asp Phe Gly Thr Tyr Tyr Cys Gln Gln Ala Asp Ser Phe Pro Arg	
85 90 95	
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys	
100 105	
<210> SEQ ID NO 137	
<211> LENGTH: 121	
<212> TYPE: PRT	
<213> ORGANISM: Artificial sequence	
<220> FEATURE:	
<223> OTHER INFORMATION: Synthetic polypeptide	
<400> SEQUENCE: 137	
Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg	
1 5 10 15	
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr	
20 25 30	
Gly Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val	
35 40 45	
Ala Val Ile Ser Tyr Asp Gly Ser Asp Asn Thr Tyr Ala Pro Phe Val	
50 55 60	
Asn Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr	
65 70 75 80	
Leu Gln Met Asn Ser Leu Arg Ala Asp Asp Thr Ala Val Tyr Tyr Cys	
85 90 95	
Ala Arg Arg Arg Pro Gly Ser Phe Pro Gly Leu Cys Asp Tyr Trp Gly	
100 105 110	
Gln Gly Ala Leu Val Thr Val Ser Ser	
115 120	
<210> SEQ ID NO 138	
<211> LENGTH: 107	
<212> TYPE: PRT	
<213> ORGANISM: Artificial sequence	
<220> FEATURE:	
<223> OTHER INFORMATION: Synthetic polypeptide	
<400> SEQUENCE: 138	
Asp Ile Gln Met Thr Gln Ser Pro Ser Thr Met Ser Ala Ser Val Gly	
1 5 10 15	

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Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Lys Trp
      20              25              30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
      35              40              45
Tyr Lys Ala Ser Thr Leu Gln Thr Gly Val Pro Ser Arg Phe Ser Gly
      50              55              60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Asn Ser Leu Gln Pro
      65              70              75              80
Asp Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Asn Ser Tyr Ser Tyr
      85              90              95
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
      100              105

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<210> SEQ ID NO 139
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 139

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Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1          5          10          15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Gly Ser Phe Ser Arg Leu
      20          25          30
Thr Ile Ile Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
      35          40          45
Gly Gly His Ile Pro Ile Phe Gly Thr Thr Asn Tyr Ala Leu Lys Phe
      50          55          60
Gln Gly Arg Val Thr Ile Thr Ala Asp Lys Thr Thr Ser Thr Ala Tyr
      65          70          75          80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Ile Tyr Tyr Cys
      85          90          95
Ala Arg Met Tyr Ser Gly His Asp Gly Val Asp Val Trp Gly Gln Gly
      100          105          110
Thr Leu Val Thr Val Ser Ser
      115

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<210> SEQ ID NO 140
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 140

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Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1          5          10          15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Arg Ser Tyr
      20          25          30
Leu Ala Trp Tyr Gln His Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
      35          40          45
Tyr Asp Ala Ser Asn Arg Ala Lys Gly Ile Pro Ala Arg Phe Ser Gly
      50          55          60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
      65          70          75          80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Arg Ser Thr Trp Pro Pro

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-continued

85 90 95
 Gly Met Phe Gly Gln Gly Thr Arg Val Glu Ile Lys
 100 105

<210> SEQ ID NO 141
 <211> LENGTH: 127
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 141

Gln Leu Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys His Ser Glu
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Ser Gly
 20 25 30
 Phe Tyr Tyr Trp Gly Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu
 35 40 45
 Trp Ile Gly Thr Ile Tyr Asp Ser Gly Arg Thr Tyr Asp Asn Pro Ser
 50 55 60
 Leu Lys Ser Arg Val Thr Ile Ser Ala Asp Thr Ser Lys Lys Gln Phe
 65 70 75 80
 Ser Leu Thr Leu Arg Ser Ala Thr Ala Ala Asp Thr Ala Val Tyr Phe
 85 90 95
 Cys Ala Arg His Leu Thr His Leu Tyr Gly Asp Tyr Val Thr Pro Asp
 100 105 110
 Ala Leu Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 142
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 142

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Ser Ser
 20 25 30
 Phe Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45
 Ile Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Arg
 50 55 60
 Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
 65 70 75 80
 Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Ser Asn Ser Arg
 85 90 95
 Leu Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 143
 <211> LENGTH: 126
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 143

-continued

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Thr Thr Tyr
 20 25 30
 Ser Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
 35 40 45
 Ser Tyr Ile Ser Ser Gly Ser Ser Asn Ile Tyr Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Asn
 65 70 75 80
 Leu Gln Met Ser Ser Leu Arg Asp Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Ala His Gly Arg Ile Val Asn Ser Gly Val Val Ile Ser Arg
 100 105 110
 Phe Asp Pro Trp Gly Gln Gly Ile Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 144
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 144

Gln Ser Ala Leu Thr Gln Pro Ala Ser Val Ser Gly Ser Pro Gly Gln
 1 5 10 15
 Ser Ile Thr Ile Ser Cys Thr Gly Thr Ser Ser Asp Val Gly Gly Tyr
 20 25 30
 Asn Tyr Val Ser Trp Tyr Gln Leu His Pro Gly Lys Ala Pro Lys Leu
 35 40 45
 Met Ile Phe Glu Val Thr Tyr Arg Pro Ser Gly Val Ser Asn Arg Phe
 50 55 60
 Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Thr Ile Ser Gly Leu
 65 70 75 80
 Gln Ala Glu Asp Glu Ala Asp Tyr Phe Cys Ser Ser Tyr Thr Thr Ser
 85 90 95
 Asn Thr Leu Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 145
 <211> LENGTH: 126
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 145

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Arg Thr Ser Gly Tyr Thr Phe Thr Ala Tyr
 20 25 30
 Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Pro Glu Trp Met
 35 40 45
 Gly Arg Ile Asn Pro Ser Ser Gly Gly Ala Gln Tyr Ala Gln Lys Phe
 50 55 60
 Gln Gly Arg Val Thr Met Thr Arg Asp Thr Ser Ile Ser Thr Thr Tyr

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65	70	75	80
Met Thr Leu Ser Gly Leu Thr Ser Asp Asp Thr Ala Val Phe Tyr Cys			
	85	90	95
Ala Arg Met Gly Cys Arg Ser Asp Arg Cys Tyr Ser Thr Asn Tyr Asn			
	100	105	110
Phe Asp Gln Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser			
	115	120	125

<210> SEQ ID NO 146
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 146

Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Gln			
1	5	10	15
Thr Ala Arg Ile Pro Cys Gly Gly Asn Asn Ile Gly Thr Lys Ser Val			
	20	25	30
His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Val Ser			
	35	40	45
Asn Asp Ser Asp Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser			
	50	55	60
Lys Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Arg Val Glu Ala Gly			
65	70	75	80
Asp Glu Ala Asp Tyr Tyr Cys Gln Val Trp Asp Ser Gly Ile Asp Val			
	85	90	95
Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu			
	100	105	

<210> SEQ ID NO 147
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 147

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly			
1	5	10	15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Ile Ser Pro Tyr			
	20	25	30
Gly Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val			
	35	40	45
Ser Phe Ile Ser Ser Ser Ser Arg Tyr Thr Tyr Tyr Ala Asp Ser Val			
	50	55	60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Ser			
65	70	75	80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys			
	85	90	95
Ala Arg Glu Arg Gly His Ser Thr Ser Ser Ser Tyr Phe Asp Ser Trp			
	100	105	110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser			
	115	120	

<210> SEQ ID NO 148
 <211> LENGTH: 108

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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 148

Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Leu Ala Pro Gly Lys
1           5           10           15

Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Ile Gly Thr Lys Thr Val
20          25          30

Ser Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Met Tyr
35          40          45

Tyr Asp Ser Asp Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
50          55          60

Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Asn Arg Val Glu Ala Gly
65          70          75          80

Asp Glu Ala Asp Tyr Tyr Cys Arg Val Trp Asp Ser Asp Thr Asp His
85          90          95

Arg Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100         105

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<210> SEQ ID NO 149
<211> LENGTH: 127
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 149

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
1           5           10           15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Ala
20          25          30

Trp Ile Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45

Gly Arg Ile Gln Thr Lys Thr Asp Gly Gly Thr Thr Asp Tyr Ala Ala
50          55          60

Pro Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
65          70          75          80

Leu Tyr Leu Gln Met Asn Ser Leu Lys Thr Glu Asp Thr Ala Leu Tyr
85          90          95

Tyr Cys Ser Thr Gly Pro Tyr Tyr Tyr Asp Thr Ser Gly Tyr Pro Gln
100         105         110

Pro Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115         120         125

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<210> SEQ ID NO 150
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 150

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1           5           10           15

Thr Ala Ser Ile Thr Cys Ser Gly Asp Lys Leu Gly Asp Lys Tyr Ala
20          25          30

Cys Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Val Leu Val Ile Tyr

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35	40	45
Gln Asp Thr Lys Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser		
50	55	60
Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Gly Thr Gln Ala Met		
65	70	75
Asp Glu Ala Asp Tyr Tyr Cys Gln Ala Trp Asp Ser Ser Thr Val Val		
85	90	95
Phe Gly Gly Gly Thr Lys Leu Thr Val Leu		
100	105	

<210> SEQ ID NO 151
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 151

Glu Val Gln Met Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly		
1	5	10
Ser Leu Arg Leu Ser Cys Ser Val Ser Gly Phe Asp Phe Ser Arg Tyr		
20	25	30
Thr Met Asn Trp Phe Arg Gln Ala Pro Gly Glu Gly Leu Lys Trp Val		
35	40	45
Ser Ser Ile Ser Ser Thr Ser Leu Tyr Thr Phe Tyr Ala Asp Ser Val		
50	55	60
Lys Gly Arg Phe Ser Ile Ser Arg Asp Asn Ala Gln Gly Ser Leu Ser		
65	70	75
Leu Gln Met Ser Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys		
85	90	95
Ala Arg Val Val Gly Pro Ala Glu Leu Asp Tyr Trp Gly Gln Gly Val		
100	105	110
Leu Val Thr Val Ser Ser		
115		

<210> SEQ ID NO 152
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 152

Val Thr Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly		
1	5	10
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Ile Gly Val Asp		
20	25	30
Leu Gly Trp Phe Gln Gln Arg Pro Gly Lys Ala Pro Lys Leu Leu Ile		
35	40	45
Tyr Gly Ala Ser Arg Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly		
50	55	60
Arg Gly Ser Gly Thr Phe Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro		
65	70	75
Glu Asp Phe Thr Thr Tyr Phe Cys Leu Gln Asp Tyr Asn Tyr Pro Trp		
85	90	95
Thr Leu Gly Gln Gly Thr Thr Val Gly Val Lys		
100	105	

-continued

<210> SEQ ID NO 153
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 153

Gln Val His Leu Val Gln Ser Gly Ser Ala Val Arg Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asp Tyr
 20 25 30
 Tyr Ile His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45
 Gly Trp Ile Asn Pro Lys Thr Gly Gly Ser Asn Tyr Thr Gln Arg Phe
 50 55 60
 Gln Ala Arg Val Thr Met Thr Trp Asp Thr Ser Ile Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Ala Gly Arg Asn Gly Tyr Asp Tyr Trp Gly Gln Gly Thr Leu
 100 105 110
 Val Thr Val Ser Ser
 115

<210> SEQ ID NO 154
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 154

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
 1 5 10 15
 Thr Ala Arg Ile Thr Cys Ser Ala Asp Ala Leu Pro Lys Gln Tyr Ala
 20 25 30
 Tyr Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Met Ile Tyr
 35 40 45
 Gln Asp Thr Glu Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
 50 55 60
 Ser Ser Gly Thr Thr Val Thr Leu Thr Ile Ser Gly Val Gln Ala Glu
 65 70 75 80
 Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Gly Asp Ser Ser Gly Thr Tyr
 85 90 95
 Leu Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105

<210> SEQ ID NO 155
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 155

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Lys Phe Arg Asn Tyr

-continued

20	25	30
Ala Met Thr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val		
35	40	45
Ser Thr Ile Thr Ser Gly Gly Ser Thr Glu Tyr Ala Asp Ala Val Lys		
50	55	60
Gly Arg Phe Ile Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu		
65	70	80
Gln Met Asn Ser Leu Arg Ala Asp Asp Thr Ala Val Tyr Tyr Cys Thr		
85	90	95
Val Pro Trp Gly Asn Tyr Asn Asp Tyr Val Ser Asp Tyr Trp Gly Gln		
100	105	110
Gly Thr Leu Val Pro Val Ser Ser		
115	120	

<210> SEQ ID NO 156
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 156

Glu Val Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1 5 10 15
Gln Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Arg Val Gly Asn Ser
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Ser Leu Leu Ile
35 40 45
Tyr Asp Ala Ser Lys Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ile Ser Leu Glu Ser
65 70 75 80
Glu Asp Phe Ala Val Tyr Tyr Cys His Gln His Ser Thr Trp Pro Arg
85 90 95
Gly Thr Phe Gly Gln Gly Thr Arg Leu Glu Ile Lys
100 105

<210> SEQ ID NO 157
 <211> LENGTH: 124
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 157

Gln Val His Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ile Phe Ser Arg Tyr
20 25 30
Pro Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ala Leu Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Phe
65 70 75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

-continued

Ala Arg His Phe Leu Pro Tyr Ser Ser Ser Trp Tyr Gln Gly Phe Asn
 100 105 110

Tyr Trp Gly Gln Gly Ile Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 158
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 158

Asn Phe Met Leu Thr Gln Pro His Ser Val Ser Glu Ser Pro Gly Lys
 1 5 10 15

Thr Val Thr Ile Ser Cys Thr Arg Ser Ser Gly Ser Ile Ala Thr Asn
 20 25 30

Tyr Val Gln Trp Tyr Gln Gln Arg Pro Gly Ser Ser Pro Thr Pro Ile
 35 40 45

Ile Phe Glu Asp Ser Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Ile Asp Ser Ser Ser Asn Ser Ala Ser Leu Thr Ile Ser Gly
 65 70 75 80

Leu Arg Thr Asp Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Asp Asn
 85 90 95

Ser Asn Arg Ala Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 159
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 159

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
 1 5 10 15

Ser Leu Arg Leu Ser Cys Glu Ala Ser Gly Phe Leu Phe Ser Arg Tyr
 20 25 30

Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Asp Trp Val
 35 40 45

Ala Val Ile Ser Tyr Asp Gly Asn Lys Lys Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Val Asn Ser Leu Arg Val Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Gly Val Pro Tyr Gly Asp Thr Leu Thr Gly Leu Val Tyr Trp
 100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 160
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

-continued

<400> SEQUENCE: 160

```

Asn Phe Met Leu Thr Gln Pro His Ser Val Ser Glu Ser Pro Gly Lys
1           5           10           15
Thr Val Thr Ile Ser Cys Thr Arg Ser Ser Gly Thr Ile Ala Ser Asn
          20           25           30
Tyr Val Gln Trp Tyr Gln Gln Arg Pro Gly Ser Ala Pro Thr Thr Val
          35           40           45
Ile Tyr Glu Asp Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
          50           55           60
Gly Ser Ile Asp Ser Ser Ser Asn Ser Ala Ser Leu Thr Ile Ser Gly
65           70           75           80
Leu Lys Thr Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Asp Asn
          85           90           95
Ser Asp Arg Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
          100          105          110

```

<210> SEQ ID NO 161

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 161

```

Gln Val Thr Leu Lys Glu Ser Gly Pro Val Leu Val Lys Pro Thr Glu
1           5           10           15
Thr Leu Thr Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Arg Asn Ala
          20           25           30
Arg Met Gly Val Ser Trp Ile Arg Gln Pro Pro Gly Lys Ala Leu Glu
          35           40           45
Trp Leu Ala His Ile Phe Ser Asn Asp Glu Lys Ser Tyr Asn Thr Ser
          50           55           60
Leu Lys Ser Arg Leu Ser Ile Ser Lys Asp Thr Ser Lys Ser Gln Val
65           70           75           80
Val Leu Thr Met Thr Ser Met Asp Pro Leu Asp Thr Ala Thr Tyr Phe
          85           90           95
Cys Ala Arg Leu Leu Val Ala Gly Thr Phe Leu Pro Ser His Tyr Phe
          100          105          110
Asp Tyr Trp Gly Gln Gly Ile Leu Val Thr Val Ser Ser
          115          120          125

```

<210> SEQ ID NO 162

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 162

```

Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Val Thr Pro Gly Lys
1           5           10           15
Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Ile Gly Leu Lys Ser Val
          20           25           30
Phe Trp Tyr Gln Glu Arg Pro Asp Gln Ala Pro Val Val Val Ile Tyr
          35           40           45
Tyr Asp Ser Ala Arg Pro Ser Gly Ile Pro Glu Arg Ile Ser Gly Ser
          50           55           60

```

-continued

Lys Ser Gly Asn Thr Ala Thr Leu Thr Ile Thr Arg Val Glu Ala Gly
65 70 75 80

Asp Glu Ala Asp Tyr Phe Cys Gln Val Trp Asp Ser Ser Arg Asn His
85 90 95

Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105

<210> SEQ ID NO 163
<211> LENGTH: 122
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 163

Glu Leu Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr
20 25 30

Ser Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Tyr Ile Ser Ser Ser Ser Ser Thr Ile Gln Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Thr Arg Gln Val Gly Ala Asp Phe Ser Gly Arg Gly Phe Asp Tyr Trp
100 105 110

Gly Gln Gly Thr Leu Leu Thr Val Ser Ser
115 120

<210> SEQ ID NO 164
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 164

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1 5 10 15

Thr Ala Thr Ile Thr Cys Phe Gly Asp Lys Leu Gly Asp Lys Tyr Ala
20 25 30

Cys Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Val Leu Val Ile Tyr
35 40 45

Gln Asp Ser Lys Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
50 55 60

Lys Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Gly Thr Gln Ala Met
65 70 75 80

Asp Glu Ala Asp Tyr Tyr Cys Gln Ala Trp Asp Ser Ser Thr Ala Val
85 90 95

Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105

<210> SEQ ID NO 165
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence

-continued

<220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 165

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Lys Phe Ser Val Tyr
 20 25 30
 Ala Leu Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
 35 40 45
 Ser Tyr Ile Ser Ser Ser Gly Ser Thr Ile Tyr Tyr Ser Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Val Gly Asn Ser Leu Phe
 65 70 75 80
 Val Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Gly Ile Tyr Tyr Cys
 85 90 95
 Ala Thr Ala Arg His Ile Thr Asn Asp Gly Phe Asp Ile Trp Gly Gln
 100 105 110
 Gly Thr Met Val Ile Val Ser Ser
 115 120

<210> SEQ ID NO 166
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 166

Asp Ile Gln Leu Thr Gln Ser Pro Ser Phe Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Arg Phe
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
 35 40 45
 Tyr Ser Ala Ser Thr Leu Gln Arg Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Phe Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Leu Asn Ser His Pro Arg
 85 90 95
 Met Phe Thr Phe Gly Pro Gly Thr Thr Val Asp Ile Lys
 100 105

<210> SEQ ID NO 167
 <211> LENGTH: 127
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 167

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Ala Val Ser Gly Tyr Leu Ile Ser Asn Gly
 20 25 30
 Tyr Tyr Trp Gly Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp
 35 40 45

-continued

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Ile Gly Ser Ile Tyr His Thr Arg Ser Thr Tyr Tyr Asn Pro Ser Leu
 50                      55                      60

Lys Ser Arg Val Ser Ile Ser Val Asp Thr Ser Lys Asn Arg Phe Ser
 65                      70                      75                      80

Leu Arg Leu Arg Ser Val Thr Ala Ala Asp Thr Ala Phe Tyr Tyr Cys
                      85                      90                      95

Ala Arg Gly Pro Gly His Cys Tyr Gly Asp Asp Asp Cys Tyr Ala Tyr
100                      105                      110

Tyr Phe Asp Gln Trp Gly Gln Gly Thr Pro Val Thr Val Ser Pro
115                      120                      125

```

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<210> SEQ ID NO 168
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 168

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Val Ser Ser Ser Val Gly
 1                      5                      10                      15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Asn Trp
20                      25                      30

Leu Ala Trp Tyr Gln Gln Asn Pro Gly Lys Ala Pro Lys Leu Leu Ile
35                      40                      45

Tyr Asp Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
50                      55                      60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn Ser Leu Gln Pro
65                      70                      75                      80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ala Asn Ser Phe Pro Phe
85                      90                      95

Thr Phe Gly Pro Gly Thr Lys Val Asp Ile Lys
100                      105

```

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<210> SEQ ID NO 169
<211> LENGTH: 127
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 169

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```

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1                      5                      10                      15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Gly Thr Phe Ser Arg Phe
20                      25                      30

Ala Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Gln Trp Met
35                      40                      45

Gly Gly Ile Leu Pro Ile Phe Gly Thr Ala Asn Tyr Ala Gln Lys Phe
50                      55                      60

Gln Gly Arg Val Thr Ile Thr Ala Asp Thr Ser Thr Ser Thr Ala Tyr
65                      70                      75                      80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85                      90                      95

Ala Arg Ser Leu Pro Tyr Cys Thr Asn Asp Val Cys Ser Asn Gln Asn
100                      105                      110

Thr Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115                      120                      125

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-continued

<210> SEQ ID NO 170
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 170

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
 1 5 10 15
 Ala Ala Arg Ile Thr Cys Ser Gly Asp Ala Leu Pro Lys Gln Tyr Ala
 20 25 30
 Tyr Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr
 35 40 45
 Glu Asp Asn Lys Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
 50 55 60
 Ser Ser Gly Thr Thr Val Thr Leu Thr Ile Ser Gly Val Gln Ala Glu
 65 70 75 80
 Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Ala Asp Ser Ser Gly Thr Tyr
 85 90 95
 Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105

<210> SEQ ID NO 171
 <211> LENGTH: 123
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 171

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Ser Gly
 20 25 30
 Asp Tyr Tyr Trp Ser Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu
 35 40 45
 Trp Ile Gly Tyr Ile Tyr Tyr Ser Gly Ser Thr Tyr Tyr Asn Pro Ser
 50 55 60
 Leu Lys Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln Phe
 65 70 75 80
 Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr
 85 90 95
 Cys Ala Ser Arg Tyr Gly Asp Pro Ile Gly Asp Asn Trp Phe Asp Pro
 100 105 110
 Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 172
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 172

Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Val Thr Pro Gly Lys
 1 5 10 15

-continued

Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Ile Gly Leu Lys Ser Val
 20 25 30

Phe Trp Tyr Gln Glu Arg Pro Asp Gln Ala Pro Val Val Val Ile Tyr
 35 40 45

Tyr Asp Ser Ala Arg Pro Ser Gly Ile Pro Glu Arg Ile Ser Gly Ser
 50 55 60

Lys Ser Gly Asn Thr Ala Thr Leu Thr Ile Thr Arg Val Glu Ala Gly
 65 70 75 80

Asp Glu Ala Asp Tyr Phe Cys Gln Val Trp Asp Ser Ser Arg Asn His
 85 90 95

Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105

<210> SEQ ID NO 173
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 173

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Arg Phe Ser Phe Tyr
 20 25 30

Gly Met Thr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Thr Gly Ala Thr Arg Asn Cys Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asp Ser Leu Arg Val Asp Asp Thr Ala Val Phe Tyr Cys
 85 90 95

Val Arg Arg Phe Pro Met Thr Thr Val Thr Ser Phe Asp Ser Trp Gly
 100 105 110

Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 174
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 174

Gln Ser Ala Leu Thr Gln Pro Arg Ser Val Ser Gly Ser Pro Gly Gln
 1 5 10 15

Ser Val Thr Ile Ser Cys Thr Gly Thr Ser Ser Asp Val Gly Gly Tyr
 20 25 30

Asn Phe Val Ser Trp Tyr Gln Gln His Pro Gly Lys Ala Pro Lys Leu
 35 40 45

Met Ile Phe Asp Val Thr Gly Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60

Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Thr Ile Ala Gly Leu
 65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gly Ala Tyr Ala Gly Phe
 85 90 95

-continued

Asn Ala Leu Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105

<210> SEQ ID NO 175
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 175

Glu Val Gln Leu Val Gln Ser Gly Gly Gly Leu Val Arg Pro Gly Gly
 1 5 10 15
 Ser Val Arg Leu Ser Cys Val Ala Ser Gly Phe Pro Phe Asn Met Phe
 20 25 30
 Trp Met Gly Trp Val Arg Gln Thr Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ala Asn Ile Lys Gln Asp Gly Ser Glu Lys Tyr Tyr Val Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Ala Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Phe
 65 70 75 80
 Leu Gln Met Asp Ser Leu Ser Val Gly Asp Thr Ala Ile Tyr Tyr Cys
 85 90 95
 Val Arg Glu Gly Val Arg Arg Val Val Val Arg Ser Thr Gly Tyr Phe
 100 105 110
 Asp Phe Trp Gly Gln Gly Gln Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 176
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 176

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Met Ser Val Ser Pro Gly Gln
 1 5 10 15
 Thr Ala Arg Ile Thr Cys Ser Gly Asp Ala Val Pro Ile Lys Tyr Val
 20 25 30
 Tyr Trp Tyr Gln Gln Arg Ser Gly Gln Ala Pro Val Leu Val Ile Tyr
 35 40 45
 Glu Asp Asp Arg Arg Pro Ser Gly Ile Ser Glu Arg Phe Ser Gly Ser
 50 55 60
 Ser Ser Gly Thr Thr Ala Thr Leu Thr Ile Thr Gly Ala Gln Val Glu
 65 70 75 80
 Asp Glu Gly Asp Tyr Tyr Cys Tyr Ser Thr Asp Ser Ser Gly Tyr Gln
 85 90 95
 Arg Ala Phe Gly Gly Gly Thr Thr Leu Thr Val Leu
 100 105

<210> SEQ ID NO 177
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 177

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Arg Val Gln Leu Gln Glu Ser Ala Pro Gly Leu Val Arg Pro Ser Glu
1           5           10           15

Thr Leu Ser Leu Thr Cys Ser Val Ser Gly Gly Pro Ile Ser Asn Gly
           20           25           30

Pro Tyr Tyr Trp Ser Trp Ile Arg Gln His Pro Gly Lys Gly Leu Glu
           35           40           45

Trp Ile Gly Phe Ile Phe Tyr Ser Gly Ser Thr Asn Tyr Asn Pro Ser
50           55           60

Leu Arg Gly Arg Val Thr Met Ala Val Asp Thr Ser Lys Asn Gln Phe
65           70           75           80

Ser Leu Arg Leu Asn Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr
           85           90           95

Cys Ala Arg His Val Val Thr Ala Ser Gly Trp Phe Asp Pro Trp Gly
           100          105          110

Gln Gly Thr Leu Val Thr Val Ser Ser
115           120

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<210> SEQ ID NO 178
<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 178

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Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1           5           10           15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Thr Asp
           20           25           30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Val Leu Ile
           35           40           45

Tyr Asp Ala Phe Lys Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly
50           55           60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
65           70           75           80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Arg Ser Arg Trp Pro Pro
           85           90           95

Pro Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100          105

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<210> SEQ ID NO 179
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 179

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Arg Val Gln Leu Gln Glu Ser Ala Pro Gly Leu Val Arg Pro Ser Glu
1           5           10           15

Thr Leu Ser Leu Thr Cys Ser Val Ser Gly Gly Pro Ile Ser Asn Gly
           20           25           30

Pro Tyr Tyr Trp Ser Trp Ile Arg Gln His Pro Gly Lys Gly Leu Glu
           35           40           45

Trp Ile Gly Phe Ile Phe Tyr Ser Gly Ser Thr Asn Tyr Asn Pro Ser
50           55           60

Leu Arg Gly Arg Val Thr Met Ala Val Asp Thr Ser Lys Asn Gln Phe
65           70           75           80

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-continued

Ser Leu Arg Leu Asn Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr
85 90 95

Cys Ala Arg His Val Val Thr Ala Ser Gly Trp Phe Asp Pro Trp Gly
100 105 110

Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 180

<211> LENGTH: 109

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 180

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Thr Asp
20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Val Leu Ile
35 40 45

Tyr Asp Ala Phe Lys Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Arg Ser Arg Trp Pro Pro
85 90 95

Pro Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 181

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 181

Gln Val His Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Asp Ser Gly Val Thr Phe Ser Asp Asn
20 25 30

Ala Leu Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Val Ile Ser Tyr Asp Gly Ser Ser Arg Tyr Tyr Ala Asp Ser Val
50 55 60

Arg Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asp Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Arg Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
85 90 95

Ala Arg Val Thr Ala Asp Tyr Tyr Glu Ser Ser Gly Lys Val Phe Trp
100 105 110

Gly Gln Gly Ala Leu Val Val Val Ser Ser
115 120

<210> SEQ ID NO 182

<211> LENGTH: 107

<212> TYPE: PRT

-continued

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 182
Asp Ile Gln Met Thr Gln Ser Pro Ser Thr Leu Ser Ala Ser Val Gly
1             5             10            15
Asp Arg Val Ser Ile Thr Cys Arg Ala Ser Gln Ser Val Arg Ser Trp
20            25            30
Leu Ala Trp Tyr Gln His Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35            40            45
Tyr Lys Ala Ser Ser Leu Glu Ser Gly Val Pro Ser Arg Phe Ser Gly
50            55            60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ala
65            70            75            80
Asp Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Gln Thr Phe Ser Trp
85            90            95
Thr Phe Gly Gln Gly Thr Thr Val Glu Val Lys
100           105

<210> SEQ ID NO 183
<211> LENGTH: 124
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 183
Gln Val Gln Leu Gln Glu Ser Gly Pro Ala Leu Val Lys Pro Ser Glu
1             5             10            15
Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Asp His
20            25            30
Tyr Trp Ser Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Ile
35            40            45
Gly Tyr Ile Tyr Thr Ser Gly Thr Thr Asn Tyr Asn Pro Ser Leu Lys
50            55            60
Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Lys Gln Phe Ser Leu
65            70            75            80
Asn Leu Arg Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
85            90            95
Arg Ser Leu Glu Thr Val Ile Arg Phe Tyr Tyr Tyr His Tyr Met Asp
100           105           110
Val Trp Gly Lys Gly Thr Thr Val Ile Val Ser Ser
115           120

<210> SEQ ID NO 184
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 184
Asp Ile Val Met Thr Gln Ser Pro Leu Ser Leu Pro Val Thr Pro Gly
1             5             10            15
Glu Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Leu Gln Ser
20            25            30
Asp Gly Tyr Ser Tyr Leu Asp Trp Tyr Leu Gln Lys Pro Gly Gln Ser
35            40            45

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-continued

Pro Gln Leu Leu Ile Tyr Leu Gly Ser Asn Arg Ala Ser Gly Val Pro
 50 55 60
 Asp Arg Phe Ser Val Ile Gly Ser Gly Thr Tyr Phe Thr Leu Lys Ile
 65 70 75 80
 Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Phe Cys Met Gln Ala
 85 90 95
 Leu Gln Thr Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105 110

<210> SEQ ID NO 185
 <211> LENGTH: 114
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 185

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Thr Val Ala Gly Gly Ser Ile Gly Asp Tyr
 20 25 30
 His Trp Asn Trp Ile Arg Gln Pro Ala Gly Lys Gly Leu Glu Trp Ile
 35 40 45
 Gly Arg Ile His Ser Ser Gly Asn Thr Asp Tyr Asn Pro Ser Leu Lys
 50 55 60
 Ser Arg Val Thr Met Ser Val Asp Thr Ser Lys Asn Gln Phe Ser Leu
 65 70 75 80
 Lys Leu Arg Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95
 Arg Gln Asn Val Phe Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val
 100 105 110
 Ser Ser

<210> SEQ ID NO 186
 <211> LENGTH: 113
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 186

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Leu Ser Leu Gly
 1 5 10 15
 Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Val Leu Phe Ser
 20 25 30
 Ser Asn Asn Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
 35 40 45
 Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
 50 55 60
 Pro Asp Arg Phe Ser Gly Ser Gly Ser Glu Thr Asp Phe Thr Leu Thr
 65 70 75 80
 Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Phe Cys Gln Gln
 85 90 95
 Phe Tyr Thr Thr Pro Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile
 100 105 110
 Lys

-continued

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<210> SEQ ID NO 187
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 187

Gln Val Gln Met Gln Glu Ser Gly Pro Gly Leu Val Lys Ala Ser Glu
1             5             10             15

Thr Leu Ser Leu Thr Cys Ser Val Ser Gly Ile Ser Ile Asn Asn Tyr
20            25            30

Tyr Trp Ser Trp Phe Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Ile
35            40            45

Gly Tyr Val Tyr Ser Thr Gly Ser Ser Lys Tyr Asn Pro Ser Leu Glu
50            55            60

Arg Arg Ala Thr Met Ser Val Asp Thr Ser Asn Asn Asn Phe Ser Leu
65            70            75            80

Arg Leu Thr Ser Val Thr Thr Ala Asp Thr Ala Val Tyr Tyr Cys Ala
85            90            95

Arg Gly Ser Met Pro His Ile Trp Gly Gln Gly Leu Leu Val Thr Val
100           105           110

Ser Ser

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<210> SEQ ID NO 188
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 188

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1             5             10             15

Arg Val Thr Ile Ser Cys Ser Gly Ser Thr Ser Asn Ile Glu Thr Asn
20            25            30

Tyr Val Tyr Trp Tyr Gln Gln Val Pro Gly Thr Ala Pro Lys Pro Leu
35            40            45

Val Tyr Arg Asn Asp Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50            55            60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Val Ile Ser Gly Leu Arg
65            70            75            80

Thr Glu Asp Glu Ala Ala Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85            90            95

Lys Ala Pro Val Phe Gly Ala Gly Thr Lys Val Ala Val Leu
100           105           110

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<210> SEQ ID NO 189
<211> LENGTH: 130
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 189

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Ile Lys Pro Gly Gly
1             5             10             15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ile Thr Phe Ser Asn Ala
20            25            30

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-continued

Trp Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Gly Arg Ile Glu Ser Lys Ile Asp Gly Gly Thr Ile Asp Tyr Ala Thr
 50 55 60
 Pro Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
 65 70 75 80
 Leu Tyr Leu Gln Met Asn Ser Leu Lys Thr Glu Asp Thr Ala Val Tyr
 85 90 95
 Tyr Cys Thr Thr Asp Gln Gly Tyr Tyr Asp Arg Ser Gly Tyr Trp Val
 100 105 110
 Val Gly Asn His Phe Asp Tyr Trp Gly Gln Gly Ile Leu Val Thr Val
 115 120 125
 Ser Ser
 130

<210> SEQ ID NO 190
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 190

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Leu
 1 5 10 15
 Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Thr Asn Ile Gly Ala Gly
 20 25 30
 Tyr Asp Val His Trp Tyr Gln His Leu Pro Gly Thr Ala Pro Lys Leu
 35 40 45
 Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60
 Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
 65 70 75 80
 Gln Ala Asp Asp Ala Ala Asp Tyr Tyr Cys Gln Ser Tyr Asp Arg Ser
 85 90 95
 Leu Ser Thr Tyr Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu
 100 105 110

<210> SEQ ID NO 191
 <211> LENGTH: 123
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 191

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Ala Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Gly Ile Ser Gly Thr Thr Gly Ser Thr Tyr Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Val His
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys

-continued

85	90	95
Ala Lys Asp Ser His Ser Met Ile Val Val Asp His Ala Phe Asp Ile		
100	105	110
Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser		
115	120	

<210> SEQ ID NO 192
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 192

Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Gln		
1	5	10
Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Ile Gly Thr Lys Ser Val		
20	25	30
His Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Val Leu Val Val Tyr		
35	40	45
Asp Asp Ser Asp Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser		
50	55	60
Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Arg Val Glu Ala Gly		
65	70	75
Asp Glu Ala Asp Tyr Tyr Cys Gln Val Trp Asp Ser Tyr Asn Val His		
85	90	95
Tyr Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu		
100	105	

<210> SEQ ID NO 193
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 193

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly		
1	5	10
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr		
20	25	30
Ile Met Thr Trp Val Arg Gln Ala Pro Gly Arg Gly Leu Glu Trp Val		
35	40	45
Ser Ser Ile Ser Thr Ser Ser Val Tyr Thr Phe Tyr Ala Asp Ser Leu		
50	55	60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Val Tyr		
65	70	75
Leu Gln Met Asn Ser Leu Arg Ala Asp Asp Thr Ala Val Tyr Tyr Cys		
85	90	95
Ala Arg Glu Glu Gly Phe Arg Ala Tyr Asn Leu Tyr Trp Gly Gln Gly		
100	105	110
Thr Leu Val Thr Val Ser Ser		
115		

<210> SEQ ID NO 194
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 194

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Glu Tyr Asn
 20 25 30

Tyr Val Tyr Trp Tyr Gln Lys Phe Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45

Ile Tyr Lys Asn Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Arg
 65 70 75 80

Ser Glu Asp Glu Gly Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ile Leu
 85 90 95

Ser Gly Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 195

<211> LENGTH: 121

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 195

Gln Val Gln Leu Val Glu Ser Gly Gly Val Val Gln Pro Gly Arg
 1 5 10 15

Ser Gln Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Arg Phe
 20 25 30

Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Val Ile Ser Phe Asp Gly Ser Asn Arg Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Thr Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Asn Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Asp Trp Asp Arg Leu Val Arg Ser Ala Val Gly Tyr Trp Gly
 100 105 110

Gln Gly Thr Leu Val Ser Val Ser
 115 120

<210> SEQ ID NO 196

<211> LENGTH: 110

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 196

Gln Ser Ala Leu Thr Gln Pro Arg Ser Val Ser Gly Ser Pro Gly Gln
 1 5 10 15

Ser Val Thr Ile Ser Cys Thr Gly Thr Ser Asn Asp Val Gly Gly Tyr
 20 25 30

Asn Phe Val Ser Trp Tyr Gln Gln His Pro Gly Lys Ala Pro Lys Leu
 35 40 45

Met Ile Phe Asp Val Ile Arg Arg Pro Ser Gly Val Pro Gly Arg Phe

-continued

50	55	60	
Ser Gly Ser Lys Ser Gly Asp Thr Ala Ser Leu Ile Ile Ser Gly Leu			
65	70	75	80
Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Cys Ser Tyr Ala Gly Thr			
	85	90	95
Tyr Thr Trp Val Phe Gly Ala Gly Thr Thr Leu Thr Val Leu			
	100	105	110

<210> SEQ ID NO 197
 <211> LENGTH: 126
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 197

Gln Val His Leu Gln Glu Ser Gly Pro Arg Leu Val Lys Pro Ser Glu		
1	5	10
Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Val Ser Thr Ala		
	20	25
Thr Tyr Tyr Trp Ser Trp Ile Arg Gln Ser Pro Gly Arg Gly Leu Glu		
	35	40
Trp Ile Gly Tyr Ile Tyr Ser Ser Gly Asn Thr Asn Tyr Asn Pro Ser		
	50	55
Leu Lys Ser Arg Val Thr Ile Ser Leu Asp Thr Pro Asn Asn Gln Leu		
65	70	75
Ser Leu Thr Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Ile Tyr Tyr		
	85	90
Cys Glu Arg Arg Leu Arg Ile Leu Ser Ile Glu Arg Asn Tyr Tyr Ala		
	100	105
Met Asp Val Trp Gly Gln Gly Thr Pro Val Thr Val Ser Ser		
	115	120

<210> SEQ ID NO 198
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 198

Glu Val Val Leu Thr Gln Thr Pro Gly Thr Leu Ser Leu Ser Pro Gly		
1	5	10
Glu Gly Ala Thr Leu Ser Cys Arg Ala Ser Gln Arg Val Val Asn Asn		
	20	25
Tyr Leu Ala Trp Tyr Gln Gln Arg Ala Gly Gln Ala Pro Arg Leu Leu		
	35	40
Ile Phe Gly Ala Ser Asn Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser		
	50	55
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Arg Lys Leu Glu		
65	70	75
Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Pro Trp		
	85	90
Thr Phe Gly His Gly Thr Lys Val Glu Met Lys		
	100	105

<210> SEQ ID NO 199
 <211> LENGTH: 130

-continued

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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 199

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Ala Met His Trp Val Arg Gln Ala Pro Gly Arg Gly Leu Glu Trp Val
35          40          45
Ala Val Ile Ser Tyr Asp Ala Ser Lys Lys Tyr His Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Ser Ser Lys Asn Thr Leu Phe
65          70          75          80
Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Asp His Val Pro Pro Lys Asp Cys Ser Asp Gly Asn Cys His
100         105         110
Ser Asp Tyr Gly Met Asp Val Trp Gly Gln Gly Thr Thr Val Thr Val
115         120         125

Ser Ser
130

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<210> SEQ ID NO 200
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 200

Asp Ile Gln Met Thr Gln Ser Pro Ser Thr Met Ser Ala Ser Val Gly
1          5          10          15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Lys Trp
20          25          30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35          40          45
Tyr Lys Ala Ser Thr Leu Gln Thr Gly Val Pro Ser Arg Phe Ser Gly
50          55          60
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Asn Ser Leu Gln Pro
65          70          75          80
Asp Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Asn Ser Tyr Ser Tyr
85          90          95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100         105

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<210> SEQ ID NO 201
<211> LENGTH: 122
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 201

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1          5          10          15
Ser Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Asn Phe

```

-continued

20	25	30
Ala Val His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met		
35	40	45
Gly Trp Ile Asn Pro Gly Asn Arg Asn Thr Lys Tyr Ser His Asn Phe		
50	55	60
Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Asn Thr Ala Tyr		
65	70	75
Met Glu Leu Ser Ser Leu Arg Ser Gln Asp Thr Ala Val Tyr Tyr Cys		
85	90	95
Ala Arg Leu Pro Ile Ala Ala Ala Gly Arg Gly Trp Phe Asp Pro Trp		
100	105	110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser		
115	120	

<210> SEQ ID NO 202
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 202

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
1 5 10 15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ile Ser Thr
20 25 30
Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Val Leu
35 40 45
Ile Tyr Asp Val Ser Thr Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
50 55 60
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
65 70 75 80
Pro Glu Asp Phe Ala Val Tyr Phe Cys His Gln Tyr Gly Ser Ser Pro
85 90 95
Ala Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 203
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 203

Glu Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Thr Tyr
20 25 30
Gly Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Tyr Ile Ser Ser Ala Thr Thr Thr Phe Tyr Tyr Ala Asp Ser Val
50 55 60
Arg Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Phe
65 70 75 80
Leu His Met Lys Ser Leu Arg Asp Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

-continued

Ala Arg Val Tyr Thr Met Leu Arg Gly Ala Ser Met Asp Val Trp Gly
 100 105 110

His Gly Thr Thr Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 204
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 204

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Thr Tyr
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Asp Ser Ala Asn Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
 65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Leu Arg Ile Thr Trp Pro Pro
 85 90 95

Ile Phe Thr Phe Gly Pro Gly Thr Lys Val Asp Val Lys
 100 105

<210> SEQ ID NO 205
 <211> LENGTH: 123
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 205

Gln Val His Leu Val Gln Ser Gly Ala Glu Val Lys Lys Ser Gly Ala
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Thr Phe Gly Tyr Thr Phe Thr Ala Tyr
 20 25 30

Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Pro Glu Trp Met
 35 40 45

Gly Trp Ile Asn Pro Ile Ser Gly Gly Thr Asn Tyr Ala Gln Lys Phe
 50 55 60

Gln Gly Arg Val Ser Met Thr Arg Asp Thr Ser Ile Ser Thr Ala Tyr
 65 70 75 80

Met Gly Leu Ser Arg Leu Arg Pro Asp Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Val Lys Cys Ser Ser Ala Asn Cys Tyr Gly Asn Phe Asp Tyr
 100 105 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 206
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

-continued

<400> SEQUENCE: 206

Gln Ser Ala Leu Thr Gln Pro Pro Ser Ala Ser Gly Ser Pro Gly Glu
 1 5 10 15
 Ser Val Thr Ile Ser Cys Thr Gly Thr Ser Ser Asp Val Gly Gly Tyr
 20 25 30
 Asn Tyr Val Ser Trp Tyr Gln Gln His Pro Gly Lys Ala Pro Lys Leu
 35 40 45
 Met Ile Tyr Glu Val Ser Lys Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60
 Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Thr Val Ser Gly Leu
 65 70 75 80
 Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Ala Gly Ser
 85 90 95
 Asn Asn Leu Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 207

<211> LENGTH: 123

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 207

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
 1 5 10 15
 Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Arg Phe Thr Asn Tyr
 20 25 30
 Arg Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
 35 40 45
 Gly Ile Ile Tyr Pro Gly Gly Ser Asp Thr Arg Tyr Ser Pro Ser Leu
 50 55 60
 Gln Gly Gln Val Thr Met Ser Val Asp Lys Ser Ile Ser Thr Ala Tyr
 65 70 75 80
 Leu Met Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys
 85 90 95
 Ala Arg Gln Thr Thr Gln Asn Ser Gly Tyr Asp Arg Trp Phe Asp Ser
 100 105 110
 Trp Gly Gln Gly Thr His Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 208

<211> LENGTH: 110

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 208

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15
 Arg Val Thr Ile Ser Cys Ser Gly Ser Thr Ser Ser Ile Gly Ser Asn
 20 25 30
 Ile Val Asn Trp Tyr Gln His Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45
 Ile Tyr Ile Asn Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60

-continued

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85 90 95

Asn Gly Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 209
<211> LENGTH: 126
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 209

Glu Val Gln Val Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ile Thr Phe Ser Arg His
20 25 30

Thr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Ala Ile Ser Gly Ser Gly Gly Ser Thr Tyr His Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Ser Ser Lys Ser Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Ile Ser Val Pro Leu Leu Arg Phe Leu Glu Trp Phe Gln His Pro
100 105 110

Phe Asp Phe Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120 125

<210> SEQ ID NO 210
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 210

Glu Ile Val Met Thr Gln Ser Pro Ala Ser Leu Ser Val Ser Pro Gly
1 5 10 15

Glu Arg Val Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Thr
20 25 30

Leu Ala Trp Tyr Gln His Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Ser Gly Ala Ser Thr Arg Ala Thr Gly Val Pro Ala Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys His Gln Tyr Ile Asn Trp Pro Pro
85 90 95

Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 211
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence

-continued

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 211

Glu Val Arg Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Thr Tyr
20           25           30
Ser Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35           40           45
Ala Ser Ile Ser Ser Thr Gly Ser Tyr Ile Tyr Asn Ala Asp Ser Leu
50           55           60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Phe
65           70           75           80
Leu Gln Met Asn Ser Leu Arg Val Glu Asp Thr Ala Val Tyr Tyr Cys
85           90           95
Val Arg Phe Thr Met Thr Thr Val Thr Asn Phe Asp Ser Trp Gly Gln
100          105          110
Gly Thr Leu Val Thr Val Ser Ser
115          120

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<210> SEQ ID NO 212
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 212

Gln Ser Ala Leu Thr Gln Pro Arg Ser Val Ser Gly Ser Pro Gly Gln
1           5           10           15
Ser Val Thr Ile Ser Cys Thr Gly Thr Ser Ser Asp Val Gly Ala Tyr
20           25           30
Ser Tyr Val Ser Trp Tyr Gln Gln His Pro Gly Lys Ala Pro Lys Leu
35           40           45
Met Ile Tyr Asp Val Tyr Arg Arg Pro Ser Gly Val Pro Gly Arg Phe
50           55           60
Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Thr Val Ser Gly Leu
65           70           75           80
Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Cys Ser His Ala Gly Ser
85           90           95
His Thr Trp Val Phe Gly Gly Gly Thr Lys Val Thr Val Leu
100          105          110

```

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<210> SEQ ID NO 213
<211> LENGTH: 126
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 213

Gln Leu Gln Val Val Ala Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1           5           10           15
Ser Leu Arg Leu Ser Cys Lys Ala Ser Gly Phe Thr Phe Thr Asn Tyr
20           25           30
Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35           40           45

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-continued

Ala Phe Ile Ser Tyr Asp Gly Gly Asn Lys Phe Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Arg Asn Thr Val Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Val Ala Asp Thr Ala Met Tyr Tyr Cys
85 90 95

Pro Lys Val Ile Pro His Pro Tyr Tyr Asp Ser Ser Gly Asp Asp Ala
100 105 110

Phe Asp Ile Trp Gly Gln Gly Thr Met Val Ala Ile Ser Ser
115 120 125

<210> SEQ ID NO 214
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 214

Glu Ile Val Met Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Trp Ala Ser Gln Ser Ile Ser Arg Asn
20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Tyr Gly Ala Ser Thr Arg Ala Thr Gly Ile Pro Ala Lys Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Val Ser Ser Leu Gln Ser
65 70 75 80

Glu Asp Leu Ala Val Tyr Tyr Cys Gln Gln Tyr Ser Lys Leu Pro Ile
85 90 95

Thr Phe Gly Gln Gly Thr Arg Leu Glu Ile Lys
100 105

<210> SEQ ID NO 215
<211> LENGTH: 122
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 215

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Pro Phe Ser Ser Tyr
20 25 30

Ser Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Tyr Ile Ser Gly Ser Gly Gly Asp Ile Tyr Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Arg Asn Ser Leu Ser
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Asp Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Gly Leu Val Ala Thr Thr Gly Thr Arg Tyr Phe Asp Tyr Trp
100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

-continued

<210> SEQ ID NO 216
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 216

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Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1           5           10           15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Arg Ser Tyr
20          25          30
Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile
35          40          45
Tyr Asp Ala Ser Asn Arg Ala Thr Gly Ile Pro Val Arg Phe Ser Gly
50          55          60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
65          70          75          80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Arg Ser Tyr Trp Pro Pro
85          90          95
Phe Thr Phe Gly Gly Thr Lys Val Glu Ile Lys
100         105

```

<210> SEQ ID NO 217
 <211> LENGTH: 124
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 217

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Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1           5           10           15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Gly Thr Phe Arg Arg Phe
20          25          30
Ala Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
35          40          45
Gly Gly Ile Ile Pro Ile Leu Gly Arg Gly Lys Tyr Ala Gln Lys Phe
50          55          60
Gln Gly Arg Val Arg Ile Thr Ala Asp Glu Ser Thr Ser Thr Ala Tyr
65          70          75          80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Phe Ile Ser Thr Ala Ser Tyr Val Pro Gly Thr Phe Glu Asp
100         105         110
Val Trp Gly Gln Gly Thr Thr Val Thr Val Ser Ser
115         120

```

<210> SEQ ID NO 218
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 218

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Asp Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1           5           10           15

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-continued

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Ser Tyr
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Tyr Asp Ala Ser Asn Arg Ala Ala Gly Ile Pro Ala Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Glu Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
 65 70 75 80

Glu Asp Val Ala Val Tyr Tyr Cys Gln Gln Arg Ser Asp Trp Pro Pro
 85 90 95

Gly Thr Phe Gly Gln Gly Thr Asn Val Glu Ile Lys
 100 105

<210> SEQ ID NO 219
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 219

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Asn Tyr
 20 25 30

Asn Ile Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ser Ile Ser Ser Thr Gly Ser Tyr Ile His Tyr Ala Asp Leu Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Val Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Met Val Arg Asn Thr Val Thr Ala Phe Asp Tyr Trp Gly Gln
 100 105 110

Gly Thr Leu Val Ser Val Ser Ser
 115 120

<210> SEQ ID NO 220
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 220

Gln Ser Ala Leu Thr Gln Pro Ala Ser Val Ser Gly Ser Pro Gly Gln
 1 5 10 15

Ser Ile Thr Ile Ser Cys Thr Gly Thr Ser Ser Asp Val Gly Gly Tyr
 20 25 30

Asn Phe Val Ser Trp Tyr Gln Gln Gln Pro Gly Arg Ala Pro Lys Leu
 35 40 45

Leu Ile Tyr Glu Val Ile Lys Arg Pro Ser Gly Val Ser Asp Arg Phe
 50 55 60

Ser Gly Ser Lys Ser Gly Asp Thr Ala Ser Leu Thr Ile Ser Gly Leu
 65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Cys Ser Tyr Gly Gly Asn
 85 90 95

-continued

Asn Ser Trp Met Phe Gly Gly Gly Thr Met Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 221
 <211> LENGTH: 124
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 221

Gln Val Gln Val Val Gln Ser Gly Ala Glu Met Lys Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Val Ser Gly Tyr Arg Leu Ile Asp Leu
 20 25 30
 Pro Leu His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Met
 35 40 45
 Gly Leu Phe Asp Pro Glu Lys Ala Glu Ala Ile Tyr Ser Gln Lys Phe
 50 55 60
 Gln Asp Lys Val Thr Ile Ser Glu Asp Thr Ser Ile Asp Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Asn Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Thr Trp Gly Val Glu Val Val Asn Gly Arg Arg Asp Tyr Phe Asp
 100 105 110
 Ser Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 222
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 222

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
 1 5 10 15
 Thr Ala Arg Ile Thr Cys Tyr Ala Asp Val Leu Ser Asn Gln Tyr Thr
 20 25 30
 Tyr Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr
 35 40 45
 Lys Asp Thr Glu Arg Pro Ser Gly Ile Pro Glu Arg Phe Ala Gly Ser
 50 55 60
 Ser Ser Gly Thr Thr Val Thr Leu Val Ile Asn Gly Val Arg Ala Glu
 65 70 75 80
 Asp Glu Ala Tyr Tyr Tyr Cys Gln Ser Ala Asp Asn Thr Arg Ile Thr
 85 90 95
 Val Phe Gly Gly Gly Thr Lys Leu Ser Val Leu
 100 105

<210> SEQ ID NO 223
 <211> LENGTH: 128
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 223

-continued

Gly Val Gln Leu Val Glu Ser Gly Gly Leu Ile Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30
 Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Tyr Ile Ser Thr Ser Gly Ser Thr Ile Tyr Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Arg Asn Ser Leu Ser
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Asp Val Arg Asp Cys Ser Ala Leu Thr Cys Pro Arg Arg Gly
 100 105 110
 Asp Ala Phe Asp Phe Trp Gly Arg Gly Thr Arg Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 224
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 224

Asp Val Val Met Thr Gln Ser Pro Leu Ser Leu Pro Val Thr Leu Gly
 1 5 10 15
 Gln Ser Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Val Tyr Ser
 20 25 30
 Asp Gly Asn Thr Tyr Leu Asn Trp Phe Gln Gln Arg Pro Gly Gln Ser
 35 40 45
 Pro Arg Arg Leu Ile Tyr Lys Val Ser Asn Arg Asp Ser Gly Val Pro
 50 55 60
 Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
 65 70 75 80
 Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr Cys Met Gln Gly
 85 90 95
 Thr His Trp Pro Arg Thr Phe Gly Pro Gly Thr Lys Val Asp Ile Lys
 100 105 110

<210> SEQ ID NO 225
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 225

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Val Tyr
 20 25 30
 Pro Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ser Ile Ser Ser Ser Ser Arg Tyr Ile Ser Tyr Ala Asp Ser Leu
 50 55 60
 Arg Gly Arg Ile Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
 65 70 75 80

-continued

Leu Gln Met Asn Ser Leu Arg Val Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Val Lys Val Gly Gly Ser Lys His Gln Tyr Tyr Phe Asp Tyr Trp Gly
 100 105 110

Gln Gly Ser Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 226
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 226

Gln Ser Val Leu Thr Gln Pro Pro Ser Thr Ser Gly Ile Pro Gly Gln
 1 5 10 15

Thr Val Thr Ile Ser Cys Ser Gly Ser Arg Ser Asn Ile Gly Ser Tyr
 20 25 30

Thr Val Asn Trp Tyr Glu Gln Leu Pro Gly Thr Ala Pro Lys Leu Val
 35 40 45

Ile Phe Asn Asn Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
 65 70 75 80

Ser Glu Gly Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
 85 90 95

Asn Gly Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 227
 <211> LENGTH: 124
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 227

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Arg Pro Gly Ala
 1 5 10 15

Ser Val Lys Val Phe Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
 20 25 30

Asp Ile Ile Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Val
 35 40 45

Gly Trp Ile Ser Thr Tyr Asn Gly Asn Thr Asn Tyr Glu Gln Asn Leu
 50 55 60

Gln Gly Arg Val Thr Met Thr Thr Asp Thr Ser Thr Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Arg Ser Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Glu Arg Cys Ser Thr Ser Thr Cys Tyr Ser Arg Tyr Ala Asp
 100 105 110

Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 228
 <211> LENGTH: 107
 <212> TYPE: PRT

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 228
Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1             5             10            15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Asn Ile Tyr
20            25            30
Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Val Leu Ile
35            40            45
Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Arg Gly
50            55            60
Ser Gly Ser Gly Thr Asp Phe Ile Leu Thr Ile Ser Ser Leu Gln Pro
65            70            75            80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Tyr Arg Ser Pro Arg
85            90            95
Thr Phe Gly Gln Gly Thr Glu Val Glu Ile Lys
100           105

<210> SEQ ID NO 229
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 229
Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Ala Gln Pro Gly Glu
1             5             10            15
Ser Met Arg Leu Ser Cys Val Ala Ser Gly Phe Thr Leu Ser Arg Tyr
20            25            30
Glu Val Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
35            40            45
Ser Tyr Ile Ser Ser Gly Gly Pro Ser Ile Tyr Tyr Ala Asp Ser Val
50            55            60
Lys Gly Arg Phe Thr Ile Ser Arg Asn Ser Ala Glu Asn Ser Leu Glu
65            70            75            80
Leu Gln Met Ser Thr Leu Arg Thr Glu Asp Thr Ala Val Tyr Tyr Cys
85            90            95
Met Arg Glu Gly Leu Thr Tyr Tyr Asp Ser Thr Ile Trp Gly Gln Gly
100           105           110
Thr Leu Val Ala Val Ser Ser
115

<210> SEQ ID NO 230
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 230
Gln Asn Val Leu Thr Gln Ser Pro Ser Ala Ser Ala Ser Leu Gly Ala
1             5             10            15
Ser Val Lys Leu Thr Cys Thr Leu Asn Ser Gly His Ser Arg Tyr Ala
20            25            30
Ile Ala Trp His Gln His Gln Pro Gln Arg Gly Pro Arg Phe Leu Met
35            40            45

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Lys Ile Asn Ser Asp Gly Arg His Ile Arg Gly Asp Gly Ile Ser Asp
50 55 60

Arg Phe Ser Gly Ser Ala Ser Gly Ala Glu Arg His Leu Thr Ile Ser
65 70 75 80

Ser Leu Gln Pro Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Thr Trp Gly
85 90 95

Thr Gly Phe Arg Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 231
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 231

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Glu Tyr
20 25 30

Ala Met His Trp Val Arg Gln Val Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Gly Ile Ser Trp Asn Gly Gly Ser Lys Gly Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Arg Tyr Ser Leu Ser
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Thr Glu Asp Thr Ala Leu Tyr Tyr Cys
85 90 95

Ala Lys Asp Asp Tyr Glu Gly Ala Gly Phe Asp Ile Trp Gly Gln Gly
100 105 110

Thr Val Val Thr Val Ser Ser
115

<210> SEQ ID NO 232
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 232

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Ser Tyr
20 25 30

Leu Gly Trp Tyr Gln Gln Lys Ser Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
65 70 75 80

Glu Asp Phe Ala Ile Tyr Tyr Cys Gln Gln Arg Ser Asn Trp Pro Ile
85 90 95

Thr Phe Gly Gln Gly Thr Arg Leu Glu Ile Lys
100 105

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<210> SEQ ID NO 233
<211> LENGTH: 126
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 233

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Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
1             5             10             15

Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Tyr Leu Ile Ser Asn Gly
                20             25             30

Tyr Tyr Trp Gly Trp Ile Arg Gln Ser Pro Gly Lys Gly Leu Glu Trp
            35             40             45

Ile Gly Ser Ile Tyr Tyr Thr Arg Asp Thr Tyr Tyr Asn Trp Ser Leu
50             55             60

Lys Ser Arg Ile Thr Ile Ser Val Asp Thr Ser Lys Lys Gln Phe Ser
65             70             75             80

Leu Lys Leu Tyr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys
            85             90             95

Val Arg His Glu Gly Ser Cys Asn Asp Gly Ser Cys Tyr Gly Ser Phe
100            105            110

Val Asp Asn Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115            120            125

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<210> SEQ ID NO 234
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 234

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Val Ser Ala Ser Val Gly
1             5             10             15

Asp Arg Val Thr Leu Thr Cys Arg Ala Ser Gln Asp Ile Ser Ser Trp
20             25             30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35             40             45

Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
50             55             60

Ser Gly Ser Gly Thr His Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65             70             75             80

Glu Asp Phe Ala Thr Tyr Phe Cys Gln Gln Ala Asp Ser Phe Ile Thr
85             90             95

Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100            105

```

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<210> SEQ ID NO 235
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 235

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1             5             10             15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Asp Phe Thr Phe Ser Ser Tyr
20             25             30

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-continued

Thr Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Asp Gly Val Ser Thr Lys Asn Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Ser Val Ser Arg Asp Asn Ser Lys Asn Thr Leu Phe
 65 70 75 80

Leu Gln Leu Asn Ser Leu Arg Ala Glu Asp Thr Ala Phe Tyr Tyr Cys
 85 90 95

Ala Arg Gly Gly Thr Phe His Asn Trp Tyr Phe Asp Leu Trp Gly Arg
 100 105 110

Gly Val Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 236
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 236

Glu Ile Val Met Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly
 1 5 10 15

Glu Thr Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Asp Asn
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Ser Gly Ala Ser Thr Arg Ala Thr Asp Phe Pro Ala Arg Phe Arg Gly
 50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser
 65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Lys Asn Trp Pro Arg
 85 90 95

Thr Phe Gly Arg Gly Thr Lys Val Glu Val Arg
 100 105

<210> SEQ ID NO 237
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 237

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr
 20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Pro Glu Trp Val
 35 40 45

Ser Gly Ile Ser Gly Ser Gly Gly Ser Thr Asn Tyr Ala Asp Ser Ala
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Lys Gly Thr Ile Thr Tyr Ser Tyr Tyr Tyr Met Ala Val Trp Gly

-continued

100	105	110
Lys Gly Thr Thr Val Thr Val Ser Ser		
115	120	
<210> SEQ ID NO 238 <211> LENGTH: 107 <212> TYPE: PRT <213> ORGANISM: Artificial sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic polypeptide		
<400> SEQUENCE: 238		
Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly		
1	5	10 15
Glu Arg Gly Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Arg Ser Ser		
20	25	30
Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu		
35	40	45
Ile Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser		
50	55	60
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Gly Arg Leu Glu		
65	70	75 80
Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Thr Ser Ile		
85	90	95
Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys		
100	105	
<210> SEQ ID NO 239 <211> LENGTH: 124 <212> TYPE: PRT <213> ORGANISM: Artificial sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic polypeptide		
<400> SEQUENCE: 239		
Gln Val Gln Val Val Gln Ser Gly Ala Glu Met Arg Lys Pro Gly Ala		
1	5	10 15
Ser Val Arg Val Ser Cys Lys Val Ser Gly Tyr Arg Leu Thr Asp Leu		
20	25	30
Pro Leu His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile		
35	40	45
Gly Phe Val Asp Leu Glu Lys Arg Glu Ile Ile Tyr Ala Gln Lys Phe		
50	55	60
Gln Gly Lys Val Thr Ile Thr Glu Asp Thr Ser Ala Asp Thr Ala Tyr		
65	70	75 80
Met Glu Leu Asn Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys		
85	90	95
Ala Thr Trp Gly Ile Glu Val Val Asn Gly Arg Asp Glu Phe Phe Asp		
100	105	110
Ser Trp Gly Gln Gly Thr Leu Val Ser Val Ser Ser		
115	120	
<210> SEQ ID NO 240 <211> LENGTH: 106 <212> TYPE: PRT <213> ORGANISM: Artificial sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic polypeptide		
<400> SEQUENCE: 240		

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Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1          5          10          15
Thr Ala Arg Ile Thr Cys Tyr Ala Asp Val Leu Ser Lys Gln Tyr Thr
20          25          30
Tyr Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr
35          40          45
Lys Asp Thr Glu Arg Pro Ser Gly Ile Pro Glu Arg Phe Ala Gly Ser
50          55          60
Ser Ser Gly Thr Thr Val Thr Leu Ile Ile Asn Gly Val Arg Thr Glu
65          70          75          80
Asp Glu Ala Tyr Tyr Tyr Cys Gln Ser Ala Asp Thr Arg Ile Thr Val
85          90          95
Phe Gly Gly Gly Thr Lys Leu Ser Val Leu
100          105

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<210> SEQ ID NO 241
<211> LENGTH: 124
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 241

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Gln Val Gln Val Val Gln Ser Gly Ala Glu Met Lys Lys Pro Gly Ala
1          5          10          15
Ser Val Lys Val Ser Cys Lys Val Ser Gly Tyr Ser Leu Ser Asp Leu
20          25          30
Pro Leu His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Met
35          40          45
Gly Leu Phe Asp Pro Ile Asn Gly Glu Ile Ile Tyr Ala Gln Thr Phe
50          55          60
Gln Gly Lys Val Thr Ile Ser Glu Asp Thr Ser Ile Asp Thr Ala Tyr
65          70          75          80
Met Glu Leu Asn Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Thr Trp Gly Val Ala Val Val Ser Gly Arg Arg Asp Tyr Phe Asp
100          105          110
Ser Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115          120

```

```

<210> SEQ ID NO 242
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polypeptide

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<400> SEQUENCE: 242

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Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1          5          10          15
Thr Ala Arg Ile Thr Cys Tyr Ala Ala Val Leu Ser Asn Gln Tyr Thr
20          25          30
Tyr Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr
35          40          45
Lys Asp Thr Glu Arg Pro Ser Gly Ile Pro Glu Arg Phe Ala Gly Ser
50          55          60
Ser Ser Gly Thr Thr Val Thr Leu Ile Ile Asn Gly Val Arg Thr Glu

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-continued

65	70	75	80
Asp Glu Ala Tyr Tyr Tyr Cys Gln Thr Ala Asp Thr Lys Tyr Thr Val			
	85	90	95
Phe Gly Gly Gly Thr Lys Leu Ser Val Leu			
	100	105	

<210> SEQ ID NO 243
 <211> LENGTH: 126
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 243

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln			
1	5	10	15
Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Thr Ser Gly			
	20	25	30
Asp Tyr Tyr Trp Asn Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu			
	35	40	45
Trp Ile Gly Tyr Ile Tyr His Ser Gly Thr Thr Tyr Tyr Asn Pro Ser			
	50	55	60
Leu Lys Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Arg Phe			
65	70	75	80
Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe			
	85	90	95
Cys Ala Arg Ala Tyr Ala Tyr Glu Phe Trp Ser Gly Tyr Pro Asn Trp			
	100	105	110
Phe Asp Pro Trp Gly Leu Gly Thr Leu Val Thr Val Ser Ser			
	115	120	125

<210> SEQ ID NO 244
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 244

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly			
1	5	10	15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Arg Ile Ser Thr Tyr			
	20	25	30
Val Asn Trp Tyr Gln Val Lys Ala Gly Thr Ala Pro Lys Val Leu Ile			
	35	40	45
Tyr Ala Ala Ser Ser Leu Gln Thr Gly Val Pro Ser Arg Phe Ser Gly			
	50	55	60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Val Ser Leu Gln Pro			
65	70	75	80
Glu Asp Phe Thr Thr Tyr Phe Cys Gln Gln Ser Tyr Ser Pro Pro Trp			
	85	90	95
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys			
	100	105	

<210> SEQ ID NO 245
 <211> LENGTH: 126
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 245

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ile Phe Asn Arg Tyr
 20 25 30

Ala Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Leu Ile Ser Tyr Asp Gly Ile Asn Lys Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Ser Ile Ser Arg Asp Asn Ser Lys Ser Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Phe Tyr Cys
 85 90 95

Ala Arg Gly Leu Gly Tyr Cys Ser Gly Thr Gly Gly Ser Cys Thr Pro
 100 105 110

Phe Glu Tyr Trp Gly Gln Gly Ile Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 246

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 246

Asp Ile Gln Met Thr Gln Ser Pro Pro Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Lys Lys Tyr
 20 25 30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Asn Ala Pro Lys Leu Leu Ile
 35 40 45

Tyr Gly Ala Ser Asn Leu Gln Thr Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Ala Leu Ser Ile Ser Ser Leu Glu Thr
 65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Asp Ser Ala Pro Pro
 85 90 95

Thr Phe Gly Gly Gly Thr Lys Val Glu Phe Lys
 100 105

<210> SEQ ID NO 247

<211> LENGTH: 135

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 247

Gln Val Gln Leu Gln Gln Arg Gly Ala Gly Leu Leu Lys Pro Ser Glu
 1 5 10 15

Thr Leu Ser Leu Thr Cys Glu Ile Tyr Gly Ala Ser Leu Asn Asp Tyr
 20 25 30

Asp Trp Thr Trp Ile Arg Gln Pro Pro Gly Lys Arg Leu Glu Trp Ile
 35 40 45

Gly Val Ile Asn Arg Arg Asp Thr Val Asp Tyr Asn Pro Ser Leu Lys

-continued

50	55	60
Ser Arg Val Thr Leu Ser Leu Asp Thr Ser Lys Asn Gln Leu Ser Leu		
65	70	75 80
Ser Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Val		
	85	90 95
Arg Val Pro Arg Arg Gly Phe Glu Gly Ser Phe Gly Phe Cys Asp Asp		
	100	105 110
Thr Ala Cys Arg Tyr Gly His Thr Trp Phe Asp Pro Trp Gly Gln Gly		
	115	120 125
Thr Leu Val Thr Val Ser Ser		
	130	135

<210> SEQ ID NO 248
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 248

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Arg Asp Tyr
20 25 30
Leu Asn Trp Tyr Gln Gln Arg Pro Gly Lys Ala Pro Lys Val Leu Ile
35 40 45
Phe Ala Gly Ser Arg Leu Glu Ser Gly Val Pro Ser Arg Phe Arg Gly
50 55 60
Arg Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Asp Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Tyr Leu Thr Pro Pro
85 90 95
Thr Phe Gly Gln Gly Thr Thr Val Asp Ile Lys
100 105

<210> SEQ ID NO 249
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 249

Gln Val Thr Leu Lys Glu Ser Gly Pro Val Leu Val Lys Pro Thr Glu
1 5 10 15
Thr Leu Thr Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Arg Asn Gly
20 25 30
Arg Met Gly Val Ser Trp Ile Arg Gln Pro Pro Gly Lys Ala Leu Glu
35 40 45
Trp Leu Ala His Ile Phe Ala Ser Asp Glu Lys Ser Tyr Ser Thr Ser
50 55 60
Gln Arg Thr Arg Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln Val
65 70 75 80
Val Leu Ser Met Thr Asp Met Asp Pro Val Asp Thr Ala Thr Tyr Tyr
85 90 95
Cys Ala Arg Ile Leu Lys Phe Gly Thr Met Arg Ala Ala Tyr Tyr Phe
100 105 110

-continued

Asp Tyr Trp Gly Gln Gly Ala Leu Val Pro Val Ser Ser
 115 120 125

<210> SEQ ID NO 250
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 250

Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Lys
 1 5 10 15
 Thr Ala Arg Ile Thr Cys Gly Gly Ile Asn Ile Gly Ile Arg Thr Val
 20 25 30
 His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Met Leu Val Ile Tyr
 35 40 45
 Tyr Asp Ser Asp Arg Pro Leu Gly Ile Pro Glu Arg Phe Ser Gly Ser
 50 55 60
 Lys Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Arg Val Glu Ala Gly
 65 70 75 80
 Asp Glu Ala Asp Tyr Tyr Cys Gln Val Trp Asp Ser Ser Ser Asp His
 85 90 95
 Val Val Phe Gly Gly Gly Thr Lys Leu Thr Ala Leu
 100 105

<210> SEQ ID NO 251
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 251

Gln Val His Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Val Ser Gly Phe Thr Phe Ser Asp Tyr
 20 25 30
 Tyr Met Ser Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45
 Ser Tyr Ile Ser Ser Ser Gly Ser Thr Ile Tyr Tyr Ala Asp Ser Met
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Arg Asn Ser Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Val Glu Asp Thr Ala Phe Tyr Tyr Cys
 85 90 95
 Ala Gly Ser Lys Val Gly Tyr Thr Thr Gly Arg Arg Asn Trp Tyr Phe
 100 105 110
 Asp Leu Trp Gly Arg Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 252
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 252

Gln Ser Ala Leu Thr Gln Pro Pro Ser Ala Ser Gly Ser Pro Gly Gln

-continued

1	5	10	15
Ser Val Thr	Ile Ser Cys Thr	Gly Thr Ser Ser	Asp Val Gly Ala Tyr
	20	25	30
Asn Tyr Val	Ser Trp Tyr	Gln Gln His Pro	Gly Lys Ala Pro Lys Leu
	35	40	45
Met Ile Tyr	Glu Val Thr	Lys Arg Pro Ser	Gly Val Pro Asp Arg Phe
	50	55	60
Ser Gly Ser	Lys Ser Gly	Asn Thr Ala Ser	Leu Thr Val Ser Gly Leu
	65	70	75
Gln Ala Glu	Asp Glu Ala Asp	Tyr Tyr Cys Ser	Ser Tyr Ala Gly Asn
	85	90	95
Asn Asn Leu	Val Phe Gly Gly	Gly Thr Lys Leu	Thr Val Leu
	100	105	110

<210> SEQ ID NO 253
 <211> LENGTH: 126
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 253

Gln Val Gln	Leu Leu Gln	Ser Gly Ser	Glu Val Arg	Lys Pro Gly	Ala
1	5	10	15		
Ser Val Asn	Ile His Cys	Lys Ala Ser	Gly Phe Thr	Phe Thr Asp	Phe
	20	25	30		
Tyr Leu His	Trp Val Arg	Gln Ala Pro	Gly Gln Gly	Leu Glu Trp	Met
	35	40	45		
Gly Ile Ile	Asn Pro Glu	Thr Gly Glu	Thr Thr Tyr	Ser Gln Lys	Phe
	50	55	60		
Gln Gly Arg	Val Thr Met	Thr Arg Asp	Thr Ser Thr	Ser Val Val	Asn
	65	70	75		80
Leu Glu Val	Arg Ser Leu	Arg Ser Glu	Asp Thr Ala	Ile Tyr Tyr	Cys
	85	90	95		
Ala Arg Asp	Leu Val Val	Val Val Pro	Val Glu Met	Ser Arg Arg	Ala
	100	105	110		
Phe Asp Ile	Trp Gly Gln	Gly Ile Met	Val Thr Val	Ser Ser	
	115	120	125		

<210> SEQ ID NO 254
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 254

Ser Tyr Val	Leu Thr Gln	Pro Pro Ser	Val Ser Val	Ala Pro Gly	Gln
1	5	10	15		
Thr Ala Arg	Ile Pro Cys	Gly Gly Asn	Asn Ile Glu	Arg Lys Ser	Val
	20	25	30		
His Trp Tyr	Gln Gln Arg	Pro Gly Gln	Ala Pro Val	Leu Val Val	Tyr
	35	40	45		
Asp Asp Thr	Val Arg Pro	Ser Gly Ile	Pro Glu Arg	Phe Ser Gly	Ser
	50	55	60		
Asn Ser Gly	Ser Thr Ala	Thr Leu Thr	Ile Ser Arg	Val Gly Ala	Gly
	65	70	75		80

-continued

Asp Glu Gly Asp Tyr Tyr Cys Gln Val Trp Asp Ser Thr Thr Asp His
85 90 95

Gly Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105

<210> SEQ ID NO 255
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 255

Arg Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1 5 10 15

Ser Val Lys Val Ser Cys Gln Thr Ser Gly Tyr Ile Phe Ser Ala Tyr
20 25 30

Tyr Ile Tyr Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
35 40 45

Gly Arg Met Asn Ala Lys Ser Gly Gly Ala Asn Thr Ala Gln Gln Phe
50 55 60

Gln Gly Arg Leu Thr Met Thr Arg Asp Met Ser Val Ser Thr Ala Tyr
65 70 75 80

Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Asp Tyr Arg Asp Asp Tyr Met Trp Gly Ser Tyr Arg Pro Leu
100 105 110

Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120 125

<210> SEQ ID NO 256
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic polypeptide

<400> SEQUENCE: 256

Glu Val Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1 5 10 15

Glu Arg Ala Ser Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Gly Tyr
20 25 30

Leu Ala Trp Tyr Gln His Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Tyr Asp Ala Ser Asn Arg Ala Ala Gly Ile Pro Ala Arg Phe Arg Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
65 70 75 80

Glu Asp Phe Ala Val Tyr Phe Cys Gln Gln Arg Ser Asn Gly Leu Thr
85 90 95

Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 257
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic peptide

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<400> SEQUENCE: 257

Gly Phe Ile Phe Ser Arg Tyr Ala
1 5

<210> SEQ ID NO 258

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 258

Ile Ser Tyr Asp Ala Arg Asn Ser
1 5

<210> SEQ ID NO 259

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 259

Ala Arg Pro Thr Leu Pro Tyr Ser Asn Asn Trp Tyr Ala Pro Glu Tyr
1 5 10 15

<210> SEQ ID NO 260

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 260

Gly Phe Ser Phe Ser Ser Tyr Ala
1 5

<210> SEQ ID NO 261

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 261

Ile Ser Gly Asn Gly Asn Gly Arg
1 5

<210> SEQ ID NO 262

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 262

Ala Lys Val Val Arg Ile Ala Ala Val Leu Tyr Tyr Phe Asp Tyr
1 5 10 15

<210> SEQ ID NO 263

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 263

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Gly Phe Thr Phe Ser Asn Tyr Ala
1 5

<210> SEQ ID NO 264
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 264

Ile Ser Gly Ser Gly Gly Leu Thr
1 5

<210> SEQ ID NO 265
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 265

Ala Arg Val Lys Ser Thr Thr Gly Thr Thr Ala Leu Val Phe Asp Ile
1 5 10 15

<210> SEQ ID NO 266
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 266

Gly Phe Thr Phe Ile Asn Tyr Gly
1 5

<210> SEQ ID NO 267
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 267

Ile Ser Asn Asp Gly Ser Tyr Asn
1 5

<210> SEQ ID NO 268
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 268

Ala Lys Asp Lys His Gly Asp Phe Asp Tyr Tyr Gly Val Asp Val
1 5 10 15

<210> SEQ ID NO 269
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 269

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Gly Phe Thr Phe Ser Ser Tyr Gly
1 5

<210> SEQ ID NO 270
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 270

Ile Ser Tyr Asp Gly Ser Asp Asn
1 5

<210> SEQ ID NO 271
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 271

Ala Arg Arg Arg Pro Gly Ser Phe Pro Gly Leu Cys Asp Tyr
1 5 10

<210> SEQ ID NO 272
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 272

Gly Gly Ser Phe Ser Arg Leu Thr
1 5

<210> SEQ ID NO 273
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 273

His Ile Pro Ile Phe Gly Thr Thr
1 5

<210> SEQ ID NO 274
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 274

Ala Arg Met Tyr Ser Gly His Asp Gly Val Asp Val
1 5 10

<210> SEQ ID NO 275
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 275

Gly Gly Ser Ile Ser Ser Gly Phe Tyr Tyr

-continued

1	5	10
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<210> SEQ ID NO 276
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 276

Ile Tyr Asp Ser Gly Arg Thr
1 5

<210> SEQ ID NO 277
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 277

Ala Arg His Leu Thr His Leu Tyr Gly Asp Tyr Val Thr Pro Asp Ala
1 5 10 15

Leu Asp Ile

<210> SEQ ID NO 278
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 278

Gly Phe Thr Phe Thr Thr Tyr Ser
1 5

<210> SEQ ID NO 279
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 279

Ile Ser Ser Gly Ser Ser Asn Ile
1 5

<210> SEQ ID NO 280
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 280

Ala Arg Ala His Gly Arg Ile Val Asn Ser Gly Val Val Ile Ser Arg
1 5 10 15

Phe Asp Pro

<210> SEQ ID NO 281
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

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<400> SEQUENCE: 281

Gly Tyr Thr Phe Thr Ala Tyr Tyr
1 5

<210> SEQ ID NO 282

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 282

Ile Asn Pro Ser Ser Gly Gly Ala
1 5

<210> SEQ ID NO 283

<211> LENGTH: 19

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 283

Ala Arg Met Gly Cys Arg Ser Asp Arg Cys Tyr Ser Thr Asn Tyr Asn
1 5 10 15

Phe Asp Gln

<210> SEQ ID NO 284

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 284

Gly Phe Thr Ile Ser Pro Tyr Gly
1 5

<210> SEQ ID NO 285

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 285

Ile Ser Ser Ser Ser Arg Tyr Thr
1 5

<210> SEQ ID NO 286

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 286

Ala Arg Glu Arg Gly His Ser Thr Ser Ser Ser Tyr Phe Asp Ser
1 5 10 15

<210> SEQ ID NO 287

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

-continued

<400> SEQUENCE: 287

Gly Phe Thr Phe Ser Asn Ala Trp
1 5

<210> SEQ ID NO 288

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 288

Ile Gln Thr Lys Thr Asp Gly Gly Thr Thr
1 5 10

<210> SEQ ID NO 289

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 289

Ser Thr Gly Pro Tyr Tyr Tyr Asp Thr Ser Gly Tyr Pro Gln Pro Phe
1 5 10 15

Asp Tyr

<210> SEQ ID NO 290

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 290

Gly Phe Asp Phe Ser Arg Tyr Thr
1 5

<210> SEQ ID NO 291

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 291

Ile Ser Ser Thr Ser Leu Tyr Thr
1 5

<210> SEQ ID NO 292

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 292

Ala Arg Val Val Gly Pro Ala Glu Leu Asp Tyr
1 5 10

<210> SEQ ID NO 293

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 293

Gly Tyr Thr Phe Thr Asp Tyr Tyr
1 5

<210> SEQ ID NO 294

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 294

Ile Asn Pro Lys Thr Gly Gly Ser
1 5

<210> SEQ ID NO 295

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 295

Ala Arg Ala Gly Arg Asn Gly Tyr Asp Tyr
1 5 10

<210> SEQ ID NO 296

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 296

Gly Phe Lys Phe Arg Asn Tyr Ala
1 5

<210> SEQ ID NO 297

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 297

Ile Thr Ser Gly Gly Ser Thr
1 5

<210> SEQ ID NO 298

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 298

Thr Val Pro Trp Gly Asn Tyr Asn Asp Tyr Val Ser Asp Tyr
1 5 10

<210> SEQ ID NO 299

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

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<400> SEQUENCE: 299

Gly Phe Ile Phe Ser Arg Tyr Pro
1 5

<210> SEQ ID NO 300

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 300

Ile Ser Tyr Asp Gly Asn Asn Lys
1 5

<210> SEQ ID NO 301

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 301

Ala Arg His Phe Leu Pro Tyr Ser Ser Ser Trp Tyr Gln Gly Phe Asn
1 5 10 15

Tyr

<210> SEQ ID NO 302

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 302

Gly Phe Leu Phe Ser Arg Tyr Gly
1 5

<210> SEQ ID NO 303

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 303

Ile Ser Tyr Asp Gly Asn Lys Lys
1 5

<210> SEQ ID NO 304

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 304

Ala Arg Gly Val Pro Tyr Gly Asp Thr Leu Thr Gly Leu Val Tyr
1 5 10 15

<210> SEQ ID NO 305

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 305

Gly Phe Ser Leu Arg Asn Ala Arg Met Gly
1 5 10

<210> SEQ ID NO 306

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 306

Ile Phe Ser Asn Asp Glu Lys
1 5

<210> SEQ ID NO 307

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 307

Ala Arg Leu Leu Val Ala Gly Thr Phe Leu Pro Ser His Tyr Phe Asp
1 5 10 15

Tyr

<210> SEQ ID NO 308

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 308

Gly Phe Thr Phe Ser Thr Tyr Ser
1 5

<210> SEQ ID NO 309

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 309

Ile Ser Ser Ser Ser Ser Thr Ile
1 5

<210> SEQ ID NO 310

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 310

Thr Arg Gln Val Gly Ala Asp Phe Ser Gly Arg Gly Phe Asp Tyr
1 5 10 15

<210> SEQ ID NO 311

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 311

Gly Phe Lys Phe Ser Val Tyr Ala
1 5

<210> SEQ ID NO 312
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 312

Ile Ser Ser Ser Gly Ser Thr Ile
1 5

<210> SEQ ID NO 313
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 313

Ala Thr Ala Arg His Ile Thr Asn Asp Gly Phe Asp Ile
1 5 10

<210> SEQ ID NO 314
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 314

Gly Tyr Leu Ile Ser Asn Gly Tyr Tyr
1 5

<210> SEQ ID NO 315
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 315

Ile Tyr His Thr Arg Ser Thr
1 5

<210> SEQ ID NO 316
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 316

Ala Arg Gly Pro Gly His Cys Tyr Gly Asp Asp Asp Cys Tyr Ala Tyr
1 5 10 15

Tyr Phe Asp Gln
20

<210> SEQ ID NO 317
<211> LENGTH: 8

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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 317

Gly Gly Thr Phe Ser Arg Phe Ala
1 5

<210> SEQ ID NO 318
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 318

Ile Leu Pro Ile Phe Gly Thr Ala
1 5

<210> SEQ ID NO 319
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 319

Ala Arg Ser Leu Pro Tyr Cys Thr Asn Asp Val Cys Ser Asn Gln Asn
1 5 10 15

Thr Phe Asp Tyr
20

<210> SEQ ID NO 320
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 320

Gly Gly Ser Ile Ser Ser Gly Asp Tyr Tyr
1 5 10

<210> SEQ ID NO 321
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 321

Ile Tyr Tyr Ser Gly Ser Thr
1 5

<210> SEQ ID NO 322
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 322

Ala Ser Arg Tyr Gly Asp Pro Ile Gly Asp Asn Trp Phe Asp Pro
1 5 10 15

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<210> SEQ ID NO 323
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 323

Gly Phe Arg Phe Ser Phe Tyr Gly
1 5

<210> SEQ ID NO 324
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 324

Ile Ser Gly Thr Gly Ala Thr Arg
1 5

<210> SEQ ID NO 325
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 325

Val Arg Arg Phe Pro Met Thr Thr Val Thr Ser Phe Asp Ser
1 5 10

<210> SEQ ID NO 326
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 326

Gly Phe Pro Phe Asn Met Phe Trp
1 5

<210> SEQ ID NO 327
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 327

Ile Lys Gln Asp Gly Ser Glu Lys
1 5

<210> SEQ ID NO 328
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 328

Val Arg Glu Gly Val Arg Arg Val Val Val Arg Ser Thr Gly Tyr Phe
1 5 10 15

Asp Phe

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<210> SEQ ID NO 329
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 329

Gly Gly Pro Ile Ser Asn Gly Pro Tyr Tyr
1 5 10

<210> SEQ ID NO 330
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 330

Ile Phe Tyr Ser Gly Ser Thr
1 5

<210> SEQ ID NO 331
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 331

Ala Arg His Val Val Thr Ala Ser Gly Trp Phe Asp Pro
1 5 10

<210> SEQ ID NO 332
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 332

Gly Gly Pro Ile Ser Asn Gly Pro Tyr Tyr
1 5 10

<210> SEQ ID NO 333
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 333

Ile Phe Tyr Ser Gly Ser Thr
1 5

<210> SEQ ID NO 334
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 334

Ala Arg His Val Val Thr Ala Ser Gly Trp Phe Asp Pro
1 5 10

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<210> SEQ ID NO 335
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 335

Gly Val Thr Phe Ser Asp Asn Ala
1 5

<210> SEQ ID NO 336
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 336

Ile Ser Tyr Asp Gly Ser Ser Arg
1 5

<210> SEQ ID NO 337
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 337

Ala Arg Val Thr Ala Asp Tyr Tyr Glu Ser Ser Gly Lys Val Phe
1 5 10 15

<210> SEQ ID NO 338
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 338

Gly Gly Ser Ile Ser Asp His Tyr
1 5

<210> SEQ ID NO 339
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 339

Ile Tyr Thr Ser Gly Thr Thr
1 5

<210> SEQ ID NO 340
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 340

Ala Arg Ser Leu Glu Thr Val Ile Arg Phe Tyr Tyr Tyr His Tyr Met
1 5 10 15

Asp Val

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<210> SEQ ID NO 341
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 341

Gly Gly Ser Ile Gly Asp Tyr His
1 5

<210> SEQ ID NO 342
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 342

Ile His Ser Ser Gly Asn Thr
1 5

<210> SEQ ID NO 343
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 343

Ala Arg Gln Asn Val Phe Asp Ile
1 5

<210> SEQ ID NO 344
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 344

Gly Ile Ser Ile Asn Asn Tyr Tyr
1 5

<210> SEQ ID NO 345
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 345

Val Tyr Ser Thr Gly Ser Ser
1 5

<210> SEQ ID NO 346
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 346

Ala Arg Gly Ser Met Pro His Ile
1 5

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<210> SEQ ID NO 347
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 347

Gly Ile Thr Phe Ser Asn Ala Trp
1 5

<210> SEQ ID NO 348
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 348

Ile Glu Ser Lys Ile Asp Gly Gly Thr Ile
1 5 10

<210> SEQ ID NO 349
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 349

Thr Thr Asp Gln Gly Tyr Tyr Asp Arg Ser Gly Tyr Trp Val Val Gly
1 5 10 15

Asn His Phe Asp Tyr
20

<210> SEQ ID NO 350
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 350

Gly Phe Thr Phe Ser Ser Tyr Ala
1 5

<210> SEQ ID NO 351
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 351

Ile Ser Gly Thr Thr Gly Ser Thr
1 5

<210> SEQ ID NO 352
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 352

Ala Lys Asp Ser His Ser Met Ile Val Val Asp His Ala Phe Asp Ile
1 5 10 15

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<210> SEQ ID NO 353
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 353

Gly Phe Thr Phe Ser Thr Tyr Ile
1 5

<210> SEQ ID NO 354
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 354

Ile Ser Thr Ser Ser Val Tyr Thr
1 5

<210> SEQ ID NO 355
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 355

Ala Arg Glu Glu Gly Phe Arg Ala Tyr Asn Leu Tyr
1 5 10

<210> SEQ ID NO 356
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 356

Gly Phe Thr Phe Ser Arg Phe Gly
1 5

<210> SEQ ID NO 357
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 357

Ile Ser Phe Asp Gly Ser Asn Arg
1 5

<210> SEQ ID NO 358
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 358

Ala Arg Asp Trp Asp Arg Leu Val Arg Ser Ala Val Gly Tyr
1 5 10

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<210> SEQ ID NO 359
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 359

Gly Gly Ser Val Ser Thr Ala Thr Tyr Tyr
1 5 10

<210> SEQ ID NO 360
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 360

Ile Tyr Ser Ser Gly Asn Thr
1 5

<210> SEQ ID NO 361
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 361

Glu Arg Arg Leu Arg Ile Leu Ser Ile Glu Arg Asn Tyr Tyr Ala Met
1 5 10 15

Asp Val

<210> SEQ ID NO 362
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 362

Gly Phe Thr Phe Ser Ser Tyr Ala
1 5

<210> SEQ ID NO 363
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 363

Ile Ser Tyr Asp Ala Ser Lys Lys
1 5

<210> SEQ ID NO 364
<211> LENGTH: 23
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 364

Ala Arg Asp His Val Pro Pro Lys Asp Cys Ser Asp Gly Asn Cys His
1 5 10 15

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Ser Asp Tyr Gly Met Asp Val
20

<210> SEQ ID NO 365
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 365

Gly Tyr Ser Phe Thr Asn Phe Ala
1 5

<210> SEQ ID NO 366
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 366

Ile Asn Pro Gly Asn Arg Asn Thr
1 5

<210> SEQ ID NO 367
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 367

Ala Arg Leu Pro Ile Ala Ala Ala Gly Arg Gly Trp Phe Asp Pro
1 5 10 15

<210> SEQ ID NO 368
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 368

Gly Phe Thr Phe Asn Thr Tyr Gly
1 5

<210> SEQ ID NO 369
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 369

Ile Ser Ser Ala Thr Thr Thr Phe
1 5

<210> SEQ ID NO 370
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 370

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Ala Arg Val Tyr Thr Met Leu Arg Gly Ala Ser Met Asp Val
1 5 10

<210> SEQ ID NO 371
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 371

Gly Tyr Thr Phe Thr Ala Tyr Tyr
1 5

<210> SEQ ID NO 372
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 372

Ile Asn Pro Ile Ser Gly Gly Thr
1 5

<210> SEQ ID NO 373
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 373

Ala Arg Val Lys Cys Ser Ser Ala Asn Cys Tyr Gly Asn Phe Asp Tyr
1 5 10 15

<210> SEQ ID NO 374
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 374

Gly Tyr Arg Phe Thr Asn Tyr Arg
1 5

<210> SEQ ID NO 375
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 375

Ile Tyr Pro Gly Gly Ser Asp Thr
1 5

<210> SEQ ID NO 376
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 376

Ala Arg Gln Thr Thr Gln Asn Ser Gly Tyr Asp Arg Trp Phe Asp Ser

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1	5	10	15
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<210> SEQ ID NO 377
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic peptide

 <400> SEQUENCE: 377

 Gly Ile Thr Phe Ser Arg His Thr
 1 5

<210> SEQ ID NO 378
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic peptide

 <400> SEQUENCE: 378

 Ile Ser Gly Ser Gly Gly Ser Thr
 1 5

<210> SEQ ID NO 379
 <211> LENGTH: 19
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic peptide

 <400> SEQUENCE: 379

 Ala Ile Ser Val Pro Leu Leu Arg Phe Leu Glu Trp Phe Gln His Pro
 1 5 10 15

 Phe Asp Phe

<210> SEQ ID NO 380
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic peptide

 <400> SEQUENCE: 380

 Gly Phe Thr Phe Asn Thr Tyr Ser
 1 5

<210> SEQ ID NO 381
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic peptide

 <400> SEQUENCE: 381

 Ile Ser Ser Thr Gly Ser Tyr Ile
 1 5

<210> SEQ ID NO 382
 <211> LENGTH: 13
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic peptide

 <400> SEQUENCE: 382

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Val	Arg	Phe	Thr	Met	Thr	Thr	Val	Thr	Asn	Phe	Asp	Ser
1				5					10			

<210> SEQ ID NO 383
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 383

Gly	Phe	Thr	Phe	Thr	Asn	Tyr	Gly
1				5			

<210> SEQ ID NO 384
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 384

Ile	Ser	Tyr	Asp	Gly	Asn	Lys
1				5		

<210> SEQ ID NO 385
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 385

Pro	Lys	Val	Ile	Pro	His	Pro	Tyr	Tyr	Asp	Ser	Ser	Gly	Asp	Asp	Ala
1				5					10					15	

Phe Asp Ile

<210> SEQ ID NO 386
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 386

Gly	Phe	Pro	Phe	Ser	Ser	Tyr	Ser
1				5			

<210> SEQ ID NO 387
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 387

Ile	Ser	Gly	Ser	Gly	Gly	Asp	Ile
1				5			

<210> SEQ ID NO 388
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 388

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Ala Arg Gly Leu Val Ala Thr Thr Gly Thr Arg Tyr Phe Asp Tyr
1 5 10 15

<210> SEQ ID NO 389
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 389

Gly Gly Thr Phe Arg Arg Phe Ala
1 5

<210> SEQ ID NO 390
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 390

Ile Ile Pro Ile Leu Gly Arg Gly
1 5

<210> SEQ ID NO 391
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 391

Ala Arg Phe Ile Ser Thr Ala Ser Tyr Val Pro Gly Thr Phe Glu Asp
1 5 10 15

Val

<210> SEQ ID NO 392
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 392

Gly Phe Thr Phe Arg Asn Tyr Asn
1 5

<210> SEQ ID NO 393
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 393

Ile Ser Ser Thr Gly Ser Tyr Ile
1 5

<210> SEQ ID NO 394
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

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<400> SEQUENCE: 394

Ala Arg Met Val Arg Asn Thr Val Thr Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 395

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 395

Gly Tyr Arg Leu Ile Asp Leu Pro
1 5

<210> SEQ ID NO 396

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 396

Phe Asp Pro Glu Lys Ala Glu Ala
1 5

<210> SEQ ID NO 397

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 397

Ala Thr Trp Gly Val Glu Val Val Asn Gly Arg Arg Asp Tyr Phe Asp
1 5 10 15

Ser

<210> SEQ ID NO 398

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 398

Gly Phe Thr Phe Ser Ser Phe Glu
1 5

<210> SEQ ID NO 399

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 399

Ile Ser Thr Ser Gly Ser Thr Ile
1 5

<210> SEQ ID NO 400

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

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<400> SEQUENCE: 400

Ala Arg Asp Val Arg Asp Cys Ser Ala Leu Thr Cys Pro Arg Arg Gly
1 5 10 15Asp Ala Phe Asp Phe
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<210> SEQ ID NO 401

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 401

Gly Phe Ser Phe Ser Val Tyr Pro
1 5

<210> SEQ ID NO 402

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 402

Ile Ser Ser Ser Ser Arg Tyr Ile
1 5

<210> SEQ ID NO 403

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 403

Val Lys Val Gly Gly Ser Lys His Gln Tyr Tyr Phe Asp Tyr
1 5 10

<210> SEQ ID NO 404

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 404

Gly Tyr Thr Phe Thr Asn Tyr Asp
1 5

<210> SEQ ID NO 405

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 405

Ile Ser Thr Tyr Asn Gly Asn Thr
1 5

<210> SEQ ID NO 406

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 406

Ala Arg Glu Arg Cys Ser Thr Ser Thr Cys Tyr Ser Arg Tyr Ala Asp
1 5 10 15

Tyr

<210> SEQ ID NO 407
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 407

Gly Phe Thr Leu Ser Arg Tyr Glu
1 5

<210> SEQ ID NO 408
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 408

Ile Ser Ser Gly Gly Pro Ser Ile
1 5

<210> SEQ ID NO 409
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 409

Met Arg Glu Gly Leu Thr Tyr Tyr Asp Ser Thr Ile
1 5 10

<210> SEQ ID NO 410
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 410

Gly Phe Thr Phe Asp Glu Tyr Ala
1 5

<210> SEQ ID NO 411
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 411

Ile Ser Trp Asn Gly Gly Ser Lys
1 5

<210> SEQ ID NO 412
<211> LENGTH: 12
<212> TYPE: PRT

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 412

Ala Lys Asp Asp Tyr Glu Gly Ala Gly Phe Asp Ile
1 5 10

<210> SEQ ID NO 413
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 413

Gly Tyr Leu Ile Ser Asn Gly Tyr Tyr
1 5

<210> SEQ ID NO 414
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 414

Ile Tyr Tyr Thr Arg Asp Thr
1 5

<210> SEQ ID NO 415
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 415

Val Arg His Glu Gly Ser Cys Asn Asp Gly Ser Cys Tyr Gly Ser Phe
1 5 10 15

Val Asp Asn

<210> SEQ ID NO 416
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 416

Asp Phe Thr Phe Ser Ser Tyr Thr
1 5

<210> SEQ ID NO 417
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 417

Ile Ser Gly Asp Gly Val Ser Thr
1 5

<210> SEQ ID NO 418
<211> LENGTH: 13

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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 418

Ala Arg Gly Gly Thr Phe His Asn Trp Tyr Phe Asp Leu
1 5 10

<210> SEQ ID NO 419
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 419

Gly Phe Thr Phe Ser Thr Tyr Ala
1 5

<210> SEQ ID NO 420
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 420

Ile Ser Gly Ser Gly Gly Ser Thr
1 5

<210> SEQ ID NO 421
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 421

Ala Lys Gly Thr Ile Thr Tyr Ser Tyr Tyr Tyr Met Ala Val
1 5 10

<210> SEQ ID NO 422
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 422

Gly Tyr Arg Leu Thr Asp Leu Pro
1 5

<210> SEQ ID NO 423
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 423

Val Asp Leu Glu Lys Arg Glu Ile
1 5

<210> SEQ ID NO 424
<211> LENGTH: 17
<212> TYPE: PRT

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 424

Ala Thr Trp Gly Ile Glu Val Val Asn Gly Arg Asp Glu Phe Phe Asp
1 5 10 15

Ser

<210> SEQ ID NO 425
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 425

Gly Tyr Ser Leu Ser Asp Leu Pro
1 5

<210> SEQ ID NO 426
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 426

Phe Asp Pro Ile Asn Gly Glu Ile
1 5

<210> SEQ ID NO 427
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 427

Ala Thr Trp Gly Val Ala Val Val Ser Gly Arg Arg Asp Tyr Phe Asp
1 5 10 15

Ser

<210> SEQ ID NO 428
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 428

Gly Gly Ser Ile Thr Ser Gly Asp Tyr Tyr
1 5 10

<210> SEQ ID NO 429
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 429

Ile Tyr His Ser Gly Thr Thr
1 5

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<210> SEQ ID NO 430
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 430

Ala Arg Ala Tyr Ala Tyr Glu Phe Trp Ser Gly Tyr Pro Asn Trp Phe
1 5 10 15

Asp Pro

<210> SEQ ID NO 431
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 431

Gly Phe Ile Phe Asn Arg Tyr Ala
1 5

<210> SEQ ID NO 432
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 432

Ile Ser Tyr Asp Gly Ile Asn Lys
1 5

<210> SEQ ID NO 433
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 433

Ala Arg Gly Leu Gly Tyr Cys Ser Gly Thr Gly Gly Ser Cys Thr Pro
1 5 10 15

Phe Glu Tyr

<210> SEQ ID NO 434
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 434

Gly Ala Ser Leu Asn Asp Tyr Asp
1 5

<210> SEQ ID NO 435
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 435

Ile Asn Arg Arg Asp Thr Val

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1 5

<210> SEQ ID NO 436
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 436

Val Arg Val Pro Arg Arg Gly Phe Glu Gly Ser Phe Gly Phe Cys Asp
1 5 10 15

Asp Thr Ala Cys Arg Tyr Gly His Thr Trp Phe Asp Pro
 20 25

<210> SEQ ID NO 437
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 437

Gly Phe Ser Leu Arg Asn Gly Arg Met Gly
1 5 10

<210> SEQ ID NO 438
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 438

Ile Phe Ala Ser Asp Glu Lys
1 5

<210> SEQ ID NO 439
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 439

Ala Arg Ile Leu Lys Phe Gly Thr Met Arg Ala Ala Tyr Tyr Phe Asp
1 5 10 15

Tyr

<210> SEQ ID NO 440
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 440

Gly Phe Thr Phe Ser Asp Tyr Tyr
1 5

<210> SEQ ID NO 441
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

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<400> SEQUENCE: 441

Ile Ser Ser Ser Gly Ser Thr Ile
1 5

<210> SEQ ID NO 442

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 442

Ala Gly Ser Lys Val Gly Tyr Thr Thr Gly Arg Arg Asn Trp Tyr Phe
1 5 10 15

Asp Leu

<210> SEQ ID NO 443

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 443

Gly Phe Thr Phe Thr Asp Phe Tyr
1 5

<210> SEQ ID NO 444

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 444

Ile Asn Pro Glu Thr Gly Glu Thr
1 5

<210> SEQ ID NO 445

<211> LENGTH: 19

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 445

Ala Arg Asp Leu Val Val Val Val Pro Val Glu Met Ser Arg Arg Ala
1 5 10 15

Phe Asp Ile

<210> SEQ ID NO 446

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 446

Gly Tyr Ile Phe Ser Ala Tyr Tyr
1 5

<210> SEQ ID NO 447

<211> LENGTH: 8

<212> TYPE: PRT

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 447

Met Asn Ala Lys Ser Gly Gly Ala
1 5

<210> SEQ ID NO 448
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 448

Ala Arg Asp Tyr Arg Asp Asp Tyr Met Trp Gly Ser Tyr Arg Pro Leu
1 5 10 15

Asp Tyr

<210> SEQ ID NO 449
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 449

Ala Leu Pro Lys Lys Tyr
1 5

<210> SEQ ID NO 450

<400> SEQUENCE: 450

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<210> SEQ ID NO 451
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 451

Ser Ser Thr Asp Ser Ser Gly Asn Pro Val Leu
1 5 10

<210> SEQ ID NO 452
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 452

Gln Ser Val Ser Thr Tyr
1 5

<210> SEQ ID NO 453

<400> SEQUENCE: 453

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<210> SEQ ID NO 454

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<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 454

Gln Gln Arg Ser Ser Trp Pro Ile Thr
1 5

<210> SEQ ID NO 455
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 455

Ser Gly Ser Val Ser Ser Ser Tyr Tyr
1 5

<210> SEQ ID NO 456

<400> SEQUENCE: 456

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<210> SEQ ID NO 457
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 457

Gly Leu Tyr Met Gly Ser Gly Ile Trp Ile
1 5 10

<210> SEQ ID NO 458
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 458

Gln Gly Ile Ser Ser Trp
1 5

<210> SEQ ID NO 459

<400> SEQUENCE: 459

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<210> SEQ ID NO 460
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 460

Gln Gln Ala Asp Ser Phe Pro Arg Thr
1 5

<210> SEQ ID NO 461

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<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 461

Gln Ser Ile Ser Lys Trp
1 5

<210> SEQ ID NO 462

<400> SEQUENCE: 462

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<210> SEQ ID NO 463
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 463

Gln Gln His Asn Ser Tyr Ser Tyr Thr
1 5

<210> SEQ ID NO 464
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 464

Gln Ser Val Arg Ser Tyr
1 5

<210> SEQ ID NO 465

<400> SEQUENCE: 465

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<210> SEQ ID NO 466
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 466

Gln Gln Arg Ser Thr Trp Pro Pro Gly Met
1 5 10

<210> SEQ ID NO 467
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 467

Gln Ser Val Ser Ser Ser Phe
1 5

<210> SEQ ID NO 468

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<400> SEQUENCE: 468

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<210> SEQ ID NO 469

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 469

Gln Gln Tyr Ser Asn Ser Arg Leu Thr
1 5

<210> SEQ ID NO 470

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 470

Ser Ser Asp Val Gly Gly Tyr Asn Tyr
1 5

<210> SEQ ID NO 471

<400> SEQUENCE: 471

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<210> SEQ ID NO 472

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 472

Ser Ser Tyr Thr Thr Ser Asn Thr Leu Val Val
1 5 10

<210> SEQ ID NO 473

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 473

Asn Ile Gly Thr Lys Ser
1 5

<210> SEQ ID NO 474

<400> SEQUENCE: 474

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<210> SEQ ID NO 475

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

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<400> SEQUENCE: 475

Gln Val Trp Asp Ser Gly Ile Asp Val Val
1 5 10

<210> SEQ ID NO 476

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 476

Asn Ile Gly Thr Lys Thr
1 5

<210> SEQ ID NO 477

<400> SEQUENCE: 477

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<210> SEQ ID NO 478

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 478

Arg Val Trp Asp Ser Asp Thr Asp His Arg Val
1 5 10

<210> SEQ ID NO 479

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 479

Lys Leu Gly Asp Lys Tyr
1 5

<210> SEQ ID NO 480

<400> SEQUENCE: 480

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<210> SEQ ID NO 481

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 481

Gln Ala Trp Asp Ser Ser Thr Val Val
1 5

<210> SEQ ID NO 482

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

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<400> SEQUENCE: 482

Gln Asp Ile Gly Val Asp
1 5

<210> SEQ ID NO 483

<400> SEQUENCE: 483

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<210> SEQ ID NO 484

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 484

Leu Gln Asp Tyr Asn Tyr Pro Trp Thr
1 5

<210> SEQ ID NO 485

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 485

Ala Leu Pro Lys Gln Tyr
1 5

<210> SEQ ID NO 486

<400> SEQUENCE: 486

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<210> SEQ ID NO 487

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 487

Gln Ser Gly Asp Ser Ser Gly Thr Tyr Leu Val
1 5 10

<210> SEQ ID NO 488

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 488

Gln Arg Val Gly Asn Ser
1 5

<210> SEQ ID NO 489

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<210> SEQ ID NO 490
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 490

His Gln His Ser Thr Trp Pro Arg Gly Thr
1 5 10

<210> SEQ ID NO 491
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 491

Ser Gly Ser Ile Ala Thr Asn Tyr
1 5

<210> SEQ ID NO 492

<400> SEQUENCE: 492

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<210> SEQ ID NO 493
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 493

Gln Ser Tyr Asp Asn Ser Asn Arg Ala Val Val
1 5 10

<210> SEQ ID NO 494
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 494

Ser Gly Thr Ile Ala Ser Asn Tyr
1 5

<210> SEQ ID NO 495

<400> SEQUENCE: 495

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<210> SEQ ID NO 496
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 496

Gln Ser Tyr Asp Asn Ser Asp Arg Val
1 5

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<210> SEQ ID NO 497
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 497

Asn Ile Gly Leu Lys Ser
1 5

<210> SEQ ID NO 498

<400> SEQUENCE: 498

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<210> SEQ ID NO 499
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 499

Gln Val Trp Asp Ser Ser Arg Asn His Pro Val
1 5 10

<210> SEQ ID NO 500
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 500

Lys Leu Gly Asp Lys Tyr
1 5

<210> SEQ ID NO 501

<400> SEQUENCE: 501

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<210> SEQ ID NO 502
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 502

Gln Ala Trp Asp Ser Ser Thr Ala Val
1 5

<210> SEQ ID NO 503
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 503

Gln Gly Ile Ser Arg Phe
1 5

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<210> SEQ ID NO 504

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 505

Gln Gln Leu Asn Ser His Pro Arg Met Phe Thr

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<211> LENGTH: 6

<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 506

Gln Gly Ile Ser Asn Trp

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<210> SEQ ID NO 507

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<210> SEQ ID NO 508

<211> LENGTH: 9

<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 508

Gln Gln Ala Asn Ser Phe Pro Phe Thr

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<211> LENGTH: 6

<212> TYPE: PRT

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<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

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Ala Leu Pro Lys Gln Tyr

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<223> OTHER INFORMATION: Synthetic peptide

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Gln Ser Ala Asp Ser Ser Gly Thr Tyr Val Val
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<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

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Asn Ile Gly Leu Lys Ser
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Gln Val Trp Asp Ser Ser Arg Asn His Pro Val
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<223> OTHER INFORMATION: Synthetic peptide

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<223> OTHER INFORMATION: Synthetic peptide

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<223> OTHER INFORMATION: Synthetic peptide

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<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

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<210> SEQ ID NO 522

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<210> SEQ ID NO 523

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<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

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<400> SEQUENCE: 526

Gln Gln Arg Ser Arg Trp Pro Pro Pro Tyr Thr
1 5 10

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Gln Ser Val Arg Ser Trp
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Gln Gln Tyr Gln Thr Phe Ser Trp Thr
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<212> TYPE: PRT
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Gln Ser Leu Leu Gln Ser Asp Gly Tyr Ser Tyr
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<210> SEQ ID NO 531

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Met Gln Ala Leu Gln Thr Pro Trp Thr
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<400> SEQUENCE: 533

Gln Ser Val Leu Phe Ser Ser Asn Asn Lys Asn Tyr
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<210> SEQ ID NO 534

<400> SEQUENCE: 534

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<212> TYPE: PRT
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<400> SEQUENCE: 535

Gln Gln Phe Tyr Thr Thr Pro Leu Thr
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<210> SEQ ID NO 536
<211> LENGTH: 8
<212> TYPE: PRT
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<400> SEQUENCE: 536

Thr Ser Asn Ile Glu Thr Asn Tyr
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<210> SEQ ID NO 537

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<212> TYPE: PRT
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<400> SEQUENCE: 538

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<400> SEQUENCE: 539

Ser Thr Asn Ile Gly Ala Gly Tyr Asp
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<212> TYPE: PRT

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<211> LENGTH: 6

<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 542

Asn Ile Gly Thr Lys Ser

1 5

<210> SEQ ID NO 543

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<210> SEQ ID NO 544

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

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Gln Val Trp Asp Ser Tyr Asn Val His Tyr Val

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Ser Ser Asn Ile Glu Tyr Asn Tyr

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<213> ORGANISM: Artificial sequence

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<223> OTHER INFORMATION: Synthetic peptide

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<210> SEQ ID NO 548

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<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 548

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<211> LENGTH: 10

<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 550

Cys Ser Tyr Ala Gly Thr Tyr Thr Trp Val
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<210> SEQ ID NO 551

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 551

Gln Arg Val Val Asn Asn Tyr
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<210> SEQ ID NO 552

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<210> SEQ ID NO 553

<211> LENGTH: 8

<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 553

Gln Gln Tyr Gly Ser Pro Trp Thr
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<210> SEQ ID NO 554

<211> LENGTH: 6

<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

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Gln Ser Ile Ser Lys Trp
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<210> SEQ ID NO 555

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<211> LENGTH: 9

<212> TYPE: PRT

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 556

Gln Gln His Asn Ser Tyr Ser Tyr Thr
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<210> SEQ ID NO 557

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 557

Gln Ser Val Ile Ser Thr Tyr
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<210> SEQ ID NO 558

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<210> SEQ ID NO 559

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 559

His Gln Tyr Gly Ser Ser Pro Ala Thr
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<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 560

Gln Ser Val Gly Thr Tyr
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<210> SEQ ID NO 561

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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 562

Gln Leu Arg Ile Thr Trp Pro Pro Ile Phe Thr
1 5 10

<210> SEQ ID NO 563
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 563

Ser Ser Asp Val Gly Gly Tyr Asn Tyr
1 5

<210> SEQ ID NO 564

<400> SEQUENCE: 564

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<212> TYPE: PRT
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<400> SEQUENCE: 565

Ser Ser Tyr Ala Gly Ser Asn Asn Leu Val
1 5 10

<210> SEQ ID NO 566
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 566

Thr Ser Ser Ile Gly Ser Asn Ile
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<210> SEQ ID NO 567

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<210> SEQ ID NO 568
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<212> TYPE: PRT
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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 568

Ala Ala Trp Asp Asp Ser Leu Asn Gly Trp Val
1 5 10

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<210> SEQ ID NO 569
<211> LENGTH: 6
<212> TYPE: PRT
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Gln Ser Val Gly Ser Thr
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<210> SEQ ID NO 570

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<210> SEQ ID NO 571
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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 571

His Gln Tyr Ile Asn Trp Pro Pro Trp Thr
1 5 10

<210> SEQ ID NO 572
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 572

Ser Ser Asp Val Gly Ala Tyr Ser Tyr
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<210> SEQ ID NO 573

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<212> TYPE: PRT
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<400> SEQUENCE: 574

Cys Ser His Ala Gly Ser His Thr Trp Val
1 5 10

<210> SEQ ID NO 575
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<212> TYPE: PRT
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<400> SEQUENCE: 575

Gln Ser Ile Ser Arg Asn
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<210> SEQ ID NO 576

<400> SEQUENCE: 576

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<210> SEQ ID NO 577

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

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Gln Gln Tyr Ser Lys Leu Pro Ile Thr

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<210> SEQ ID NO 578

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 578

Gln Ser Val Arg Ser Tyr

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<210> SEQ ID NO 579

<400> SEQUENCE: 579

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Gln Gln Arg Ser Tyr Trp Pro Pro Phe Thr

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<210> SEQ ID NO 581

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 581

Gln Ser Val Ser Ser Tyr

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<210> SEQ ID NO 582

<400> SEQUENCE: 582

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<210> SEQ ID NO 583

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 583

Gln Gln Arg Ser Asp Trp Pro Pro Gly Thr
1 5 10

<210> SEQ ID NO 584
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 584

Ser Ser Asp Val Gly Gly Tyr Asn Phe
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<210> SEQ ID NO 585

<400> SEQUENCE: 585

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<210> SEQ ID NO 586
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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 586

Cys Ser Tyr Gly Gly Asn Asn Ser Trp Met
1 5 10

<210> SEQ ID NO 587
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 587

Val Leu Ser Asn Gln Tyr
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<210> SEQ ID NO 588

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<210> SEQ ID NO 589
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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 589

Gln Ser Ala Asp Asn Thr Arg Ile Thr Val
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<210> SEQ ID NO 590
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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 590

Gln Ser Leu Val Tyr Ser Asp Gly Asn Thr Tyr
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<210> SEQ ID NO 591

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<210> SEQ ID NO 592
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<212> TYPE: PRT
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<400> SEQUENCE: 592

Met Gln Gly Thr His Trp Pro Arg Thr
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<210> SEQ ID NO 593
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<212> TYPE: PRT
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<400> SEQUENCE: 593

Arg Ser Asn Ile Gly Ser Tyr Thr
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<210> SEQ ID NO 594

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<210> SEQ ID NO 595
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<212> TYPE: PRT
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 595

Ala Ala Trp Asp Asp Ser Leu Asn Gly Val Val
1 5 10

<210> SEQ ID NO 596
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 596

Gln Ser Ile Asn Ile Tyr
1 5

<210> SEQ ID NO 597

<400> SEQUENCE: 597

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<210> SEQ ID NO 598
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<212> TYPE: PRT
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<400> SEQUENCE: 598

Gln Gln Ser Tyr Arg Ser Pro Arg Thr
1 5

<210> SEQ ID NO 599
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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 599

Ser Gly His Ser Arg Tyr Ala
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<210> SEQ ID NO 600
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 600

Ile Asn Ser Asp Gly Arg His
1 5

<210> SEQ ID NO 601
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<212> TYPE: PRT
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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 601

Gln Thr Trp Gly Thr Gly Phe Arg Val
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<210> SEQ ID NO 602
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 602

Gln Ser Val Ser Ser Tyr
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<210> SEQ ID NO 603

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<210> SEQ ID NO 604
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<212> TYPE: PRT
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 604

Gln Gln Arg Ser Asn Trp Pro Ile Thr
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<210> SEQ ID NO 605
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<212> TYPE: PRT
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<400> SEQUENCE: 605

Gln Asp Ile Ser Ser Trp
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<210> SEQ ID NO 606

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<212> TYPE: PRT
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<400> SEQUENCE: 607

Gln Gln Ala Asp Ser Phe Ile Thr
1 5

<210> SEQ ID NO 608
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 608

Gln Ser Ile Gly Asp Asn
1 5

<210> SEQ ID NO 609

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<212> TYPE: PRT
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<400> SEQUENCE: 610

Gln Gln Tyr Lys Asn Trp Pro Arg Thr
1 5

<210> SEQ ID NO 611
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<212> TYPE: PRT
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 611

Gln Ser Val Arg Ser Ser Tyr
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<210> SEQ ID NO 612

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<212> TYPE: PRT
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<400> SEQUENCE: 613

Gln Gln Tyr Gly Thr Ser Ile Thr
1 5

<210> SEQ ID NO 614
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 614

Val Leu Ser Lys Gln Tyr
1 5

<210> SEQ ID NO 615

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<210> SEQ ID NO 616
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<212> TYPE: PRT
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<400> SEQUENCE: 616

Gln Ser Ala Asp Thr Arg Ile Thr Val
1 5

<210> SEQ ID NO 617
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 617

Val Leu Ser Asn Gln Tyr
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<210> SEQ ID NO 618

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<210> SEQ ID NO 619
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<400> SEQUENCE: 619

Gln Thr Ala Asp Thr Lys Tyr Thr Val
1 5

<210> SEQ ID NO 620
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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 620

Gln Arg Ile Ser Thr Tyr
1 5

<210> SEQ ID NO 621

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<210> SEQ ID NO 622
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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 622

Gln Gln Ser Tyr Ser Pro Pro Trp Thr
1 5

<210> SEQ ID NO 623
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 623

Gln Ser Ile Lys Lys Tyr
1 5

<210> SEQ ID NO 624

<400> SEQUENCE: 624

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<210> SEQ ID NO 625
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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 625

Gln Gln Ser Asp Ser Ala Pro Pro Thr

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1 5

<210> SEQ ID NO 626
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 626

Gln Ser Ile Arg Asp Tyr
1 5

<210> SEQ ID NO 627

<400> SEQUENCE: 627

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<210> SEQ ID NO 628
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 628

Gln Gln Ser Tyr Leu Thr Pro Pro Thr
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<210> SEQ ID NO 629
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 629

Asn Ile Gly Ile Arg Thr
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<210> SEQ ID NO 630

<400> SEQUENCE: 630

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<210> SEQ ID NO 631
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 631

Gln Val Trp Asp Ser Ser Ser Asp His Val Val
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<210> SEQ ID NO 632
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 632

Ser Ser Asp Val Gly Ala Tyr Asn Tyr

-continued

1 5

<210> SEQ ID NO 633

<400> SEQUENCE: 633

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<210> SEQ ID NO 634

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 634

Ser Ser Tyr Ala Gly Asn Asn Asn Leu Val

1 5 10

<210> SEQ ID NO 635

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 635

Asn Ile Glu Arg Lys Ser

1 5

<210> SEQ ID NO 636

<400> SEQUENCE: 636

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<210> SEQ ID NO 637

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 637

Gln Val Trp Asp Ser Thr Thr Asp His Gly Val

1 5 10

<210> SEQ ID NO 638

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic peptide

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<210> SEQ ID NO 639

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<210> SEQ ID NO 640

<211> LENGTH: 8

<212> TYPE: PRT

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<213> ORGANISM: Artificial sequence
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 <223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 640

Gln Gln Arg Ser Asn Gly Leu Thr
 1 5

<210> SEQ ID NO 641
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 <223> OTHER INFORMATION: Synthetic primer

<400> SEQUENCE: 641

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<210> SEQ ID NO 642
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 <212> TYPE: DNA
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic primer

<400> SEQUENCE: 642

ccattgaatc cctgggcctt 20

<210> SEQ ID NO 643
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 643

Ala Arg Asp Arg Pro Ile Met Glu Gly Glu Gly Leu Asp Glu Leu Thr
 1 5 10 15
 Gly Tyr Tyr Val Tyr Gln Tyr Tyr Ala Met Asp Val
 20 25

<210> SEQ ID NO 644
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 644

Ala Ala Trp Asp Asp Arg Leu Ser Gly Val Val
 1 5 10

<210> SEQ ID NO 645
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 645

Gln Gln Tyr Asn Ser Tyr Pro Leu Thr
 1 5

<210> SEQ ID NO 646
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 646

Gln Gln Arg Ser Asn Trp Pro Pro Leu Thr
 1 5 10

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<210> SEQ ID NO 647
 <211> LENGTH: 9
 <212> TYPE: PRT
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<400> SEQUENCE: 647

Gln Gln Tyr Asn Ser Leu Pro Trp Thr
 1 5

What is claimed is:

1. A method of treating a subject infected with enterovirus D68 (EV-D68) or reducing the likelihood of infection of a subject at risk of contracting EV-D68, comprising delivering to said subject an antibody or antibody fragment comprising heavy chain CDR1-3 of SEQ ID NOS: 257, 258 and 259, and light chain CDR1-3 of SEQ ID NOS: 449, 450 and 451; heavy chain CDR1-3 of SEQ ID NOS: 260, 261 and 262, and light chain CDR1-3 of SEQ ID NOS: 452, 453 and 454; heavy chain CDR1-3 of SEQ ID NOS: 263, 264 and 265, and light chain CDR1-3 of SEQ ID NOS: 455, 456 and 457; heavy chain CDR1-3 of SEQ ID NOS: 266, 267 and 268, and light chain CDR1-3 of SEQ ID NOS: 458, 459 and 460; heavy chain CDR1-3 of SEQ ID NOS: 269, 270 and 271, and light chain CDR1-3 of SEQ ID NOS: 461, 462 and 463; heavy chain CDR1-3 of SEQ ID NOS: 272, 273 and 274, and light chain CDR1-3 of SEQ ID NOS: 464, 465 and 466; heavy chain CDR1-3 of SEQ ID NOS: 275, 276 and 277, and light chain CDR1-3 of SEQ ID NOS: 467, 468 and 469; heavy chain CDR1-3 of SEQ ID NOS: 278, 279 and 280, and light chain CDR1-3 of SEQ ID NOS: 470, 471 and 472; heavy chain CDR1-3 of SEQ ID NOS: 281, 282 and 283, and light chain CDR1-3 of SEQ ID NOS: 473, 474 and 475; heavy chain CDR1-3 of SEQ ID NOS: 284, 285 and 286, and light chain CDR1-3 of SEQ ID NOS: 476, 477 and 478; heavy chain CDR1-3 of SEQ ID NOS: 287, 288 and 289, and light chain CDR1-3 of SEQ ID NOS: 479, 480 and 481; heavy chain CDR1-3 of SEQ ID NOS: 290, 291 and 292, and light chain CDR1-3 of SEQ ID NOS: 482, 483 and 484; heavy chain CDR1-3 of SEQ ID NOS: 293, 294 and 295, and light chain CDR1-3 of SEQ ID NOS: 485, 486 and 487; heavy chain CDR1-3 of SEQ ID NOS: 296, 297 and 298, and light chain CDR1-3 of SEQ ID NOS: 488, 489 and 490; heavy chain CDR1-3 of SEQ ID NOS: 299, 300 and 301, and light chain CDR1-3 of SEQ ID NOS: 491, 492 and 493; heavy chain CDR1-3 of SEQ ID NOS: 302, 303 and 304, and light chain CDR1-3 of SEQ ID NOS: 494, 495 and 496; heavy chain CDR1-3 of SEQ ID NOS: 305, 306 and 307, and light chain CDR1-3 of SEQ ID NOS: 497, 498 and 499; heavy chain CDR1-3 of SEQ ID NOS: 308, 309 and 310, and light chain CDR1-3 of SEQ ID NOS: 500, 501 and 502; heavy chain CDR1-3 of SEQ ID NOS: 311, 312 and 313, and light chain CDR1-3 of SEQ ID NOS: 503, 504 and 505; heavy chain CDR1-3 of SEQ ID NOS: 314, 315 and 316, and light chain CDR1-3 of SEQ ID NOS: 506, 507 and 508; heavy chain CDR1-3 of SEQ ID NOS: 317, 318 and 319, and light chain CDR1-3 of SEQ ID NOS: 509, 510 and 511; heavy chain CDR1-3 of SEQ ID NOS: 320, 321 and 322, and light chain CDR1-3 of SEQ ID NOS: 512, 513 and 514; heavy chain CDR1-3 of SEQ ID NOS: 323, 324 and 325, and light chain CDR1-3 of SEQ ID NOS: 515, 516 and 517; heavy chain CDR1-3 of SEQ ID NOS: 326, 327 and 328, and light chain CDR1-3 of SEQ ID NOS: 518, 519 and 520;

heavy chain CDR1-3 of SEQ ID NOS: 329, 330 and 331, and light chain CDR1-3 of SEQ ID NOS: 521, 522 and 523; heavy chain CDR1-3 of SEQ ID NOS: 332, 333 and 334, and light chain CDR1-3 of SEQ ID NOS: 524, 525 and 526; heavy chain CDR1-3 of SEQ ID NOS: 335, 336 and 337, and light chain CDR1-3 of SEQ ID NOS: 527, 528 and 529; heavy chain CDR1-3 of SEQ ID NOS: 338, 339 and 340, and light chain CDR1-3 of SEQ ID NOS: 530, 531 and 532; heavy chain CDR1-3 of SEQ ID NOS: 341, 342 and 343, and light chain CDR1-3 of SEQ ID NOS: 533, 534 and 535; heavy chain CDR1-3 of SEQ ID NOS: 344, 345 and 346, and light chain CDR1-3 of SEQ ID NOS: 536, 537 and 538; heavy chain CDR1-3 of SEQ ID NOS: 347, 348 and 349, and light chain CDR1-3 of SEQ ID NOS: 539, 540 and 541; heavy chain CDR1-3 of SEQ ID NOS: 350, 351 and 352, and light chain CDR1-3 of SEQ ID NOS: 542, 543 and 544; heavy chain CDR1-3 of SEQ ID NOS: 353, 354 and 355, and light chain CDR1-3 of SEQ ID NOS: 545, 546 and 547; heavy chain CDR1-3 of SEQ ID NOS: 356, 357 and 358, and light chain CDR1-3 of SEQ ID NOS: 548, 549 and 550; heavy chain CDR1-3 of SEQ ID NOS: 359, 360 and 361, and light chain CDR1-3 of SEQ ID NOS: 551, 552 and 553; heavy chain CDR1-3 of SEQ ID NOS: 362, 363 and 364, and light chain CDR1-3 of SEQ ID NOS: 554, 555 and 556; heavy chain CDR1-3 of SEQ ID NOS: 365, 366 and 367, and light chain CDR1-3 of SEQ ID NOS: 557, 558 and 559; heavy chain CDR1-3 of SEQ ID NOS: 368, 369 and 370, and light chain CDR1-3 of SEQ ID NOS: 560, 561 and 562; heavy chain CDR1-3 of SEQ ID NOS: 371, 372 and 373, and light chain CDR1-3 of SEQ ID NOS: 563, 564 and 565; heavy chain CDR1-3 of SEQ ID NOS: 374, 375 and 376, and light chain CDR1-3 of SEQ ID NOS: 566, 567 and 568; heavy chain CDR1-3 of SEQ ID NOS: 377, 378 and 379, and light chain CDR1-3 of SEQ ID NOS: 569, 570 and 571; heavy chain CDR1-3 of SEQ ID NOS: 380, 381 and 382, and light chain CDR1-3 of SEQ ID NOS: 572, 573 and 574; heavy chain CDR1-3 of SEQ ID NOS: 383, 384 and 385, and light chain CDR1-3 of SEQ ID NOS: 575, 576 and 577; heavy chain CDR1-3 of SEQ ID NOS: 386, 387 and 388, and light chain CDR1-3 of SEQ ID NOS: 578, 579 and 580; heavy chain CDR1-3 of SEQ ID NOS: 389, 390 and 391, and light chain CDR1-3 of SEQ ID NOS: 581, 582 and 583; heavy chain CDR1-3 of SEQ ID NOS: 392, 393 and 394, and light chain CDR1-3 of SEQ ID NOS: 584, 585 and 586; heavy chain CDR1-3 of SEQ ID NOS: 395, 396 and 397, and light chain CDR1-3 of SEQ ID NOS: 587, 588 and 589; heavy chain CDR1-3 of SEQ ID NOS: 398, 399 and 400, and light chain CDR1-3 of SEQ ID NOS: 590, 591 and 592; heavy chain CDR1-3 of SEQ ID NOS: 401, 402 and 403, and light chain CDR1-3 of SEQ ID NOS: 593, 594 and 595; heavy chain CDR1-3 of SEQ ID NOS: 404, 405 and 406, and light chain CDR1-3 of SEQ ID NOS: 596, 597 and 598; heavy chain CDR1-3 of SEQ ID NOS: 407, 408 and 409,

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and light chain CDR1-3 of SEQ ID NOS: 599, 600 and 601; heavy chain CDR1-3 of SEQ ID NOS: 410, 411 and 412, and light chain CDR1-3 of SEQ ID NOS: 602, 603 and 604; heavy chain CDR1-3 of SEQ ID NOS: 413, 414 and 415, and light chain CDR1-3 of SEQ ID NOS: 605, 606 and 607; heavy chain CDR1-3 of SEQ ID NOS: 416, 417 and 418, and light chain CDR1-3 of SEQ ID NOS: 608, 609 and 610; heavy chain CDR1-3 of SEQ ID NOS: 419, 420 and 421, and light chain CDR1-3 of SEQ ID NOS: 611, 612 and 613; heavy chain CDR1-3 of SEQ ID NOS: 422, 423 and 424, and light chain CDR1-3 of SEQ ID NOS: 614, 615 and 616; heavy chain CDR1-3 of SEQ ID NOS: 425, 426 and 427, and light chain CDR1-3 of SEQ ID NOS: 617, 618 and 619; heavy chain CDR1-3 of SEQ ID NOS: 428, 429 and 430, and light chain CDR1-3 of SEQ ID NOS: 620, 621 and 622; heavy chain CDR1-3 of SEQ ID NOS: 431, 432 and 433, and light chain CDR1-3 of SEQ ID NOS: 623, 624 and 625; heavy chain CDR1-3 of SEQ ID NOS: 434, 435 and 436, and light chain CDR1-3 of SEQ ID NOS: 626, 627 and 628; heavy chain CDR1-3 of SEQ ID NOS: 437, 438 and 439, and light chain CDR1-3 of SEQ ID NOS: 629, 630 and 631; heavy chain CDR1-3 of SEQ ID NOS: 440, 441 and 442, and light chain CDR1-3 of SEQ ID NOS: 632, 633 and 634; heavy chain CDR1-3 of SEQ ID NOS: 443, 444 and 445, and light chain CDR1-3 of SEQ ID NOS: 635, 636 and 637; or heavy chain CDR1-3 of SEQ ID NOS: 446, 447 and 448, and light chain CDR1-3 of SEQ ID NOS: 638, 639 and 640.

2. A method of detecting enterovirus D68 (EV-D68) infection in a subject comprising:

- (a) contacting a sample from said subject with an antibody or antibody fragment comprising heavy chain CDR1-3 of SEQ ID NOS: 257, 258 and 259, and light chain CDR1-3 of SEQ ID NOS: 449, 450 and 451; heavy chain CDR1-3 of SEQ ID NOS: 260, 261 and 262, and light chain CDR1-3 of SEQ ID NOS: 452, 453 and 454; heavy chain CDR1-3 of SEQ ID NOS: 263, 264 and 265, and light chain CDR1-3 of SEQ ID NOS: 455, 456 and 457; heavy chain CDR1-3 of SEQ ID NOS: 266, 267 and 268, and light chain CDR1-3 of SEQ ID NOS: 458, 459 and 460; heavy chain CDR1-3 of SEQ ID NOS: 269, 270 and 271, and light chain CDR1-3 of SEQ ID NOS: 461, 462 and 463; heavy chain CDR1-3 of SEQ ID NOS: 272, 273 and 274, and light chain CDR1-3 of SEQ ID NOS: 464, 465 and 466; heavy chain CDR1-3 of SEQ ID NOS: 275, 276 and 277, and light chain CDR1-3 of SEQ ID NOS: 467, 468 and 469; heavy chain CDR1-3 of SEQ ID NOS: 278, 279 and 280, and light chain CDR1-3 of SEQ ID NOS: 470, 471 and 472; heavy chain CDR1-3 of SEQ ID NOS: 281, 282 and 283, and light chain CDR1-3 of SEQ ID NOS: 473, 474 and 475; heavy chain CDR1-3 of SEQ ID NOS: 284, 285 and 286, and light chain CDR1-3 of SEQ ID NOS: 476, 477 and 478; heavy chain CDR1-3 of SEQ ID NOS: 287, 288 and 289, and light chain CDR1-3 of SEQ ID NOS: 479, 480 and 481; heavy chain CDR1-3 of SEQ ID NOS: 290, 291 and 292, and light chain CDR1-3 of SEQ ID NOS: 482, 483 and 484; heavy chain CDR1-3 of SEQ ID NOS: 293, 294 and 295, and light chain CDR1-3 of SEQ ID NOS: 485, 486 and 487; heavy chain CDR1-3 of SEQ ID NOS: 296, 297 and 298, and light chain CDR1-3 of SEQ ID NOS: 488, 489 and 490; heavy chain CDR1-3 of SEQ ID NOS: 299, 300 and 301, and light chain CDR1-3 of SEQ ID NOS: 491, 492 and 493; heavy chain CDR1-3 of SEQ ID NOS: 302, 303 and 304, and light chain CDR1-3 of SEQ ID NOS: 494, 495 and 496; heavy chain CDR1-3 of SEQ ID NOS: 305, 306 and 307, and

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light chain CDR1-3 of SEQ ID NOS: 497, 498 and 499; heavy chain CDR1-3 of SEQ ID NOS: 308, 309 and 310, and light chain CDR1-3 of SEQ ID NOS: 500, 501 and 502; heavy chain CDR1-3 of SEQ ID NOS: 311, 312 and 313, and light chain CDR1-3 of SEQ ID NOS: 503, 504 and 505; heavy chain CDR1-3 of SEQ ID NOS: 314, 315 and 316, and light chain CDR1-3 of SEQ ID NOS: 506, 507 and 508; heavy chain CDR1-3 of SEQ ID NOS: 317, 318 and 319, and light chain CDR1-3 of SEQ ID NOS: 509, 510 and 511; heavy chain CDR1-3 of SEQ ID NOS: 320, 321 and 322, and light chain CDR1-3 of SEQ ID NOS: 512, 513 and 514; heavy chain CDR1-3 of SEQ ID NOS: 323, 324 and 325, and light chain CDR1-3 of SEQ ID NOS: 515, 516 and 517; heavy chain CDR1-3 of SEQ ID NOS: 326, 327 and 328, and light chain CDR1-3 of SEQ ID NOS: 518, 519 and 520; heavy chain CDR1-3 of SEQ ID NOS: 329, 330 and 331, and light chain CDR1-3 of SEQ ID NOS: 521, 522 and 523; heavy chain CDR1-3 of SEQ ID NOS: 332, 333 and 334, and light chain CDR1-3 of SEQ ID NOS: 524, 525 and 526; heavy chain CDR1-3 of SEQ ID NOS: 335, 336 and 347, and light chain CDR1-3 of SEQ ID NOS: 527, 528 and 529; heavy chain CDR1-3 of SEQ ID NOS: 338, 339 and 340, and light chain CDR1-3 of SEQ ID NOS: 530, 531 and 532; heavy chain CDR1-3 of SEQ ID NOS: 341, 342 and 343, and light chain CDR1-3 of SEQ ID NOS: 533, 534 and 535; heavy chain CDR1-3 of SEQ ID NOS: 344, 345 and 346, and light chain CDR1-3 of SEQ ID NOS: 536, 537 and 538; heavy chain CDR1-3 of SEQ ID NOS: 337, 348 and 349, and light chain CDR1-3 of SEQ ID NOS: 539, 540 and 541; heavy chain CDR1-3 of SEQ ID NOS: 350, 351 and 352, and light chain CDR1-3 of SEQ ID NOS: 542, 543 and 544; heavy chain CDR1-3 of SEQ ID NOS: 353, 354 and 355, and light chain CDR1-3 of SEQ ID NOS: 545, 546 and 547; heavy chain CDR1-3 of SEQ ID NOS: 356, 357 and 358, and light chain CDR1-3 of SEQ ID NOS: 548, 549 and 550; heavy chain CDR1-3 of SEQ ID NOS: 359, 360 and 361, and light chain CDR1-3 of SEQ ID NOS: 551, 552 and 553; heavy chain CDR1-3 of SEQ ID NOS: 362, 363 and 364, and light chain CDR1-3 of SEQ ID NOS: 554, 555 and 556; heavy chain CDR1-3 of SEQ ID NOS: 365, 366 and 367, and light chain CDR1-3 of SEQ ID NOS: 557, 558 and 559; heavy chain CDR1-3 of SEQ ID NOS: 368, 369 and 370, and light chain CDR1-3 of SEQ ID NOS: 560, 561 and 562; heavy chain CDR1-3 of SEQ ID NOS: 371, 372 and 373, and light chain CDR1-3 of SEQ ID NOS: 563, 564 and 565; heavy chain CDR1-3 of SEQ ID NOS: 374, 375 and 376, and light chain CDR1-3 of SEQ ID NOS: 566, 567 and 568; heavy chain CDR1-3 of SEQ ID NOS: 377, 378 and 379, and light chain CDR1-3 of SEQ ID NOS: 569, 570 and 571; heavy chain CDR1-3 of SEQ ID NOS: 380, 381 and 382, and light chain CDR1-3 of SEQ ID NOS: 572, 573 and 574; heavy chain CDR1-3 of SEQ ID NOS: 383, 384 and 385, and light chain CDR1-3 of SEQ ID NOS: 575, 576 and 577; heavy chain CDR1-3 of SEQ ID NOS: 386, 387 and 388, and light chain CDR1-3 of SEQ ID NOS: 578, 579 and 580; heavy chain CDR1-3 of SEQ ID NOS: 389, 390 and 391, and light chain CDR1-3 of SEQ ID NOS: 581, 582 and 583; heavy chain CDR1-3 of SEQ ID NOS: 392, 393 and 394, and light chain CDR1-3 of SEQ ID NOS: 584, 585 and 586; heavy chain CDR1-3 of SEQ ID NOS: 395, 396 and 397, and light chain CDR1-3 of SEQ ID NOS: 587, 588 and 589;

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heavy chain CDR1-3 of SEQ ID NOS: 398, 399 and 400, and light chain CDR1-3 of SEQ ID NOS: 590, 591 and 592; heavy chain CDR1-3 of SEQ ID NOS: 401, 402 and 403, and light chain CDR1-3 of SEQ ID NOS: 593, 594 and 595; heavy chain CDR1-3 of SEQ ID NOS: 404, 405 and 406, and light chain CDR1-3 of SEQ ID NOS: 596, 597 and 598; heavy chain CDR1-3 of SEQ ID NOS: 407, 408 and 409, and light chain CDR1-3 of SEQ ID NOS: 599, 600 and 601; heavy chain CDR1-3 of SEQ ID NOS: 410, 411 and 412, and light chain CDR1-3 of SEQ ID NOS: 602, 603 and 604; heavy chain CDR1-3 of SEQ ID NOS: 413, 414 and 415, and light chain CDR1-3 of SEQ ID NOS: 605, 606 and 607; heavy chain CDR1-3 of SEQ ID NOS: 416, 417 and 418, and light chain CDR1-3 of SEQ ID NOS: 608, 609 and 610; heavy chain CDR1-3 of SEQ ID NOS: 419, 420 and 421, and light chain CDR1-3 of SEQ ID NOS: 611, 612 and 613; heavy chain CDR1-3 of SEQ ID NOS: 422, 423 and 424, and light chain CDR1-3 of SEQ ID NOS: 614, 615 and 616; heavy chain CDR1-3 of SEQ ID NOS: 425, 426 and 427, and light chain CDR1-3 of SEQ ID NOS: 617, 618 and 619; heavy chain CDR1-3 of SEQ ID NOS: 428, 429 and 430, and light chain CDR1-3 of SEQ ID NOS: 620, 621 and 622; heavy chain CDR1-3 of SEQ ID NOS: 431, 432 and 433, and light chain CDR1-3 of SEQ ID NOS: 623, 624 and 625; heavy chain CDR1-3 of SEQ ID NOS: 434, 435 and 436, and light chain CDR1-3 of SEQ ID NOS: 626, 627 and 628; heavy chain CDR1-3 of SEQ ID NOS: 437, 438 and 439, and light chain CDR1-3 of SEQ ID NOS: 629, 630 and 631; heavy chain CDR1-3 of SEQ ID NOS: 440, 441 and 442, and light chain CDR1-3 of SEQ ID NOS: 632, 633 and 634; heavy chain CDR1-3 of SEQ ID NOS: 443, 444 and 445, and light chain CDR1-3 of SEQ ID NOS: 635, 636 and 637; or heavy chain CDR1-3 of SEQ ID NOS: 446, 447 and 448, and light chain CDR1-3 of SEQ ID NOS: 638, 639 and 640; and

- (b) detecting EV-D68 in said sample by binding of said antibody or antibody fragment to a EV-D68 antigen in said sample.

3. The method of claim 2, wherein said sample is a body fluid.

4. The method of claim 2, wherein said sample is blood, sputum, tears, saliva, mucous or serum, semen, cervical or vaginal secretions, amniotic fluid, placental tissues, urine, exudate, transudate, respiratory droplets or aerosol, tissue scrapings or feces.

5. The method of claim 2, wherein detection comprises ELISA, RIA, lateral flow assay or Western blot.

6. The method of claim 2, further comprising performing steps (a) and (b) a second time and determining a change in EV-D68 antigen levels as compared to a previous assay.

7. The method of claim 2, wherein the antibody or antibody fragment is encoded by heavy and light chain variable regions comprising SEQ ID NOS: 1 and 2; SEQ ID NOS: 3 and 4; SEQ ID NOS: 5 and 6; SEQ ID NOS: 7 and 8; SEQ ID NOS: 9 and 10; SEQ ID NOS: 11 and 12; SEQ ID NOS: 13 and 14; SEQ ID NOS: 15 and 16; SEQ ID NOS: 17 and 18; SEQ ID NOS: 19 and 20; SEQ ID NOS: 21 and 22; SEQ ID NOS: 23 and 24; SEQ ID NOS: 25 and 26; SEQ ID NOS: 27 and 28; SEQ ID NOS: 29 and 30; SEQ ID NOS: 31 and 32; SEQ ID NOS: 33 and 34; SEQ ID NOS: 35 and 36; SEQ ID NOS: 37 and 38; SEQ ID NOS: 39 and 40; SEQ ID NOS: 41 and 42; SEQ ID NOS: 43 and 44; SEQ ID NOS: 45 and 46; SEQ ID NOS: 47 and 48; SEQ ID NOS: 49 and 50; SEQ ID NOS: 51 and 52; SEQ ID NOS: 53 and 54; SEQ

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ID NOS: 55 and 56; SEQ ID NOS: 57 and 58; SEQ ID NOS: 59 and 60; SEQ ID NOS: 61 and 62; SEQ ID NOS: 63 and 64; SEQ ID NOS: 65 and 66; SEQ ID NOS: 67 and 68; SEQ ID NOS: 69 and 70; SEQ ID NOS: 71 and 72; SEQ ID NOS: 73 and 74; SEQ ID NOS: 75 and 76; SEQ ID NOS: 77 and 78; SEQ ID NOS: 79 and 80; SEQ ID NOS: 81 and 82; SEQ ID NOS: 83 and 84; SEQ ID NOS: 85 and 86; SEQ ID NOS: 87 and 88; SEQ ID NOS: 89 and 90; SEQ ID NOS: 91 and 92; SEQ ID NOS: 93 and 94; SEQ ID NOS: 95 and 96; SEQ ID NOS: 97 and 98; SEQ ID NOS: 99 and 100; SEQ ID NOS: 101 and 102; SEQ ID NOS: 103 and 104; SEQ ID NOS: 105 and 106; SEQ ID NOS: 107 and 108; SEQ ID NOS: 109 and 110; SEQ ID NOS: 111 and 112; SEQ ID NOS: 113 and 114; SEQ ID NOS: 115 and 116; SEQ ID NOS: 117 and 118; SEQ ID NOS: 119 and 120; SEQ ID NOS: 121 and 122; SEQ ID NOS: 123 and 124; SEQ ID NOS: 125 and 126; or SEQ ID NOS: 127 and 128.

8. The method of claim 2, wherein said antibody or antibody fragment is encoded by heavy and light chain variable regions comprising SEQ ID NOS: 1 and 2; SEQ ID NOS: 3 and 4; SEQ ID NOS: 5 and 6; SEQ ID NOS: 7 and 8; SEQ ID NOS: 9 and 10; SEQ ID NOS: 11 and 12; SEQ ID NOS: 13 and 14; SEQ ID NOS: 15 and 16; SEQ ID NOS: 17 and 18; SEQ ID NOS: 19 and 20; SEQ ID NOS: 21 and 22; SEQ ID NOS: 23 and 24; SEQ ID NOS: 25 and 26; SEQ ID NOS: 27 and 28; SEQ ID NOS: 29 and 30; SEQ ID NOS: 31 and 32; SEQ ID NOS: 33 and 34; SEQ ID NOS: 35 and 36; SEQ ID NOS: 37 and 38; SEQ ID NOS: 39 and 40; SEQ ID NOS: 41 and 42; SEQ ID NOS: 43 and 44; SEQ ID NOS: 45 and 46; SEQ ID NOS: 47 and 48; SEQ ID NOS: 49 and 50; SEQ ID NOS: 51 and 52; SEQ ID NOS: 53 and 54; SEQ ID NOS: 55 and 56; SEQ ID NOS: 57 and 58; SEQ ID NOS: 59 and 60; SEQ ID NOS: 61 and 62; SEQ ID NOS: 63 and 64; SEQ ID NOS: 65 and 66; SEQ ID NOS: 67 and 68; SEQ ID NOS: 69 and 70; SEQ ID NOS: 71 and 72; SEQ ID NOS: 73 and 74; SEQ ID NOS: 75 and 76; SEQ ID NOS: 77 and 78; SEQ ID NOS: 79 and 80; SEQ ID NOS: 81 and 82; SEQ ID NOS: 83 and 84; SEQ ID NOS: 85 and 86; SEQ ID NOS: 87 and 88; SEQ ID NOS: 89 and 90; SEQ ID NOS: 91 and 92; SEQ ID NOS: 93 and 94; SEQ ID NOS: 95 and 96; SEQ ID NOS: 97 and 98; SEQ ID NOS: 99 and 100; SEQ ID NOS: 101 and 102; SEQ ID NOS: 103 and 104; SEQ ID NOS: 105 and 106; SEQ ID NOS: 107 and 108; SEQ ID NOS: 109 and 110; SEQ ID NOS: 111 and 112; SEQ ID NOS: 113 and 114; SEQ ID NOS: 115 and 116; SEQ ID NOS: 117 and 118; SEQ ID NOS: 119 and 120; SEQ ID NOS: 121 and 122; SEQ ID NOS: 123 and 124; SEQ ID NOS: 125 and 126; or SEQ ID NOS: 127 and 128.

9. The method of claim 2, wherein said antibody or antibody fragment comprises heavy and light chain variable sequences comprising SEQ ID NOS: 129 and 130; SEQ ID NOS: 131 and 132; SEQ ID NOS: 133 and 134; SEQ ID NOS: 135 and 136; SEQ ID NOS: 137 and 138; SEQ ID NOS: 139 and 140; SEQ ID NOS: 141 and 142; SEQ ID NOS: 143 and 144; SEQ ID NOS: 145 and 146; SEQ ID NOS: 147 and 148; SEQ ID NOS: 149 and 150; SEQ ID NOS: 151 and 152; SEQ ID NOS: 153 and 154; SEQ ID NOS: 155 and 156; SEQ ID NOS: 157 and 158; SEQ ID NOS: 159 and 160; SEQ ID NOS: 161 and 162; SEQ ID NOS: 163 and 164; SEQ ID NOS: 165 and 166; SEQ ID NOS: 167 and 168; SEQ ID NOS: 169 and 170; SEQ ID NOS: 171 and 172; SEQ ID NOS: 173 and 174; SEQ ID NOS: 175 and 176; SEQ ID NOS: 177 and 178; SEQ ID NOS: 179 and 180; SEQ ID NOS: 181 and 182; SEQ ID NOS: 183 and 184; SEQ ID NOS: 185 and 186; SEQ ID NOS: 187 and 188; SEQ ID NOS: 189 and 190; SEQ ID NOS: 191 and 192; SEQ ID NOS: 193 and 194; SEQ ID

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NOS: 195 and 196; SEQ ID NOS: 197 and 198; SEQ ID NOS: 199 and 200; SEQ ID NOS: 201 and 202; SEQ ID NOS: 203 and 204; SEQ ID NOS: 205 and 206; SEQ ID NOS: 207 and 208; SEQ ID NOS: 209 and 210; SEQ ID NOS: 211 and 212; SEQ ID NOS: 213 and 214; SEQ ID NOS: 215 and 216; SEQ ID NOS: 217 and 218; SEQ ID NOS: 219 and 220; SEQ ID NOS: 221 and 222; SEQ ID NOS: 223 and 224; SEQ ID NOS: 225 and 226; SEQ ID NOS: 227 and 228; SEQ ID NOS: 229 and 230; SEQ ID NOS: 231 and 232; SEQ ID NOS: 233 and 234; SEQ ID NOS: 235 and 236; SEQ ID NOS: 237 and 238; SEQ ID NOS: 239 and 240; SEQ ID NOS: 241 and 242; SEQ ID NOS: 243 and 244; SEQ ID NOS: 245 and 246; SEQ ID NOS: 247 and 248; SEQ ID NOS: 249 and 250; SEQ ID NOS: 251 and 252; SEQ ID NOS: 253 and 254; or SEQ ID NOS: 255 and 256.

10. The method of claim 2, wherein said antibody or antibody fragment comprises heavy and light chain variable sequences having 95% identity to SEQ ID NOS: 129 and 130; SEQ ID NOS: 131 and 132; SEQ ID NOS: 133 and 134; SEQ ID NOS: 135 and 136; SEQ ID NOS: 137 and 138; SEQ ID NOS: 139 and 140; SEQ ID NOS: 141 and 142; SEQ ID NOS: 143 and 144; SEQ ID NOS: 145 and 146; SEQ ID NOS: 147 and 148; SEQ ID NOS: 149 and 150; SEQ ID NOS: 151 and 152; SEQ ID NOS: 153 and 154; SEQ ID NOS: 155 and 156; SEQ ID NOS: 157 and 158; SEQ ID NOS: 159 and 160; SEQ ID NOS: 161 and 162; SEQ ID NOS: 163 and 164; SEQ ID NOS: 165 and 166; SEQ ID NOS: 167 and 168; SEQ ID NOS: 169 and 170; SEQ ID NOS: 171 and 172; SEQ ID NOS: 173 and 174; SEQ ID NOS: 175 and 176; SEQ ID NOS: 177 and 178; SEQ ID NOS: 179 and 180; SEQ ID NOS: 181 and 182; SEQ ID NOS: 183 and 184; SEQ ID NOS: 185 and 186; SEQ ID NOS: 187 and 188; SEQ ID NOS: 189 and 190; SEQ ID NOS: 191 and 192; SEQ ID NOS: 193 and 194; SEQ ID NOS: 195 and 196; SEQ ID NOS: 197 and 198; SEQ ID NOS: 199 and 200; SEQ ID NOS: 201 and 202; SEQ ID NOS: 203 and 204; SEQ ID NOS: 205 and 206; SEQ ID NOS: 207 and 208; SEQ ID NOS: 209 and 210; SEQ ID NOS: 211 and 212; SEQ ID NOS: 213 and 214; SEQ ID NOS: 215 and 216; SEQ ID NOS: 217 and 218; SEQ ID NOS: 219 and 220; SEQ ID NOS: 221 and 222; SEQ ID NOS: 223 and 224; SEQ ID NOS: 225 and 226; SEQ ID NOS: 227 and 228; SEQ ID NOS: 229 and 230; SEQ ID NOS: 231 and 232; SEQ ID NOS: 233 and 234; SEQ ID NOS: 235 and 236; SEQ ID NOS: 237 and 238; SEQ ID NOS: 239 and 240; SEQ ID NOS: 241 and 242; SEQ ID NOS: 243 and 244; SEQ ID NOS: 245 and 246; SEQ ID NOS: 247 and 248; SEQ ID NOS: 249 and 250; SEQ ID NOS: 251 and 252; SEQ ID NOS: 253 and 254; or SEQ ID NOS: 255 and 256.

11. The method of claim 2, wherein the antibody fragment is a recombinant scFv (single chain fragment variable) antibody, Fab fragment, F(ab')₂ fragment, or Fv fragment.

12. The method of claim 2, wherein the antibody or antibody fragment comprises heavy chain CDR1-3 of SEQ ID NOS: 314-316 and light chain CDR1-3 of SEQ ID NOS: 506-508 or heavy chain CDR1-3 of SEQ ID NOS: 413-415 and light chain CDR1-3 of SEQ ID NOS: 605-607.

13. The method of claim 2, wherein the antibody or antibody fragment comprises a heavy chain variable region of SEQ ID NO: 165 and a light chain variable region of SEQ ID NO: 166 or a heavy chain variable region of SEQ ID NO: 233 and a light chain variable region of SEQ ID NO: 234.

14. The method of claim 2, wherein the antibody or antibody fragment comprises heavy and light chain variable sequences having 95% identity to a heavy chain variable

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region of SEQ ID NO: 165 and a light chain variable region of SEQ ID NO: 166 or a heavy chain variable region of SEQ ID NO: 233 and a light chain variable region of SEQ ID NO: 234.

15. A monoclonal antibody or fragment thereof, wherein the antibody or antibody fragment comprises heavy chain CDR1-3 of SEQ ID NOS: 257, 258 and 259, and light chain CDR1-3 of SEQ ID NOS: 449, 450 and 451; heavy chain CDR1-3 of SEQ ID NOS: 260, 261 and 262, and light chain CDR1-3 of SEQ ID NOS: 452, 453 and 454; heavy chain CDR1-3 of SEQ ID NOS: 263, 264 and 265, and light chain CDR1-3 of SEQ ID NOS: 455, 456 and 457; heavy chain CDR1-3 of SEQ ID NOS: 266, 267 and 268, and light chain CDR1-3 of SEQ ID NOS: 458, 459 and 460; heavy chain CDR1-3 of SEQ ID NOS: 269, 270 and 271, and light chain CDR1-3 of SEQ ID NOS: 461, 462 and 463; heavy chain CDR1-3 of SEQ ID NOS: 272, 273 and 274, and light chain CDR1-3 of SEQ ID NOS: 464, 465 and 466; heavy chain CDR1-3 of SEQ ID NOS: 275, 276 and 277, and light chain CDR1-3 of SEQ ID NOS: 467, 468 and 469; heavy chain CDR1-3 of SEQ ID NOS: 278, 279 and 280, and light chain CDR1-3 of SEQ ID NOS: 470, 471 and 472; heavy chain CDR1-3 of SEQ ID NOS: 281, 282 and 283, and light chain CDR1-3 of SEQ ID NOS: 473, 474 and 475; heavy chain CDR1-3 of SEQ ID NOS: 284, 285 and 286, and light chain CDR1-3 of SEQ ID NOS: 476, 477 and 478; heavy chain CDR1-3 of SEQ ID NOS: 287, 288 and 289, and light chain CDR1-3 of SEQ ID NOS: 479, 480 and 481; heavy chain CDR1-3 of SEQ ID NOS: 290, 291 and 292, and light chain CDR1-3 of SEQ ID NOS: 482, 483 and 484; heavy chain CDR1-3 of SEQ ID NOS: 293, 294 and 295, and light chain CDR1-3 of SEQ ID NOS: 485, 486 and 487; heavy chain CDR1-3 of SEQ ID NOS: 296, 297 and 298, and light chain CDR1-3 of SEQ ID NOS: 488, 489 and 490; heavy chain CDR1-3 of SEQ ID NOS: 299, 300 and 301, and light chain CDR1-3 of SEQ ID NOS: 491, 492 and 493; heavy chain CDR1-3 of SEQ ID NOS: 302, 303 and 304, and light chain CDR1-3 of SEQ ID NOS: 494, 495 and 496; heavy chain CDR1-3 of SEQ ID NOS: 305, 306 and 307, and light chain CDR1-3 of SEQ ID NOS: 497, 498 and 499; heavy chain CDR1-3 of SEQ ID NOS: 308, 309 and 310, and light chain CDR1-3 of SEQ ID NOS: 500, 501 and 502; heavy chain CDR1-3 of SEQ ID NOS: 311, 312 and 313, and light chain CDR1-3 of SEQ ID NOS: 503, 504 and 505; heavy chain CDR1-3 of SEQ ID NOS: 314, 315 and 316, and light chain CDR1-3 of SEQ ID NOS: 506, 507 and 508; heavy chain CDR1-3 of SEQ ID NOS: 317, 318 and 319, and light chain CDR1-3 of SEQ ID NOS: 509, 510 and 511; heavy chain CDR1-3 of SEQ ID NOS: 320, 321 and 322, and light chain CDR1-3 of SEQ ID NOS: 512, 513 and 514; heavy chain CDR1-3 of SEQ ID NOS: 323, 324 and 325, and light chain CDR1-3 of SEQ ID NOS: 515, 516 and 517; heavy chain CDR1-3 of SEQ ID NOS: 326, 327 and 328, and light chain CDR1-3 of SEQ ID NOS: 518, 519 and 520; heavy chain CDR1-3 of SEQ ID NOS: 329, 330 and 331, and light chain CDR1-3 of SEQ ID NOS: 521, 522 and 523; heavy chain CDR1-3 of SEQ ID NOS: 332, 333 and 334, and light chain CDR1-3 of SEQ ID NOS: 524, 525 and 526; heavy chain CDR1-3 of SEQ ID NOS: 335, 336 and 337, and light chain CDR1-3 of SEQ ID NOS: 527, 528 and 529; heavy chain CDR1-3 of SEQ ID NOS: 338, 339 and 340, and light chain CDR1-3 of SEQ ID NOS: 530, 531 and 532; heavy chain CDR1-3 of SEQ ID NOS: 341, 342 and 343, and light chain CDR1-3 of SEQ ID NOS: 533, 534 and 535; heavy chain CDR1-3 of SEQ ID NOS: 344, 345 and 346, and light chain CDR1-3 of SEQ ID NOS: 536, 537 and 538; heavy chain CDR1-3 of SEQ ID NOS: 347, 348 and 349, and light chain

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CDR1-3 of SEQ ID NOS: 539, 540 and 541; heavy chain CDR1-3 of SEQ ID NOS: 350, 351 and 352, and light chain CDR1-3 of SEQ ID NOS: 542, 543 and 544; heavy chain CDR1-3 of SEQ ID NOS: 353, 354 and 355, and light chain CDR1-3 of SEQ ID NOS: 545, 546 and 547; heavy chain CDR1-3 of SEQ ID NOS: 356, 357 and 358, and light chain CDR1-3 of SEQ ID NOS: 548, 549 and 550; heavy chain CDR1-3 of SEQ ID NOS: 359, 360 and 361, and light chain CDR1-3 of SEQ ID NOS: 551, 552 and 553; heavy chain CDR1-3 of SEQ ID NOS: 362, 363 and 364, and light chain CDR1-3 of SEQ ID NOS: 554, 555 and 556; heavy chain CDR1-3 of SEQ ID NOS: 365, 366 and 367, and light chain CDR1-3 of SEQ ID NOS: 557, 558 and 559; heavy chain CDR1-3 of SEQ ID NOS: 368, 369 and 370, and light chain CDR1-3 of SEQ ID NOS: 560, 561 and 562; heavy chain CDR1-3 of SEQ ID NOS: 371, 372 and 373, and light chain CDR1-3 of SEQ ID NOS: 563, 564 and 565; heavy chain CDR1-3 of SEQ ID NOS: 374, 375 and 376, and light chain CDR1-3 of SEQ ID NOS: 566, 567 and 568; heavy chain CDR1-3 of SEQ ID NOS: 377, 378 and 379, and light chain CDR1-3 of SEQ ID NOS: 569, 570 and 571; heavy chain CDR1-3 of SEQ ID NOS: 380, 381 and 382, and light chain CDR1-3 of SEQ ID NOS: 572, 573 and 574; heavy chain CDR1-3 of SEQ ID NOS: 383, 384 and 385, and light chain CDR1-3 of SEQ ID NOS: 575, 576 and 577; heavy chain CDR1-3 of SEQ ID NOS: 386, 387 and 388, and light chain CDR1-3 of SEQ ID NOS: 578, 579 and 580; heavy chain CDR1-3 of SEQ ID NOS: 389, 390 and 391, and light chain CDR1-3 of SEQ ID NOS: 581, 582 and 583; heavy chain CDR1-3 of SEQ ID NOS: 392, 393 and 394, and light chain CDR1-3 of SEQ ID NOS: 584, 585 and 586; heavy chain CDR1-3 of SEQ ID NOS: 395, 396 and 397, and light chain CDR1-3 of SEQ ID NOS: 587, 588 and 589; heavy chain CDR1-3 of SEQ ID NOS: 398, 399 and 400, and light chain CDR1-3 of SEQ ID NOS: 590, 591 and 592; heavy chain CDR1-3 of SEQ ID NOS: 401, 402 and 403, and light chain CDR1-3 of SEQ ID NOS: 593, 594 and 595; heavy chain CDR1-3 of SEQ ID NOS: 404, 405 and 406, and light chain CDR1-3 of SEQ ID NOS: 596, 597 and 598; heavy chain CDR1-3 of SEQ ID NOS: 407, 408 and 409, and light chain CDR1-3 of SEQ ID NOS: 599, 600 and 601; heavy chain CDR1-3 of SEQ ID NOS: 410, 411 and 412, and light chain CDR1-3 of SEQ ID NOS: 602, 603 and 604; heavy chain CDR1-3 of SEQ ID NOS: 413, 414 and 415, and light chain CDR1-3 of SEQ ID NOS: 605, 606 and 607; heavy chain CDR1-3 of SEQ ID NOS: 416, 417 and 418, and light chain CDR1-3 of SEQ ID NOS: 608, 609 and 610; heavy chain CDR1-3 of SEQ ID NOS: 419, 420 and 421, and light chain CDR1-3 of SEQ ID NOS: 611, 612 and 613; heavy chain CDR1-3 of SEQ ID NOS: 422, 423 and 424, and light chain CDR1-3 of SEQ ID NOS: 614, 615 and 616; heavy chain CDR1-3 of SEQ ID NOS: 425, 426 and 427, and light chain CDR1-3 of SEQ ID NOS: 617, 618 and 619; heavy chain CDR1-3 of SEQ ID NOS: 428, 429 and 430, and light chain CDR1-3 of SEQ ID NOS: 620, 621 and 622; heavy chain CDR1-3 of SEQ ID NOS: 431, 432 and 433, and light chain CDR1-3 of SEQ ID NOS: 623, 624 and 625; heavy chain CDR1-3 of SEQ ID NOS: 434, 435 and 436, and light chain CDR1-3 of SEQ ID NOS: 626, 627 and 628; heavy chain CDR1-3 of SEQ ID NOS: 437, 438 and 439, and light chain CDR1-3 of SEQ ID NOS: 629, 630 and 631; heavy chain CDR1-3 of SEQ ID NOS: 440, 441 and 442, and light chain CDR1-3 of SEQ ID NOS: 632, 633 and 634; heavy chain CDR1-3 of SEQ ID NOS: 443, 444 and 445, and light chain CDR1-3 of SEQ ID NOS: 635, 636 and 637; or heavy chain CDR1-3 of SEQ ID NOS: 446, 447 and 448, and light chain CDR1-3 of SEQ ID NOS: 638, 639 and 640, wherein said

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antibody is a chimeric antibody or a bispecific antibody, and said antibody fragment is an scFv (single chain fragment variable).

16. The monoclonal antibody or fragment thereof of claim **15**, wherein said antibody or antibody fragment is encoded by heavy and light chain variable sequences comprising SEQ ID NOS: 1 and 2; SEQ ID NOS: 3 and 4; SEQ ID NOS: 5 and 6; SEQ ID NOS: 7 and 8; SEQ ID NOS: 9 and 10; SEQ ID NOS: 11 and 12; SEQ ID NOS: 13 and 14; SEQ ID NOS: 15 and 16; SEQ ID NOS: 17 and 18; SEQ ID NOS: 19 and 20; SEQ ID NOS: 21 and 22; SEQ ID NOS: 23 and 24; SEQ ID NOS: 25 and 26; SEQ ID NOS: 27 and 28; SEQ ID NOS: 29 and 30; SEQ ID NOS: 31 and 32; SEQ ID NOS: 33 and 34; SEQ ID NOS: 35 and 36; SEQ ID NOS: 37 and 38; SEQ ID NOS: 39 and 40; SEQ ID NOS: 41 and 42; SEQ ID NOS: 43 and 44; SEQ ID NOS: 45 and 46; SEQ ID NOS: 47 and 48; SEQ ID NOS: 49 and 50; SEQ ID NOS: 51 and 52; SEQ ID NOS: 53 and 54; SEQ ID NOS: 55 and 56; SEQ ID NOS: 57 and 58; SEQ ID NOS: 59 and 60; SEQ ID NOS: 61 and 62; SEQ ID NOS: 63 and 64; SEQ ID NOS: 65 and 66; SEQ ID NOS: 67 and 68; SEQ ID NOS: 69 and 70; SEQ ID NOS: 71 and 72; SEQ ID NOS: 73 and 74; SEQ ID NOS: 75 and 76; SEQ ID NOS: 77 and 78; SEQ ID NOS: 79 and 80; SEQ ID NOS: 81 and 82; SEQ ID NOS: 83 and 84; SEQ ID NOS: 85 and 86; SEQ ID NOS: 87 and 88; SEQ ID NOS: 89 and 90; SEQ ID NOS: 91 and 92; SEQ ID NOS: 93 and 94; SEQ ID NOS: 95 and 96; SEQ ID NOS: 97 and 98; SEQ ID NOS: 99 and 100; SEQ ID NOS: 101 and 102; SEQ ID NOS: 103 and 104; SEQ ID NOS: 105 and 106; SEQ ID NOS: 107 and 108; SEQ ID NOS: 109 and 110; SEQ ID NOS: 111 and 112; SEQ ID NOS: 113 and 114; SEQ ID NOS: 115 and 116; SEQ ID NOS: 117 and 118; SEQ ID NOS: 119 and 120; SEQ ID NOS: 121 and 122; SEQ ID NOS: 123 and 124; SEQ ID NOS: 125 and 126; or SEQ ID NOS: 127 and 128.

17. The monoclonal antibody or fragment thereof of claim **15**, wherein said antibody or antibody fragment is encoded by heavy and light chain variable sequences having at least 95% identity to SEQ ID NOS: 1 and 2; SEQ ID NOS: 3 and 4; SEQ ID NOS: 5 and 6; SEQ ID NOS: 7 and 8; SEQ ID NOS: 9 and 10; SEQ ID NOS: 11 and 12; SEQ ID NOS: 13 and 14; SEQ ID NOS: 15 and 16; SEQ ID NOS: 17 and 18; SEQ ID NOS: 19 and 20; SEQ ID NOS: 21 and 22; SEQ ID NOS: 23 and 24; SEQ ID NOS: 25 and 26; SEQ ID NOS: 27 and 28; SEQ ID NOS: 29 and 30; SEQ ID NOS: 31 and 32; SEQ ID NOS: 33 and 34; SEQ ID NOS: 35 and 36; SEQ ID NOS: 37 and 38; SEQ ID NOS: 39 and 40; SEQ ID NOS: 41 and 42; SEQ ID NOS: 43 and 44; SEQ ID NOS: 45 and 46; SEQ ID NOS: 47 and 48; SEQ ID NOS: 49 and 50; SEQ ID NOS: 51 and 52; SEQ ID NOS: 53 and 54; SEQ ID NOS: 55 and 56; SEQ ID NOS: 57 and 58; SEQ ID NOS: 59 and 60; SEQ ID NOS: 61 and 62; SEQ ID NOS: 63 and 64; SEQ ID NOS: 65 and 66; SEQ ID NOS: 67 and 68; SEQ ID NOS: 69 and 70; SEQ ID NOS: 71 and 72; SEQ ID NOS: 73 and 74; SEQ ID NOS: 75 and 76; SEQ ID NOS: 77 and 78; SEQ ID NOS: 79 and 80; SEQ ID NOS: 81 and 82; SEQ ID NOS: 83 and 84; SEQ ID NOS: 85 and 86; SEQ ID NOS: 87 and 88; SEQ ID NOS: 89 and 90; SEQ ID NOS: 91 and 92; SEQ ID NOS: 93 and 94; SEQ ID NOS: 95 and 96; SEQ ID NOS: 97 and 98; SEQ ID NOS: 99 and 100; SEQ ID NOS: 101 and 102; SEQ ID NOS: 103 and 104; SEQ ID NOS: 105 and 106; SEQ ID NOS: 107 and 108; SEQ ID NOS: 109 and 110; SEQ ID NOS: 111 and 112; SEQ ID NOS: 113 and 114; SEQ ID NOS: 115 and 116; SEQ ID NOS: 117 and 118; SEQ ID NOS: 119 and 120; SEQ ID NOS: 121 and 122; SEQ ID NOS: 123 and 124; SEQ ID NOS: 125 and 126; or SEQ ID NOS: 127 and 128.

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18. The monoclonal antibody or fragment thereof of claim 15, wherein said antibody or antibody fragment comprises heavy and light chain variable sequences comprising SEQ ID NOS: 129 and 130; SEQ ID NOS: 131 and 132; SEQ ID NOS: 133 and 134; SEQ ID NOS: 135 and 136; SEQ ID NOS: 137 and 138; SEQ ID NOS: 139 and 140; SEQ ID NOS: 141 and 142; SEQ ID NOS: 143 and 144; SEQ ID NOS: 145 and 146; SEQ ID NOS: 147 and 148; SEQ ID NOS: 149 and 150; SEQ ID NOS: 151 and 152; SEQ ID NOS: 153 and 154; SEQ ID NOS: 155 and 156; SEQ ID NOS: 157 and 158; SEQ ID NOS: 159 and 160; SEQ ID NOS: 161 and 162; SEQ ID NOS: 163 and 164; SEQ ID NOS: 165 and 166; SEQ ID NOS: 167 and 168; SEQ ID NOS: 169 and 170; SEQ ID NOS: 171 and 172; SEQ ID NOS: 173 and 174; SEQ ID NOS: 175 and 176; SEQ ID NOS: 177 and 178; SEQ ID NOS: 179 and 180; SEQ ID NOS: 181 and 182; SEQ ID NOS: 183 and 184; SEQ ID NOS: 185 and 186; SEQ ID NOS: 187 and 188; SEQ ID NOS: 189 and 190; SEQ ID NOS: 191 and 192; SEQ ID NOS: 193 and 194; SEQ ID NOS: 195 and 196; SEQ ID NOS: 197 and 198; SEQ ID NOS: 199 and 200; SEQ ID NOS: 201 and 202; SEQ ID NOS: 203 and 204; SEQ ID NOS: 205 and 206; SEQ ID NOS: 207 and 208; SEQ ID NOS: 209 and 210; SEQ ID NOS: 211 and 212; SEQ ID NOS: 213 and 214; SEQ ID NOS: 215 and 216; SEQ ID NOS: 217 and 218; SEQ ID NOS: 219 and 220; SEQ ID NOS: 221 and 222; SEQ ID NOS: 223 and 224; SEQ ID NOS: 225 and 226; SEQ ID NOS: 227 and 228; SEQ ID NOS: 229 and 230; SEQ ID NOS: 231 and 232; SEQ ID NOS: 233 and 234; SEQ ID NOS: 235 and 236; SEQ ID NOS: 237 and 238; SEQ ID NOS: 239 and 240; SEQ ID NOS: 241 and 242; SEQ ID NOS: 243 and 244; SEQ ID NOS: 245 and 246; SEQ ID NOS: 247 and 248; SEQ ID NOS: 249 and 250; SEQ ID NOS: 251 and 252; SEQ ID NOS: 253 and 254; or SEQ ID NOS: 255 and 256.

19. The monoclonal antibody or fragment thereof of claim 15, wherein said antibody or antibody fragment comprises heavy and light chain variable sequences having 95% identity to SEQ ID NOS: 129 and 130; SEQ ID NOS: 131 and 132; SEQ ID NOS: 133 and 134; SEQ ID NOS: 135 and 136; SEQ ID NOS: 137 and 138; SEQ ID NOS: 139 and 140; SEQ ID NOS: 141 and 142; SEQ ID NOS: 143 and 144; SEQ ID NOS: 145 and 146; SEQ ID NOS: 147 and 148; SEQ ID NOS: 149 and 150; SEQ ID NOS: 151 and 152; SEQ ID NOS: 153 and 154; SEQ ID NOS: 155 and 156; SEQ ID NOS: 157 and 158; SEQ ID NOS: 159 and 160; SEQ ID NOS: 161 and 162; SEQ ID NOS: 163 and 164; SEQ ID NOS: 165 and 166; SEQ ID NOS: 167 and 168; SEQ ID NOS: 169 and 170; SEQ ID NOS: 171 and 172; SEQ ID NOS: 173 and 174; SEQ ID NOS: 175 and 176; SEQ ID NOS: 177 and 178; SEQ ID NOS: 179 and 180; SEQ ID NOS: 181 and 182; SEQ ID NOS: 183 and 184; SEQ ID NOS: 185 and 186; SEQ ID NOS: 187 and 188; SEQ ID NOS: 189 and 190; SEQ ID NOS: 191 and 192; SEQ ID NOS: 193 and 194; SEQ ID NOS: 195 and 196; SEQ ID NOS: 197 and 198; SEQ ID NOS: 199 and 200; SEQ ID NOS: 201 and 202; SEQ ID NOS: 203 and 204; SEQ ID NOS: 205 and 206; SEQ ID NOS: 207 and 208; SEQ ID NOS: 209 and 210; SEQ ID NOS: 211 and 212; SEQ ID NOS: 213 and 214; SEQ ID NOS: 215 and 216; SEQ ID NOS: 217 and 218; SEQ ID NOS: 219 and 220; SEQ ID NOS: 221 and 222; SEQ ID NOS: 223 and 224; SEQ ID NOS: 225 and 226; SEQ ID NOS: 227 and 228; SEQ ID NOS: 229 and 230; SEQ ID NOS: 231 and 232; SEQ ID NOS: 233 and 234; SEQ ID NOS: 235 and 236; SEQ ID NOS: 237 and 238; SEQ ID NOS: 239 and 240; SEQ ID NOS: 241 and 242; SEQ ID NOS: 243 and

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244; SEQ ID NOS: 245 and 246; SEQ ID NOS: 247 and 248; SEQ ID NOS: 249 and 250; SEQ ID NOS: 251 and 252; SEQ ID NOS: 253 and 254; or SEQ ID NOS: 255 and 256.

20. The monoclonal antibody or fragment thereof of claim 15, wherein the antibody or antibody fragment comprises heavy chain CDR1-3 of SEQ ID NOS: 314-316 and light chain CDR1-3 of SEQ ID NOS: 506-508 or heavy chain CDR1-3 of SEQ ID NOS: 413-415 and light chain CDR1-3 of SEQ ID NOS: 605-607.

21. The monoclonal antibody or fragment thereof of claim 15, wherein the antibody or antibody fragment comprises a heavy chain variable region of SEQ ID NO: 165 and a light chain variable region of SEQ ID NO: 166 or a heavy chain variable region of SEQ ID NO: 233 and a light chain variable region of SEQ ID NO: 234.

22. The monoclonal antibody or fragment thereof of claim 15, wherein the antibody or antibody fragment comprises heavy and light chain variable sequences having 95% identity to a heavy chain variable region of SEQ ID NO: 165 and a light chain variable region of SEQ ID NO: 166 or a heavy chain variable region of SEQ ID NO: 233 and a light chain variable region of SEQ ID NO: 234.

23. A hybridoma or engineered cell encoding an antibody or antibody fragment, wherein the antibody or antibody fragment comprises heavy chain CDR1-3 of SEQ ID NOS: 257, 258 and 259, and light chain CDR1-3 of SEQ ID NOS: 449, 450 and 451; heavy chain CDR1-3 of SEQ ID NOS: 260, 261 and 262, and light chain CDR1-3 of SEQ ID NOS: 452, 453 and 454; heavy chain CDR1-3 of SEQ ID NOS: 263, 264 and 265, and light chain CDR1-3 of SEQ ID NOS: 455, 456 and 457; heavy chain CDR1-3 of SEQ ID NOS: 266, 267 and 268, and light chain CDR1-3 of SEQ ID NOS: 458, 459 and 460; heavy chain CDR1-3 of SEQ ID NOS: 269, 270 and 271, and light chain CDR1-3 of SEQ ID NOS: 461, 462 and 463; heavy chain CDR1-3 of SEQ ID NOS: 272, 273 and 274, and light chain CDR1-3 of SEQ ID NOS: 464, 465 and 466; heavy chain CDR1-3 of SEQ ID NOS: 275, 276 and 277, and light chain CDR1-3 of SEQ ID NOS: 467, 468 and 469; heavy chain CDR1-3 of SEQ ID NOS: 278, 279 and 280, and light chain CDR1-3 of SEQ ID NOS: 470, 471 and 472; heavy chain CDR1-3 of SEQ ID NOS: 281, 282 and 283, and light chain CDR1-3 of SEQ ID NOS: 473, 474 and 475; heavy chain CDR1-3 of SEQ ID NOS: 284, 285 and 286, and light chain CDR1-3 of SEQ ID NOS: 476, 477 and 478; heavy chain CDR1-3 of SEQ ID NOS: 287, 288 and 289, and light chain CDR1-3 of SEQ ID NOS: 479, 480 and 481; heavy chain CDR1-3 of SEQ ID NOS: 290, 291 and 292, and light chain CDR1-3 of SEQ ID NOS: 482, 483 and 484; heavy chain CDR1-3 of SEQ ID NOS: 293, 294 and 295, and light chain CDR1-3 of SEQ ID NOS: 485, 486 and 487; heavy chain CDR1-3 of SEQ ID NOS: 296, 297 and 298, and light chain CDR1-3 of SEQ ID NOS: 488, 489 and 490; heavy chain CDR1-3 of SEQ ID NOS: 299, 300 and 301, and light chain CDR1-3 of SEQ ID NOS: 491, 492 and 493; heavy chain CDR1-3 of SEQ ID NOS: 302, 303 and 304, and light chain CDR1-3 of SEQ ID NOS: 494, 495 and 496; heavy chain CDR1-3 of SEQ ID NOS: 305, 306 and 307, and light chain CDR1-3 of SEQ ID NOS: 497, 498 and 499; heavy chain CDR1-3 of SEQ ID NOS: 308, 309 and 310, and light chain CDR1-3 of SEQ ID NOS: 500, 501 and 502; heavy chain CDR1-3 of SEQ ID NOS: 311, 312 and 313, and light chain CDR1-3 of SEQ ID NOS: 503, 504 and 505; heavy chain CDR1-3 of SEQ ID NOS: 314, 315 and 316, and light chain CDR1-3 of SEQ ID NOS: 506, 507 and 508; heavy chain CDR1-3 of SEQ ID NOS: 317, 318 and 319, and light chain CDR1-3 of SEQ ID NOS:

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509, 510 and 511; heavy chain CDR1-3 of SEQ ID NOS: 320, 321 and 322, and light chain CDR1-3 of SEQ ID NOS: 512, 513 and 514; heavy chain CDR1-3 of SEQ ID NOS: 323, 324 and 325, and light chain CDR1-3 of SEQ ID NOS: 515, 516 and 517; heavy chain CDR1-3 of SEQ ID NOS: 326, 327 and 328, and light chain CDR1-3 of SEQ ID NOS: 518, 519 and 520; heavy chain CDR1-3 of SEQ ID NOS: 329, 330 and 331, and light chain CDR1-3 of SEQ ID NOS: 521, 522 and 523; heavy chain CDR1-3 of SEQ ID NOS: 332, 333 and 334, and light chain CDR1-3 of SEQ ID NOS: 524, 525 and 526; heavy chain CDR1-3 of SEQ ID NOS: 335, 336 and 337, and light chain CDR1-3 of SEQ ID NOS: 527, 528 and 529; heavy chain CDR1-3 of SEQ ID NOS: 338, 339 and 340, and light chain CDR1-3 of SEQ ID NOS: 530, 531 and 532; heavy chain CDR1-3 of SEQ ID NOS: 341, 342 and 343, and light chain CDR1-3 of SEQ ID NOS: 533, 534 and 535; heavy chain CDR1-3 of SEQ ID NOS: 344, 345 and 346, and light chain CDR1-3 of SEQ ID NOS: 536, 537 and 538; heavy chain CDR1-3 of SEQ ID NOS: 347, 348 and 349, and light chain CDR1-3 of SEQ ID NOS: 539, 540 and 541; heavy chain CDR1-3 of SEQ ID NOS: 350, 351 and 352, and light chain CDR1-3 of SEQ ID NOS: 542, 543 and 544; heavy chain CDR1-3 of SEQ ID NOS: 353, 354 and 355, and light chain CDR1-3 of SEQ ID NOS: 545, 546 and 547; heavy chain CDR1-3 of SEQ ID NOS: 356, 357 and 358, and light chain CDR1-3 of SEQ ID NOS: 548, 549 and 550; heavy chain CDR1-3 of SEQ ID NOS: 359, 360 and 361, and light chain CDR1-3 of SEQ ID NOS: 551, 552 and 553; heavy chain CDR1-3 of SEQ ID NOS: 362, 363 and 364, and light chain CDR1-3 of SEQ ID NOS: 554, 555 and 556; heavy chain CDR1-3 of SEQ ID NOS: 365, 366 and 367, and light chain CDR1-3 of SEQ ID NOS: 557, 558 and 559; heavy chain CDR1-3 of SEQ ID NOS: 368, 369 and 370, and light chain CDR1-3 of SEQ ID NOS: 560, 561 and 562; heavy chain CDR1-3 of SEQ ID NOS: 371, 372 and 373, and light chain CDR1-3 of SEQ ID NOS: 563, 564 and 565; heavy chain CDR1-3 of SEQ ID NOS: 374, 375 and 376, and light chain CDR1-3 of SEQ ID NOS: 566, 567 and 568; heavy chain CDR1-3 of SEQ ID NOS: 377, 378 and 379, and light chain CDR1-3 of SEQ ID NOS: 569, 570 and 571; heavy chain CDR1-3 of SEQ ID NOS: 380, 381 and 382, and light chain CDR1-3 of SEQ ID NOS: 572, 573 and 574; heavy chain CDR1-3 of SEQ ID NOS: 383, 384 and 385, and light chain CDR1-3 of SEQ ID NOS: 575, 576 and 577; heavy chain CDR1-3 of SEQ ID NOS: 386, 387 and 388, and light chain CDR1-3 of SEQ ID NOS: 578, 579 and 580; heavy chain CDR1-3 of SEQ ID NOS: 389, 390 and 391, and light chain CDR1-3 of SEQ ID NOS: 581, 582 and 583; heavy chain CDR1-3 of SEQ ID NOS: 392, 393 and 394, and light chain CDR1-3 of SEQ ID NOS: 584, 585 and 586; heavy chain CDR1-3 of SEQ ID NOS: 395, 396 and 397, and light chain CDR1-3 of SEQ ID NOS: 587, 588 and 589; heavy chain CDR1-3 of SEQ ID NOS: 398, 399 and 400, and light chain CDR1-3 of SEQ ID NOS:

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590, 591 and 592; heavy chain CDR1-3 of SEQ ID NOS: 401, 402 and 403, and light chain CDR1-3 of SEQ ID NOS: 593, 594 and 595; heavy chain CDR1-3 of SEQ ID NOS: 404, 405 and 406, and light chain CDR1-3 of SEQ ID NOS: 596, 597 and 598; heavy chain CDR1-3 of SEQ ID NOS: 407, 408 and 409, and light chain CDR1-3 of SEQ ID NOS: 599, 600 and 601; heavy chain CDR1-3 of SEQ ID NOS: 410, 411 and 412, and light chain CDR1-3 of SEQ ID NOS: 602, 603 and 604; heavy chain CDR1-3 of SEQ ID NOS: 413, 414 and 415, and light chain CDR1-3 of SEQ ID NOS: 605, 606 and 607; heavy chain CDR1-3 of SEQ ID NOS: 416, 417 and 418, and light chain CDR1-3 of SEQ ID NOS: 608, 609 and 610; heavy chain CDR1-3 of SEQ ID NOS: 419, 420 and 421, and light chain CDR1-3 of SEQ ID NOS: 611, 612 and 613; heavy chain CDR1-3 of SEQ ID NOS: 422, 423 and 424, and light chain CDR1-3 of SEQ ID NOS: 614, 615 and 616; heavy chain CDR1-3 of SEQ ID NOS: 425, 426 and 427, and light chain CDR1-3 of SEQ ID NOS: 617, 618 and 619; heavy chain CDR1-3 of SEQ ID NOS: 428, 429 and 430, and light chain CDR1-3 of SEQ ID NOS: 620, 621 and 622; heavy chain CDR1-3 of SEQ ID NOS: 431, 432 and 433, and light chain CDR1-3 of SEQ ID NOS: 623, 624 and 625; heavy chain CDR1-3 of SEQ ID NOS: 434, 435 and 436, and light chain CDR1-3 of SEQ ID NOS: 626, 627 and 628; heavy chain CDR1-3 of SEQ ID NOS: 437, 438 and 439, and light chain CDR1-3 of SEQ ID NOS: 629, 630 and 631; heavy chain CDR1-3 of SEQ ID NOS: 440, 441 and 442, and light chain CDR1-3 of SEQ ID NOS: 632, 633 and 634; heavy chain CDR1-3 of SEQ ID NOS: 443, 444 and 445, and light chain CDR1-3 of SEQ ID NOS: 635, 636 and 637; or heavy chain CDR1-3 of SEQ ID NOS: 446, 447 and 448, and light chain CDR1-3 of SEQ ID NOS: 638, 639 and 640.

24. The hybridoma or engineered cell of claim **23**, wherein the antibody or antibody fragment comprises heavy chain CDR1-3 of SEQ ID NOS: 314-316 and light chain CDR1-3 of SEQ ID NOS: 506-508 or heavy chain CDR1-3 of SEQ ID NOS: 413-415 and light chain CDR1-3 of SEQ ID NOS: 605-607.

25. The hybridoma or engineered cell of claim **23**, wherein the antibody or antibody fragment comprises a heavy chain variable region of SEQ ID NO: 165 and a light chain variable region of SEQ ID NO: 166 or a heavy chain variable region of SEQ ID NO: 233 and a light chain variable region of SEQ ID NO: 234.

26. The hybridoma or engineered cell of claim **23**, wherein the antibody or antibody fragment comprises heavy and light chain variable sequences having 95% identity to a heavy chain variable region of SEQ ID NO: 165 and a light chain variable region of SEQ ID NO: 166 or a heavy chain variable region of SEQ ID NO: 233 and a light chain variable region of SEQ ID NO: 234.

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